

2022 ESC Guidelines on cardio-oncology developed in collaboration with the European Hematology Association (EHA), the European Society for Therapeutic Radiology and Oncology (ESTRO) and the International Cardio-Oncology Society (IC-OS): Developed by the task force on cardio-oncology of the European Society of Cardiology (ESC) *Free*

Alexander R Lyon

(Chairperson) (United Kingdom)

Corresponding authors: Alexander R. Lyon, National Heart and Lung Institute, Imperial College London and Cardio-Oncology Service, Royal Brompton Hospital, London, United Kingdom. Tel: +44 207 352 8121, E-mail: a.lyon@imperial.ac.uk.

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Teresa LÃ³pez-FernÃ¡ndez

(Chairperson) (Spain)

Teresa LÃ³pez-FernÃ¡ndez, Cardiology Department, La Paz University Hospital, IdiPAZ Research Institute, Madrid, Spain. Tel: +34 619 227 076, E-mail: tlfernandez8@gmail.com.

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Liam S Couch

(Task Force Coordinator) (United Kingdom)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Riccardo Asteggiano

(Italy)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Marianne C Aznar

(United Kingdom)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Jutta Bergler-Klein

(Austria)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Giuseppe Boriani

(Italy)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Daniela Cardinale

(Italy)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Raul Cordoba

(Spain)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

Bernard Cosyns

(Belgium)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[... Show more](#)

[David J Cutter](#)

David J Cutter

(United Kingdom)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Evandro de Azambuja](#)

Evandro de Azambuja

(Belgium)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Rudolf A de Boer](#)

Rudolf A de Boer

(Netherlands)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Susan F Dent](#)

Susan F Dent

(United States of America)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Dimitrios Farmakis](#)

Dimitrios Farmakis

(Cyprus)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Sofie A Gevaert](#)

Sofie A Gevaert

(Belgium)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Diana A Gorog](#)

Diana A Gorog

(United Kingdom)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Joerg Herrmann](#)

Joerg Herrmann

(United States of America)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Daniel Lenihan](#)

Daniel Lenihan

(United States of America)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Javid Moslehi](#)

Javid Moslehi

(United States of America)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Brenda Moura](#)

Brenda Moura

(Portugal)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Sonja S Salinger](#)

Sonja S Salinger

(Serbia)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Richard Stephens](#)

Richard Stephens

(United Kingdom)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Thomas M Suter](#)

Thomas M Suter

(Switzerland)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Sebastian Szmit](#)

Sebastian Szmit

(Poland)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Juan Tamargo](#)

Juan Tamargo

(Spain)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Paaladinesh Thavendiranathan](#)

Paaladinesh Thavendiranathan

(Canada)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Carlo G Tocchetti](#)

Carlo G Tocchetti

(Italy)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Peter van der Meer](#)

Peter van der Meer

(Netherlands)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[Helena J H van der Pal](#)

Helena J H van der Pal

(Netherlands)

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)

[ESC Scientific Document Group](#)

ESC Scientific Document Group

Search for other works by this author on:

[Oxford Academic](#) [PubMed](#) [Google Scholar](#)



The two chairpersons contributed equally to the document and are joint corresponding authors.

Author/Task Force Member affiliations are listed in Author information.

¹Representing the European Society for Therapeutic Radiology and Oncology (ESTRO); ²representing the European Hematology Association (EHA); ³representing the International Cardio-Oncology Society (IC-OS).

ESC Clinical Practice Guidelines (CPG) Committee: listed in the Appendix.

ESC subspecialty communities having participated in the development of this document:

Associations: Association for Acute CardioVascular Care (ACVC), European Association of Cardiovascular Imaging (EACVI), European Association of Preventive Cardiology (EAPC), European Association of Percutaneous Cardiovascular Interventions (EAPCI), European Heart Rhythm Association (EHRA), Heart Failure Association (HFA).

Councils: Council of Cardio-Oncology, Council on Hypertension, Council on Valvular Heart Disease.

Working Groups: Aorta and Peripheral Vascular Diseases, Cardiovascular Pharmacotherapy, e-Cardiology, Myocardial Function, Pulmonary Circulation and Right Ventricular Function, Thrombosis.

Patient Forum

The content of these European Society of Cardiology (ESC) Guidelines has been published for personal and educational use only. No commercial use is authorized. No part of the ESC Guidelines may be translated or reproduced in any form without written permission from the ESC. Permission can be obtained upon submission of a written request to Oxford University Press, the publisher of the *European Heart Journal* and the party authorized to handle such permissions on behalf of the ESC (journals.permissions@oup.com).

Disclaimer: The ESC Guidelines represent the views of the ESC and were produced after careful consideration of the scientific and medical knowledge and the evidence available at the time of their publication. The ESC is not responsible in the event of any contradiction, discrepancy and/or ambiguity between the ESC Guidelines and any other official recommendations or guidelines issued by the relevant public health authorities, in particular in relation to good use of healthcare or therapeutic strategies. Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgement, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies; however, the ESC Guidelines do not override, in any way whatsoever, the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient and, where appropriate and/or necessary, the patient's caregiver. Nor do the ESC Guidelines exempt health professionals from taking into full and careful consideration the relevant official updated recommendations or guidelines issued by the competent public health authorities, in order to manage each patient's case in light of the scientifically accepted data pursuant to their respective ethical and professional obligations. It is also the health professional's responsibility to verify the applicable rules and regulations relating to drugs and medical devices at the time of prescription.

This article has been co-published with permission in the *European Heart Journal* and the *European Heart Journal* *Cardiovascular Imaging*. © The European Society of Cardiology 2022. All rights reserved. The articles are identical except for minor stylistic and spelling differences in keeping with each journal's

style. Either citation can be used when citing this article. For permissions please e-mail:

journals.permissions@oup.com.

All experts involved in the development of these Guidelines have submitted declarations of interest. These have been compiled in a report and published in a supplementary document simultaneously to the Guidelines. The report is also available on the ESC website www.escardio.org/Guidelines

[Author Notes](#)

European Heart Journal, Volume 43, Issue 41, 1 November 2022, Pages 4229–4361,

<https://doi.org/10.1093/eurheartj/ehac244>

Published:

26 August 2022

[Guidelines](#), [Androgen deprivation therapy](#), [Anthracycline](#), [Atrial fibrillation](#), [Arrhythmias](#), [Biomarkers](#), [Cancer](#), [Cancer survivors](#), [Carcinoid syndrome](#), [Amyloid light-chain cardiac amyloidosis](#), [Cardiac magnetic resonance](#), [Cardiac tumour](#), [Cardio-oncology](#), [Cardiotoxicity](#), [Coronary artery disease](#), [Chemotherapy](#), [Echocardiography](#), [Fluoropyrimidine](#), [Heart failure](#), [Haematopoietic stem cell transplantation](#), [Hormone therapy](#), [Hypertension](#), [Immunotherapy](#), [Ischaemic heart disease](#), [Myocarditis](#), [Pericardial disease](#), [Pulmonary hypertension](#), [Thrombosis](#), [Risk stratification](#), [Trastuzumab](#), [Valvular heart disease](#), [Vascular endothelial growth factor inhibitors \(VEGFi\)](#), [Venous thromboembolism](#), [Pericardial disease](#), [Proteasome inhibitors](#), [QTc prolongation](#), [Radiotherapy](#), [Strain](#)

All experts involved in the development of these Guidelines have submitted declarations of interest. These have been compiled in a report and published in a supplementary document simultaneously to the Guidelines. The report is also available on the ESC website www.escardio.org/Guidelines



See the *European Heart Journal* online for supplementary data that includes background information and detailed discussion of the data that have provided the basis of the guidelines.

Table of contents

- 1. Preamble 4236
- 2. Introduction 4238
 - 2.1. Cancer and cardiovascular needs of patients with cancer 4238
 - 2.2. Role of cardio-oncology services 4238
 - 2.3. General principles of cardio-oncology 4239
- 3. Cancer therapy-related cardiovascular toxicity definitions 4242
- 4. Cardiovascular toxicity risk stratification before anticancer therapy 4244
 - 4.1. General approach to cardiovascular toxicity risk in patients with cancer 4244
 - 4.2. History and clinical examination 4249

- 4.3. Electrocardiogram 4249
- 4.4. Cardiac serum biomarkers 4251
- 4.5. Cardiovascular imaging 4252
- 4.6. Cardiopulmonary fitness assessment 4253
- 4.7. Cardiovascular risk evaluation before cancer surgery 4254
- 4.8. Genetic testing 4254
- 5. Prevention and monitoring of cardiovascular complications during cancer therapy 4254
- 5.1. General principles 4254
- 5.2. Primary prevention strategies 4256
 - 5.2.1. Primary prevention of cancer therapy-related cardiovascular toxicity during anthracycline chemotherapy 4256
 - 5.2.2. Primary prevention of radiation-induced cardiovascular toxicity 4256
- 5.3. Secondary prevention strategies 4257
- 5.4. Cardiovascular surveillance during cancer therapies 4257
 - 5.4.1. Cardiac serum biomarkers 4257
 - 5.4.2. Cardiac imaging 4257
- 5.5. Cancer therapy-related cardiovascular toxicity monitoring protocols 4259
 - 5.5.1. Anthracycline chemotherapy 4259
 - 5.5.2. HER2-targeted therapies 4259
 - 5.5.3. Fluoropyrimidines 4261
 - 5.5.4. Vascular endothelial growth factor inhibitors 4261
 - 5.5.5. Multitargeted kinase inhibitors targeting BCR-ABL 4264
 - 5.5.6. Bruton tyrosine kinase inhibitors 4267
 - 5.5.7. Multiple myeloma therapies 4268
 - 5.5.8. Rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor treatment 4271
 - 5.5.9. Immune checkpoint inhibitors 4273

- 5.5.10. Androgen deprivation therapies for prostate cancer 4274
- 5.5.11. Endocrine therapies for breast cancer 4276
- 5.5.12. Cyclin-dependent kinase 4/6 inhibitors 4276
- 5.5.13. Anaplastic lymphoma kinase inhibitors 4277
- 5.5.14. Epidermal growth factor receptor inhibitors 4277
- 5.5.15. Chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies 4278
- 5.5.16. Radiotherapy 4279
- 5.5.17. Haematopoietic stem cell transplantation 4279
- 5.5.18. Other cancer treatments 4282
- 6. Diagnosis and management of acute and subacute cardiovascular toxicity in patients receiving anticancer treatment 4282
- 6.1. Cancer therapy-related cardiac dysfunction 4282
- 6.1.1. Anthracycline chemotherapy-related cardiac dysfunction 4282
- 6.1.2. Human epidermal receptor 2-targeted therapy- related cardiac dysfunction 4284
- 6.1.3. Immune checkpoint inhibitor-associated myocarditis and non-inflammatory heart failure 4286
- 6.1.4. Chimeric antigen receptor T cell and tumour- infiltrating lymphocytes therapies and heart dysfunction 4290
- 6.1.5. Heart failure during haematopoietic stem cell transplantation 4291
- 6.1.6. Takotsubo syndrome and cancer 4291
- 6.2. Coronary artery disease 4291
- 6.2.1. Acute coronary syndromes 4291
- 6.2.2. Chronic coronary syndromes 4293
- 6.3. Valvular heart disease 4293
- 6.4. Cardiac arrhythmias 4294
- 6.4.1. Atrial fibrillation 4294
- 6.4.2. Long corrected QT interval and ventricular arrhythmias 4298

- 6.4.3. Bradyarrhythmias 4300
- 6.5. Arterial hypertension 4300
- 6.6. Thrombosis and thromboembolic events 4303
 - 6.6.1. Venous thromboembolism 4303
 - 6.6.2. Arterial thromboembolism 4304
 - 6.6.3. Intracardiac thrombosis 4304
 - 6.6.4. Anticoagulation therapy 4305
- 6.7. Bleeding complications 4306
 - 6.7.1. High-risk patients 4306
 - 6.7.2. Antiplatelet therapy 4306
 - 6.7.3. Management of bleeding 4306
- 6.8. Peripheral artery disease 4307
- 6.9. Pulmonary hypertension 4307
- 6.10. Pericardial diseases 4308
 - 6.10.1. Pericarditis 4308
 - 6.10.2. Pericardial effusion 4308
- 7. End-of-cancer therapy cardiovascular risk assessment 4309
 - 7.1. Cardiovascular evaluation during the first year after cardiotoxic anticancer therapy 4309
 - 7.2. Which cancer survivors require cardiovascular surveillance in the first year after cancer treatment? 4309
 - 7.3. Management of cancer therapy-related cardiac dysfunction at the end-of-therapy assessment 4310
 - 7.4. Cardiopulmonary exercise testing and fitness during the end-of-therapy assessment 4310
 - 7.5. The role of cardiac rehabilitation 4310
- 8. Long-term follow-up and chronic cardiovascular complications in cancer survivors 4312
 - 8.1. Cancer survivors 4312
 - 8.1.1. Adult survivors of childhood and adolescent cancer 4312

- 8.1.2. Adult cancer survivors 4313
- 8.2. Myocardial dysfunction and heart failure 4315
- 8.3. Coronary artery disease 4316
- 8.4. Valvular heart disease 4317
- 8.5. Peripheral artery disease and stroke 4317
- 8.6. Pericardial complications 4317
- 8.7. Arrhythmias and autonomic disease 4318
- 8.8. Metabolic syndrome, lipid abnormalities, diabetes mellitus, and hypertension 4318
- 8.9. Pregnancy in cancer survivors 4318
- 8.10. Pulmonary hypertension 4319
- 9. Special populations 4319
- 9.1. Cardiac tumours 4319
- 9.2. Pregnant patients with cancer 4319
- 9.2.1. Left ventricular dysfunction and heart failure 4320
- 9.2.2. Venous thromboembolism and pulmonary embolism 4320
- 9.3. Carcinoid valvular heart disease 4323
- 9.4. Amyloid light-chain cardiac amyloidosis 4324
- 9.5. Cardiac implantable electronic devices 4326
- 10. Patient information, communication, and self-management 4330
- 11. The role of scientific societies in the promotion and development of cardio-oncology in modern medicine 4331
- 12. Key messages 4331
- 13. Future needs 4332
- 14. Gaps in evidence 4333
- 15. What to do and what not to do messages from the Guidelines 4333
- 16. Quality indicators for cardio-oncology 4342
- 17. Supplementary data 4342

- 18. Data availability statementâ^{FADEPH}â^{FADEPH}4342
- 19. Author informationâ^{FADEPH}â^{FADEPH}4342
- 20. Appendixâ^{FADEPH}â^{FADEPH}4342
- 21. Referencesâ^{FADEPH}â^{FADEPH}4343

Tables of Recommendations

- Recommendation Table 1 â Recommendations for a general approach to cardiovascular toxicity risk categorizationâ4249
- Recommendation Table 2 â Recommendations for electrocardiogram baseline assessmentâ4251
- Recommendation Table 3 â Recommendation for cardiac biomarker assessment prior to potentially cardiotoxic therapiesâ4252
- Recommendation Table 4 â Recommendations for cardiac imaging modalities in patients with cancerâ4253
- Recommendation Table 5 â Recommendations for primary prevention of cancer therapy-related cardiovascular toxicityâ4256
- Recommendation Table 6 â Recommendation for secondary prevention of cancer therapy-related cardiovascular toxicityâ4257
- Recommendation Table 7 â Recommendations for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapyâ4259
- Recommendation Table 8 â Recommendations for baseline risk assessment and monitoring during human epidermal receptor 2-targeted therapies and in the first 12 months after therapyâ4260
- Recommendation Table 9 â Recommendations for baseline risk assessment and monitoring during fluoropyrimidine therapyâ4261
- Recommendation Table 10 â Recommendations for baseline risk assessment and monitoring during vascular endothelial growth factor inhibitorsâ4264
- Recommendation Table 11 â Recommendations for baseline risk assessment and monitoring during second- and third-generation breakpoint cluster region/Abelson oncogene locus tyrosine kinase inhibitorsâ4267
- Recommendation Table 12 â Recommendations for baseline risk assessment and monitoring during Bruton tyrosine kinase inhibitor therapyâ4267

- Recommendation Table 13 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during multiple myeloma therapies [FADBPFA](#) [FADBPFA](#) 4271
- Recommendation Table 14 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during combined rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor therapy [FADBPFA](#) [FADBPFA](#) 4272
- Recommendation Table 15 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during immunotherapy [FADBPFA](#) [FADBPFA](#) 4274
- Recommendation Table 16 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer [FADBPFA](#) [FADBPFA](#) 4276
- Recommendation Table 17 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during endocrine therapy for breast cancer [FADBPFA](#) [FADBPFA](#) 4276
- Recommendation Table 18 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy [FADBPFA](#) [FADBPFA](#) 4277
- Recommendation Table 19 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring during anaplastic lymphoma kinase and epidermal growth factor receptor inhibitors [FADBPFA](#) [FADBPFA](#) 4277
- Recommendation Table 20 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies [FADBPFA](#) [FADBPFA](#) 4279
- Recommendation Table 21 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment of patients before radiotherapy to a volume including the heart [FADBPFA](#) [FADBPFA](#) 4279
- Recommendation Table 22 [â](#) [FADSCCH](#) Recommendations for baseline risk assessment in haematopoietic stem cell transplantation patients [FADBPFA](#) [FADBPFA](#) 4281
- Recommendation Table 23 [â](#) [FADSCCH](#) Recommendation for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment [FADBPFA](#) [FADBPFA](#) 4282
- Recommendation Table 24 [â](#) [FADSCCH](#) Recommendations for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy [FADBPFA](#) [FADBPFA](#) 4284
- Recommendation Table 25 [â](#) [FADSCCH](#) Recommendations for the management of cancer treatment-related cardiac dysfunction during human epidermal receptor 2-targeted therapies [FADBPFA](#) [FADBPFA](#) 4286
- Recommendation Table 26 [â](#) [FADSCCH](#) Recommendations for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis [FADBPFA](#) [FADBPFA](#) 4289
- Recommendation Table 27 [â](#) [FADSCCH](#) Recommendations for the diagnosis and management of Takotsubo syndrome in patients with cancer [FADBPFA](#) [FADBPFA](#) 4291
- Recommendation Table 28 [â](#) [FADSCCH](#) Recommendations for the management of acute coronary syndromes in patients receiving anticancer treatment [FADBPFA](#) [FADBPFA](#) 4292

- Recommendation Table 29 â€” Recommendation for the management of chronic coronary syndromes in patients receiving anticancer treatmentâ€”4293
- Recommendation Table 30 â€” Recommendations for the management of valvular heart disease in patients receiving anticancer treatmentâ€”4293
- Recommendation Table 31 â€” Recommendations for the management of atrial fibrillation in patients receiving anticancer treatmentâ€”4297
- Recommendation Table 32 â€” Recommendations for the management of long corrected QT interval and ventricular arrhythmias in patients receiving anticancer treatmentâ€”4300
- Recommendation Table 33 â€” Recommendations for the management of arterial hypertension in patients receiving anticancer treatmentâ€”4302
- Recommendation Table 34 â€” Recommendations for the management of venous thromboembolism in patients receiving anticancer treatmentâ€”4305
- Recommendation Table 35 â€” Recommendations for venous thromboembolism prophylaxis during anticancer treatmentâ€”4306
- Recommendation Table 36 â€” Recommendation for management of peripheral artery disease during anticancer treatmentâ€”4307
- Recommendation Table 37 â€” Recommendations for the management of pulmonary hypertension during anticancer treatmentâ€”4308
- Recommendation Table 38 â€” Recommendations for the management of pericardial diseases in patients receiving anticancer treatmentâ€”4309
- Recommendation Table 39 â€” Recommendations for end-of-cancer therapy cardiovascular risk assessmentâ€”4312
- Recommendation Table 40 â€” Recommendations for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivorsâ€”4313
- Recommendation Table 41 â€” Recommendations for cardiovascular surveillance in asymptomatic adult cancer survivorsâ€”4314
- Recommendation Table 42 â€” Recommendations for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapyâ€”4316
- Recommendation Table 43 â€” Recommendations for adult cancer survivors with coronary artery diseaseâ€”4316
- Recommendation Table 44 â€” Recommendations for adult cancer survivors with valvular heart diseaseâ€”4317

- Recommendation Table 45 â€” Recommendation for adult cancer survivors with pericardial complicationsâ€”4318
- Recommendation Table 46 â€” Recommendations for cardiovascular monitoring in cancer survivors during pregnancyâ€”4319
- Recommendation Table 47 â€” Recommendations for cardiovascular assessment and monitoring of pregnant women with cancerâ€”4323
- Recommendation Table 48 â€” Recommendations for carcinoid valvular heart diseasesâ€”4323
- Recommendation Table 49 â€” Recommendations for amyloid light-chain cardiac amyloidosis diagnosis and monitoringâ€”4326
- Recommendation Table 50 â€” Recommendations for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapyâ€”4330

List of tables

- Table 1 Classes of recommendationsâ€”4237
- Table 2 Levels of evidenceâ€”4237
- Table 3 Cancer therapy-related cardiovascular toxicity definitionsâ€”4243
- Table 4 Heart Failure Associationâ€”International Cardio-Oncology Society baseline cardiovascular toxicity risk stratificationâ€”4246
- Table 5 Anthracycline equivalence doseâ€”4247
- Table 6 Factors that could influence peri-operative risk during cancer surgery and preventive strategiesâ€”4254
- Table 7 Cancer treatments that predispose to acute coronary syndromesâ€”4291
- Table 8 Risk factors for drug-induced QT prolongation and torsade de pointesâ€”4298
- Table 9 Classification of corrected QT interval prolongation induced by cancer drug therapyâ€”4298
- Table 10 Risk factors for future cardiovascular disease at the end-of-cancer therapy cardiovascular risk assessmentâ€”4310
- Table 11 Risk categories for asymptomatic adults who are childhood and adolescent cancer survivorsâ€”4313
- Table 12 Risk categories for asymptomatic adult cancer survivorsâ€”4314
- Table 13 Management strategies and surgery indications for symptomatic and asymptomatic patients with benign and malignant cardiac tumoursâ€”4321

List of figures

- Figure 1 Video 1 Central Illustration: Dynamics of cardiovascular toxicity risk of patients with cancer over their therapy continuumâ^{FADEPH}â^{FADEPH}4239
- Figure 2 Cardio-oncology care pathwaysâ^{FADEPH}â^{FADEPH}4240
- Figure 3 Baseline cardiovascular toxicity risk assessment checklistâ^{FADEPH}â^{FADEPH}4241
- Figure 4 Dimensions of cancer therapy-related cardiovascular toxicity risk and disease severityâ^{FADEPH}â^{FADEPH}4242
- Figure 5 Baseline cardiovascular toxicity risk assessment before anticancer therapyâ^{FADEPH}â^{FADEPH}4245
- Figure 6 General cardio-oncology approach after Heart Failure Associationâ^{FADEPH} International Cardio-Oncology Society cardiovascular toxicity risk assessmentâ^{FADEPH}â^{FADEPH}4248
- Figure 7 Baseline screening recommendations for patients with cancer treated with potentially cardiotoxic drugsâ^{FADEPH}â^{FADEPH}4250
- Figure 8 Recommended transthoracic echocardiography and cardiac magnetic resonance imaging parameters in the evaluation of patients with cancerâ^{FADEPH}â^{FADEPH}4252
- Figure 9 Primary and secondary cancer therapy-related cardiovascular toxicity preventionâ^{FADEPH}â^{FADEPH}4255
- Figure 10 Cardiovascular toxicity monitoring in patients receiving anthracycline chemotherapyâ^{FADEPH}â^{FADEPH}4258
- Figure 11 Cardiovascular toxicity monitoring in patients receiving human epidermal receptor 2-targeted therapiesâ^{FADEPH}â^{FADEPH}4260
- Figure 12 Vascular endothelial growth factor inhibitors-related cardiovascular toxicitiesâ^{FADEPH}â^{FADEPH}4262
- Figure 13 Cardiovascular toxicity monitoring in patients receiving vascular endothelial growth factor inhibitorsâ^{FADEPH}â^{FADEPH}4263
- Figure 14 Breakpoint cluster regionâ^{FADEPH} Abelson oncogene locus tyrosine kinase inhibitor-related cardiovascular toxicitiesâ^{FADEPH}â^{FADEPH}4265
- Figure 15 Second- and third-generation breakpoint cluster regionâ^{FADEPH} Abelson oncogene locus tyrosine kinase inhibitors surveillance protocolâ^{FADEPH}â^{FADEPH}4266
- Figure 16 Multiple myeloma drug-related cardiovascular toxicitiesâ^{FADEPH}â^{FADEPH}4268
- Figure 17 Cardiovascular monitoring in patients with multiple myeloma receiving proteasome inhibitorsâ^{FADEPH}â^{FADEPH}4269
- Figure 18 Risk factors for venous thromboembolic events in patients with multiple myelomaâ^{FADEPH}â^{FADEPH}4270

- Figure 19 Rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor-related cardiovascular toxicitiesâ^{FADBP}â^{FADBP}4272
- Figure 20 Cardiovascular surveillance in patients treated with immune checkpoint inhibitorsâ^{FADBP}â^{FADBP}4273
- Figure 21 Androgen deprivation therapy-related cardiovascular toxicitiesâ^{FADBP}â^{FADBP}4275
- Figure 22 Anaplastic lymphoma kinase and epidermal growth factor receptor inhibitor-related cardiovascular toxicitiesâ^{FADBP}â^{FADBP}4278
- Figure 23 Radiotherapy mean heart dose and associated cardiovascular toxicity riskâ^{FADBP}â^{FADBP}4280
- Figure 24 Risk factors and cardiovascular surveillance in patients referred for haematopoietic stem cell transplantationâ^{FADBP}â^{FADBP}4281
- Figure 25 Management of anthracycline chemotherapy-related cardiac dysfunctionâ^{FADBP}â^{FADBP}4283
- Figure 26 Management of human epidermal receptor 2-targeted therapy-related cardiac dysfunctionâ^{FADBP}â^{FADBP}4285
- Figure 27 Direct and indirect immune checkpoint inhibitor-related cardiovascular toxicityâ^{FADBP}â^{FADBP}4287
- Figure 28 Diagnosis and management of immune checkpoint inhibitor-related myocarditisâ^{FADBP}â^{FADBP}4288
- Figure 29 Diagnosis and management workup in cancer-related Takotsubo syndromeâ^{FADBP}â^{FADBP}4290
- Figure 30 Pathophysiology of atrial fibrillation associated with cancerâ^{FADBP}â^{FADBP}4295
- Figure 31 Structured approach to anticoagulation for atrial fibrillation in patients with cancerâ^{FADBP}â^{FADBP}4296
- Figure 32 Corrected QT interval monitoring before and during treatment with corrected QT interval-prolonging anticancer drugsâ^{FADBP}â^{FADBP}4299
- Figure 33 Recommended threshold for asymptomatic hypertension treatment in different clinical scenariosâ^{FADBP}â^{FADBP}4301
- Figure 34 Treatment of arterial hypertension in patients with cancerâ^{FADBP}â^{FADBP}4302
- Figure 35 Risk factors for venous thromboembolism in patients with cancerâ^{FADBP}â^{FADBP}4303
- Figure 36 Structured approach to anticoagulation for venous thromboembolism in patients with active cancerâ^{FADBP}â^{FADBP}4304
- Figure 37 Management of cancer therapy-related cardiac dysfunction after cancer therapyâ^{FADBP}â^{FADBP}4311
- Figure 38 Long-term follow-up in cancer survivorsâ^{FADBP}â^{FADBP}4315
- Figure 39 Location of primary and secondary cardiac tumoursâ^{FADBP}â^{FADBP}4320
- Figure 40 Diagnostic algorithm for cardiac massesâ^{FADBP}â^{FADBP}4321

- Figure 41 Cardiac monitoring protocol for pregnant women receiving anthracycline-based chemotherapyâ^{FADEPH}â^{FADEPH}4322
- Figure 42 Carcinoid heart disease: clinical features and diagnostic testsâ^{FADEPH}â^{FADEPH}4324
- Figure 43 Non-invasive diagnosis of amyloid light-chain cardiac amyloidosisâ^{FADEPH}â^{FADEPH}4325
- Figure 44 Risk stratification in patients with a cardiac implantable electronic device undergoing radiotherapyâ^{FADEPH}â^{FADEPH}4327
- Figure 45 Management of patients with a cardiac implantable electronic device located in the radiotherapy treatment beamâ^{FADEPH}â^{FADEPH}4328
- Figure 46 Management of patients with a cardiac implantable electronic device located outside the radiotherapy treatment volumeâ^{FADEPH}â^{FADEPH}4329
- Figure 47 Patient information, communication, and self-managementâ^{FADEPH}â^{FADEPH}4330
- Figure 48 The role of scientific societies in the promotion and development of cardio-oncologyâ^{FADEPH}â^{FADEPH}4331

Abbreviations and acronyms

- [2D](#)
Two-dimensional
- [3D](#)
Three-dimensional
- [5-FU](#)
5-fluorouracil
- [5HIAA](#)
5-hydroxyindoleacetic acid
- [aa^{FADEPH}](#)
Late diastolic velocity of mitral annulus obtained by tissue Doppler imaging
- [ABC](#)
Atrial fibrillation Better Care

[Â](#)

- ABI
Ankle-brachial index

[Â](#)

- AC
Anthracycline chemotherapy

[Â](#)

- ACE-I
Angiotensin-converting enzyme inhibitors

[Â](#)

- ACS
Acute coronary syndromes

[Â](#)

- ADT
Androgen deprivation therapy

[Â](#)

- ADVANCE
Action in Diabetes and Vascular Disease: Preterax and Diamicron-MR Controlled Evaluation

[Â](#)

- AF
Atrial fibrillation

[Â](#)

- AI
Aromatase inhibitors

[Â](#)

- AL-CA
Amyloid light-chain cardiac amyloidosis

[Â](#)

- ALK
Anaplastic lymphoma kinase

[Â](#)

- ANS
Autonomic nervous system

[Â](#)

- ARB
Angiotensin receptor blockers

[Â](#)

- ARISTOTLE
Apixaban for Reduction in Stroke and Other Thromboembolic Events in Atrial Fibrillation

[Â](#)

- ASCVD
AtheroSclerotic Cardiovascular Disease

[Â](#)

- ASPIRE
Carfilzomib, Lenalidomide, and Dexamethasone vs. Lenalidomide and Dexamethasone for the Treatment of Patients with Relapsed Multiple Myeloma

[Â](#)

- ASTCT
American Society for Transplantation and Cellular Therapy

[Â](#)

- ATAC
Arimidex[®] and Tamoxifen Alone or in Combination

[Â](#)

- ATE
Arterial thromboembolism

[Â](#)

- AV
Atrioventricular

[Â](#)

- BB
Beta-blockers

[Â](#)

- BC
Breast cancer

[Â](#)

- BCR-ABL
Breakpoint cluster region[†]Abelson oncogene locus

[Â](#)

- BIG
Breast International Group

[Â](#)

- BLEED

Increased bleeding risk

[Â](#)

- BMI

Body mass index

[Â](#)

- BNP

B-type natriuretic peptide

[Â](#)

- BP

Blood pressure

[Â](#)

- BTK

Bruton tyrosine kinase

[Â](#)

- C

Chemotherapy cycle

[Â](#)

- CABG

Coronary artery bypass graft

[Â](#)

- CAD

Coronary artery disease

[Â](#)

- CARDIOTOX

CARDIOvascular TOXicity induced by cancer-related therapies

[Â](#)

- CAR-T

Chimeric antigen receptor T cell

[Â](#)

- CCB

Calcium channel blockers

[Â](#)

- CCS

Chronic coronary syndromes

[Â](#)

- CCTA

Coronary computed tomography angiography

[Â](#)

- CCU

Coronary care unit

[Â](#)

- CDK

Cyclin-dependent kinase

[Â](#)

- CHA₂DS₂-VASc

Congestive heart failure, Hypertension, Age 75 years (2 points), Diabetes mellitus, Stroke (2 points), Vascular disease, Age 65 years, Sex category (female)

[Â](#)

- CIED

Cardiac implantable electronic device

[Â](#)

- CML

Chronic myeloid leukaemia

[Â](#)

- CMR

Cardiac magnetic resonance

[Â](#)

- COMPASS-CAT

Prospective COMparison of Methods for thromboembolic risk assessment with clinical Perceptions and AwareneSS in real-life patients, Cancer Associated Thrombosis

[Â](#)

- CPET

Cardiopulmonary exercise testing

[Â](#)

- CrCl

Creatinine clearance

[Â](#)

- CRF

Cardiorespiratory fitness

[Â](#)

- CRS

Cytokine release syndrome

[Â](#)

- CS
Cancer survivors

[Â](#)

- CT
Computed tomography

[Â](#)

- CTLA-4
Cytotoxic T lymphocyte-associated antigen-4

[Â](#)

- cTn
Cardiac troponin

[Â](#)

- CTRCD
Cancer therapy-related cardiac dysfunction

[Â](#)

- CTR-CVT
Cancer therapy-related cardiovascular toxicity

[Â](#)

- CV
Cardiovascular

[Â](#)

- CVD
Cardiovascular disease

[Â](#)

- CVRF
Cardiovascular risk factors

[Â](#)

- DAPT
Dual antiplatelet therapy

[Â](#)

- DASISION
DASatinib vs. Imatinib Study In treatment- $\text{Na}^{\text{+}}$ -ve chronic myeloid leukaemia patients

[Â](#)

- DL
Dyslipidaemia

[Â](#)

- DM
Diabetes mellitus

[Â](#)

- DNR
Do not resuscitate

[Â](#)

- DVT
Deep vein thrombosis

[Â](#)

- E
Mitral inflow early diastolic velocity obtained by pulsed wave

[Â](#)

- $e\hat{a}_{PRD2}$
Early diastolic velocity of the mitral annulus obtained by tissue Doppler imaging

[Â](#)

- EACTS
European Association for Cardio-Thoracic Surgery

[Â](#)

- EBC
Early breast cancer

[Â](#)

- ECG
Electrocardiogram

[Â](#)

- Echo
Echocardiography

[Â](#)

- ECV
Extracellular volume fraction

[Â](#)

- eGFR
Estimated glomerular filtration rate

[Â](#)

- EGFR
Epidermal growth factor receptor

[Â](#)

- EMA
European Medicines Agency

[Â](#)

- EMB
Endomyocardial biopsy

[Â](#)

- ENGAGE AF-TIMI 48
Effective Anticoagulation with Factor Xa Next Generation in Atrial Fibrillation-Thrombolysis in Myocardial Infarction 48

[Â](#)

- ENOXACAN
Enoxaparin and Cancer

[Â](#)

- EoL
End of life

[Â](#)

- ERS
European Respiratory Society

[Â](#)

- ESC
European Society of Cardiology

[Â](#)

- ESC-CCO
European Society of Cardiology Council of Cardio-Oncology

[Â](#)

- ESH
European Society of Hypertension

[Â](#)

- EuroSCORE
European System for Cardiac Operative Risk Evaluation

[Â](#)

- FAC
Fractional area change

[Â](#)

- FDA

Food and Drug Administration

[Â](#)

- FLT3
FMS-like tyrosine kinase 3

[Â](#)

- FWLS
Free wall longitudinal strain

[Â](#)

- GI
Gastrointestinal

[Â](#)

- GLS
Global longitudinal strain

[Â](#)

- GnRH
Gonadotropin-releasing hormone

[Â](#)

- GU
Genitourinary

[Â](#)

- GVHD
Graft vs. host disease

[Â](#)

- Gy
Gray

[Â](#)

- HAS-BLED
Hypertension, Abnormal renal and liver function, Stroke, Bleeding Labile international normalized ratio, Elderly, Drugs or alcohol

[Â](#)

- HbA1c
Glycated haemoglobin

[Â](#)

- HDU
High-dependency unit

[Â](#)

- HER2
Human epidermal receptor 2

[Â](#)

- HF
Heart failure

[Â](#)

- HFA
Heart Failure Association

[Â](#)

- HFmrEF
Heart failure with mildly reduced ejection fraction

[Â](#)

- HFpEF
Heart failure with preserved ejection fraction

[Â](#)

- HFrEF
Heart failure with reduced ejection fraction

[Â](#)

- HG
Hyperglycaemia

[Â](#)

- HIIT
High-intensity interval training

[Â](#)

- HSCT
Haematopoietic stem cell transplantation

[Â](#)

- hs-cTn
High-sensitivity cardiac troponin

[Â](#)

- HTN
Hypertension

[Â](#)

- ICD
Implantable cardioverter defibrillator

[Â](#)

- ICI
Immune checkpoint inhibitors

[Â](#)

- ICOS
International Cardio-Oncology Society

[Â](#)

- ICU
Intensive care unit

[Â](#)

- IHD
Ischaemic heart disease

[Â](#)

- IMiD
Immunomodulatory drugs

[Â](#)

- i.v.
Intravenous

[Â](#)

- IVC
Inferior vena cava

[Â](#)

- IVS
Intraventricular septum

[Â](#)

- LA
Left atrial

[Â](#)

- LAA
Left atrial appendage

[Â](#)

- LGE
Late gadolinium enhancement

[Â](#)

- LIMA
Left internal mammary artery

[Â](#)

- LMWH
Low-molecular-weight heparins

[Â](#)

- LQTS
Long QT syndrome

[Â](#)

- LS
Longitudinal strain

[Â](#)

- LV
Left ventricular

[Â](#)

- LVD
Left ventricular dysfunction

[Â](#)

- LVEDD
Left ventricular end diastolic diameter

[Â](#)

- LVEF
Left ventricular ejection fraction

[Â](#)

- LVV
Left ventricular volume

[Â](#)

- M
Months

[Â](#)

- MACE
Major adverse cardiovascular events

[Â](#)

- MCS
Mechanical circulatory support

[Â](#)

- MDT
Multidisciplinary team

[Â](#)

- MedDRA
Medical dictionary for regulatory activities

[Â](#)

- MEK
Mitogen-activated extracellular signal-regulated kinase

[Â](#)

- MHD
Mean heart dose

[Â](#)

- MI
Myocardial infarction

[Â](#)

- MM
Multiple myeloma

[Â](#)

- MUGA
Multigated acquisition nuclear imaging

[Â](#)

- N
No

[Â](#)

- NOAC
Non-vitamin K antagonist oral anticoagulants

[Â](#)

- NP
Natriuretic peptides

[Â](#)

- NSTE-ACS
Non-ST-segment elevation acute coronary syndromes

[Â](#)

- NT-proBNP
N-terminal pro-B-type natriuretic peptide

[Â](#)

- PAD
Peripheral artery disease

[Â](#)

- PAH
Pulmonary arterial hypertension

[Â](#)

- PAP
Pulmonary arterial pressure

[Â](#)

- PCI
Percutaneous coronary intervention

[Â](#)

- PD-1
Programmed death-1

[Â](#)

- PD-L1
Programmed death-ligand 1

[Â](#)

- PE
Pulmonary embolism

[Â](#)

- Peric-E
Pericardial effusion

[Â](#)

- PET
Positron emission tomography

[Â](#)

- PH
Pulmonary hypertension

[Â](#)

- PI
Proteasome inhibitors

[Â](#)

- Pleu-E
Pleural effusion

[Â](#)

- PRECISE-DAPT

PREdicting bleeding Complications In patients undergoing Stent implantation and subsEquent Dual Anti Platelet Therapy

[Â](#)

- PRONOUNCE

A Trial Comparing Cardiovascular Safety of Degarelix Versus Leuprolide in Patients With Advanced Prostate Cancer and Cardiovascular Disease

[Â](#)

- PW

Left ventricular posterior wall

[Â](#)

- QI

Quality indicator

[Â](#)

- ~~â~~QTc

Corrected QT interval prolongation

[Â](#)

- QTc

Corrected QT interval

[Â](#)

- QTcF

Corrected QT interval using Fridericia correction

[Â](#)

- RA

Right atrial

[Â](#)

- RAF

Rapidly accelerated fibrosarcoma

[Â](#)

- RCT

Randomized controlled trial

[Â](#)

- RIMA

Right internal mammary artery

[Â](#)

- ROCKET AF

Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation

[Â](#)

- RT
Radiotherapy

[Â](#)

- RV
Right ventricular

[Â](#)

- RVEF
Right ventricular ejection fraction

[Â](#)

- RVV
Right ventricular volume

[Â](#)

- s_a^{PAD2}
Systolic velocity of tricuspid annulus obtained by doppler tissue imaging

[Â](#)

- SBr
Sinus bradycardia

[Â](#)

- SCORE2
Systematic Coronary Risk Estimation 2

[Â](#)

- SCORE2-OP
Systematic Coronary Risk Estimation 2 a^{PAD2CH} Older Persons

[Â](#)

- SEER
Surveillance, Epidemiology, and End Results

[Â](#)

- SMART
Second manifestations of arterial disease

[Â](#)

- sPAP
Systolic pulmonary artery pressure

[Â](#)

- SPEP
Serum protein electrophoresis

[Â](#)

- STEMI
ST-segment elevation myocardial infarction

[Â](#)

- STIR
Short tau inversion recovery

[Â](#)

- STS PROM
Society of Thoracic Surgeons ^{PADSTS} Predicted Risk of Mortality

[Â](#)

- SVT
Supraventricular tachycardia

[Â](#)

- SYNTAX
SYNergy between percutaneous coronary intervention with TAXus and cardiac surgery

[Â](#)

- TAPSE
Tricuspid annular plane systolic excursion

[Â](#)

- TAVI
Transcatheter aortic valve implantation

[Â](#)

- TBIP
Thromboembolic risk, Bleeding risk, drugs ^{PADSTS} drug Interactions, Patient preferences

[Â](#)

- TdP
Torsade de pointes

[Â](#)

- TIL
Tumour-infiltrating lymphocytes

[Â](#)

- TKI
Tyrosine kinase inhibitors

[Â](#)

- TRV
Tricuspid regurgitation velocity

[Â](#)

- TTE
Transthoracic echocardiography

[Â](#)

- TTS
Takotsubo syndrome

[Â](#)

- tx
Treatment

[Â](#)

- ULN
Upper limit of normal

[Â](#)

- UPEP
Urine protein electrophoresis

[Â](#)

- VA
Ventricular arrhythmias

[Â](#)

- VascTox
Vascular toxicity

[Â](#)

- VEGF
Vascular endothelial growth factor

[Â](#)

- VEGFi
Vascular endothelial growth factor inhibitors

[Â](#)

- VH
Very high risk

[Â](#)

- VHD
Valvular heart disease

[Â](#)

- VKA
Vitamin K antagonists

[Â](#)

- VTE
Venous thromboembolism

[Â](#)

- Y
Yes

1. Preamble

Guidelines summarize and evaluate available evidence with the aim of assisting health professionals in proposing the best management strategies for an individual patient with a given condition. Guidelines and their recommendations should facilitate decision making of health professionals in their daily practice. However, guidelines are not a substitute for the patient² relationship with their practitioner. The final decisions concerning an individual patient must be made by the responsible health professional(s), based on what they consider to be the most appropriate in the circumstances. These decisions are made in consultation with the patient and caregiver as appropriate.

Guidelines are intended for use by health professionals. To ensure that all users have access to the most recent recommendations, the ESC makes its Guidelines freely available. The ESC warns readers that the technical language may be misinterpreted and declines any responsibility in this respect.

A great number of guidelines have been issued in recent years by the ESC. Because of their impact on clinical practice, quality criteria for the development of guidelines have been established to make all decisions transparent to the user. The recommendations for formulating and issuing ESC Guidelines can be found on the ESC website (<https://www.escardio.org/Guidelines>). The ESC Guidelines represent the official position of the ESC on a given topic and are regularly updated.

In addition to the publication of Clinical Practice Guidelines, the ESC carries out the EURObservational Research Programme of international registries of cardiovascular diseases and interventions, which are essential to assess diagnostic/therapeutic processes, use of resources and adherence to guidelines. These registries aim at providing a better understanding of medical practice in Europe and around the world, based on high-quality data collected during routine clinical practice.

Furthermore, the ESC develops sets of quality indicators (QIs), which are tools to evaluate the level of implementation of the guidelines and may be used by the ESC, hospitals, healthcare providers and professionals to measure clinical practice, and in educational programmes, alongside the key messages from the guidelines, to improve quality of care and clinical outcomes.

The Members of this Task Force were selected by the ESC to represent professionals involved with the medical care of patients with this pathology. The selection procedure aimed to ensure that there is a

representative mix of members predominantly from across the whole of the ESC region and from relevant ESC Subspecialty Communities. Consideration was given to diversity and inclusion, notably with respect to gender and country of origin. A critical evaluation of diagnostic and therapeutic procedures was performed, including assessment of the risk¹benefit ratio. The level of evidence and the strength of the recommendation of particular management options were weighed and scored according to predefined scales, as outlined below. The Task Force followed the ESC voting procedures. All recommendations subject to a vote achieved at least 75% among voting members.

Table 1
[Open in new tab](#)

Classes of recommendations

Classes of recommendations	Definition		Wording to use
	Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
	Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
	Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
	Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
	Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

Classes of recommendations	Definition		Wording to use
	Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
	Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
	Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
	Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
	Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

Table 1
[Open in new tab](#)

Classes of recommendations

Classes of recommendations	Definition		Wording to use
	Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
	Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
	Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
	Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
	Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

©ESC 2022

Classes of recommendations	Definition		Wording to use
	Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
	Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
	Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
	Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
	Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

©ESC 2022

Table 2

[Open in new tab](#)

Levels of evidence

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2022

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2022

Table 2

[Open in new tab](#)

Levels of evidence

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2022

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2022

The experts of the writing and reviewing panels provided declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. Their declarations of interest were reviewed according to the ESC declaration of interest rules and can be found on the ESC website

(<http://www.escardio.org/Guidelines>) and have been compiled in a report and published in a supplementary document simultaneously to the guidelines.

This process ensures transparency and prevents potential biases in the development and review processes. Any changes in declarations of interest that arise during the writing period were notified to the ESC and updated. The Task Force received its entire financial support from the ESC without any involvement from the healthcare industry.

The ESC CPG supervises and coordinates the preparation of new guidelines. The Committee is also responsible for the approval process of these guidelines. The ESC Guidelines undergo extensive review by the CPG and external experts, including a mix of members from across the whole of the ESC region and from relevant ESC Subspecialty Communities and National Cardiac Societies. After appropriate revisions, the guidelines are signed off by all the experts involved in the Task Force. The finalized document is signed off by the CPG for publication in the *European Heart Journal*. The guidelines were developed after careful consideration of the scientific and medical knowledge and the evidence available at the time of their writing.

The task of developing ESC Guidelines also includes the creation of educational tools and implementation programmes for the recommendations, including condensed pocket guideline versions, summary slides, summary cards for non-specialists and an electronic version for digital applications (smartphones, etc.). These versions are abridged and thus, for more detailed information, the user should always access the full-text version of the guidelines, which is freely available via the ESC website and the *European Heart Journal*. The National Cardiac Societies of the ESC are encouraged to endorse, adopt, translate and implement all ESC Guidelines. Implementation programmes are needed because it has been shown that the outcome of disease may be favourably influenced by the thorough application of clinical recommendations.

Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgement, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies. However, the ESC Guidelines do not override in any way whatsoever the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient or the patient's caregiver where appropriate and/or necessary. It is also the health professional's responsibility to verify the rules and regulations applicable in each country to drugs and devices at the time of prescription and to respect the ethical rules of their profession.

Off-label use of medication may be presented in this guideline if sufficient level of evidence shows that it can be considered medically appropriate to a given condition and if patients could benefit from the recommended therapy. However, the final decisions concerning an individual patient must be made by the responsible health professional giving special consideration to:

1. the specific situation of the patient. In this respect, it is specified that, unless otherwise provided for by national regulations, off-label use of medication should be limited to situations where it is in the patient's interest to do so, with regard to the quality, safety and efficacy of care, and only after the patient has been informed and has provided consent;

2. country-specific health regulations, indications by governmental drug regulatory agencies and the ethical rules to which health professionals are subject, where applicable.

2. Introduction

This is the first European Society of Cardiology (ESC) guideline on cardio-oncology. The aim of this guideline is to help all the healthcare professionals providing care to oncology patients before, during, and after their cancer treatments with respect to their cardiovascular (CV) health and wellness. This guideline provides guidance on the definitions, diagnosis, treatment, and prevention of cancer therapy-related CV toxicity (CTR-CVT), and the management of CV disease (CVD) caused directly or indirectly by cancer. This area of medicine has limited trials and evidence on which to base decision-making and, where evidence is limited, this guideline provides the consensus of expert opinion to guide healthcare professionals.

This guideline includes the definitions of CTR-CVT ([Section 3](#)),¹ and provides a personalized approach to care based upon the baseline CV toxicity risk assessment ([Section 4](#)) and new protocols for CV surveillance during cancer treatment ([Section 5](#)). The management of acute CTR-CVT is addressed in [Section 6](#), where patients with active cancer are those receiving anticancer treatment. Throughout these sections, decision-making depends upon the risk/benefit balance of oncology treatment efficacy and the severity and impact of CTR-CVT. Guidance is provided for the first 12 months after completion of cardiotoxic treatments ([Section 7](#)), when subacute CVD can emerge, and when patients who developed CTR-CVT during cancer treatment are reviewed. Diagnosis and management of the long-term CV complications of previous oncology treatments, beyond 12 months after completing the cardiotoxic treatments, and integration into the overall survivorship strategy for cancer survivors (CS) is presented in [Section 8](#) with new long-term surveillance recommendations for high-risk patients.

In [Section 9](#), we address special populations where CVDs are directly caused by the cancer, or where special considerations are required. [Section 10](#) provides information for patients'  involvement in their own care. The final section highlights the role of the ESC and the ESC Council of Cardio-Oncology (ESC-CCO).

CTR-CVT risk is a dynamic variable, and the risk changes throughout the pathway of care ([Figure 1](#), [Video 1](#)). Absolute risk of CTR-CVT is important to understand and balance against the absolute benefit of the cancer treatment before and during treatment. However, CTR-CVT risk can be influenced by several variables, including implementation of primary prevention treatments, optimization of pre-existing CVD, dose, frequency, and duration of oncology treatment, emergence of CV complications during treatment and their severity, and in CS, the overall cumulative treatment received, the time since treatment, and the interaction with other CVDs.

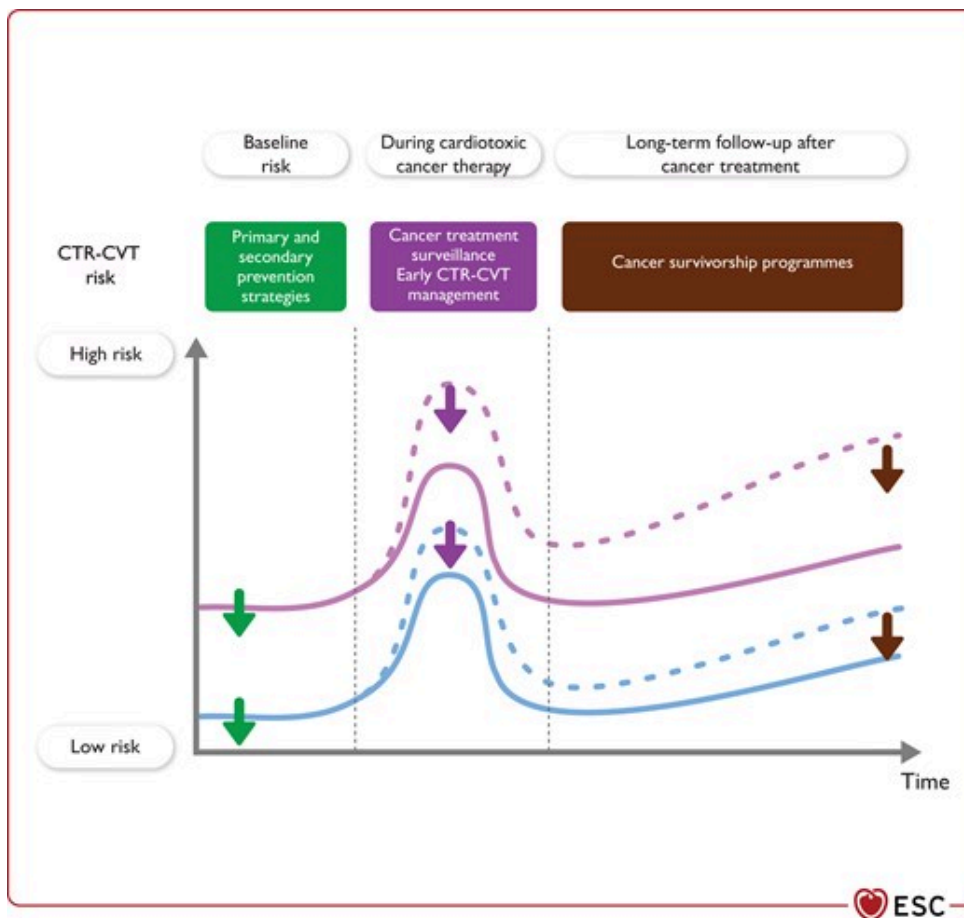


Figure 1

[Video 1](#) Central Illustration: Dynamics of cardiovascular toxicity risk of patients with cancer over their therapy continuum.

CS, cancer survivors; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; CTR-CVT risk is a dynamic variable that changes through the pathway of care, and is influenced by several conditions including age, cancer history, pre-existing CVRF or CVD, and previous cardiotoxic cancer therapy. The CTR-CVT risk changes during and after treatment according to type, dose, frequency, and duration of oncology treatment (blue solid line). Pre-existing CVRF, CVD, or previous cancer treatments may increase the magnitude of acute and long-term CV toxicity risk (purple solid line). CTR-CVT risk remains variable in extent during anticancer treatment and may or may not gradually increase over time (dotted lines). Cardio-oncology strategy may reduce the magnitude of CTR-CVT by: (1) optimizing CVD and CVRF management (green arrows); (2) considering cardioprotective strategies in high-risk patients (green arrows); (3) organizing cancer treatment surveillance; and (4) introducing early cardioprotection after the detection of subclinical CTR-CVT (purple arrows). CV risk assessment within the first year after completion of cardiotoxic cancer therapy identifies CS who require long-term follow-up. Cancer survivorship programmes that include annual CV risk assessment and CVRF/CVD management are recommended to minimize long-term CV adverse events (brown arrows).

[Open in new tab](#)[Download slide](#)

2.1. Cancer and cardiovascular needs of patients with cancer

Since the 1990s, there has been a steady decline in cancer-related mortality mirrored by a steady increase in CS.^{2,3} In this context, treatment-related side effects have gained more significance. Management of CTR-CVT has a tremendous impact on the type of anticancer therapies that patients can receive as well as the long-term morbidity and mortality outcomes of patients with cancer. Effective management of patients with both cancer and CVD requires the unique interest and expertise of healthcare providers, which has led to the formation of a new discipline: cardio-oncology.^{4,5} A recently published ESC-CCO document describes appropriate criteria for the organization and implementation of cardio-oncology services.⁵

2.2. Role of cardio-oncology services

The overarching goal of the cardio-oncology discipline is to allow patients with cancer to receive the best possible cancer treatments safely, minimizing CTR-CVT across the entire continuum of cancer care.⁵ Before initiation of cancer therapies with a known CV toxicity profile, the cardio-oncology team should identify and treat CV risk factors (CVRF) and pre-existing CVDs and define an appropriate prevention and surveillance plan for early identification and appropriate management of potential CV complications (*FigureÂ 2*). Another important aspect is the participation in interdisciplinary discussions regarding the benefits and risks of certain cancer treatments and their continuation or interruption should side effects become apparent. After cancer treatment has been completed, the focus shifts to co-ordination of long-term follow-up and treatment. For patients on long-term cancer therapies with CV toxicity risk, surveillance should continue until the treatment is finished.^{6,7} There is also the need for re-assessment of CV risks in patients requiring treatment for secondary malignancies.

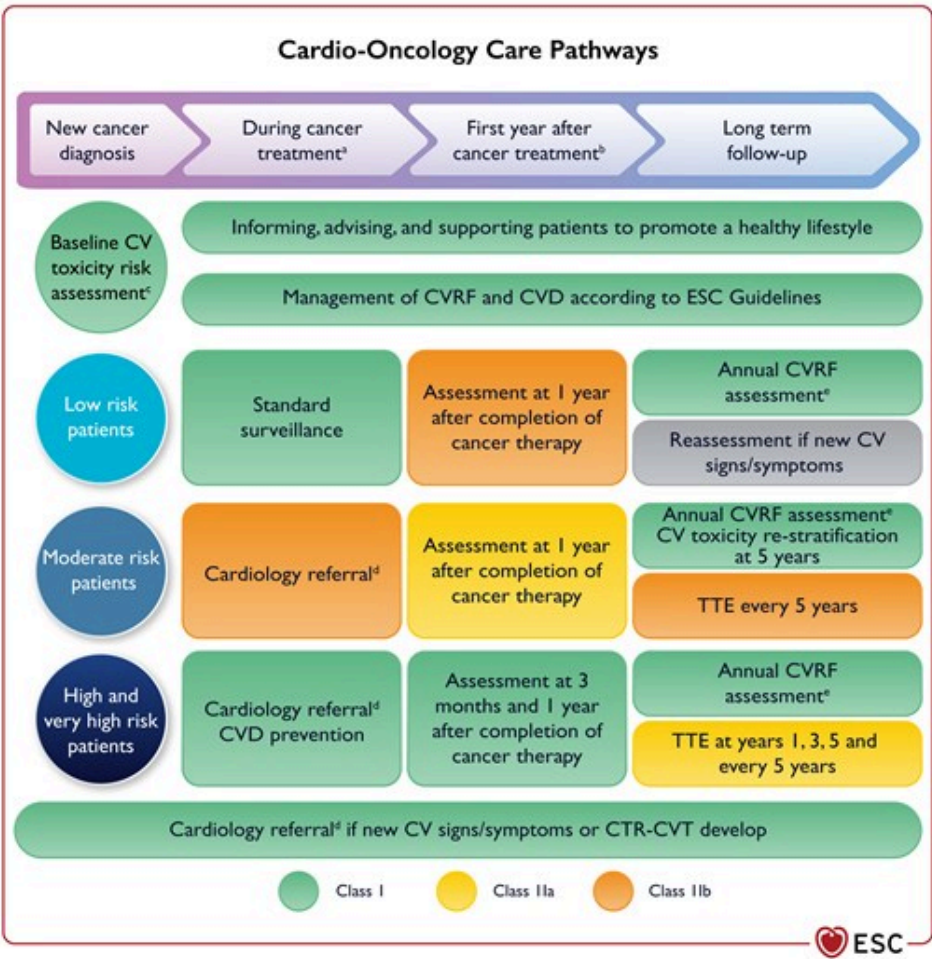


Figure 2

Cardio-oncology care pathways.

BP, blood pressure; CS, cancer survivors; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; ESC, European Society of Cardiology; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; NP, natriuretic peptides; RT, radiotherapy; TTE, transthoracic echocardiography. ^aCV surveillance according to baseline CV toxicity risk, type of cancer, cancer stage, and cancer therapy. ^bCTR-CVT risk assessment is recommended during the first year after cardiotoxic cancer treatment to establish a long-term follow-up care plan. ^cThe use of HFA-ICOS risk assessment tools should be considered to assess CTR-CVT risk in patients with cancer scheduled to receive cardiotoxic anticancer therapy. Clinical assessment and ECG are recommended at baseline in all patients with cancer and echocardiography, cardiac biomarkers, or other cardiac imaging tests in selected patients according to baseline CV toxicity risk and cancer treatment type (see [Figure 7](#)). ^dCardio-oncology referral is recommended when available, alternatively patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer. ^eAnnual CV risk assessment (including clinical review, BP, lipid profile, HbA1c, ECG, and NP) and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT to a volume exposing the heart.

[Open in new tab](#)[Download slide](#)

2.3. General principles of cardio-oncology

A guiding principle of cardio-oncology is the integration of clinical disciplines. Cardio-oncology providers must have knowledge of the broad scope of cardiology, oncology, and haematology management.⁵ Recommendations are formed regarding the most permissible (from a CVD perspective) and the most effective (from an oncological perspective) cancer treatment. Adjudication of CV events occurring in patients on active therapy is another important aspect of cardio-oncology practice.^{1,3} This is in addition to recommendations on best treatment and management practices. This includes the full scope of CV therapies, including healthy lifestyle promotion and pharmacological, device, and surgical treatments.^{4,9,10}

The principle underlying the dynamic course of CTR-CVT development in patients with cancer is that the absolute risk depends on their baseline risk and changes with exposure to cardiotoxic therapies over time ([Figure 3](#)).¹¹ This has been recognized in conceptual models, with risk stratification tools designed to grade patients with cancer into low, moderate, high, and very high risk of CV complications prior to starting treatment. These have been published by the Heart Failure Association (HFA) of the ESC in collaboration with the International Cardio-Oncology Society (ICOS) (see [Section 4](#)).^{12,13} Severity, duration, and type of manifestation of CTR-CVT vary by type of malignancy and cancer treatment. The risk itself can be understood in two ways: (1) the likelihood of its occurrence and (2) the severity of the complication ([Figure 4](#)). For example, a patient could be very likely to experience a CTR-CVT, but if this event is mild, oncology treatment should continue. Conversely, a patient at low likelihood could still be at high risk according to the severity of the event, which would lead to interruption of cancer treatment, e.g. a significant decline in left ventricular (LV) ejection fraction (LVEF) to < 40% with anthracycline chemotherapy. The timeline of these developments

may also be rather different. After the cardiotoxic cancer treatment has been completed, a new risk assessment is recommended to establish different long-term trajectories of CV health. These trajectories are impacted by the permanent CV toxic effects and cardiac or vascular injury of some cancer therapies, patient-related CVRF, environmental factors, and stressors (e.g. acute viral infections). The aim should be to personalize approaches to minimize CTR-CVT and improve both cancer and CV outcomes.

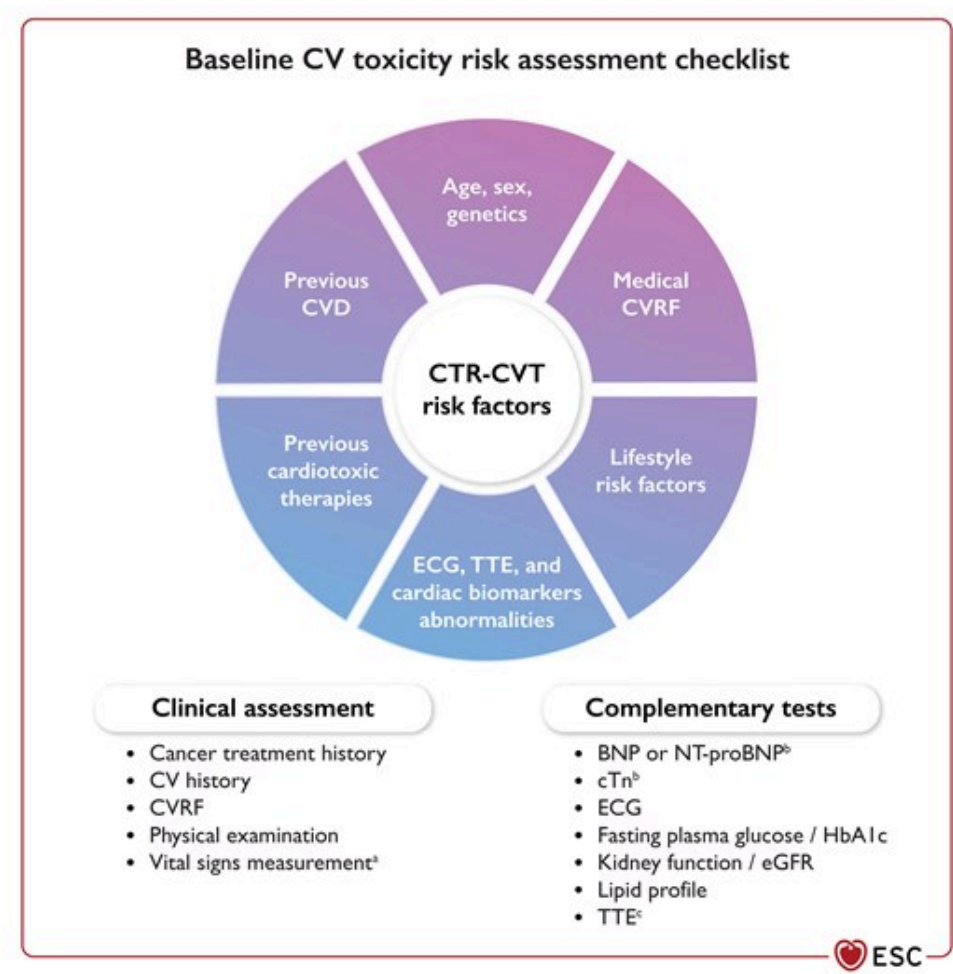


Figure 3
Baseline cardiovascular toxicity risk assessment checklist.

BNP, B-type natriuretic peptide; cTn, cardiac troponin; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, CV disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; HbA1c, glycated haemoglobin; NT-proBNP, N-terminal pro-BNP; TTE, transthoracic echocardiography. ^aIncluding blood pressure, heart rate, height, weight, and body mass index. ^bCardiac biomarkers (troponin and NP) should be measured in patients at risk of CTRCD where available and results should be interpreted according to the patient clinical status, type of cancer treatment, and kidney function. ^cConsider other CV complementary tests in selected patients: cardiac magnetic resonance, coronary computed tomography angiography, CPET (in selected patients for pre-operative [lung, colon, and rectal cancers] risk stratification). See [Section 4.6](#).

[Open in new tab](#)[Download slide](#)

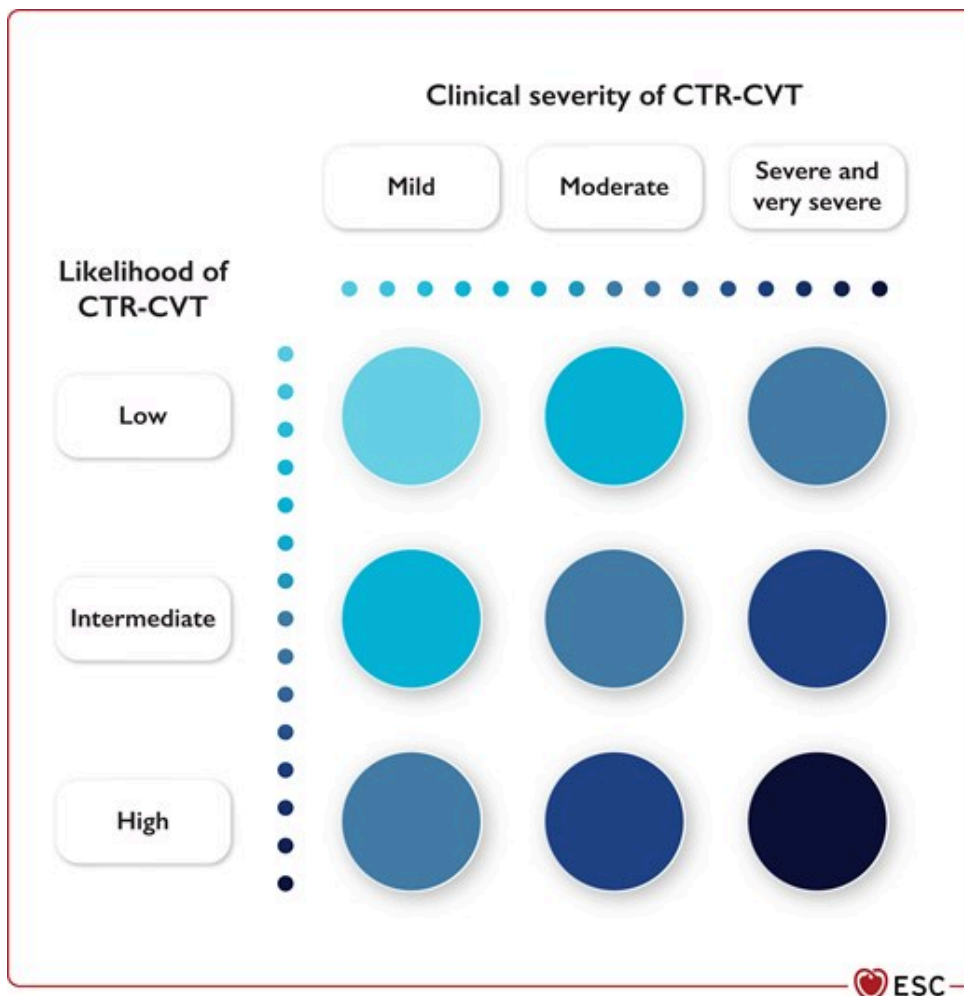


Figure 4
Dimensions of cancer therapy-related cardiovascular toxicity risk and disease severity.

CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular. The ultimate risk is the combination of the likelihood (based on reported incidence) and degree (severity or grade) of the adverse event. The most vulnerable patient groups are those at high likelihood of experiencing a severe adverse event. The level of attention that needs to be devoted to these patients varies accordingly. The risk and type of CTR-CVT, as well as the potential for reversibility, depends on different factors, listed in [Figure 3](#), that should be considered to define global CV and oncological prognosis and to individualize CTR-CVT surveillance. Additional factors that add to the complexity of CTR-CVT risk assessment are the cancer type and prognosis, and type, duration, and intensity of cancer treatment.

[Open in new tab](#)[Download slide](#)

3. Cancer therapy-related cardiovascular toxicity definitions

Several terminologies and definitions have previously been proposed to describe the spectrum of CTR-CVT, leading to inconsistencies in diagnosis and management. The need to harmonize these definitions has frequently been stated and recognized, and resulted in the recent international definitions of CTR-CVT¹ supported by this guideline ([Table 3](#); [Supplementary data, Table S1](#)). This document will focus on consensus definitions for cardiomyopathy and heart failure (HF), myocarditis, vascular toxicities, hypertension, cardiac arrhythmias, and corrected QT interval (QTc) prolongation. The definitions of other

CTR-CVT, including pericardial and valvular heart diseases (VHDs), are the same as those used for the general cardiology population. For cardiac injury, cardiomyopathy, and HF, the descriptive term cancer therapy-related cardiac dysfunction (CTRCD) is recommended as it captures the broad spectrum of possible presentations and the aetiological link with the broad scope of various cancer therapies, including chemotherapy, targeted agents, immune therapies, and radiation therapy.

Table 3
[Open in new tab](#)

Cancer therapy-related cardiovascular toxicity definitions

CTRCD		
Symptomatic CTRCD (HF) ^{a,b}	Very severe	HF requiring inotropic support, mechanical circulatory support, or consideration of transplantation
	Severe	HF hospitalization
	Moderate	Need for outpatient intensification of diuretic and HF therapy
	Mild	Mild HF symptoms, no intensification of therapy required
Asymptomatic CTRCD	Severe	New LVEF reduction to <40%
	Moderate	New LVEF reduction by ≥10 percentage points to an LVEF of 40–49% OR New LVEF reduction by <10 percentage points to an LVEF of 40–49% AND either new relative decline in GLS by >15% from baseline OR new rise in cardiac biomarkers ^c
	Mild	LVEF ≥50% AND new relative decline in GLS by >15% from baseline AND/OR new rise in cardiac biomarkers ^c
ICI myocarditis (either pathohistological diagnosis or clinical diagnosis)		
Pathohistological diagnosis (EMB)	Multifocal inflammatory cell infiltrates with overt cardiomyocyte loss by light microscopy	
Clinical diagnosis ^d	cTn elevation (new or significant change from baseline) ^e with 1 major criterion or 2 minor criteria, after exclusion of ACS and acute infectious myocarditis based on clinical suspicion ^f	

	Major criterion: • CMR diagnostic for acute myocarditis (modified Lake Louise criteria)^g
	Minor criteria: • Clinical syndrome (including any one of the following: fatigue, myalgias, chest pain, diplopia, ptosis, shortness of breath, orthopnoea, lower-extremity oedema, palpitations, light-headedness/dizziness, syncope, muscle weakness, cardiogenic shock) • Ventricular arrhythmia (including cardiac arrest) and/or new conduction system disease • Decline in LV systolic function, with or without regional wall motion abnormalities in a non-Takotsubo pattern • Other immune-related adverse events, particularly myositis, myopathy, myasthenia gravis • Suggestive CMR^h
Severity of myocarditis	• Fulminant: Haemodynamic instability, HF requiring non-invasive or invasive ventilation, complete or high-grade heart block, and/or significant ventricular arrhythmia • Non-fulminant: including symptomatic but haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immuno-related adverse events. Patients may have reduced LVEF but no features of severe disease • Steroid refractory: non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone
Recovery from myocarditis	• Complete recovery: Patients with complete resolution of acute symptoms, normalization of biomarkers, and recovery of LVEF after discontinuation of immunosuppression. CMR may still show LGE or elevated T1 due to fibrosis, but any suggestion of acute oedema should be absent • Recovering: Ongoing improvement in patient clinical symptoms, signs, biomarkers, and imaging parameters, but not yet normalized, while on tapering doses of immunosuppression
Vascular toxicity (for general cardiology definitions, see Supplementary data, Table S1)	

Asymptomatic vascular toxicity	CAD	Symptomatic vascular toxicity	Stroke
	PAD		Transient ischaemic attack
	Carotid artery disease		MI
	Venous thrombosis		ACS
	Arterial thrombosis		CCS
	Peripheral vasoreactivity		PAD
	Coronary epicardial vasoreactivity		Vasospastic angina
	Coronary microvascular vasoreactivity		Microvascular angina
			Raynaud's phenomenon
Arterial hypertension			
Treatment threshold for hypertension before, during, and after therapy	In patients with high CV risk ⁱ : ≥ 130 mmHg systolic and/or ≥ 80 mmHg diastolic		
	Otherwise: ≥ 140 mmHg systolic and/or ≥ 90 mmHg diastolic		
Cancer therapy holding threshold	≥ 180 mmHg systolic and/or ≥ 110 mmHg diastolic		
Hypertensive emergency	(Very high) BP elevation associated with acute hypertension-mediated organ damage (heart, retina, brain, kidneys, and large arteries), requiring immediate BP reduction to limit extension or promote regression of target organ damage		
Cardiac arrhythmias			
QT prolongation	Prolonged: QTcF ≥ 500 ms ^j		
Bradycardia	For general cardiology definitions, see Supplementary data, Table S1		

Supraventricular tachycardia	
Ventricular arrhythmias	
AF	

CTRCD		
Symptomatic CTRCD (HF) ^{a,b}	Very severe	HF requiring inotropic support, mechanical circulatory support, or consideration of transplantation
	Severe	HF hospitalization
	Moderate	Need for outpatient intensification of diuretic and HF therapy
	Mild	Mild HF symptoms, no intensification of therapy required
Asymptomatic CTRCD	Severe	New LVEF reduction to <40%
	Moderate	New LVEF reduction by ≥ 10 percentage points to an LVEF of 40% ^{FADSTS} 49% OR New LVEF reduction by <10 percentage points to an LVEF of 40% ^{FADSTS} 49% AND either new relative decline in GLS by >15% from baseline OR new rise in cardiac biomarkers ^c
	Mild	LVEF $\geq 50\%$ AND new relative decline in GLS by >15% from baseline AND/OR new rise in cardiac biomarkers ^c
ICI myocarditis (either pathohistological diagnosis or clinical diagnosis)		
Pathohistological diagnosis (EMB)	Multifocal inflammatory cell infiltrates with overt cardiomyocyte loss by light microscopy	
Clinical diagnosis ^d	cTn elevation (new or significant change from baseline) ^e $\hat{=}$ with 1 major criterion or 2 minor criteria, after exclusion of ACS and acute infectious myocarditis based on clinical suspicion ^f	
	Major criterion:	

	• CMR diagnostic for acute myocarditis (modified Lake Louise criteria)^g
	Minor criteria: • Clinical syndrome (including any one of the following: fatigue, myalgias, chest pain, diplopia, ptosis, shortness of breath, orthopnoea, lower-extremity oedema, palpitations, light-headedness/dizziness, syncope, muscle weakness, cardiogenic shock) • Ventricular arrhythmia (including cardiac arrest) and/or new conduction system disease • Decline in LV systolic function, with or without regional wall motion abnormalities in a non-Takotsubo pattern • Other immune-related adverse events, particularly myositis, myopathy, myasthenia gravis • Suggestive CMR^h
Severity of myocarditis	• Fulminant: Haemodynamic instability, HF requiring non-invasive or invasive ventilation, complete or high-grade heart block, and/or significant ventricular arrhythmia • Non-fulminant: including symptomatic but haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immuno-related adverse events. Patients may have reduced LVEF but no features of severe disease • Steroid refractory: non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone
Recovery from myocarditis	• Complete recovery: Patients with complete resolution of acute symptoms, normalization of biomarkers, and recovery of LVEF after discontinuation of immunosuppression. CMR may still show LGE or elevated T1 due to fibrosis, but any suggestion of acute oedema should be absent • Recovering: Ongoing improvement in patient clinical symptoms, signs, biomarkers, and imaging parameters, but not yet normalized, while on tapering doses of immunosuppression
Vascular toxicity (for general cardiology definitions, see Supplementary data, Table S1)	

Asymptomatic vascular toxicity	CAD	Symptomatic vascular toxicity	Stroke
	PAD		Transient ischaemic attack
	Carotid artery disease		MI
	Venous thrombosis		ACS
	Arterial thrombosis		CCS
	Peripheral vasoreactivity		PAD
	Coronary epicardial vasoreactivity		Vasospastic angina
	Coronary microvascular vasoreactivity		Microvascular angina
			Raynaudâs phenomenon
Arterial hypertension			
Treatment threshold for hypertension before, during, and after therapy	In patients with high CV risk ⁱ : ââ		

Supraventricular tachycardia	
Ventricular arrhythmias	
AF	

ACS, acute coronary syndromes; AF, atrial fibrillation; BNP, B-type natriuretic peptide; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECV, extracellular volume fraction; EMB, endomyocardial biopsy; ESC, European Society of Cardiology; GLS, global longitudinal strain; HF, heart failure; HFmrEF, HF with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; ICI, immune checkpoint inhibitors; LGE, late gadolinium enhancement; LV, left ventricular; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PAD, peripheral artery disease; QTcF, corrected QT interval using Fridericia correction; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2âOlder Persons.

See [Supplementary data, Table S1](#) for expanded definitions.

a

With LVEF and supportive diagnostic biomarkers based on the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF.¹⁴

b

Symptomatic CTRCD represents HF, which is a clinical syndrome consisting of cardinal symptoms (e.g. breathlessness, ankle swelling, and fatigue) that may be accompanied by signs (e.g. elevated jugular venous pressure, pulmonary crackles, and peripheral oedema) and has traditionally been divided into distinct phenotypes based on the measurement of LVEF: $\leq 40\%$ = HFrEF; $41\text{--}49\%$ = HFmrEF; $\geq 50\%$ = HFpEF.

c

cTnI/cTnT > 99th percentile, BNP $\geq 35\text{ ng/mL}$, NT-proBNP $\geq 125\text{ ng/mL}$ or new significant rise from baseline beyond the biological and analytical variation of the assay used.

d

Clinical diagnoses should be confirmed with magnetic resonance imaging or EMB if possible and without causing treatment delays. Treatment with immunosuppression should be promptly initiated while awaiting further confirmatory testing in symptomatic patients.

e

Both troponin I and troponin T can be used; however, clinical observations suggest that troponin T may be falsely elevated in patients with concomitant myositis and without myocarditis.^{15,17}

f According to local protocols.

g Diagnostic CMR: Based on updated Lake Louise criteria¹⁸: T2-based criterion + T1-based criterion ± supportive criteria (T2-based criteria: regional or global increase of native T2, or T2 signal intensity; T1-based criteria: regional or global increase of native T1, or regional or global increase in the ECV, or presence of LGE; supportive criteria: pericarditis and/or regional or global LV systolic dysfunction).

h *Suggestive CMR*: meeting some but not all of the modified Lake Louise criteria. The presence of T2- or T1-based criteria may support a diagnosis of acute myocardial inflammation in the appropriate clinical scenario.

i SCORE2 (<70 years), SCORE2-OP (≥70 years) or equivalent.¹⁹ CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

j QTcF 480–500 ms: correct reversible causes, minimize other QT prolonging medications, close QTcF monitoring. Fridericia correction is recommended ($QTcF = QT / \sqrt[3]{RR}$).²⁰

Table 3

[Open in new tab](#)

Cancer therapy-related cardiovascular toxicity definitions

CTRCD		
Symptomatic CTRCD (HF) ^{a,b}	Very severe	HF requiring inotropic support, mechanical circulatory support, or consideration of transplantation
	Severe	HF hospitalization
	Moderate	Need for outpatient intensification of diuretic and HF therapy
	Mild	Mild HF symptoms, no intensification of therapy required
Asymptomatic CTRCD	Severe	New LVEF reduction to <40%
	Moderate	New LVEF reduction by ≥10 percentage points to an LVEF of 40–49% OR New LVEF reduction by <10 percentage points to an LVEF of 40–49% AND either new relative decline

		in GLS by >15% from baseline OR new rise in cardiac biomarkers ^c
	Mild	LVEF $\geq$ 50% AND new relative decline in GLS by >15% from baseline AND/OR new rise in cardiac biomarkers ^c
ICI myocarditis (either pathohistological diagnosis or clinical diagnosis)		
Pathohistological diagnosis (EMB)	Multifocal inflammatory cell infiltrates with overt cardiomyocyte loss by light microscopy	
Clinical diagnosis^d	cTn elevation (new or significant change from baseline) ^e AND with 1 major criterion or 2 minor criteria , after exclusion of ACS and acute infectious myocarditis based on clinical suspicion ^f	
	Major criterion: • CMR diagnostic for acute myocarditis (modified Lake Louise criteria)^g	
	Minor criteria: • Clinical syndrome (including any one of the following: fatigue, myalgias, chest pain, diplopia, ptosis, shortness of breath, orthopnoea, lower-extremity oedema, palpitations, light-headedness/dizziness, syncope, muscle weakness, cardiogenic shock) • Ventricular arrhythmia (including cardiac arrest) and/or new conduction system disease • Decline in LV systolic function, with or without regional wall motion abnormalities in a non-Takotsubo pattern • Other immune-related adverse events, particularly myositis, myopathy, myasthenia gravis • Suggestive CMR^h	
Severity of myocarditis	• Fulminant: Haemodynamic instability, HF requiring non-invasive or invasive ventilation, complete or high-grade heart block, and/or significant ventricular arrhythmia • Non-fulminant: including symptomatic but haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immuno-related adverse events. Patients may have reduced LVEF but no features of severe disease	

	• Steroid refractory: non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone		
Recovery from myocarditis	• Complete recovery: Patients with complete resolution of acute symptoms, normalization of biomarkers, and recovery of LVEF after discontinuation of immunosuppression. CMR may still show LGE or elevated T1 due to fibrosis, but any suggestion of acute oedema should be absent • Recovering: Ongoing improvement in patient clinical symptoms, signs, biomarkers, and imaging parameters, but not yet normalized, while on tapering doses of immunosuppression		
Vascular toxicity (for general cardiology definitions, see Supplementary data, Table S1)			
Asymptomatic vascular toxicity	CAD	Symptomatic vascular toxicity	Stroke
	PAD		Transient ischaemic attack
	Carotid artery disease		MI
	Venous thrombosis		ACS
	Arterial thrombosis		CCS
	Peripheral vasoreactivity		PAD
	Coronary epicardial vasoreactivity		Vasospastic angina
	Coronary microvascular vasoreactivity		Microvascular angina
			Raynaud’s phenomenon
Arterial hypertension			
Treatment threshold for hypertension	In patients with high CV risk: $\geq 130/80$ mmHg systolic and/or ≥ 80 mmHg diastolic		

before, during, and after therapy	Otherwise: $\hat{a}_{HTJ} \leq 140 \hat{a}_{PADNEL}$ mmHg systolic and/or $\hat{a}_{HTJ} \leq 90 \hat{a}_{PADNEL}$ mmHg diastolic
Cancer therapy holding threshold	$\hat{a}_{HTJ} \leq 180 \hat{a}_{PADNEL}$ mmHg systolic and/or $\hat{a}_{HTJ} \leq 110 \hat{a}_{PADNEL}$ mmHg diastolic
Hypertensive emergency	(Very high) BP elevation associated with acute hypertension-mediated organ damage (heart, retina, brain, kidneys, and large arteries), requiring immediate BP reduction to limit extension or promote regression of target organ damage
Cardiac arrhythmias	
QT prolongation	Prolonged: $QTcF \hat{A} > \hat{A} 500 \hat{a}_{PADNEL}$ ms ^j
Bradycardia	For general cardiology definitions, see Supplementary data, Table S1
Supraventricular tachycardia	
Ventricular arrhythmias	
AF	

CTRCD		
Symptomatic CTRCD (HF) ^{a,b}	Very severe	HF requiring inotropic support, mechanical circulatory support, or consideration of transplantation
	Severe	HF hospitalization
	Moderate	Need for outpatient intensification of diuretic and HF therapy
	Mild	Mild HF symptoms, no intensification of therapy required
Asymptomatic CTRCD	Severe	New LVEF reduction to <40%
	Moderate	New LVEF reduction by $\hat{a}_{HTJ} \leq 10$ percentage points to an LVEF of $40 \hat{a}_{PADSTS} \leq 49\%$ OR New LVEF reduction by <10 percentage points to an LVEF of $40 \hat{a}_{PADSTS} \leq 49\%$ AND either new relative decline in GLS by >15% from baseline OR new rise in cardiac biomarkers ^c

	Mild	LVEF $\leq$ 50% AND new relative decline in GLS by >15% from baseline AND/OR new rise in cardiac biomarkers^c
ICI myocarditis (either pathohistological diagnosis or clinical diagnosis)		
Pathohistological diagnosis (EMB)	Multifocal inflammatory cell infiltrates with overt cardiomyocyte loss by light microscopy	
Clinical diagnosis^d	cTn elevation (new or significant change from baseline)^e with 1 major criterion or 2 minor criteria, after exclusion of ACS and acute infectious myocarditis based on clinical suspicion^f	
	Major criterion: CMR diagnostic for acute myocarditis (modified Lake Louise criteria)^g	
	Minor criteria: - Clinical syndrome (including any one of the following: fatigue, myalgias, chest pain, diplopia, ptosis, shortness of breath, orthopnoea, lower-extremity oedema, palpitations, light-headedness/dizziness, syncope, muscle weakness, cardiogenic shock) - Ventricular arrhythmia (including cardiac arrest) and/or new conduction system disease - Decline in LV systolic function, with or without regional wall motion abnormalities in a non-Takotsubo pattern - Other immune-related adverse events, particularly myositis, myopathy, myasthenia gravis - Suggestive CMR^h	
Severity of myocarditis	Fulminant: Haemodynamic instability, HF requiring non-invasive or invasive ventilation, complete or high-grade heart block, and/or significant ventricular arrhythmia Non-fulminant: including symptomatic but haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immuno-related adverse events. Patients may have reduced LVEF but no features of severe disease Steroid refractory: non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies)	

	despite high-dose methylprednisolone		
Recovery from myocarditis	• Complete recovery: Patients with complete resolution of acute symptoms, normalization of biomarkers, and recovery of LVEF after discontinuation of immunosuppression. CMR may still show LGE or elevated T1 due to fibrosis, but any suggestion of acute oedema should be absent • Recovering: Ongoing improvement in patient clinical symptoms, signs, biomarkers, and imaging parameters, but not yet normalized, while on tapering doses of immunosuppression		
Vascular toxicity (for general cardiology definitions, see Supplementary data, Table S1)			
Asymptomatic vascular toxicity	CAD	Symptomatic vascular toxicity	Stroke
	PAD		Transient ischaemic attack
	Carotid artery disease		MI
	Venous thrombosis		ACS
	Arterial thrombosis		CCS
	Peripheral vasoreactivity		PAD
	Coronary epicardial vasoreactivity		Vasospastic angina
	Coronary microvascular vasoreactivity		Microvascular angina
			Raynaud’s phenomenon
Arterial hypertension			
Treatment threshold for hypertension before, during, and after therapy	In patients with high CV risk: ≥ 130 mmHg systolic and/or ≥ 80 mmHg diastolic		
	Otherwise: ≥ 140 mmHg systolic and/or ≥ 90 mmHg diastolic		

Cancer therapy holding threshold	â€¥180â€¥mmHg systolic and/or â€¥110â€¥mmHg diastolic
Hypertensive emergency	(Very high) BP elevation associated with acute hypertension-mediated organ damage (heart, retina, brain, kidneys, and large arteries), requiring immediate BP reduction to limit extension or promote regression of target organ damage
Cardiac arrhythmias	
QT prolongation	Prolonged: QTcF $\hat{A}$ > $\hat{A}$ 500â€¥ms^j
Bradycardia	For general cardiology definitions, see Supplementary data, Table S1
Supraventricular tachycardia	
Ventricular arrhythmias	
AF	

ACS, acute coronary syndromes; AF, atrial fibrillation; BNP, B-type natriuretic peptide; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECV, extracellular volume fraction; EMB, endomyocardial biopsy; ESC, European Society of Cardiology; GLS, global longitudinal strain; HF, heart failure; HFmrEF, HF with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; ICI, immune checkpoint inhibitors; LGE, late gadolinium enhancement; LV, left ventricular; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PAD, peripheral artery disease; QTcF, corrected QT interval using Fridericia correction; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2â€¥Older Persons.

See [Supplementary data, Table S1](#) for expanded definitions.

a

With LVEF and supportive diagnostic biomarkers based on the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF.¹⁴

b

Symptomatic CTRCD represents HF, which is a clinical syndrome consisting of cardinal symptoms (e.g. breathlessness, ankle swelling, and fatigue) that may be accompanied by signs (e.g. elevated jugular venous pressure, pulmonary crackles, and peripheral oedema) and has traditionally been divided into distinct phenotypes based on the measurement of LVEF: â€¥40% = HFrEF; 41â€¥49% = HFmrEF; â€¥50% = HFpEF.

c

cTnI/cTnT > 99th percentile, BNP ≥ 350 pg/mL, NT-proBNP ≥ 1250 pg/mL or new significant rise from baseline beyond the biological and analytical variation of the assay used.

d

Clinical diagnoses should be confirmed with magnetic resonance imaging or EMB if possible and without causing treatment delays. Treatment with immunosuppression should be promptly initiated while awaiting further confirmatory testing in symptomatic patients.

e

Both troponin I and troponin T can be used; however, clinical observations suggest that troponin T may be falsely elevated in patients with concomitant myositis and without myocarditis.^{15,17}

f

According to local protocols.

g

Diagnostic CMR: Based on updated Lake Louise criteria¹⁸: T2-based criterion + T1-based criterion $\hat{+}$ supportive criteria (T2-based criteria: regional or global increase of native T2, or T2 signal intensity; T1-based criteria: regional or global increase of native T1, or regional or global increase in the ECV, or presence of LGE; supportive criteria: pericarditis and/or regional or global LV systolic dysfunction).

h

Suggestive CMR: meeting some but not all of the modified Lake Louise criteria. The presence of T2- or T1-based criteria may support a diagnosis of acute myocardial inflammation in the appropriate clinical scenario.

i

SCORE2 (<70 years), SCORE2-OP (≥ 70 years) or equivalent.¹⁹ CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk $\geq 7.5\%$; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk $\geq 10\%$; ≥ 70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk $\geq 15\%$.

j

QTcF 480–500 ms: correct reversible causes, minimize other QT prolonging medications, close QTcF monitoring. Fridericia correction is recommended ($QTcF = QT/3\sqrt{RR}$).²⁰

4. Cardiovascular toxicity risk stratification before anticancer therapy

The optimal time to consider CVD prevention strategies in patients with cancer is at the time of cancer diagnosis and prior to the initiation of cancer treatment.^{4,5} This enables the oncology team to consider CV risk while making cancer treatment choices, educating patients regarding their CV risk, personalizing CV surveillance and follow-up strategies, and making appropriate referrals of high-risk patients to cardio-oncology services. These strategies are needed to mitigate CVD risk, and improve the adherence to effective cancer treatments and the overall survival.

CVD prevention strategies require a personalized approach. Risk assessment is a challenging task and it is vital that clinicians adopt a systematic approach without delaying oncological treatment.^{12,21,22} [Figure 5](#) provides a comprehensive approach to risk assessment. The choice of the cardiac tests (electrocardiogram [ECG], biomarkers, and imaging) should be individualized based on CV risk and the planned cancer treatments.

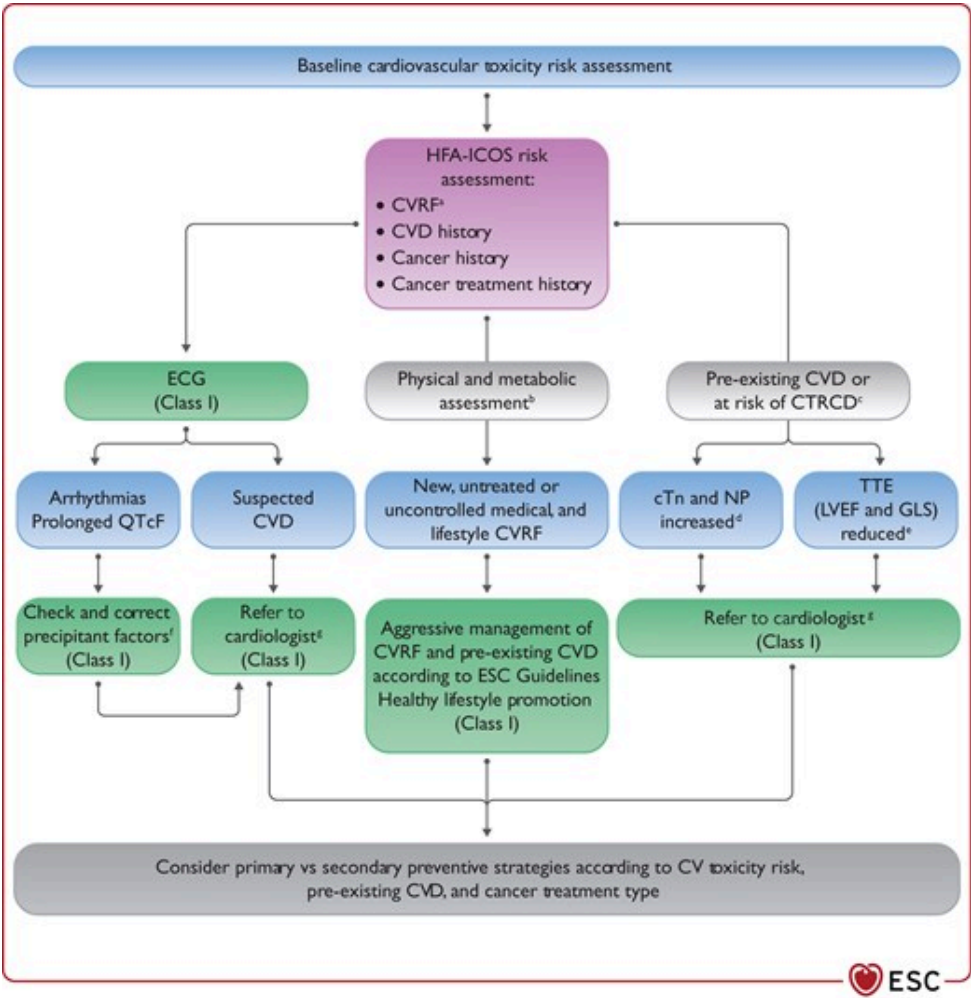


Figure 5
Baseline cardiovascular toxicity risk assessment before anticancer therapy.

BNP, B-type natriuretic peptide; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ECG, electrocardiogram; ESC, European Society of Cardiology; GLS, global longitudinal strain; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LVEF, left ventricular ejection fraction; NP, natriuretic peptides (including BNP and NT-proBNP); NT-proBNP, N-terminal pro-BNP peptide; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TTE, transthoracic echocardiography. ^aWhen assessing CVRF, include information about unhealthy lifestyle including sedentary behaviour, smoking, and alcohol intake. ^bSee [Figure 3](#). ^cAccording to cancer treatment and HFA-ICOS risk assessment. ^dcTnI/T > 99th percentile, BNP > 350 pg/mL, NT-proBNP > 1250 pg/mL. ^ePatients with baseline LVEF < 50% or in the low normal range (LVEF 50–54%) should be referred to a specialized cardiologist or cardio-oncologist. When TTE is used, ideally three-dimensional-LVEF and GLS should be measured. If GLS assessment is not available, other markers of longitudinal function (e.g. annular Doppler velocity) should be considered. Cardiac

magnetic resonance should be considered if echocardiography is of non-diagnostic quality.^fAnaemia, infections, electrolyte abnormalities, metabolic problems, other QTc-prolonging drugs.^gCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

[Open in new tabDownload slide](#)

4.1. General approach to cardiovascular toxicity risk in patients with cancer

Pre-treatment CTR-CVT risk assessment should ideally be performed using a recognized risk stratification method where multiple risk factors are incorporated to determine patient-specific risk.²³ Only a limited number of retrospective risk scores have been published in patients with cancer. Most of these scores have been developed for specific cancer-patient groups and cannot be readily applied or extrapolated to other type of malignancies.^{24aPADSTS29} While further validation is needed, HFA-ICOS risk assessment tools should be considered to determine pre-treatment risk of CTR-CVT as they are easy to use and implement in oncology and haematology services ([TableÂ 4; Supplementary data, Tables S2aPADSTS57](#)).^{12,13} Other CV risk calculators (e.g. SMART [Second manifestations of arterial disease] risk score, ADVANCE [Action in Diabetes and Vascular Disease: Preterax and Diamicron-MR Controlled Evaluation] risk score, SCORE2 [Systematic Coronary Risk Estimation 2], SCORE2-OP [Systematic Coronary Risk Estimation 2aPADSTCH Older Persons], ASCVD [AtheroSclerotic Cardiovascular Disease] risk score, U-Prevent, and lifetime risk calculators) may be considered at baseline for the assessment of CV risk, considering that cancer itself may increase the likelihood of CVD.^{19,23,30,31}

TableÂ 4

[Open in new tab](#)

Heart Failure AssociationaPADSTSInternational Cardio-Oncology Society baseline cardiovascular toxicity risk stratification

Baseline CV toxicity risk factors	Anthracycline chemotherapy	HER2-targeted therapies	VEGF inhibitors	BCR-ABL inhibitors	Multiple myeloma therapies	RAF and MEK inhibitors
Previous CVD						
HF/cardiomyopathy/CTRCD	VH	VH	VH	H	VH	VH
Severe VHD	H	H	aPADSTS	aPADSTS	aPADSTS	H
MI or PCI or CABG	H	H	VH	aPADSTS	aPADSTS	H
Stable angina	H	H	VH	aPADSTS	aPADSTS	H
Arterial vascular disease	aPADSTS	aPADSTS	VH	VH	VH	aPADSTS
Abnormal ankle-brachial pressure index	aPADSTS	aPADSTS	aPADSTS	H	aPADSTS	aPADSTS

PH	â€	â€	â€	H	â€	â€
Arterial thrombosis with TKI	â€	â€	â€	VH	â€	â€
Venous thrombosis (DVT/PE)	â€	â€	H	M2	VH	â€
Arrhythmia ^a	â€	M2	M2	M2	M2	M1
QTc â€ 480â€ms	â€	â€	H	H	â€	â€
450 â€ QTc < 480â€ms (men); 460 â€ QTc < 480â€ms (women)	â€	â€	M2	M2	â€	â€
Prior PI CV toxicity	â€	â€	â€	â€	VH	â€
Prior IMiD CV toxicity	â€	â€	â€	â€	H	â€
Cardiac imaging						
LVEF < 50%	H	H	H	H	H	H
LVEF 50â€54%	M2	M2	M2	â€	M2	M2
LV hypertrophy	â€	â€	â€	â€	M1	â€
Cardiac amyloidosis	â€	â€	â€	â€	VH	â€
Cardiac biomarkers						
Elevated baseline cTn ^b	M1	M2	M1	â€	M2	M2
Elevated baseline NP ^b	M1	M2	M1	â€	H	M2
Age and CVRF						
Age â€ 80 years	H	H	â€	â€	â€	M1
Age 65â€79 years	M2	M2	â€	â€	â€	M1
Age â€ 75 years	â€	â€	H	H	H	M1
Age 65â€74 years	â€	â€	M1	M2	M1	M1
Age â€ 60 years	â€	â€	â€	M1	â€	â€

CVD 10-year risk score > 20%	â	â	â	H	â	â
Hypertension ^c	M1	M1	H	M2	M1	M2
Chronic kidney disease ^d	M1	M1	M1	M1	M1	M1
Proteinuria	â	â	M1	â	â	â
DM ^e	M1	M1	M1	M1	M1	M1
Hyperlipidaemia ^f	â	â	M1	M1	M1	â
Family history of thrombophilia	â	â	â	M1	M1	â
Current cancer treatment						
Dexamethasone > 160â ^{g,h,i,j,k} mg/month	â	â	â	â	M1	â
Includes anthracycline before HER2-targeted therapy	â	M1 ^g	â	â	â	â
Previous exposure to						
Anthracycline	H	M2 ^h	H	â	H	H
Trastuzumab	â	VH	â	â	â	â
RT to left chest or mediastinum	H	M2	M1	â	M1	M2
Non-anthracycline chemotherapy	M1	â	â	â	â	â
Lifestyle risk factors						
Current smoker or significant smoking history	M1	M1	M1	H	M1	M1
Obesity (BMI > 30â ^{j,k} kg/m ²)	M1	M1	M1	M1	M1	M1
Baseline CV toxicity risk factors	Anthracycline chemotherapy	HER2-targeted	VEGF inhibitors	BCR-ABL	Multiple myeloma	RAF and MEK

		therapies		inhibitors	therapies	inhibitors
Previous CVD						
HF/cardiomyopathy/CTRCD	VH	VH	VH	H	VH	VH
Severe VHD	H	H	â€ADSTS	â€ADSTS	â€ADSTS	H
MI or PCI or CABG	H	H	VH	â€ADSTS	â€ADSTS	H
Stable angina	H	H	VH	â€ADSTS	â€ADSTS	H
Arterial vascular disease	â€ADSTS	â€ADSTS	VH	VH	VH	â€ADSTS
Abnormal ankle-brachial pressure index	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS	â€ADSTS
PH	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS	â€ADSTS
Arterial thrombosis with TKI	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS	â€ADSTS
Venous thrombosis (DVT/PE)	â€ADSTS	â€ADSTS	H	M2	VH	â€ADSTS
Arrhythmia ^a	â€ADSTS	M2	M2	M2	M2	M1
QTc â€HTJâ€ 480â€ADNELms	â€ADSTS	â€ADSTS	H	H	â€ADSTS	â€ADSTS
450 â€HTJâ€ QTc < 480â€ADNELms (men); 460 â€HTJâ€ QTc < 480â€ADNELms (women)	â€ADSTS	â€ADSTS	M2	M2	â€ADSTS	â€ADSTS
Prior PI CV toxicity	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS
Prior IMiD CV toxicity	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS
Cardiac imaging						
LVEF < 50%	H	H	H	H	H	H
LVEF 50â€ADSTS54%	M2	M2	M2	â€ADSTS	M2	M2
LV hypertrophy	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	M1	â€ADSTS
Cardiac amyloidosis	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS
Cardiac biomarkers						

Elevated baseline cTn ^b	M1	M2	M1	â€ADSTS	M2	M2
Elevated baseline NP ^b	M1	M2	M1	â€ADSTS	H	M2
Age and CVRF						
Age â€HTJ 80 years	H	H	â€ADSTS	â€ADSTS	â€ADSTS	M1
Age 65â€ADSTS79 years	M2	M2	â€ADSTS	â€ADSTS	â€ADSTS	M1
Age â€HTJ 75 years	â€ADSTS	â€ADSTS	H	H	H	M1
Age 65â€ADSTS74 years	â€ADSTS	â€ADSTS	M1	M2	M1	M1
Age â€HTJ 60 years	â€ADSTS	â€ADSTS	â€ADSTS	M1	â€ADSTS	â€ADSTS
CVD 10-year risk score > 20%	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS	â€ADSTS
Hypertension ^c	M1	M1	H	M2	M1	M2
Chronic kidney disease ^d	M1	M1	M1	M1	M1	M1
Proteinuria	â€ADSTS	â€ADSTS	M1	â€ADSTS	â€ADSTS	â€ADSTS
DM ^e	M1	M1	M1	M1	M1	M1
Hyperlipidaemia ^f	â€ADSTS	â€ADSTS	M1	M1	M1	â€ADSTS
Family history of thrombophilia	â€ADSTS	â€ADSTS	â€ADSTS	M1	M1	â€ADSTS
Current cancer treatment						
Dexamethasone > 160â€ADSTHEL mg/month	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	M1	â€ADSTS
Includes anthracycline before HER2-targeted therapy	â€ADSTS	M1 ^g	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS
Previous exposure to						
Anthracycline	H	M2 ^h	H	â€ADSTS	H	H
Trastuzumab	â€ADSTS	VH	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS

RT to left chest or mediastinum	H	M2	M1	â	M1	M2
Non-anthracycline chemotherapy	M1	â	â	â	â	â
Lifestyle risk factors						
Current smoker or significant smoking history	M1	M1	M1	H	M1	M1
Obesity (BMI > 30â ²)	M1	M1	M1	M1	M1	M1

AF, atrial fibrillation; BCR-ABL, breakpoint cluster regionâAbelson oncogene locus; BMI, body mass index; BNP, B-type natriuretic peptide; BP, blood pressure; CABG, coronary artery bypass graft; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; DVT, deep vein thrombosis; eGFR, estimated glomerular filtration rate; H, high risk; HbA1c, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; IMiD, immunomodulatory drugs; LV, left ventricular; LVEF, left ventricular ejection fraction; M, moderate risk; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides (including BNP and NT-proBNP); NT-proBNP, N-terminal pro-B-type natriuretic peptide; PCI, percutaneous coronary intervention; PE, pulmonary embolism; PH, pulmonary hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; RAF, rapidly accelerated fibrosarcoma; RT, radiotherapy; TKI, tyrosine kinase inhibitors; ULN, upper limit of normal; VEGFi, vascular endothelial growth factor inhibitors; VH, very high risk; VHD, valvular heart disease.

An expanded version of this table is provided in [Supplementary data, Tables S2âS7](#).

Risk level: Low risk = no risk factors OR one moderate1 risk factor; **moderate risk (M)** = moderate risk factors with a total of 2â4 points (Moderate 1 [M1] = 1 point; Moderate [M2] = 2 points); **high risk (H)** = moderate risk factors with a total of â5 points OR any high-risk factor; **very-high risk (VH)** = any very-high risk factor.

a

AF, atrial flutter, ventricular tachycardia, or ventricular fibrillation.

b

Elevated above the ULN of the local laboratory reference range.

c

Systolic BP > 140âmmHg or diastolic BP > 90âmmHg, or on treatment.

d

eGFR < 60âmL/min/1.73âm².

e

HbA1c > 7.0% or >53â€mmol/mol, or on treatment.

f

Non-high density lipoprotein cholesterol >3.8â€mmol/L (>145â€mg/dL) or on treatment.

g

High risk if anthracycline chemotherapy and trastuzumab delivered concurrently.

h

Previous malignancy (not current treatment protocol).

TableÂ 4

[Open in new tab](#)

Heart Failure Associationâ€International Cardio-Oncology Society baseline cardiovascular toxicity risk stratification

Baseline CV toxicity risk factors	Anthracycline chemotherapy	HER2-targeted therapies	VEGF inhibitors	BCR-ABL inhibitors	Multiple myeloma therapies	RAF and MEK inhibitors
Previous CVD						
HF/cardiomyopathy/CTRCD	VH	VH	VH	H	VH	VH
Severe VHD	H	H	â€	â€	â€	H
MI or PCI or CABG	H	H	VH	â€	â€	H
Stable angina	H	H	VH	â€	â€	H
Arterial vascular disease	â€	â€	VH	VH	VH	â€
Abnormal ankle-brachial pressure index	â€	â€	â€	H	â€	â€
PH	â€	â€	â€	H	â€	â€
Arterial thrombosis with TKI	â€	â€	â€	VH	â€	â€
Venous thrombosis (DVT/PE)	â€	â€	H	M2	VH	â€
Arrhythmia ^a	â€	M2	M2	M2	M2	M1
QTc â€â€480â€ms	â€	â€	H	H	â€	â€

450 ^a _{HTJ} QTc < 480 ^a _{PADNIEL} ms (men); 460 ^a _{HTJ} QTc < 480 ^a _{PADNIEL} ms (women)	^a _{PADSTS}	^a _{PADSTS}	M2	M2	^a _{PADSTS}	^a _{PADSTS}
Prior PI CV toxicity	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	VH	^a _{PADSTS}
Prior IMiD CV toxicity	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	H	^a _{PADSTS}
Cardiac imaging						
LVEF < 50%	H	H	H	H	H	H
LVEF 50 ^a _{PADSTS} –54%	M2	M2	M2	^a _{PADSTS}	M2	M2
LV hypertrophy	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	M1	^a _{PADSTS}
Cardiac amyloidosis	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	VH	^a _{PADSTS}
Cardiac biomarkers						
Elevated baseline cTn ^b	M1	M2	M1	^a _{PADSTS}	M2	M2
Elevated baseline NP ^b	M1	M2	M1	^a _{PADSTS}	H	M2
Age and CVRF						
Age ^a _{HTJ} ≥ 80 years	H	H	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	M1
Age 65 ^a _{PADSTS} –79 years	M2	M2	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	M1
Age ^a _{HTJ} ≥ 75 years	^a _{PADSTS}	^a _{PADSTS}	H	H	H	M1
Age 65 ^a _{PADSTS} –74 years	^a _{PADSTS}	^a _{PADSTS}	M1	M2	M1	M1
Age ^a _{HTJ} ≥ 60 years	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	M1	^a _{PADSTS}	^a _{PADSTS}
CVD 10-year risk score > 20%	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}	H	^a _{PADSTS}	^a _{PADSTS}
Hypertension ^c	M1	M1	H	M2	M1	M2
Chronic kidney disease ^d	M1	M1	M1	M1	M1	M1
Proteinuria	^a _{PADSTS}	^a _{PADSTS}	M1	^a _{PADSTS}	^a _{PADSTS}	^a _{PADSTS}
DM ^e	M1	M1	M1	M1	M1	M1
Hyperlipidaemia ^f	^a _{PADSTS}	^a _{PADSTS}	M1	M1	M1	^a _{PADSTS}

Family history of thrombophilia	â€ADSTs	â€ADSTs	â€ADSTs	M1	M1	â€ADSTs
Current cancer treatment						
Dexamethasone > 160â€ADSTs mg/month	â€ADSTs	â€ADSTs	â€ADSTs	â€ADSTs	M1	â€ADSTs
Includes anthracycline before HER2-targeted therapy	â€ADSTs	M1 ^g	â€ADSTs	â€ADSTs	â€ADSTs	â€ADSTs
Previous exposure to						
Anthracycline	H	M2 ^h	H	â€ADSTs	H	H
Trastuzumab	â€ADSTs	VH	â€ADSTs	â€ADSTs	â€ADSTs	â€ADSTs
RT to left chest or mediastinum	H	M2	M1	â€ADSTs	M1	M2
Non-anthracycline chemotherapy	M1	â€ADSTs	â€ADSTs	â€ADSTs	â€ADSTs	â€ADSTs
Lifestyle risk factors						
Current smoker or significant smoking history	M1	M1	M1	H	M1	M1
Obesity (BMI > 30â€ADSTs kg/m ²)	M1	M1	M1	M1	M1	M1

Baseline CV toxicity risk factors	Anthracycline chemotherapy	HER2-targeted therapies	VEGF inhibitors	BCR-ABL inhibitors	Multiple myeloma therapies	RAF and MEK inhibitors
Previous CVD						
HF/cardiomyopathy/CTRCD	VH	VH	VH	H	VH	VH
Severe VHD	H	H	â€ADSTs	â€ADSTs	â€ADSTs	H
MI or PCI or CABG	H	H	VH	â€ADSTs	â€ADSTs	H
Stable angina	H	H	VH	â€ADSTs	â€ADSTs	H
Arterial vascular disease	â€ADSTs	â€ADSTs	VH	VH	VH	â€ADSTs

Abnormal ankle-brachial pressure index	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS	â€ADSTS
PH	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS	â€ADSTS
Arterial thrombosis with TKI	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS	â€ADSTS
Venous thrombosis (DVT/PE)	â€ADSTS	â€ADSTS	H	M2	VH	â€ADSTS
Arrhythmia ^a	â€ADSTS	M2	M2	M2	M2	M1
QTc â€HTJâ€ 480â€ADNELms	â€ADSTS	â€ADSTS	H	H	â€ADSTS	â€ADSTS
450 â€HTJâ€ QTc < 480â€ADNELms (men); 460 â€HTJâ€ QTc < 480â€ADNELms (women)	â€ADSTS	â€ADSTS	M2	M2	â€ADSTS	â€ADSTS
Prior PI CV toxicity	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS
Prior IMiD CV toxicity	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	H	â€ADSTS
Cardiac imaging						
LVEF < 50%	H	H	H	H	H	H
LVEF 50â€ADSTS54%	M2	M2	M2	â€ADSTS	M2	M2
LV hypertrophy	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	M1	â€ADSTS
Cardiac amyloidosis	â€ADSTS	â€ADSTS	â€ADSTS	â€ADSTS	VH	â€ADSTS
Cardiac biomarkers						
Elevated baseline cTn ^b	M1	M2	M1	â€ADSTS	M2	M2
Elevated baseline NP ^b	M1	M2	M1	â€ADSTS	H	M2
Age and CVRF						
Age â€HTJâ€ 80 years	H	H	â€ADSTS	â€ADSTS	â€ADSTS	M1
Age 65â€ADSTS79 years	M2	M2	â€ADSTS	â€ADSTS	â€ADSTS	M1
Age â€HTJâ€ 75 years	â€ADSTS	â€ADSTS	H	H	H	M1
Age 65â€ADSTS74 years	â€ADSTS	â€ADSTS	M1	M2	M1	M1

Age $\geq$ 60 years	â	â	â	M1	â	â
CVD 10-year risk score > 20%	â	â	â	H	â	â
Hypertension ^c	M1	M1	H	M2	M1	M2
Chronic kidney disease ^d	M1	M1	M1	M1	M1	M1
Proteinuria	â	â	M1	â	â	â
DM ^e	M1	M1	M1	M1	M1	M1
Hyperlipidaemia ^f	â	â	M1	M1	M1	â
Family history of thrombophilia	â	â	â	M1	M1	â
Current cancer treatment						
Dexamethasone > 160â mg/month	â	â	â	â	M1	â
Includes anthracycline before HER2-targeted therapy	â	M1 ^g	â	â	â	â
Previous exposure to						
Anthracycline	H	M2 ^h	H	â	H	H
Trastuzumab	â	VH	â	â	â	â
RT to left chest or mediastinum	H	M2	M1	â	M1	M2
Non-anthracycline chemotherapy	M1	â	â	â	â	â
Lifestyle risk factors						
Current smoker or significant smoking history	M1	M1	M1	H	M1	M1
Obesity (BMI > 30â kg/m ²)	M1	M1	M1	M1	M1	M1

AF, atrial fibrillation; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BMI, body mass index; BNP, B-type natriuretic peptide; BP, blood pressure; CABG, coronary artery bypass graft; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; DVT, deep vein thrombosis; eGFR, estimated glomerular filtration rate; H, high risk; HbA1c, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; IMiD, immunomodulatory drugs; LV, left ventricular; LVEF, left ventricular ejection fraction; M, moderate risk; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides (including BNP and NT-proBNP); NT-proBNP, N-terminal pro-B-type natriuretic peptide; PCI, percutaneous coronary intervention; PE, pulmonary embolism; PH, pulmonary hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; RAF, rapidly accelerated fibrosarcoma; RT, radiotherapy; TKI, tyrosine kinase inhibitors; ULN, upper limit of normal; VEGFi, vascular endothelial growth factor inhibitors; VH, very high risk; VHD, valvular heart disease.

An expanded version of this table is provided in [Supplementary data, Tables S2âS7](#).

Risk level: **Low risk** = no risk factors OR one moderate risk factor; **moderate risk (M)** = moderate risk factors with a total of 2â4 points (Moderate 1 [M1] = 1 point; Moderate [M2] = 2 points); **high risk (H)** = moderate risk factors with a total of â¥5 points OR any high-risk factor; **very-high risk (VH)** = any very-high risk factor.

a

AF, atrial flutter, ventricular tachycardia, or ventricular fibrillation.

b

Elevated above the ULN of the local laboratory reference range.

c

Systolic BP > 140â160 mmHg or diastolic BP > 90â100 mmHg, or on treatment.

d

eGFR < 60â73 mL/min/1.73 m².

e

HbA1c > 7.0% or >53â60 mmol/mol, or on treatment.

f

Non-high density lipoprotein cholesterol >3.8â4.8 mmol/L (>145â180 mg/dL) or on treatment.

g

High risk if anthracycline chemotherapy and trastuzumab delivered concurrently.

h

Previous malignancy (not current treatment protocol).

Baseline risk assessment should be considered by the treating oncology or haematology team for all patients diagnosed with cancer who are scheduled to receive a cancer treatment identified to have a clinically significant level of CRT-CVT, or by a cardiologist if appropriate. In the case of patients scheduled to receive

anthracycline chemotherapy, the total planned cumulative anthracycline dose is also relevant, and a $\geq 250\text{ mg/m}^2$ of doxorubicin or equivalent should be considered higher risk ([Table 5](#)).³²

Table 5

[Open in new tab](#)

Anthracycline equivalence dose

	Doxorubicin	Epirubicin	Daunorubicin	Mitoxantrone	Idarubicin ^a
CV toxicity dose ratio	1	0.8	0.6	10.5	5
Isoequivalent dose	100 mg/m^2	125 mg/m^2	167 mg/m^2	9.5 mg/m^2	20 mg/m^2

	Doxorubicin	Epirubicin	Daunorubicin	Mitoxantrone	Idarubicin ^a
CV toxicity dose ratio	1	0.8	0.6	10.5	5
Isoequivalent dose	100 mg/m^2	125 mg/m^2	167 mg/m^2	9.5 mg/m^2	20 mg/m^2

This table refers to anthracycline equivalence dose using doxorubicin as a reference. Note that these isoequivalent doses are derived from paediatric CS.

CS, cancer survivors; CV, cardiovascular.

a
Data for idarubicin are based upon an estimated anticancer efficacy ratio, not derived from cardiotoxicity data. The CV toxicity dose ratio provides the value that should be used to multiply the dose of the anthracycline of interest to convert to isoequivalent doses of doxorubicin; e.g. to convert 125 mg/m^2 of epirubicin to doxorubicin isoequivalent, multiply the dose by 0.8 (125 $\text{mg/m}^2 \times 0.8 = 100\text{ mg/m}^2$ of doxorubicin).

Table 5

[Open in new tab](#)

Anthracycline equivalence dose

	Doxorubicin	Epirubicin	Daunorubicin	Mitoxantrone	Idarubicin ^a
CV toxicity dose ratio	1	0.8	0.6	10.5	5

Isoequivalent dose	100 ^a mg/m ²	125 ^a mg/m ²	167 ^a mg/m ²	9.5 ^a mg/m ²	20 ^a mg/m ²
	Doxorubicin	Epirubicin	Daunorubicin	Mitoxantrone	Idarubicin^a
CV toxicity dose ratio	1	0.8	0.6	10.5	5
Isoequivalent dose	100 ^a mg/m ²	125 ^a mg/m ²	167 ^a mg/m ²	9.5 ^a mg/m ²	20 ^a mg/m ²

This table refers to anthracycline equivalence dose using doxorubicin as a reference. Note that these isoequivalent doses are derived from paediatric CS.

CS, cancer survivors; CV, cardiovascular.

a
Data for idarubicin are based upon an estimated anticancer efficacy ratio, not derived from cardiotoxicity data. The CV toxicity dose ratio provides the value that should be used to multiply the dose of the anthracycline of interest to convert to isoequivalent doses of doxorubicin; e.g. to convert 125^a mg/m² of epirubicin to doxorubicin isoequivalent, multiply the dose by 0.8 (125^a mg/m² × 0.8 = 100^a mg/m² of doxorubicin).

CV risk stratification results should be discussed with the patient and documented in clinical notes. This process will also enable future validation of these tools.

Cardiology referral (cardio-oncology programme or cardiologist with expertise in managing CVD in patients with cancer) is recommended for patients identified to be at high or very high risk for CTR-CVT at baseline ([Table 4](#)) to institute strategies to mitigate risk.³³ Patients at moderate risk can benefit from closer cardiac monitoring, strict management of traditional CVRF, and selected moderate-risk patients may also benefit from a cardio-oncology referral ([Figure 6](#)). Low-risk patients can be followed within the oncology programme with appropriate referral to cardio-oncology if a CTR-CVT emerges or new or uncontrolled CVRF appear.

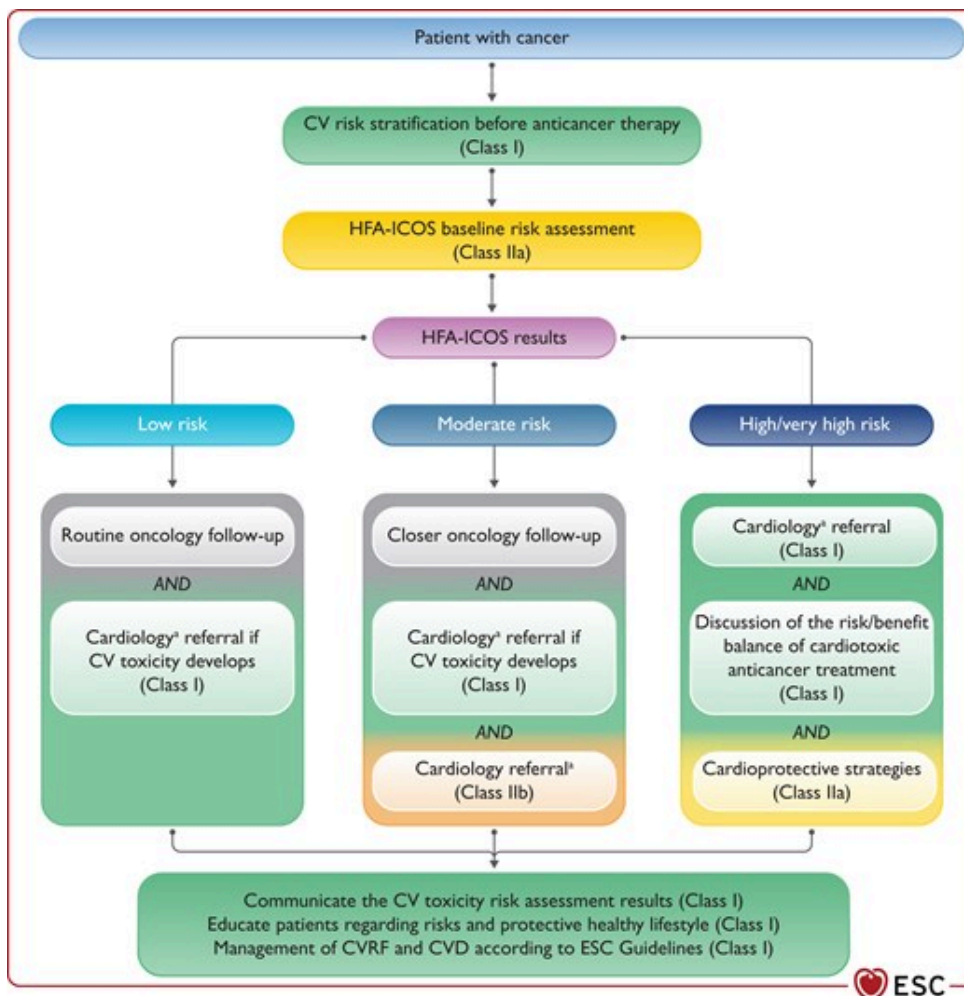


Figure 6

General cardio-oncology approach after Heart Failure Association and International Cardio-Oncology Society cardiovascular toxicity risk assessment.

CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ESC, European Society of Cardiology; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society. ^aCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

[Open in new tab](#)[Download slide](#)

Recommendation Table 1

[Open in new tab](#)

Recommendations for a general approach to cardiovascular toxicity risk categorization

Recommendations	Class ^a	Level ^b
CV toxicity risk stratification ^c before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer. ^{12,14,19,21,25,28,31}	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
The use of HFA-ICOS risk assessment should be considered to stratify CV toxicity risk in patients with cancer scheduled to receive cardiotoxic anticancer therapy. ¹²	IIa	C
It is recommended that patients categorized to be at low CV toxicity risk should proceed to anticancer therapy without delay.	I	C
In patients categorized at moderate CV toxicity risk, cardiology referral ^d may be considered. ^e	IIb	C
Cardiology referral ^d is recommended in high-risk and very high-risk patients before anticancer therapy. ^f	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral ^d is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment ^g who require potentially cardiotoxic anticancer therapy.	I	C

© ESC 2022

CV, cardiovascular; CVD, CV disease; ECG, electrocardiogram; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LVEF, left ventricular ejection fraction; TTE, transthoracic echocardiography; ULN, upper limit of normal; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding clinical history and physical examination, ECG, general blood test, HbA1c, lipid profile, and cardiac serum biomarkers and/or TTE (according to cancer drug type and CV toxicity risk).

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

^eWithout delaying cancer treatments.

^fUnless there is an oncology emergency requiring immediate cancer treatment.

^gModerate-to-severe pre-existing CVDs or new abnormal findings (baseline cardiac serum biomarkers > ULN, LVEF ≤ 50%, GLS under normal local values, previously

undiagnosed moderate-to-severe myocardial, pericardial, or VHDs, abnormal baseline ECG).

Recommendations	Class ^a	Level ^b
CV toxicity risk stratification ^c before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer. ^{12,14,19,21,25,28,31}	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
The use of HFA-ICOS risk assessment should be considered to stratify CV toxicity risk in patients with cancer scheduled to receive cardiotoxic anticancer therapy. ¹²	IIa	C
It is recommended that patients categorized to be at low CV toxicity risk should proceed to anticancer therapy without delay.	I	C
In patients categorized at moderate CV toxicity risk, cardiology referral ^d may be considered. ^e	IIb	C
Cardiology referral ^d is recommended in high-risk and very high-risk patients before anticancer therapy. ^f	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral ^d is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment ^g who require potentially cardiotoxic anticancer therapy.	I	C

© ESC 2022

CV, cardiovascular; CVD, CV disease; ECG, electrocardiogram; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LVEF, left ventricular ejection fraction; TTE, transthoracic echocardiography; ULN, upper limit of normal; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding clinical history and physical examination, ECG, general blood test, HbA1c, lipid profile, and cardiac serum biomarkers and/or TTE (according to cancer drug type and CV toxicity risk).

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

^eWithout delaying cancer treatments.

^fUnless there is an oncology emergency requiring immediate cancer treatment.

^gModerate-to-severe pre-existing CVDs or new abnormal findings (baseline cardiac serum biomarkers > ULN, LVEF ≤ 50%, GLS under normal local values, previously

undiagnosed moderate-to-severe myocardial, pericardial, or VHDs, abnormal baseline ECG).

Recommendation Table 1

[Open in new tab](#)

Recommendations for a general approach to cardiovascular toxicity risk categorization

Recommendations	Class ^a	Level ^b
CV toxicity risk stratification ^c before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer. ^{12,14,19,21,25,28,31}	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
The use of HFA-ICOS risk assessment should be considered to stratify CV toxicity risk in patients with cancer scheduled to receive cardiotoxic anticancer therapy. ¹²	IIa	C
It is recommended that patients categorized to be at low CV toxicity risk should proceed to anticancer therapy without delay.	I	C
In patients categorized at moderate CV toxicity risk, cardiology referral ^d may be considered. ^e	IIb	C
Cardiology referral ^d is recommended in high-risk and very high-risk patients before anticancer therapy. ^f	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral ^d is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment ^g who require potentially cardiotoxic anticancer therapy.	I	C

© ESC 2022

CV, cardiovascular; CVD, CV disease; ECG, electrocardiogram; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LVEF, left ventricular ejection fraction; TTE, transthoracic echocardiography; ULN, upper limit of normal; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding clinical history and physical examination, ECG, general blood test, HbA1c, lipid profile, and cardiac serum biomarkers and/or TTE (according to cancer drug type and CV toxicity risk).

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

^eWithout delaying cancer treatments.

^fUnless there is an oncology emergency requiring immediate cancer treatment.

^gModerate-to-severe pre-existing CVDs or new abnormal findings (baseline cardiac serum biomarkers > ULN, LVEF ≤ 50%, GLS under normal local values, previously

undiagnosed moderate-to-severe myocardial, pericardial, or VHDs, abnormal baseline ECG).

Recommendations	Class ^a	Level ^b
CV toxicity risk stratification ^c before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer. ^{12,14,19,21,25,28,31}	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
The use of HFA-ICOS risk assessment should be considered to stratify CV toxicity risk in patients with cancer scheduled to receive cardiotoxic anticancer therapy. ¹²	IIa	C
It is recommended that patients categorized to be at low CV toxicity risk should proceed to anticancer therapy without delay.	I	C
In patients categorized at moderate CV toxicity risk, cardiology referral ^d may be considered. ^e	IIb	C
Cardiology referral ^d is recommended in high-risk and very high-risk patients before anticancer therapy. ^f	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral ^d is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment ^g who require potentially cardiotoxic anticancer therapy.	I	C

© ESC 2022

CV, cardiovascular; CVD, CV disease; ECG, electrocardiogram; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LVEF, left ventricular ejection fraction; TTE, transthoracic echocardiography; ULN, upper limit of normal; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding clinical history and physical examination, ECG, general blood test, HbA1c, lipid profile, and cardiac serum biomarkers and/or TTE (according to cancer drug type and CV toxicity risk).

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

^eWithout delaying cancer treatments.

^fUnless there is an oncology emergency requiring immediate cancer treatment.

^gModerate-to-severe pre-existing CVDs or new abnormal findings (baseline cardiac serum biomarkers > ULN, LVEF ≤ 50%, GLS under normal local values, previously

4.2. History and clinical examination

A careful clinical history and physical examination is recommended as part of the baseline risk assessment. Oncology patients can be divided into two cohorts with respect to the presence or absence of pre-existing CVD. A primary prevention strategy can be considered in patients without previous CVD or CTR-CVT while secondary prevention includes interventions in patients with prior or active CVD or previous CTR-CVT.¹²

Reviewing traditional risk factors for CVD is recommended. Where present, the efficacy of treatment and control of these modifiable risk factors should be determined to ensure optimal control during cancer therapy.^{4,34} Although recent SCORE2 and SCORE2-OP¹⁹ tables are not focused on patients with cancer, risk calculation is recommended for patients with cancer >40 years of age (unless they are automatically categorized as being at high risk or very high risk based on documented CVD, diabetes mellitus [DM], kidney disease, or a highly elevated single risk factor) as a reference to optimize CVRF treatment goals.^{19,31,35} A family history of premature CVD should be considered because genetic abnormalities associated with CVD may predispose patients with cancer to a higher risk of CTR-CVT.^{36,38} Lifestyle factors such as smoking, alcohol consumption, sedentary lifestyle, exposure to pollution, and frailty are important shared risk factors for both cancer and CVD. Information on prior history of cancer, cardiotoxic cancer therapies, and their respective doses should be collected. Patients should be asked about typical cardiac symptoms (e.g. chest pain with activity, dyspnoea on exertion, orthopnoea, palpitations, and peripheral oedema), which can guide clinical examination and investigations. Physical examination should document vital signs and look for potential indicators of undiagnosed CVD such as HF, pericardial disease, VHD, and arrhythmias.^{39,42}

The second scenario is secondary prevention in patients with a prior history of CVD. These patients with cancer are potentially at high or very high risk of future CV events,¹² and require a more comprehensive clinical evaluation of their CVD, its severity, and prior and current treatments. Depending on the type and severity of CVD, additional investigations⁴⁰ including resting or stress echocardiography, cardiac magnetic resonance (CMR), nuclear perfusion imaging, and coronary computed tomography angiography (CCTA)⁴¹ may be indicated to determine risk status. Identifying prior CVD should not automatically be a reason to withhold cancer therapy but considered an opportunity to optimize CV risk prior to and during treatment. Risk/benefit discussions should include the patient, oncologist or haematologist, and⁴² where available⁴³ a specialized cardio-oncology service.

Additional factors that add to the complexity of baseline CV risk assessment are the cancer type and prognosis, and type, duration, and intensity of cancer treatment (*FigureÂ 1*).^{4,12,43} Clinical history, physical examination features, and treatment-related risk factors that contribute to CTR-CVT for various cancer therapies are summarized in *Supplementary data, TableÂ S8*. These risk factors should be collected and considered along with baseline ECG, cardiac serum biomarkers, and cardiac imaging tests (summarized in *FigureÂ 7*) to complete baseline CTR-CVT evaluation.



Figure 7

Baseline screening recommendations for patients with cancer treated with potentially cardiotoxic drugs.

3D, three-dimensional; ADT, androgen deprivation therapy; AL-CA, amyloid light-chain cardiac amyloidosis; BC, breast cancer; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BNP, B-type natriuretic peptide; BTK, Bruton tyrosine kinase; CAR-T, chimeric antigen receptor T cell; CDK, cyclin-dependent kinase; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; LVEF, left ventricular ejection fraction; MEK, mitogen-activated extracellular signal-regulated kinase; NP, natriuretic peptides (including BNP and NT-proBNP); NT-proBNP, N-terminal pro-B-type natriuretic peptide; PI, proteasome inhibitors; RAF, rapidly accelerated fibrosarcoma; RT, radiotherapy; TIL, tumour-infiltrating lymphocytes; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography; VEGFi, vascular endothelial growth factor inhibitors.

^aIncluding patients scheduled to receive ADT for prostate cancer, CDK 4/6 inhibitors, endocrine hormone

therapy for BC and anaplastic lymphoma kinase inhibitors. ^bTTE is recommended as the first-line modality for the assessment of cardiac function. 3D echocardiography is recommended to measure LVEF. GLS is recommended in all patients with cancer having echocardiography, if available. CMR should be considered when echocardiography is unavailable or not diagnostic. ^cBaseline cTn measurement should be considered (Class IIa, Level A) in low- and moderate-risk patients post-anthracycline chemotherapy but prior to starting HER2-targeted therapies. Baseline NP and cTn measurement may be considered (Class IIb, Level C) in low- and moderate-risk patients. ^dBaseline echocardiography is recommended in patients scheduled to receive dasatinib (Class I, Level C). ^eNP and cTn measurements are recommended at baseline in patients with AL-CA (Class I, Level B).

[Open in new tab](#)[Download slide](#)

4.3. Electrocardiogram

A baseline 12-lead ECG is a readily available test that can provide important clues to underlying CVD. ECG evidence of chamber enlargement, conduction abnormalities, arrhythmias, ischaemia, or prior myocardial infarction (MI), and low voltages should be interpreted in the clinical context. A baseline ECG is recommended prior to starting a cancer treatment known to cause QTc prolongation.^{44a,49} Measurement of QTc using the Fridericia correction (QTcF) is recommended.^{44a,48} When baseline QTcF prolongation is recognized, the correction of reversible causes and the identification of genetic conditions that prolong QT is recommended (see [Section 6.4.2](#)).⁴⁵

Left atrial enlargement on baseline ECG before ibrutinib has been shown to be a predictor for the development of atrial fibrillation (AF) during chemotherapy.^{50,51} The presence of atrioventricular (AV) conduction delays and premature atrial complexes are associated with the development of atrial arrhythmias in patients undergoing autologous haematopoietic stem cell transplantation (HSCT).⁵²

Recommendation Table 2

[Open in new tab](#)

Recommendations for electrocardiogram baseline assessment

Recommendations	Class ^a	Level ^b
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, ^c referral to a cardiologist ^d is recommended.	I	C

© ESC 2022

AF, atrial fibrillation; CV, cardiovascular; ECG, electrocardiogram; LV, left ventricular; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cAdvanced conduction disease (left bundle branch block, right bundle branch block, second degree heart block, severe first degree heart block with a PR interval >300 ms); Q waves in two or more contiguous leads; LV hypertrophy; AF/atrial flutter if previously undiagnosed; QTc prolongation using Fridericia correction formula ($QTcF = QT / \sqrt[3]{RR}$) >450 ms for men and >460 ms for women or other ECG abnormality raising concern.

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

Recommendations	Class ^a	Level ^b
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, ^c referral to a cardiologist ^d is recommended.	I	C

© ESC 2022

AF, atrial fibrillation; CV, cardiovascular; ECG, electrocardiogram; LV, left ventricular; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cAdvanced conduction disease (left bundle branch block, right bundle branch block, second degree heart block, severe first degree heart block with a PR interval >300 ms); Q waves in two or more contiguous leads; LV hypertrophy; AF/atrial flutter if previously undiagnosed; QTc prolongation using Fridericia correction formula ($QTcF = QT / \sqrt[3]{RR}$) >450 ms for men and >460 ms for women or other ECG abnormality raising concern.

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

Recommendation Table 2

[Open in new tab](#)

Recommendations for electrocardiogram baseline assessment

Recommendations	Class ^a	Level ^b
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, ^c referral to a cardiologist ^d is recommended.	I	C

© ESC 2022

AF, atrial fibrillation; CV, cardiovascular; ECG, electrocardiogram; LV, left ventricular; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cAdvanced conduction disease (left bundle branch block, right bundle branch block, second degree heart block, severe first degree heart block with a PR interval >300 ms); Q waves in two or more contiguous leads; LV hypertrophy; AF/atrial flutter if previously undiagnosed; QTc prolongation using Fridericia correction formula ($QTcF = QT / \sqrt[3]{RR}$) >450 ms for men and >460 ms for women or other ECG abnormality raising concern.

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

Recommendations	Class ^a	Level ^b
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, ^c referral to a cardiologist ^d is recommended.	I	C

© ESC 2022

AF, atrial fibrillation; CV, cardiovascular; ECG, electrocardiogram; LV, left ventricular; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cAdvanced conduction disease (left bundle branch block, right bundle branch block, second degree heart block, severe first degree heart block with a PR interval >300 ms); Q waves in two or more contiguous leads; LV hypertrophy; AF/atrial flutter if previously undiagnosed; QTc prolongation using Fridericia correction formula ($QTcF = QT / \sqrt[3]{RR}$) >450 ms for men and >460 ms for women or other ECG abnormality raising concern.

^dCardio-oncology referral is recommended when available; alternatively, the patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

4.4. Cardiac serum biomarkers

The literature on the use of biomarkers for CTR-CVT risk stratification before cancer therapy is limited, and recommendations are mostly based on expert opinion.^{12,43,53a,55} Four recent position papers based on collaboration among the Cardio-Oncology Study Group of the HFA of the ESC, the ESC-CCO, and ICOS

have suggested that measurement of cardiac serum biomarkers—cardiac troponin (cTn) I or T and natriuretic peptides (NP) (e.g. B-type natriuretic peptide [BNP] or N-terminal pro-BNP [NT-proBNP])—help in baseline CV risk stratification of patients scheduled for cancer therapies including anthracyclines, human epidermal receptor 2 (HER2)-targeted therapies, vascular endothelial growth factor (VEGF) inhibitors (VEGFi), proteasome inhibitors (PI), immune checkpoint inhibitors (ICI), chimeric antigen receptor T cell (CAR-T) and tumour-infiltrating lymphocytes (TIL) therapies, allowing identification of those who may benefit from cardioprotective therapy.^{12,43,53,54} Baseline cardiac serum biomarker measurements are required if the degree of change in the biomarkers is to be used to identify subclinical cardiac injury during cancer treatment.

A few studies of paediatric and adult patients requiring anthracycline chemotherapy have reported that patients with cancer with an increased cTn before treatment were more likely to develop CTRCD.^{56a,58} However, most published studies have not reported on the prognostic value of baseline cTn measurements, possibly due to the low prevalence of patients with previous CVD or CVRF in these studies.^{55,59,60} A study of 251 women receiving trastuzumab for early HER2-positive breast cancer (BC) reported that 19% of the patients who developed cardiac dysfunction during trastuzumab therapy had positive ultrasensitive troponin I at baseline ($>80\text{a}^{\text{FADCH}}\text{ng/L}$).⁶¹ Furthermore, baseline high cTnI level was a predictor of lack of recovery despite optimal HF therapy.⁶¹ These findings have been confirmed in a subsequent study of 533 patients with BC who had serial high-sensitivity cTn (hs-cTn) I and T measurements during trastuzumab therapy.⁶² Increased baseline cTn ($>40\text{a}^{\text{FADCH}}\text{ng/L}$ and $>14\text{a}^{\text{FADCH}}\text{ng/L}$ for hs-cTnI and hs-cTnT, respectively) was associated with a four-fold risk of developing LV dysfunction (LVD).⁶² However, given the high proportion of patients with previous anthracycline exposure in both studies, these elevated cTn levels are not a true baseline as they reflect pre-trastuzumab but post-anthracycline chemotherapy. It is unclear whether pre-treatment cTn levels will be predictive of LVD in patients before any treatment, or for those BC patients treated with trastuzumab without prior anthracyclines.

NP are another potential biomarker for CV risk stratification. Several studies have shown the role of NP measurement at baseline or NP changes to predict future CTR-CVT.^{63a,65} In patients with multiple myeloma (MM), pre-treatment NP may be a predictive marker for subsequent CV adverse events. In 109 patients with relapsed MM, BNP $>100\text{a}^{\text{FADCH}}\text{pg/mL}$ or NT-proBNP $>125\text{a}^{\text{FADCH}}\text{pg/mL}$ levels before initiation of carfilzomib were associated with an odds ratio of 10.8 for subsequent CV adverse events.⁶⁶ Therefore, baseline NP measurement is recommended in high- and very high-risk patients and should be considered in low- and moderate-risk patients before PI treatment.

Baseline elevated values of CV functional peptides (including NT-proBNP) and hs-cTnT were strongly related to all-cause mortality in 555 patients with different types of tumours, suggesting that the presence of a subclinical myocardial injury might be directly linked to disease progression.⁶⁷ However, in the CARDIOTOX (CARDIOvascular TOXicity induced by cancer-related therapies) registry, in 855 patients treated with a range of oncological treatments, including radiotherapy (RT), both NT-proBNP and cTn elevation at baseline were not associated with the development of severe CTRCD (LVEF $<40\%$ or clinical HF).⁶⁸

There has also been interest in other novel biomarkers for CTR-CVT risk stratification before cancer treatment; however, the literature is limited. Candidates include myeloperoxidase, C-reactive protein, galectin-3, arginine[†], nitric oxide metabolites, growth differentiation factor-15, placental growth factor, fms-like tyrosine kinase-1, micro-ribonucleic acids, and immunoglobulin E.^{60,69a†,71} Currently, there is no evidence to support routine measurement of these novel biomarkers and more research is required.

Recommendation Table 3

[Open in new tab](#)

Recommendation for cardiac biomarker assessment prior to potentially cardiotoxic therapies

Recommendation	Class ^a	Level ^b	© ESC 2022
Baseline measurement of NP ^c and/or cTn ^d is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD. ^{e,53,55}	I	C	

cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cNPs including B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide.

^dcTn includes any of troponin I, troponin T, or hs-cTnT.

^eSpecific recommendations for baseline cardiac biomarkers in patients with cancer at low, moderate, high, and very high risk of cancer therapy-related cardiovascular toxicity are included in [Section 5](#).

Recommendation	Class ^a	Level ^b
Baseline measurement of NP ^c and/or cTn ^d is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD. ^{e,53,55}	I	C

© ESC 2022

cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cNPs including B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide.

^dcTn includes any of troponin I, troponin T, or hs-cTnT.

^eSpecific recommendations for baseline cardiac biomarkers in patients with cancer at low, moderate, high, and very high risk of cancer therapy-related cardiovascular toxicity are included in [Section 5](#).

Recommendation Table 3

[Open in new tab](#)

Recommendation	Class ^a	Level ^b
Baseline measurement of NP ^c and/or cTn ^d is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD. ^{e,53,55}	I	C

© ESC 2022

cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cNPs including B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide.

^dcTn includes any of troponin I, troponin T, or hs-cTnT.

^eSpecific recommendations for baseline cardiac biomarkers in patients with cancer at low, moderate, high, and very high risk of cancer therapy-related cardiovascular toxicity are included in [Section 5](#).

Recommendation	Class ^a	Level ^b
Baseline measurement of NP ^c and/or cTn ^d is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD. ^{e,53,55}	I	C

© ESC 2022

cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cNPs including B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide.

^dcTn includes any of troponin I, troponin T, or hs-cTnT.

^eSpecific recommendations for baseline cardiac biomarkers in patients with cancer at low, moderate, high, and very high risk of cancer therapy-related cardiovascular toxicity are included in [Section 5](#).

4.5. Cardiovascular imaging

CV imaging has an important role in identifying patients with subclinical CVD, determining the degree of pre-existing cardiac comorbidity prior to decisions regarding cancer therapy, and serves as a reference for identification of changes during treatment and long-term follow-up.^{12,54,72a,74} Transthoracic echocardiography (TTE) is the preferred imaging technique for baseline risk stratification as it provides quantitative assessment of LV and right ventricular (RV) function, chamber dilation, LV hypertrophy, regional wall motion abnormalities, diastolic function, VHD, pulmonary arterial pressure (PAP), and pericardial disease, which may influence the therapeutic decision.^{22,72} Suggestions for the components of a baseline echocardiography study are provided in [Figure 8](#).

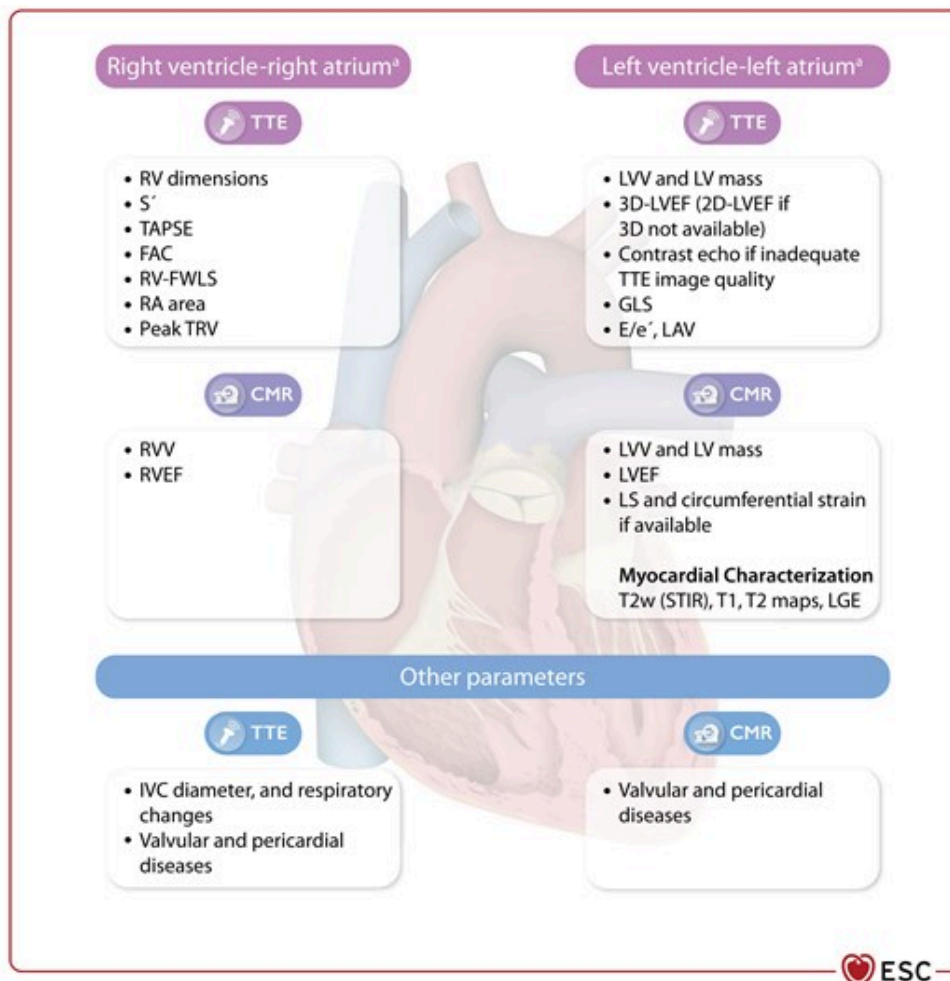


Figure 8

Recommended transthoracic echocardiography and cardiac magnetic resonance imaging parameters in the evaluation of patients with cancer.

2D, two-dimensional; 3D, three-dimensional; BP, blood pressure; CMR, cardiac magnetic resonance; E, mitral inflow early diastolic velocity obtained by pulsed wave; ea_{FAC2} , early diastolic velocity of the mitral annulus obtained by tissue doppler imaging; echo, echocardiography; FAC, fractional area change; FWLS, free wall longitudinal strain; GLS, global longitudinal strain; IVC, inferior vena cava; LAV, left atrial volume; LGE, late gadolinium enhancement; LS, longitudinal strain; LV, left ventricular; LVEF, left ventricular ejection fraction; LVV, left ventricular volume; RA, right atrial; RV, right ventricular; RVEF, right ventricular ejection fraction; RVV, right ventricular volume; sa_{FAC2} , systolic velocity of tricuspid annulus obtained by doppler tissue imaging; STIR, short tau inversion recovery; TAPSE, tricuspid annular plane systolic excursion; TTE, transthoracic echocardiography; TRV, tricuspid regurgitation velocity. ^aChanges in systemic arterial BP and loading conditions may influence cardiac function measurements.

[Open in new tab](#)[Download slide](#)

Current definitions of CTRCD are based on a reduction of LVEF and/or relative changes in global longitudinal strain (GLS) ([Table 3](#)). Three-dimensional (3D) echocardiography is the preferred echocardiography modality for the assessment of LVEF and cardiac volumes.^{54,75a}^{PARAGS:79} If 3D echocardiography is not feasible (e.g. unavailable or poor tracking), the modified two-dimensional (2D) Simpson's^{PARAGS} biplane method is recommended.^{80,81} In patients with inadequate TTE image quality, ultrasound-enhancing contrast agents should be added to improve evaluation of LV function and volumes if two or more LV segments are not well

visualized.⁸² Alternatively, in subjects with poor-quality echocardiography windows, when available, CMR should be considered ([Figure 8](#)).^{14,72,83,84} If TTE and CMR are both unavailable for the assessment of LVEF, multigated acquisition nuclear imaging (MUGA) can be considered as a third-line modality. MUGA scans should be avoided whenever possible due to radiation exposure and the inability to obtain other important information (e.g. VHD, PAP, or GLS).

Baseline LVEF and GLS are recommended in all patients evaluated with TTE before cardiotoxic cancer treatment initiation to stratify CTR-CVT risk and to identify significant changes during treatment.^{8,64} Changes in loading conditions occur frequently during chemotherapy (e.g. volume increase due to intravenous [i.v.] fluids, volume loss due to vomiting or diarrhoea, blood pressure [BP] and heart rate changes with pain or stress) and may affect cardiac volumes, LVEF, and GLS quantification. Systemic arterial BP measurement is recommended with all resting TTE as it can influence cardiac function measurements and should be recorded on the TTE report. A baseline borderline (50–54%) or reduced (<50%) LVEF is a risk factor for future CTR-CVT from most cardiotoxic cancer therapies, in particular with anthracyclines or trastuzumab.^{12,24,74} Increased baseline indexed LV end-diastolic volume can be a predictor of major CV events (symptomatic HF or cardiac death) during anthracycline chemotherapy in patients with preserved LVEF.⁸⁵

A normal LVEF does not exclude CTRCD and deformation parameters can detect early systolic impairment with sufficient test reliability.^{86,89} Determination of GLS using speckle tracking is recommended at baseline, using three apical views,⁹⁰ particularly in moderate- and high-risk patients. Baseline GLS can predict LVD^{89a,94} in patients receiving anthracyclines and/or trastuzumab. Strain measurements may be subject to inter-vendor variability⁹⁵ and serial GLS measurement for each patient is recommended to be performed using the same machine/software. A median GLS change of 13.6% predicted a future fall in LVEF with a 95% upper limit of GLS reduction of 15%.⁹³ Using the 15% cut-off improves specificity and is therefore the threshold recommended when monitoring GLS during cancer therapy. Global circumferential strain⁹⁶ has been reported to identify patients at risk of CTRCD, but data are currently insufficient to recommend its use routinely. Baseline LV diastolic function may be associated with a small risk of subsequent systolic dysfunction, especially with anthracyclines and trastuzumab, although the evidence is not consistent.^{97,98} Chest computed tomography (CT) or CMR may be helpful for identifying subclinical CVD such as coronary calcium or intracardiac masses on readily available routine imaging performed for cancer staging.⁹⁹

In the secondary prevention setting or patients with symptoms or signs of pre-existing CVD, a careful evaluation should begin with a comprehensive TTE.⁷³ This is both to obtain baseline assessment as in the primary prevention setting and to determine the severity of the underlying CVD. In case of poor-quality or uninterpretable TTE images, or if a specific CVD is identified (e.g. hypertrophic cardiomyopathy), CMR should be considered for further risk assessment.

Functional imaging tests for myocardial ischaemia¹⁰⁰ including stress echocardiography, perfusion CMR, or nuclear myocardial perfusion imaging¹⁰¹ should be performed to assess for ischaemia in symptomatic patients (stable angina, limiting dyspnoea) if clinical suspicion of coronary artery disease (CAD) exists, especially prior to use of cancer therapies associated with vascular toxicity (e.g. fluoropyrimidines, VEGFi, breakpoint cluster region¹⁰² Abelson oncogene locus [BCR-ABL], tyrosine kinase inhibitors [TKI]).

Alternatively, in patients with low to intermediate pre-test probability of CAD, CCTA is a robust alternate modality with high sensitivity to rule out obstructive CAD.^{[100,101](#)}

Recommendation Table 4

[Open in new tab](#)

Recommendations for cardiac imaging modalities in patients with cancer

General	Class ^a	Level ^b
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer. ^{4,12,54,94}	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF. ^{77–79,89}	I	B
GLS is recommended in all patients with cancer having echocardiography, if available. ^{75,80,81,89,90,92,93,102,103}	I	C
CMR should be considered for the assessment of cardiac function when echocardiography is unavailable or non-diagnostic. ^{83,104,105}	IIa	C
MUGA may be considered when TTE is not diagnostic and CMR is not available. ^{106–108}	IIb	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies^c		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy. ^{d,54}	I	C

© ESC 2022

3D, three-dimensional; CMR, cardiac magnetic resonance; CTR-CVT, cancer therapy-related CV toxicity; CV, cardiovascular; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction; MUGA, multigated acquisition nuclear imaging; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cSpecific recommendations for baseline CV imaging in patients with cancer at low or moderate risk of CTR-CVT are included in [Section 5](#).

^dExcept asymptomatic patients referred to breakpoint cluster region-Abelson oncogene locus therapy (BCR-ABL) where baseline TTE should be considered (see [Figure 7](#) and [Section 5.5.5](#)).

General	Class ^a	Level ^b
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer. ^{4,12,54,94}	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF. ^{77–79,89}	I	B
GLS is recommended in all patients with cancer having echocardiography, if available. ^{75,80,81,89,90,92,93,102,103}	I	C
CMR should be considered for the assessment of cardiac function when echocardiography is unavailable or non-diagnostic. ^{83,104,105}	IIa	C
MUGA may be considered when TTE is not diagnostic and CMR is not available. ^{106–108}	IIb	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies^c		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy. ^{d,54}	I	C

3D, three-dimensional; CMR, cardiac magnetic resonance; CTR-CVT, cancer therapy-related CV toxicity; CV, cardiovascular; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction; MUGA, multigated acquisition nuclear imaging; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cSpecific recommendations for baseline CV imaging in patients with cancer at low or moderate risk of CTR-CVT are included in [Section 5](#).

^dExcept asymptomatic patients referred to breakpoint cluster region-Abelson oncogene locus therapy (BCR-ABL) where baseline TTE should be considered (see [Figure 7](#) and [Section 5.5.5](#)).

© ESC 2022

Recommendation Table 4

[Open in new tab](#)

Recommendations for cardiac imaging modalities in patients with cancer

General	Class ^a	Level ^b
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer. ^{4,12,54,94}	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF. ^{77–79,89}	I	B
GLS is recommended in all patients with cancer having echocardiography, if available. ^{75,80,81,89,90,92,93,102,103}	I	C
CMR should be considered for the assessment of cardiac function when echocardiography is unavailable or non-diagnostic. ^{83,104,105}	IIa	C
MUGA may be considered when TTE is not diagnostic and CMR is not available. ^{106–108}	IIb	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies^c		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy. ^{d,54}	I	C

© ESC 2022

3D, three-dimensional; CMR, cardiac magnetic resonance; CTR-CVT, cancer therapy-related CV toxicity; CV, cardiovascular; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction; MUGA, multigated acquisition nuclear imaging; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cSpecific recommendations for baseline CV imaging in patients with cancer at low or moderate risk of CTR-CVT are included in [Section 5](#).

^dExcept asymptomatic patients referred to breakpoint cluster region-Abelson oncogene locus therapy (BCR-ABL) where baseline TTE should be considered (see [Figure 7](#) and [Section 5.5.5](#)).

General	Class ^a	Level ^b
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer. ^{4,12,54,94}	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF. ^{77–79,89}	I	B
GLS is recommended in all patients with cancer having echocardiography, if available. ^{75,80,81,89,90,92,93,102,103}	I	C
CMR should be considered for the assessment of cardiac function when echocardiography is unavailable or non-diagnostic. ^{83,104,105}	IIa	C
MUGA may be considered when TTE is not diagnostic and CMR is not available. ^{106–108}	IIb	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies^c		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy. ^{d,54}	I	C

3D, three-dimensional; CMR, cardiac magnetic resonance; CTR-CVT, cancer therapy-related CV toxicity; CV, cardiovascular; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction; MUGA, multigated acquisition nuclear imaging; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cSpecific recommendations for baseline CV imaging in patients with cancer at low or moderate risk of CTR-CVT are included in [Section 5](#).

^dExcept asymptomatic patients referred to breakpoint cluster region-Abelson oncogene locus therapy (BCR-ABL) where baseline TTE should be considered (see [Figure 7](#) and [Section 5.5.5](#)).

© ESC 2022

4.6. Cardiopulmonary fitness assessment

Maximal cardiopulmonary exercise testing (CPET) assesses the integrative capacity of the CV system to transport oxygen and energy substrate to skeletal muscle during exercise,¹⁰⁹ described as cardiorespiratory fitness (CRF). CPET can therefore provide a more global assessment of CV health than organ-specific tools. CPET-derived CRF^a is typically measured as the peak rate of oxygen consumption^{110,111} or metabolic equivalents^{111,112} during exercise.^a It is one of the most robust predictors of CV health and longevity,^{113,114} and improves risk classification.¹¹⁵ Evidence for CPET pre-treatment is limited to pre-operative risk stratification particularly for patients with lung,¹²² colon,¹²³ and rectal¹²⁴ cancers. Whether CPET performed prior to cardiotoxic cancer therapies is prognostic of future CV events is unknown.

4.7. Cardiovascular risk evaluation before cancer surgery

Cancer surgery remains the primary treatment modality for many cancers. Cardio-oncology teams should be involved in pre-operative CV risk stratification to identify and provide appropriate management and surveillance of the potential risk factors.⁵

In patients undergoing oncological surgery, peri-operative cardiac complications are determined by patient-related risk factors, the tumour type, concomitant cancer therapies, and the expected surgical risk. To ensure safe cancer surgery, consultations should be directed at: (1) patients with previous significant or symptomatic CVD; (2) patients at high and very high CV toxicity risk, according to baseline HFA-ICOS risk assessment tools,¹² when adjuvant (post-surgery) cancer treatment is planned; and (3) patients who have received neoadjuvant (prior to surgery) cancer therapy that is potentially cardiotoxic. Pre-operative clinical evaluation should not delay surgery. Complementary tests required for the patients included in groups 1 and 2 should be guided by general ESC Guidelines.¹²⁵ However, in group 3 patients, the pre-operative evaluation should be aimed at confirming that no relevant events have occurred during CV monitoring ([Section 5](#)). [TableÂ 6](#) summarizes factors that could influence peri-operative risk during cancer surgery.

TableÂ 6

[Open in new tab](#)

Factors that could influence peri-operative risk during cancer surgery and preventive strategies

	Factors that could influence peri-operative risk during cancer surgery	Preventive strategies
Patient-related factors	• Lifestyle risk factors: smoking, obesity, sedentary lifestyle • Poorly controlled CVRF: hypertension, DM • Pre-existing CVD including CTR-CVT • Cardiac medications that increase peri-operative bleeding risk (e.g. antiplatelets and anticoagulants) • Historical primary malignancy • Current cancer type, stage and location	• Optimal management of CVRF and CVD (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)
Neoadjuvant cancer therapy	• Neoadjuvant cardiotoxic cancer treatments (see Section 5; especially anthracycline chemotherapy and/or trastuzumab, ICI, VEGFi, fluoropyrimidine, and thoracic RT) • Cancer treatments that increase peri-operative bleeding risk (e.g. VEGFi, BTK inhibitors)	• Ensure optimal CV monitoring of neoadjuvant therapy (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)

	• Thrombocytopaenia caused by cancer treatment	
	Factors that could influence peri-operative risk during cancer surgery	Preventive strategies
Patient-related factors	• Lifestyle risk factors: smoking, obesity, sedentary lifestyle • Poorly controlled CVRF: hypertension, DM • Pre-existing CVD including CTR-CVT • Cardiac medications that increase peri-operative bleeding risk (e.g. antiplatelets and anticoagulants) • Historical primary malignancy • Current cancer type, stage and location	• Optimal management of CVRF and CVD (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)
Neoadjuvant cancer therapy	• Neoadjuvant cardiotoxic cancer treatments (see Section 5; especially anthracycline chemotherapy and/or trastuzumab, ICI, VEGFi, fluoropyrimidine, and thoracic RT) • Cancer treatments that increase peri-operative bleeding risk (e.g. VEGFi, BTK inhibitors) • Thrombocytopaenia caused by cancer treatment	• Ensure optimal CV monitoring of neoadjuvant therapy (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)

ATE, arterial thromboembolism; BTK, Bruton tyrosine kinase; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ICI, immune checkpoint inhibitors; RT, radiotherapy; VEGFi, vascular endothelial growth factor inhibitors; VTE, venous thromboembolism.

Table 6

[Open in new tab](#)

Factors that could influence peri-operative risk during cancer surgery and preventive strategies

	Factors that could influence peri-operative risk during cancer surgery	Preventive strategies
Patient-related factors	• Lifestyle risk factors: smoking, obesity, sedentary lifestyle • Poorly controlled CVRF: hypertension, DM • Pre-existing CVD including CTR-CVT	• Optimal management of CVRF and CVD (Section 5)

	• Cardiac medications that increase peri-operative bleeding risk (e.g. antiplatelets and anticoagulants) • Historical primary malignancy • Current cancer type, stage and location	• Optimize VTE and ATE preventive strategies (Section 6)
Neoadjuvant cancer therapy	• Neoadjuvant cardiotoxic cancer treatments (see Section 5; especially anthracycline chemotherapy and/or trastuzumab, ICI, VEGFi, fluoropyrimidine, and thoracic RT) • Cancer treatments that increase peri-operative bleeding risk (e.g. VEGFi, BTK inhibitors) • Thrombocytopaenia caused by cancer treatment	• Ensure optimal CV monitoring of neoadjuvant therapy (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)

	Factors that could influence peri-operative risk during cancer surgery	Preventive strategies
Patient-related factors	• Lifestyle risk factors: smoking, obesity, sedentary lifestyle • Poorly controlled CVRF: hypertension, DM • Pre-existing CVD including CTR-CVT • Cardiac medications that increase peri-operative bleeding risk (e.g. antiplatelets and anticoagulants) • Historical primary malignancy • Current cancer type, stage and location	• Optimal management of CVRF and CVD (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)
Neoadjuvant cancer therapy	• Neoadjuvant cardiotoxic cancer treatments (see Section 5; especially anthracycline chemotherapy and/or trastuzumab, ICI, VEGFi, fluoropyrimidine, and thoracic RT) • Cancer treatments that increase peri-operative bleeding risk (e.g. VEGFi, BTK inhibitors) • Thrombocytopaenia caused by cancer treatment	• Ensure optimal CV monitoring of neoadjuvant therapy (Section 5) • Optimize VTE and ATE preventive strategies (Section 6)

ATE, arterial thromboembolism; BTK, Bruton tyrosine kinase; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ICI, immune checkpoint inhibitors; RT, radiotherapy; VEGFi, vascular endothelial growth factor inhibitors; VTE, venous thromboembolism.

4.8. Genetic testing

Candidate gene and genome-wide association studies have resulted in the identification of 40 candidate genes and single nucleotide polymorphisms associated with anthracycline-related cardiac dysfunction.^{37,126,128} It should be noted that with the advent of immunotherapies, germline genes may not be the only genetic predispositions to CTR-CVT. A study of patients with ICI-associated myocarditis identified that the selective clonal T-cell populations infiltrating the myocardium were identical to those present in tumours and skeletal muscle, with ribonucleic acid sequencing studies revealing expression of cardiac-specific genes in the tumour,¹²⁹ raising the intriguing possibility that somatic mutations in the tumour itself could contribute to CTR-CVT. A list of genetic variants associated with CVD during cancer therapy is provided ([Supplementary data, Table S9](#)) and has recently been reviewed.³⁸

Routine use of genetic testing for the assessment of CTR-CVT risk prior to initiation of cancer therapy is not currently recommended. In the future, a personalized genetic approach may help define individual susceptibility to CVD in patients with cancer and more research is required.

5. Prevention and monitoring of cardiovascular complications during cancer therapy

5.1. General principles

CTR-CVT risk may vary according to cancer type and stage, anticancer drugs, doses, and underlying comorbidities. Certain therapy combinations (drug-drug or drug-radiation) may have a synergistically toxic effect on the heart, possibly depending on the timing of these therapies (sequential or concomitant) and previous comorbidities. The pathophysiology of CTR-CVT is out of the scope of this

guideline and is extensively reviewed in the  ESC CardioMed textbook.¹³⁰

CVD and cancer share common modifiable and non-modifiable risk factors ([Figure 3](#)).^{4,131,132} The first step is to optimize lifestyle CVRF, smoking cessation, restricting alcohol consumption to a maximum of 100 g per week, and maintaining adequate physical activity.³⁰ Exercise prescription seems to be a promising treatment to counteract anticancer treatment side effects and different types of training can be prescribed during cancer therapy according to a patient's individual characteristics.¹³³ A healthy lifestyle decreases the risks of cancer, CVD, and transition from diagnosed cancer to subsequent CVD.^{134,135}

Poor CRF is associated with a higher prevalence of acute and chronic CTR-CVT and exercise positively impacts CRF during chemotherapy, although in a recent meta-analysis, the ability of exercise to prevent CTRCD is unclear.^{136,137} CVRF must be corrected with intensive treatment of arterial hypertension,¹³⁸

DM,¹³⁹ and dyslipidaemia,¹⁴⁰ and underlying CVD and modifiable comorbidities should be managed according to appropriate 2021 ESC Guidelines on CVD prevention in clinical practice ([Figure 9](#)).¹⁹

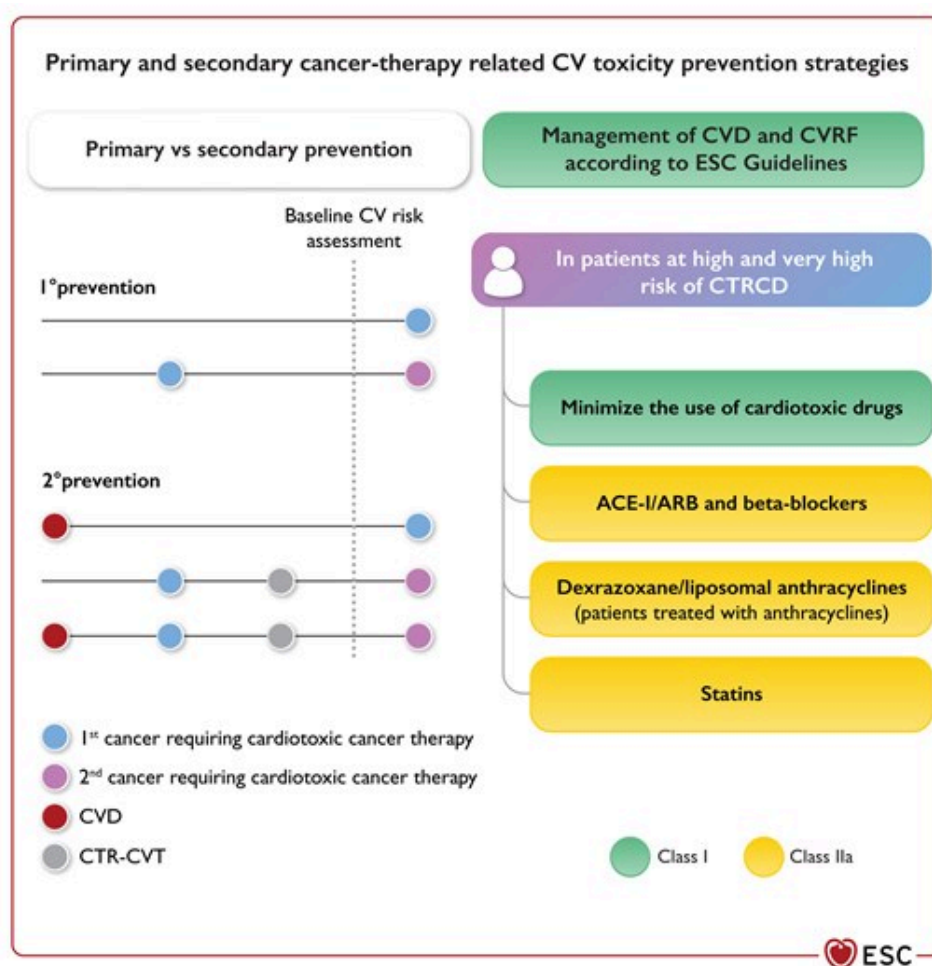


Figure 9

Primary and secondary cancer therapy-related cardiovascular toxicity prevention.

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; CTR-CVT, cancer therapy-related cardiovascular toxicity; CTRCD, cancer therapy-related cardiac dysfunction; ESC, European Society of Cardiology.¹² **Left panel** represents examples of five different primary or secondary prevention strategies definition based on the history of pre-existing CVD and/or prior CTR-CVT. **Right panel** describes general strategies to mitigate CTR-CVT risk in patients at high and very high risk of CTRCD.

[Open in new tab](#)[Download slide](#)

Special attention should also be paid to the polypharmacy frequently seen in patients with cancer, reducing the use of drugs that may interfere with cancer therapies to the essential and actively monitoring their CV side effects and drug-drug interactions.¹⁴¹ Electrolyte imbalances such as hypokalaemia and hypomagnesaemia should be corrected. The CV risk management plan should be shared with the cancer specialist team, the primary care physician, and the patient to coordinate treatment strategies.

5.2. Primary prevention strategies

Primary prevention of CTR-CVT aims to avoid or minimize the development of CV damage due to therapy in patients without CVD^{12,142} and requires a multidisciplinary team (MDT) discussion between oncologists and cardiologists for complex patients with cancer with multiple comorbidities.^{4,21,22,43,143,144}

5.2.1. Primary prevention of cancer therapy-related cardiovascular toxicity during anthracycline chemotherapy

Neurohormonal therapies during anthracycline chemotherapy (with or without subsequent trastuzumab treatment) reduced the risk of significant LVEF decline during follow-up in several small randomized controlled trials (RCTs) ([Supplementary data, Table S10](#)).¹⁴⁵ Recent meta-analyses including patients with cancer treated with anthracycline chemotherapy and HER2-targeted therapies reported that renin-angiotensin-aldosterone system blockers, beta-blockers, and mineralocorticoid receptor antagonists have a significant benefit in preventing LVEF reduction, but no statistical differences in the incidence of overt HF or other clinical outcomes were demonstrated ([Supplementary data, Table S11](#)).¹⁵⁵ This may be due, in part, to the fact that most trials included patients with a low baseline CTRCD risk and therefore larger RCTs are needed in high-risk populations.

From the oncological perspective, some strategies that have been investigated include managing anthracycline-related toxicity by adjusting the infusion time and dose intensity.¹⁶¹ Dexrazoxane and liposomal anthracyclines are currently approved in patients with high and very high CTRCD risk or who have already received high cumulative anthracycline doses.^{158,162} Dexrazoxane is protective against anthracycline-induced CTRCD. Currently, dexrazoxane is formally approved in adult patients with advanced or metastatic BC who have already received a minimum cumulative anthracycline dose of 300 mg/m² of doxorubicin or equivalent ([Table 5; Supplementary data, Table S12](#)).¹⁶³ In clinical practice, dexrazoxane infusion (dosage ratio dexrazoxane/doxorubicin is 10/1; e.g. 500 mg/m² dexrazoxane per 50 mg/m² doxorubicin) should be considered (at least 30 min prior to each anthracycline cycle) in adult patients with cancer scheduled to receive a high total cumulative anthracycline dose for curative treatment, and in patients with high and very high CTRCD risk (including those with pre-existing HF or low-normal or reduced LVEF) where anthracycline chemotherapy is deemed essential.¹⁶³

Pegylated and non-pegylated liposomal doxorubicin^{164,165,168} modify pharmacokinetics and tissue distribution without compromising antitumour efficacy. Pegylated and non-pegylated liposomal doxorubicin are approved for metastatic BC and pegylated liposomal doxorubicin is also approved for advanced ovarian cancer, acquired immune deficiency syndrome-related Kaposi sarcoma, and MM. In a recent meta-analysis of 19 trials, in both the adjuvant and metastatic context, liposomal doxorubicin was reported to be less cardiotoxic than conventional doxorubicin.¹⁶⁵ Liposomal daunorubicin is also available for acute leukaemia patients in place of daunorubicin when pre-existing LVD is present.^{164,165}

5.2.2. Primary prevention of radiation-induced cardiovascular toxicity

Primary prevention of RT-induced damage to the CV system depends on technological advances that allow improved targeting of RT delivery, thereby maintaining or increasing oncological efficacy while reducing CTR-CVT.^{169,170} Modern techniques strive to minimize the mean heart dose (MHD), either by shaping the dose distribution (intensity-modulated RT) or by using respiratory management (gating or breath-hold).^{171,172}

Proton therapy offers the potential to further decrease exposure to surrounding healthy organs.¹⁷³ However, complete cardiac avoidance is not always possible due to the proximity of the tumour (e.g. central lung tumours, mediastinal lymphomas, irradiation of the internal mammary chain in BC). In patients where RT only has a consolidating role and the risk of RT-induced CV injury is very high (e.g. due to baseline risk factors), a MDT is needed to consider the risk/benefit of RT.^{171,174}

There are no proven medical therapies to prevent RT-induced CV toxicity. One component of RT-induced CV toxicity is accelerating pre-existing CAD, and therefore tight control of CVRFs is recommended.

Recommendation Table 5

[Open in new tab](#)

Recommendations for primary prevention of cancer therapy-related cardiovascular toxicity

Recommendations	Class ^a	Level ^b
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, ^c during, and after cancer therapy. ¹⁹	I	C
Dexrazoxane should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{d,158}	IIa	B
Liposomal anthracyclines should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{e,164,165,168}	IIa	B

ACE-I or ARB and beta-blockers recommended for HF^f should be considered for primary prevention in high- and very high-risk patients receiving anthracyclines and/or anti-HER2 therapies.^{145,150,155–157,159,160,175}

Ila

B

ACE-I or ARB and beta-blockers recommended for HF^f should be considered for primary prevention in high- and very high-risk patients receiving targeted cancer therapies that may cause HF.^g

Ila

C

Statins should be considered for primary prevention in adult patients with cancer at high and very high CV toxicity risk.^{h,149,176–185}

Ila

B

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ESC, European Society of Cardiology; HER2, human epidermal receptor 2; HF, heart failure; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; MEK, mitogen-activated extracellular signal-regulated kinase; PI, proteasome inhibitors; RAF, rapidly accelerated fibrosarcoma; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

^dAs per the European Medicine Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

^eSee [Section 5.2.1](#) for specific liposomal doxorubicin type and malignancies.

^fCarvedilol (preferred beta-blocker for CV protection if there is no contraindication),¹⁸⁶ bisoprolol, controlled/extended-release metoprolol succinate and nebivolol.

^gVEGFi and bevacizumab, RAF inhibitor, MEK inhibitor, PI, dasatinib, ponatinib, and osimertinib.

^hAccording to HFA-ICOS risk assessment tools ([Section 4.1](#); [Table 4](#)).

Recommendations	Class ^a	Level ^b
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, ^c during, and after cancer therapy. ¹⁹	I	C
Dexrazoxane should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{d,158}	IIa	B
Liposomal anthracyclines should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{e,164,165,168}	IIa	B
ACE-I or ARB and beta-blockers recommended for HF ^f should be considered for primary prevention in high- and very high-risk patients receiving anthracyclines and/or anti-HER2 therapies. ^{145,150,155–157,159,160,175}	IIa	B
ACE-I or ARB and beta-blockers recommended for HF ^f should be considered for primary prevention in high- and very high-risk patients receiving targeted cancer therapies that may cause HF. ^g	IIa	C
Statins should be considered for primary prevention in adult patients with cancer at high and very high CV toxicity risk. ^{h,149,176–185}	IIa	B

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ESC, European Society of Cardiology; HER2, human epidermal receptor 2; HF, heart failure; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; MEK, mitogen-activated extracellular signal-regulated kinase; PI, proteasome inhibitors; RAF, rapidly accelerated fibrosarcoma; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

^dAs per the European Medicine Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

^eSee [Section 5.2.1](#) for specific liposomal doxorubicin type and malignancies.

^fCarvedilol (preferred beta-blocker for CV protection if there is no contraindication),¹⁸⁶ bisoprolol, controlled/extended-release metoprolol succinate and nebivolol.

^gVEGFi and bevacizumab, RAF inhibitor, MEK inhibitor, PI, dasatinib, ponatinib, and osimertinib.

^hAccording to HFA-ICOS risk assessment tools ([Section 4.1; Table 4](#)).

© ESC 2022

Recommendation Table 5

[Open in new tab](#)

Recommendations for primary prevention of cancer therapy-related cardiovascular toxicity

Recommendations	Class ^a	Level ^b
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, ^c during, and after cancer therapy. ¹⁹	I	C
Dexrazoxane should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{d,158}	IIa	B
Liposomal anthracyclines should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{e,164,165,168}	IIa	B

ACE-I or ARB and beta-blockers recommended for HF^f should be considered for primary prevention in high- and very high-risk patients receiving anthracyclines and/or anti-HER2 therapies.^{145,150,155–157,159,160,175}

Ila

B

ACE-I or ARB and beta-blockers recommended for HF^f should be considered for primary prevention in high- and very high-risk patients receiving targeted cancer therapies that may cause HF.^g

Ila

C

Statins should be considered for primary prevention in adult patients with cancer at high and very high CV toxicity risk.^{h,149,176–185}

Ila

B

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ESC, European Society of Cardiology; HER2, human epidermal receptor 2; HF, heart failure; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; MEK, mitogen-activated extracellular signal-regulated kinase; PI, proteasome inhibitors; RAF, rapidly accelerated fibrosarcoma; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

^dAs per the European Medicine Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

^eSee [Section 5.2.1](#) for specific liposomal doxorubicin type and malignancies.

^fCarvedilol (preferred beta-blocker for CV protection if there is no contraindication),¹⁸⁶ bisoprolol, controlled/extended-release metoprolol succinate and nebivolol.

^gVEGFi and bevacizumab, RAF inhibitor, MEK inhibitor, PI, dasatinib, ponatinib, and osimertinib.

^hAccording to HFA-ICOS risk assessment tools ([Section 4.1](#); [Table 4](#)).

Recommendations	Class ^a	Level ^b
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, ^c during, and after cancer therapy. ¹⁹	I	C
Dexrazoxane should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{d,158}	IIa	B
Liposomal anthracyclines should be considered in adult patients with cancer at high and very high CV toxicity risk when anthracycline chemotherapy is indicated. ^{e,164,165,168}	IIa	B
ACE-I or ARB and beta-blockers recommended for HF ^f should be considered for primary prevention in high- and very high-risk patients receiving anthracyclines and/or anti-HER2 therapies. ^{145,150,155–157,159,160,175}	IIa	B
ACE-I or ARB and beta-blockers recommended for HF ^f should be considered for primary prevention in high- and very high-risk patients receiving targeted cancer therapies that may cause HF. ^g	IIa	C
Statins should be considered for primary prevention in adult patients with cancer at high and very high CV toxicity risk. ^{h,149,176–185}	IIa	B

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CV, cardiovascular; CVD, CV disease; CVRF, CV risk factors; ESC, European Society of Cardiology; HER2, human epidermal receptor 2; HF, heart failure; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; MEK, mitogen-activated extracellular signal-regulated kinase; PI, proteasome inhibitors; RAF, rapidly accelerated fibrosarcoma; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

^dAs per the European Medicine Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

^eSee [Section 5.2.1](#) for specific liposomal doxorubicin type and malignancies.

^fCarvedilol (preferred beta-blocker for CV protection if there is no contraindication),¹⁸⁶ bisoprolol, controlled/extended-release metoprolol succinate and nebivolol.

^gVEGFi and bevacizumab, RAF inhibitor, MEK inhibitor, PI, dasatinib, ponatinib, and osimertinib.

^hAccording to HFA-ICOS risk assessment tools ([Section 4.1; Table 4](#)).

5.3. Secondary prevention strategies

Secondary prevention refers to interventions in patients with pre-existing CVD, including prior CTR-CVT, and new emerging CTR-CVT during cancer therapy. CVD and comorbidities should receive the optimal therapy before and during cancer therapy as discussed in previous sections. Regular clinical assessments, physical examinations, and CV investigations (including 12-lead ECG, TTE, and cardiac biomarkers) are recommended in patients receiving specific cardiotoxic cancer therapies, with the frequency of surveillance guided by baseline risk and the emergence of new CTR-CVT.^{5,12,33,53,54,187a,190}

Recommendation Table 6

[Open in new tab](#)

Recommendation for secondary prevention of cancer therapy-related cardiovascular toxicity

Recommendation	Class ^a	Level ^b	© ESC 2022
Management of CVD according to applicable ESC Guidelines is recommended before, ^c during, and after cancer therapy.	I	C	

CVD, cardiovascular disease; ESC, European Society of Cardiology.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

Recommendation	Class ^a	Level ^b	© ESC 2022
Management of CVD according to applicable ESC Guidelines is recommended before, ^c during, and after cancer therapy.	I	C	

CVD, cardiovascular disease; ESC, European Society of Cardiology.

^aClass of recommendation.

^bLevel of evidence.

^cWithout delaying cancer treatments.

Recommendation Table 6

[Open in new tab](#)

Recommendation for secondary prevention of cancer therapy-related cardiovascular toxicity

Recommendation	Class ^a	Level ^b
Management of CVD according to applicable ESC Guidelines is recommended before, ^c during, and after cancer therapy.	I	C

© ESC 2022

CVD, cardiovascular disease; ESC, European Society of Cardiology.
^aClass of recommendation.
^bLevel of evidence.
^cWithout delaying cancer treatments.

Recommendation	Class ^a	Level ^b
Management of CVD according to applicable ESC Guidelines is recommended before, ^c during, and after cancer therapy.	I	C

© ESC 2022

CVD, cardiovascular disease; ESC, European Society of Cardiology.
^aClass of recommendation.
^bLevel of evidence.
^cWithout delaying cancer treatments.

5.4. Cardiovascular surveillance during cancer therapies

A careful clinical evaluation and physical examination is recommended during cancer treatment to detect early signs and symptoms of CTR-CVT. ECG monitoring is required in patients at risk of cardiac arrhythmias according to specific drug protocols.

5.4.1. Cardiac serum biomarkers

During therapy, NP and cTn should be used for CTRCD screening and diagnosis and they may also serve to guide therapy.^{55,63,191aPADST:194} The release of cTn and NP differ for different cancer treatments. Therefore, an increase in biomarker level should be interpreted in the patient clinical context (cancer treatment timing and comorbidities).

It is important to consider that generally accepted cut-offs and reference values of CV biomarkers have not been established for patients with cancer or for those who receive cancer therapies. In addition, levels of NP and cTn may differ according to local laboratories and may be altered by many factors, including age, sex, renal function, obesity, infections, and comorbidities such as AF and pulmonary embolism (PE).^{53,63,195aPADST:197}

5.4.2. Cardiac imaging

Cardiac imaging plays a critical role in clinical decision-making during the cancer process.^{72,198} Imaging techniques^{aPADST:CH} particularly advanced echocardiography and CMRA^{aPADST:CH} facilitate early diagnosis and management of CTR-CVT.^{22,54,94} The frequency of cardiac imaging monitoring during therapy should be adapted according to the estimated baseline risk¹² and the expected CTR-CVT manifestation.⁵⁴ The cardiac

imaging technique used should be based on local expertise and availability, and the same imaging modality (i.e. 3D-TTE, 2D-TTE, CMR) is recommended throughout the entire treatment to decrease inter-technique variability.^{94,199,200} Cardiac imaging should be performed at any time if patients receiving cardiotoxic therapies present with new cardiac symptoms.

New definitions of CTRCD are presented in [Section 3](#).¹ Early recognition of asymptomatic CTRCD allows clinicians to incorporate cardioprotective therapy before there is a significant decline in LVEF, which may or may not be reversible, and also decreases the risk of interruptions in cancer therapy, which could otherwise affect patients' survival.^{22,43,72,94} For the diagnosis and management of asymptomatic CTRCD during cancer treatment, TTE^{22,43,72,94} including 3D-LVEF and GLS assessment^{22,43,72,94} is the preferred technique to detect and confirm cardiac dysfunction.^{72,83,93,102} GLS evaluation is particularly important in patients with low-normal LVEF to confirm or not asymptomatic myocardial damage.²⁰¹ It is recommended to use the same vendor to analyse GLS during cancer treatment to accurately compare values over time.⁷³ Therefore, a relative change in GLS has been suggested as the ideal tool to identify asymptomatic mild CTRCD.^{1,4,94} Different thresholds have been considered in the literature in recent years.^{93,202,203} Currently, a relative GLS decrease of >15% compared with baseline is the recommended threshold as it reflects the 95% upper limit in the meta-analysis of GLS to predict future significant LVEF reduction.⁹³ Using the 15% threshold will maximize specificity and minimize overdiagnosis of CTRCD and guide cardioprotective therapy.^{1,4,93}

In patients with poor TTE image quality or when TTE is not diagnostic, CMR should be considered, including fast strain-encoded CMR when available.^{105,204,205,206} MUGA can be considered as a third-line modality.

5.5. Cancer therapy-related cardiovascular toxicity monitoring protocols

5.5.1. Anthracycline chemotherapy

Anthracycline-induced CTRCD is a dose-dependent and cumulative process of variable onset that may present with symptomatic or asymptomatic CTRCD.⁴

[Figure 10](#) summarizes the recommended monitoring protocol during anthracycline therapy according to baseline CTRCD risk ([Table 4](#)). Clinical assessment combined with cardiac biomarkers (cTn and NP) and TTE (including 3D-LVEF and GLS when available) can identify both symptomatic and asymptomatic CTRCD with a reasonably high negative predictive value. This topic has been extensively reviewed in two recent HFA position statements.^{53,54} Classifying patients based on their risk of anthracycline-induced CV toxicity has allowed the early implementation of personalized preventive strategies ([Section 5.2.1](#)).¹⁴ Patients with pre-existing CVD should be treated with guideline-based medical therapy.^{14,19,207}



Figure 10

Cardiovascular toxicity monitoring in patients receiving anthracycline chemotherapy.

cTn, cardiac troponin; C, chemotherapy cycle; ECG, electrocardiogram; M, months; NP, natriuretic peptides; TTE, transthoracic echocardiography; tx, treatment. Biomarker and TTE assessment should ideally be performed before the corresponding anthracycline cycle (C1 to C6). ^aCardiac magnetic resonance should be considered for the assessment of cardiac function when TTE is unavailable or not diagnostic. In moderate-risk patients, TTE should be considered after a cumulative dose of $\geq 250 \text{ mg/m}^2$ of doxorubicin or equivalent. In low-risk patients, TTE may be considered after a cumulative dose of $\geq 250 \text{ mg/m}^2$ of doxorubicin or equivalent. ^bMeasurement of NP and/or cTn is recommended in all patients with cancer if these biomarkers are going to be used during treatment monitoring. cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion may be considered in low-risk patients (Class IIb, Level C). cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion should be considered in moderate-risk patients and in low-risk patients receiving a cumulative dose of $\geq 250 \text{ mg/m}^2$ of doxorubicin or equivalent (Class IIa, Level C).

[Open in new tab](#)[Download slide](#)

Recommendation Table 7

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended in all patients with cancer before anthracycline chemotherapy. ^{12,24,208–210}	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment. ²⁰⁸	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment. ^{24,208–210}	I	C
In moderate-risk patients, additional echocardiography should be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIa	C
In low-risk patients, additional echocardiography may be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIb	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy. ^{55,65,211}	I	B

Baseline measurement of NP and cTn should be considered in low- and moderate-risk patients prior to anthracycline chemotherapy. ²¹¹	IIa	C
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients. ^{55,175,211}	I	B
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion should be considered in moderate-risk patients and in low-risk patients receiving a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ^{55,59,212,213}	IIa	C
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion may be considered in low-risk patients. ^{55,59,212,213}	IIb	C

© ESC 2022

cTn, cardiac troponin; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended in all patients with cancer before anthracycline chemotherapy. ^{12,24,208–210}	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment. ²⁰⁸	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment. ^{24,208–210}	I	C
In moderate-risk patients, additional echocardiography should be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIa	C
In low-risk patients, additional echocardiography may be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIb	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy. ^{55,65,211}	I	B

Baseline measurement of NP and cTn should be considered in low- and moderate-risk patients prior to anthracycline chemotherapy. ²¹¹	IIa	C
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients. ^{55,175,211}	I	B
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion should be considered in moderate-risk patients and in low-risk patients receiving a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ^{55,59,212,213}	IIa	C
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion may be considered in low-risk patients. ^{55,59,212,213}	IIb	C

© ESC 2022

cTn, cardiac troponin; NP, natriuretic peptides; TTE, transthoracic echocardiography.
^aClass of recommendation.
^bLevel of evidence.
^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

Recommendation Table 7

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended in all patients with cancer before anthracycline chemotherapy. ^{12,24,208–210}	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment. ²⁰⁸	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment. ^{24,208–210}	I	C
In moderate-risk patients, additional echocardiography should be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIa	C
In low-risk patients, additional echocardiography may be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIb	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy. ^{55,65,211}	I	B

Baseline measurement of NP and cTn should be considered in low- and moderate-risk patients prior to anthracycline chemotherapy. ²¹¹	IIa	C
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients. ^{55,175,211}	I	B
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion should be considered in moderate-risk patients and in low-risk patients receiving a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ^{55,59,212,213}	IIa	C
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion may be considered in low-risk patients. ^{55,59,212,213}	IIb	C

© ESC 2022

cTn, cardiac troponin; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended in all patients with cancer before anthracycline chemotherapy. ^{12,24,208–210}	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment. ²⁰⁸	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment. ^{24,208–210}	I	C
In moderate-risk patients, additional echocardiography should be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIa	C
In low-risk patients, additional echocardiography may be considered after a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ⁷	IIb	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy. ^{55,65,211}	I	B

Baseline measurement of NP and cTn should be considered in low- and moderate-risk patients prior to anthracycline chemotherapy. ²¹¹	IIa	C
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients. ^{55,175,211}	I	B
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion should be considered in moderate-risk patients and in low-risk patients receiving a cumulative dose of ≥ 250 mg/m ² of doxorubicin or equivalent. ^{55,59,212,213}	IIa	C
cTn and NP monitoring every two cycles during anthracycline chemotherapy and within 3 months after therapy completion may be considered in low-risk patients. ^{55,59,212,213}	IIb	C

© ESC 2022

cTn, cardiac troponin; NP, natriuretic peptides; TTE, transthoracic echocardiography.
^aClass of recommendation.
^bLevel of evidence.
^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

5.5.2. HER2-targeted therapies

HER2-targeted therapies are a crucial part of the treatment of patients with HER2-positive invasive BC in both early and metastatic settings. In the neoadjuvant and/or adjuvant settings, drugs currently approved are trastuzumab, pertuzumab, trastuzumab emtansine, and neratinib. In the metastatic setting, trastuzumab, pertuzumab, trastuzumab emtansine, tucatinib, and trastuzumab deruxtecan are currently approved.^{214a,216} Trastuzumab can also be used in patients with HER2-overexpressing metastatic gastric adenocarcinomas in combination with platinum-based chemotherapy and either capecitabine or 5-fluorouracil (5-FU). It is recognized that anti-HER2 therapies may lead to LVD in up to 15a20% of patients and to overt HF if surveillance is missed, or in high- and very high-risk patients.^{217a220} LV function surveillance based on LVEF and GLS is recommended prior to and every 3 months during HER2-targeted therapies treatment surveillance ([Figure 11](#)).²² However, this single algorithm has not been tested in low- or high-risk patients and increased frequency of assessment (according to local availability) is recommended in high-risk patients.

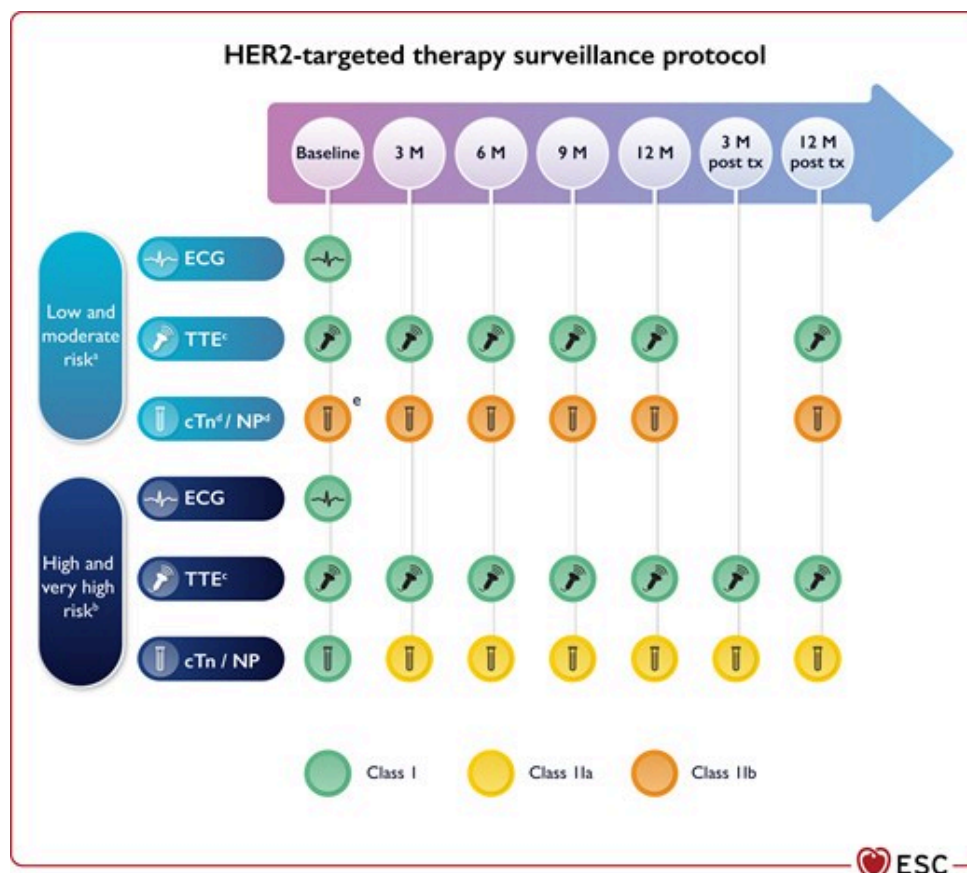


Figure 11

Cardiovascular toxicity monitoring in patients receiving human epidermal receptor 2-targeted therapies.

cTn, cardiac troponin; CV, cardiovascular; EBC, early breast cancer; ECG, electrocardiogram; HER2, human epidermal receptor 2; M, months; NP, natriuretic peptides; TTE, transthoracic echocardiography; tx, treatment. This protocol refers to CV toxicity monitoring in patients receiving neoadjuvant or adjuvant anti-HER2 targeted therapies for non-metastatic disease or first year in metastatic disease. Biomarker assessment should ideally be performed before the corresponding trastuzumab cycle. TTE should be performed in week 2 or 3 of a 3-weekly trastuzumab cycle. ^aIn low-risk HER2+ EBC patients who are asymptomatic and with a normal assessment after 3 months, reducing TTE monitoring to every 4 months may be considered (Class IIb, Level C). In low- and moderate-risk metastatic HER2+ disease, TTE surveillance can be reduced to every 6 months after the first year in asymptomatic patients with normal TTE assessment (Class I, Level C). ^bIn high- and very high-risk metastatic HER2+ disease, TTE monitoring every 2^a–3 cycles may be considered depending on the absolute risk and local availability. ^cCardiac magnetic resonance should be considered for the assessment of cardiac function when TTE is unavailable or not diagnostic. ^dMeasurement of NP and/or cTn is recommended in all patients with cancer if these biomarkers are going to be used during treatment monitoring. ^eBaseline cTn measurement should be considered in low- and moderate-risk patients after anthracycline chemotherapy but prior to starting anti-HER2 targeted therapies for CV toxicity risk prediction.

[Open in new tab](#)[Download slide](#)

The use of cardiac serum biomarkers to identify CTRCD is less well-defined during anti-HER2 treatments.²¹⁷ Measurement of cTn in BC patients after anthracycline-based chemotherapy but prior to trastuzumab should be considered, as an elevated cTn identifies patients at higher risk of trastuzumab-induced CTRCD. Serial NP

measurement was more sensitive than cTn at predicting subsequent declines in LVEF during trastuzumab treatment.⁷⁴

For patients requiring adjuvant chemotherapy and anti-HER2-targeted therapy, the use of non-anthracycline chemotherapy should be considered by the MDT according to risk of relapse, cardiac risks, and in discussion with the treating oncologist.²¹⁷ When anthracycline chemotherapy in the (neo)-adjuvant setting is necessary, sequential use (anthracyclines followed by taxanes and anti-HER2 agents) has been shown to significantly decrease the incidence of CTRCD in several adjuvant trials, compared with concomitant use in earlier trials.^{220,221,224}

Recommendation Table 8

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during human epidermal receptor 2-targeted therapies and in the first 12 months after therapy

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended before HER2-targeted therapies in all patients. ²²⁵	I	B

In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment. ^{225,226}	I	B
In low-risk HER2+ EBC patients ^d who are asymptomatic and with a normal assessment after 3 months, reducing monitoring to every 4 months may be considered.	IIb	C
In high- and very high-risk HER2+ EBC patients, ^d more frequent echocardiography monitoring ^e should be considered during treatment.	IIa	C
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment. ^f	I	C
In metastatic HER2+ disease patients at high- and very high-risk, more frequent echocardiography monitoring ^e may be considered.	IIb	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies. ^{227,228}	I	C
NP and cTn monitoring every 2–3 cycles during therapy and 3 and 12 months after the end of therapy should be considered in high- and very high-risk HER2+ EBC patients. ^{d,55}	IIa	C
Baseline cTn measurement should be considered in low- and moderate-risk patients post-anthracycline chemotherapy but prior to starting anti-HER2-targeted therapies. ^{55,62}	IIa	A
NP and cTn monitoring at baseline, every 3 months, and 12 months after therapy may be considered in low- and moderate-risk HER2+ EBC patients. ^{d,55}	IIb	C

© ESC 2022

BC, breast cancer; cTn, cardiac troponin; CV, cardiovascular; EBC, early breast cancer; HER2, human epidermal receptor 2; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

^dThese recommendations are also applicable for HER2+ non-BC patients.

^eEvery 2–3 cycles depending on the absolute risk and local availability.

^fPatients at low and moderate risk

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended before HER2-targeted therapies in all patients. ²²⁵	I	B

In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment. ^{225,226}	I	B
In low-risk HER2+ EBC patients ^d who are asymptomatic and with a normal assessment after 3 months, reducing monitoring to every 4 months may be considered.	IIb	C
In high- and very high-risk HER2+ EBC patients, ^d more frequent echocardiography monitoring ^e should be considered during treatment.	IIa	C
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment. ^f	I	C
In metastatic HER2+ disease patients at high- and very high-risk, more frequent echocardiography monitoring ^e may be considered.	IIb	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies. ^{227,228}	I	C
NP and cTn monitoring every 2–3 cycles during therapy and 3 and 12 months after the end of therapy should be considered in high- and very high-risk HER2+ EBC patients. ^{d,55}	IIa	C
Baseline cTn measurement should be considered in low- and moderate-risk patients post-anthracycline chemotherapy but prior to starting anti-HER2-targeted therapies. ^{55,62}	IIa	A
NP and cTn monitoring at baseline, every 3 months, and 12 months after therapy may be considered in low- and moderate-risk HER2+ EBC patients. ^{d,55}	IIb	C

© ESC 2022

BC, breast cancer; cTn, cardiac troponin; CV, cardiovascular; EBC, early breast cancer; HER2, human epidermal receptor 2; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

^dThese recommendations are also applicable for HER2+ non-BC patients.

^eEvery 2–3 cycles depending on the absolute risk and local availability.

^fPatients at low and moderate risk

Recommendation Table 8

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during human epidermal receptor 2-targeted therapies and in the first 12 months after therapy

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended before HER2-targeted therapies in all patients. ²²⁵	I	B

In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment. ^{225,226}	I	B
In low-risk HER2+ EBC patients ^d who are asymptomatic and with a normal assessment after 3 months, reducing monitoring to every 4 months may be considered.	IIb	C
In high- and very high-risk HER2+ EBC patients, ^d more frequent echocardiography monitoring ^e should be considered during treatment.	IIa	C
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment. ^f	I	C
In metastatic HER2+ disease patients at high- and very high-risk, more frequent echocardiography monitoring ^e may be considered.	IIb	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies. ^{227,228}	I	C
NP and cTn monitoring every 2–3 cycles during therapy and 3 and 12 months after the end of therapy should be considered in high- and very high-risk HER2+ EBC patients. ^{d,55}	IIa	C
Baseline cTn measurement should be considered in low- and moderate-risk patients post-anthracycline chemotherapy but prior to starting anti-HER2-targeted therapies. ^{55,62}	IIa	A
NP and cTn monitoring at baseline, every 3 months, and 12 months after therapy may be considered in low- and moderate-risk HER2+ EBC patients. ^{d,55}	IIb	C

© ESC 2022

BC, breast cancer; cTn, cardiac troponin; CV, cardiovascular; EBC, early breast cancer; HER2, human epidermal receptor 2; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

^dThese recommendations are also applicable for HER2+ non-BC patients.

^eEvery 2–3 cycles depending on the absolute risk and local availability.

^fPatients at low and moderate risk

Recommendations	Class ^a	Level ^b
TTE		
Baseline echocardiography ^c is recommended before HER2-targeted therapies in all patients. ²²⁵	I	B

In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment. ^{225,226}	I	B
In low-risk HER2+ EBC patients ^d who are asymptomatic and with a normal assessment after 3 months, reducing monitoring to every 4 months may be considered.	IIb	C
In high- and very high-risk HER2+ EBC patients, ^d more frequent echocardiography monitoring ^e should be considered during treatment.	IIa	C
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment. ^f	I	C
In metastatic HER2+ disease patients at high- and very high-risk, more frequent echocardiography monitoring ^e may be considered.	IIb	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies. ^{227,228}	I	C
NP and cTn monitoring every 2–3 cycles during therapy and 3 and 12 months after the end of therapy should be considered in high- and very high-risk HER2+ EBC patients. ^{d,55}	IIa	C
Baseline cTn measurement should be considered in low- and moderate-risk patients post-anthracycline chemotherapy but prior to starting anti-HER2-targeted therapies. ^{55,62}	IIa	A
NP and cTn monitoring at baseline, every 3 months, and 12 months after therapy may be considered in low- and moderate-risk HER2+ EBC patients. ^{d,55}	IIb	C

© ESC 2022

BC, breast cancer; cTn, cardiac troponin; CV, cardiovascular; EBC, early breast cancer; HER2, human epidermal receptor 2; NP, natriuretic peptides; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cIf echocardiography is unavailable or non-diagnostic, follow general cardiac imaging modalities recommendations (see [Section 4.5](#)).

^dThese recommendations are also applicable for HER2+ non-BC patients.

^eEvery 2–3 cycles depending on the absolute risk and local availability.

^fPatients at low and moderate risk

5.5.3. Fluoropyrimidines

Fluoropyrimidines such as 5-FU and its oral prodrug capecitabine are mainly used for gastrointestinal (GI) malignancies and advanced BC. The most common CTR-CVTs are angina pectoris, ischaemia-related ECG abnormalities, hypertension, Takotsubo syndrome (TTS), and MI (even in patients with normal coronary arteries),^{1,4,10,43,229,230} with rarer CTR-CVT including myocarditis, arrhythmias, and peripheral arterial toxicity (Raynaud's phenomenon and ischaemic stroke).²³¹ The incidence of myocardial ischaemia varies according to the dose, scheduling, and route of administration and is up to 10%.²³² Among the several mechanisms responsible for 5-FU-induced myocardial ischaemia are coronary vasospasm and endothelial injury.²³³ Chest pain and ischaemic ECG changes usually occur at rest (less typically during exercise) within days of drug administration and sometimes persist even after treatment cessation. CTR-CVT risk markedly increases in patients with cancer with pre-existing CAD. Aggressive control of modifiable CVRFs, according to the 2021 ESC Guidelines on CVD prevention in clinical practice,¹⁹ is recommended during and after treatment. A baseline TTE is recommended in patients with a history of symptomatic CV to confirm the presence of pre-existing regional wall motion abnormalities or LVD. Screening for CAD may be considered in selected high- and very high-risk patients before the administration of these drugs and according to local protocols and current recommendations.^{12,234,235}

Recommendation Table 9

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during fluoropyrimidine therapy

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP ^c or equivalent is recommended ¹⁹ before starting fluoropyrimidines.	I	C
A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.	I	C
Screening for CAD ^d may be considered in patients at high and very high risk of CAD ^c before fluoropyrimidines.	IIb	C

BP, blood pressure; CAD, coronary artery disease; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^dAccording to pre-existing CVD and local protocols.²³⁴

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP ^c or equivalent is recommended ¹⁹ before starting fluoropyrimidines.	I	C
A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.	I	C
Screening for CAD ^d may be considered in patients at high and very high risk of CAD ^c before fluoropyrimidines.	IIb	C

BP, blood pressure; CAD, coronary artery disease; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^dAccording to pre-existing CVD and local protocols.²³⁴

© ESC 2022

Recommendation Table 9

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during fluoropyrimidine therapy

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP ^c or equivalent is recommended ¹⁹ before starting fluoropyrimidines.	I	C
A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.	I	C
Screening for CAD ^d may be considered in patients at high and very high risk of CAD ^c before fluoropyrimidines.	IIb	C

BP, blood pressure; CAD, coronary artery disease; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^dAccording to pre-existing CVD and local protocols.²³⁴

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP ^c or equivalent is recommended ¹⁹ before starting fluoropyrimidines.	I	C
A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.	I	C
Screening for CAD ^d may be considered in patients at high and very high risk of CAD ^c before fluoropyrimidines.	IIb	C

BP, blood pressure; CAD, coronary artery disease; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^dAccording to pre-existing CVD and local protocols.²³⁴

© ESC 2022

5.5.4. Vascular endothelial growth factor inhibitors

Aberrant activation of kinases plays a critical role in both the development of numerous cancer types and in CV and metabolic homeostasis. Inhibition of the VEGF signalling pathway is achieved with either monoclonal antibodies (administered i.v.) against circulating VEGF or with small-molecule TKI (taken orally) targeting VEGF receptors.²³⁶ VEGFi are used for the treatment of numerous cancer types, including renal, thyroid, and hepatocellular carcinomas. However, their use is associated with a wide array of CV complications including hypertension, HF, QTc prolongation, and acute vascular events ([Figure 12](#)).^{131,237a,238,240} It can be challenging to assess the prognosis of patients experiencing severe CV side effects because these drugs are often used in patients with advanced cancer. The goal must be to continue VEGFi treatment for as long as possible with initiation or optimization of CV treatment if indicated.

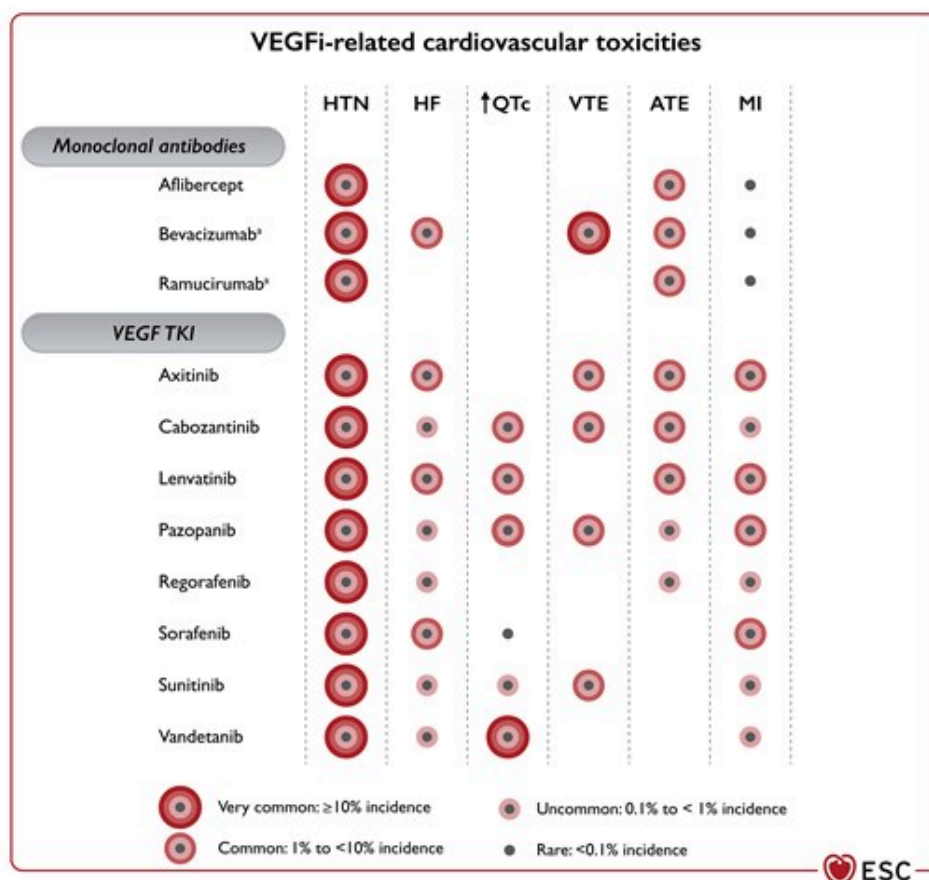


Figure 12

Vascular endothelial growth factor inhibitors-related cardiovascular toxicities.

ATE, arterial thromboembolism; EMA, European Medicines Agency; FDA, Food and Drug Administration; HF, heart failure; HTN, hypertension; MedDRA, medical dictionary for regulatory activities; MI, myocardial infarction; $\uparrow$ QTc, corrected QT interval prolongation; TKI, tyrosine kinase inhibitors; VEGF, vascular endothelial growth factor; VEGFi, vascular endothelial growth factor inhibitors; VTE, venous thromboembolism. Adverse reactions reported in multiple clinical trials or during post-marketing use are listed by system organ class (in MedDRA) and frequency. If the frequency is unknown or cannot be estimated from the available data, a blank space has been left. ^aBevacizumab: hypertension frequency 54% (EMA); 60% of the patients who received bevacizumab in combination with erlotinib. Pre-existing hypertension should be adequately controlled before starting treatment. Ramucirumab: hypertension frequency 16% (EMA/FDA); in combination with erlotinib, the incidence of hypertension was 24%. Patients with uncontrolled hypertension were excluded from the trials. Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

Hypertension is a class effect and is the most reported adverse event under VEGFi treatment. It occurs within hours or days, is dose-dependent, and is usually reversed by VEGFi discontinuation.^{131,239,241,243} The risk is higher in patients with pre-existing hypertension or CVD, previous anthracycline treatment, advanced age, history of smoking, hyperlipidaemia, and/or obesity ([Table 4](#)).^{4,244} LVD and HF occur in a minority of patients in RCTs,²⁴⁵ but are reported more frequently in routine practice²⁴⁶ and are often reversible.²⁴⁷ Acute arterial events (aortic dissection, stroke, arterial thrombosis, acute coronary events, vasospasm) and venous thromboembolism (VTE) can also complicate treatment with VEGFi.²⁴⁸ QTc prolongation has been described

with sunitinib, sorafenib, and vandetanib,²⁴⁹ but it is rarely related to severe arrhythmic events, except with vandetanib.²⁵⁰ Some small-molecule TKI (e.g. sorafenib and sunitinib) can cause AF²⁵¹ and HF.^{43,129,247}

A baseline CV risk assessment includes clinical examination, BP measurement, and an ECG with baseline QTcF measurement (see [Section 4](#)).²⁰ Especially in patients with known hypertension, BP should be controlled before VEGFi therapy. A baseline TTE is recommended for high- and very high-risk patients.¹⁴ Patients with impaired LV function and/or patients at high or very high risk of developing HF should be referred to the cardiologist before starting VEGFi therapy.¹⁴

Monitoring during and after treatment is indicated for all patients treated with a VEGFi and is based on close clinical follow-up using serial ECGs, biomarkers, and echocardiography. Early recognition and treatment of hypertension are essential to prevent other CV complications, especially HF. Home BP monitoring is recommended daily during the first cycle, after each increase of anticancer therapy dose, and every 2nd weeks thereafter.^{138,254,255} When treatment with a VEGFi is stopped, a drop in BP must be anticipated and BP-lowering therapy must be reduced and/or interrupted accordingly ([Section 6](#)).

In patients at risk of QTc prolongation, regular monitoring of the QTc interval is recommended after a dose increase, whenever other QT-prolonging agents are added, or if electrolyte imbalances occur ([Section 6](#)).

Patients treated with a VEGFi must also be screened regularly for symptoms and clinical signs of HF. Regular NP measurement and echocardiography can be useful for the detection of CTRCD, although evidence is weak ([Figure 13](#)).^{138,254,255}

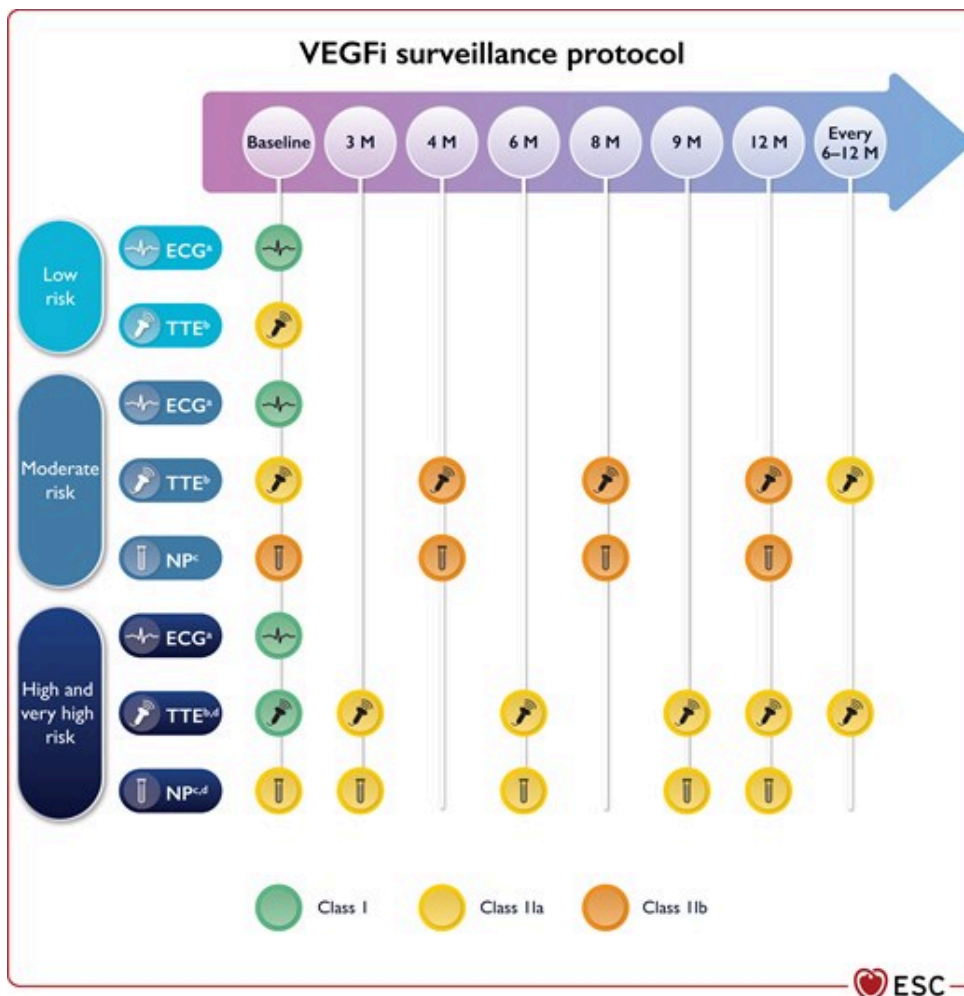


Figure 13

Cardiovascular toxicity monitoring in patients receiving vascular endothelial growth factor inhibitors.

ECG, electrocardiogram; M, months; NP, natriuretic peptides; QTc, corrected QT interval; TTE, transthoracic echocardiography; VEGFi, vascular endothelial growth factor inhibitors. ^aIn patients treated with VEGFi at moderate or high risk of QTc prolongation, ECG is recommended (Class I, Level C) monthly during the first 3 months and every 3 months thereafter (Section 6.4). Consider an ECG 2 weeks after starting treatment in high-risk patients and new monitoring in the case of any dose increase (see Section 6.4.2). ^bCardiac magnetic resonance should be considered for the assessment of cardiac function when TTE is unavailable or not diagnostic. ^cMeasurement of NP is recommended in all patients with cancer if these biomarkers are going to be used during treatment monitoring. ^dTTE and NP should be considered at 4 weeks after starting treatment in very high-risk patients.

[Open in new tab](#)[Download slide](#)

Recommendation Table 10

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during vascular endothelial growth factor inhibitors

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.	I	C
Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.	I	C
ECG monitoring		
In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc ^c monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter. ^d	I	C
Echocardiography		
Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.	I	C
Baseline echocardiography should be considered in low- and moderate-risk patients treated with a VEGFi or bevacizumab.	IIa	C

Echocardiography may be considered every 4 months during the first year in moderate-risk patients receiving VEGFi or bevacizumab.	IIb	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving a VEGFi or bevacizumab. ^e	IIa	C
Echocardiography every 6–12 months should be considered in moderate- and high-risk patients with cancer who require long-term treatment with a VEGFi.	IIa	C
Cardiac biomarker		
NP may be considered at baseline and then every 4 months during the first year in moderate-risk patients receiving a VEGFi.	IIb	C
NP should be considered at baseline, 4 weeks after starting treatment, and then every 3 months during the first year in high- and very high-risk patients receiving a VEGFi.	IIa	C

BP, blood pressure; ECG, electrocardiogram; NP, natriuretic peptides; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cQTc interval using Fridericia correction ($QTcF = QT/\sqrt{RR}$) is the preferred method.

^dConsider an ECG 2 weeks after starting treatment in high-risk patients and new monitoring in the case of any dose increase (see [Section 6.4.2](#)).

^eAn additional echocardiography 4 weeks after starting treatment should be considered in selected high- and very high-risk patients according to local availability, especially if cardiac biomarker surveillance is not available.

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.	I	C
Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.	I	C
ECG monitoring		
In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc ^c monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter. ^d	I	C
Echocardiography		
Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.	I	C
Baseline echocardiography should be considered in low- and moderate-risk patients treated with a VEGFi or bevacizumab.	IIa	C

Echocardiography may be considered every 4 months during the first year in moderate-risk patients receiving VEGFi or bevacizumab.	IIb	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving a VEGFi or bevacizumab. ^e	IIa	C
Echocardiography every 6–12 months should be considered in moderate- and high-risk patients with cancer who require long-term treatment with a VEGFi.	IIa	C
Cardiac biomarker		
NP may be considered at baseline and then every 4 months during the first year in moderate-risk patients receiving a VEGFi.	IIb	C
NP should be considered at baseline, 4 weeks after starting treatment, and then every 3 months during the first year in high- and very high-risk patients receiving a VEGFi.	IIa	C

© ESC 2022

BP, blood pressure; ECG, electrocardiogram; NP, natriuretic peptides; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cQTc interval using Fridericia correction ($QTcF = QT / \sqrt{RR}$) is the preferred method.

^dConsider an ECG 2 weeks after starting treatment in high-risk patients and new monitoring in the case of any dose increase (see [Section 6.4.2](#)).

^eAn additional echocardiography 4 weeks after starting treatment should be considered in selected high- and very high-risk patients according to local availability, especially if cardiac biomarker surveillance is not available.

Recommendation Table 10

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during vascular endothelial growth factor inhibitors

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.	I	C
Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.	I	C
ECG monitoring		
In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc ^c monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter. ^d	I	C
Echocardiography		
Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.	I	C
Baseline echocardiography should be considered in low- and moderate-risk patients treated with a VEGFi or bevacizumab.	IIa	C

Echocardiography may be considered every 4 months during the first year in moderate-risk patients receiving VEGFi or bevacizumab.	IIb	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving a VEGFi or bevacizumab. ^e	IIa	C
Echocardiography every 6–12 months should be considered in moderate- and high-risk patients with cancer who require long-term treatment with a VEGFi.	IIa	C
Cardiac biomarker		
NP may be considered at baseline and then every 4 months during the first year in moderate-risk patients receiving a VEGFi.	IIb	C
NP should be considered at baseline, 4 weeks after starting treatment, and then every 3 months during the first year in high- and very high-risk patients receiving a VEGFi.	IIa	C

© ESC 2022

BP, blood pressure; ECG, electrocardiogram; NP, natriuretic peptides; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cQTc interval using Fridericia correction ($QTcF = QT / \sqrt[3]{RR}$) is the preferred method.

^dConsider an ECG 2 weeks after starting treatment in high-risk patients and new monitoring in the case of any dose increase (see [Section 6.4.2](#)).

^eAn additional echocardiography 4 weeks after starting treatment should be considered in selected high- and very high-risk patients according to local availability, especially if cardiac biomarker surveillance is not available.

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.	I	C
Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.	I	C
ECG monitoring		
In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc ^c monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter. ^d	I	C
Echocardiography		
Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.	I	C
Baseline echocardiography should be considered in low- and moderate-risk patients treated with a VEGFi or bevacizumab.	IIa	C

Echocardiography may be considered every 4 months during the first year in moderate-risk patients receiving VEGFi or bevacizumab.	IIb	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving a VEGFi or bevacizumab. ^e	IIa	C
Echocardiography every 6–12 months should be considered in moderate- and high-risk patients with cancer who require long-term treatment with a VEGFi.	IIa	C
Cardiac biomarker		
NP may be considered at baseline and then every 4 months during the first year in moderate-risk patients receiving a VEGFi.	IIb	C
NP should be considered at baseline, 4 weeks after starting treatment, and then every 3 months during the first year in high- and very high-risk patients receiving a VEGFi.	IIa	C

BP, blood pressure; ECG, electrocardiogram; NP, natriuretic peptides; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cQTc interval using Fridericia correction ($QTcF = QT/\sqrt{RR}$) is the preferred method.

^dConsider an ECG 2 weeks after starting treatment in high-risk patients and new monitoring in the case of any dose increase (see [Section 6.4.2](#)).

^eAn additional echocardiography 4 weeks after starting treatment should be considered in selected high- and very high-risk patients according to local availability, especially if cardiac biomarker surveillance is not available.

© ESC 2022

5.5.5. Multitargeted kinase inhibitors targeting BCR-ABL

Chronic myeloid leukaemia (CML) results from aberrant activation of ABL1 kinase due to a chromosomal translocation. Small-molecule TKIs targeting BCR-ABL^a including imatinib, bosutinib, dasatinib, nilotinib, and ponatinib^a have proven effective in the treatment of CML. The toxicities associated with these TKIs are unique and due to ^aoff-target^a effects of each drug. Dasatinib is associated with group 1 pulmonary hypertension (PH), HF, and pleural and pericardial effusion, whereas nilotinib and ponatinib are generally associated with vascular events ([FigureÂ 14](#)).^{131,256a,259} Second-generation BCR-ABL TKI may induce a QTc prolongation (see [Section 6.4.2](#)). CV toxicity risk is higher in patients aged >65 years (relative risk 1.8) and in those with underlying DM (relative risk 2.5), hypertension (relative risk 3.2) or pre-existing CAD (relative risk 2.6).^{256a,258,260} Before BCR-ABL TKI therapy, it is critical to define baseline CV toxicity risk with special attention to BP, glucose, and lipids. Baseline ECG is recommended in all patients and QTc monitoring in patients treated with second-generation BCR-ABL TKI. Depending on the type of therapy used, specific CV assessments should be performed after drug initiation ([FigureÂ 15](#)).²⁵⁶

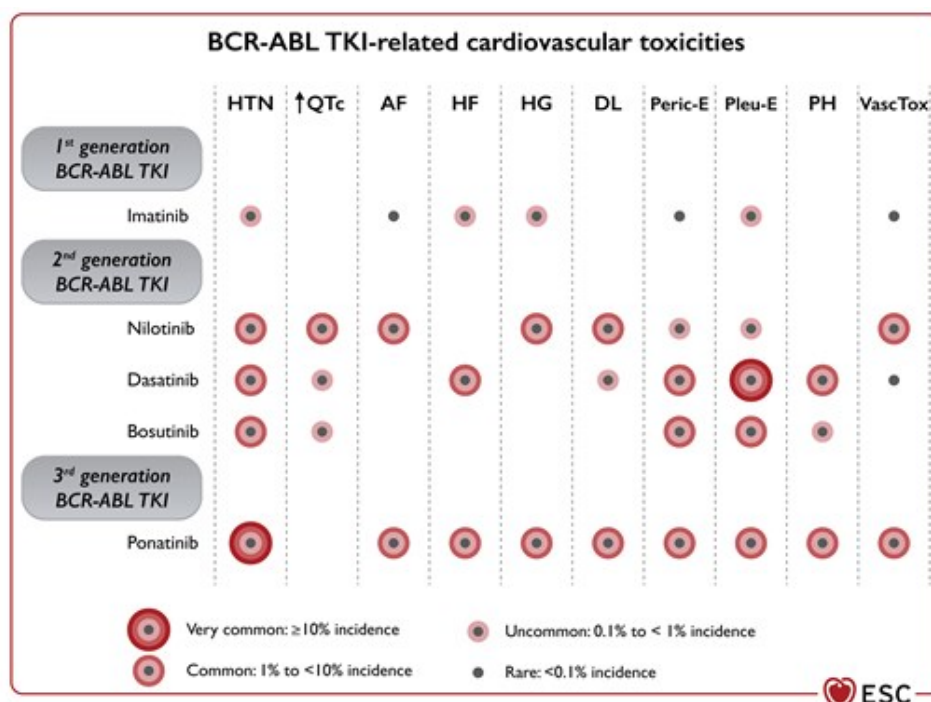


Figure 14

Breakpoint cluster regionaAbelson oncogene locus tyrosine kinase inhibitor-related cardiovascular toxicities.

AF, atrial fibrillation; BCR-ABL, breakpoint cluster regionaAbelson oncogene locus; DL, dyslipidaemia; EMA, European Medicines Agency; FDA, Food and Drug Administration; HF, heart failure; HG, hyperglycaemia; HTN, hypertension; MedDRA, medical dictionary for regulatory activities; MI, myocardial infarction; PAD, peripheral artery disease; Peric-E, pericardial effusion; PH, pulmonary hypertension; Pleu-E, pleural effusion; a↑QTc, corrected QT interval prolongation; TKI, tyrosine kinase inhibitors; VascTox, vascular toxicity (stroke, MI, PAD). Adverse reactions reported in multiple clinical trials or during post-marketing use are listed by system organ class (in MedDRA) and frequency. If the frequency is unknown or cannot be estimated from the available data, a blank space has been left.²⁶¹ Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

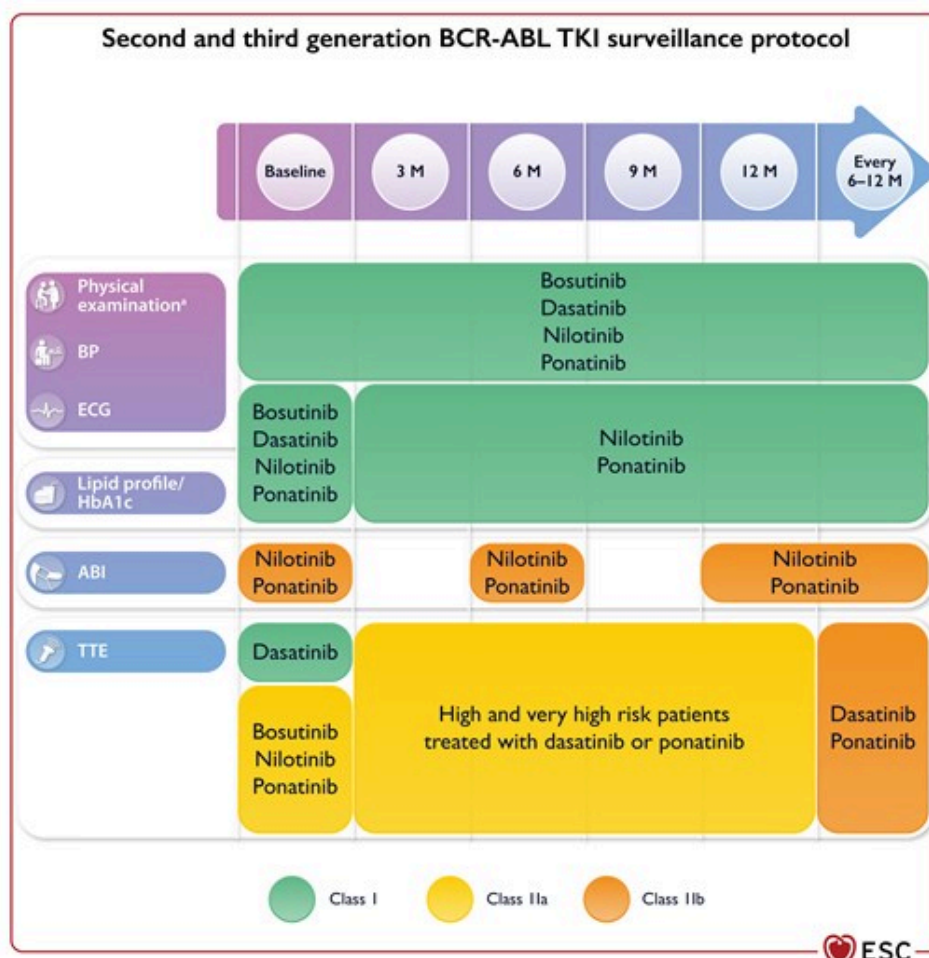


Figure 15

Second- and third-generation breakpoint cluster region-Abelson oncogene locus tyrosine kinase inhibitors surveillance protocol.

ABI, ankle-brachial index; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; M, months; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography. ^aCoronary artery calcium scoring can reclassify CV risk upwards and downwards in addition to conventional risk factors, and may be considered in men and women with calculated CV risk around decision thresholds.¹⁹

[Open in new tab](#) [Download slide](#)

Recommendation Table 11

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during second- and third-generation breakpoint cluster region-Abelson oncogene locus tyrosine kinase inhibitors

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients who require second- or third-generation BCR-ABL TKI. ^{256,261}	I	C
In patients treated with nilotinib or ponatinib, CV risk assessment ^c is recommended every 3 months during the first year and every 6–12 months thereafter. ^{256,261}	I	C
QTc ^d measurement should be considered at baseline, at 2 and 4 weeks after starting nilotinib, and 2 weeks after any dose increase. ²⁵⁹	IIa	C
Baseline echocardiography should be considered in all patients before starting second- and third-generation BCR-ABL TKI.	IIa	C
Baseline echocardiography is recommended in patients scheduled to receive dasatinib.	I	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving dasatinib or ponatinib.	IIa	C
Echocardiography may be considered every 6–12 months in patients who require long-term (>12 months) ponatinib or dasatinib.	IIb	C
Serial assessment of ankle brachial index may be considered to detect subclinical peripheral vascular disease.	IIb	C

© ESC 2022

BCR-ABL, breakpoint cluster region–Abelson oncogene locus; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement. Coronary artery calcium scoring can reclassify CV disease risk upwards and downwards in addition to conventional risk factors, and may be considered at baseline in low- and moderate-risk patients as per HFA-ICOS risk assessment tools.¹⁹

^dQTc interval using Fridericia correction ($QTcF = QT / \sqrt[3]{RR}$) is the preferred method.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients who require second- or third-generation BCR-ABL TKI. ^{256,261}	I	C
In patients treated with nilotinib or ponatinib, CV risk assessment ^c is recommended every 3 months during the first year and every 6–12 months thereafter. ^{256,261}	I	C
QTc ^d measurement should be considered at baseline, at 2 and 4 weeks after starting nilotinib, and 2 weeks after any dose increase. ²⁵⁹	IIa	C
Baseline echocardiography should be considered in all patients before starting second- and third-generation BCR-ABL TKI.	IIa	C
Baseline echocardiography is recommended in patients scheduled to receive dasatinib.	I	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving dasatinib or ponatinib.	IIa	C
Echocardiography may be considered every 6–12 months in patients who require long-term (>12 months) ponatinib or dasatinib.	IIb	C
Serial assessment of ankle brachial index may be considered to detect subclinical peripheral vascular disease.	IIb	C

© ESC 2022

BCR-ABL, breakpoint cluster region–Abelson oncogene locus; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement. Coronary artery calcium scoring can reclassify CV disease risk upwards and downwards in addition to conventional risk factors, and may be considered at baseline in low- and moderate-risk patients as per HFA-ICOS risk assessment tools.¹⁹

^dQTc interval using Fridericia correction ($QTcF = QT / \sqrt[3]{RR}$) is the preferred method.

Recommendation Table 11

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during second- and third-generation breakpoint cluster region–Abelson oncogene locus tyrosine kinase inhibitors

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients who require second- or third-generation BCR-ABL TKI. ^{256,261}	I	C
In patients treated with nilotinib or ponatinib, CV risk assessment ^c is recommended every 3 months during the first year and every 6–12 months thereafter. ^{256,261}	I	C
QTc ^d measurement should be considered at baseline, at 2 and 4 weeks after starting nilotinib, and 2 weeks after any dose increase. ²⁵⁹	IIa	C
Baseline echocardiography should be considered in all patients before starting second- and third-generation BCR-ABL TKI.	IIa	C
Baseline echocardiography is recommended in patients scheduled to receive dasatinib.	I	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving dasatinib or ponatinib.	IIa	C
Echocardiography may be considered every 6–12 months in patients who require long-term (>12 months) ponatinib or dasatinib.	IIb	C
Serial assessment of ankle brachial index may be considered to detect subclinical peripheral vascular disease.	IIb	C

© ESC 2022

BCR-ABL, breakpoint cluster region–Abelson oncogene locus; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement. Coronary artery calcium scoring can reclassify CV disease risk upwards and downwards in addition to conventional risk factors, and may be considered at baseline in low- and moderate-risk patients as per HFA-ICOS risk assessment tools.¹⁹

^dQTc interval using Fridericia correction ($QTcF = QT / \sqrt[3]{RR}$) is the preferred method.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients who require second- or third-generation BCR-ABL TKI. ^{256,261}	I	C
In patients treated with nilotinib or ponatinib, CV risk assessment ^c is recommended every 3 months during the first year and every 6–12 months thereafter. ^{256,261}	I	C
QTc ^d measurement should be considered at baseline, at 2 and 4 weeks after starting nilotinib, and 2 weeks after any dose increase. ²⁵⁹	IIa	C
Baseline echocardiography should be considered in all patients before starting second- and third-generation BCR-ABL TKI.	IIa	C
Baseline echocardiography is recommended in patients scheduled to receive dasatinib.	I	C
Echocardiography should be considered every 3 months during the first year in high- and very high-risk patients receiving dasatinib or ponatinib.	IIa	C
Echocardiography may be considered every 6–12 months in patients who require long-term (>12 months) ponatinib or dasatinib.	IIb	C
Serial assessment of ankle brachial index may be considered to detect subclinical peripheral vascular disease.	IIb	C

© ESC 2022

BCR-ABL, breakpoint cluster region–Abelson oncogene locus; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement. Coronary artery calcium scoring can reclassify CV disease risk upwards and downwards in addition to conventional risk factors, and may be considered at baseline in low- and moderate-risk patients as per HFA-ICOS risk assessment tools.¹⁹

^dQTc interval using Fridericia correction ($QTcF = QT/\sqrt[3]{RR}$) is the preferred method.

5.5.6. Bruton tyrosine kinase inhibitors

Bruton tyrosine kinase (BTK) inhibitors are increasingly used to treat lymphoid malignancies. Ibrutinib, a first-in-class irreversible oral inhibitor of BTK, has proven highly effective in chronic lymphocytic leukaemia and related B-cell malignancies including mantle cell lymphoma, Waldenström macroglobulinemia, and marginal zone lymphomas.²⁶² These disorders are usually diagnosed in elderly patients in whom frequent

comorbidities coexist at diagnosis that increase the risk of CTR-CVT.^{263,264} Ibrutinib has been associated with bleeding diathesis, infections, and an increased risk of hypertension, AF, and HF.²⁶⁵ Ibrutinib may also cause ventricular arrhythmias without prolonging QT.^{267,268} Acalabrutinib is a second-generation BTK inhibitor with greater BTK selectivity. In a recent phase III, randomized, multicentre, open-label, non-inferiority study, acalabrutinib demonstrated a non-inferior progression-free survival compared to ibrutinib in patients with previously treated chronic lymphocytic leukaemia with a lower incidence of symptomatic CV events.²⁶⁹ However, grade ≥3 AF (symptomatic AF where urgent intervention is indicated)²⁷⁰ and AF in patients ≥75 years old or with previous AF history were comparable between groups, as was the risk of CV events in patients with pre-existing CVRFs or CVD.²⁷¹ Therefore, we currently do not have enough data to establish different monitoring strategies in patients treated with these drugs.

Due to the lack of evidence-based recommendations, the management of these CV events is challenging.²⁶⁴ Antihypertensive initiation has been associated with a lower risk of a major adverse CV events (MACE).²⁶⁴ Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.²⁷²

Due to a higher bleeding risk, ibrutinib should be temporarily interrupted in patients requiring dual antiplatelet therapy (DAPT) and 3–7 days before invasive procedures. In case of emergency interventions, platelet transfusion should be considered to minimize bleeding risks.²⁶²

Recommendation Table 12

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during Bruton tyrosine kinase inhibitor therapy

Recommendations	Class ^a	Level ^b
BP monitoring and management		
BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit. ²⁶⁴	I	B
Weekly home monitoring of BP during the first 3 months and every month thereafter should be considered for patients treated with BTK inhibitors.	IIa	C
Echocardiography		
Baseline echocardiography is recommended in high-risk patients ^c scheduled to receive BTK inhibitors. ^{267,268}	I	C
TTE is recommended in all patients who develop AF during BTK inhibitor therapy.	I	C

AF

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.²⁷³

I

C

© ESC 2022

AF, atrial fibrillation; BP, blood pressure; BTK, Bruton tyrosine kinase; DM, diabetes mellitus; ECG, electrocardiogram; HF, heart failure; QTc, corrected QT interval; TTE, transthoracic echocardiography; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cMale, age ≥ 65 years, previous history of hypertension, DM, QTc ≥ 480 ms, AF, HF, cardiomyopathy, or severe VHD.^{263,274,275}

Recommendations

Class^a

Level^b

BP monitoring and management

BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit.²⁶⁴

I

B

Weekly home monitoring of BP during the first 3 months and every month thereafter should be considered for patients treated with BTK inhibitors.

IIa

C

Echocardiography

Baseline echocardiography is recommended in high-risk patients^c scheduled to receive BTK inhibitors.^{267,268}

I

C

TTE is recommended in all patients who develop AF during BTK inhibitor therapy.

I

C

AF

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.²⁷³

I

C

© ESC 2022

AF, atrial fibrillation; BP, blood pressure; BTK, Bruton tyrosine kinase; DM, diabetes mellitus; ECG, electrocardiogram; HF, heart failure; QTc, corrected QT interval; TTE, transthoracic echocardiography; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cMale, age ≥ 65 years, previous history of hypertension, DM, QTc ≥ 480 ms, AF, HF, cardiomyopathy, or severe VHD.^{263,274,275}

Recommendation Table 12

[Open in new tab](#)

Recommendations	Class ^a	Level ^b
BP monitoring and management		
BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit. ²⁶⁴	I	B
Weekly home monitoring of BP during the first 3 months and every month thereafter should be considered for patients treated with BTK inhibitors.	IIa	C
Echocardiography		
Baseline echocardiography is recommended in high-risk patients ^c scheduled to receive BTK inhibitors. ^{267,268}	I	C
TTE is recommended in all patients who develop AF during BTK inhibitor therapy.	I	C
AF		
Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy. ²⁷³	I	C

AF, atrial fibrillation; BP, blood pressure; BTK, Bruton tyrosine kinase; DM, diabetes mellitus; ECG, electrocardiogram; HF, heart failure; QTc, corrected QT interval; TTE, transthoracic echocardiography; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cMale, age ≥ 65 years, previous history of hypertension, DM, QTc ≥ 480 ms, AF, HF, cardiomyopathy, or severe VHD.^{263,274,275}

© ESC 2022

Recommendations	Class ^a	Level ^b
BP monitoring and management		
BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit. ²⁶⁴	I	B
Weekly home monitoring of BP during the first 3 months and every month thereafter should be considered for patients treated with BTK inhibitors.	IIa	C
Echocardiography		
Baseline echocardiography is recommended in high-risk patients ^c scheduled to receive BTK inhibitors. ^{267,268}	I	C
TTE is recommended in all patients who develop AF during BTK inhibitor therapy.	I	C
AF		
Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy. ²⁷³	I	C

AF, atrial fibrillation; BP, blood pressure; BTK, Bruton tyrosine kinase; DM, diabetes mellitus; ECG, electrocardiogram; HF, heart failure; QTc, corrected QT interval; TTE, transthoracic echocardiography; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cMale, age ≥ 65 years, previous history of hypertension, DM, QTc ≥ 480 ms, AF, HF, cardiomyopathy, or severe VHD.^{263,274,275}

© ESC 2022

5.5.7. Multiple myeloma therapies

There are many classes of pharmacotherapy that are approved for the treatment of MM using a range of combinations. These include immunomodulatory drugs (IMiD), dexamethasone, PI, and monoclonal antibodies (e.g. daratumumab). PIa^{PADCC} including bortezomib, carfilzomib, and ixazomib^{PADCC} have become a mainstay of therapy for newly diagnosed MM as well as relapsed disease.^{276,277} Several large studies using combination therapy for MM have demonstrated an increased risk of serious CV adverse events.^{278a^{PADCC}, 281} MM patients being treated with PI have a high incidence of coexistent CV comorbidities and increased baseline CV risk.^{282,283} PI have been associated with a variety of CV toxicities including hypertension, HF,²⁸⁴ acute coronary syndromes (ACS),⁶⁶ arrhythmias,²⁸⁵ PH,²⁸⁶ and VTE (*FigureÂ 16*).^{287,288} During therapy, cardiac biomarkers and TTE are important diagnostic and prognostic tools that can inform clinical decision-making (*FigureÂ 17*).⁶⁶

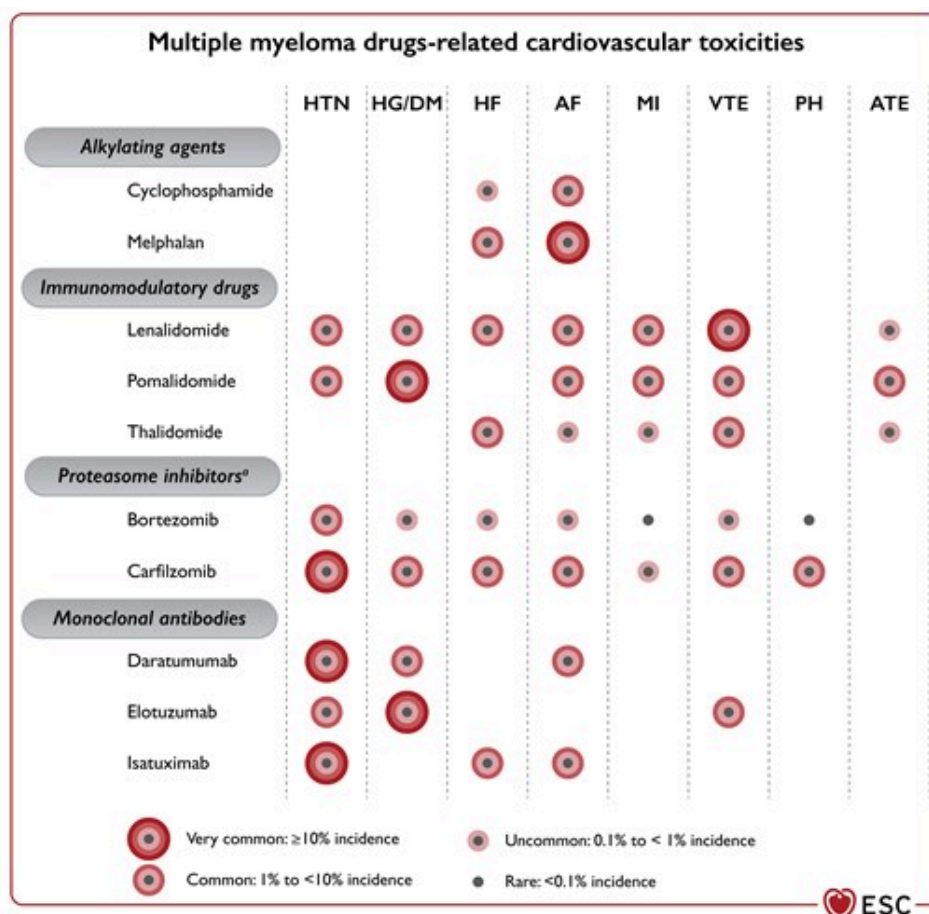


Figure 16
Multiple myeloma drug-related cardiovascular toxicities.

AF, atrial fibrillation; ATE, arterial thromboembolism; DM, diabetes mellitus; EMA, European Medicines Agency; FDA, Food and Drug Administration; HF, heart failure; HG, hyperglycaemia; HTN, hypertension; MedDRA, medical dictionary for regulatory activities; MI, myocardial infarction; PH, pulmonary hypertension; VTE, venous thromboembolism. Adverse reactions reported in multiple clinical trials or during post-marketing use are listed by system organ class (in MedDRA) and frequency. If the frequency is unknown or cannot be estimated from the available data, a blank space has been left. ^aIxazomib produces peripheral oedema in up to 18% of patients and hyperglycaemia in combination with lenalidomide or pomalidomide and dexamethasone. Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

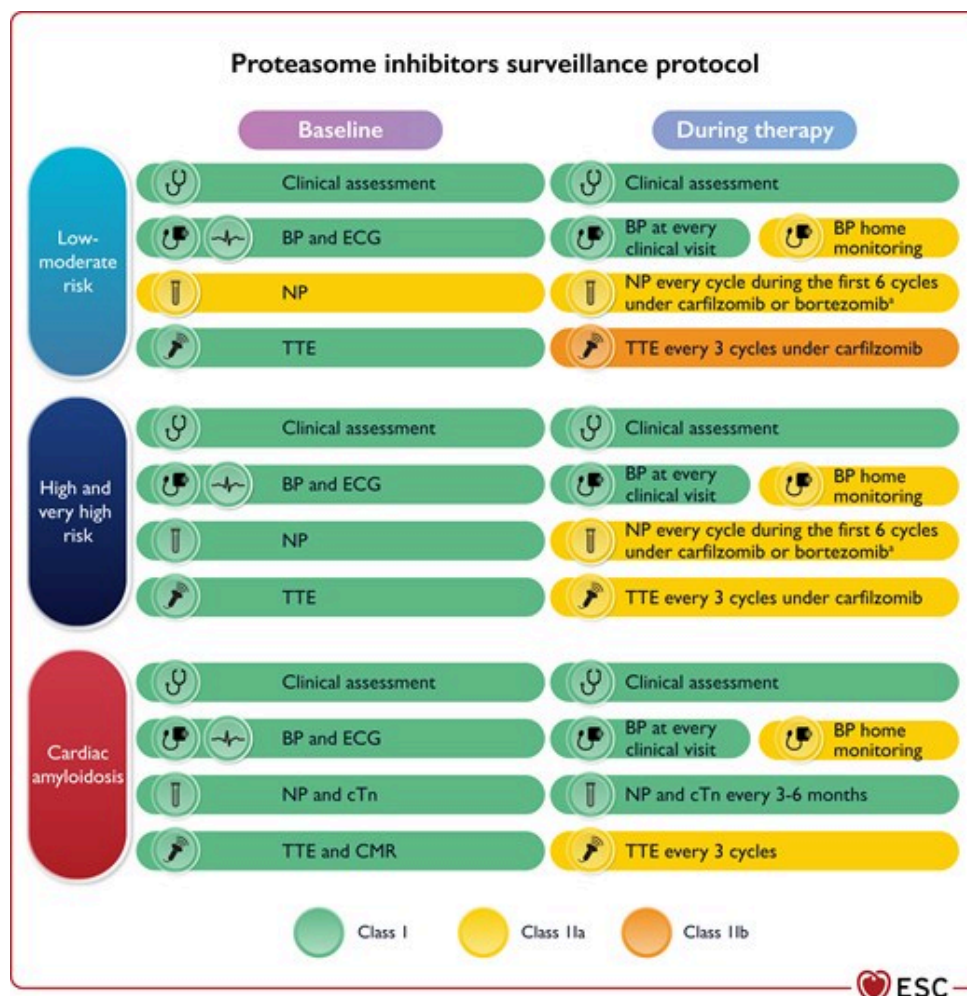


Figure 17

Cardiovascular monitoring in patients with multiple myeloma receiving proteasome inhibitors.

BP, blood pressure; CMR, cardiac magnetic resonance; cTn, cardiac troponin; ECG, electrocardiogram; NP, natriuretic peptides; TTE, transthoracic echocardiography. ^aEvery 2 months for patients treated with ixazomib.

[Open in new tab](#)[Download slide](#)

HF, especially HF with preserved ejection fraction (HFpEF) is a frequent manifestation of cardiac amyloidosis, but it is also an important adverse effect of PI therapy, especially under carfilzomib. In a safety analysis of patients with MM being treated with carfilzomib, 7.2% of patients were found to have new HF.²⁸⁴ In another study, 23% of patients with MM treated with carfilzomib developed clinical HF and/or LVD.²⁸⁹ The mechanism is not well understood but is possibly related to PI-induced oxidative stress within myocytes, inhibition of the proteasome, or transient endothelial dysfunction.^{281,283} Although no studies have yet addressed the optimal follow-up scheme in patients with MM treated with PI, a common scheme consists of 3-6-monthly visits with ECG, complete blood tests (including NP and cTn) and echocardiography surveillance during PI therapy.²⁹⁰ A recent prospective study of patients with relapsed MM confirmed the utility of NP to assist in risk stratification as well as management of CV morbidity during treatment.⁶⁶ Hypertension, another adverse effect of PI, may also contribute to the development of HFpEF.

Patients with MM are at elevated risk of thrombosis due to both patient- and myeloma-related factors, particularly the combination of PI and IMiD ([Figure 18](#)).^{279,287,291-297} In the ASPIRE (Carfilzomib, Lenalidomide, and Dexamethasone vs. Lenalidomide and Dexamethasone for the Treatment of Patients with

Relapsed Multiple Myeloma) study, patients treated with a combination of carfilzomib, lenalidomide, and dexamethasone had higher rates of VTE compared with those treated with lenalidomide and dexamethasone (6.6% vs. 3.9%).²⁷⁹ Oncological guidelines recommend the use of aspirin or prophylactic doses of low-molecular-weight heparins (LMWH) in low-risk patients receiving thalidomide- or lenalidomide-based regimens.²⁹⁸ In patients at high risk of VTE, therapeutic doses of LMWH are recommended.²⁹⁹ The role of non-vitamin K antagonist oral anticoagulants (NOAC) in MM patients needs further validation in larger trials, but recent small studies have confirmed the efficacy and safety of low doses of apixaban and rivaroxaban for VTE prevention.^{300a}³⁰²

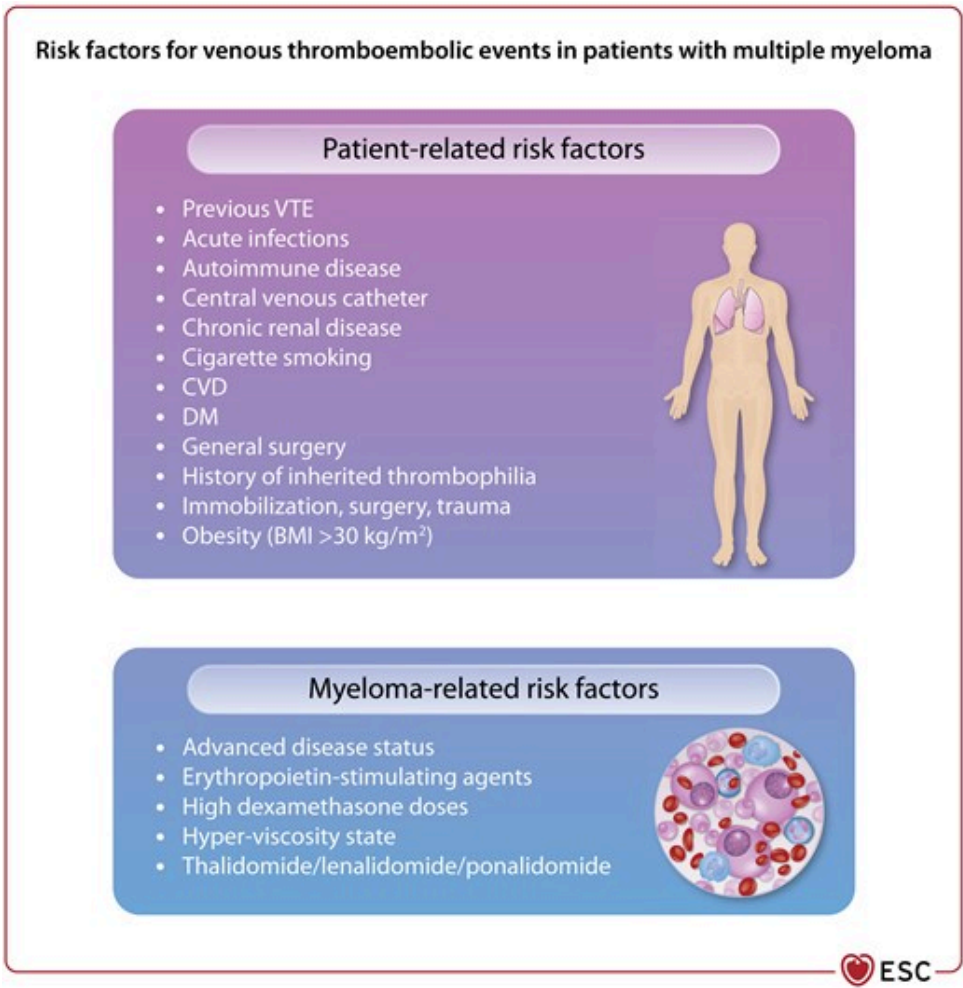


Figure 18
Risk factors for venous thromboembolic events in patients with multiple myeloma.

BMI, body mass index; CVD, cardiovascular disease; DM, diabetes mellitus; VTE, venous thromboembolism.

[Open in new tab](#)[Download slide](#)

Recommendation Table 13

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during multiple myeloma therapies

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with PI at every clinical visit.	I	C
Home monitoring of BP weekly during the first 3 months and monthly thereafter should be considered for patients treated with PI.	IIa	C
Cardiac serum biomarkers		
Measurement of NP is recommended prior to PI in high- and very high-risk patients. ^{66,303}	I	C
Measurement of NP should be considered prior to PI in low- and moderate-risk patients. ⁶⁶	IIa	C
In patients receiving carfilzomib or bortezomib, measurement of NP should be considered at baseline and every cycle during the first 6 cycles. ^{c,66}	IIa	B
NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA. ^{d,290}	I	B
TTE		
Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.	I	C
Echocardiography surveillance every 3 cycles should be considered in high- and very high-risk patients receiving carfilzomib. ²⁸⁰	IIa	B
Echocardiography surveillance every 3 cycles may be considered in low- and moderate-risk patients receiving carfilzomib.	IIb	C
Echocardiography surveillance should be considered every 3–6 months in patients with AL-CA treated with PI. ^{d,290}	IIa	C
VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE. ^{296,298,302,304,305}	I	B

Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304,305}	I	A
Aspirin should be considered as an alternative to LMWH in patients with MM with no risk factors or one VTE-related risk factor ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304–307}	IIa	B
Low doses of apixaban or rivaroxaban ^f may be considered as an alternative to LMWH or aspirin in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{300–302}	IIb	C

AL-CA, amyloid light-chain cardiac amyloidosis; BP, blood pressure; cTn, cardiac troponin; HF, heart failure; LMWH, low-molecular-weight heparins; MM, multiple myeloma; NP, natriuretic peptides; PI, proteasome inhibitors; TTE, transthoracic echocardiography; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cEvery 2 months for patients treated with oral ixazomib.

^dDepending on HF severity and treatment.

^eSee [Figure 18](#).^{295,296,299}

^fLow doses of apixaban (2.5 mg twice a day) or rivaroxaban (10 mg once a day).

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with PI at every clinical visit.	I	C
Home monitoring of BP weekly during the first 3 months and monthly thereafter should be considered for patients treated with PI.	IIa	C
Cardiac serum biomarkers		
Measurement of NP is recommended prior to PI in high- and very high-risk patients. ^{66,303}	I	C
Measurement of NP should be considered prior to PI in low- and moderate-risk patients. ⁶⁶	IIa	C
In patients receiving carfilzomib or bortezomib, measurement of NP should be considered at baseline and every cycle during the first 6 cycles. ^{c,66}	IIa	B
NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA. ^{d,290}	I	B
TTE		
Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.	I	C
Echocardiography surveillance every 3 cycles should be considered in high- and very high-risk patients receiving carfilzomib. ²⁸⁰	IIa	B
Echocardiography surveillance every 3 cycles may be considered in low- and moderate-risk patients receiving carfilzomib.	IIb	C
Echocardiography surveillance should be considered every 3–6 months in patients with AL-CA treated with PI. ^{d,290}	IIa	C
VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE. ^{296,298,302,304,305}	I	B

Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304,305}	I	A
Aspirin should be considered as an alternative to LMWH in patients with MM with no risk factors or one VTE-related risk factor ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304–307}	IIa	B
Low doses of apixaban or rivaroxaban ^f may be considered as an alternative to LMWH or aspirin in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{300–302}	IIb	C

AL-CA, amyloid light-chain cardiac amyloidosis; BP, blood pressure; cTn, cardiac troponin; HF, heart failure; LMWH, low-molecular-weight heparins; MM, multiple myeloma; NP, natriuretic peptides; PI, proteasome inhibitors; TTE, transthoracic echocardiography; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cEvery 2 months for patients treated with oral ixazomib.

^dDepending on HF severity and treatment.

^eSee [Figure 18](#).^{295,296,299}

^fLow doses of apixaban (2.5 mg twice a day) or rivaroxaban (10 mg once a day).

© ESC 2022

Recommendation Table 13

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during multiple myeloma therapies

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with PI at every clinical visit.	I	C
Home monitoring of BP weekly during the first 3 months and monthly thereafter should be considered for patients treated with PI.	IIa	C
Cardiac serum biomarkers		
Measurement of NP is recommended prior to PI in high- and very high-risk patients. ^{66,303}	I	C
Measurement of NP should be considered prior to PI in low- and moderate-risk patients. ⁶⁶	IIa	C
In patients receiving carfilzomib or bortezomib, measurement of NP should be considered at baseline and every cycle during the first 6 cycles. ^{c,66}	IIa	B
NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA. ^{d,290}	I	B
TTE		
Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.	I	C
Echocardiography surveillance every 3 cycles should be considered in high- and very high-risk patients receiving carfilzomib. ²⁸⁰	IIa	B
Echocardiography surveillance every 3 cycles may be considered in low- and moderate-risk patients receiving carfilzomib.	IIb	C
Echocardiography surveillance should be considered every 3–6 months in patients with AL-CA treated with PI. ^{d,290}	IIa	C
VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE. ^{296,298,302,304,305}	I	B

Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304,305}	I	A
Aspirin should be considered as an alternative to LMWH in patients with MM with no risk factors or one VTE-related risk factor ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304–307}	IIa	B
Low doses of apixaban or rivaroxaban ^f may be considered as an alternative to LMWH or aspirin in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{300–302}	IIb	C

AL-CA, amyloid light-chain cardiac amyloidosis; BP, blood pressure; cTn, cardiac troponin; HF, heart failure; LMWH, low-molecular-weight heparins; MM, multiple myeloma; NP, natriuretic peptides; PI, proteasome inhibitors; TTE, transthoracic echocardiography; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cEvery 2 months for patients treated with oral ixazomib.

^dDepending on HF severity and treatment.

^eSee [Figure 18](#).^{295,296,299}

^fLow doses of apixaban (2.5 mg twice a day) or rivaroxaban (10 mg once a day).

Recommendations	Class ^a	Level ^b
BP monitoring		
BP measurement is recommended for patients treated with PI at every clinical visit.	I	C
Home monitoring of BP weekly during the first 3 months and monthly thereafter should be considered for patients treated with PI.	IIa	C
Cardiac serum biomarkers		
Measurement of NP is recommended prior to PI in high- and very high-risk patients. ^{66,303}	I	C
Measurement of NP should be considered prior to PI in low- and moderate-risk patients. ⁶⁶	IIa	C
In patients receiving carfilzomib or bortezomib, measurement of NP should be considered at baseline and every cycle during the first 6 cycles. ^{c,66}	IIa	B
NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA. ^{d,290}	I	B
TTE		
Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.	I	C
Echocardiography surveillance every 3 cycles should be considered in high- and very high-risk patients receiving carfilzomib. ²⁸⁰	IIa	B
Echocardiography surveillance every 3 cycles may be considered in low- and moderate-risk patients receiving carfilzomib.	IIb	C
Echocardiography surveillance should be considered every 3–6 months in patients with AL-CA treated with PI. ^{d,290}	IIa	C
VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE. ^{296,298,302,304,305}	I	B

Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304,305}	I	A
Aspirin should be considered as an alternative to LMWH in patients with MM with no risk factors or one VTE-related risk factor ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{296,304–307}	IIa	B
Low doses of apixaban or rivaroxaban ^f may be considered as an alternative to LMWH or aspirin in patients with MM with VTE-related risk factors ^e (excluding previous VTE) at least during the first 6 months of therapy. ^{300–302}	IIb	C

AL-CA, amyloid light-chain cardiac amyloidosis; BP, blood pressure; cTn, cardiac troponin; HF, heart failure; LMWH, low-molecular-weight heparins; MM, multiple myeloma; NP, natriuretic peptides; PI, proteasome inhibitors; TTE, transthoracic echocardiography; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cEvery 2 months for patients treated with oral ixazomib.

^dDepending on HF severity and treatment.

^eSee [Figure 18](#).^{295,296,299}

^fLow doses of apixaban (2.5 mg twice a day) or rivaroxaban (10 mg once a day).

© ESC 2022

5.5.8. Rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor treatment

The rapidly accelerated fibrosarcoma (RAF) inhibitors vemurafenib, dabrafenib, and encorafenib are approved for the treatment of metastatic melanoma with a *BRAF* V600 mutation. The mitogen-activated extracellular signal-regulated kinase (MEK) inhibitors trametinib, cobimetinib, binimetinib, and selumetinib have also shown significant clinical activity in melanoma patients whose tumour contains a *BRAF* V600 mutation, and are now largely used in combination with RAF inhibitors. The main CV effects to be considered are hypertension, PE, and CTRCD, which are associated with all combinations of RAF and MEK inhibitors, and QTc prolongation, associated solely with the coadministration of cobimetinib and vemurafenib ([Figure 19](#)).^{12,308,309} RAF inhibitor treatment alone or in combination with a MEK inhibitor is associated with an increased risk of MI and AF.³⁰⁸

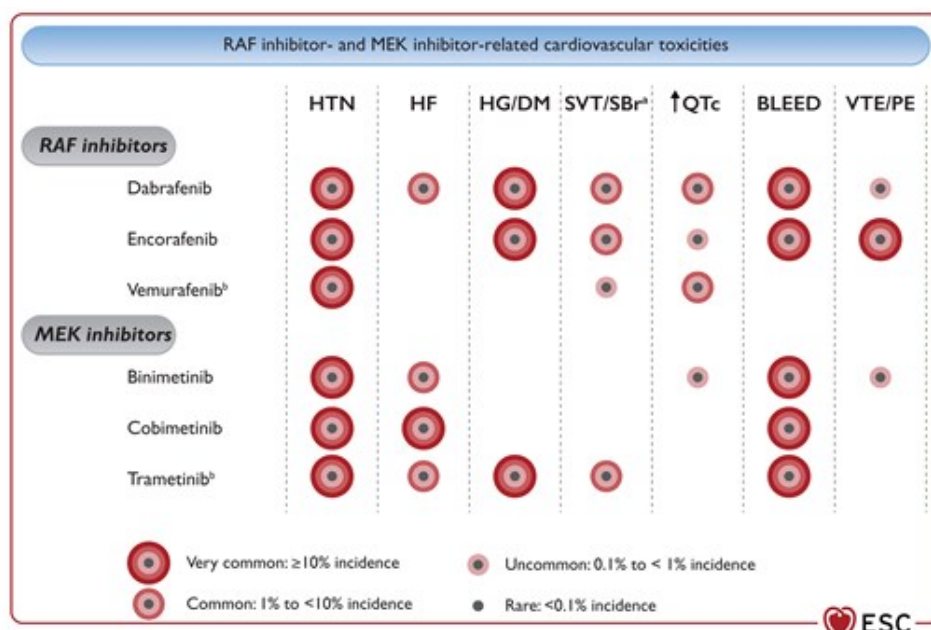


Figure 19

Rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor-related cardiovascular toxicities.

AF, atrial fibrillation; BLEED, increased bleeding risk; DM, diabetes mellitus; EMA, European Medicines Agency; FDA, Food and Drug Administration; HF, heart failure; HG, hyperglycaemia; HTN, hypertension; MedDRA, medical dictionary for regulatory activities; MEK, mitogen-activated extracellular signal-regulated kinase; PE, pulmonary embolism; ^aSVT, sinus bradycardia; SVT, supraventricular tachycardia; VTE, venous thromboembolism. Adverse reactions reported in multiple clinical trials or during post-marketing use are listed by system organ class (in MedDRA) and frequency. If the frequency is unknown or cannot be estimated from the available data, a blank space has been left. ^aDabrafenib is related with SBr. Encorafenib is related with SVT. Vemurafenib rarely causes AF. Trametinib is related with bradycardia in some post-marketing reports.

^bPeripheral oedema is very common. Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

Patients with cancer with pre-existing CVD have an increased frequency of CV adverse events during treatment with MEK and RAF inhibitors, and therefore baseline risk stratification is recommended.¹² Most cardiac complications induced by administration of MEK and RAF inhibitors seem to be attributable to the MEK inhibitor, with the RAF inhibitor enhancing the toxic effects of the MEK inhibitor.^{310a}^{FADTS}³¹³ Hypertension and LVD were twice as frequent when MEK and RAF inhibitors were coadministered compared with single therapy with RAF inhibitor alone.³¹⁴

CTRCD can manifest any time from the first month of therapy to 2 years after the end of the oncological treatment.³¹⁵ Baseline TTE is recommended in patients at moderate to high risk of CTR-CVT. During treatment, it is necessary to monitor BP at each visit and promote weekly outpatient monitoring during the first 3 months and monthly thereafter. In patients treated with cobimetinib/vemurafenib, an ECG is recommended

at 2 and 4 weeks after initiation of treatment and every 3 months thereafter. In high-risk patients, periodic monitoring of ventricular function with echocardiography should be considered every 6 months to 12 months.

CV protective medications (such as angiotensin-converting enzyme inhibitors [ACE-I], angiotensin receptor blockers [ARB], and beta-blockers) have not been evaluated in patients treated with MEK and RAF inhibitors but, from a mechanistic perspective, beta-blockers might prevent CTRCD induced by MEK inhibitors. The MEK/ERK pathway has a cardiac protective effect, regulated by beta-adrenergic signalling, which also controls the p38 mitogen-activated protein kinases pathway, associated with cardiotoxic effects. Beta-blockers might exert their cardioprotective effects by reducing p38 signalling.³¹⁵

Recommendation Table 14

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during combined rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor therapy

Recommendations	Class ^a	Level ^b
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter. ^c	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Baseline echocardiography may be considered in low- and moderate-risk patients scheduled to receive combined RAF and MEK inhibitors.	IIb	C
Echocardiography should be considered every 4 months during the first year in high- and very high-risk patients receiving combined RAF and MEK inhibitors.	IIa	C

BP, blood pressure; ECG, electrocardiogram; MEK, mitogen-activated extracellular signal-regulated kinase; RAF, rapidly accelerated fibrosarcoma.
^aClass of recommendation.
^bLevel of evidence.
^cConsider an ECG and new monitoring in the case of any dose increase (see Section 6.4.2).

© ESC 2022

Recommendations	Class ^a	Level ^b
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter. ^c	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Baseline echocardiography may be considered in low- and moderate-risk patients scheduled to receive combined RAF and MEK inhibitors.	IIb	C
Echocardiography should be considered every 4 months during the first year in high- and very high-risk patients receiving combined RAF and MEK inhibitors.	IIa	C

BP, blood pressure; ECG, electrocardiogram; MEK, mitogen-activated extracellular signal-regulated kinase; RAF, rapidly accelerated fibrosarcoma.
^aClass of recommendation.
^bLevel of evidence.
^cConsider an ECG and new monitoring in the case of any dose increase (see Section 6.4.2).

© ESC 2022

Recommendation Table 14

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during combined rapidly accelerated fibrosarcoma and mitogen-activated extracellular signal-regulated kinase inhibitor therapy

Recommendations	Class ^a	Level ^b
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter. ^c	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Baseline echocardiography may be considered in low- and moderate-risk patients scheduled to receive combined RAF and MEK inhibitors.	IIb	C
Echocardiography should be considered every 4 months during the first year in high- and very high-risk patients receiving combined RAF and MEK inhibitors.	IIa	C

BP, blood pressure; ECG, electrocardiogram; MEK, mitogen-activated extracellular signal-regulated kinase; RAF, rapidly accelerated fibrosarcoma.
^aClass of recommendation.
^bLevel of evidence.
^cConsider an ECG and new monitoring in the case of any dose increase (see Section 6.4.2).

© ESC 2022

Recommendations	Class ^a	Level ^b
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter. ^c	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Baseline echocardiography may be considered in low- and moderate-risk patients scheduled to receive combined RAF and MEK inhibitors.	IIb	C
Echocardiography should be considered every 4 months during the first year in high- and very high-risk patients receiving combined RAF and MEK inhibitors.	IIa	C

BP, blood pressure; ECG, electrocardiogram; MEK, mitogen-activated extracellular signal-regulated kinase; RAF, rapidly accelerated fibrosarcoma.
^aClass of recommendation.
^bLevel of evidence.
^cConsider an ECG and new monitoring in the case of any dose increase (see Section 6.4.2).

© ESC 2022

5.5.9. Immune checkpoint inhibitors

Immunotherapies, which harness the immune system to destroy cancer cells, come in different forms but the most widely used are ICI.³¹⁶ The immune checkpoints are proteins expressed in the T cells that inhibit their activation when they contact a body cell. ICI include monoclonal antibodies that block the immune brakes or regulators, cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) (ipilimumab, tremelimumab), programmed death-1 (PD-1) (nivolumab, cemiplimab, pembrolizumab), and programmed death-ligand 1 (PD-L1) (atezolizumab, avelumab, durvalumab) expressed in the cancer cells, with the consequent cytotoxic immune response. By blocking these checkpoints from binding with their partner proteins, ICI inhibit the $\text{CD}28/\text{B7}$ signal, activating T cells and promoting killing of cancer cells. Although their pathophysiology is not clearly defined, ICI may also trigger an overactivation of T cells against non-cancerous tissues, leading to immune-related adverse events.³¹⁷ Immune-related CV side effects may lead to life-threatening CV complications such as fulminant myocarditis, myopericarditis, cardiac dysfunction, arrhythmias, or MI, which often results in the discontinuation of ICI.^{318,319}

The largest case series of 122 patients with ICI-associated myocarditis had early onset of symptoms (median of 30 days after initial exposure to ICI), and up to 50% died.³²⁰ Late CV events (>90 days) are less well characterized but generally exhibit a higher risk of non-inflammatory HF, progressive atherosclerosis, hypertension, and mortality rates.³²¹ Other CV toxicities described during ICI therapy are MI, AV block, supraventricular and ventricular arrhythmias, sudden death, Takotsubo-like syndrome, non-inflammatory HF, hypercholesterolaemia, pericarditis, pericardial effusion, ischaemic stroke, and VTE.³²² A meta-analysis including 32 518 patients receiving ICI treatment reported an increased risk of myocarditis, pericardial diseases, HF, dyslipidaemia, MI, and cerebral arterial ischaemia.³²³ Conditions related with high baseline ICI-related CV toxicity risk include dual ICI therapy (e.g. ipilimumab and nivolumab), combination ICI therapy with other cardiotoxic therapies, and patients with ICI-related non-CV events or prior CTRCD or CVD

(Figure 20).^{324,325} All patients on ICI treatment should have an ECG and troponin assay at baseline (Figure 20).^{326,327,329} High-risk patients should additionally have a TTE evaluation at baseline. Due to the lack of evidence-based recommendations, the monitoring of ICI therapy is challenging. Once started on therapy, ECG, cTn, and NP should be checked.^{330,331,332} In the JAVELIN trial, which assessed avelumab plus axitinib vs. sunitinib, no clinical value was observed for on-treatment routine TTE monitoring in asymptomatic patients.³³³ However, in high-risk patients, and in those with high baseline cTn levels, TTE monitoring may be considered. In patients who develop ECG abnormalities, new biomarker changes, or new cardiac symptoms at any time, prompt cardio-oncology evaluation is strongly recommended, including TTE for the evaluation of LVEF and GLS, and CMR when myocarditis is suspected (Table 3).³³⁴

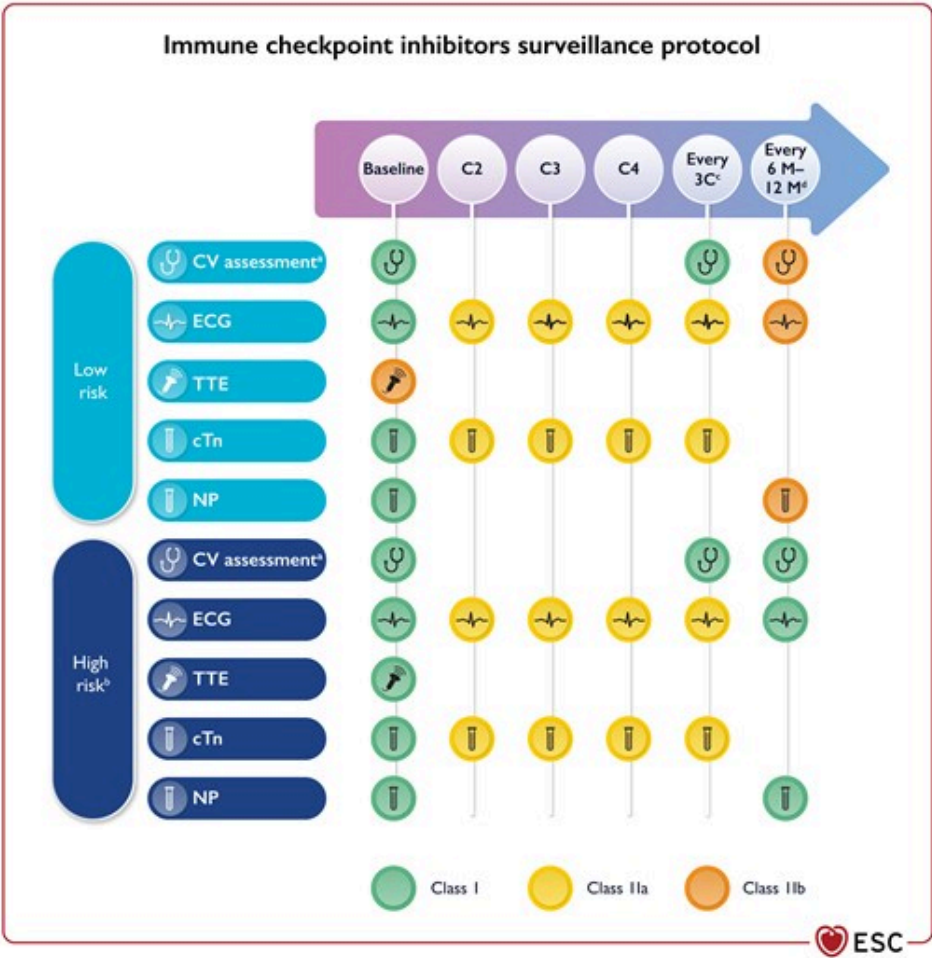


Figure 20
Cardiovascular surveillance in patients treated with immune checkpoint inhibitors.

BNP, B-type natriuretic peptide; BP, blood pressure; C, chemotherapy cycle; cTn, cardiac troponin; CV, cardiovascular; CVD, cardiovascular disease; CTRCD, cancer therapy-related cardiac dysfunction; ECG, electrocardiogram; HbA1c, glycated haemoglobin; ICI, immune checkpoint inhibitors; M, months; NP, natriuretic peptides (including BNP and NT-proBNP); NT-proBNP, N-terminal pro-B-type natriuretic peptide; TTE, transthoracic echocardiography. ^aIncluding physical examination, BP, lipid profile, and HbA1c. ^bDual ICI, combination ICI-cardiotoxic therapy, ICI-related non-CV events, prior CTRCD or CVD. ^cEvery three cycles until completion of therapy to detect subclinical ICI-related CV toxicity. ^dIn patients who require long-term (>12 months) ICI treatment.

Recommendations	Class ^a	Level ^b
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy. ³³³	I	B
Baseline echocardiography is recommended in high-risk patients ^c before starting ICI therapy. ³³³	I	B
Baseline echocardiography may be considered in all patients before starting ICI therapy.	IIb	C
Serial ECG and cTn measurements should be considered before ICI doses 2, 3, and 4, and if normal, reduce to every three doses until completion of therapy to detect subclinical ICI-related CV toxicity. ³³³	IIa	B
CV assessment ^d is recommended every 6–12 months in high-risk patients ^c who require long-term (>12 months) ICI treatment. ^{321–323,335,336}	I	C
CV assessment ^d may be considered every 6–12 months in all patients who require long-term (>12 months) ICI treatment.	IIb	C

BNP, B-type natriuretic peptide; BP, blood pressure; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; ICI, immune checkpoint inhibitors; NP, natriuretic peptides; NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^aClass of recommendation.

^bLevel of evidence.

^cDual ICI, combination ICI-cardiotoxic therapy; ICI-related non-CV events, prior CTRCD, or CVD.

^dPhysical examination, BP, NP (BNP or NT-proBNP), lipid profile, HbA1c, and ECG.

© ESC 2022

Recommendations	Class ^a	Level ^b
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy. ³³³	I	B
Baseline echocardiography is recommended in high-risk patients ^c before starting ICI therapy. ³³³	I	B
Baseline echocardiography may be considered in all patients before starting ICI therapy.	IIb	C

Serial ECG and cTn measurements should be considered before ICI doses 2, 3, and 4, and if normal, reduce to every three doses until completion of therapy to detect subclinical ICI-related CV toxicity. ³³³	IIa	B
CV assessment ^d is recommended every 6–12 months in high-risk patients ^c who require long-term (>12 months) ICI treatment. ^{321–323,335,336}	I	C
CV assessment ^d may be considered every 6–12 months in all patients who require long-term (>12 months) ICI treatment.	IIb	C

© ESC 2022

BNP, B-type natriuretic peptide; BP, blood pressure; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; ICI, immune checkpoint inhibitors; NP, natriuretic peptides; NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^aClass of recommendation.

^bLevel of evidence.

^cDual ICI, combination ICI-cardiotoxic therapy; ICI-related non-CV events, prior CTRCD, or CVD.

^dPhysical examination, BP, NP (BNP or NT-proBNP), lipid profile, HbA1c, and ECG.

Recommendation Table 15

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during immunotherapy

Recommendations	Class ^a	Level ^b
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy. ³³³	I	B
Baseline echocardiography is recommended in high-risk patients ^c before starting ICI therapy. ³³³	I	B
Baseline echocardiography may be considered in all patients before starting ICI therapy.	IIb	C

Serial ECG and cTn measurements should be considered before ICI doses 2, 3, and 4, and if normal, reduce to every three doses until completion of therapy to detect subclinical ICI-related CV toxicity. ³³³	IIa	B
CV assessment ^d is recommended every 6–12 months in high-risk patients ^c who require long-term (>12 months) ICI treatment. ^{321–323,335,336}	I	C
CV assessment ^d may be considered every 6–12 months in all patients who require long-term (>12 months) ICI treatment.	IIb	C

© ESC 2022

BNP, B-type natriuretic peptide; BP, blood pressure; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; ICI, immune checkpoint inhibitors; NP, natriuretic peptides; NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^aClass of recommendation.

^bLevel of evidence.

^cDual ICI, combination ICI-cardiotoxic therapy; ICI-related non-CV events, prior CTRCD, or CVD.

^dPhysical examination, BP, NP (BNP or NT-proBNP), lipid profile, HbA1c, and ECG.

Recommendations	Class ^a	Level ^b
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy. ³³³	I	B
Baseline echocardiography is recommended in high-risk patients ^c before starting ICI therapy. ³³³	I	B
Baseline echocardiography may be considered in all patients before starting ICI therapy.	IIb	C

Serial ECG and cTn measurements should be considered before ICI doses 2, 3, and 4, and if normal, reduce to every three doses until completion of therapy to detect subclinical ICI-related CV toxicity. ³³³	IIa	B
CV assessment ^d is recommended every 6–12 months in high-risk patients ^c who require long-term (>12 months) ICI treatment. ^{321–323,335,336}	I	C
CV assessment ^d may be considered every 6–12 months in all patients who require long-term (>12 months) ICI treatment.	IIb	C

BNP, B-type natriuretic peptide; BP, blood pressure; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; ICI, immune checkpoint inhibitors; NP, natriuretic peptides; NT-proBNP, N-terminal pro-B-type natriuretic peptide.

^aClass of recommendation.

^bLevel of evidence.

^cDual ICI, combination ICI-cardiotoxic therapy; ICI-related non-CV events, prior CTRCD, or CVD.

^dPhysical examination, BP, NP (BNP or NT-proBNP), lipid profile, HbA1c, and ECG.

© ESC 2022

5.5.10. Androgen deprivation therapies for prostate cancer

Androgen deprivation therapy (ADT) is prescribed in 40% of men with prostate cancer as neoadjuvant and/or adjuvant therapy to RT or for biochemical relapse following prostate cancer surgery. Gonadotropin-releasing hormone (GnRH) agonists are the most frequently prescribed ADT. However, GnRH agonists are associated with an increased CV risk and mortality, particularly in patients with prostate cancer aged >60 years.^{337,338} Baseline risk stratification in patients requiring GnRH agonists depends on vascular disease risk (Figure 21).^{339,340} No dedicated CV toxicity risk calculators have been developed for patients receiving ADT. It was the consensus of the authors to recommend SCORE2 or SCORE2-OP to stratify CV risk in patients receiving ADT without previous CVD.¹⁹

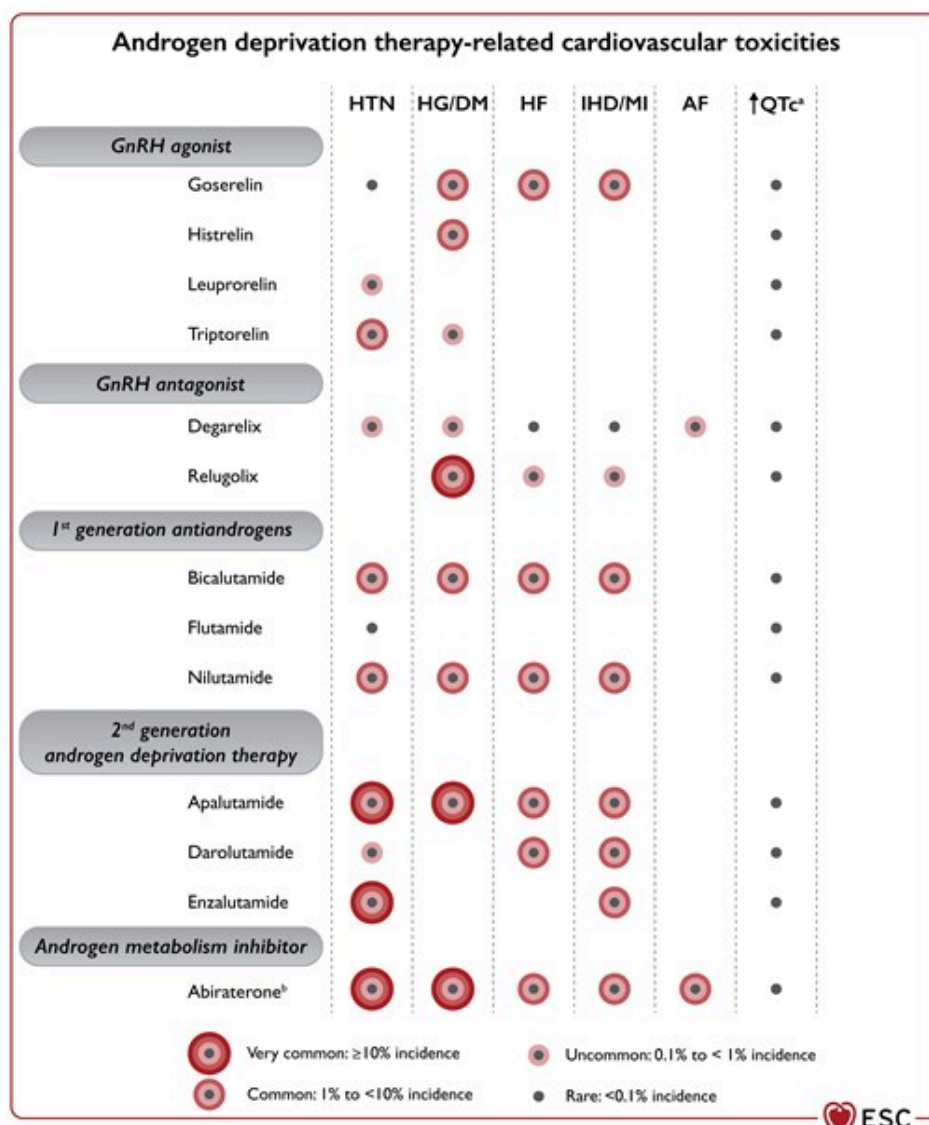


Figure 21

Androgen deprivation therapy-related cardiovascular toxicities.

ADT, androgen deprivation therapy; AF, atrial fibrillation; DM, diabetes mellitus; EMA, European Medicines Agency; FDA, Food and Drug Administration; GnRH, gonadotropin-releasing hormone; HF, heart failure; HG, hyperglycaemia; HTN, hypertension; IHD, ischaemic heart disease; MedDRA, medical dictionary for regulatory activities; MI, myocardial infarction; ↑QTc, corrected QT interval prolongation; TdP, torsade de pointes. Adverse reactions reported in multiple clinical trials or during post-marketing use are listed by system organ class (in MedDRA) and frequency. If the frequency is unknown or cannot be estimated from the available data, a blank space has been left. ^aADT may prolong the QTc interval. In patients with a history of risk factors for QT prolongation and in patients receiving concomitant medicinal products that might prolong the QT interval, physicians should assess the benefit/risk ratio including the potential for TdP prior to initiating the treatment. ^bIncreased risk of QTc prolongation in combination with ADT.^{49,339,340,349,350} Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

The use of GnRH antagonists represents an alternative in the treatment of prostate cancer, and preclinical and clinical (HERO trial)³⁴¹ data suggest that GnRH antagonist use is associated with significantly lower overall mortality and CV events compared with agonists.³⁴² However, more research is needed in this field. In the

PRONOUNCE trial, no difference in MACE at 1 year was observed between degarelix (a GnRH antagonist) and leuprolide (a GnRH agonist), although the trial was stopped early.³⁴³ Lower CV event rates were detected compared with previous studies and all patients were reviewed by a cardiologist at enrolment (leading to optimal CVRF management).³⁴³

The main CV effects to be considered are hypertension, DM, ischaemic heart disease (IHD) and CTRCD.^{339,344} ADT is uncommonly associated with QTc prolongation and rarely causes torsade de pointes (TdP) through blockade of testosterone effects on ventricular repolarization.^{345,346} ECG monitoring and correction of QT prolongation precipitant factors (see [Section 6.4.2](#); [TableÂ 9](#); [Supplementary data, TableÂ S13](#)) is recommended^{340,347,348} during prostate cancer treatment if the baseline QTc interval is prolonged.^{49,339,340,347,349,350}

Recommendation Table 16

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended in patients treated with ADT without pre-existing CVD. ^{19,341,342}	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy. ^{e,339–342}	I	B
A GnRH antagonist should be considered in patients with pre-existing symptomatic CAD ^f who require ADT. ^{341,342}	IIa	B
Annual CV risk assessment ^c is recommended during ADT. ^{19,339,341,342}	I	B

© ESC 2022

ACS, acute coronary syndromes; ADT, androgen deprivation therapy; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; GnRH, gonadotropin-releasing hormone; HbA1c, glycated haemoglobin; QTc, corrected QT interval; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG, and patient education on healthy lifestyle and lifestyle risk factor control is recommended.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%, moderate risk 5% to <10%, high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^eSee Table 9.

^fCCS and ACS.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended in patients treated with ADT without pre-existing CVD. ^{19,341,342}	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy. ^{e,339–342}	I	B
A GnRH antagonist should be considered in patients with pre-existing symptomatic CAD ^f who require ADT. ^{341,342}	IIa	B
Annual CV risk assessment ^c is recommended during ADT. ^{19,339,341,342}	I	B

ACS, acute coronary syndromes; ADT, androgen deprivation therapy; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; GnRH, gonadotropin-releasing hormone; HbA1c, glycated haemoglobin; QTc, corrected QT interval; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG, and patient education on healthy lifestyle and lifestyle risk factor control is recommended.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^eSee Table 9.

^fCCS and ACS.

© ESC 2022

Recommendation Table 16

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended in patients treated with ADT without pre-existing CVD. ^{19,341,342}	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy. ^{e,339–342}	I	B
A GnRH antagonist should be considered in patients with pre-existing symptomatic CAD ^f who require ADT. ^{341,342}	IIa	B
Annual CV risk assessment ^c is recommended during ADT. ^{19,339,341,342}	I	B

© ESC 2022

ACS, acute coronary syndromes; ADT, androgen deprivation therapy; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; GnRH, gonadotropin-releasing hormone; HbA1c, glycated haemoglobin; QTc, corrected QT interval; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG, and patient education on healthy lifestyle and lifestyle risk factor control is recommended.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%, moderate risk 5% to <10%, high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^eSee Table 9.

^fCCS and ACS.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended in patients treated with ADT without pre-existing CVD. ^{19,341,342}	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy. ^{e,339–342}	I	B
A GnRH antagonist should be considered in patients with pre-existing symptomatic CAD ^f who require ADT. ^{341,342}	IIa	B
Annual CV risk assessment ^c is recommended during ADT. ^{19,339,341,342}	I	B

ACS, acute coronary syndromes; ADT, androgen deprivation therapy; BP, blood pressure; CAD, coronary artery disease; CCS, chronic coronary syndromes; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; GnRH, gonadotropin-releasing hormone; HbA1c, glycated haemoglobin; QTc, corrected QT interval; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG, and patient education on healthy lifestyle and lifestyle risk factor control is recommended.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%, moderate risk 5% to <10%, high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.

^eSee Table 9.

^fCCS and ACS.

© ESC 2022

5.5.11. Endocrine therapies for breast cancer

Endocrine therapy is a common treatment as 65–70% of all early and metastatic BC patients develop hormone receptor-positive disease.²² Selective oestrogen receptor modulators (tamoxifen, toremifene) or aromatase inhibitors (AI) (letrozole, anastrozole, or exemestane) are recommended in early BC (EBC) according to menopausal status, comorbidities, and the risk of disease relapse. The use of AI in combination with cyclin-dependent kinase (CDK) 4/6 inhibitors is recommended as first- or second-line therapy in patient with hormone receptor-positive/HER2-negative metastatic BC.

The use of AI increases the risk of dyslipidaemia, metabolic syndrome, hypertension, HF, and MI.³³⁹ In the ATAC (Arimidex and Tamoxifen Alone or in Combination) trial, anastrozole-treated patients with pre-existing CAD experienced more CV events (17% vs. 10%) and cholesterol level elevation (9% vs. 5%) than those treated with tamoxifen.^{351,352} Similarly, HF was significantly more common with letrozole compared with tamoxifen in the BIG (Breast International Group) 1–98 trial.³⁵³ Longer AI treatment duration was associated with increased odds of developing CVD in two large meta-analyses.^{354,355}

Significantly increased VTE risk has been consistently demonstrated with tamoxifen^{351,353} and it is not recommended in patients with thrombotic risks. Toremifene and high-dose tamoxifen were found to prolong QTc interval^{339,340}; however, no risk data have been published in patients treated with the standard tamoxifen dose used in BC (20^{mg}/day).

The risks of VTE, hypercholesterolaemia, and CVD should be discussed with patients, while recognizing that the absolute benefits of preventing BC recurrence usually outweigh the CV risks.³³⁹ In patients <70 years old without clinical manifestations of atherosclerotic disease, estimation of 10-year fatal and non-fatal CVD risk with SCORE2 (if ≥ 70 years, SCORE2-OP) is recommended.¹⁹ Cholesterol levels and BP should be monitored regularly in patients receiving AI.³⁵⁶ Physical activity and healthy diet are also advised to reduce weight and cholesterol levels. Smoking cessation is strongly recommended to reduce CV risk (e.g. CAD during AI therapy and VTE during tamoxifen therapy).

Recommendation Table 17

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during endocrine therapy for breast cancer

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^{d,e} is recommended in BC patients receiving endocrine therapies without pre-existing CVD. ¹⁹	I	C
Annual CV risk assessment ^c is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	I	C
CV risk assessment ^c should be considered every 5 years in BC patients with low or moderate 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	IIa	C

© ESC 2022

BC, breast cancer; BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dOr other validated CV risk scores.

^eSCORE2 (<70 years) or SCORE2-OP (≥ 70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk $\geq 7.5\%$; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk $\geq 10\%$; ≥ 70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk $\geq 15\%$.¹⁹

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^{d,e} is recommended in BC patients receiving endocrine therapies without pre-existing CVD. ¹⁹	I	C
Annual CV risk assessment ^c is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	I	C
CV risk assessment ^c should be considered every 5 years in BC patients with low or moderate 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	IIa	C

BC, breast cancer; BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dOr other validated CV risk scores.

^eSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

© ESC 2022

Recommendation Table 17

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during endocrine therapy for breast cancer

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^{d,e} is recommended in BC patients receiving endocrine therapies without pre-existing CVD. ¹⁹	I	C
Annual CV risk assessment ^c is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	I	C
CV risk assessment ^c should be considered every 5 years in BC patients with low or moderate 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	IIa	C

© ESC 2022

BC, breast cancer; BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dOr other validated CV risk scores.

^eSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^{d,e} is recommended in BC patients receiving endocrine therapies without pre-existing CVD. ¹⁹	I	C
Annual CV risk assessment ^c is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	I	C
CV risk assessment ^c should be considered every 5 years in BC patients with low or moderate 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP. ^{d,e}	IIa	C

© ESC 2022

BC, breast cancer; BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dOr other validated CV risk scores.

^eSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

5.5.12. Cyclin-dependent kinase 4/6 inhibitors

The use of CDK 4/6 inhibitors (palbociclib, ribociclib, and abemaciclib) in combination with endocrine therapy is approved for the treatment of patients with hormone receptor-positive/HER2-negative metastatic BC. This combination has resulted in improvements in progression-free survival and, in some trials, overall survival.^{357a,359} CDK 4/6 inhibitors have demonstrated a potential for QT prolongation,^{339,360} particularly with ribociclib. The phase III trials of ribociclib incorporated routine ECG monitoring.^{361a,368} Baseline ECG is recommended and ECGs should be repeated at day 14 of the first cycle, before the second cycle, with any dose increase and as clinically indicated.³⁵⁷

In patients who already have, or are at significant risk of developing, QT prolongation ([Section 6.4.2](#)), the risks/benefits for ribociclib should be discussed by a MDT. Importantly, the use of ribociclib should be avoided in combination with drugs known to prolong QT interval and/or strong CYP3A inhibitors.³⁵⁷

The prescribing information does not recommend ribociclib in combination with tamoxifen due to a higher risk of QTc prolongation.^{252,367}

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy

Recommendations	Class ^a	Level ^b
QTc ^{c,d} monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib. ^{361,365,367,368}	I	A
QTc ^{c,d} monitoring is recommended in patients treated with ribociclib with any dose increase. ^{361,365,367,368}	I	B
QTc ^c monitoring should be considered in patients treated with palbociclib or abemaciclib who have a baseline QTc above the normal range ^c or other conditions that may prolong the QTc interval. ^e	IIa	C

© ESC 2022

QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cQT interval using Fridericia correction ($QTcF = QT/\sqrt{RR}$) is the preferred method in patients with cancer. Upper 99% limit of normal for QTc values in the general population are 450 ms for men and 460 ms for women.³⁶⁹

^dAccording to the European Medicines Agency: (1) ribociclib should be interrupted when QTcF > 480 ms; (2) if QTcF prolongation resolves to <481 ms, resume treatment at the same dose level; (3) if QTcF ≥ 481 ms recurs, interrupt dose until QTcF resolves to <481 ms and then resume ribociclib at next lower dose level.

^eSee [Section 6.4.2](#) and [Table 8](#).

Recommendations	Class ^a	Level ^b
QTc ^{c,d} monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib. ^{361,365,367,368}	I	A
QTc ^{c,d} monitoring is recommended in patients treated with ribociclib with any dose increase. ^{361,365,367,368}	I	B
QTc ^c monitoring should be considered in patients treated with palbociclib or abemaciclib who have a baseline QTc above the normal range ^c or other conditions that may prolong the QTc interval. ^e	IIa	C

© ESC 2022

QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cQT interval using Fridericia correction ($QTcF = QT/\sqrt{RR}$) is the preferred method in patients with cancer. Upper 99% limit of normal for QTc values in the general population are 450 ms for men and 460 ms for women.³⁶⁹

^dAccording to the European Medicines Agency: (1) ribociclib should be interrupted when $QTcF > 480$ ms; (2) if QTcF prolongation resolves to < 481 ms, resume treatment at the same dose level; (3) if $QTcF \geq 481$ ms recurs, interrupt dose until QTcF resolves to < 481 ms and then resume ribociclib at next lower dose level.

^eSee [Section 6.4.2](#) and [Table 8](#).

Recommendation Table 18

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy

Recommendations	Class ^a	Level ^b
QTc ^{c,d} monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib. ^{361,365,367,368}	I	A
QTc ^{c,d} monitoring is recommended in patients treated with ribociclib with any dose increase. ^{361,365,367,368}	I	B
QTc ^c monitoring should be considered in patients treated with palbociclib or abemaciclib who have a baseline QTc above the normal range ^c or other conditions that may prolong the QTc interval. ^e	IIa	C

© ESC 2022

QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.

^aClass of recommendation.

^bLevel of evidence.

^cQT interval using Fridericia correction ($QTcF = QT/\sqrt[3]{RR}$) is the preferred method in patients with cancer. Upper 99% limit of normal for QTc values in the general population are 450 ms for men and 460 ms for women.³⁶⁹

^dAccording to the European Medicines Agency: (1) ribociclib should be interrupted when $QTcF > 480$ ms; (2) if QTcF prolongation resolves to < 481 ms, resume treatment at the same dose level; (3) if $QTcF \geq 481$ ms recurs, interrupt dose until QTcF resolves to < 481 ms and then resume ribociclib at next lower dose level.

^eSee [Section 6.4.2](#) and [Table 8](#).

Recommendations	Class ^a	Level ^b
QTc ^{c,d} monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib. ^{361,365,367,368}	I	A
QTc ^{c,d} monitoring is recommended in patients treated with ribociclib with any dose increase. ^{361,365,367,368}	I	B
QTc ^c monitoring should be considered in patients treated with palbociclib or abemaciclib who have a baseline QTc above the normal range ^c or other conditions that may prolong the QTc interval. ^e	IIa	C

QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction.
^aClass of recommendation.
^bLevel of evidence.
^cQT interval using Fridericia correction ($QTcF = QT/\sqrt{RR}$) is the preferred method in patients with cancer. Upper 99% limit of normal for QTc values in the general population are 450 ms for men and 460 ms for women.³⁶⁹
^dAccording to the European Medicines Agency: (1) ribociclib should be interrupted when QTcF > 480 ms; (2) if QTcF prolongation resolves to <481 ms, resume treatment at the same dose level; (3) if QTcF ≥ 481 ms recurs, interrupt dose until QTcF resolves to <481 ms and then resume ribociclib at next lower dose level.
^eSee [Section 6.4.2](#) and [Table 8](#).

© ESC 2022

5.5.13. Anaplastic lymphoma kinase inhibitors

Patients with cancer treated with anaplastic lymphoma kinase (ALK) inhibitors may develop adverse CV events including sinus bradycardia, AV block, QTc prolongation, hypertension, hyperglycaemia, and dyslipidaemia.^{370,371} ACS and HF have rarely been described under crizotinib.³⁷² A baseline ECG is recommended in patients prior to starting an ALK inhibitor, especially crizotinib, and patients may have an ECG 4 weeks after the start of treatment and every 3â€¦6 months thereafter, particularly if the baseline ECG is abnormal. Home BP monitoring should be considered in patients treated with brigatinib or lorlatinib. Patients receiving lorlatinib or crizotinib treatment should have cholesterol levels checked every 3â€¦6 months and treated if elevated.

5.5.14. Epidermal growth factor receptor inhibitors

Osimertinib is an oral irreversible, epidermal growth factor receptor (EGFR)-TKI approved for patients with non-small cell lung cancer expressing *EGFR* mutations. Recent data have shown that osimertinib is associated with an increased risk of QTc prolongation, AF, VTE, LVD, and HF ([FigureÂ 22](#)).^{373,374} A study of 123 patients with *EGFR*-mutant non-small cell lung cancer treated with osimertinib reported a 4.9% incidence of HF or MI and a significant decrease in LVEF <53% in 11% of patients with TTE surveillance.³⁷⁵ Pre-existing hypertension and older age are risk factors for LVD and HF (3.9% and 2.6% incidence, respectively).³⁷⁶ LVD and HF were more common during the first year of therapy.³⁷⁶

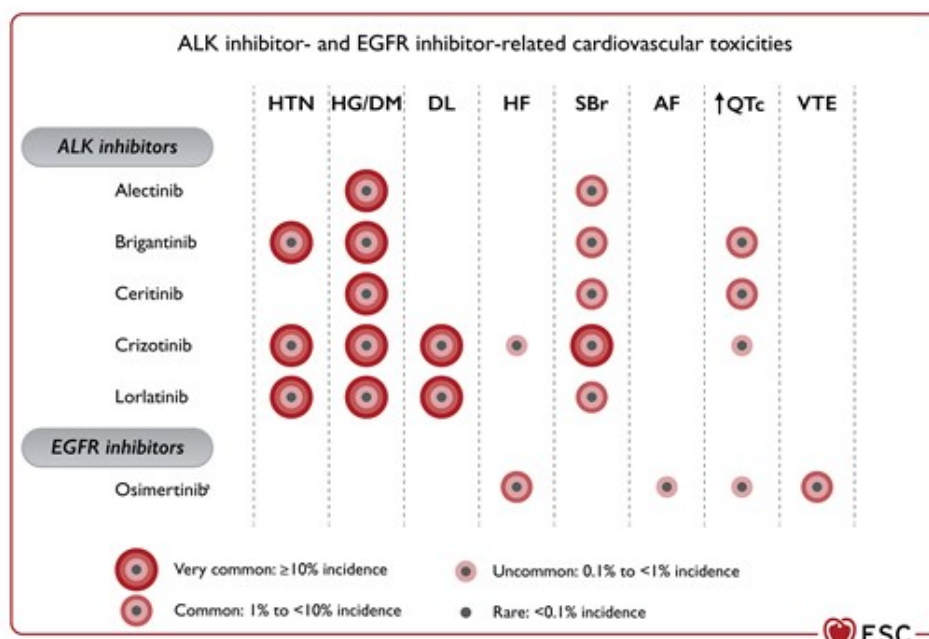


Figure 22

Anaplastic lymphoma kinase and epidermal growth factor receptor inhibitor-related cardiovascular toxicities.

AF, atrial fibrillation; ALK, anaplastic lymphoma kinase; DL, dyslipidaemia; DM, diabetes mellitus; EGFR, epidermal growth factor receptor; EMA, European Medicines Agency; FDA, Food and Drug Administration; HF, heart failure; HG, hyperglycaemia; HTN, hypertension; MedDRA, medical dictionary for regulatory activities; ^aOsimertinib increases the risk of hypomagnesaemia. Figure developed from EMA prescribing information,²⁵² FDA prescribing information.²⁵³

[Open in new tab](#)[Download slide](#)

Baseline CV risk stratification, ECG and TTE prior to starting osimertinib is recommended. Three-monthly echocardiographic surveillance for new LVD during osimertinib treatment should be considered. Close monitoring of magnesium levels is also recommended to minimize the risk of osimertinib-induced hypomagnesaemia and QTc prolongation.

Recommendation Table 19

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during anaplastic lymphoma kinase and epidermal growth factor receptor inhibitors

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib. ³⁷⁶	I	B
Home BP monitoring should be considered for patients treated with brigatinib, crizotinib, or lorlatinib.	IIa	C
Cholesterol profile assessment every 3–6 months should be considered for patients on crizotinib and lorlatinib.	IIa	C
Echocardiography should be considered every 3 months in patients during osimertinib therapy. ³⁷⁶	IIa	B
ECG should be considered 4 weeks after starting therapy and every 3–6 months in patients during ALK inhibitor therapy.	IIa	C

© ESC 2022

ALK, anaplastic lymphoma kinase; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; HbA1c, glycated haemoglobin.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib. ³⁷⁶	I	B
Home BP monitoring should be considered for patients treated with brigatinib, crizotinib, or lorlatinib.	IIa	C
Cholesterol profile assessment every 3–6 months should be considered for patients on crizotinib and lorlatinib.	IIa	C
Echocardiography should be considered every 3 months in patients during osimertinib therapy. ³⁷⁶	IIa	B
ECG should be considered 4 weeks after starting therapy and every 3–6 months in patients during ALK inhibitor therapy.	IIa	C

© ESC 2022

ALK, anaplastic lymphoma kinase; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; HbA1c, glycated haemoglobin.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement.

Recommendation Table 19

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring during anaplastic lymphoma kinase and epidermal growth factor receptor inhibitors

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib. ³⁷⁶	I	B
Home BP monitoring should be considered for patients treated with brigatinib, crizotinib, or lorlatinib.	IIa	C
Cholesterol profile assessment every 3–6 months should be considered for patients on crizotinib and lorlatinib.	IIa	C
Echocardiography should be considered every 3 months in patients during osimertinib therapy. ³⁷⁶	IIa	B
ECG should be considered 4 weeks after starting therapy and every 3–6 months in patients during ALK inhibitor therapy.	IIa	C

© ESC 2022

ALK, anaplastic lymphoma kinase; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; HbA1c, glycated haemoglobin.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement.

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib. ³⁷⁶	I	B
Home BP monitoring should be considered for patients treated with brigatinib, crizotinib, or lorlatinib.	IIa	C
Cholesterol profile assessment every 3–6 months should be considered for patients on crizotinib and lorlatinib.	IIa	C
Echocardiography should be considered every 3 months in patients during osimertinib therapy. ³⁷⁶	IIa	B
ECG should be considered 4 weeks after starting therapy and every 3–6 months in patients during ALK inhibitor therapy.	IIa	C

© ESC 2022

ALK, anaplastic lymphoma kinase; BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; HbA1c, glycated haemoglobin.

^aClass of recommendation.

^bLevel of evidence.

^cPhysical examination, BP measurement, ECG, lipid profile, and HbA1c measurement.

5.5.15. Chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies

CAR-T therapy is used for the treatment of acute lymphocytic leukaemia and aggressive B-cell lymphomas.³⁷⁷ Although the reported incidence is variable, there is a growing recognition of the association between CAR-T therapy and CTR-CVT, including LVD, HF, cardiac arrhythmias, pericardial effusion, TTS, and cardiac arrest.^{378,383} The majority of the described CV toxicities have been shown to be associated with the occurrence of cytokine release syndrome (CRS).^{377,384} Baseline CV evaluation including ECG, NP, and cTn is recommended in all patients. Baseline TTE should also be considered, especially in patients with pre-existing CVRF and CVD. After receiving CAR-T therapy, patients may develop systemic inflammatory syndromes.³⁸⁵ CRS should be suspected when a patient develops fever, with or without tachypnoea, tachycardia, hypotension, hypoxia, and/or other end-organ dysfunction hours to days after treatment.³⁸⁵ A high index of suspicion is necessary to diagnose CRS and to distinguish it from other conditions that occur in these settings (infections, HF, drug reactions, and PE).^{378,386} Among adults, there was a relationship between CRS and CV events. An elevation in cTn is commonly seen in patients with CRS and is associated with an increased risk for subsequent CV events.³⁷⁸ In a recent retrospective pharmacovigilance study, CAR-T was associated with tachyarrhythmias (AF the most common, followed by ventricular arrhythmias), cardiomyopathy, and pleural and pericardial diseases.³⁷⁹ Globally, the fatality rate of CV and pulmonary

adverse events was 30.9%.^{378,379,387} Early cardiac evaluation in patients with cTn increase should include NP, ECG, and echocardiography (see [Section 6.1.4](#) for management).³⁸⁸

Adoptive cellular therapy with TIL has emerged as an effective treatment option for unresectable stage III/IV metastatic melanoma. With TILs, the CV toxicity appears to be related to direct myocardial and vascular toxicity.³⁸⁰ Baseline assessment and CV surveillance in patients before TIL therapies is the same pathway recommended for CAR-T therapies.

Recommendation Table 20

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies

Recommendations	Class ^a	Level ^b
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography should be considered before starting CAR-T and TIL therapies. ³⁸⁸	IIa	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 . ^{c,378,388}	I	C

© ESC 2022

ASTCT, American Society for Transplantation and Cellular Therapy; CAR-T, chimeric antigen receptor T cell; CRS, cytokine release syndrome; cTn, cardiac troponin; CVD, cardiovascular disease; ECG, electrocardiogram; NP, natriuretic peptides; TIL, tumour-infiltrating lymphocytes.

^aClass of recommendation.

^bLevel of evidence.

^cDetermine CRS grade according to ASTCT grading: Grade 1: fever; Grade 2: fever AND hypotension not requiring vasopressors AND/OR hypoxia requiring low-flow nasal oxygen; Grade 3: fever AND hypotension requiring one vasopressor $\pm$ vasopressin AND/OR hypoxia requiring high-flow nasal cannula or facemask or non-rebreather mask or Venturi mask; Grade 4: fever AND hypotension requiring multiple vasopressors, not including vasopressin AND/OR hypoxia requiring positive airway pressure.

Recommendations	Class ^a	Level ^b
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography should be considered before starting CAR-T and TIL therapies. ³⁸⁸	IIa	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 . ^{c,378,388}	I	C

© ESC 2022

ASTCT, American Society for Transplantation and Cellular Therapy; CAR-T, chimeric antigen receptor T cell; CRS, cytokine release syndrome; cTn, cardiac troponin; CVD, cardiovascular disease; ECG, electrocardiogram; NP, natriuretic peptides; TIL, tumour-infiltrating lymphocytes.

^aClass of recommendation.

^bLevel of evidence.

^cDetermine CRS grade according to ASTCT grading: Grade 1: fever; Grade 2: fever AND hypotension not requiring vasopressors AND/OR hypoxia requiring low-flow nasal oxygen; Grade 3: fever AND hypotension requiring one vasopressor $\pm$ vasopressin AND/OR hypoxia requiring high-flow nasal cannula or facemask or non-rebreather mask or Venturi mask; Grade 4: fever AND hypotension requiring multiple vasopressors, not including vasopressin AND/OR hypoxia requiring positive airway pressure.

Recommendation Table 20

[Open in new tab](#)

Recommendations for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies

Recommendations	Class ^a	Level ^b
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography should be considered before starting CAR-T and TIL therapies. ³⁸⁸	IIa	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 . ^{c,378,388}	I	C

© ESC 2022

ASTCT, American Society for Transplantation and Cellular Therapy; CAR-T, chimeric antigen receptor T cell; CRS, cytokine release syndrome; cTn, cardiac troponin; CVD, cardiovascular disease; ECG, electrocardiogram; NP, natriuretic peptides; TIL, tumour-infiltrating lymphocytes.

^aClass of recommendation.

^bLevel of evidence.

^cDetermine CRS grade according to ASTCT grading: Grade 1: fever; Grade 2: fever AND hypotension not requiring vasopressors AND/OR hypoxia requiring low-flow nasal oxygen; Grade 3: fever AND hypotension requiring one vasopressor $\pm$ vasopressin AND/OR hypoxia requiring high-flow nasal cannula or facemask or non-rebreather mask or Venturi mask; Grade 4: fever AND hypotension requiring multiple vasopressors, not including vasopressin AND/OR hypoxia requiring positive airway pressure.

Recommendations	Class ^a	Level ^b
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies. ³⁸⁸	I	C
A baseline echocardiography should be considered before starting CAR-T and TIL therapies. ³⁸⁸	IIa	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 . ^{c,378,388}	I	C

© ESC 2022

ASTCT, American Society for Transplantation and Cellular Therapy; CAR-T, chimeric antigen receptor T cell; CRS, cytokine release syndrome; cTn, cardiac troponin; CVD, cardiovascular disease; ECG, electrocardiogram; NP, natriuretic peptides; TIL, tumour-infiltrating lymphocytes.

^aClass of recommendation.

^bLevel of evidence.

^cDetermine CRS grade according to ASTCT grading: Grade 1: fever; Grade 2: fever AND hypotension not requiring vasopressors AND/OR hypoxia requiring low-flow nasal oxygen; Grade 3: fever AND hypotension requiring one vasopressor $\pm$ vasopressin AND/OR hypoxia requiring high-flow nasal cannula or facemask or non-rebreather mask or Venturi mask; Grade 4: fever AND hypotension requiring multiple vasopressors, not including vasopressin AND/OR hypoxia requiring positive airway pressure.

5.5.16. Radiotherapy

RT increases the risk of developing subsequent CVD and peripheral artery disease (PAD).^{173,389,394} There is ongoing debate regarding the safest radiation dose, which cardiac substructures are most sensitive to RT-induced injury, and the most appropriate strategies to minimize RT-related CVD.^{395,396} The heart is considered a radiosensitive organ at risk during RT and radiation exposure to the heart should be kept as low as reasonably achievable because there is no safe dose ([Figure 23](#)).^{389,390} RT-induced CV toxicity risk categorization based on MHD^{389,397} is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure (e.g. 35 Gray [Gy] prescribed dose to approximately 70% of the heart is equivalent to approximately 25 Gy MHD, whereas 35 Gy prescribed dose to approximately 40% of the heart is equivalent to approximately 15 Gy MHD). However, MHD is not a perfect metric, and in some patients, a very small portion of the heart might be irradiated to a very high dose, still conveying a substantial risk despite a low MHD.³⁹⁸ Therefore, depending on dose distribution and exposure of specific cardiac substructures and CVRFs, the cancer treatment team may judge the patient to belong to a higher-risk category.^{397,399,401}

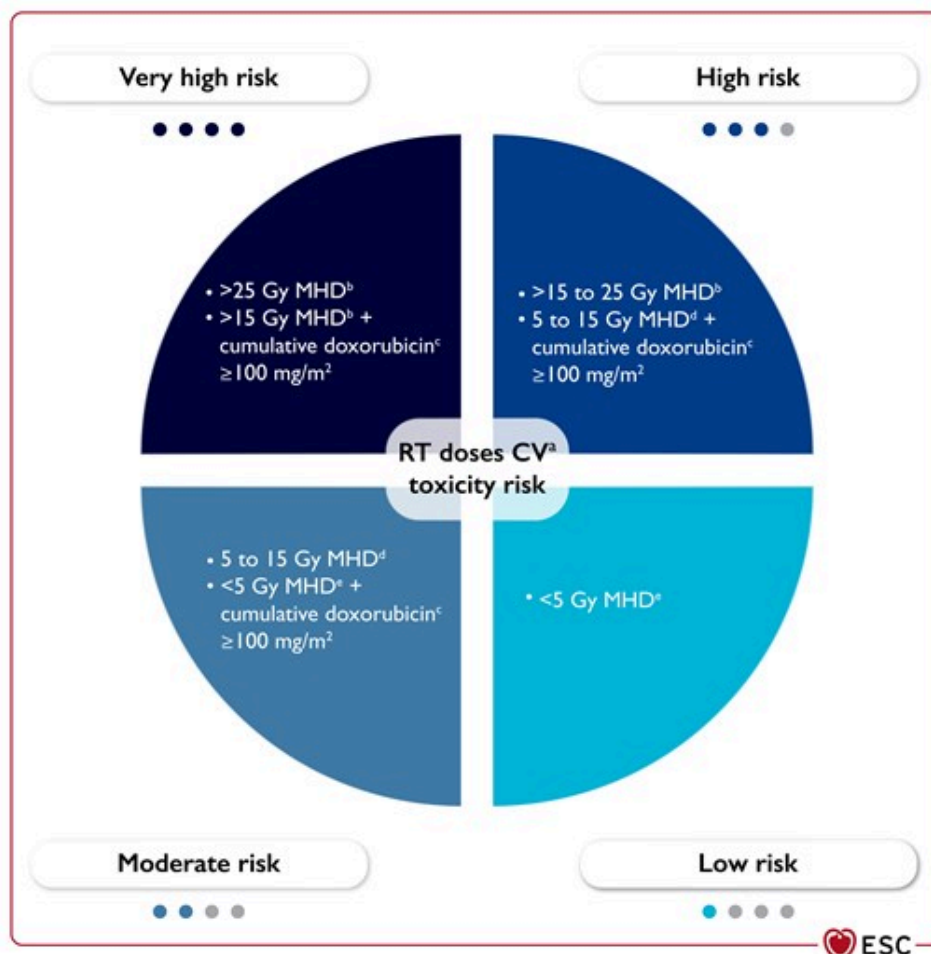


Figure 23

Radiotherapy mean heart dose and associated cardiovascular toxicity risk.

CV, cardiovascular; Gy, Gray; MHD, mean heart dose; RT, radiotherapy. ^aRT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors) the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category if only a small part of the heart was exposed to a relatively low prescribed dose. ^bOr prescribed RT 35 Gy to a volume exposing the heart if MHD is not available. Note that in this case, the limited information about cardiac exposure does not allow one to distinguish between high- and very high-risk categories. ^cOr equivalent. ^dOr prescribed RT 15 Gy to a volume exposing the heart if MHD is not available. ^eOr prescribed RT < 15 Gy to a volume exposing the heart if MHD is not available.

[Open in new tab](#) [Download slide](#)

Strategies to prevent and attenuate CV complications of RT have focused on reducing radiation exposure of the heart and CV substructures during cancer treatment and include the following.

1. Modification of cancer management to omit RT. This emphasizes the importance of integrating a personalized cardio-oncology evaluation. ^{402a}
2. Modification of the dose and volume of RT treatments where possible. RT protocols should target the minimum volume required to the minimum dose needed to obtain the desired clinical benefit.

3. Modification of delivery techniques to reduce cardiac radiation exposure should lead to a considerable reduction in risk. Modern heart-sparing RT strategies include: the optimal use of modern intensity-modulated photon RT technologies; the use of deep inspiration breath-hold or respiratory-gated techniques in BC,⁴⁰⁵ lymphoma,⁴⁰⁶ and lung cancer⁴⁰⁷; or the use of image-guided RT to ensure accuracy of delivery and proton beam therapy.⁴⁰⁸

The incidence of cardiac events following RT may vary according to patient risk factors and synergistic effects of radiation with other cardiotoxic cancer treatments.^{12,173}

There are no known RT-specific secondary preventative measures (e.g. drug treatments) to reduce the risk of CV events following RT. However, given the known importance of conventional CVRF on the incidence of RT-related events, optimization of modifiable CVRF is recommended in all patients before and after RT.

Recommendation Table 21

[Open in new tab](#)

Recommendations for baseline risk assessment of patients before radiotherapy to a volume including the heart

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended. ^{19,389}	I	B
Baseline echocardiography should be considered in patients with previous CVD before RT to a volume including the heart.	Ila	C

© ESC 2022

BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended. ^{19,389}	I	B
Baseline echocardiography should be considered in patients with previous CVD before RT to a volume including the heart.	Ila	C

© ESC 2022

BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

Recommendation Table 21

[Open in new tab](#)

Recommendations for baseline risk assessment of patients before radiotherapy to a volume including the heart

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended. ^{19,389}	I	B
Baseline echocardiography should be considered in patients with previous CVD before RT to a volume including the heart.	Ila	C

© ESC 2022

BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

Recommendations	Class ^a	Level ^b
Baseline CV risk assessment ^c and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP ^d is recommended. ^{19,389}	I	B
Baseline echocardiography should be considered in patients with previous CVD before RT to a volume including the heart.	Ila	C

© ESC 2022

BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons.

^aClass of recommendation.

^bLevel of evidence.

^cBP, lipids, fasting glucose, HbA1c, ECG and patient education on healthy lifestyle and lifestyle risk factor control.

^dSCORE2 (<70 years) or SCORE2-OP (≥70 years) CV risk stratification: <50 years: low risk <2.5%, moderate risk 2.5% to <7.5%, high risk ≥7.5%; 50–69 years: low risk <5%; moderate risk 5% to <10%; high risk ≥10%; ≥70 years: low risk <7.5%, moderate risk 7.5% to <15%, high risk ≥15%.¹⁹

5.5.17. Haematopoietic stem cell transplantation

HSCT constitutes a potentially curative therapeutic option for many haematological malignancies. Improvements in HSCT techniques and supportive strategies have markedly decreased treatment-related mortality ([Supplementary data, TableÂ S14](#)).^{409,410} There is a growing recognition of HSCT-related CV toxicities and HSCT survivors constitute a population at high future CV risk. Several factors contribute to define the risk of HSCT-related CV toxicities, including the HSCT type (higher risk after allogeneic HSCT), multiple uncontrolled CVRF, pre-existing CV conditions (AF or atrial flutter, sick sinus syndrome, ventricular arrhythmias, CAD, MI, moderate-to-severe VHD, and HF or LVEF <50%),⁴¹¹ direct cardiotoxic effects of anticancer therapies received prior to and during HSCT (anthracycline-combined induction regimen, mediastinal RT, total body irradiation, or cyclophosphamide-based conditioning regimen) ([Supplementary data, TableÂ S14](#)) and the development of graft vs. host disease (GVHD), thrombotic microangiopathy, or sepsis.^{410,412} In the early phase following HSCT (<100 days), the most frequent CV event is AF, although some patients may experience HF, hypertension, hypotension, pericardial effusion, or VTE.^{413,414} Late toxicities include DM, dyslipidaemia, metabolic syndrome, hypertension, HF, CAD, conduction disorders, and pericardial effusion.⁴¹⁰ Acute GVHD is associated with thrombosis and inflammatory myocardial damage (myocarditis, HF, conduction abnormalities, arrhythmias, and pericardial effusions), and chronic GVHD has been linked with increasing risk of hypertension, DM, and dyslipidaemia.^{415,416}

A comprehensive CV evaluation, including NP assessment, ECG, and TTE, has become a core component of the pre-HSCT assessment^{409,410} to detect undiagnosed CVD, stratify CTR-CVT risk, and optimize pre-existing CV conditions.^{411,417a}^{FASTS:420} In early surveillance, TTE monitoring is recommended in high-risk HSCT recipients at 3 and 12 months as LVEF and GLS can decrease after transplant (see [Section 7](#)). Independent factors associated with long-term CVD in HSCT survivors are allogeneic HSCT, pre-existing CVD or multiple uncontrolled CVRF, cancer treatment history (mediastinal or mantle field radiation, alkylating

agents, $>250\text{ mg/m}^2$ doxorubicin or equivalent), high-risk conditioning schemes (total body irradiation, alkylating agents), and GVHD.⁴¹⁰ Figure 24 summarizes strategies for the prevention and attenuation of CV complications in patients undergoing HSCT.

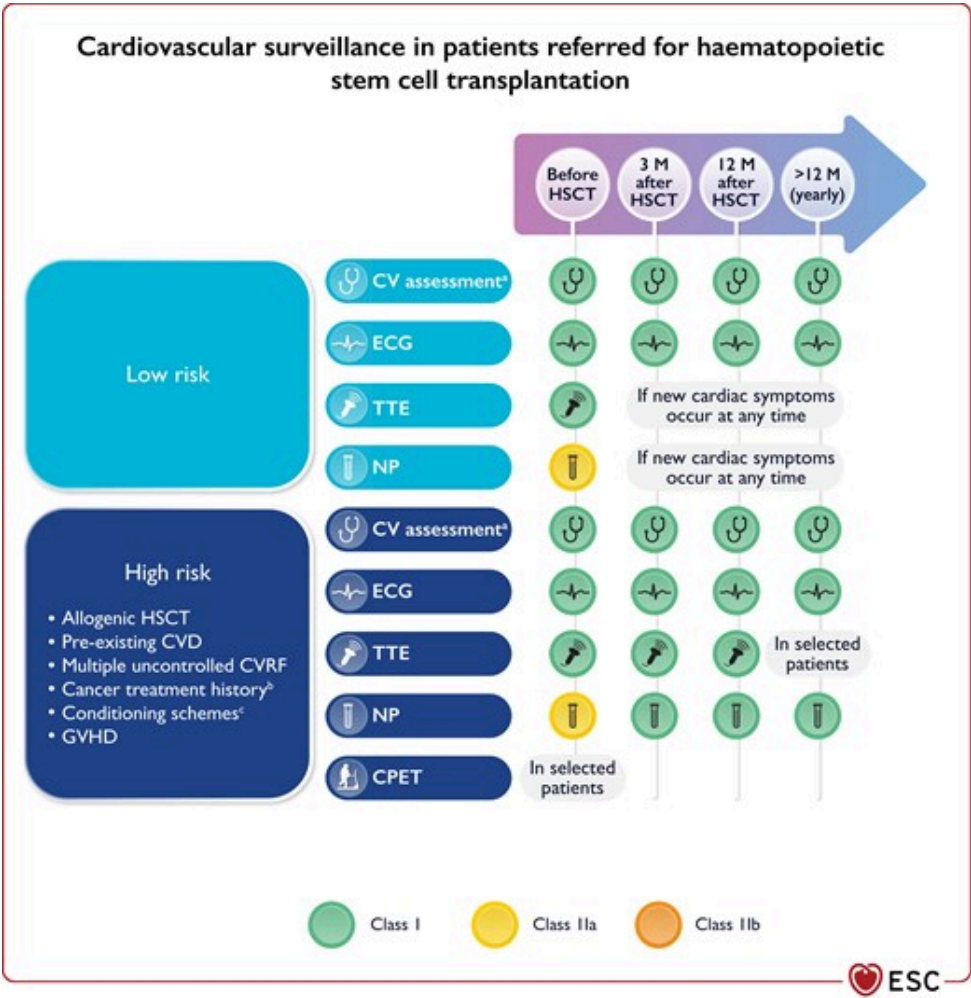


Figure 24 Risk factors and cardiovascular surveillance in patients referred for haematopoietic stem cell transplantation.

BNP, B-type natriuretic peptide; BP, blood pressure; CPET, cardiopulmonary exercise testing; CV, cardiovascular; CVD, CV disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; GVHD, graft vs. host disease; HbA1c, glycated haemoglobin; HSCT, haematopoietic stem cell transplantation; M, months; NP, natriuretic peptides (including BNP or NT-proBNP); NT-proBNP, N-terminal pro-BNP; TTE, transthoracic echocardiography. ^aIncluding physical examination, BP, lipid profile, and HbA1c. ^bMediastinal or mantle field radiation, alkylating agents, $>250\text{ mg/m}^2$ doxorubicin or equivalent. ^cTotal body irradiation, alkylating agents.

[Open in new tab](#)[Download slide](#)

Recommendation Table 22

[Open in new tab](#)

Recommendations for baseline risk assessment in haematopoietic stem cell transplantation patients

Recommendations	Class ^a	Level ^b	© ESC 2022
Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.	I	C	
Echocardiography is recommended in all patients before HSCT.	I	C	
Baseline NP measurement should be considered before HSCT. ^{417,418}	Ila	C	

BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HSCT, haematopoietic stem cell transplantation; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

Recommendations	Class ^a	Level ^b	© ESC 2022
Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.	I	C	
Echocardiography is recommended in all patients before HSCT.	I	C	
Baseline NP measurement should be considered before HSCT. ^{417,418}	Ila	C	

BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HSCT, haematopoietic stem cell transplantation; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

Recommendation Table 22

[Open in new tab](#)

Recommendations for baseline risk assessment in haematopoietic stem cell transplantation patients

Recommendations	Class ^a	Level ^b
Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.	I	C
Echocardiography is recommended in all patients before HSCT.	I	C
Baseline NP measurement should be considered before HSCT. ^{417,418}	IIa	C

BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HSCT, haematopoietic stem cell transplantation; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

Recommendations	Class ^a	Level ^b
Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.	I	C
Echocardiography is recommended in all patients before HSCT.	I	C
Baseline NP measurement should be considered before HSCT. ^{417,418}	IIa	C

BP, blood pressure; CV, cardiovascular; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HSCT, haematopoietic stem cell transplantation; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

5.5.18. Other cancer treatments

Several other cancer therapies may also induce clinically relevant CV events. Cyclophosphamide, cisplatin, ifosfamide, and taxanes (paclitaxel and docetaxel) can induce myocardial dysfunction and HF.⁴

Cyclophosphamide CV toxicity is primarily seen in patients receiving high doses (>140 mg/kg) before HSCT and typically occurs within days of drug administration.⁴¹⁰

Platinum-containing chemotherapy (cisplatin, carboplatin, oxaliplatin) may cause vascular disease (vasospasm, MI, and venous and arterial thrombosis). These may occur during treatment and also contribute to increased long-term risk of CAD in survivors. Patients with testicular cancer treated with cisplatin have a higher risk for vascular disease at long-term follow-up.⁴²¹ The risk of the individual patient is still hard to predict, but lifestyle interventions, a high degree of clinical suspicion in patients who experience chest pain, and close CVRF monitoring is recommended during and after therapy.⁴²² Cisplatin⁴²² infrequently causes HF;

however, because it requires the administration of a high i.v. volume to avoid renal toxicity, patients with pre-existing CVD may develop symptomatic HF.

Arsenic trioxide is used to treat some leukaemias and myelomas. Arsenic trioxide frequently prolongs the QT interval (26a93% of patients), and life-threatening ventricular tachyarrhythmias have been reported.^{45,259} QTc prolongation was observed 1a5 weeks after arsenic trioxide infusion and then returned towards baseline by the end of 8 weeks. Patients receiving treatment with arsenic trioxide should be monitored weekly with ECG during the first 8 weeks of therapy. Electrolyte monitoring is also required as arsenic trioxide may induce hypokalaemia, hypomagnesaemia, and renal dysfunction. Risk factors for QT prolongation should be controlled before, during, and after cancer treatment ([Section 6.4.2](#)).

Several FMS-like tyrosine kinase 3 (FLT3) inhibitors (first-generation: midostaurin; second-generation: gilteritinib) have been tested for the treatment of acute myeloid leukaemias. Gilteritinib-induced differentiation syndrome (fever, dyspnoea, pleuropericardial effusion, pulmonary oedema, peripheral oedema, hypotension, renal dysfunction, and rash) requires early corticosteroid therapy and haemodynamic monitoring until resolution of symptoms. Midostaurin and gilteritinib may prolong QTc interval and close electrolyte surveillance and minimizing drugdrug interactions are required (see [Section 6.4.2](#); [Table 9](#); [Supplementary data, Tables S15 and S16](#)).⁴²³

6. Diagnosis and management of acute and subacute cardiovascular toxicity in patients receiving anticancer treatment

A coordinated MDT is recommended to discuss patients with cancer who develop acute CV complications of their cancer treatment.⁵ Referral to a specialized cardio-oncology service is recommended for patients with cancer who present with new CTR-CVT during and after cancer treatment.¹² The prevention and management of CVD in patients with cancer should generally follow published ESC Guidelines for specific CVD. This chapter provides guidance on the management of CTR-CVT that occur during cancer treatment, and highlights where management differs for patients with cancer compared with those without. The decision to initiate CV treatment (medication, devices) needs to include consideration of a range of factors including both cancer and CV symptom burden, cancer prognosis, ongoing cancer treatment requirements including alternative options, possible adverse drug reactions, drugdrug interactions, and patient preferences. An extensive list of drugdrug interactions is provided in [Supplementary data, Tables S15a](#)¹²[S17](#).

Recommendation Table 23

[Open in new tab](#)

Recommendation for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment

Recommendation	Class ^a	Level ^b
A specialist CV assessment ^c is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment. ⁵	I	C

CV, cardiovascular; CVD, cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.
^cCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

© ESC 2022

Recommendation	Class ^a	Level ^b
A specialist CV assessment ^c is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment. ⁵	I	C

CV, cardiovascular; CVD, cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.
^cCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

© ESC 2022

Recommendation Table 23

[Open in new tab](#)

Recommendation for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment

Recommendation	Class ^a	Level ^b
A specialist CV assessment ^c is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment. ⁵	I	C

CV, cardiovascular; CVD, cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.
^cCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

© ESC 2022

Recommendation	Class ^a	Level ^b
A specialist CV assessment ^c is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment. ⁵	I	C

CV, cardiovascular; CVD, cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.
^cCardio-oncology referral is recommended when available; alternatively, patients should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer.

© ESC 2022

6.1. Cancer therapy-related cardiac dysfunction

6.1.1. Anthracycline chemotherapy-related cardiac dysfunction

CTRCD during anthracycline chemotherapy may present clinically or be detected in asymptomatic patients during surveillance ([FigureÂ 10](#); [TableÂ 3](#)).⁴ The diagnosis of anthracycline chemotherapy-related cardiac dysfunction includes new CV symptoms, new abnormalities in cardiac function on CV imaging, and/or new increases in cardiac biomarkers ([TableÂ 3](#)). A MDT discussion is recommended to consider the risk/benefit ratio of continuing anthracycline chemotherapy in patients who develop new CTRCD.

Discontinuation of anthracycline chemotherapy is recommended in patients with cancer who develop severe symptomatic CTRCD.²² There are rare exceptions where rechallenge with further anthracycline chemotherapy may be considered after a MDT discussion, using prevention strategies described below and under close monitoring with each cycle of anthracycline chemotherapy. Temporary interruption of anthracycline chemotherapy is recommended in patients who develop moderate symptomatic CTRCD, and in patients who develop moderate or severe asymptomatic CTRCD. A MDT approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD.

Guideline-based HF therapy is recommended in patients who develop symptomatic CTRCD or asymptomatic moderate or severe CTRCD during anthracycline chemotherapy. The use of an ACE-I/ARB or angiotensin receptor neprilysin inhibitor, a beta-blocker, a sodium glucose co-transporter 2 inhibitor, and a mineralocorticoid receptor antagonist is recommended unless the drugs are contraindicated or not tolerated. Up-titration to target doses as described in the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF is recommended.¹⁴ ACE-I, ARB, and/or beta-blockers should be considered in mild asymptomatic CTRCD while anthracycline chemotherapy continues uninterrupted ([FigureÂ 25](#)).^{1,14,102,424} The beneficial effects of aerobic exercise before and during anthracycline chemotherapy have been demonstrated and is recommended for patients with cancer who develop CTRCD.¹¹

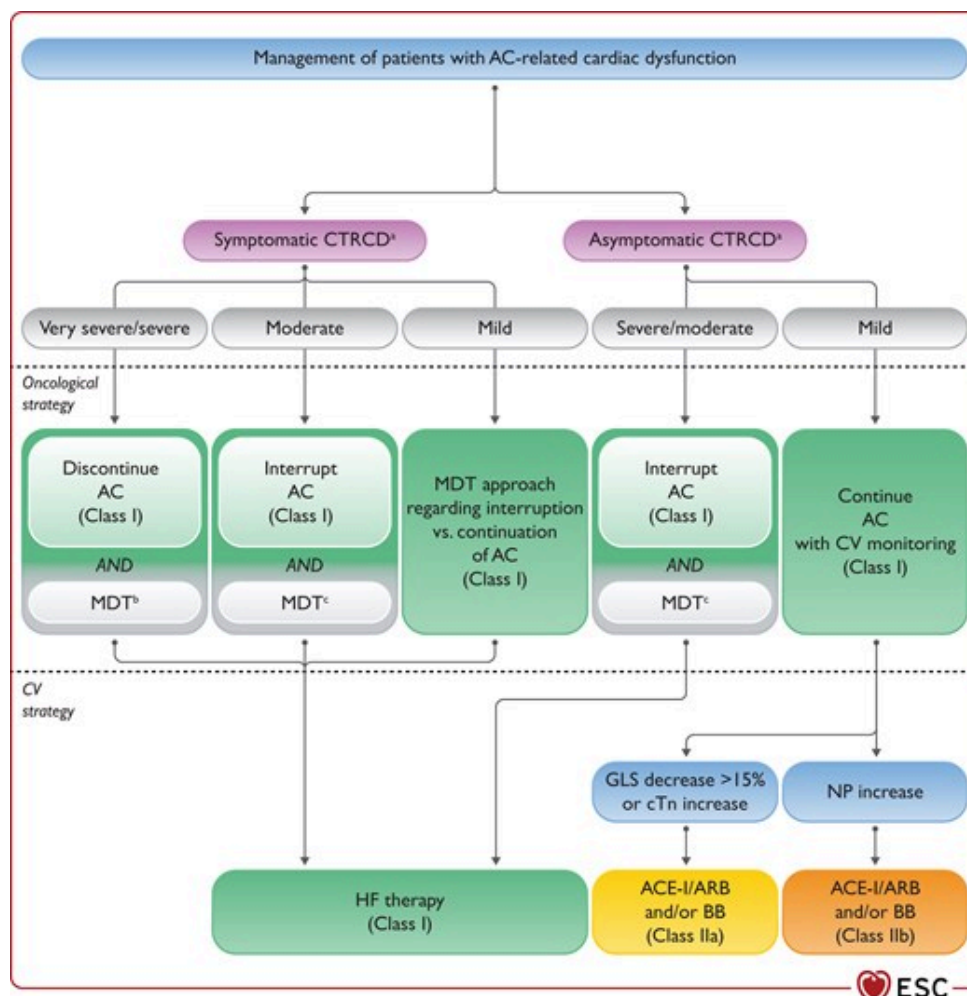


Figure 25

Management of anthracycline chemotherapy-related cardiac dysfunction.

AC, anthracycline chemotherapy; ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BB, beta-blockers; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; NP, natriuretic peptides. ^aSee [Table 3 \(Section 3\)](#) for complete definition (symptomatic CTRCD: symptomatic confirmed HF syndrome; asymptomatic severe CTRCD: LVEF < 40%; asymptomatic moderate CTRCD: LVEF 40–49%; asymptomatic mild CTRCD: LVEF > 50%). ^bIn rare exceptions, anthracycline chemotherapy may be restarted after recovery of LV function with optimal HF therapy. ^cA MDT discussion is recommended before restarting anthracycline chemotherapy after recovery of LV function.

[Open in new tab](#) [Download slide](#)

A MDT is recommended to discuss restarting anthracycline chemotherapy in patients who developed mild or moderate symptomatic CTRCD, or moderate or severe asymptomatic CTRCD, after recovery of LV function under HF treatment. If there is a compelling reason to continue anthracycline chemotherapy, three other strategies exist in addition to continuing ACE-I/ARB and beta-blockers at target doses for HF.¹⁴ First, minimizing the dose of anthracycline chemotherapy administered. Second, switching to liposomal anthracycline preparations. Third, pre-treatment with dexrazoxane before each further cycle of anthracycline chemotherapy ([Section 5.2.1](#)).

Close cardiac monitoring every 1â2 cycles is recommended in patients who restart anthracycline chemotherapy following an episode of CTRCD and in patients with mild asymptomatic CTRCD while they continue anthracycline chemotherapy.

Recommendation Table 24

[Open in new tab](#)

Recommendations for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy

Recommendations	Class ^a	Level ^b
Anthracycline chemotherapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy. ^{c,208,425}	I	B
Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD. ^c	I	C
Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD ^c and a multidisciplinary approach regarding the decision to restart is recommended.	I	C
A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD. ^c	I	C
Anthracycline chemotherapy-induced asymptomatic CTRCD		
Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD. ^{c,22}	I	C
A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD. ^{c,22}	I	C
Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c or a troponin or a NP elevation $> \text{ULN}$.	I	C
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c should be considered for ACE-I/ARB and/or beta-blockers. ^{d,75,93,102}	IIa	B
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a troponin elevation $> \text{ULN}$ should be considered for ACE-I/ARB and/or beta-blockers. ^{d,147,211}	IIa	B

Asymptomatic patients who have LVEF $\geq$ 50% and who have developed NP >ULN may be considered for ACE-I/ARB and/or beta-blockers.^{d,211}

IIb

C

Strategies for restarting anthracycline chemotherapy in patients with CTRCD

Liposomal anthracycline^e may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.

IIb

C

Dexrazoxane^f may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.

IIb

C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; LVEF, left ventricular ejection fraction; NP, natriuretic peptides; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#). Significant fall in GLS = relative reduction >15%.

^dAvoid hypotension.

^eSee text for specific liposomal doxorubicin type and malignancies ([Section 5.2](#)).

^fAs per the European Medicines Agency: $\geq$ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: $\geq$ 300 mg/m² doxorubicin or equivalent.

Recommendations	Class ^a	Level ^b
Anthracycline chemotherapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy. ^{c,208,425}	I	B
Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD. ^c	I	C
Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD ^c and a multidisciplinary approach regarding the decision to restart is recommended.	I	C
A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD. ^c	I	C
Anthracycline chemotherapy-induced asymptomatic CTRCD		
Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD. ^{c,22}	I	C
A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD. ^{c,22}	I	C
Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c or a troponin or a NP elevation $> \text{ULN}$.	I	C
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c should be considered for ACE-I/ARB and/or beta-blockers. ^{d,75,93,102}	IIa	B
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a troponin elevation $> \text{ULN}$ should be considered for ACE-I/ARB and/or beta-blockers. ^{d,147,211}	IIa	B

Asymptomatic patients who have LVEF $\geq$ 50% and who have developed NP >ULN may be considered for ACE-I/ARB and/or beta-blockers. ^{d,211}	IIb	C
Strategies for restarting anthracycline chemotherapy in patients with CTRCD		
Liposomal anthracycline ^e may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD ^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.	IIb	C
Dexrazoxane ^f may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD ^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.	IIb	C

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; LVEF, left ventricular ejection fraction; NP, natriuretic peptides; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#). Significant fall in GLS = relative reduction >15%.

^dAvoid hypotension.

^eSee text for specific liposomal doxorubicin type and malignancies ([Section 5.2](#)).

^fAs per the European Medicines Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

© ESC 2022

Recommendation Table 24

[Open in new tab](#)

Recommendations for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy

Recommendations	Class ^a	Level ^b
Anthracycline chemotherapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy. ^{c,208,425}	I	B
Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD. ^c	I	C
Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD ^c and a multidisciplinary approach regarding the decision to restart is recommended.	I	C
A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD. ^c	I	C
Anthracycline chemotherapy-induced asymptomatic CTRCD		
Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD. ^{c,22}	I	C
A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD. ^{c,22}	I	C
Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c or a troponin or a NP elevation $> \text{ULN}$.	I	C
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c should be considered for ACE-I/ARB and/or beta-blockers. ^{d,75,93,102}	IIa	B
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a troponin elevation $> \text{ULN}$ should be considered for ACE-I/ARB and/or beta-blockers. ^{d,147,211}	IIa	B

Asymptomatic patients who have LVEF $\geq$ 50% and who have developed NP >ULN may be considered for ACE-I/ARB and/or beta-blockers.^{d,211}

IIb

C

Strategies for restarting anthracycline chemotherapy in patients with CTRCD

Liposomal anthracycline^e may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.

IIb

C

Dexrazoxane^f may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.

IIb

C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; LVEF, left ventricular ejection fraction; NP, natriuretic peptides; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#). Significant fall in GLS = relative reduction >15%.

^dAvoid hypotension.

^eSee text for specific liposomal doxorubicin type and malignancies ([Section 5.2](#)).

^fAs per the European Medicines Agency: $\geq$ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: $\geq$ 300 mg/m² doxorubicin or equivalent.

Recommendations	Class ^a	Level ^b
Anthracycline chemotherapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy. ^{c,208,425}	I	B
Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD. ^c	I	C
Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD ^c and a multidisciplinary approach regarding the decision to restart is recommended.	I	C
A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD. ^c	I	C
Anthracycline chemotherapy-induced asymptomatic CTRCD		
Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD. ^{c,22}	I	C
A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD. ^{c,22}	I	C
Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c or a troponin or a NP elevation $> \text{ULN}$.	I	C
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a significant fall in GLS ^c should be considered for ACE-I/ARB and/or beta-blockers. ^{d,75,93,102}	IIa	B
Asymptomatic patients who have LVEF $\geq 50\%$ and who have developed a troponin elevation $> \text{ULN}$ should be considered for ACE-I/ARB and/or beta-blockers. ^{d,147,211}	IIa	B

Asymptomatic patients who have LVEF $\geq$ 50% and who have developed NP >ULN may be considered for ACE-I/ARB and/or beta-blockers. ^{d,211}	IIb	C
Strategies for restarting anthracycline chemotherapy in patients with CTRCD		
Liposomal anthracycline ^e may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD ^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.	IIb	C
Dexrazoxane ^f may be considered in patients with moderate or severe symptomatic or asymptomatic CTRCD ^c who require further anthracycline chemotherapy to reduce the risk of further CV toxicity.	IIb	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; LVEF, left ventricular ejection fraction; NP, natriuretic peptides; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cSee Table 3. Significant fall in GLS = relative reduction >15%.

^dAvoid hypotension.

^eSee text for specific liposomal doxorubicin type and malignancies (Section 5.2).

^fAs per the European Medicines Agency: ≥ 350 mg/m² doxorubicin or equivalent; as per the United States Food and Drug Administration: ≥ 300 mg/m² doxorubicin or equivalent.

6.1.2. Human epidermal receptor 2-targeted therapy-related cardiac dysfunction

The diagnosis of HER2-targeted therapy-related CTRCD can be made using the combination of new CV symptoms, imaging, and biomarkers. Patients may present with symptomatic CTRCD or may be asymptomatic.⁴²⁶ Early treatment of symptomatic and asymptomatic severe CTRCD (LVEF < 40%), according to the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF,¹⁴ is recommended to prevent worsening HF,⁴²⁵ particularly when targeted cancer therapy is continued.⁴²⁷ In patients who develop CTRCD, a MDT is recommended to guide clinical decisions. Temporary interruption is recommended in patients who develop moderate or severe symptomatic CTRCD or severe asymptomatic CTRCD (LVEF < 40%) during HER2-targeted therapy. In patients with mild symptomatic CTRCD, a MDT approach is recommended to continue vs. interrupt HER2-targeted therapy. In patients with asymptomatic moderate CTRCD (LVEF 40–49%), HER2-targeted treatment should be continued, and cardioprotective therapy (ACE-I/ARB and beta-blockers) is recommended with frequent cardiac monitoring.^{22,33,189} In patients with asymptomatic mild CTRCD (LVEF $\geq$ 50% with a significant new GLS reduction and/or cardiac biomarker increase), continuing HER2-targeted treatment is recommended and cardioprotective therapy (ACE-I/ARB and/or beta-blockers) should be considered.^{22,211,428,429}

Frequent cardiac surveillance with cardiac imaging and cardiac serum biomarkers is recommended in all patients with CTRCD who continue HER2-targeted cancer therapies and in those who restart after an interruption following resolution of HF signs and symptoms and recovery of LVEF $\geq$ 40% (and ideally recovery to LVEF $\geq$ 50%) (Figure 26).^{22,33,189} Echocardiography and cardiac serum biomarker measurement every two cycles for the first four cycles after restarting HER2-targeted therapy is recommended, and then the frequency can be reduced if cardiac function and biomarker levels remain stable.

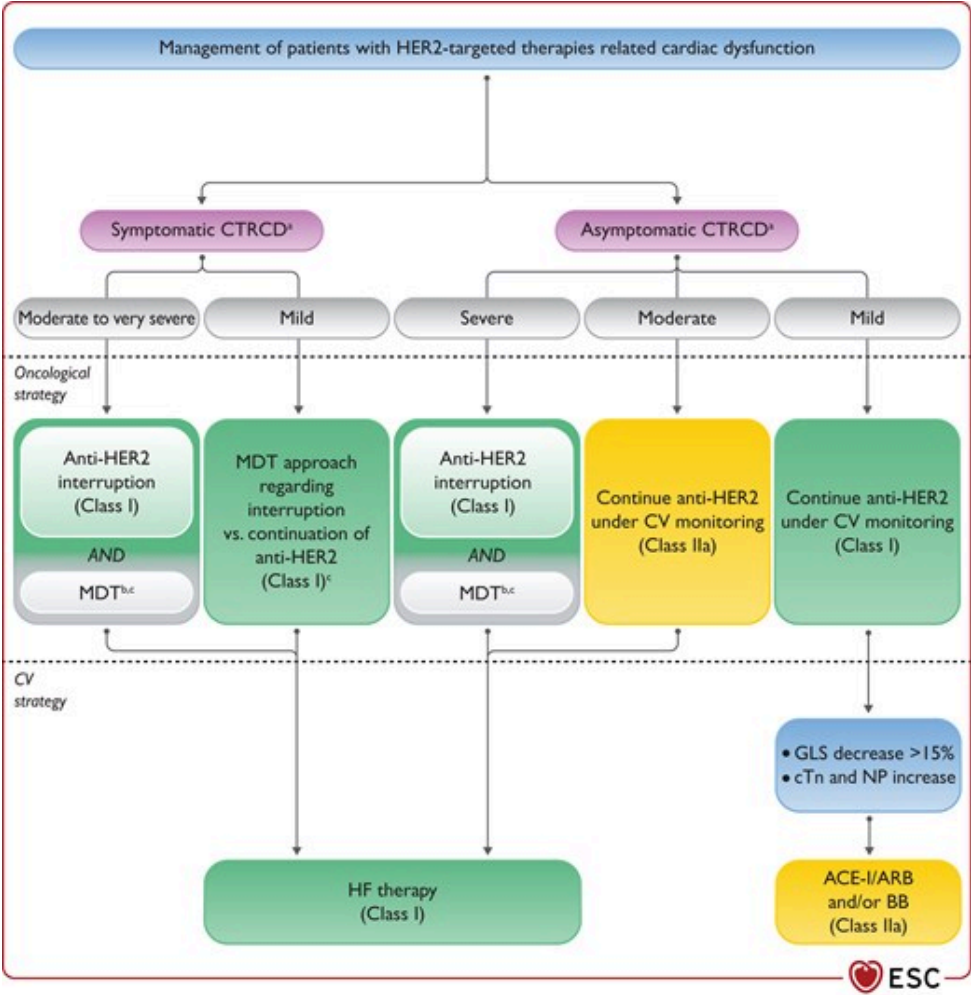


Figure 26
Management of human epidermal receptor 2-targeted therapy-related cardiac dysfunction.

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BB, beta-blockers; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HF, heart failure; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; NP, natriuretic peptides. ^aSee Table 3 (Section 3) (symptomatic CTRCD: symptomatic confirmed HF syndrome; asymptomatic severe CTRCD: LVEF < 40%; asymptomatic moderate CTRCD: LVEF 40–49%; asymptomatic mild CTRCD: LVEF > 50%). ^bFor patients in whom HER2-targeted therapy has been interrupted, whose signs and symptoms of HF do not resolve and/or LVEF remains <40%, resumption of HER2-targeted therapy may be considered if no alternative therapeutic option exists. In advanced cancer that only responds well to trastuzumab, the risk/benefit ratio may warrant continued therapy if other options remain limited. ^cFor patients where HER2-targeted therapy has been interrupted and who have recovered LVEF $\geq$ 40% and are now asymptomatic, resumption of HER2-targeted therapy

should be considered, supported by HF therapy, and echocardiography and cardiac biomarker assessment every two cycles for the first four cycles after restarting and then the frequency can be reduced.²²

[Open in new tab](#)[Download slide](#)

Recommendation Table 25

[Open in new tab](#)

Recommendations for the management of cancer treatment-related cardiac dysfunction during human epidermal receptor 2-targeted therapies

Recommendations	Class ^a	Level ^b
HER2-targeted therapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF < 50% ^c during HER2-targeted treatment. ^{14,61,430,431}	I	B
Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD ^c and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved. ^d	I	C
In patients who develop mild symptomatic CTRCD, ^c HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended. ^{d,431,432}	I	C
HER2-targeted therapy-induced asymptomatic CTRCD		
Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD. ^c	I	C
A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD. ^c	I	C
Continuation of HER2-targeted therapy should be considered in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c with more frequent cardiac monitoring. ^{33,189,428,433}	IIa	B
Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF ≥ 50%) CTRCD ^c with more frequent cardiac monitoring. ⁴²⁸	I	C
ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c during HER2-targeted treatment. ^{e,189}	I	C
ACE-I/ARB and/or beta-blockers should be		

considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a significant fall in GLS^c while continuing HER2-targeted therapy.^{e,22,428}

Ila

B

ACE-I/ARB and/or beta-blockers should be considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a new troponin or NP rise while continuing HER2-targeted therapy.^{e,22,211,428}

Ila

B

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HF, heart failure; LV, left ventricular; LVEF, LV ejection fraction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#).

^dFor patients where HER2-targeted therapy has been interrupted and who have recovered LVEF $\geq$ 40% and are now asymptomatic, resumption of HER2-targeted therapy should be considered supported by HF therapy and echocardiography and cardiac biomarkers assessment every two cycles for the first four cycles after restarting and then frequency can be reduced.

^eAvoid hypotension.

Recommendations	Class ^a	Level ^b
HER2-targeted therapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF < 50% ^c during HER2-targeted treatment. ^{14,61,430,431}	I	B
Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD ^c and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved. ^d	I	C
In patients who develop mild symptomatic CTRCD, ^c HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended. ^{d,431,432}	I	C
HER2-targeted therapy-induced asymptomatic CTRCD		
Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD. ^c	I	C
A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD. ^c	I	C
Continuation of HER2-targeted therapy should be considered in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c with more frequent cardiac monitoring. ^{33,189,428,433}	IIa	B
Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF ≥ 50%) CTRCD ^c with more frequent cardiac monitoring. ⁴²⁸	I	C
ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c during HER2-targeted treatment. ^{e,189}	I	C
ACE-I/ARB and/or beta-blockers should be		

considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a significant fall in GLS ^c while continuing HER2-targeted therapy. ^{e,22,428}	Ila	B
ACE-I/ARB and/or beta-blockers should be considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a new troponin or NP rise while continuing HER2-targeted therapy. ^{e,22,211,428}	Ila	B

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HF, heart failure; LV, left ventricular; LVEF, LV ejection fraction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#).

^dFor patients where HER2-targeted therapy has been interrupted and who have recovered LVEF $\geq$ 40% and are now asymptomatic, resumption of HER2-targeted therapy should be considered supported by HF therapy and echocardiography and cardiac biomarkers assessment every two cycles for the first four cycles after restarting and then frequency can be reduced.

^eAvoid hypotension.

Recommendation Table 25

[Open in new tab](#)

Recommendations for the management of cancer treatment-related cardiac dysfunction during human epidermal receptor 2-targeted therapies

Recommendations	Class ^a	Level ^b
HER2-targeted therapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF < 50% ^c during HER2-targeted treatment. ^{14,61,430,431}	I	B
Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD ^c and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved. ^d	I	C
In patients who develop mild symptomatic CTRCD, ^c HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended. ^{d,431,432}	I	C
HER2-targeted therapy-induced asymptomatic CTRCD		
Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD. ^c	I	C
A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD. ^c	I	C
Continuation of HER2-targeted therapy should be considered in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c with more frequent cardiac monitoring. ^{33,189,428,433}	IIa	B
Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF ≥ 50%) CTRCD ^c with more frequent cardiac monitoring. ⁴²⁸	I	C
ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c during HER2-targeted treatment. ^{e,189}	I	C
ACE-I/ARB and/or beta-blockers should be		

considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a significant fall in GLS^c while continuing HER2-targeted therapy.^{e,22,428}

Ila

B

ACE-I/ARB and/or beta-blockers should be considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a new troponin or NP rise while continuing HER2-targeted therapy.^{e,22,211,428}

Ila

B

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HF, heart failure; LV, left ventricular; LVEF, LV ejection fraction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#).

^dFor patients where HER2-targeted therapy has been interrupted and who have recovered LVEF $\geq$ 40% and are now asymptomatic, resumption of HER2-targeted therapy should be considered supported by HF therapy and echocardiography and cardiac biomarkers assessment every two cycles for the first four cycles after restarting and then frequency can be reduced.

^eAvoid hypotension.

Recommendations	Class ^a	Level ^b
HER2-targeted therapy-induced symptomatic CTRCD		
HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF < 50% ^c during HER2-targeted treatment. ^{14,61,430,431}	I	B
Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD ^c and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved. ^d	I	C
In patients who develop mild symptomatic CTRCD, ^c HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended. ^{d,431,432}	I	C
HER2-targeted therapy-induced asymptomatic CTRCD		
Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD. ^c	I	C
A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD. ^c	I	C
Continuation of HER2-targeted therapy should be considered in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c with more frequent cardiac monitoring. ^{33,189,428,433}	IIa	B
Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF ≥ 50%) CTRCD ^c with more frequent cardiac monitoring. ⁴²⁸	I	C
ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD ^c during HER2-targeted treatment. ^{e,189}	I	C
ACE-I/ARB and/or beta-blockers should be		

considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a significant fall in GLS ^c while continuing HER2-targeted therapy. ^{e,22,428}	Ila	B
ACE-I/ARB and/or beta-blockers should be considered in asymptomatic patients receiving HER2-targeted therapies who have LVEF $\geq$ 50% but develop a new troponin or NP rise while continuing HER2-targeted therapy. ^{e,22,211,428}	Ila	B

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; HER2, human epidermal receptor 2; HF, heart failure; LV, left ventricular; LVEF, LV ejection fraction; NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#).

^dFor patients where HER2-targeted therapy has been interrupted and who have recovered LVEF $\geq$ 40% and are now asymptomatic, resumption of HER2-targeted therapy should be considered supported by HF therapy and echocardiography and cardiac biomarkers assessment every two cycles for the first four cycles after restarting and then frequency can be reduced.

^eAvoid hypotension.

© ESC 2022

6.1.3. Immune checkpoint inhibitor-associated myocarditis and non-inflammatory heart failure

Myocarditis is a severe complication of ICI with a high fatality rate that most frequently develops during the first 12 weeks of treatment, although late cases (after week 20) may occur.³⁸⁶ Other ICI-related CV toxicities include dyslipidaemia, ACS, vasculitis, AV block, supraventricular and ventricular arrhythmias, sudden death, TTS, non-inflammatory LVD, pericarditis, pericardial effusion, and ischaemic stroke, with higher risks for myocarditis (odds ratio 4.42) and dyslipidaemia (odds ratio 3.68) ([FigureÂ 27](#)).^{323,325}

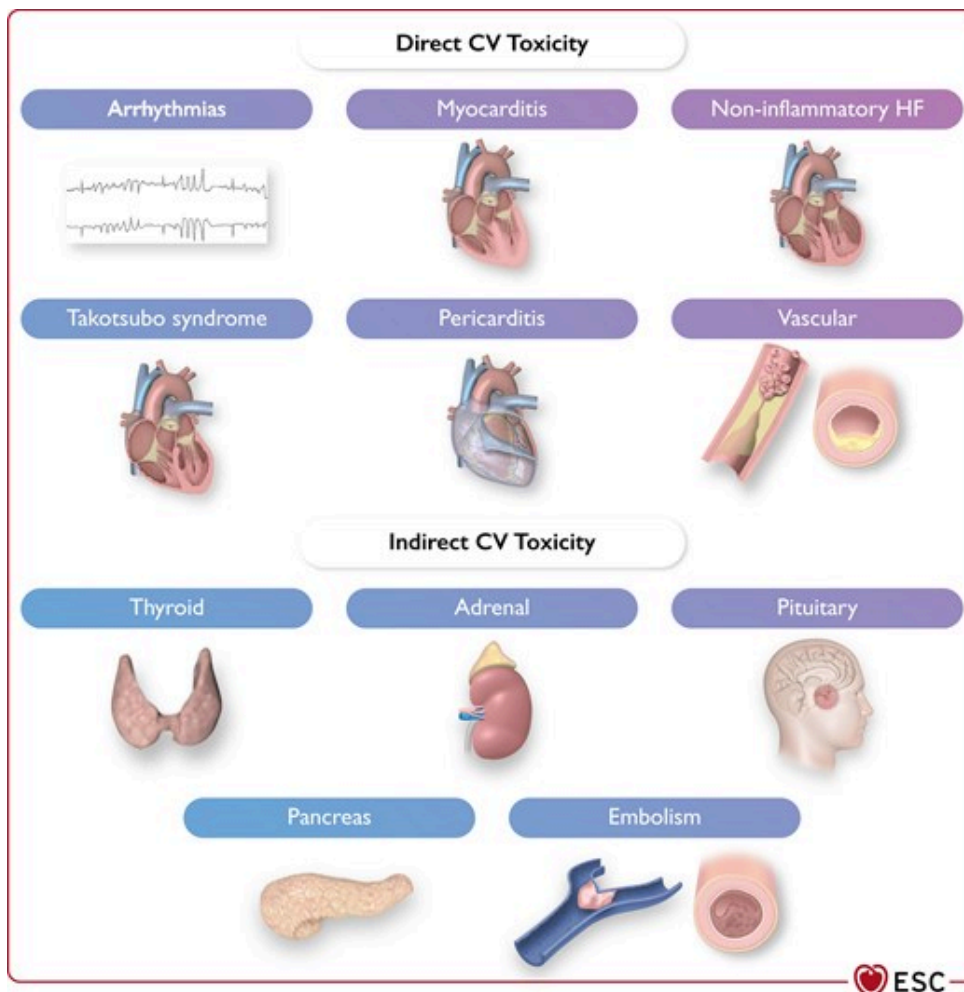


Figure 27

Direct and indirect immune checkpoint inhibitor-related cardiovascular toxicity.

CV, cardiovascular; HF, heart failure.

[Open in new tab](#)[Download slide](#)

The diagnosis of ICI-associated myocarditis is initially based on the presence of symptoms, a new increase in troponin (associated with either CV symptoms or non-CV immuno-related adverse events), and new ECG abnormalities (AV or intraventricular conduction disorders, bradycardia, tachyarrhythmias) (see [Section 3; Table 3](#)).^{17,434,435} Any abnormal finding should prompt urgent CV imaging and other causes of myocardial injury (e.g. ACS, acute infectious myocarditis) should be excluded. Treatment with high-dose methylprednisolone should be promptly initiated in haemodynamically unstable patients (including those with ventricular arrhythmias [VA] or complete AV block) while awaiting further confirmatory testing.⁴³⁶ TTE and CMR are recommended in all patients with suspected ICI-associated myocarditis. Currently, specific CMR features for ICI-induced myocarditis are not well described and modified Lake Louise criteria are recommended ([Table 3](#)).¹⁸ Cardiac fluorodeoxyglucose positron emission tomography (PET) may be considered^{437,438} if CMR is not available or contraindicated, although PET sensitivity is low and requires a strict 18-h carbohydrate-free fast.⁴³⁹ Endomyocardial biopsy (EMB) should be considered in cases where the diagnosis is suspected but not confirmed non-invasively (e.g. conflicting results of cardiac imaging and biomarkers or clinically unstable patients).⁴⁴⁰ All cases of ICI-associated myocarditis should be classified according to the severity of the myocarditis (fulminant or non-fulminant, including symptomatic but

haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immune-related adverse events) to guide the management pathway (Figure 28).³³¹

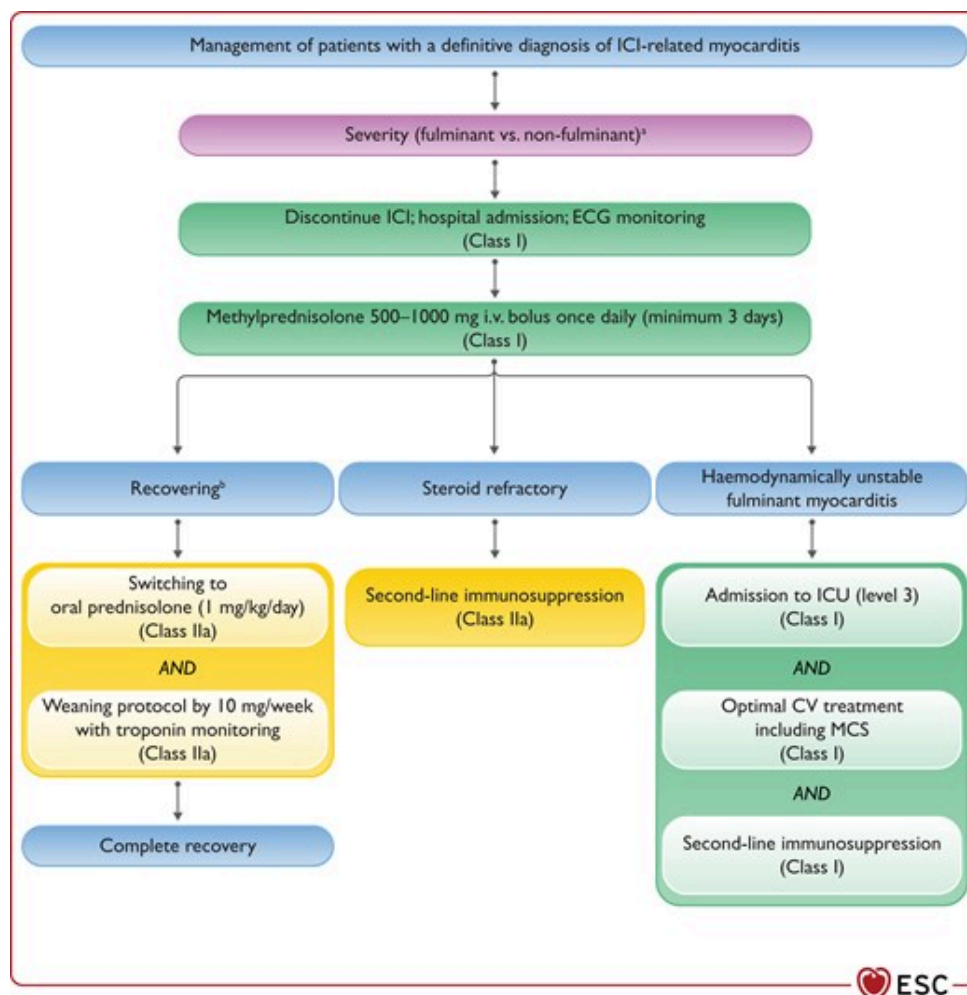


Figure 28

Diagnosis and management of immune checkpoint inhibitor-related myocarditis.

CMR, cardiac magnetic resonance; CV, cardiovascular; ECG, electrocardiogram; HF, heart failure; ICI, immune checkpoint inhibitor; ICU, intensive care unit; i.v., intravenous; LGE, late gadolinium enhancement; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support. ^a**Fulminant**: haemodynamic instability, HF requiring non-invasive or invasive ventilation, complete or high-grade heart block, and/or significant ventricular arrhythmia. **Non-fulminant**: including symptomatic but haemodynamically and electrically stable patients and incidental cases diagnosed at the same time as other immuno-related adverse events. Patients may have reduced LVEF but no features of severe disease. ^b**Recovering**: ongoing improvement in patient clinical symptoms, signs, biomarkers, and imaging parameters, but not yet normalized, while on tapering doses of immunosuppression. **Complete recovery**: patients with complete resolution of acute symptoms, normalization of biomarkers, and recovery of LVEF after discontinuation of immunosuppression. CMR may still show LGE or elevated T1 due to fibrosis, but any suggestion of acute oedema should be absent.

[Open in new tab](#)[Download slide](#)

Interruption of ICI treatment is recommended in all cases of suspected ICI-associated myocarditis (any patient developing new cardiac symptoms, new cardiac arrhythmias, new heart blocks, or new troponin increase who

has received an ICI therapy in the past 12 weeks) while investigations are performed. Once the abnormal findings have resolved, a MDT discussion is recommended to determine the risk/benefit to permanent stopping vs. resuming ICI treatment in patients with suspected but not confirmed myocarditis.

Cessation of ICI treatment is recommended in patients with cancer with fulminant or non-fulminant ICI-associated myocarditis and the patient should be admitted to hospital and a level 2 or 3 bed with continuous ECG monitoring is required. CV complications should be treated as per specific ESC Guidelines (HF,¹⁴ tachyarrhythmias,^{441,442} AV block,⁴⁴³ or pericardial effusion⁴⁴⁴).

Treatment of both non-fulminant and fulminant ICI-associated myocarditis with methylprednisolone 500–1000 mg i.v. bolus once daily for the first 3–5 days should be started as soon as possible, once the diagnosis is considered likely, to reduce MACE including mortality.^{386,436} If clinical improvement is observed (cTn reduced by >50% from peak level within 24–72 h and any LVD, AV block, and arrhythmias resolved), switching to oral prednisolone is recommended starting at 1 mg/kg up to 80 mg/day. Although the most appropriate weaning off protocol is not confirmed, a weekly reduction of oral prednisolone (most commonly by 10 mg per week) under clinical, ECG, and cTn surveillance should be considered (*Figure 28*). A reassessment of LV function and cTn should be considered when the prednisolone dose is reduced to 20 mg/day and then continue weaning the prednisolone by 5 mg per week to 5 mg/day, and a final reduction from 5 mg/day in 1-mg per week steps.

If the troponin does not reduce significantly (>50% reduction from peak) and/or AV block, ventricular arrhythmias, or LVD persist despite 3 days of i.v. methylprednisolone plus cardiac treatments, then steroid-resistant ICI-associated myocarditis is confirmed and second-line immunosuppression should be considered.^{22,445,446} There is a lack of data to recommend a specific second-line immunosuppression regimen and MDT discussion is recommended. Several agents are currently being investigated with promising results from case series including i.v. mycophenolate mofetil, anti-thymocyte globulin (anti-CD3 antibody), i.v. immunoglobulin, plasma exchange, tocilizumab, abatacept (CTLA-4 agonist), alemtuzumab (anti-CD52 antibody), and tofacitinib. Caution is advised against the use of infliximab for steroid-refractory myocarditis and HF.^{447,448} Patients with fulminant ICI-associated myocarditis, complicated by haemodynamic and/or electrical instability, require admission to the intensive care unit (ICU) and cardiogenic shock should be managed according to the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF.¹⁴ A single dose of i.v. methylprednisolone should be considered in clinically unstable patients with cancer where ICI-induced myocarditis is suspected at presentation but before definitive diagnosis can be confirmed.

Following recovery from ICI-associated myocarditis and weaning of oral steroid therapy, MDT discussion is recommended to review the decision on whether to restart ICI treatment. This depends on various factors including the severity of the ICI-associated myocarditis (fulminant vs. non-fulminant vs. asymptomatic), alternative oncology treatment options, metastatic vs. adjuvant/neoadjuvant indication, and reducing from dual ICI to single ICI treatment if triggered by combination ICI treatment.⁴⁴⁹

Non-inflammatory HF syndromes have also been observed in patients treated with ICI. These include TTS, non-inflammatory HF or LVD,⁴⁵⁰ and post-MI HF.^{451,452} Non-inflammatory HF is generally a late event and the diagnostic workflow should be based on defining the HF phenotype and excluding myocarditis, TTS, and

ACS.¹⁴ There is also evidence that vasculitis and CAD can occur after ICI treatment.³³⁵ HF treatment as per the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF is indicated,¹⁴ but there is no indication for immunosuppression if myocarditis has been excluded. Interruption vs. continuing ICI therapy depends on the severity of the HF syndrome and each case should be reviewed by a MDT. Arrhythmias, such as AF, can be seen in patients with ICI therapy without myocarditis (e.g. ICI-associated thyroiditis with thyrotoxicosis, ICI-associated pericarditis, or ICI-associated severe systemic inflammatory syndromes). ICI treatment can be continued after excluding myocarditis.

Recommendation Table 26

[Open in new tab](#)

Recommendations for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis

Recommendations	Class ^a	Level ^b
cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis. ^{320,434,435,453}	I	B
In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.	I	C

EMB should be considered to confirm the diagnosis of ICI-associated myocarditis if the diagnosis is suspected but not confirmed after cardiac imaging and biomarkers. ^c	IIa	C
Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids ^d are recommended in patients with cancer and confirmed ICI-associated myocarditis. ^{22,436,454}	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction. ^e	I	C
Switching from i.v. to oral prednisolone should be considered after clinical improvement (resolution of: symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction ^e). ^f	IIa	C
Second-line immunosuppression treatment should be considered in patients with steroid-refractory ICI-associated myocarditis. ^g	IIa	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis. ¹⁴	I	C
A single dose of i.v. methylprednisolone ^d should be considered in unstable ^h patients with cancer where ICI-induced myocarditis is suspected.	IIa	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C

© ESC 2022

AV, atrioventricular; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; EMB, endomyocardial biopsy; HF, heart failure; ICI, immune checkpoint inhibitors; ICU, intensive care unit; i.v., intravenous; LGE, late gadolinium enhancement; LV, left ventricular; LVD, LV dysfunction; LVEF, LV ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#) for ICI-related myocarditis definition. EMB should be considered in unstable patients or when CMR is contraindicated.

^dEarly: ≤24 h; high-dose corticosteroids (methylprednisolone 500–1000 mg/day).

^eReduction of cTn by >50% from peak level.

^f**Complete recovery:** Patients with complete resolution of acute symptoms,

normalization of biomarkers, or reduction of cTn by >50% from peak level and recovery of LVEF after discontinuation of immunosuppression are considered to have achieved complete recovery. CMR may still show LGE or elevated T1 due to fibrosis but any suggestion of acute oedema should be absent. **Incomplete recovery:** (1) an increase in symptoms or biomarkers of myocarditis or an inability to taper immunosuppression without a clinical or biomarker flare; (2) patients with persistent LVD despite resolution of acute symptoms with immunosuppression.

^g**Steroid refractory:** non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone ([Table 3](#); [Supplementary data, Table S1](#)).

^h**Unstable:** patients with symptomatic HF, ventricular arrhythmias, new complete heart block.

Recommendations	Class ^a	Level ^b
cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis. 320,434,435,453	I	B
In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.	I	C

EMB should be considered to confirm the diagnosis of ICI-associated myocarditis if the diagnosis is suspected but not confirmed after cardiac imaging and biomarkers. ^c	IIa	C
Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids ^d are recommended in patients with cancer and confirmed ICI-associated myocarditis. ^{22,436,454}	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction. ^e	I	C
Switching from i.v. to oral prednisolone should be considered after clinical improvement (resolution of: symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction ^e). ^f	IIa	C
Second-line immunosuppression treatment should be considered in patients with steroid-refractory ICI-associated myocarditis. ^g	IIa	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis. ¹⁴	I	C
A single dose of i.v. methylprednisolone ^d should be considered in unstable ^h patients with cancer where ICI-induced myocarditis is suspected.	IIa	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C

© ESC 2022

AV, atrioventricular; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; EMB, endomyocardial biopsy; HF, heart failure; ICI, immune checkpoint inhibitors; ICU, intensive care unit; i.v., intravenous; LGE, late gadolinium enhancement; LV, left ventricular; LVD, LV dysfunction; LVEF, LV ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#) for ICI-related myocarditis definition. EMB should be considered in unstable patients or when CMR is contraindicated.

^dEarly: ≤24 h; high-dose corticosteroids (methylprednisolone 500–1000 mg/day).

^eReduction of cTn by >50% from peak level.

^f**Complete recovery:** Patients with complete resolution of acute symptoms,

normalization of biomarkers, or reduction of cTn by >50% from peak level and recovery of LVEF after discontinuation of immunosuppression are considered to have achieved complete recovery. CMR may still show LGE or elevated T1 due to fibrosis but any suggestion of acute oedema should be absent. **Incomplete recovery:** (1) an increase in symptoms or biomarkers of myocarditis or an inability to taper immunosuppression without a clinical or biomarker flare; (2) patients with persistent LVD despite resolution of acute symptoms with immunosuppression.

^g**Steroid refractory:** non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone ([Table 3](#); [Supplementary data, Table S1](#)).

^h**Unstable:** patients with symptomatic HF, ventricular arrhythmias, new complete heart block.

Recommendation Table 26

[Open in new tab](#)

Recommendations for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis

Recommendations	Class ^a	Level ^b
cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis. 320,434,435,453	I	B
In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.	I	C

EMB should be considered to confirm the diagnosis of ICI-associated myocarditis if the diagnosis is suspected but not confirmed after cardiac imaging and biomarkers. ^c	IIa	C
Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids ^d are recommended in patients with cancer and confirmed ICI-associated myocarditis. ^{22,436,454}	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction. ^e	I	C
Switching from i.v. to oral prednisolone should be considered after clinical improvement (resolution of: symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction ^e). ^f	IIa	C
Second-line immunosuppression treatment should be considered in patients with steroid-refractory ICI-associated myocarditis. ^g	IIa	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis. ¹⁴	I	C
A single dose of i.v. methylprednisolone ^d should be considered in unstable ^h patients with cancer where ICI-induced myocarditis is suspected.	IIa	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C

© ESC 2022

AV, atrioventricular; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; EMB, endomyocardial biopsy; HF, heart failure; ICI, immune checkpoint inhibitors; ICU, intensive care unit; i.v., intravenous; LGE, late gadolinium enhancement; LV, left ventricular; LVD, LV dysfunction; LVEF, LV ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#) for ICI-related myocarditis definition. EMB should be considered in unstable patients or when CMR is contraindicated.

^dEarly: ≤24 h; high-dose corticosteroids (methylprednisolone 500–1000 mg/day).

^eReduction of cTn by >50% from peak level.

^f**Complete recovery:** Patients with complete resolution of acute symptoms,

normalization of biomarkers, or reduction of cTn by >50% from peak level and recovery of LVEF after discontinuation of immunosuppression are considered to have achieved complete recovery. CMR may still show LGE or elevated T1 due to fibrosis but any suggestion of acute oedema should be absent. **Incomplete recovery:** (1) an increase in symptoms or biomarkers of myocarditis or an inability to taper immunosuppression without a clinical or biomarker flare; (2) patients with persistent LVD despite resolution of acute symptoms with immunosuppression.

^g**Steroid refractory:** non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone ([Table 3](#); [Supplementary data, Table S1](#)).

^h**Unstable:** patients with symptomatic HF, ventricular arrhythmias, new complete heart block.

Recommendations	Class ^a	Level ^b
cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis. 320,434,435,453	I	B
In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.	I	C

EMB should be considered to confirm the diagnosis of ICI-associated myocarditis if the diagnosis is suspected but not confirmed after cardiac imaging and biomarkers. ^c	IIa	C
Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids ^d are recommended in patients with cancer and confirmed ICI-associated myocarditis. ^{22,436,454}	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction. ^e	I	C
Switching from i.v. to oral prednisolone should be considered after clinical improvement (resolution of: symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction ^e). ^f	IIa	C
Second-line immunosuppression treatment should be considered in patients with steroid-refractory ICI-associated myocarditis. ^g	IIa	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis. ¹⁴	I	C
A single dose of i.v. methylprednisolone ^d should be considered in unstable ^h patients with cancer where ICI-induced myocarditis is suspected.	IIa	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C

© ESC 2022

AV, atrioventricular; CMR, cardiac magnetic resonance; cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; EMB, endomyocardial biopsy; HF, heart failure; ICI, immune checkpoint inhibitors; ICU, intensive care unit; i.v., intravenous; LGE, late gadolinium enhancement; LV, left ventricular; LVD, LV dysfunction; LVEF, LV ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Table 3](#) for ICI-related myocarditis definition. EMB should be considered in unstable patients or when CMR is contraindicated.

^dEarly: ≤24 h; high-dose corticosteroids (methylprednisolone 500–1000 mg/day).

^eReduction of cTn by >50% from peak level.

^f**Complete recovery:** Patients with complete resolution of acute symptoms,

normalization of biomarkers, or reduction of cTn by >50% from peak level and recovery of LVEF after discontinuation of immunosuppression are considered to have achieved complete recovery. CMR may still show LGE or elevated T1 due to fibrosis but any suggestion of acute oedema should be absent. **Incomplete recovery:** (1) an increase in symptoms or biomarkers of myocarditis or an inability to taper immunosuppression without a clinical or biomarker flare; (2) patients with persistent LVD despite resolution of acute symptoms with immunosuppression.

^gSteroid refractory: non-resolving or worsening myocarditis (clinical worsening or persistent troponin elevation after exclusion of other aetiologies) despite high-dose methylprednisolone ([Table 3](#); [Supplementary data, Table S1](#)).

^hUnstable: patients with symptomatic HF, ventricular arrhythmias, new complete heart block.

6.1.4. Chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies and heart dysfunction

Although no large-scale studies on the multiple CV complications among adults treated with CAR-T therapies exist, small studies and case reports have shown that CV complications represent around 20% of adverse events.³⁷⁸ CV complications are associated with high mortality rates, and are secondary to CRS and immune effector cell-associated neurotoxicity syndrome. The most common CV complications in patients receiving CAR-T therapies are arrhythmias (77.6%), including QTc prolongation, ventricular arrhythmias, and AF; HF (14.3%); and MI and VTE (0.5%).⁴⁵⁵ When suspected, a resting 12-lead ECG, continuous ECG monitoring, TTE, and cTn and NP are recommended. Admission to ICU (level 3) is recommended in severe cases due to the risk of malignant cardiac arrhythmias, circulatory collapse, and multiorgan system failure. In general, the degree of elevation of cytokines correlates with the severity of CRS. C-reactive protein is not specific for CRS and changes in C-reactive protein may lag behind clinical changes by a median of 12 h. A dramatic elevation of interleukin-6 is a supportive finding for the diagnosis of CRS. Management of the specific CV complication should follow ESC Guidelines, with additional management of the CRS (e.g. the anti-interleukin-6 receptor antibody, tocilizumab, and dexamethasone).³⁸¹

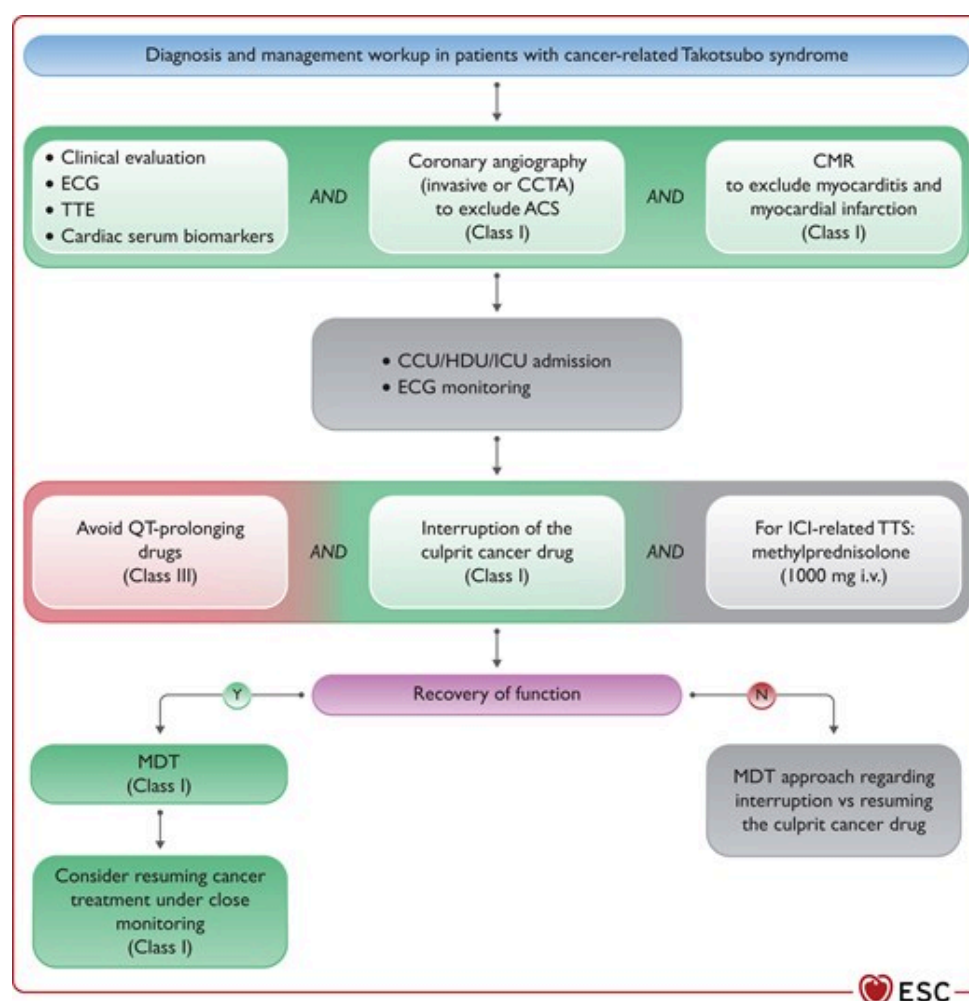
Although CV complications are common with TIL therapies, survival does not appear to be significantly affected. The most frequent CV events are hypotension that may require treatment with i.v. fluids and pressors, AF, and to a lesser extent, cTn elevation suggestive of myocardial damage.³⁸⁰ Further research is needed to define mechanisms and potential prevention strategies to help clinicians with the management of these CV events.

6.1.5. Heart failure during haematopoietic stem cell transplantation

CV complications during HSCT, including congestive HF,⁴⁵⁶ arterial events, tamponade, and rhythm disturbances (AF, atrial flutter, and supraventricular tachycardia),⁴⁵⁷ are uncommon but clinically relevant, and should be treated as per specific ESC Guidelines (HF,¹⁴ tachyarrhythmias,^{273,441} pericardial effusion,⁴⁴⁴ or acute coronary syndrome⁴⁵⁸). Studies of treatments during HSCT to prevent both acute and late CV toxicity are limited.¹⁴⁵ ACE-I and beta-blockers may be effective, but this requires further confirmation. Outpatient and home-based exercise and education programmes instituted after HSCT can improve exercise capacity and quality of life,⁴⁵⁹ and the role of exercise pre-habilitation prior to HSCT is being investigated.^{460,461}

6.1.6. Takotsubo syndrome and cancer

The prevalence of malignant diseases is high in patients with TTS and is a risk factor for worse outcomes. Malignancy itself, some cancer treatments (5-FU, ICI, VEGFi), and the stress associated with the diagnosis, investigations, and treatment are recognized triggers or predisposing factors for TTS.^{462,463,466} Diagnosis using general TTS criteria is recommended.^{467,468} Investigations in a patient with cancer with suspected TTS should include clinical examination, ECG, TTE, cardiac biomarkers (cTn and NP), and CMR (FigureÂ 29).^{468,469} Most patients require invasive coronary angiography to exclude acute MI. In patients with advanced malignancy or significant thrombocytopaenia where invasive coronary angiography is contraindicated, a CCTA is recommended. Cardiac imaging studies should be performed as early as possible when the diagnosis is suspected as LVD can be transient, and if significant LVD is detected then repeat imaging to confirm recovery is recommended.



FigureÂ 29

Diagnosis and management workup in cancer-related Takotsubo syndrome.

ACS, acute coronary syndromes; CCTA, coronary computed tomography angiography; CCU, coronary care unit; CMR, cardiac magnetic resonance; ECG, electrocardiogram; HDU, high-dependency unit; ICI, immune checkpoint inhibitor; ICU, intensive care unit; i.v., intravenous; MDT, multidisciplinary team; N, no TTE, transthoracic echocardiography; TTS, Takotsubo syndrome; Y, yes.

[Open in new tab](#)[Download slide](#)

Interruption of the culprit cancer drug in patients with TTS is recommended. QT-prolonging drugs should be avoided.⁴⁶⁷ In cases of ICI-associated TTS, the role of immunosuppression is unknown and if myocardial

inflammation is present in a TTS pattern on CMR then i.v. methylprednisolone is recommended given the overlap between ICI-induced TTS and ICI-induced myocarditis. Limited information exists regarding the feasibility of ICI rechallenge following TTS and after recovery of LV function.

A MDT discussion is recommended after recovery from the acute phase of TTS and, if restarting the culprit cancer drug is required from an oncology perspective, regular cardiac biomarker monitoring is recommended (e.g. cTn and NP measured before every ICI cycle, and TTE if a new rise in cardiac biomarkers occurs) ([Figure 29](#)).

Recommendation Table 27

[Open in new tab](#)

Recommendations for the diagnosis and management of Takotsubo syndrome in patients with cancer

Recommendations	Class ^a	Level ^b
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI. ⁴⁵⁸	I	B
QT-prolonging drugs are not recommended during the acute TTS phase. ^c	III	C

© ESC 2022

ACS, acute coronary syndromes; CMR, cardiac magnetic resonance; CCTA, coronary computed tomography angiography; LV, left ventricular; MI, myocardial infarction; QTc, corrected QT interval; TTS, Takotsubo syndrome.

^aClass of recommendation.

^bLevel of evidence.

^cUntil full recovery and normalization of LV function and QTc.

Recommendations	Class ^a	Level ^b
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI. ⁴⁵⁸	I	B
QT-prolonging drugs are not recommended during the acute TTS phase. ^c	III	C

© ESC 2022

ACS, acute coronary syndromes; CMR, cardiac magnetic resonance; CCTA, coronary computed tomography angiography; LV, left ventricular; MI, myocardial infarction; QTc, corrected QT interval; TTS, Takotsubo syndrome.

^aClass of recommendation.

^bLevel of evidence.

^cUntil full recovery and normalization of LV function and QTc.

Recommendation Table 27

[Open in new tab](#)

Recommendations	Class ^a	Level ^b
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI. ⁴⁵⁸	I	B
QT-prolonging drugs are not recommended during the acute TTS phase. ^c	III	C

© ESC 2022

ACS, acute coronary syndromes; CMR, cardiac magnetic resonance; CCTA, coronary computed tomography angiography; LV, left ventricular; MI, myocardial infarction; QTc, corrected QT interval; TTS, Takotsubo syndrome.

^aClass of recommendation.

^bLevel of evidence.

^cUntil full recovery and normalization of LV function and QTc.

Recommendations	Class ^a	Level ^b
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI. ⁴⁵⁸	I	B
QT-prolonging drugs are not recommended during the acute TTS phase. ^c	III	C

© ESC 2022

ACS, acute coronary syndromes; CMR, cardiac magnetic resonance; CCTA, coronary computed tomography angiography; LV, left ventricular; MI, myocardial infarction; QTc, corrected QT interval; TTS, Takotsubo syndrome.

^aClass of recommendation.

^bLevel of evidence.

^cUntil full recovery and normalization of LV function and QTc.

6.2. Coronary artery disease

6.2.1. Acute coronary syndromes

Patients with cancer are at increased risk of CAD because of shared CVRFs³⁴ and CV toxicity of cancer therapy¹² compounded by a cancer-induced pro-inflammatory and prothrombotic state (Table 7).^{467,468,470a,473}

Table 7

[Open in new tab](#)

Cancer treatments that predispose to acute coronary syndromes

Accelerated atherosclerosis and plaque rupture	ADT (GnRH agonists), ICI, nilotinib, ponatinib, radiation therapy, VEGFi
Vasospasm	Bleomycin, fluoropyrimidines, taxanes, VEGFi, vinca alkaloids
Coronary thrombosis	Alkylating agents (cisplatin, cyclophosphamide), erlotinib, ICI, IMiD (lenalidomide, thalidomide), monoclonal antibodies (VEGFi, anti-CD20), nilotinib, platinum chemotherapy, PI, ponatinib, VEGFi.

Accelerated atherosclerosis and plaque rupture	ADT (GnRH agonists), ICI, nilotinib, ponatinib, radiation therapy, VEGFi
Vasospasm	Bleomycin, fluoropyrimidines, taxanes, VEGFi, vinca alkaloids
Coronary thrombosis	Alkylating agents (cisplatin, cyclophosphamide), erlotinib, ICI, IMiD (lenalidomide, thalidomide), monoclonal antibodies (VEGFi, anti-CD20), nilotinib, platinum chemotherapy, PI, ponatinib, VEGFi.

ADT, androgen deprivation therapy; GnRH, gonadotropin-releasing hormone; ICI, immune checkpoint inhibitors; IMiD, immunomodulatory drugs; PI, proteasome inhibitors; VEGFi, vascular endothelial growth factor inhibitors.

Table 7

[Open in new tab](#)

Cancer treatments that predispose to acute coronary syndromes

Accelerated atherosclerosis and plaque rupture	ADT (GnRH agonists), ICI, nilotinib, ponatinib, radiation therapy, VEGFi
Vasospasm	Bleomycin, fluoropyrimidines, taxanes, VEGFi, vinca alkaloids
Coronary thrombosis	Alkylating agents (cisplatin, cyclophosphamide), erlotinib, ICI, IMiD (lenalidomide, thalidomide), monoclonal antibodies (VEGFi, anti-CD20), nilotinib, platinum chemotherapy, PI, ponatinib, VEGFi.

Accelerated atherosclerosis and plaque rupture	ADT (GnRH agonists), ICI, nilotinib, ponatinib, radiation therapy, VEGFi
Vasospasm	Bleomycin, fluoropyrimidines, taxanes, VEGFi, vinca alkaloids

Coronary thrombosis	Alkylating agents (cisplatin, cyclophosphamide), erlotinib, ICI, IMiD (lenalidomide, thalidomide), monoclonal antibodies (VEGFi, anti-CD20), nilotinib, platinum chemotherapy, PI, ponatinib, VEGFi.
---------------------	--

ADT, androgen deprivation therapy; GnRH, gonadotropin-releasing hormone; ICI, immune checkpoint inhibitors; IMiD, immunomodulatory drugs; PI, proteasome inhibitors; VEGFi, vascular endothelial growth factor inhibitors.

Current knowledge on ACS in patients with cancer is based on observational data and registries demonstrating that, especially when diagnosed within 1 year, they are at increased risk for major CV events, bleeding, and cardiac and non-cardiac mortality.^{474,480} The proportion of ACS patients with a diagnosis of cancer is rising and constitutes about 3% of large series.⁴⁷⁵

Diagnosis of ACS is based on the same principles as in patients without cancer, including symptoms, an early 12-lead ECG, and serial measurements of hs-cTn for patients presenting with possible non-ST-segment elevation ACS (NSTEMI-ACS).⁴⁵⁸ Clinical presentation can be atypical⁴⁸¹ or masked by cancer or therapy-related side effects; therefore, diagnostic suspicion should be increased in patients at high CV risk or treated with vascular cardiotoxic therapies ([Table 7](#)). Echocardiography improves the diagnostic precision in patients with atypical symptoms and assesses for other cardiac causes of chest pain.

Management of ACS in patients with cancer can be challenging because of frailty, increased bleeding risk, thrombocytopenia, increased thrombotic risk, and the possible need for future surgery/interventions.⁴⁸²

Cancer treatment should be temporarily interrupted, and an urgent multidisciplinary approach⁵ is indicated to plan an individualized guideline-based management, taking into account cancer status, prognosis, and the patient's preferences regarding invasive management. As in patients without cancer, admission to a monitored unit and initiation of appropriate anti-ischaemic and antithrombotic treatment are indicated, in the absence of contraindications.

A large retrospective propensity score-matching analysis found that percutaneous coronary intervention (PCI), despite its lower use, was strongly associated with lower adjusted MACE and all-cause mortality in patients with cancer (Hodgkin and non-Hodgkin lymphomas and breast, lung, colon, and prostate cancers).⁴⁸³

Therefore, immediate coronary angiography and PCI are recommended in patients with cancer and ACS if cancer prognosis is ≤ 6 months or if they have acute complications of ACS (cardiogenic shock, pulmonary oedema, ventricular tachyarrhythmias), where PCI offers palliation of symptoms.⁴⁸³ When stenting is indicated, third-generation drug-eluting stents are preferred because of the lower risk of in-stent thrombosis.

Balloon angioplasty is associated with worse outcome⁴⁷⁴ and should only be used in case of severe thrombocytopenia or need for urgent surgery. Fractional flow reserve or instantaneous free wave ratio are advised by experts⁴⁸⁴ to avoid unnecessary interventions while intravascular ultrasound and optical coherence tomography can be used to ensure optimal stent apposition and expansion, to avoid thrombotic complications.⁴⁸⁵

Retrospective data have demonstrated a lower use of invasive management in patients with cancer with ST-segment elevation MI (STEMI), with a better outcome for invasively treated patients.^{475,480,483} PCI has not demonstrated a mortality benefit in patients with advanced cancer and NSTEMI-ACS compared with optimal

medical therapy.⁴⁷⁹ Therefore, a non-invasive approach can be attempted in low-risk (without signs or symptoms of ongoing ischaemia or haemodynamic instability) NSTEMI-ACS patients with poor cancer prognosis (<6 months).

Due to a potentially higher bleeding risk (especially in patients with active GI cancer),⁴⁷⁷ the preferred antithrombotic strategy after drug-eluting stent consists of DAPT with aspirin and clopidogrel instead of newer P2Y₁₂ antagonists. The duration of DAPT should be kept as short as possible (1–3 months).⁴⁵⁸ In patients with need for therapeutic anticoagulation and antiplatelet therapy, a NOAC and single oral antiplatelet (preferably clopidogrel) is the default strategy after a short period of triple antithrombotic therapy (up to 1 week in hospital).⁴⁵⁸ Coronary artery bypass graft (CABG) surgery can be considered in patients with extensive CAD who are not amenable with PCI, after MDT discussion and where cancer prognosis is >12 months.

Thrombocytopenia (platelet count < 100 000/ÅµL) is present in about 10% of patients with cancer and may complicate ACS management. Based on a small series, coronary angiography can be safely performed in these patients when preventative measures to avoid bleeding are taken: platelet transfusion before catheterization (for platelets <20 000/ÅµL), radial access, careful haemostasis, and the use of a lower heparin dose (30–50 Åµg/kg).⁴⁸⁶ Antiplatelets should not be withheld unless platelet count is <10 000/ÅµL for aspirin or <30 000/ÅµL for clopidogrel. For PCI and CABG, experts advise minimum platelet counts of 30 000/ÅµL and 50 000/ÅµL, respectively.⁴⁸⁴

In case of MI with non-obstructive coronary arteries, CMR may be considered to detect other causes of myocardial injury, especially myocarditis and TTS.

When acute ischaemia is provoked by cancer therapy, alternative cancer therapies should be considered after a MDT discussion. In the case of coronary vasospasm secondary to fluoropyrimidines, and in the absence of an alternative therapy, a rechallenge, although controversial, can be considered in a monitored unit after exclusion of severe CAD (CT or coronary angiography) and after initiation of prophylactic therapy with long-acting nitrates and calcium channel blockers (CCB).^{487,489}

Following ACS, a review of the cancer medications is recommended, and any cancer drug associated with thrombosis and MI should be stopped. Restarting cancer drugs associated with acute thrombosis and MI after ACS ([Table 7](#)) should occur only after a MDT to explore other cancer therapies, with appropriate patient education and consent. Cancer therapies not associated with MI can be restarted once revascularization, where indicated, has been completed and the patient is stabilized on ACS medical therapy without complications.

Recommendation Table 28

[Open in new tab](#)

Recommendations for the management of acute coronary syndromes in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months. ^{475,479,483}	I	B
A conservative non-invasive strategy should be considered in patients with poor cancer prognosis ^c (with life expectancy < 6 months) and/or very high bleeding risk presenting with STEMI or NSTEMI-ACS. ⁴⁷⁹	IIa	C
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause. ^{d,10,490}	I	C
A short DAPT strategy should be considered in patients with cancer with very high bleeding risk treated with PCI for an ACS. ^e	IIa	C
In patients with cancer, thrombocytopenia, and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia, and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Ticagrelor or prasugrel may be considered in patients with cancer with low bleeding risk and excessive thrombotic risk who are treated with PCI for ACS.	IIb	C

ACS, acute coronary syndromes; CrCl, creatinine clearance; CV, cardiovascular; DAPT, dual antiplatelet therapy; GI, gastrointestinal; GU, genitourinary; NSTEMI-ACS, non-ST-segment elevation acute coronary syndromes; PCI, percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cRelated to advanced cancer stage and/or severe irreversible non-CV comorbidities.

^dAnticancer therapies associated with high risk of ACS (very common [$> 10\%$]): capecitabine, paclitaxel, cisplatin, carfilzomib, bevacizumab, ramucirumab, aflibercept, axitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, ponatinib, erlotinib.

^eHigh risk of GI or GU bleeding, significant drug–drug interactions, severe renal dysfunction (CrCl < 30 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase $> 2 \times$ ULN), or significant thrombocytopenia (platelet count $< 50\,000/\mu\text{L}$).

© ESC 2022

Recommendations	Class ^a	Level ^b
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months. ^{475,479,483}	I	B
A conservative non-invasive strategy should be considered in patients with poor cancer prognosis ^c (with life expectancy < 6 months) and/or very high bleeding risk presenting with STEMI or NSTEMI-ACS. ⁴⁷⁹	IIa	C
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause. ^{d,10,490}	I	C
A short DAPT strategy should be considered in patients with cancer with very high bleeding risk treated with PCI for an ACS. ^e	IIa	C
In patients with cancer, thrombocytopenia, and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia, and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Ticagrelor or prasugrel may be considered in patients with cancer with low bleeding risk and excessive thrombotic risk who are treated with PCI for ACS.	IIb	C

ACS, acute coronary syndromes; CrCl, creatinine clearance; CV, cardiovascular; DAPT, dual antiplatelet therapy; GI, gastrointestinal; GU, genitourinary; NSTEMI-ACS, non-ST-segment elevation acute coronary syndromes; PCI, percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cRelated to advanced cancer stage and/or severe irreversible non-CV comorbidities.

^dAnticancer therapies associated with high risk of ACS (very common [$> 10\%$]): capecitabine, paclitaxel, cisplatin, carfilzomib, bevacizumab, ramucirumab, aflibercept, axitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, ponatinib, erlotinib.

^eHigh risk of GI or GU bleeding, significant drug–drug interactions, severe renal dysfunction (CrCl < 30 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase $> 2 \times$ ULN), or significant thrombocytopenia (platelet count $< 50\,000/\mu\text{L}$).

© ESC 2022

Recommendation Table 28

[Open in new tab](#)

Recommendations for the management of acute coronary syndromes in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-Acute Coronary Syndrome (ACS) with life expectancy ≥ 6 months. ^{475,479,483}	I	B
A conservative non-invasive strategy should be considered in patients with poor cancer prognosis ^c (with life expectancy < 6 months) and/or very high bleeding risk presenting with STEMI or NSTEMI-Acute Coronary Syndrome (ACS). ⁴⁷⁹	IIa	C
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause. ^{d,10,490}	I	C
A short DAPT strategy should be considered in patients with cancer with very high bleeding risk treated with PCI for an ACS. ^e	IIa	C

In patients with cancer, thrombocytopaenia, and ACS, aspirin is not recommended if platelets <10 000/ μ L.	III	C
In patients with cancer, thrombocytopaenia, and ACS, clopidogrel is not recommended if platelets <30 000/ μ L and prasugrel or ticagrelor are not recommended if platelets <50 000/ μ L.	III	C
Ticagrelor or prasugrel may be considered in patients with cancer with low bleeding risk and excessive thrombotic risk who are treated with PCI for ACS.	IIb	C

© ESC 2022

ACS, acute coronary syndromes; CrCl, creatinine clearance; CV, cardiovascular; DAPT, dual antiplatelet therapy; GI, gastrointestinal; GU, genitourinary; NSTEMI-ACS, non-ST-segment elevation acute coronary syndromes; PCI, percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cRelated to advanced cancer stage and/or severe irreversible non-CV comorbidities.

^dAnticancer therapies associated with high risk of ACS (very common [$>10\%$]): capecitabine, paclitaxel, cisplatin, carfilzomib, bevacizumab, ramucirumab, aflibercept, axitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, ponatinib, erlotinib.

^eHigh risk of GI or GU bleeding, significant drug–drug interactions, severe renal dysfunction (CrCl < 30 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase $>2 \times$ ULN), or significant thrombocytopaenia (platelet count < 50 000/ μ L).

Recommendations	Class ^a	Level ^b
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months. ^{475,479,483}	I	B
A conservative non-invasive strategy should be considered in patients with poor cancer prognosis ^c (with life expectancy < 6 months) and/or very high bleeding risk presenting with STEMI or NSTEMI-ACS. ⁴⁷⁹	IIa	C
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause. ^{d,10,490}	I	C
A short DAPT strategy should be considered in patients with cancer with very high bleeding risk treated with PCI for an ACS. ^e	IIa	C
In patients with cancer, thrombocytopenia, and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia, and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Ticagrelor or prasugrel may be considered in patients with cancer with low bleeding risk and excessive thrombotic risk who are treated with PCI for ACS.	IIb	C

ACS, acute coronary syndromes; CrCl, creatinine clearance; CV, cardiovascular; DAPT, dual antiplatelet therapy; GI, gastrointestinal; GU, genitourinary; NSTEMI-ACS, non-ST-segment elevation acute coronary syndromes; PCI, percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction; ULN, upper limit of normal.

^aClass of recommendation.

^bLevel of evidence.

^cRelated to advanced cancer stage and/or severe irreversible non-CV comorbidities.

^dAnticancer therapies associated with high risk of ACS (very common [$> 10\%$]): capecitabine, paclitaxel, cisplatin, carfilzomib, bevacizumab, ramucirumab, aflibercept, axitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, ponatinib, erlotinib.

^eHigh risk of GI or GU bleeding, significant drug–drug interactions, severe renal dysfunction (CrCl < 30 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase $> 2 \times$ ULN), or significant thrombocytopenia (platelet count $< 50\,000/\mu\text{L}$).

© ESC 2022

6.2.2. Chronic coronary syndromes

Several cancer treatments are associated with an increased risk of stable angina and chronic coronary syndromes (CCS).⁴⁹¹ 5-FU and capecitabine can precipitate effort angina in some cases.^{4,482,492} Platinum-containing chemotherapy-induced ischaemia usually occurs after one of the first three cycles and in patients with underlying CAD.⁴⁹³ The incidence of cardiac ischaemia is 1â5% with antimicrotubule agents, 2â3% with small-molecule VEGF-TKI, and 0.6â1.5% with VEGFi monoclonal antibody therapies.⁴⁹² Nilotinib, ponatinib,⁴⁹⁴ and ICI³³⁵ also accelerate atherosclerosis, which can lead to stable angina.

Patients receiving cancer therapy who present with new stable angina should have careful clinical evaluation, with aggressive CVRF modification and an initial medical management of their symptoms.⁴⁸⁴ The diagnosis and management of CAD should follow the 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes.¹⁰⁰

The management of CCS is similar in patients with and without cancer, in accordance with guideline recommendations.¹⁰⁰ However, in the setting of CCS, decisions regarding coronary revascularization should be undertaken by a MDT that includes cardio-oncology, intervention, and oncology specialists.⁵ PCI in patients with cancer is associated with an increased risk of bleeding, 90-day readmissions for acute MI, in-hospital and long-term mortality, and the need for repeat revascularization, with the magnitude of risk depending on both cancer type and stage.^{495,496} The excess bleeding risk should be mitigated by keeping the duration of DAPT as short as possible.^{497,498} The risk is higher in patients with a cancer diagnosis within the preceding year.⁴⁷⁷

Recommendation Table 29

[Open in new tab](#)

Recommendation for the management of chronic coronary syndromes in patients receiving anticancer treatment

Recommendation	Class ^a	Level ^b
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment. ^{100,498}	I	C

CCS, chronic coronary syndromes; DAPT, dual antiplatelet therapy.
^aClass of recommendation.
^bLevel of evidence.

© ESC 2022

Recommendation	Class ^a	Level ^b
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment. ^{100,498}	I	C

CCS, chronic coronary syndromes; DAPT, dual antiplatelet therapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendation Table 29

[Open in new tab](#)

Recommendation for the management of chronic coronary syndromes in patients receiving anticancer treatment

Recommendation	Class ^a	Level ^b
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment. ^{100,498}	I	C

CCS, chronic coronary syndromes; DAPT, dual antiplatelet therapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendation	Class ^a	Level ^b
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment. ^{100,498}	I	C

CCS, chronic coronary syndromes; DAPT, dual antiplatelet therapy.
^aClass of recommendation.
^bLevel of evidence.

6.3. Valvular heart disease

New or worsening VHD in patients with cancer may be related to coexisting conditions, including CTRCD, ACS, PH, endocarditis, cardiac tumours, and mechanical prosthetic valve thrombosis.^{499,500}

Pre-existing severe VHD is associated with an increased risk of CTRCD,^{12,501a,503} and may also pose a risk for cancer surgery outcomes. In patients with mechanical prosthetic valves, the risk of thrombosis vs. bleeding should be carefully balanced during chemotherapy treatment. In patients with severe VHD diagnosed at baseline assessment, a MDT is required before cancer therapy to decide the best treatment option. Cardiac surgery is frequently challenging in patients with cancer because of comorbidities, frailty, mediastinal fibrosis due to prior RT, impaired wound healing, and the need for urgent oncology treatment (surgery, chemotherapy, targeted cancer therapies that effect wound healing). Transcatheter aortic valve implantation (TAVI) may be a viable option for patients with cancer with severe aortic stenosis to limit recovery time and delays in starting cancer treatment.^{504a,506}

Patients with cancer suspected of new or worsening VHD, such as dyspnoea or a new cardiac murmur, or those with fever and positive blood cultures, should be screened for endocarditis and managed according to the recommendations from the 2021 ESC/European Association for Cardio-Thoracic Surgery (EACTS) Guidelines for the management of VHD,⁵⁰⁷ while considering the cancer-related prognosis. If valve surgery or percutaneous valve treatment is indicated in a patient receiving cancer treatment, then a MDT is recommended regarding type of valve treatment and periprocedural management of cancer treatments.

Recommendation Table 30

[Open in new tab](#)

Recommendations for the management of valvular heart disease in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/ EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences. ⁵⁰⁷	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD ⁵⁰⁷ is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C

© ESC 2022

EACTS, European Association for Cardio-Thoracic Surgery; ESC, European Society of Cardiology; VHD, valvular heart disease.
^aClass of recommendation.
^bLevel of evidence.

Recommendations	Class ^a	Level ^b
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences. ⁵⁰⁷	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD ⁵⁰⁷ is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C

EACTS, European Association for Cardio-Thoracic Surgery; ESC, European Society of Cardiology; VHD, valvular heart disease.
^aClass of recommendation.
^bLevel of evidence.

Recommendation Table 30

[Open in new tab](#)

Recommendations for the management of valvular heart disease in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences. ⁵⁰⁷	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD ⁵⁰⁷ is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C

EACTS, European Association for Cardio-Thoracic Surgery; ESC, European Society of Cardiology; VHD, valvular heart disease.
^aClass of recommendation.
^bLevel of evidence.

Recommendations	Class ^a	Level ^b
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences. ⁵⁰⁷	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD ⁵⁰⁷ is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C

EACTS, European Association for Cardio-Thoracic Surgery; ESC, European Society of Cardiology; VHD, valvular heart disease.
^aClass of recommendation.
^bLevel of evidence.

6.4. Cardiac arrhythmias

6.4.1. Atrial fibrillation

AF may occur in patients with cancer in different settings: it may be a marker of cancer type or occult cancer, or it may develop in patients undergoing surgery, chemotherapy, or RT.^{508,509} All types of cancer show an increased risk of AF compared with the control group, but the risk of AF depends on the cancer type and stage.^{510,511} AF during a cancer treatment may be caused by a specific therapy or interaction with a pre-existing substrate in older patients with cancer.

During cancer therapy AF may occur with a frequency ranging from 2% to 16%, according to a variety of factors,^{4,490,508,512a,513,514} and may present either as first-diagnosed AF or as recurrence of paroxysmal AF. The risk of developing AF is greater in patients older than 65 years and/or with pre-existing CVD.^{4,509,512,515} Cancer surgery is associated with a variable rate of AF occurrence, with the highest incidence reported for lung surgery, ranging from 6% to 32%, but with occurrence also in cases of non-thoracic surgery (e.g. 4a⁵¹⁶ 5% after colectomy).⁵⁰⁹

Many anticancer drugs have been associated with an increased risk of AF both in terms of incident and recurrent AF ([Supplementary data, Table S18](#)).²⁵¹ AF may occur shortly after treatment⁵¹⁶ or weeks or months after starting treatment.^{517,518} The pathophysiology of AF associated with cancer is complex and has been extensively reviewed elsewhere ([Figure 30](#)).⁵⁰⁹

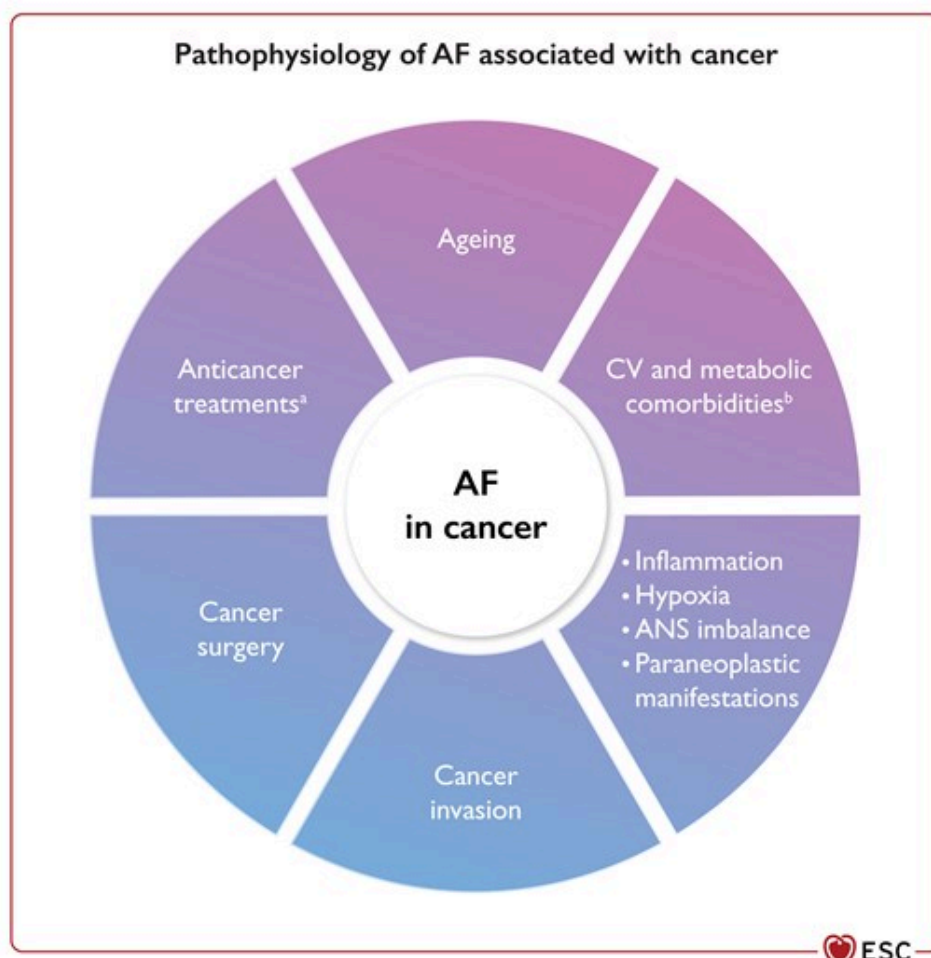


Figure 30

Pathophysiology of atrial fibrillation associated with cancer.

AF, atrial fibrillation; ANS, autonomic nervous system; CV, cardiovascular; DM, diabetes mellitus; HF, heart failure; IHD, ischaemic heart disease; VHD, valvular heart disease. ^a[Supplementary data, Table S18](#).

^bObesity, hypertension, DM, CVDs (HF, VHD, IHD, cardiomyopathies, cardiac amyloidosis), thyroid diseases, obstructive sleep apnoea, chronic obstructive pulmonary disease, chronic kidney disease, autonomic dysfunction, alcohol consumption, genetic predisposition.

[Open in new tab](#)[Download slide](#)

In patients with cancer, the occurrence of AF is associated with a two-fold higher risk of systemic thromboembolism/stroke and a six-fold increase in the risk of HF.^{4,509,512} The coexistence of cancer increases the risk of all-cause mortality, major bleeding, and intracranial haemorrhage in patients with AF. The association between cancer and ischaemic stroke differs between cancer types, and in some types, the risk of bleeding seems to exceed the thromboembolic risk.⁵¹⁹ The management of AF in patients with cancer should follow the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation²⁷³ and the [AFSSG](#) (Atrial fibrillation Better Care) approach should be applied (A: Anticoagulation to avoid stroke/systemic embolism, B: Better symptom control with rate- and/or rhythm-control drugs and interventions, and C: Comorbidities and CVRF management, including lifestyle changes).^{273,520}

The acute management of AF in patients with cancer should consider electrical cardioversion in cases of haemodynamic instability,⁵²¹ while in others, the alternative between rate and rhythm control has several

important considerations specific to patients with cancer. Drugs for rhythm control may lead to QT-interval prolongation,³⁶⁹ frequently have drug^{FADSTS} drug interactions with cancer therapies, or may have a limited efficacy if a cancer therapy is the specific cause of the AF.⁵⁰⁸ Among rate-control drugs, beta-blockers are preferred, especially if the cancer therapies have potential CTRCD risk, whereas diltiazem and verapamil should be avoided where possible due to their drug^{FADSTS} drug interactions and negative inotropic effects.⁵⁰⁸ The possibility of AF ablation should be discussed in selected patients with HF/LVD and/or uncontrolled symptoms, taking into consideration cancer status and prognosis in the context of a MDT approach.⁵²²

A complex issue in patients with cancer with new AF is risk stratification for stroke/systemic embolism, which according to guidelines, should be based on the CHA₂DS₂-VASc score (Congestive heart failure, Hypertension, Age ^{HTC} ≥ 75 years [2 points], Diabetes mellitus, Stroke [2 points]^{FADSTS} Vascular disease, Age 65^{FADSTS}–74 years, Sex category [female]).^{273,523,524} The CHA₂DS₂-VASc score has not been extensively validated in patients with cancer.⁵²⁵ In a large cohort of patients with AF, the predictive value of the CHA₂DS₂-VASc score was lower in patients with cancer than in those without, but a progressive increase in the risk of ischaemic stroke according to the CHA₂DS₂-VASc score was also found in AF patients with cancer (from 0.9% per year to 8.9% per year).⁵¹⁹ However, the scope of this score is not to identify high-risk patients, but rather to identify low-risk individuals in whom anticoagulation can be avoided. A study based on the Danish healthcare system data set found that CHA₂DS₂-VASc scores of 0 and 1 in patients with recent cancer were linked with higher risk of stroke/thromboembolism at 2 years than in patients without recent cancer.⁵²⁶ This concept should be considered in defining the risk/benefit ratio of anticoagulation in individual patients with cancer. Therefore, the decision for anticoagulation in patients with an active malignancy should take into account the enhanced thrombotic and/or bleeding risk and other risk prediction scores used for general AF populations.⁵⁰⁹ For bleeding risk assessment, the HAS-BLED (Hypertension, Abnormal renal and liver function, Stroke, Bleeding Labile international normalized ratio, Elderly, Drugs or alcohol) score may be considered. A proposed approach to anticoagulation therapy in cancer, based on the acronym T (thrombotic risk), B (bleeding risk), I (interactions among drugs), P (patient access and preferences), is outlined in [Figure 31](#).^{519,527}

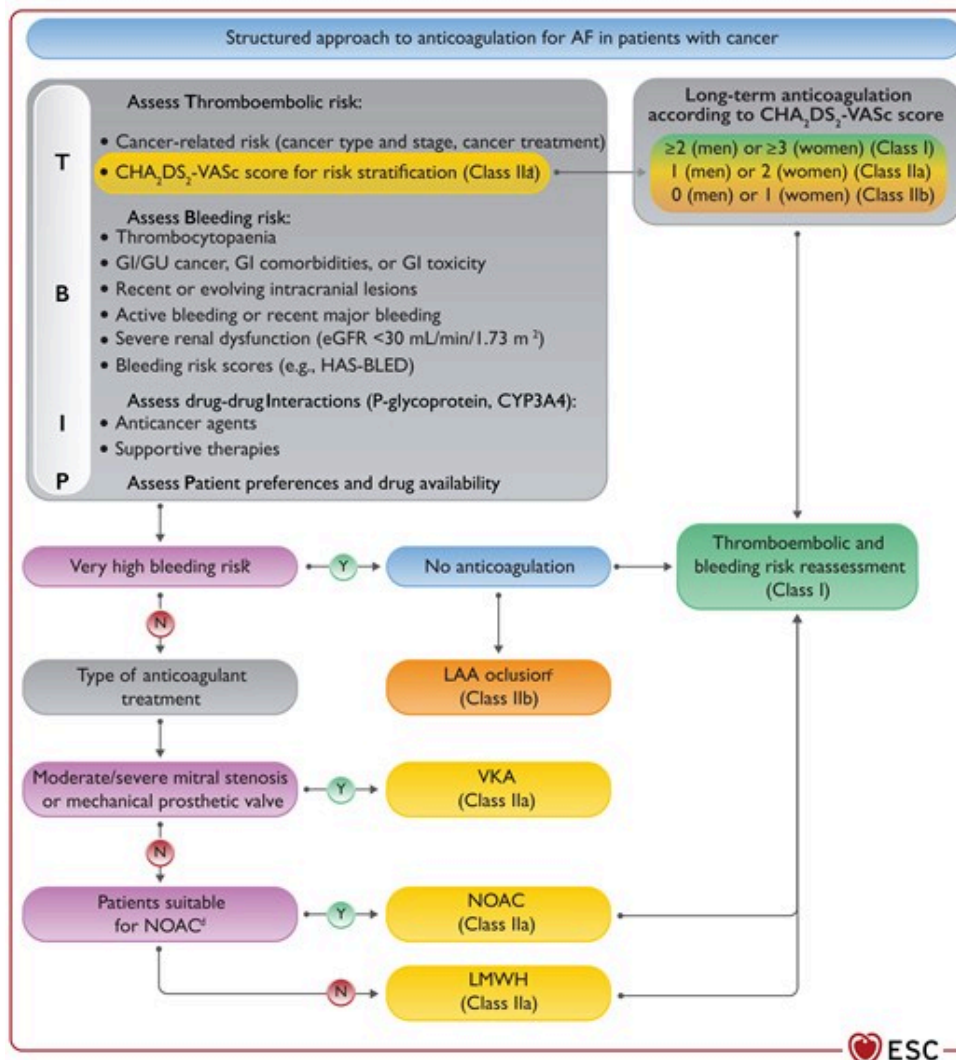


Figure 31

Structured approach to anticoagulation for atrial fibrillation in patients with cancer.

AF, atrial fibrillation; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age ≥75 years (2 points), Diabetes mellitus, Stroke (2 points), Atrial fibrillation, Vascular disease, Age 65–74 years, Sex category (female); CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; GI, gastrointestinal; GU, genitourinary; HAS-BLED, Hypertension, Abnormal renal and liver function, Stroke, Bleeding Labile international normalized ratio, Elderly, Drugs or alcohol; LA, left atrial; LAA, left atrial appendage; LMWH, low-molecular-weight heparins; N, no; NOAC, non-vitamin K antagonist oral anticoagulant; VKA, vitamin K antagonists; Y, yes. ^aIn selected patients, cardiac imaging parameters related to increased thromboembolic risk should be considered (LAA thrombus, severely dilated left atrium, severely impaired LA strain⁵²⁸). ^b**Very high bleeding risk:** active or recent major bleeding (<1 month previously); recent/evolving intracranial lesions; platelet count <25 000/μL. According to the International Society on Thrombosis and Haemostasis,⁵²⁹ major bleeding is defined as a fall in haemoglobin level ≥2 g/dL and/or transfusion of ≥2 units of red blood cells and/or fatal bleeding and/or bleeding in a critical area (intracranial, intraspinal, intraocular, pericardial, intra-articular, intramuscular with compartment syndrome, or retroperitoneal). ^c**Percutaneous left appendage closure** may be considered in patients with a life expectancy of >1 year who are at high thromboembolic and bleeding risk and in whom anticoagulation is contraindicated. ^d**Conditions favouring LMWH:** unoperated

GI/GU cancer; GI comorbidities or toxicity; severe renal dysfunction ($\text{CrCl} < 15 \text{ mL/min}$); NOAC major drug-drug interactions, platelet count $< 50,000/\mu\text{L}$.

[Open in new tab](#)[Download slide](#)

Long-term anticoagulation is recommended in adult patients with $\text{CHA}_2\text{DS}_2\text{-VASc}$ score ≥ 2 in men or ≥ 3 in women and must be considered also when the score is 1 in men and 2 in women.²⁷³ The clinical pattern of AF (i.e. first detected, paroxysmal, persistent, long-standing persistent, permanent, post-operative) should not influence the indication of thromboprophylaxis.²⁷³ The same approach can be proposed for patients with cancer and AF, also considering that the $\text{CHA}_2\text{DS}_2\text{-VASc}$ score likely underestimates their thromboembolic risk.⁵³⁰ In the specific setting of cancer, decision-making on long-term oral anticoagulation should also consider the cancer-related type, stage, prognosis and the potentially changing thromboembolic or bleeding risk.^{508,509} The use of vitamin K antagonists (VKA) in cancer is limited by their drawbacks in this setting; however, they remain the only indicated anticoagulants in patients with moderate to severe mitral stenosis or a mechanical prosthetic valve. LMWH constitute a viable short-term anticoagulation option, particularly in hospitalized patients with a recent cancer diagnosis, advanced cancer disease, or during some cancer treatments (e.g. patients receiving myelosuppressive chemotherapy or with recent active bleeding). However, LMWH efficacy for stroke or systemic embolism prevention in AF has not been established and their use is only based on their proven efficacy and safety in VTE. The use of a NOAC for AF has not been evaluated in a dedicated RCT in patients with cancer. However, secondary analyses of seminal NOAC trials using direct factor Xa inhibitors (ROCKET AF [Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation], ARISTOTLE [Apixaban for Reduction in Stroke and Other Thromboembolic Events in Atrial Fibrillation], ENGAGE AF-TIMI 48 [Effective Anticoagulation with Factor Xa Next Generation in Atrial Fibrillation]) and observational data suggest better safety and at least similar effectiveness of the NOAC when compared with VKA in patients with AF and active cancer.⁵³¹⁻⁵³⁸ NOAC use in cancer is limited by drug-drug interactions,⁵⁰⁸ severe renal dysfunction, increased risk of bleeding in patients with unoperated or residual GI or genitourinary (GU) malignancies, or impaired GI absorption.

Left atrial appendage (LAA) occluder devices are used in very selected patients with cancer in clinical practice. The potential complications related to the implantation, including device-related thrombosis and the lack of prospective data in the setting of patients with cancer have to be taken into consideration for this option. In a recent retrospective analysis of patients referred to LAA occlusion the risk of in-hospital ischaemic stroke/transient ischaemic attack was higher in patients with active cancer than in those with no cancer or prior history of cancer. The rate of in-hospital composite outcome (in-hospital death, ischaemic stroke/transient ischaemic attack, systemic embolism, bleeding requiring blood transfusion, pericardial effusion/cardiac tamponade treated with pericardiocentesis or surgically, and removal of embolized device) and 30-day/180-day readmission outcomes were not significantly different between the groups.⁵³⁹

The onset of AF may be related to transient factors, such as the peri-operative period or the effect of drugs known to facilitate AF onset. The traditional assumption that in these cases, AF may occur as an isolated event without recurrence may not be valid as the occurrence of AF may often be related to a pre-existing atrial substrate with vulnerability to AF.⁵⁴⁰ Post-operative AF has been associated with a four- to five-fold risk of

AF recurrence in the following 5 years, along with a comparable long-term thromboembolic risk with AF not related to surgery.^{273,540,541} Anticoagulation therapy yielded a similarly lower risk of thromboembolic events and all-cause death in both groups.⁵⁴¹ In the absence of direct evidence, anticoagulation to prevent thromboembolic events should be considered in patients at risk for stroke with AF after cancer surgery considering the anticipated net clinical benefit and informed patient preferences.²⁷³ Similarly, in patients with AF apparently related to transient factors⁵⁴² such as chemotherapy, other drugs, or electrolyte disturbances⁵⁴³ a careful clinical assessment of the propensity to further develop AF is recommended, with need to revisit the risk/benefit ratio of long-term prescription of anticoagulation after a period of 3 months.

In patients with cancer and newly detected or recurrence of AF, decision making on anticancer treatment requires a cardio-oncology MDT management,⁵ taking into account that neither the presence nor the risk of AF constitutes contraindications to anticancer treatment.^{508,517}

Recommendation Table 31

[Open in new tab](#)

Recommendations for the management of atrial fibrillation in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
CHA ₂ DS ₂ -VASc score should be considered for risk stratification for stroke/systemic thromboembolism taking into account that it may underestimate the actual thromboembolic risk. ^{519,526}	IIa	C
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥2 (men) or ≥3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	I	C

Long-term anticoagulation should be considered for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score = 1 (men) or = 2 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	IIa	C
Patients with cancer, ^c AF, and CHA ₂ DS ₂ -VASc score 0 (men) or 1 (women) may have a higher thrombotic risk than patients without cancer and may be considered for therapeutic anticoagulation after consideration of the bleeding risk. ⁵²⁶	IIb	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF. ^{d,273}	I	C
NOAC should be considered for stroke prevention in preference to LMWH and VKA (excluding patients with mechanical heart valves or moderate-to-severe mitral stenosis) in patients without a high bleeding risk, significant drug–drug interactions, or severe renal dysfunction. ^{531–537}	IIa	B
LMWH should be considered in patients with active cancer ^e and AF who are not suitable for NOAC. ^{f,525}	IIa	C
LAA occlusion may be considered for stroke prevention in patients with cancer with AF and contraindications for long-term anticoagulation with a life expectancy >12 months. ^{273,539}	IIb	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer. ²⁷³	III	C
Heart rate control strategy, preferably with beta-blockers, should be considered in patients who develop well-tolerated AF while they are receiving active cancer treatment ^g .	IIa	C

© ESC 2022

5-FU, 5-fluorouracil; AF, atrial fibrillation; BMI, body mass index; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; HF, heart failure; LAA, left atrial appendage; LMWH, low-molecular-weight heparins; LV, left ventricular; MM, multiple myeloma; NOAC, non-vitamin K antagonist oral anticoagulants; VKA, vitamin K antagonists.

^aClass of recommendation.

^bLevel of evidence.

^cFactors that may increase thromboembolic risk in patients with cancer including comorbidities (proteinuria > 150 mg/24 h, eGFR < 45 mL/min/1.73 m², BMI ≥ 30 kg/m², thrombophilia), cancer type (pancreatic, gastric, ovarian, brain, lung, MM), cancer stage (metastatic disease) anticancer therapies: alkylating agents, aflibercept, bevacizumab, anthracyclines, capecitabine, 5-FU, gemcitabine, methotrexate, EGFR inhibitors, bleomycin, axitinib, lenvatinib, pazopanib, sorafenib, sunitinib, carfilzomib, irinotecan, taxanes, tasonermin, tretinoin.

^dStroke and bleeding risk may change during both cancer treatment and the course of the

underlying disease; reassessment is important to inform treatment decisions and address potentially modifiable bleeding risk factors.

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

^fHigh bleeding risk, severe renal dysfunction (CrCl < 15 mL/min); NOAC major drug–drug interactions.

^gAsymptomatic or mild symptomatic patients without HF signs or symptoms or deterioration of LV function. The optimal heart rate target in AF patients is unclear. A resting heart rate <110 bpm (i.e. lenient rate control) should be considered as the initial heart rate target for rate control therapy. A review of rate vs. rhythm strategy should be made at the end of cancer treatment.²⁷³

Recommendations	Class ^a	Level ^b
CHA ₂ DS ₂ -VASc score should be considered for risk stratification for stroke/systemic thromboembolism taking into account that it may underestimate the actual thromboembolic risk. ^{519,526}	IIa	C
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥2 (men) or ≥3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	I	C

Long-term anticoagulation should be considered for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score = 1 (men) or = 2 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	IIa	C
Patients with cancer, ^c AF, and CHA ₂ DS ₂ -VASc score 0 (men) or 1 (women) may have a higher thrombotic risk than patients without cancer and may be considered for therapeutic anticoagulation after consideration of the bleeding risk. ⁵²⁶	IIb	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF. ^{d,273}	I	C
NOAC should be considered for stroke prevention in preference to LMWH and VKA (excluding patients with mechanical heart valves or moderate-to-severe mitral stenosis) in patients without a high bleeding risk, significant drug–drug interactions, or severe renal dysfunction. ^{531–537}	IIa	B
LMWH should be considered in patients with active cancer ^e and AF who are not suitable for NOAC. ^{f,525}	IIa	C
LAA occlusion may be considered for stroke prevention in patients with cancer with AF and contraindications for long-term anticoagulation with a life expectancy >12 months. ^{273,539}	IIb	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer. ²⁷³	III	C
Heart rate control strategy, preferably with beta-blockers, should be considered in patients who develop well-tolerated AF while they are receiving active cancer treatment ^g .	IIa	C

© ESC 2022

5-FU, 5-fluorouracil; AF, atrial fibrillation; BMI, body mass index; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; HF, heart failure; LAA, left atrial appendage; LMWH, low-molecular-weight heparins; LV, left ventricular; MM, multiple myeloma; NOAC, non-vitamin K antagonist oral anticoagulants; VKA, vitamin K antagonists.

^aClass of recommendation.

^bLevel of evidence.

^cFactors that may increase thromboembolic risk in patients with cancer including comorbidities (proteinuria > 150 mg/24 h, eGFR < 45 mL/min/1.73 m², BMI ≥ 30 kg/m², thrombophilia), cancer type (pancreatic, gastric, ovarian, brain, lung, MM), cancer stage (metastatic disease) anticancer therapies: alkylating agents, aflibercept, bevacizumab, anthracyclines, capecitabine, 5-FU, gemcitabine, methotrexate, EGFR inhibitors, bleomycin, axitinib, lenvatinib, pazopanib, sorafenib, sunitinib, carfilzomib, irinotecan, taxanes, tasonermin, tretinoin.

^dStroke and bleeding risk may change during both cancer treatment and the course of the

underlying disease; reassessment is important to inform treatment decisions and address potentially modifiable bleeding risk factors.

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

^fHigh bleeding risk, severe renal dysfunction (CrCl < 15 mL/min); NOAC major drug–drug interactions.

^gAsymptomatic or mild symptomatic patients without HF signs or symptoms or deterioration of LV function. The optimal heart rate target in AF patients is unclear. A resting heart rate <110 bpm (i.e. lenient rate control) should be considered as the initial heart rate target for rate control therapy. A review of rate vs. rhythm strategy should be made at the end of cancer treatment.²⁷³

Recommendation Table 31

[Open in new tab](#)

Recommendations for the management of atrial fibrillation in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
CHA ₂ DS ₂ -VASc score should be considered for risk stratification for stroke/systemic thromboembolism taking into account that it may underestimate the actual thromboembolic risk. ^{519,526}	IIa	C
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥2 (men) or ≥3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	I	C

Long-term anticoagulation should be considered for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score = 1 (men) or = 2 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	IIa	C
Patients with cancer, ^c AF, and CHA ₂ DS ₂ -VASc score 0 (men) or 1 (women) may have a higher thrombotic risk than patients without cancer and may be considered for therapeutic anticoagulation after consideration of the bleeding risk. ⁵²⁶	IIb	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF. ^{d,273}	I	C
NOAC should be considered for stroke prevention in preference to LMWH and VKA (excluding patients with mechanical heart valves or moderate-to-severe mitral stenosis) in patients without a high bleeding risk, significant drug–drug interactions, or severe renal dysfunction. ^{531–537}	IIa	B
LMWH should be considered in patients with active cancer ^e and AF who are not suitable for NOAC. ^{f,525}	IIa	C
LAA occlusion may be considered for stroke prevention in patients with cancer with AF and contraindications for long-term anticoagulation with a life expectancy >12 months. ^{273,539}	IIb	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer. ²⁷³	III	C
Heart rate control strategy, preferably with beta-blockers, should be considered in patients who develop well-tolerated AF while they are receiving active cancer treatment ^g .	IIa	C

© ESC 2022

5-FU, 5-fluorouracil; AF, atrial fibrillation; BMI, body mass index; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; HF, heart failure; LAA, left atrial appendage; LMWH, low-molecular-weight heparins; LV, left ventricular; MM, multiple myeloma; NOAC, non-vitamin K antagonist oral anticoagulants; VKA, vitamin K antagonists.

^aClass of recommendation.

^bLevel of evidence.

^cFactors that may increase thromboembolic risk in patients with cancer including comorbidities (proteinuria > 150 mg/24 h, eGFR < 45 mL/min/1.73 m², BMI ≥ 30 kg/m², thrombophilia), cancer type (pancreatic, gastric, ovarian, brain, lung, MM), cancer stage (metastatic disease) anticancer therapies: alkylating agents, aflibercept, bevacizumab, anthracyclines, capecitabine, 5-FU, gemcitabine, methotrexate, EGFR inhibitors, bleomycin, axitinib, lenvatinib, pazopanib, sorafenib, sunitinib, carfilzomib, irinotecan, taxanes, tasonermin, tretinoin.

^dStroke and bleeding risk may change during both cancer treatment and the course of the

underlying disease; reassessment is important to inform treatment decisions and address potentially modifiable bleeding risk factors.

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

^fHigh bleeding risk, severe renal dysfunction (CrCl < 15 mL/min); NOAC major drug–drug interactions.

^gAsymptomatic or mild symptomatic patients without HF signs or symptoms or deterioration of LV function. The optimal heart rate target in AF patients is unclear. A resting heart rate <110 bpm (i.e. lenient rate control) should be considered as the initial heart rate target for rate control therapy. A review of rate vs. rhythm strategy should be made at the end of cancer treatment.²⁷³

Recommendations	Class ^a	Level ^b
CHA ₂ DS ₂ -VASc score should be considered for risk stratification for stroke/systemic thromboembolism taking into account that it may underestimate the actual thromboembolic risk. ^{519,526}	IIa	C
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥2 (men) or ≥3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	I	C

Long-term anticoagulation should be considered for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score = 1 (men) or = 2 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. ²⁷³	IIa	C
Patients with cancer, ^c AF, and CHA ₂ DS ₂ -VASc score 0 (men) or 1 (women) may have a higher thrombotic risk than patients without cancer and may be considered for therapeutic anticoagulation after consideration of the bleeding risk. ⁵²⁶	IIb	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF. ^{d,273}	I	C
NOAC should be considered for stroke prevention in preference to LMWH and VKA (excluding patients with mechanical heart valves or moderate-to-severe mitral stenosis) in patients without a high bleeding risk, significant drug–drug interactions, or severe renal dysfunction. ^{531–537}	IIa	B
LMWH should be considered in patients with active cancer ^e and AF who are not suitable for NOAC. ^{f,525}	IIa	C
LAA occlusion may be considered for stroke prevention in patients with cancer with AF and contraindications for long-term anticoagulation with a life expectancy >12 months. ^{273,539}	IIb	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer. ²⁷³	III	C
Heart rate control strategy, preferably with beta-blockers, should be considered in patients who develop well-tolerated AF while they are receiving active cancer treatment ^g .	IIa	C

© ESC 2022

5-FU, 5-fluorouracil; AF, atrial fibrillation; BMI, body mass index; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; HF, heart failure; LAA, left atrial appendage; LMWH, low-molecular-weight heparins; LV, left ventricular; MM, multiple myeloma; NOAC, non-vitamin K antagonist oral anticoagulants; VKA, vitamin K antagonists.

^aClass of recommendation.

^bLevel of evidence.

^cFactors that may increase thromboembolic risk in patients with cancer including comorbidities (proteinuria > 150 mg/24 h, eGFR < 45 mL/min/1.73 m², BMI ≥ 30 kg/m², thrombophilia), cancer type (pancreatic, gastric, ovarian, brain, lung, MM), cancer stage (metastatic disease) anticancer therapies: alkylating agents, aflibercept, bevacizumab, anthracyclines, capecitabine, 5-FU, gemcitabine, methotrexate, EGFR inhibitors, bleomycin, axitinib, lenvatinib, pazopanib, sorafenib, sunitinib, carfilzomib, irinotecan, taxanes, tasonermin, tretinoin.

^dStroke and bleeding risk may change during both cancer treatment and the course of the

underlying disease; reassessment is important to inform treatment decisions and address potentially modifiable bleeding risk factors.

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

^fHigh bleeding risk, severe renal dysfunction (CrCl < 15 mL/min); NOAC major drug–drug interactions.

^gAsymptomatic or mild symptomatic patients without HF signs or symptoms or deterioration of LV function. The optimal heart rate target in AF patients is unclear. A resting heart rate <110 bpm (i.e. lenient rate control) should be considered as the initial heart rate target for rate control therapy. A review of rate vs. rhythm strategy should be made at the end of cancer treatment.²⁷³

6.4.2. Long corrected QT interval and ventricular arrhythmias

VA are not common during cancer, although their incidence increases in patients with advanced cancer and CV comorbidities.^{49,259,516,542} Mechanisms proposed to explain cancer therapy-induced VA include: (1) direct effects of cancer drugs on the activity/expression of ionic channels that regulate the ventricular action potential,^{4,369,442,516,542,543} and (2) a permanent arrhythmogenic substrate created by cancer and systemic inflammation caused by cancer, pre-existing CV comorbidities, and/or a new CTR-CVT.^{4,9,259,369,442,516,542,543}

Treatment of cancer therapy-induced VA should follow general clinical guidelines.^{22,442,544} In patients with asymptomatic self-terminating VA, drug discontinuation is not required unless they have additional CVRF or persistent ECG abnormalities.²⁷⁰ Symptomatic VA require cancer drug dose reduction or discontinuation and patients should be referred to the cardiologist for evaluation and treatment.^{4,442}

Recurrent symptomatic life-threatening VA require urgent intervention.^{4,270,442,544} The administration of class IA, IC, and III anti-arrhythmic drugs is limited by the risk of drug–drug interactions and QTc prolongation. Beta-blockers and class IB drugs are less likely to cause drug interactions or QTc prolongation. Beta-blockers are the preferred choice if the cancer drug is also associated with CTRCD. Amiodarone is the antiarrhythmic drug of choice in patients with structural heart disease and haemodynamic instability. Decisions on the use of antiarrhythmic drugs or device therapy (cardioverter defibrillators, catheter ablation) should consider life expectancy, quality of life, and complication risks.³⁴⁹

Most cancer therapy-induced VA are related to a prolongation of QTc leading to the development of TdP.^{259,516,542} Risk factors for QTc prolongation and TdP are summarized in [TableÂ 8](#).^{4,22,45,48,516,543}

TableÂ 8

[Open in new tab](#)

Risk factors for drug-induced QT prolongation and torsade de pointes

Correctable	Non-correctable
QT-prolonging drugs ^a • Antiarrhythmics • Antibiotics • Antidepressants	Acute myocardial ischaemia Age > 65 years Baseline QTc interval prolongation ^b Family history of sudden death (congenital LQTS or genetic polymorphism)

• Antifungals • Antiemetics • Antihistamines • Antipsychotics • Loop diuretics • Opioids (methadone) Bradyarrhythmia Electrolyte imbalance/abnormalities • Hypokalaemia ($\hat{a}_{HTJ} \square 3.5 \hat{a}_{PADNEL}$ mEq/L) • Hypomagnesaemia ($\hat{a}_{HTJ} \square 1.6 \hat{a}_{PADNEL}$ mEq/L) • Hypocalcaemia ($\hat{a}_{HTJ} \square 8.5 \hat{a}_{PADNEL}$ mEq/L) Inadequate dose adjustment of renal or hepatic cleared QT-prolonging drugs	Female sex Impaired renal function (for renally excreted drugs) Liver disease (for hepatically excreted drugs) Personal history of syncope or drug-induced TdP Pre-existing CVD (CAD, HF, LV hypertrophy)
--	--

Correctable	Non-correctable
QT-prolonging drugs^a • Antiarrhythmics • Antibiotics • Antidepressants • Antifungals • Antiemetics • Antihistamines • Antipsychotics • Loop diuretics • Opioids (methadone) Bradyarrhythmia Electrolyte imbalance/abnormalities • Hypokalaemia ($\hat{a}_{HTJ} \square 3.5 \hat{a}_{PADNEL}$ mEq/L)	Acute myocardial ischaemia Age > 65 years Baseline QTc interval prolongation^b Family history of sudden death (congenital LQTS or genetic polymorphism) Female sex Impaired renal function (for renally excreted drugs) Liver disease (for hepatically excreted drugs) Personal history of syncope or drug-induced TdP Pre-existing CVD (CAD, HF, LV hypertrophy)

- Hypomagnesaemia ($\hat{a}_{HTJ} \square 1.6 \hat{a}_{PADNEL}$ mEq/L)

- Hypocalcaemia ($\hat{a}_{HTJ} \square 8.5 \hat{a}_{PADNEL}$ mEq/L)

Inadequate dose adjustment of renal or hepatic cleared QT-prolonging drugs

CAD, coronary artery disease; CVD, cardiovascular disease; HF, heart failure; LQTS, long QT syndrome; LV, left ventricular; QTc, corrected QT interval; TdP, torsade de pointes.

a

See <https://www.crediblemeds.org>.

b

QTc using Fridericia correction ($QTcF = QT/3 \hat{a}_{HTJSCI} RR$) is recommended in patients with cancer.

Table 8

[Open in new tab](#)

Risk factors for drug-induced QT prolongation and torsade de pointes

Correctable	Non-correctable
QT-prolonging drugs ^a • Antiarrhythmics • Antibiotics • Antidepressants • Antifungals • Antiemetics • Antihistamines • Antipsychotics • Loop diuretics • Opioids (methadone) Bradyarrhythmia Electrolyte imbalance/abnormalities • Hypokalaemia ($\hat{a}_{HTJ} \square 3.5 \hat{a}_{PADNEL}$ mEq/L) • Hypomagnesaemia ($\hat{a}_{HTJ} \square 1.6 \hat{a}_{PADNEL}$ mEq/L) • Hypocalcaemia ($\hat{a}_{HTJ} \square 8.5 \hat{a}_{PADNEL}$ mEq/L)	Acute myocardial ischaemia Age > 65 years Baseline QTc interval prolongation ^b Family history of sudden death (congenital LQTS or genetic polymorphism) Female sex Impaired renal function (for renally excreted drugs) Liver disease (for hepatically excreted drugs) Personal history of syncope or drug-induced TdP Pre-existing CVD (CAD, HF, LV hypertrophy)

Inadequate dose adjustment of renal or hepatic cleared QT-prolonging drugs	
--	--

Correctable	Non-correctable
QT-prolonging drugs ^a • Antiarrhythmics • Antibiotics • Antidepressants • Antifungals • Antiemetics • Antihistamines • Antipsychotics • Loop diuretics • Opioids (methadone) Bradyarrhythmia Electrolyte imbalance/abnormalities • Hypokalaemia ($\hat{a}_{HTJ} \square 3.5 \hat{a}_{PADNEL}$ mEq/L) • Hypomagnesaemia ($\hat{a}_{HTJ} \square 1.6 \hat{a}_{PADNEL}$ mEq/L) • Hypocalcaemia ($\hat{a}_{HTJ} \square 8.5 \hat{a}_{PADNEL}$ mEq/L) Inadequate dose adjustment of renal or hepatic cleared QT-prolonging drugs	Acute myocardial ischaemia Age > 65 years Baseline QTc interval prolongation ^b Family history of sudden death (congenital LQTS or genetic polymorphism) Female sex Impaired renal function (for renally excreted drugs) Liver disease (for hepatically excreted drugs) Personal history of syncope or drug-induced TdP Pre-existing CVD (CAD, HF, LV hypertrophy)

CAD, coronary artery disease; CVD, cardiovascular disease; HF, heart failure; LQTS, long QT syndrome; LV, left ventricular; QTc, corrected QT interval; TdP, torsade de pointes.

a

See <https://www.crediblemeds.org>.

b

QTc using Fridericia correction ($QTcF = QT / \sqrt[3]{\hat{a}_{HTJ} \square RR}$) is recommended in patients with cancer.

The upper 99% limits of normal for QTc values in the general population are $450 \hat{a}_{PADNEL}$ ms for men and $460 \hat{a}_{PADNEL}$ ms for women.⁵⁴⁵ Although there is no threshold of QTc prolongation at which TdP can occur, a QTc $\hat{a}_{HTJ} \square 500 \hat{a}_{PADNEL}$ ms is associated with a two- to three-fold higher risk for TdP, while TdP rarely occurs

when QTc is <500 ms.⁴⁴² Although the incidence of QTc prolongation ≥500 ms and TdP is low during cancer therapy, QTc prolongation to levels that require closer monitoring (QTc ≥480 ms) is more common (Table 9).^{4,9,22,45,48,49,259,369,516,543,546} Changes in the QT interval of >60 ms from baseline should not routinely affect treatment decisions if the QTc remains <500 ms.¹ Cardiology consultation is advised in patients with an abnormal baseline QTc interval, patients treated with drugs that prolong the QT interval, those who develop new cardiac symptoms (syncope or pre-syncope, rapid palpitations or QTc prolongation with new-onset bradycardia, high degree of heart block), and/or those with known inherited arrhythmia disorders.^{4,45,48,442,544} The challenges for the cardio-oncology teams are to identify patients more susceptible to developing VA, determine whether a VA is directly due to CTR-CVT, individualize the treatment strategy, and optimize clinical monitoring during treatment.

Table 9

[Open in new tab](#)

Classification of corrected QT interval prolongation induced by cancer drug therapy

Classification	Drugs
High risk: QTcF prolongation ≥10 ms and risk of TdP	• Aclarubicin • Arsenic trioxide • Glasdegib • Nilotinib • Oxaliplatin • Pazopanib • Ribociclib • Sunitinib • Toremifene • Vandetanib
Moderate risk: QTcF prolongation ≥10 ms and low or no risk of TdP (or uncertain)	• Abarelix • Belinostat • Brigantini • Carbozantinib • Ceritinib • Crizotinib

	• Dovitinib • Entrectinib • Eribulin • Gilteritinib • Ivosidenib • Lapatinib • Lenvatinib • Osimertinib • Panobinostat • Rucaparib • Selpercatinib • Sorafenib • Tipiracil/trifluridine • Vemurafenib
Low risk: QTcF prolongation < 10a3ms ^a	• ADT • Afatinib • Axitinib • Binimetinib • Bortezomib • Bosutinib • Carfilzomib • Dabrafenib • Dasatinib • Encorafenib • Midostaurin • Pertuzumab

	• Ponatinib • Romidepsin • Quizartinib • Tamoxifen • Vorinostat
--	---

Classification	Drugs
High risk: QTcF prolongation ≥ 10 ms and risk of TdP	• Aclarubicin • Arsenic trioxide • Glasdegib • Nilotinib • Oxaliplatin • Pazopanib • Ribociclib • Sunitinib • Toremifene • Vandetanib
Moderate risk: QTcF prolongation ≥ 10 ms and low or no risk of TdP (or uncertain)	• Abarelix • Belinostat • Brigantiniib • Carbozantinib • Ceritinib • Crizotinib • Dovitinib • Entrectinib • Eribulin

	• Gilteritinib • Ivosidenib • Lapatinib • Lenvatinib • Osimertinib • Panobinostat • Rucaparib • Selpercatinib • Sorafenib • Tipiracil/trifluridine • Vemurafenib
Low risk: QTcF prolongation < 10aFADNELms ^a	• ADT • Afatinib • Axitinib • Binimetinib • Bortezomib • Bosutinib • Carfilzomib • Dabrafenib • Dasatinib • Encorafenib • Midostaurin • Pertuzumab • Ponatinib • Romidepsin • Quizartinib

	• Tamoxifen • Vorinostat
--	---

ADT, androgen deprivation therapy; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.

a
ADT may prolong the QTc interval (GnRH agonist, GnRH antagonist, bicalutamide, flutamide, apalutamide, darolutamide, enzalutamide, and abiraterone) (see [FigureÂ 21](#)). Developed from EMA prescribing information,^{[252](#)} FDA prescribing information,^{[253](#)} and AZCERT.^{[547](#)}

TableÂ 9

[Open in new tab](#)

Classification of corrected QT interval prolongation induced by cancer drug therapy

Classification	Drugs
High risk: QTcF prolongation ≥ 10 ms and risk of TdP	• Aclarubicin • Arsenic trioxide • Glasdegib • Nilotinib • Oxaliplatin • Pazopanib • Ribociclib • Sunitinib • Toremifene • Vandetanib
Moderate risk: QTcF prolongation ≥ 10 ms and low or no risk of TdP (or uncertain)	• Abarelix • Belinostat • Brigantini • Carbozantinib • Ceritinib • Crizotinib • Dovitinib

	• Entrectinib • Eribulin • Gilteritinib • Ivosidenib • Lapatinib • Lenvatinib • Osimertinib • Panobinostat • Rucaparib • Selpercatinib • Sorafenib • Tipiracil/trifluridine • Vemurafenib
Low risk: QTcF prolongation < 10a ^a ms	• ADT • Afatinib • Axitinib • Binimetinib • Bortezomib • Bosutinib • Carfilzomib • Dabrafenib • Dasatinib • Encorafenib • Midostaurin • Pertuzumab • Ponatinib

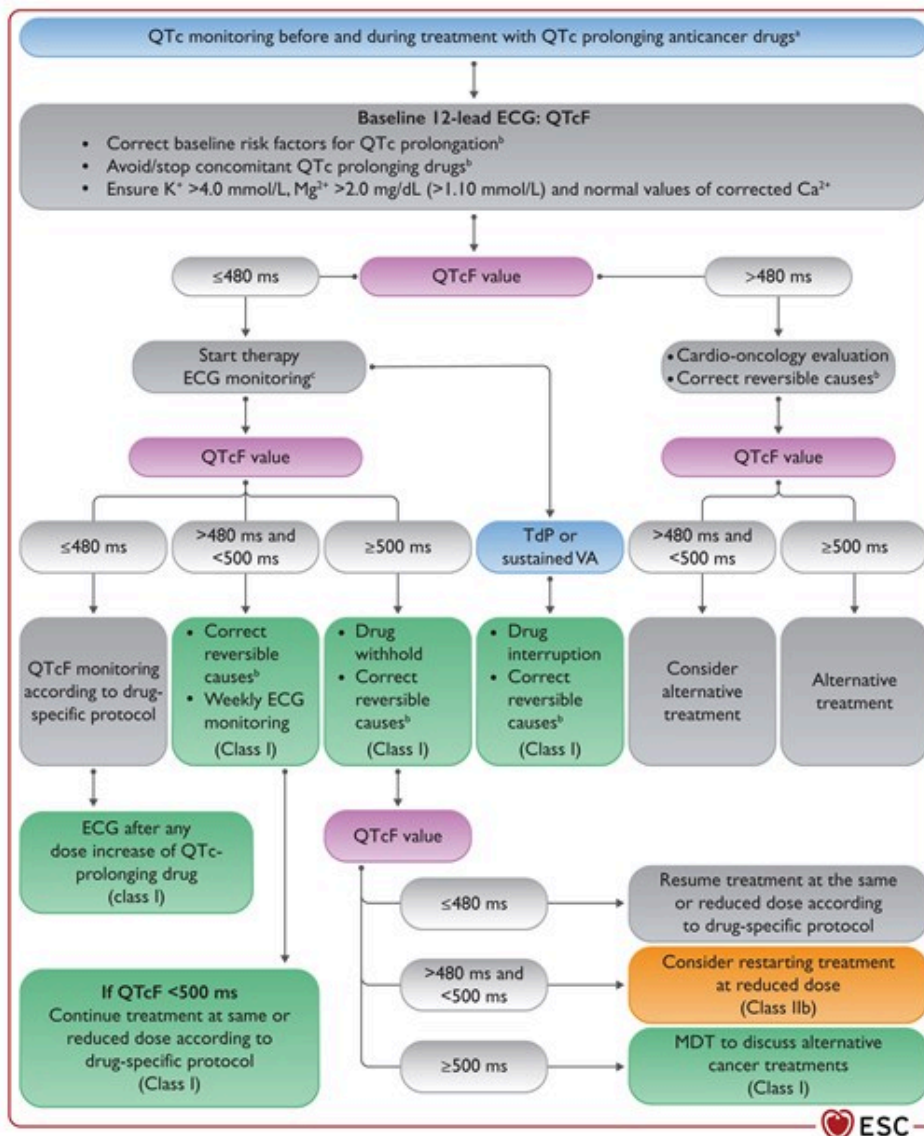
	• Romidepsin • Quizartinib • Tamoxifen • Vorinostat
Classification	Drugs
High risk: QTcF prolongation $\geq 10a_{\text{HTJ}} \geq 10a_{\text{PADNEL}}$ ms and risk of TdP	• Aclarubicin • Arsenic trioxide • Glasdegib • Nilotinib • Oxaliplatin • Pazopanib • Ribociclib • Sunitinib • Toremifene • Vandetanib
Moderate risk: QTcF prolongation $\geq 10a_{\text{HTJ}} \geq 10a_{\text{PADNEL}}$ ms and low or no risk of TdP (or uncertain)	• Abarelix • Belinostat • Brigantiniib • Carbozantinib • Ceritinib • Crizotinib • Dovitinib • Entrectinib • Eribulin • Gilteritinib

	• Ivosidenib • Lapatinib • Lenvatinib • Osimertinib • Panobinostat • Rucaparib • Selpercatinib • Sorafenib • Tipiracil/trifluridine • Vemurafenib
Low risk: QTcF prolongation < 10a ^a ms	• ADT • Afatinib • Axitinib • Binimetinib • Bortezomib • Bosutinib • Carfilzomib • Dabrafenib • Dasatinib • Encorafenib • Midostaurin • Pertuzumab • Ponatinib • Romidepsin • Quizartinib • Tamoxifen

ADT, androgen deprivation therapy; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.

a

ADT may prolong the QTc interval (GnRH agonist, GnRH antagonist, bicalutamide, flutamide, apalutamide, darolutamide, enzalutamide, and abiraterone) (see [FigureÂ 21](#)). Developed from EMA prescribing information,²⁵² FDA prescribing information,²⁵³ and AZCERT.⁵⁴⁷ [FigureÂ 32](#) shows the algorithm for the management of QTc prolongation during cancer therapy. In patients with cancer, the Fridericia formula is recommended and has demonstrated less error than other correction methods such as Bazzett at both high and low heart rates.⁴⁴ In patients treated with QTc-prolonging drugs, serum electrolytes and other risk factors should be closely monitored and corrected, and concomitant QT-prolonging drugs avoided if possible.^{4,22,45,369,543} For selected cancer drugs, there are specific manufacturer recommendations for ECG monitoring during treatment, dosage adjustments, or discontinuation of therapy in case of QTc prolongation.⁵⁴⁸



FigureÂ 32

Corrected QT interval monitoring before and during treatment with corrected QT interval-prolonging anticancer drugs.

Ca²⁺, calcium; ECG, electrocardiogram; K⁺, potassium; MDT, multidisciplinary team; Mg²⁺, magnesium; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TdP, Torsade de pointes; VA, ventricular arrhythmias. QT interval using Fridericia correction ($QTcF = QT / \sqrt[3]{RR}$) is recommended in patients with cancer. Upper 99% limits of normal for QTc values in the general population are 450 ms for men and 460 ms for women.³⁶⁹ ^a [Table 9](#). ^b [Table 8](#) and <https://www.crediblemeds.org>. ^cECG monitoring at baseline, once steady-state anticancer drug levels have been achieved, after each dose modification, or any treatment interruption >2 weeks; monthly for the first 3 months, and then periodically during treatment depending on patient-specific risk factors and cancer treatment.

[Open in new tab](#)[Download slide](#)

Although there are no recommendations, patients with cancer with QTc prolongation associated with severe bradycardia or sinus pauses may benefit from isoprenaline infusion or temporary pacing. Despite present restrictions, the improved prognosis for many malignancies is increasing the number of patients with cancer who are candidates for an implantable cardioverter defibrillator (ICD), particularly when life expectancy is >1 year (including patients who experienced resuscitated sudden cardiac death or severe VA from a QTc-prolonging drug with no alternative treatment available).

Recommendation Table 32

[Open in new tab](#)

Recommendations for the management of long corrected QT interval and ventricular arrhythmias in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment. ⁵⁴⁹	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation. ⁵⁴⁹	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors ^c is recommended in patients with cancer who develop QTcF ≥ 500 ms. ^{349,442,546}	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy. ^{349,442,546}	I	C
A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy. ^{270,442,544}	I	C

Restarting QTc-prolonging cancer therapy

A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.^{4,22,259,349,442,546}

I

C

In patients who experienced significant QTcF prolongation, restarting the culprit QTc-prolonging cancer treatment may be considered, ideally at a reduced dose according to each drug recommendation.^{45,259,349,442,546,549}

IIb

C

Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.⁵⁴⁹

I

C

© ESC 2022

ECG, electrocardiogram; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.

^aClass of recommendation.

^bLevel of evidence.

^cSee <https://www.crediblemeds.org> and [Table 8](#).

Recommendations	Class ^a	Level ^b
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment. ⁵⁴⁹	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation. ⁵⁴⁹	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors ^c is recommended in patients with cancer who develop QTcF ≥ 500 ms. ^{349,442,546}	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy. ^{349,442,546}	I	C
A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy. ^{270,442,544}	I	C

Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments. ^{4,22,259,349,442,546}	I	C
In patients who experienced significant QTcF prolongation, restarting the culprit QTc-prolonging cancer treatment may be considered, ideally at a reduced dose according to each drug recommendation. ^{45,259,349,442,546,549}	IIb	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy. ⁵⁴⁹	I	C

ECG, electrocardiogram; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.
^aClass of recommendation.
^bLevel of evidence.
^cSee <https://www.crediblemeds.org> and [Table 8](#).

© ESC 2022

Recommendation Table 32

[Open in new tab](#)

Recommendations for the management of long corrected QT interval and ventricular arrhythmias in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment. ⁵⁴⁹	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation. ⁵⁴⁹	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors ^c is recommended in patients with cancer who develop QTcF ≥ 500 ms. ^{349,442,546}	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy. ^{349,442,546}	I	C
A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy. ^{270,442,544}	I	C

Restarting QTc-prolonging cancer therapy

A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.^{4,22,259,349,442,546}

I

C

In patients who experienced significant QTcF prolongation, restarting the culprit QTc-prolonging cancer treatment may be considered, ideally at a reduced dose according to each drug recommendation.^{45,259,349,442,546,549}

IIb

C

Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.⁵⁴⁹

I

C

© ESC 2022

ECG, electrocardiogram; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.

^aClass of recommendation.

^bLevel of evidence.

^cSee <https://www.crediblemeds.org> and [Table 8](#).

Recommendations	Class ^a	Level ^b
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment. ⁵⁴⁹	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation. ⁵⁴⁹	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors ^c is recommended in patients with cancer who develop QTcF ≥ 500 ms. ^{349,442,546}	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy. ^{349,442,546}	I	C
A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy. ^{270,442,544}	I	C

Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments. ^{4,22,259,349,442,546}	I	C
In patients who experienced significant QTcF prolongation, restarting the culprit QTc-prolonging cancer treatment may be considered, ideally at a reduced dose according to each drug recommendation. ^{45,259,349,442,546,549}	IIb	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy. ⁵⁴⁹	I	C

ECG, electrocardiogram; QTc, corrected QT interval; QTcF, corrected QT interval using Fridericia correction; TdP, torsade de pointes.
^aClass of recommendation.
^bLevel of evidence.
^cSee <https://www.crediblemeds.org> and [Table 8](#).

© ESC 2022

6.4.3. Bradyarrhythmias

AV conduction disease can be caused by ICI in the presence or absence of myocarditis. If the PR interval increases (new first-degree heart block) in patients treated with ICI, serial ECG monitoring is recommended, and if PR prolongation to >300 ms develops, the patient should be hospitalized under close ECG monitoring and i.v. methylprednisolone is recommended.⁵⁵⁰

IMiD (thalidomide, pomalidomide)²⁸⁵ and ALK inhibitors (crizotinib, alectinib, brigatinib, or ceritinib)⁵⁵¹ are associated with sinus bradycardia. Holter ECG monitoring is recommended to exclude significant sinus pauses in symptomatic patients. In asymptomatic patients with normal LV function, sinus bradycardia is usually well tolerated and treatment can continue. If patients are symptomatic (syncope, pre-syncope or reduced exercise tolerance from chronotropic incompetence) then a trial of cancer drug withdrawal to confirm causation with the symptoms is recommended. A MDT discussion is needed to analyse risks/benefits of alternative cancer therapies vs. restarting the culprit cancer therapy at a lower dose with heart rate monitoring. In selected cases, when no cancer treatment alternative is available, pacing is indicated.

6.5. Arterial hypertension

Arterial hypertension in patients with cancer may be caused by their cancer treatments (e.g. VEGFi, second- and third-generation BCR-ABL TKI, brigatinib, ibrutinib, fluoropyrimidines, cisplatin, abiraterone, bicalutamide, enzalutamide), non-cancer drugs (e.g. corticosteroids, non-steroidal anti-inflammatory drugs), and other factors including stress, pain, excessive alcohol consumption, renal impairment, untreated sleep apnoea, obesity, and reduced exercise.⁵⁵² In all patients with cancer with new hypertension assessment, correction of these other factors is important before considering interruption of a cancer treatment.

Untreated hypertension³⁴⁴ is a confirmed risk factor of HF during treatment with anthracyclines,⁵⁵³ ibrutinib,²⁶⁴ and VEGFi.⁵⁵⁴ Given that many of the cancer therapies that cause hypertension also cause CTRCD, treatment of hypertension with ACE-I or ARB as first-line therapy is recommended to reduce the risk of CTRCD. Combination therapy with an ACE-I or ARB and a dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg due to the more rapid onset of BP control with the combination compared with ACE-I/ARB monotherapy ([Figures 33 and 34](#)).

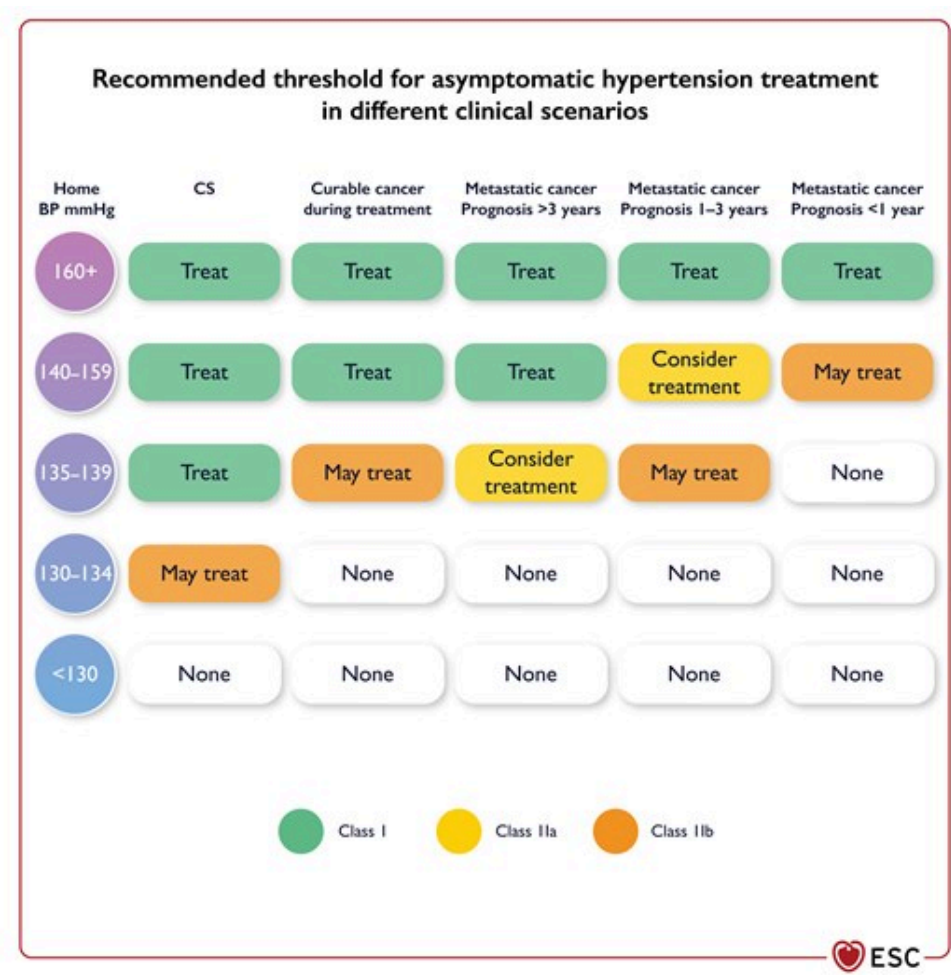


Figure 33
Recommended threshold for asymptomatic hypertension treatment in different clinical scenarios.

BP, blood pressure; CS, cancer survivors.

[Open in new tab](#)[Download slide](#)

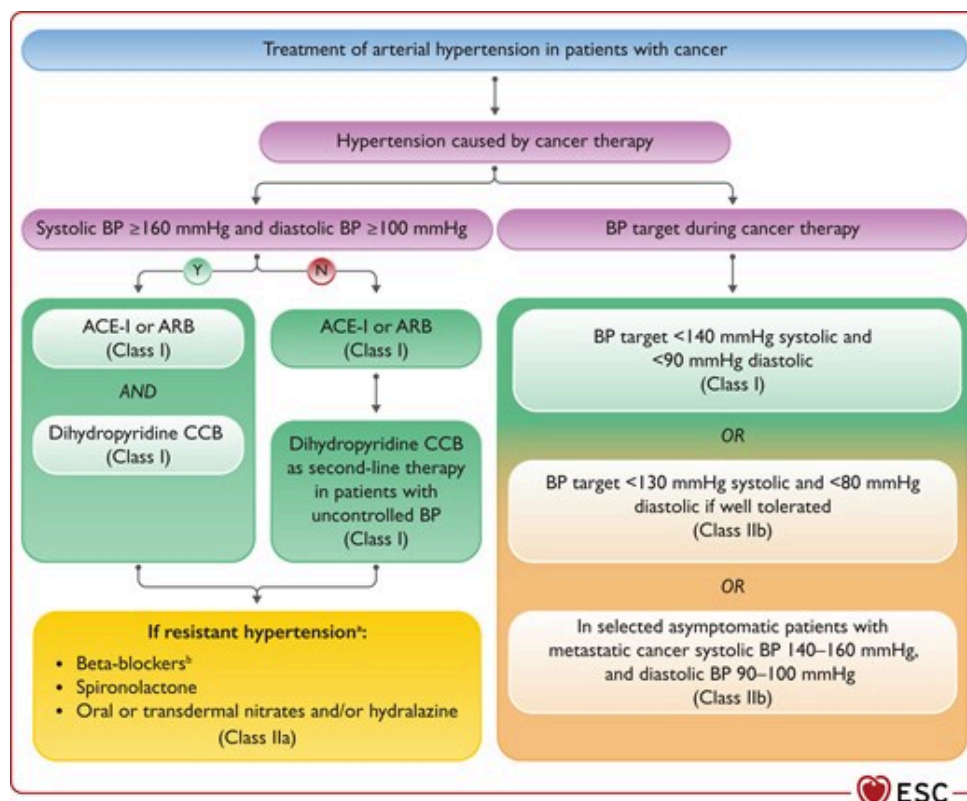


Figure 34
Treatment of arterial hypertension in patients with cancer.

AF, atrial fibrillation; ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BP, blood pressure; CCB, calcium channel blockers; HF, heart failure; MI, myocardial infarction; N, no; VEGFi, vascular endothelial growth factor inhibitors; Y, yes. ^aResistant hypertension is defined as BP being uncontrolled despite treatment with optimal or best-tolerated doses of three or more drugs including a diuretic, and confirmed by ambulatory and home BP monitoring. ^bConsider beta-blockers (nebivolol or carvedilol are preferred in patients on VEGFi) at any treatment step, when there is a specific indication for their use, e.g. HF, angina, post-MI, or AF.

[Open in new tab](#)[Download slide](#)

If severe hypertension is diagnosed (systolic BP ≥ 180 mmHg or diastolic BP ≥ 110 mmHg), the competing cancer and CV risks should be evaluated by a MDT, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic BP) and <100 mmHg (diastolic BP). Culprit cancer therapy can be restarted once BP is controlled, with consideration for dose reduction.

In patients with resistant cancer therapy-related hypertension, spironolactone, oral or transdermal nitrates, and/or hydralazine should be considered. In patients with cancer with evidence of high sympathetic tone, stress, and/or pain, beta-blockers including carvedilol or nebivolol should be considered. Diuretics, preferably spironolactone, may be considered in patients with cancer with hypertension and evidence of increased fluid retention, with monitoring of BP, electrolytes, and renal function.

The decisions to initiate BP treatment and BP targets during the management of cancer-drug induced hypertension depend upon the context of the cancer and prognosis ([Figure 34](#)). CS should be treated

according to the 2018 ESC/European Society of Hypertension (ESH) Guidelines for the management of arterial hypertension.¹³⁸

Recommendation Table 33

[Open in new tab](#)

Recommendations for the management of arterial hypertension in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
A BP target <130 mmHg systolic and <80 mmHg diastolic may be considered during cancer therapy provided that the treatment is well tolerated.	IIb	C
In selected asymptomatic patients with metastatic cancer, a systolic BP 140–160 mmHg and diastolic BP 90–100 mmHg treatment threshold may be considered provided there is ongoing BP monitoring.	IIb	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C

Cancer therapy-induced arterial hypertension treatment

ACE-I or ARB are the first-line antihypertensive drugs^c recommended for BP management in patients with cancer.^{555–557}

I

B

Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.

I

C

Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.

I

C

Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.^d

III

C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BP, blood pressure; CCB, calcium channel blockers; CV, cardiovascular.

^aClass of recommendation.

^bLevel of evidence.

^cUnless contraindicated.

^dIn selected patients with cancer, who are intolerant to multiple other antihypertensive drugs, diltiazem and verapamil may be considered with close monitoring of drug–drug interactions.

Recommendations	Class ^a	Level ^b
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
A BP target <130 mmHg systolic and <80 mmHg diastolic may be considered during cancer therapy provided that the treatment is well tolerated.	IIb	C
In selected asymptomatic patients with metastatic cancer, a systolic BP 140–160 mmHg and diastolic BP 90–100 mmHg treatment threshold may be considered provided there is ongoing BP monitoring.	IIb	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C

Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs ^c recommended for BP management in patients with cancer. ^{555–557}	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions. ^d	III	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BP, blood pressure; CCB, calcium channel blockers; CV, cardiovascular.

^aClass of recommendation.

^bLevel of evidence.

^cUnless contraindicated.

^dIn selected patients with cancer, who are intolerant to multiple other antihypertensive drugs, diltiazem and verapamil may be considered with close monitoring of drug–drug interactions.

Recommendation Table 33

[Open in new tab](#)

Recommendations for the management of arterial hypertension in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
A BP target <130 mmHg systolic and <80 mmHg diastolic may be considered during cancer therapy provided that the treatment is well tolerated.	IIb	C
In selected asymptomatic patients with metastatic cancer, a systolic BP 140–160 mmHg and diastolic BP 90–100 mmHg treatment threshold may be considered provided there is ongoing BP monitoring.	IIb	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C

Cancer therapy-induced arterial hypertension treatment

ACE-I or ARB are the first-line antihypertensive drugs^c recommended for BP management in patients with cancer.^{555–557}

I

B

Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.

I

C

Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.

I

C

Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.^d

III

C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BP, blood pressure; CCB, calcium channel blockers; CV, cardiovascular.

^aClass of recommendation.

^bLevel of evidence.

^cUnless contraindicated.

^dIn selected patients with cancer, who are intolerant to multiple other antihypertensive drugs, diltiazem and verapamil may be considered with close monitoring of drug–drug interactions.

Recommendations	Class ^a	Level ^b
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
A BP target <130 mmHg systolic and <80 mmHg diastolic may be considered during cancer therapy provided that the treatment is well tolerated.	IIb	C
In selected asymptomatic patients with metastatic cancer, a systolic BP 140–160 mmHg and diastolic BP 90–100 mmHg treatment threshold may be considered provided there is ongoing BP monitoring.	IIb	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C

Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs ^c recommended for BP management in patients with cancer. ^{555–557}	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP $\geq$ 160 mmHg and diastolic BP $\geq$ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions. ^d	III	C

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BP, blood pressure; CCB, calcium channel blockers; CV, cardiovascular.

^aClass of recommendation.

^bLevel of evidence.

^cUnless contraindicated.

^dIn selected patients with cancer, who are intolerant to multiple other antihypertensive drugs, diltiazem and verapamil may be considered with close monitoring of drug–drug interactions.

© ESC 2022

6.6. Thrombosis and thromboembolic events

Thromboembolic events that develop during cancer and its treatment encompass both VTE and arterial thromboembolism (ATE) and are collectively referred to as cancer-associated thrombosis. Cancer-associated thrombosis is determined by the prothrombotic milieu induced by cancer, the prothrombotic properties of certain anticancer and adjunctive therapies, and patient-related risk factors, including demographics, genetic predisposition, and comorbidities.⁵¹³

6.6.1. Venous thromboembolism

VTE, including deep vein thrombosis (DVT) and PE, is the second-leading cause of death in patients with malignancies.⁵⁵⁸ Cancer confers a five-fold higher risk of VTE and cancer-associated VTE represents 30% of all VTE cases.^{559,560} The risk of VTE varies in the course of cancer, with the highest risk occurring in the period following cancer diagnosis, during hospitalization and chemotherapy, and upon development of metastatic disease.^{561,562} Unprovoked VTE may be the first clinical sign of a malignancy, followed by a 5% incidence of cancer diagnosis during the subsequent 12 months.⁵⁶³

The risk factors for VTE in cancer are summarized in [FigureÂ 35](#).^{564,565} Patients with symptoms or signs suggestive of VTE, such as unilateral lower limb oedema or unexplained dyspnoea, should be screened with lower-extremity venous ultrasonography or contrast-enhanced CT for DVT and CT pulmonary angiography for PE, according to the 2019 ESC Guidelines for the diagnosis and management of acute pulmonary

embolism recommendations⁵⁶⁶ and the second consensus document on diagnosis and management of acute deep vein thrombosis.⁵⁶⁷

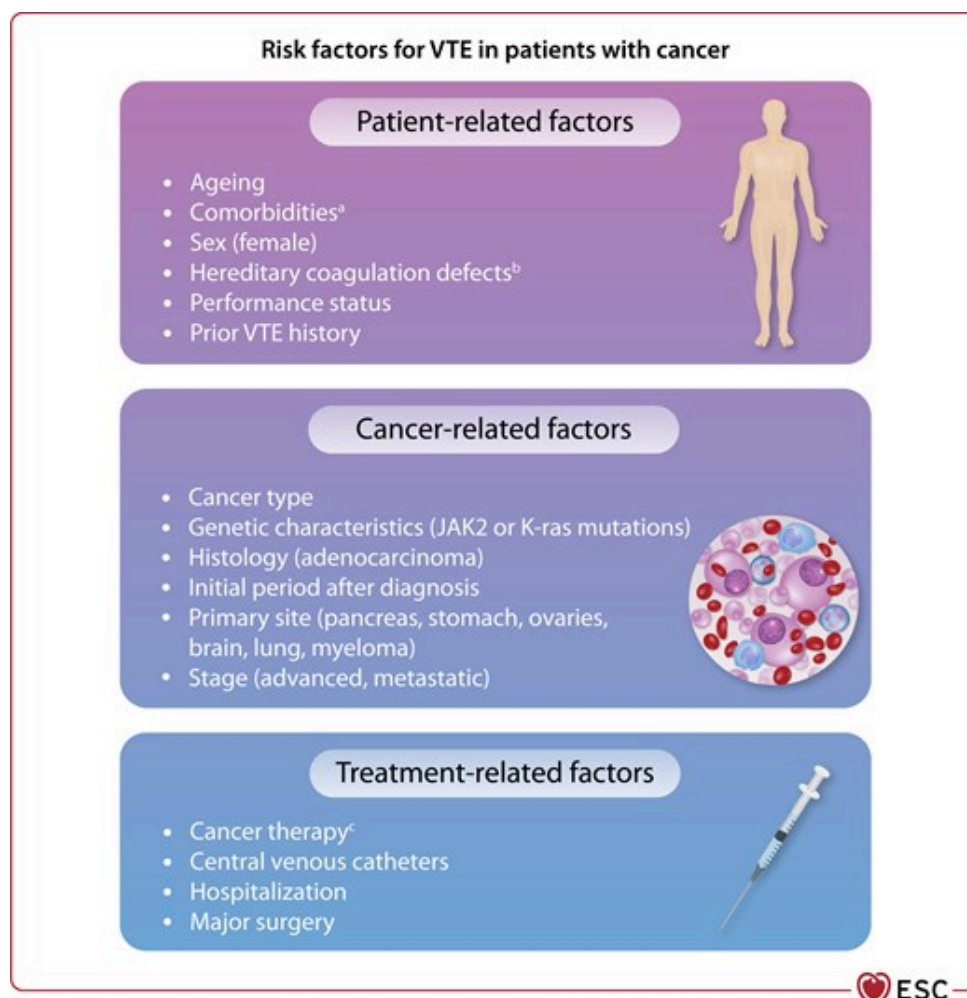


Figure 35

Risk factors for venous thromboembolism in patients with cancer.

ATE, arterial thromboembolism; BMI, body mass index; CrCl, creatinine clearance; IMiD, immunomodulatory drugs; PI, proteasome inhibitors; VTE, venous thromboembolism. ^aAcute infection, chronic kidney disease (CrCl < 45 mL/min), pulmonary disease, obesity (BMI ≥ 30 kg/m²), ATE. ^bFactor V Leiden, prothrombin gene mutation. ^cChemotherapy (carboplatin, cyclophosphamide, anthracyclines, antimetabolites, irinotecan, taxanes, tasonermin), anti-angiogenic agents (bevacizumab, axitinib, lenvatinib, pazopanib, sorafenib, sunitinib), IMiD (thalidomide, lenalidomide), PI (carfilzomib), hormonal therapy, erythropoiesis-stimulating agents.

[Open in new tab](#)[Download slide](#)

6.6.2. Arterial thromboembolism

Cancer carries a two-fold higher risk of ATE, including MI and ischaemic stroke.⁵⁶⁸ ATE risk is higher in men, with advanced age, and in patients with lung or kidney cancer. Pathologies related to ATE in cancer include ischaemic stroke induced by AF or RT-induced carotid artery disease, embolization by tumour cells or non-bacterial thrombotic endocarditis, disseminated intravascular coagulation-related peripheral microcirculatory thromboembolism, paradoxical cerebral embolism in the course of VTE, and cerebral sinus thrombosis.⁵⁶⁹

6.6.3. Intracardiac thrombosis

Intracardiac thrombus in patients with malignancies may result from the prothrombotic properties of cancer and its treatment and the use of central venous catheters. Thrombus is the most common intracardiac mass and it can occur within any cardiac chamber. Right atrial thrombi are often related to a venous catheter where the line has inappropriately advanced into the right atrium. Intraventricular thrombi usually occur in the setting of CTRCD. LAA thrombi are most commonly associated with AF, which may also be related to cancer or its therapy.

Patients with systemic embolization should be screened for cardiac origin of thrombus initially with TTE and/or transoesophageal echocardiography.⁵²⁸ CMR is more sensitive and specific than TTE for detecting intracardiac thrombi and late gadolinium enhancement (LGE) CMR with the long inversion time technique is currently considered the gold standard.^{570,571}

6.6.4. Anticoagulation therapy

Patients with cancer frequently have both an increased thrombotic risk and an increased bleeding risk associated with certain cancer locations (e.g. GI, intracranial), thrombocytopaenia, and other coagulation defects (secondary to bone marrow invasion, cancer therapies, or cancer itself) and associated comorbidities (e.g. renal or hepatic dysfunction, GI toxicities). Several anticancer agents are further characterized by drug^{PADSTS} drug interactions with anticoagulants. All these factors may render anticoagulation in cancer quite challenging. A proposed approach to anticoagulation therapy in cancer-associated venous thrombosis, based on the TBIP acronym (Thromboembolic risk, Bleeding risk, drug^{PADSTS} drug Interactions, Patient preferences), is outlined in [Figure 36](#).⁵²⁷

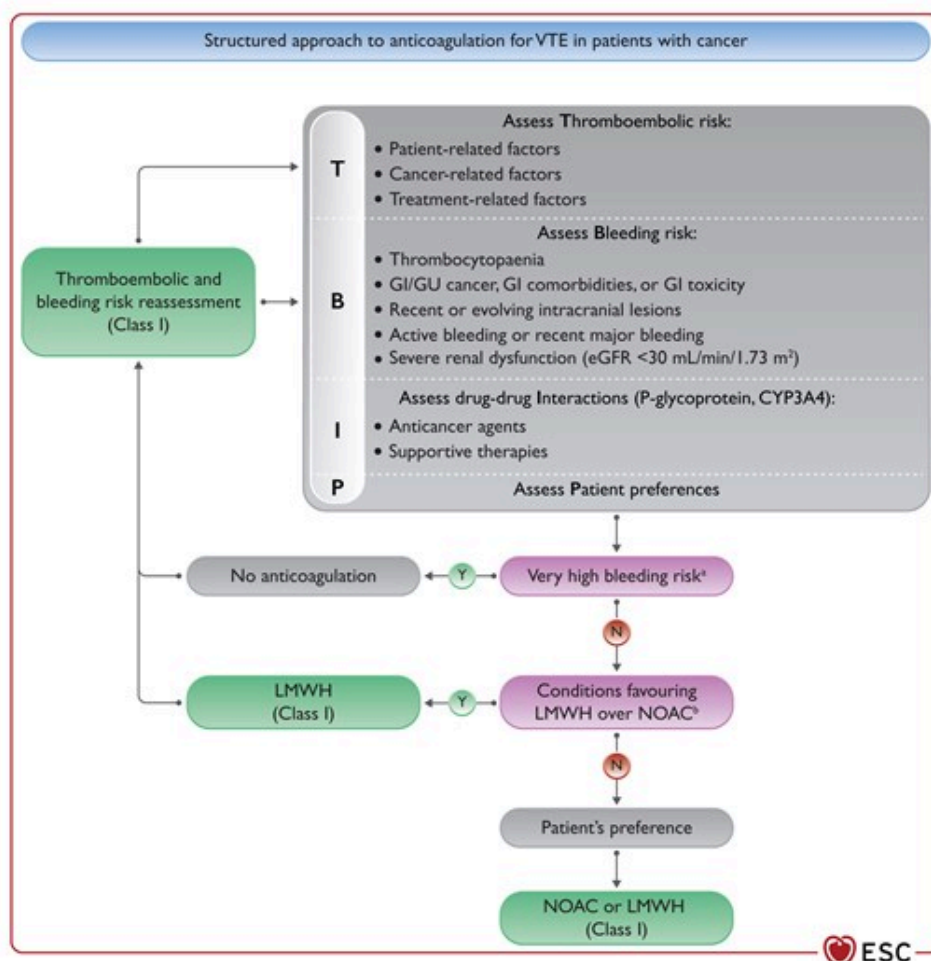


Figure 36

Structured approach to anticoagulation for venous thromboembolism in patients with active cancer.

CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; GI, gastrointestinal; GU, genitourinary; LMWH, low-molecular-weight heparins; N, no; NOAC, non-vitamin K antagonist oral anticoagulants; VTE, venous thromboembolism; Y, yes. ^a**Very high bleeding risk:** active or recent major bleeding (<1 month); recent/evolving intracranial lesions; platelet count <25 000/ μ L. According to the International Society on Thrombosis and Haemostasis,⁵²⁹ major bleeding is defined as: fall in haemoglobin level ≥ 2 g/dL, transfusion of ≥ 2 units of red blood cells, fatal bleeding, or bleeding in a critical area (intracranial, intraspinal, intraocular, pericardial, intra-articular, intramuscular with compartment syndrome, or retroperitoneal). ^b**Conditions favouring LMWH:** unoperated GI/GU cancer; GI comorbidities or toxicity; severe renal dysfunction (CrCl < 15 mL/min); NOAC major drug-drug interactions, platelet count < 50 000/ μ L.

[Open in new tab](#)[Download slide](#)

6.6.4.1. Treatment and secondary prevention of venous thromboembolism

Several large RCTs and meta-analyses have shown that LMWH decrease the risk of recurrent VTE by 40% compared to VKA, with a similar risk of major bleeding.⁵⁷²⁻⁵⁷⁶ However, VKA are characterized by an unpredictable anticoagulation effect and low time in therapeutic range in patients with malignancies due to multiple drug-drug interactions, GI toxicity, malnutrition, and liver dysfunction.⁵⁷⁷

NOAC have been assessed as potential alternatives to LMWH for cancer-associated VTE, based on RCTs that compared edoxaban, rivaroxaban or apixaban to dalteparin.⁵⁷⁸⁻⁵⁸³ The totality of evidence derived by these trials and subsequent meta-analyses⁵⁸⁴⁻⁵⁸⁶ shows that NOAC are non-inferior to dalteparin in reducing the risk of VTE recurrence. The risk of major bleeding was similar, although NOAC were associated with an increased risk of clinically relevant non-major bleeding, particularly in patients with luminal GI and GU malignancies.⁵⁸⁶ As a result, edoxaban, rivaroxaban, and apixaban are recommended for the treatment of VTE (DVT and PE) in patients with cancer without any of the following bleeding risk factors: unoperated GI or GU malignancies, history of recent bleeding or within 7 days of major surgery, significant thrombocytopenia (platelet count < 50 000/ μ L), severe renal dysfunction (creatinine clearance (CrCl < 15 mL/min), or GI comorbidities.^{582,586} In addition, drug-drug interactions between NOAC, cancer therapies, and other concomitant treatments should be checked.⁵⁸⁷ There are also concerns about NOAC in patients with GI toxicity such as vomiting or those having undergone gastrectomy or extensive intestine resection, as well as those with severely impaired renal function. Shared decision-making considering informed patient preferences should guide the choice of anticoagulation.

Incidentally encountered proximal DVT or PE should be treated in the same manner as symptomatic VTE as they bear similar rates of recurrence and mortality.⁵⁸⁸

The minimal duration of anticoagulation is 6 months and extended anticoagulation is suggested in the presence of active malignancy, metastatic disease, or chemotherapy use. Cohort studies have shown that extended LMWH therapy beyond 6 and up to 12 months is safe in cancer-associated VTE.^{589,590} However, patients

with cancer are also at high risk of bleeding during anticoagulant treatment and a periodic assessment of the risk/benefit ratio should be performed.

In VTE relapse under anticoagulation, the patient should be investigated for treatment adherence, cancer progression or relapse, while a different anticoagulation strategy should be endorsed (e.g. replacement of NOAC with LMWH). The management of patients with VTE and a platelet count $<25\,000/\mu\text{L}$ should be individualized by a MDT.²⁹⁹

The duration of anticoagulation in patients with catheter-associated thrombosis depends upon whether the catheter is removed or remains *in situ*. If removed, then anticoagulation should continue for a minimum of 3 months and until follow-up cardiac imaging confirms resolution of the thrombus. If the catheter remains *in situ*, then long-term therapeutic anticoagulation should continue.

Recommendation Table 34

[Open in new tab](#)

Recommendations for the management of venous thromboembolism in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
Apixaban, edoxaban, or rivaroxaban ^c are recommended for the treatment of symptomatic or incidental VTE in patients with cancer <i>without</i> contraindications. ^{d,578–581,584,585}	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count >50 000/μL. ^{298,299,578–581,584,585}	I	A
In patients with cancer with platelet counts of 25 000–50 000/μL, anticoagulation with half-dose LMWH may be considered after a multidisciplinary discussion. ⁵⁹¹	IIb	C
Prolongation of anticoagulation therapy beyond 6 months should be considered in selected patients with active cancer ^e including metastatic disease. ^{589,590}	IIa	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains <i>in situ</i> .	I	C

CrCl, creatinine clearance; GI, gastrointestinal; GU, genitourinary; LMWH, low-molecular-weight heparins; ULN, upper limit of normal; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cDrugs are listed in alphabetical order

^dHigh risk of GI or GU bleeding, GI absorption concerns, significant drug–drug interactions, severe renal dysfunction (CrCl < 15 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase > 2 × ULN), or significant thrombocytopenia (platelet count < 50 000/μL). In addition, patients with primary brain tumours or brain metastases and acute leukaemia were excluded from the seminal apixaban trial.⁵⁸⁰

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

Recommendations	Class ^a	Level ^b
Apixaban, edoxaban, or rivaroxaban ^c are recommended for the treatment of symptomatic or incidental VTE in patients with cancer <i>without</i> contraindications. ^{d,578–581,584,585}	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count >50 000/μL. ^{298,299,578–581,584,585}	I	A
In patients with cancer with platelet counts of 25 000–50 000/μL, anticoagulation with half-dose LMWH may be considered after a multidisciplinary discussion. ⁵⁹¹	IIb	C
Prolongation of anticoagulation therapy beyond 6 months should be considered in selected patients with active cancer ^e including metastatic disease. ^{589,590}	IIa	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains <i>in situ</i> .	I	C

CrCl, creatinine clearance; GI, gastrointestinal; GU, genitourinary; LMWH, low-molecular-weight heparins; ULN, upper limit of normal; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cDrugs are listed in alphabetical order

^dHigh risk of GI or GU bleeding, GI absorption concerns, significant drug–drug interactions, severe renal dysfunction (CrCl < 15 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase > 2 × ULN), or significant thrombocytopenia (platelet count < 50 000/μL). In addition, patients with primary brain tumours or brain metastases and acute leukaemia were excluded from the seminal apixaban trial.⁵⁸⁰

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

© ESC 2022

Recommendation Table 34

[Open in new tab](#)

Recommendations for the management of venous thromboembolism in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
Apixaban, edoxaban, or rivaroxaban ^c are recommended for the treatment of symptomatic or incidental VTE in patients with cancer <i>without</i> contraindications. ^{d,578–581,584,585}	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count >50 000/μL. ^{298,299,578–581,584,585}	I	A
In patients with cancer with platelet counts of 25 000–50 000/μL, anticoagulation with half-dose LMWH may be considered after a multidisciplinary discussion. ⁵⁹¹	IIb	C
Prolongation of anticoagulation therapy beyond 6 months should be considered in selected patients with active cancer ^e including metastatic disease. ^{589,590}	IIa	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains <i>in situ</i> .	I	C

CrCl, creatinine clearance; GI, gastrointestinal; GU, genitourinary; LMWH, low-molecular-weight heparins; ULN, upper limit of normal; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cDrugs are listed in alphabetical order

^dHigh risk of GI or GU bleeding, GI absorption concerns, significant drug–drug interactions, severe renal dysfunction (CrCl < 15 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase > 2 × ULN), or significant thrombocytopenia (platelet count < 50 000/μL). In addition, patients with primary brain tumours or brain metastases and acute leukaemia were excluded from the seminal apixaban trial.⁵⁸⁰

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

Recommendations	Class ^a	Level ^b
Apixaban, edoxaban, or rivaroxaban ^c are recommended for the treatment of symptomatic or incidental VTE in patients with cancer <i>without</i> contraindications. ^{d,578–581,584,585}	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count >50 000/μL. ^{298,299,578–581,584,585}	I	A
In patients with cancer with platelet counts of 25 000–50 000/μL, anticoagulation with half-dose LMWH may be considered after a multidisciplinary discussion. ⁵⁹¹	IIb	C
Prolongation of anticoagulation therapy beyond 6 months should be considered in selected patients with active cancer ^e including metastatic disease. ^{589,590}	IIa	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains <i>in situ</i> .	I	C

© ESC 2022

CrCl, creatinine clearance; GI, gastrointestinal; GU, genitourinary; LMWH, low-molecular-weight heparins; ULN, upper limit of normal; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cDrugs are listed in alphabetical order

^dHigh risk of GI or GU bleeding, GI absorption concerns, significant drug–drug interactions, severe renal dysfunction (CrCl < 15 mL/min), significant liver disease (alanine aminotransferase/aspartate aminotransferase > 2 × ULN), or significant thrombocytopenia (platelet count < 50 000/μL). In addition, patients with primary brain tumours or brain metastases and acute leukaemia were excluded from the seminal apixaban trial.⁵⁸⁰

^ePatients receiving cancer treatment, patients diagnosed with cancer in the past 6 months, and patients with progressive or advanced disease.

6.6.4.2. Primary prevention of venous thromboembolism

Patients undergoing surgery and those who are hospitalized or in prolonged bed rest require thromboprophylaxis with low-dose anticoagulation.^{298,299,592a,593,594} The ENOXACAN (Enoxaparin and Cancer) II study showed favourable outcomes with LMWH as primary thromboprophylaxis for 4 weeks after major abdominal or pelvic cancer surgery.⁵⁹⁵ For ambulatory patients, VTE risk should be individually determined and proposed scores such as the Khorana or the COMPASS-CAT (prospective COMparison of Methods for thromboembolic risk assessment with clinical Perceptions and Awareness in real-life

patients' (Cancer Associated Thrombosis) score may be useful.^{596,597} Further trials and a meta-analysis have shown that LMWH significantly reduced the incidence of symptomatic VTE in ambulatory patients with cancer receiving chemotherapy with acceptable safety.^{598,599,600} Two randomized, placebo-controlled, double-blind clinical trials have assessed the role of NOAC in primary prevention of VTE in high-risk ambulatory patients receiving systemic cancer therapy (Khorana score ≥ 2).^{601,602} Over a follow-up period of 180 days, apixaban (2.5 mg twice a day)⁶⁰¹ therapy resulted in a significantly lower rate of VTE, although the rate of major bleeding episodes was higher than with placebo. Rivaroxaban (10 mg once a day)⁶⁰² treatment resulted in a non-significantly lower incidence of VTE or death due to VTE with low bleeding risk (no significant differences with placebo). Further data on the use of NOAC in this setting are warranted. Consideration of such therapy should be accompanied by a discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of prophylaxis.

Recommendation Table 35

[Open in new tab](#)

Recommendations for venous thromboembolism prophylaxis during anticancer treatment

Recommendation	Class ^a	Level ^b
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk. ^{c,298,299,595}	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bedrest or reduced mobility in the absence of bleeding or other contraindications. ^{298,299,592,594}	I	B
For ambulatory patients with cancer at high risk of thrombosis receiving systemic therapy, ^d primary thromboprophylaxis with a NOAC (apixaban or rivaroxaban) or LMWH may be considered, provided there are no significant contraindications. ^{e,298,593,594,601,602}	IIb	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C

LMWH, low-molecular-weight heparins; NOAC, non-vitamin K antagonist oral anticoagulants; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cReduced mobility, obesity, VTE history.

^dLocally advanced or metastatic pancreas or lung cancer or Khorana score ≥ 2 .

^eRisk factors for bleeding, significant drug–drug interactions, or severe renal dysfunction.

© ESC 2022

Recommendation	Class ^a	Level ^b
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk. ^{c,298,299,595}	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bedrest or reduced mobility in the absence of bleeding or other contraindications. ^{298,299,592,594}	I	B
For ambulatory patients with cancer at high risk of thrombosis receiving systemic therapy, ^d primary thromboprophylaxis with a NOAC (apixaban or rivaroxaban) or LMWH may be considered, provided there are no significant contraindications. ^{e,298,593,594,601,602}	IIb	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C

LMWH, low-molecular-weight heparins; NOAC, non-vitamin K antagonist oral anticoagulants; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cReduced mobility, obesity, VTE history.

^dLocally advanced or metastatic pancreas or lung cancer or Khorana score ≥ 2 .

^eRisk factors for bleeding, significant drug–drug interactions, or severe renal dysfunction.

© ESC 2022

Recommendation Table 35

[Open in new tab](#)

Recommendations for venous thromboembolism prophylaxis during anticancer treatment

Recommendation	Class ^a	Level ^b
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk. ^{c,298,299,595}	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bedrest or reduced mobility in the absence of bleeding or other contraindications. ^{298,299,592,594}	I	B
For ambulatory patients with cancer at high risk of thrombosis receiving systemic therapy, ^d primary thromboprophylaxis with a NOAC (apixaban or rivaroxaban) or LMWH may be considered, provided there are no significant contraindications. ^{e,298,593,594,601,602}	IIb	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C

LMWH, low-molecular-weight heparins; NOAC, non-vitamin K antagonist oral anticoagulants; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cReduced mobility, obesity, VTE history.

^dLocally advanced or metastatic pancreas or lung cancer or Khorana score ≥ 2 .

^eRisk factors for bleeding, significant drug–drug interactions, or severe renal dysfunction.

© ESC 2022

Recommendation	Class ^a	Level ^b
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk. ^{c,298,299,595}	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bedrest or reduced mobility in the absence of bleeding or other contraindications. ^{298,299,592,594}	I	B
For ambulatory patients with cancer at high risk of thrombosis receiving systemic therapy, ^d primary thromboprophylaxis with a NOAC (apixaban or rivaroxaban) or LMWH may be considered, provided there are no significant contraindications. ^{e,298,593,594,601,602}	IIb	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C

LMWH, low-molecular-weight heparins; NOAC, non-vitamin K antagonist oral anticoagulants; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

^cReduced mobility, obesity, VTE history.

^dLocally advanced or metastatic pancreas or lung cancer or Khorana score ≥ 2 .

^eRisk factors for bleeding, significant drug–drug interactions, or severe renal dysfunction.

© ESC 2022

6.7. Bleeding complications

Bleeding complications are more common in patients with cancer than in patients without cancer. This may be directly related to the tumour itself, or indirectly related to chemotherapy- or RT-induced weakening of mucosal barriers.⁵³⁰

6.7.1. High-risk patients

GI and GU cancers are associated with a significant excess bleeding risk compared with other solid tumours.⁶⁰³ Thrombocytopaenia and platelet dysfunction due to haematological malignancies or bone marrow suppression can exacerbate bleeding. Other bleeding risk factors include advancing age, renal or hepatic impairment, metastatic disease, low body mass index, and treatment with ibrutinib, VEGFi, cetuximab, or

bevacizumab.^{578,603a} Gastric protection with routine proton pump inhibitor use should be considered in all patients with cancer on DAPT^{606,607} or anticoagulation.⁵³⁰

6.7.2. Antiplatelet therapy

Antiplatelet therapy, in particular DAPT, increases the risk of bleeding in patients with cancer.⁴⁷⁷ Following ACS and/or PCI, the risk of bleeding is approximately 1.6-fold greater in patients with cancer than in those without.^{477,605} The risk is greatest in those diagnosed with cancer in the preceding year, whereas more remote cancers carry lower excess risk.⁴⁷⁷ The PRECISE-DAPT (PREdicting bleeding Complications In patients undergoing Stent implantation and subSEquent Dual Anti Platelet Therapy) score appears not to perform well for predicting bleeding in patients with cancer.⁴⁷⁷ In order to reduce bleeding risk, the duration and intensity of DAPT should be minimized^{477,607} and triple therapy avoided whenever possible. At the same time, DAPT^a if indicated^a should not be withheld without good reason. A recent expert consensus statement suggests reduced platelet count thresholds for CV therapies, recommending aspirin initiation for platelet counts >10 000/ μ L and DAPT initiation (with aspirin and clopidogrel) for platelet counts >30 000/ μ L.⁶⁰⁸ In patients with platelet counts <50 000/ μ L, clopidogrel is preferred over prasugrel or ticagrelor, and glycoprotein IIb/IIIa inhibitors should be avoided.⁶⁰⁸ To reduce peri-procedural bleeding, PCI should preferably be undertaken via the radial approach⁴⁸⁴ and prophylactic platelet transfusion may be considered for patients with platelet count <20 000/ μ L.⁶⁰⁹

6.7.3. Management of bleeding

Basic principles of bleeding management should be followed with control of the bleeding source whenever possible. Platelet transfusions for significant thrombocytopaenia and withholding and reversal of anticoagulation for life-threatening bleeding may be needed as in the general population.^{530,610} Antifibrinolytic agents, such as tranexamic acid or e-aminocaproic acid, can be considered. Non-specific support of haemostasis using coagulation factor concentrates and specific reversal agents may be needed for patients on a NOAC with life-threatening bleeding.⁵³⁰ Recombinant activated factor VII or activated prothrombin complex concentrates should be avoided in patients with recent thrombosis.

6.8. Peripheral artery disease

There is growing evidence that cancer therapy affects the vasculature. A recent meta-analysis showed a significantly increased arterial stiffness after anthracycline and non-anthracycline treatment.⁶¹¹ Paraneoplastic acral vascular syndrome was described after initiation of nivolumab, with first symptoms 3 weeks after initiation of therapy.⁶¹² Raynaud phenomenon has been associated with the use of bleomycin, cyclophosphamide, platinum compounds, vinca alkaloids, and fluoropyrimidines.⁴⁹¹ Usual treatment of Raynaud^a includes non-pharmacological measures to help prevent an episode (avoidance of provoking factors such as cold temperature, vasoconstricting drugs) and a long-acting dihydropyridine CCB (amlodipine, modified-release nifedipine).

Treatment with nilotinib or ponatinib may be associated with an increased risk of vascular adverse events, including arterial stiffness and PAD development.⁴⁹⁴ In a subgroup of patients, these events are severe or even life-threatening.⁶¹³ Although the exact mechanisms remain unknown, we recommend screening for pre-

existing PAD and for vascular risk factors such as DM in all patients before and during nilotinib or ponatinib therapy. Pooled data from three clinical trials showed arterial occlusive disease to be related to dose intensity in ponatinib-treated patients,⁶¹⁴ but PAD was not addressed separately. If rapidly progressive PAD occurs with second-generation TKI, it may be advisable to switch to an alternative lower-risk TKI (e.g. imatinib). Platelet aggregation inhibitors or anticoagulation and statins should be considered. Despite lack of evidence, all risk factors should be corrected.⁶¹⁵

Recommendation Table 36

[Open in new tab](#)

Recommendation for management of peripheral artery disease during anticancer treatment

Recommendation	Class ^a	Level ^b
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy ^c is recommended.	I	C

© ESC 2022

PAD, peripheral artery disease; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cVEGFi, nilotinib, ponatinib, platins, etc.

Recommendation	Class ^a	Level ^b
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy ^c is recommended.	I	C

© ESC 2022

PAD, peripheral artery disease; VEGFi, vascular endothelial growth factor inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cVEGFi, nilotinib, ponatinib, platins, etc.

Recommendation Table 36

[Open in new tab](#)

Recommendation for management of peripheral artery disease during anticancer treatment

Recommendation	Class ^a	Level ^b
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy ^c is recommended.	I	C

PAD, peripheral artery disease; VEGFi, vascular endothelial growth factor inhibitors.
^aClass of recommendation.
^bLevel of evidence.
^cVEGFi, nilotinib, ponatinib, platins, etc.

© ESC 2022

Recommendation	Class ^a	Level ^b
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy ^c is recommended.	I	C

PAD, peripheral artery disease; VEGFi, vascular endothelial growth factor inhibitors.
^aClass of recommendation.
^bLevel of evidence.
^cVEGFi, nilotinib, ponatinib, platins, etc.

© ESC 2022

6.9. Pulmonary hypertension

All five groups of the PH classification can be observed in patients with cancer. Several cancer drugs can cause group 1 PH (pulmonary arterial hypertension [PAH]), including carfilzomib, bosutinib, dasatinib,⁶¹⁶ ponatinib, interferon alpha, and alkylating agents (e.g. mitomycin C and cyclophosphamide, which mostly cause pulmonary veno-occlusive disease).⁶¹⁷ PH associated with left heart disease (group 2) is related to drugs causing HF (e.g. anthracyclines). PH associated with lung disease (group 3) is related to drugs and therapies causing pulmonary fibrosis (e.g. bleomycin, thoracic radiation). The most common pulmonary vascular disease complicating cancer is VTE, which can cause chronic thromboembolic PH (group 4). Of note, central venous catheters are important causes of group 4 PH complicating cancer management. Other group 4 PH due to pulmonary artery obstructions includes angiosarcoma and other malignant tumours (e.g. renal carcinoma, uterine carcinoma, germ cell tumours of the testis).⁶¹⁸

PH with unclear and/or multifactorial mechanisms (group 5) includes several conditions that may be complicated by complex and sometimes overlapping pulmonary vascular involvement. Tumoral PH includes pulmonary tumour micro-embolism and pulmonary tumour thrombotic microangiopathy.⁶¹⁹ Multiple causes of PH have been described in patients with chronic myeloproliferative disorders. In chronic myelogenous leukaemia, spleen enlargement and anaemia can give rise to hyperkinetic syndrome. In polycythaemia vera and essential thrombocythemia, there is an increased risk of VTE and chronic thromboembolic PH. Moreover, formation of a blood clot within the hepatic veins can lead to Budd-Chiari syndrome and subsequent

porto-PH. Pulmonary extramedullary haematopoiesis complicating idiopathic or secondary myelofibrosis may also contribute to dyspnoea and PH.⁶²⁰

Symptoms of PH are non-specific, such as shortness of breath and fatigue. In later stages, symptoms of right-sided HF may develop. An ECG should be performed and examined for RV hypertrophy, but a normal ECG does not exclude PH. Echocardiography is the first choice for assessing PH probability in patients who develop symptoms and/or signs suggestive of PH during cancer treatment. When peak tricuspid regurgitation velocity (TRV) is ≥ 2.8 m/s (equating to an estimated systolic PAP [sPAP] of ≥ 35 mmHg) and no other signs of PH are present, then the probability of PH is low. In the absence of a tricuspid regurgitant jet, other echocardiography signs may increase suspicion of PH (e.g. RV/LV basal diameter ratio > 1 , RV outflow tract acceleration time < 105 ms, inferior vena cava diameter > 21 mm with decreased inspiratory collapse).⁶²⁰ Baseline TTE should be considered in patients receiving cancer drugs that can cause PH; however, a right-heart catheterization is required for definitive diagnosis of PH and to support PAH treatment decisions. In the DASISION (DASatinib vs. Imatinib Study In treatment-Na⁺ve chronic myeloid leukemia patients) trial, 5% of patients randomized to dasatinib were diagnosed with PH, compared with 0.4% of those randomized to imatinib.⁶²¹ In patients who develop PH, dasatinib treatment should be interrupted and an alternative TKI should be used.⁶¹⁶

Overall management of PH in oncology patients should be based on the 2022 ESC/European Respiratory Society (ERS) Guidelines for the diagnosis and treatment of pulmonary hypertension.⁶²⁰ Referral to a PH centre is recommended for multidisciplinary management with the oncology team. In patients with CML treated with drugs causing PH-induced PAH, discontinuation of the potential culprit therapy is recommended if there is a high probability of new PH (peak TRV > 3.4 m/s, equating to an estimated sPAP of ≥ 50 mmHg) until the diagnosis is confirmed or ruled out by a right-heart catheterization. In CML patients on dasatinib, an alternative BCR-ABL TKI is recommended if they develop symptomatic PAH or an asymptomatic increase in peak TRV > 3.4 m/s. Dasatinib dose reduction and close monitoring of peak TRV with TTE every 4 weeks should be considered in CML patients who develop new asymptomatic peak TRV ranging from 2.9 to 3.4 m/s.⁶²⁰ If peak TRV remains normal or mildly elevated on serial monitoring, then dasatinib can continue, with reduced TTE monitoring to every 3 months. If peak TRV continues to rise, then right-heart catheterization should be performed, dasatinib treatment should be stopped, and PAH drugs should be considered if PAH is confirmed.

Recommendation Table 37

[Open in new tab](#)

Recommendations for the management of pulmonary hypertension during anticancer treatment

Recommendations	Class ^a	Level ^b
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV >3.4 m/s.	I	C
Dasatinib dose reduction and close monitoring of peak TRV with echocardiography should be considered in patients who develop new asymptomatic peak TRV ranging from 2.9 to 3.4 m/s.	IIa	C
In patients with confirmed dasatinib-induced PAH ^c or new asymptomatic peak TRV >3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C

© ESC 2022

BCL-ABL, breakpoint cluster region–Abelson oncogene locus; PAH, pulmonary arterial hypertension; TRV, tricuspid regurgitation velocity.

^aClass of recommendation.

^bLevel of evidence.

^cDefinite diagnosis of PAH requires a right-heart catheterization.

Recommendations	Class ^a	Level ^b
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV >3.4 m/s.	I	C
Dasatinib dose reduction and close monitoring of peak TRV with echocardiography should be considered in patients who develop new asymptomatic peak TRV ranging from 2.9 to 3.4 m/s.	IIa	C
In patients with confirmed dasatinib-induced PAH ^c or new asymptomatic peak TRV >3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C

© ESC 2022

BCL-ABL, breakpoint cluster region–Abelson oncogene locus; PAH, pulmonary arterial hypertension; TRV, tricuspid regurgitation velocity.

^aClass of recommendation.

^bLevel of evidence.

^cDefinite diagnosis of PAH requires a right-heart catheterization.

Recommendation Table 37

[Open in new tab](#)

Recommendations	Class ^a	Level ^b
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV >3.4 m/s.	I	C
Dasatinib dose reduction and close monitoring of peak TRV with echocardiography should be considered in patients who develop new asymptomatic peak TRV ranging from 2.9 to 3.4 m/s.	IIa	C
In patients with confirmed dasatinib-induced PAH ^c or new asymptomatic peak TRV >3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C

© ESC 2022

BCL-ABL, breakpoint cluster region–Abelson oncogene locus; PAH, pulmonary arterial hypertension; TRV, tricuspid regurgitation velocity.

^aClass of recommendation.

^bLevel of evidence.

^cDefinite diagnosis of PAH requires a right-heart catheterization.

Recommendations	Class ^a	Level ^b
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV >3.4 m/s.	I	C
Dasatinib dose reduction and close monitoring of peak TRV with echocardiography should be considered in patients who develop new asymptomatic peak TRV ranging from 2.9 to 3.4 m/s.	IIa	C
In patients with confirmed dasatinib-induced PAH ^c or new asymptomatic peak TRV >3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C

© ESC 2022

BCL-ABL, breakpoint cluster region–Abelson oncogene locus; PAH, pulmonary arterial hypertension; TRV, tricuspid regurgitation velocity.

^aClass of recommendation.

^bLevel of evidence.

^cDefinite diagnosis of PAH requires a right-heart catheterization.

6.10. Pericardial diseases

Pericarditis and pericardial effusion can be related to a wide range of cancer treatments including chest radiation, cytotoxic therapies (anthracyclines, bleomycin, cyclophosphamide, cytarabine), targeted therapies (all-*trans* retinoic acid, arsenic trioxide, dasatinib), and immune-based therapies (interleukin-2, interferon- γ ±ICI). A combination of therapies may have a synergistic effect on the pericardium. These therapy-induced complications must be differentiated from progressive cancer (local invasion, metastatic involvement, or mediastinal lymphatic drainage obstruction) and non-cancer-related causes such as infection, especially in immune-compromised patients.⁶²² A careful history and clinical examination are of help to determine the cause. TTE plays a central role in diagnosis and management. CT and CMR can provide additional information on pericardial inflammation and constrictive physiology. The principles for the diagnosis and management should follow the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases,⁴⁴⁴ but there are some specific issues to consider in patients with cancer.⁴⁸²

6.10.1. Pericarditis

The diagnosis of pericarditis in patients with cancer follows the same principles as in those without, but symptoms can be atypical.⁴⁴⁴ Acute pericarditis caused by radiation has become rare due to lower doses and improved radiation techniques. It occurs within days to weeks after treatment and is usually self-limiting, but can evolve towards constrictive pericarditis many years later ([Section 8.6](#)). Pericarditis caused by conventional cancer therapies often resolves with standard therapy or after discontinuation of the treatment.⁴⁴⁴ Cancer treatment interruption should be discussed with the cardio-oncology team. Treatment with anti-inflammatory drugs (e.g. ibuprofen) and colchicine, in the absence of contraindications, is recommended as it reduces the rate of recurrence requiring repeat intervention.⁶²³ Low-to-moderate doses of steroids are only indicated for resistant cases except ICI-related pericarditis.⁴⁴⁴

ICI-associated pericarditis has a median time of onset of 30 days in retrospective surveillance studies and is associated with a poor prognosis, especially in case of concomitant myocarditis.^{444,624}

In patients with severe ICI-associated pericarditis with moderate or severe effusion, ICI discontinuation and high-dose steroids (methylprednisolone 1â€mg/kg/day) with or without colchicine are recommended, as well as pericardiocentesis in case of cardiac tamponade.^{624,625} In case of refractory pericarditis, immunosuppressive drugs should be considered. For uncomplicated ICI-related pericarditis, the ICI might be continued and colchicine or non-steroidal anti-inflammatory drugs could be considered.^{326,444} For patients requiring ICI discontinuation, restarting ICI can be considered in a MDT discussion after resolution of pericardial disease and under close monitoring.

6.10.2. Pericardial effusion

Pericardial effusions are often observed as an incidental finding in patients with cancer. Cancer therapy is the cause of a pericardial effusion in <30% of cases, although this may increase with the expanding use of ICI in cancer. Malignancy-related pericardial effusions caused by direct (lung, oesophageal, breast) or metastatic invasion (haematological malignancies, ovarian, melanoma) or by lymph node obstruction are generally associated with poor prognosis.

Clinical presentation depends on the size of the effusion and the speed of its growth.⁴⁴⁴ Malignancy-related pericardial effusions make up >30% of patients presenting with cardiac tamponade⁶²⁶ and usually develop slowly, resulting in larger pericardial effusions at the time of diagnosis compared with non-malignant pericardial effusions. Management consists of determination of the cause and evaluation of the haemodynamic impact. Small-to-medium-sized effusions (>4 and <20 mm) can be monitored with a reassessment 7–14 days after initial diagnosis and at further 4–6-weekly intervals.^{444,627} In unstable patients with signs of tamponade, immediate echocardiographic-guided percutaneous pericardiocentesis is preferred over surgical pericardiotomy to minimize potential complications.⁶²⁸ In patients with cardiac tamponade due to malignant pericardial effusions, colchicine may be useful to improve clinical outcomes and reduce the rate of repeat intervention.⁶²³ Drainage of a pericardial effusion related to ICI is rarely required⁶²⁹ and corticosteroids should be considered.⁶³⁰ Intrapericardial instillation of cytostatic/sclerosing agents, colchicine,⁶²³ and radiation for radiation-sensitive tumours can reduce recurrence after drainage. The creation of a pleuropericardial or pleuroperitoneal window with balloon pericardiotomy or surgery should be considered in case of recurrent malignant pericardial effusions after emergency pericardiocentesis.⁴⁴⁴

A surgical pericardial window should be considered if the percutaneous approach is not feasible and in stable patients with large (>20 mm) or rapidly expanding malignant pericardial effusions prior to the development of cardiac tamponade.

Recommendation Table 38

[Open in new tab](#)

Recommendations for the management of pericardial diseases in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy. ⁴⁴⁴	I	C
A surgical pericardial window should be considered if the percutaneous approach is not feasible or in cases of recurrent malignant pericardial effusions.	IIa	C
Intrapericardial instillation of cytostatic or sclerosing agents may be considered for prevention of recurrence.	IIb	C
Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis. ^{326,624,625,630}	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C

CMR, cardiac magnetic resonance; CT, computed tomography; CV, cardiovascular; ECG, electrocardiogram; ESC, European Society of Cardiology; ICI, immune checkpoint inhibitors.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

Recommendations	Class ^a	Level ^b
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy. ⁴⁴⁴	I	C
A surgical pericardial window should be considered if the percutaneous approach is not feasible or in cases of recurrent malignant pericardial effusions.	IIa	C
Intrapericardial instillation of cytostatic or sclerosing agents may be considered for prevention of recurrence.	IIb	C
Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis. ^{326,624,625,630}	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C

CMR, cardiac magnetic resonance; CT, computed tomography; CV, cardiovascular; ECG, electrocardiogram; ESC, European Society of Cardiology; ICI, immune checkpoint inhibitors.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

Recommendation Table 38

[Open in new tab](#)

Recommendations for the management of pericardial diseases in patients receiving anticancer treatment

Recommendations	Class ^a	Level ^b
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy. ⁴⁴⁴	I	C
A surgical pericardial window should be considered if the percutaneous approach is not feasible or in cases of recurrent malignant pericardial effusions.	IIa	C
Intrapericardial instillation of cytostatic or sclerosing agents may be considered for prevention of recurrence.	IIb	C
Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis. ^{326,624,625,630}	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C

CMR, cardiac magnetic resonance; CT, computed tomography; CV, cardiovascular; ECG, electrocardiogram; ESC, European Society of Cardiology; ICI, immune checkpoint inhibitors.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

Recommendations	Class ^a	Level ^b
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy. ⁴⁴⁴	I	C
A surgical pericardial window should be considered if the percutaneous approach is not feasible or in cases of recurrent malignant pericardial effusions.	IIa	C
Intrapericardial instillation of cytostatic or sclerosing agents may be considered for prevention of recurrence.	IIb	C
Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis. ^{326,624,625,630}	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C

CMR, cardiac magnetic resonance; CT, computed tomography; CV, cardiovascular; ECG, electrocardiogram; ESC, European Society of Cardiology; ICI, immune checkpoint inhibitors.

^aClass of recommendation.

^bLevel of evidence.

© ESC 2022

7. End-of-cancer therapy cardiovascular risk assessment

7.1. Cardiovascular evaluation during the first year after cardiotoxic anticancer therapy

End-of-cancer therapy CV risk assessment covers the first 12 months after the last cardiotoxic cancer treatment. These recommendations are where cardiotoxic cancer therapy has been successfully completed with good long-term prognosis. These recommendations are not indicated when cancer therapies are discontinued due to cancer progression and prognosis is poor, or where end-of-life care is indicated. Selected patients with cancer continue on long-term oncology therapies, e.g. women with oestrogen receptor-positive early invasive BC. In this example, the end-of-therapy risk assessment refers to the timepoint from the last anthracycline or trastuzumab dose.

High-risk patients can be identified at completion of their cardiotoxic cancer therapies by their clinical characteristics, history of CTR-CVT during treatment, and by elevated cardiac biomarkers and/or abnormal CV imaging at follow up.^{53,54,92} Cardiac serum biomarkers (NP and cTn) are useful given their high negative predictive value for future CV events.^{197,631} In a prospective study of 2625 adult patients with cancer that assessed LVEF after anthracycline-based chemotherapy, the overall incidence of CTRCD was 9%; 98% of cases could be detected within 12 months after chemotherapy and the median time from chemotherapy to CTRCD detection was 3.5 months (interquartile range 3â6 months).²⁰⁸ The response to ACE-I treatment declined when the interval between the end of chemotherapy and CTRCD detection lengthened; complete LVEF recovery was not observed in patients where treatment was delayed by >6 months.⁴²⁵

Measurement of cTn after completion of anthracycline chemotherapy during the end-of-treatment assessment should be considered. Rises in cTnI after anthracycline chemotherapy identify patients at risk of future cardiac dysfunction who then benefit from CV protection.⁴ Educating patients with cancer of their potential increased CVD risk and supporting them to make appropriate healthy lifestyle choices is recommended. CS should also be advised to promptly report early signs and symptoms of possible CVD and inform medical teams of their previous cardiotoxic cancer therapies. CVRF including hypertension, DM, and dyslipidaemia correlate with the probability of future CV events in CS and should be well controlled after completion of cancer therapy.^{31,632,633}

7.2. Which cancer survivors require cardiovascular surveillance in the first year after cancer treatment?

The end-of-treatment risk assessment ideally identifies those high-risk CS, who require long-term CV surveillance, based on the following criteria ([TableÂ 10](#)):

1. Baseline high or very high risk based on HFA-ICOS risk assessment tools¹² ([Section 4](#)).
2. Cardiotoxic cancer therapy with a high risk of long-term CV complications^{7,21} ([Section 8](#)).
3. Moderate or severe CTR-CVT diagnosed during cancer treatment ([TableÂ 3](#)).⁶⁸
4. New abnormalities in cardiac function detected by echocardiography, new elevated cardiac serum biomarkers, or newly CV symptoms detected at the end-of-therapy assessment (3 or 12 months after treatment).^{68,208}

Table 10

[Open in new tab](#)

Risk factors for future cardiovascular disease at the end-of-cancer therapy cardiovascular risk assessment

High-risk conditions
High- and very-high baseline CV toxicity risk based on HFA-ICOS assessment
Specific anticancer treatment proven to have a high risk of long-term CV complications^a • $\geq 250 \text{ mg/m}^2$ Doxorubicin^b • RT > 15 Gy MHD^c • Both doxorubicin $\geq 100 \text{ mg/m}^2$ and RT ≥ 15 Gy MHD^d • High-risk HSCT patients^e
Moderate or severe CTR-CVT during cancer treatment (especially CTRCD), ICI-related myocarditis, cardiac arrhythmias, or severe vascular toxicities (ACS, stroke, PVD)
New CV symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment

High-risk conditions
High- and very-high baseline CV toxicity risk based on HFA-ICOS assessment
Specific anticancer treatment proven to have a high risk of long-term CV complications^a • $\geq 250 \text{ mg/m}^2$ Doxorubicin^b • RT > 15 Gy MHD^c • Both doxorubicin $\geq 100 \text{ mg/m}^2$ and RT ≥ 15 Gy MHD^d • High-risk HSCT patients^e
Moderate or severe CTR-CVT during cancer treatment (especially CTRCD), ICI-related myocarditis, cardiac arrhythmias, or severe vascular toxicities (ACS, stroke, PVD)
New CV symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment

ACS, acute coronary syndromes; CTRCD, cancer therapy-related cardiac dysfunction; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; GVHD, graft vs. host disease; Gy, Gray; HFA, Heart Failure Association; HSCT,

haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICOS, International Cardio-Oncology Society; MHD, mean heart dose; PVD, peripheral vascular disease; RT, radiotherapy.

a
RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category if only a small part of the heart is exposed to a relatively high prescribed dose (i.e. RT to left breast or left chest wall only).

b
Or doxorubicin equivalent.

c
Or prescribed RT 35 Gy to a volume exposing the heart if MHD is not available.

d
Or prescribed RT 34 Gy to a volume exposing the heart if MHD is not available.

e
High-risk HSCT patients: allogenic HSCT; pre-existing CVD or multiple uncontrolled CVRF; cancer treatment history (mediastinal or mantle field radiation, alkylating agents, >250 mg/m² doxorubicin or equivalent); conditioning schemes (total body irradiation, alkylating agents); development of GVHD.

Table 10

[Open in new tab](#)

Risk factors for future cardiovascular disease at the end-of-cancer therapy cardiovascular risk assessment

High-risk conditions
High- and very-high baseline CV toxicity risk based on HFA-ICOS assessment
Specific anticancer treatment proven to have a high risk of long-term CV complications ^a • Doxorubicin^b 250 mg/m² • RT > 15 Gy MHD^c • Both doxorubicin^b 100 mg/m² and RT 5 Gy MHD^d • High-risk HSCT patients^e
Moderate or severe CTR-CVT during cancer treatment (especially CTRCD), ICI-related myocarditis, cardiac arrhythmias, or severe vascular toxicities (ACS, stroke, PVD)
New CV symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment

High-risk conditions
High- and very-high baseline CV toxicity risk based on HFA-ICOS assessment
Specific anticancer treatment proven to have a high risk of long-term CV complications^a • ^a ≥ 250 mg/m² Doxorubicin^b • ^a RT > 15 Gy MHD^c • ^a Both doxorubicin^b ≥ 100 mg/m² and RT ≥ 15 Gy MHD^d • ^a High-risk HSCT patients^e
Moderate or severe CTR-CVT during cancer treatment (especially CTRCD), ICI-related myocarditis, cardiac arrhythmias, or severe vascular toxicities (ACS, stroke, PVD)
New CV symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment

ACS, acute coronary syndromes; CTRCD, cancer therapy-related cardiac dysfunction; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; GVHD, graft vs. host disease; Gy, Gray; HFA, Heart Failure Association; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICOS, International Cardio-Oncology Society; MHD, mean heart dose; PVD, peripheral vascular disease; RT, radiotherapy.

^a RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category if only a small part of the heart is exposed to a relatively high prescribed dose (i.e. RT to left breast or left chest wall only).

^b Or doxorubicin equivalent.

^c Or prescribed RT ≥ 35 Gy to a volume exposing the heart if MHD is not available.

^d Or prescribed RT ≥ 34 Gy to a volume exposing the heart if MHD is not available.

^e High-risk HSCT patients: allogenic HSCT; pre-existing CVD or multiple uncontrolled CVRF; cancer treatment history (mediastinal or mantle field radiation, alkylating agents, ≥ 250 mg/m² doxorubicin or equivalent); conditioning schemes (total body irradiation, alkylating agents); development of GVHD.

The timing of the first CV assessment after cardiotoxic cancer treatment depends on the risk defined by baseline CV assessment, the type of cancer therapy, and whether CTR-CVT was diagnosed during treatment.

In asymptomatic high-risk patients ([TableÂ 10](#)), echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy.[53,54,59,61,68,148,208,425](#) In asymptomatic moderate-risk patients (according to CV toxicity baseline risk stratification), echocardiography and cardiac serum biomarkers should be considered within 12 months after completion of cancer therapy.[53,54,59,61,68,148,208](#) In asymptomatic low-risk patients (according to CV toxicity baseline risk stratification), echocardiography and cardiac serum biomarkers may be considered within 12 months after completion of cancer therapy.[634](#)

All patients started on CV therapies (ACE-I/ARB/angiotensin receptor-neprilysin inhibitors, beta-blockers, mineralocorticoid receptor antagonists, sodium-glucose co-transporter 2 inhibitors, antihypertensive medications, antiarrhythmic medications, antiplatelet therapies, statins) for any CTR-CVT (especially CTRCD) should have a clinical assessment, ECG, echocardiography, and cardiac serum biomarkers (if LV systolic dysfunction/HF is a potential risk) at 3, 6, and 12 months after completing cancer treatment. A MDT-based approach to palliative and end-of-life care for patients with cancer with HF or other CTR-CVT should be focused on symptom relief according to general ESC Guidelines.

7.3. Management of cancer therapy-related cardiac dysfunction at the end-of-therapy assessment

During this end-of-treatment assessment, a review of cardioprotective medications initiated during cancer therapy to treat CTRCD is recommended ([FigureÂ 37](#)). In selected patients with asymptomatic mild or moderate CTRCD who have fully recovered with normal TTE and cardiac serum biomarkers, a trial of weaning off CV medication should be considered after MDT discussion. This is most common after asymptomatic mild or moderate CTRCD secondary to trastuzumab, particularly in younger otherwise healthy HER2+ BC survivors with no exposure to anthracycline chemotherapy. Further assessment of cardiac function with TTE and cardiac serum biomarkers is recommended following withdrawal of CV medication in patients with previous CTRCD to ensure cardiac function remains normal.

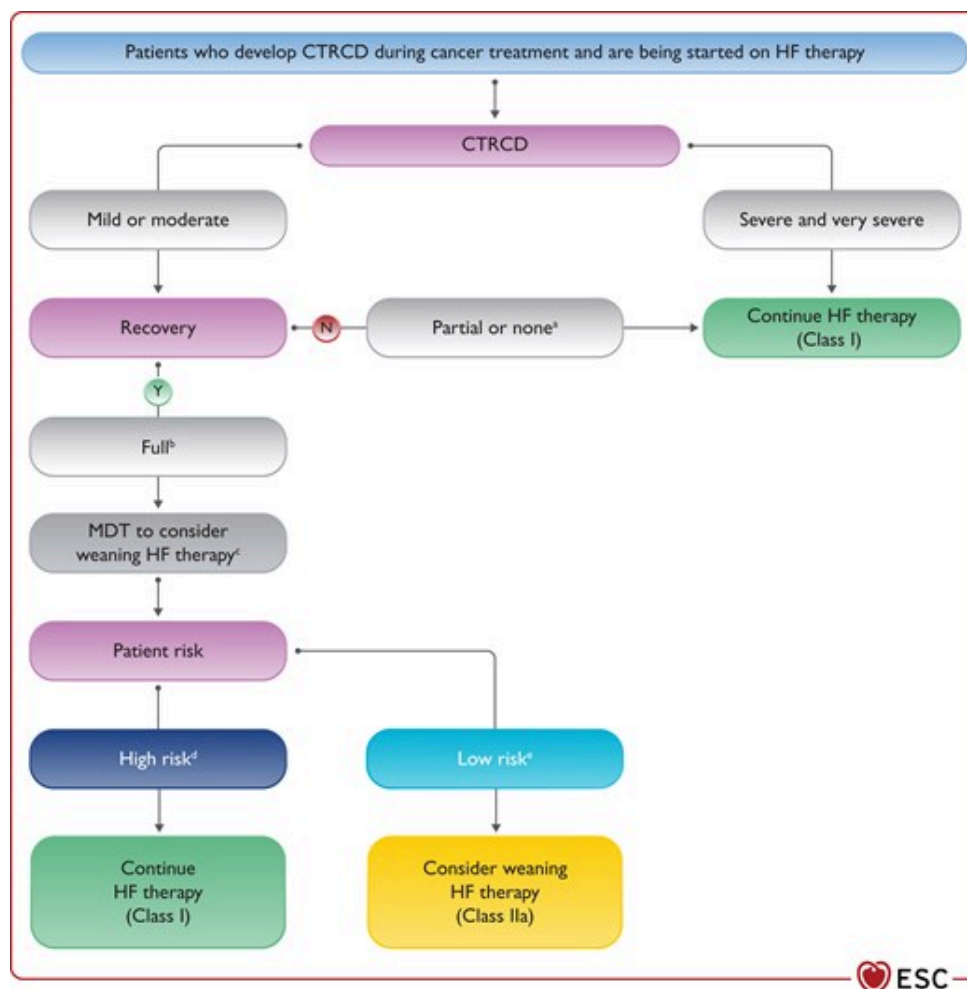


Figure 37

Management of cancer therapy-related cardiac dysfunction after cancer therapy.

CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; GLS, global longitudinal strain; HF, heart failure; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; N, no; Y, yes. ^a**Partial or no recovery**: patients who do not meet all of the criteria for full recovery. ^b**Full recovery**: no signs or symptoms of HF + LVEF > 50% + GLS within normal range or similar to baseline measurements + cardiac serum biomarkers within the normal range or similar to baseline measurements ^cThe **CTRCD trajectory** of each patient is unique and dynamic and withdrawal of HF therapy requires a MDT to consider several key points that help to stratify patients into low- or high-risk categories. **Key points to consider during a MDT discussion are**: HFA-ICOS baseline CV toxicity risk assessment, pre-existing indications for CV medication, class of cancer treatment causing CTRCD (generally reversible vs. generally irreversible), magnitude and duration of CTRCD before recovery, intensity of HF therapy needed to recover LV function, family history of cardiomyopathy or known cardiomyopathy gene carrier (see [Section 4.8](#)). ^dSee [Table 10](#). ^e**Low-risk patient characteristics**: low to moderate baseline CV toxicity risk (HFA-ICOS risk assessment), no pre-existing indications for CV medication, cancer treatment generally associated with reversible myocardial damage, asymptomatic mild CTRCD, early cardiac function recovery (3â6 months) under HF therapy, no family history of cardiomyopathy.

[Open in new tab](#)[Download slide](#)

Continuing long-term CV medication is generally recommended in patients with moderate and severe symptomatic or severe asymptomatic CTRCD due to the high rate of recurrent HF. Long-term treatment is also recommended in CS with mild or moderate CTRCD who fail to recover normal LV function at their end-of-therapy assessment ([Figure 37](#)).

7.4. Cardiopulmonary exercise testing and fitness during the end-of-therapy assessment

CRF impairment is a strong predictor of patient outcome following cancer treatment and an intervention target in CS. Low CRF is associated with poor quality of life, increased morbidity, reduced exercise cardiac function and worse CVD risk profile, and is a robust independent predictor of all-cause, cancer-related, and CVD-related mortality in CS.^{119,120} Recent evidence suggests the risk of CVD-related mortality in CS decreases by 14% per 1 metabolic equivalent (3.5 mL O₂/kg/min) increase in CRF.¹²⁰

CPET may be considered for CS with exertional limitation, who may have substantial benefit from cardiac rehabilitation. Eligible patients include those treated with higher doses of anthracycline chemotherapy and/or RT to a volume including the heart, high CV toxicity risk at baseline, patients who developed CTRCD during cancer therapy, and those identified with new abnormalities in LV function at their end-of-therapy assessment.¹¹ CPET can be an objective tool in the diagnosis of decreased physical capacity and identify CV vs. non-CV causes.⁶³⁵

7.5. The role of cardiac rehabilitation

Exercise is a potent multitargeted therapy that prevents and treats multiple competing mechanisms of CTR-CVT in CS, including CRF impairment,⁶³⁶ CV injury, and pre-existing and new CVRF.¹³⁷ Prescribing exercise facilitates the delivery of therapeutic exercise that is individualized to a person's fitness level and systematically progressed to optimize physiological adaptation.⁶³⁷ Current evidence demonstrates that supervised exercise therapy (including high-intensity interval training [HIIT]) is safe and well tolerated,⁶³⁸ attenuates CTR-CVT risk, and improves CRF. Furthermore, HIIT reduces CVRF⁴⁶⁰ and CV risk⁶³⁹ in patients with cancer in the pre-, active-, and post-treatment settings. HIIT-related benefits on CRF, physical activity behaviour, fatigue, and quality of life persist months post-intervention.^{640,641} HIIT may not be feasible in elderly and frail patients.⁶⁴² Dedicated cardio-oncology rehabilitation programmes are currently under development.¹¹

Recommendation Table 39

[Open in new tab](#)

Recommendations for end-of-cancer therapy cardiovascular risk assessment

Recommendations	Class ^a	Level ^b
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended. ^c	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy ^{c,12,22,31,632,643} and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice. ¹⁹	I	B
In asymptomatic high-risk patients, ^d echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208,425}	I	B
In asymptomatic moderate-risk patients, ^e echocardiography and cardiac serum biomarkers should be considered within 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208}	IIa	B
In asymptomatic low-risk patients, ^e echocardiography and cardiac serum biomarkers may be considered within 12 months after completion of cancer therapy. ⁶³⁴	IIb	C
Cardiology referral ^f is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment. ¹¹	I	C
In selected patients with exercise intolerance persisting at 12 months after cancer treatment and with normal resting echocardiogram and cardiac biomarkers, exercise stress echocardiography and/or CPET may be considered.	IIb	C
Targeted cardiac rehabilitation should be considered in CS with high CV risk. ^{638–640}	IIa	B
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is		

recommended in patients who developed TKI-mediated hypertension during cancer therapy. ^{644,645}	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy. ^{10,237}	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy. ⁶⁴⁶	I	C

© ESC 2022

CPET, cardiopulmonary exercise testing; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; ESC, European Society of Cardiology; LQTS, long QT syndrome; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding regulation of hypertension, DM, dyslipidaemia, smoking cessation, weight loss in case of obesity, and an adequate amount of exercise.

^dHigh-risk patients: see [Table 10](#).

^eModerate- or low-risk patients: according to CV toxicity baseline risk stratification.

^fCardio-oncology referral is recommended when available; alternatively, the patient should be referred to a cardiologist with expertise in managing CVD in patients with cancer.

Recommendations	Class ^a	Level ^b
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended. ^c	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy ^{c,12,22,31,632,643} and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice. ¹⁹	I	B
In asymptomatic high-risk patients, ^d echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208,425}	I	B
In asymptomatic moderate-risk patients, ^e echocardiography and cardiac serum biomarkers should be considered within 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208}	IIa	B
In asymptomatic low-risk patients, ^e echocardiography and cardiac serum biomarkers may be considered within 12 months after completion of cancer therapy. ⁶³⁴	IIb	C
Cardiology referral ^f is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment. ¹¹	I	C
In selected patients with exercise intolerance persisting at 12 months after cancer treatment and with normal resting echocardiogram and cardiac biomarkers, exercise stress echocardiography and/or CPET may be considered.	IIb	C
Targeted cardiac rehabilitation should be considered in CS with high CV risk. ^{638–640}	IIa	B
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is		

recommended in patients who developed TKI-mediated hypertension during cancer therapy. ^{644,645}	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy. ^{10,237}	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy. ⁶⁴⁶	I	C

© ESC 2022

CPET, cardiopulmonary exercise testing; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; ESC, European Society of Cardiology; LQTS, long QT syndrome; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding regulation of hypertension, DM, dyslipidaemia, smoking cessation, weight loss in case of obesity, and an adequate amount of exercise.

^dHigh-risk patients: see [Table 10](#).

^eModerate- or low-risk patients: according to CV toxicity baseline risk stratification.

^fCardio-oncology referral is recommended when available; alternatively, the patient should be referred to a cardiologist with expertise in managing CVD in patients with cancer.

Recommendation Table 39

[Open in new tab](#)

Recommendations for end-of-cancer therapy cardiovascular risk assessment

Recommendations	Class ^a	Level ^b
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended. ^c	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy ^{c,12,22,31,632,643} and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice. ¹⁹	I	B
In asymptomatic high-risk patients, ^d echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208,425}	I	B
In asymptomatic moderate-risk patients, ^e echocardiography and cardiac serum biomarkers should be considered within 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208}	IIa	B
In asymptomatic low-risk patients, ^e echocardiography and cardiac serum biomarkers may be considered within 12 months after completion of cancer therapy. ⁶³⁴	IIb	C
Cardiology referral ^f is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment. ¹¹	I	C
In selected patients with exercise intolerance persisting at 12 months after cancer treatment and with normal resting echocardiogram and cardiac biomarkers, exercise stress echocardiography and/or CPET may be considered.	IIb	C
Targeted cardiac rehabilitation should be considered in CS with high CV risk. ^{638–640}	IIa	B
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is		

recommended in patients who developed TKI-mediated hypertension during cancer therapy. ^{644,645}	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy. ^{10,237}	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy. ⁶⁴⁶	I	C

© ESC 2022

CPET, cardiopulmonary exercise testing; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; ESC, European Society of Cardiology; LQTS, long QT syndrome; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding regulation of hypertension, DM, dyslipidaemia, smoking cessation, weight loss in case of obesity, and an adequate amount of exercise.

^dHigh-risk patients: see [Table 10](#).

^eModerate- or low-risk patients: according to CV toxicity baseline risk stratification.

^fCardio-oncology referral is recommended when available; alternatively, the patient should be referred to a cardiologist with expertise in managing CVD in patients with cancer.

Recommendations	Class ^a	Level ^b
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended. ^c	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy ^{c,12,22,31,632,643} and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice. ¹⁹	I	B
In asymptomatic high-risk patients, ^d echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208,425}	I	B
In asymptomatic moderate-risk patients, ^e echocardiography and cardiac serum biomarkers should be considered within 12 months after completion of cancer therapy. ^{53,54,59,61,68,148,208}	IIa	B
In asymptomatic low-risk patients, ^e echocardiography and cardiac serum biomarkers may be considered within 12 months after completion of cancer therapy. ⁶³⁴	IIb	C
Cardiology referral ^f is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment. ¹¹	I	C
In selected patients with exercise intolerance persisting at 12 months after cancer treatment and with normal resting echocardiogram and cardiac biomarkers, exercise stress echocardiography and/or CPET may be considered.	IIb	C
Targeted cardiac rehabilitation should be considered in CS with high CV risk. ^{638–640}	IIa	B
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is		

recommended in patients who developed TKI-mediated hypertension during cancer therapy. ^{644,645}	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy. ^{10,237}	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy. ⁶⁴⁶	I	C

© ESC 2022

CPET, cardiopulmonary exercise testing; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; ESC, European Society of Cardiology; LQTS, long QT syndrome; TKI, tyrosine kinase inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding regulation of hypertension, DM, dyslipidaemia, smoking cessation, weight loss in case of obesity, and an adequate amount of exercise.

^dHigh-risk patients: see [Table 10](#).

^eModerate- or low-risk patients: according to CV toxicity baseline risk stratification.

^fCardio-oncology referral is recommended when available; alternatively, the patient should be referred to a cardiologist with expertise in managing CVD in patients with cancer.

8. Long-term follow-up and chronic cardiovascular complications in cancer survivors

8.1. Cancer survivors

8.1.1. Adult survivors of childhood and adolescent cancer

The survival of children and adolescents with cancer has increased considerably in recent decades, with 5-year survival rates currently exceeding 80%.⁶⁴⁷ However, the long-term health effects in the growing population of childhood and adolescent CS are a major concern.⁶⁴⁸ CTR-CVT, as a consequence of treatment with anthracyclines, mitoxantrone, and/or chest-directed RT can manifest as CTRCD but also as VHD, CAD, arrhythmias, autonomic dysfunction, pericardial disease, and premature CV mortality, depending on the type of cardiotoxic treatment.^{643,649}

CTRCD is one of the most frequent late effects in childhood CS who received cardiotoxic cancer treatment and contributes to significant morbidity and non-cancer-related mortality later in life.⁶⁵⁰ The cumulative incidence of CTRCD varies depending on the diagnostic criteria applied and the population studied and ranges from 4.8% to 10.6% at 40aFADST545 years of age.⁶⁵¹ RT to a field involving the heart increases the risk of CTRCD and valvular and vascular complications.⁶⁵²

Follow-up of paediatric CS according to the International Late Effects of Childhood Cancer Guideline Harmonization Group is recommended.⁶⁵³ This includes risk stratification based upon the total cumulative dose of anthracycline chemotherapy and MHD delivered ([TableÂ 11](#)). Annual review of CVRF and education to promote a healthy lifestyle is recommended. The frequency of CV review with TTE depends upon risk. A CV review should be considered every 5 years for moderate-risk childhood and adolescent adult CS and every 2 years for high-risk childhood and adolescent adult CS. A recent retrospective analysis has shown that quantification of LVEF >5 years after cancer diagnosis improves long-term childhood CS risk stratification. A LVEF of 40â49% is associated with an almost eight-fold increased risk for LVEF <40% at 10-year follow-up compared with patients with a preserved LVEF (â¥50%).⁶⁵⁴ Lifelong surveillance for high-risk survivors is recommended.⁷

TableÂ 11
[Open in new tab](#)

Risk categories for asymptomatic adults who are childhood and adolescent cancer survivors

Risk category	RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)	Combination therapy	
			RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)
Very high risk	>25 ^c	â¥400	>15 ^c	â¥100
High risk	>15 to 25 ^c	250â¥399	5â¥15 ^d	â¥100
Moderate risk	5â¥15 ^d	100â¥249	<5 ^e	â¥100
Low risk	<5 ^e	<100	â¥	

Risk category	RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)	Combination therapy	
			RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)
Very high risk	>25 ^c	â¥400	>15 ^c	â¥100
High risk	>15 to 25 ^c	250â¥399	5â¥15 ^d	â¥100
Moderate risk	5â¥15 ^d	100â¥249	<5 ^e	â¥100

Low risk	<5 ^e	<100	â _{FADSTS}
----------	-----------------	------	---------------------

Gy, Gray; MHD, mean heart dose; RT, radiotherapy.

a
RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category if only a small part of the heart is exposed to a relatively high prescribed dose.

b
Or doxorubicin equivalent.

c
Or prescribed RT â_{HTJ} 35â_{FADNEL} Gy to a volume exposing the heart if MHD is not available. Note that in this case, the limited information about cardiac exposure does not allow one to distinguish between high- and very high-risk categories.

d
Or prescribed RT 15â_{FADSTS} 34â_{FADNEL} Gy to a volume exposing the heart if MHD is not available.

e
Or prescribed RT < 15â_{FADNEL} Gy to a volume exposing the heart if MHD is not available.

TableÂ 11

[Open in new tab](#)

Risk categories for asymptomatic adults who are childhood and adolescent cancer survivors

Risk category	RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)	Combination therapy	
			RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)
Very high risk	>25 ^c	â _{HTJ} 400	>15 ^c	â _{HTJ} 100
High risk	>15 to 25 ^c	250â _{FADSTS} 399	5â _{FADSTS} 15 ^d	â _{HTJ} 100
Moderate risk	5â _{FADSTS} 15 ^d	100â _{FADSTS} 249	<5 ^e	â _{HTJ} 100
Low risk	<5 ^e	<100	â _{FADSTS}	

Risk category	RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)	Combination therapy	
			RT dose ^a (Gy MHD)	Total cumulative doxorubicin ^b dose (mg/m ²)
Very high risk	>25 ^c	≥400	>15 ^c	≥100
High risk	>15 to 25 ^c	250–399	5–15 ^d	≥100
Moderate risk	5–15 ^d	100–249	<5 ^e	≥100
Low risk	<5 ^e	<100		

Gy, Gray; MHD, mean heart dose; RT, radiotherapy.

^a RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category if only a small part of the heart is exposed to a relatively high prescribed dose.

^b Or doxorubicin equivalent.

^c Or prescribed RT ≥35 Gy to a volume exposing the heart if MHD is not available. Note that in this case, the limited information about cardiac exposure does not allow one to distinguish between high- and very high-risk categories.

^d Or prescribed RT 15–34 Gy to a volume exposing the heart if MHD is not available.

^e Or prescribed RT <15 Gy to a volume exposing the heart if MHD is not available.

Recommendation Table 40

[Open in new tab](#)

Recommendations for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors

Recommendations	Class ^a	Level ^b
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended. ^{655–657}	I	B
Annual screening for modifiable CVRF ^c is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment ^d is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Echocardiography surveillance should be considered every 2 years in adults who are high-risk childhood and adolescent ^e CS. ⁷	Ila	B
Echocardiography surveillance should be considered every 5 years in adults who are moderate-risk childhood and adolescent ^e CS. ^{7,654}	Ila	B

© ESC 2022

BP, blood pressure; CS, cancer survivors; CV, cardiovascular; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cObesity, sedentary lifestyle, cigarette smoking, alcohol intake, unhealthy diet, dyslipidaemia, hypertension, DM.

^dBP, lipids, fasting glucose, HbA1c, ECG, and TTE.

^eSee [Table 11](#).

Recommendations	Class ^a	Level ^b
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended. ^{655–657}	I	B
Annual screening for modifiable CVRF ^c is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment ^d is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Echocardiography surveillance should be considered every 2 years in adults who are high-risk childhood and adolescent ^e CS. ⁷	IIa	B
Echocardiography surveillance should be considered every 5 years in adults who are moderate-risk childhood and adolescent ^e CS. ^{7,654}	IIa	B

BP, blood pressure; CS, cancer survivors; CV, cardiovascular; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cObesity, sedentary lifestyle, cigarette smoking, alcohol intake, unhealthy diet, dyslipidaemia, hypertension, DM.

^dBP, lipids, fasting glucose, HbA1c, ECG, and TTE.

^eSee [Table 11](#).

© ESC 2022

Recommendation Table 40

[Open in new tab](#)

Recommendations for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors

Recommendations	Class ^a	Level ^b
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended. ^{655–657}	I	B
Annual screening for modifiable CVRF ^c is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment ^d is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Echocardiography surveillance should be considered every 2 years in adults who are high-risk childhood and adolescent ^e CS. ⁷	IIa	B
Echocardiography surveillance should be considered every 5 years in adults who are moderate-risk childhood and adolescent ^e CS. ^{7,654}	IIa	B

© ESC 2022

BP, blood pressure; CS, cancer survivors; CV, cardiovascular; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cObesity, sedentary lifestyle, cigarette smoking, alcohol intake, unhealthy diet, dyslipidaemia, hypertension, DM.

^dBP, lipids, fasting glucose, HbA1c, ECG, and TTE.

^eSee [Table 11](#).

Recommendations	Class ^a	Level ^b
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended. ^{655–657}	I	B
Annual screening for modifiable CVRF ^c is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment ^d is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Echocardiography surveillance should be considered every 2 years in adults who are high-risk childhood and adolescent ^e CS. ⁷	IIa	B
Echocardiography surveillance should be considered every 5 years in adults who are moderate-risk childhood and adolescent ^e CS. ^{7,654}	IIa	B

BP, blood pressure; CS, cancer survivors; CV, cardiovascular; CVRF, cardiovascular risk factors; DM, diabetes mellitus; ECG, electrocardiogram; HbA1c, glycated haemoglobin; RT, radiotherapy; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cObesity, sedentary lifestyle, cigarette smoking, alcohol intake, unhealthy diet, dyslipidaemia, hypertension, DM.

^dBP, lipids, fasting glucose, HbA1c, ECG, and TTE.

^eSee [Table 11](#).

© ESC 2022

8.1.2. Adult cancer survivors

Long-term cancer survivorship care is an advancing field of research. Many survivors will experience several cancer- and treatment-related late effects throughout their lives, including CTR-CVT. Besides affecting their physical and psychosocial health status, these might reduce life expectancy and quality of life. This is relevant in some cancer types, when CVD risk⁶⁶⁰ especially CTRCD risk⁶⁶² exceeds cancer mortality.^{658,659} The risk of fatal heart disease is increased more than two-fold in survivors of several solid cancers and lymphoma compared with the general population.^{660,662}

CV risk assessment at the end of therapy ([Section 7](#)) identifies CS who require long-term cardiology follow-up beyond the first 12 months after completing their cancer treatment. Asymptomatic CS with new or persisting abnormalities at their end-of-therapy assessment will be identified as at high risk for future CV events and require long-term surveillance.

Specific cancer treatments carry the highest risk of long-term CV toxicity including anthracycline chemotherapy and RT where the heart is within the RT treatment volume. Progressive RT-related CV toxicity typically develops 5â10 years after the initial treatment, and may cause CAD and HF at an incidence up to six-fold higher than in the general population. An increased CV mortality compared with the general population has been attributed to radiation-associated heart disease in Hodgkin lymphoma, non-Hodgkin lymphoma, BC, and patients with lung cancer.^{663a,665} The incidence and progression of the radiation-related CV complications depends on the dose to the CV tissue and on concomitant cancer therapies and patient characteristics, such as pre-existing CVD, CVRF, and age.^{389,400}

Late CV complications are also observed in CS who required HSCT. The incidence of HF increases up to 14.5% in women 15 years after HSCT. Risk factors for CVD following HSCT include age, anthracycline dose, chest radiation exposure, hypertension, DM, and smoking.⁶⁶⁶

Long-term follow-up surveillance, based on CV toxicity risks ([Table 12](#)), includes patient education and CVRF optimization. An annual clinical CV risk assessment is recommended for all adult CS to optimize CVRF control, promote a healthy lifestyle, and symptom review. This can be done in collaboration with primary care or a specialist in CV medicine with expertise in CVRF management. CS at high or very high risk of future CVD can be divided into those at high early risk (within 5 years of completing cancer therapy) and those at high late risk (>30 years from completing treatment). The timing and frequency of other complementary tests depends upon the risk for CTR-CVT ([Figure 38](#)).

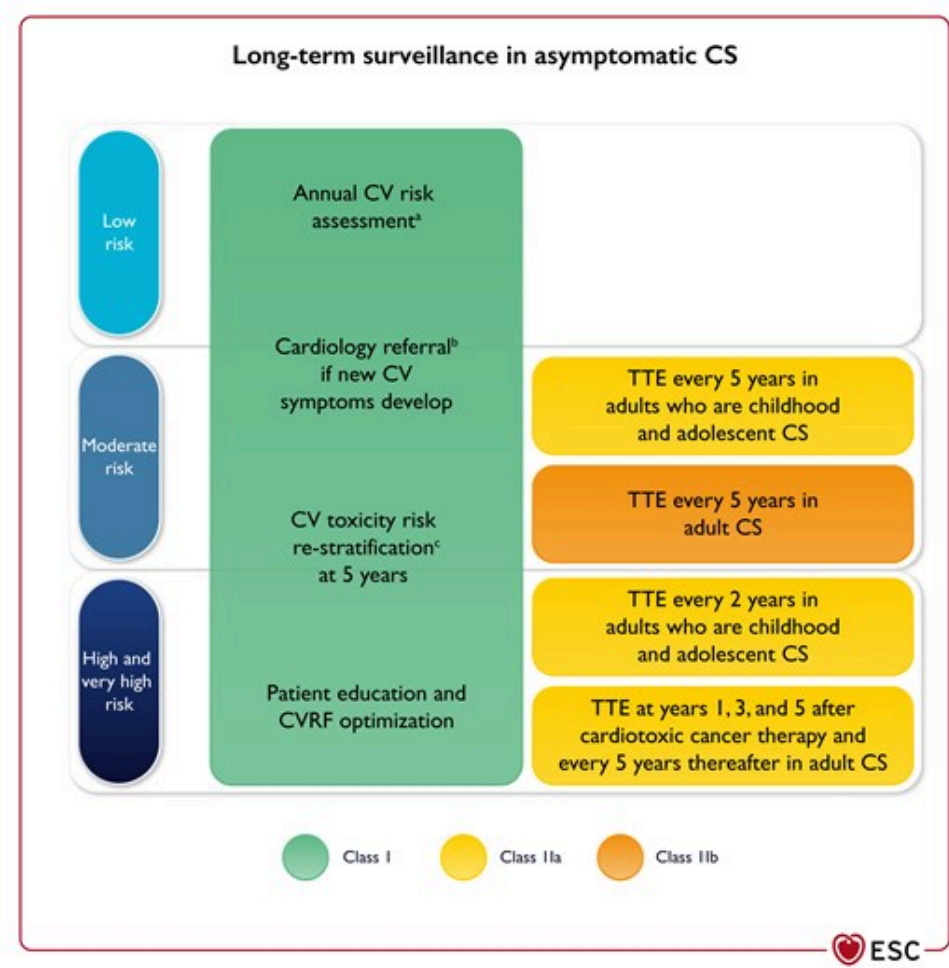


Figure 38
Long-term follow-up in cancer survivors.

BP, blood pressure; CAD, coronary artery disease; CS, cancer survivors; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; HbA1c, glycated haemoglobin; NP, natriuretic peptides; TTE, transthoracic echocardiography. ^aClinical review, BP, lipid profile, HbA1c, ECG, NP. In selected patients, non-invasive screening for CAD and carotid or renal diseases every 5â10 years, starting at 5 years after radiation may be considered. ^bCardio-oncology referral is recommended when available; alternatively, the patient should be referred to a specialized cardiologist with expertise in managing CVD in patients with cancer. ^cRestratification includes evaluation of new or pre-existing CVRF and CVD (including CTR-CVT).

[Open in new tab](#)[Download slide](#)

Table 12

[Open in new tab](#)

Risk categories for asymptomatic adult cancer survivors

Risk category ^a	Patient characteristics
Very high risk	• Very high baseline CV toxicity risk pre-treatment • Doxorubicin^b 400âmg/m² • RT > 25âGy MHD^c • RT > 15â25âGy MHD^c + doxorubicin^b 100âmg/m²
Early high risk (<5 years after therapy)	• High baseline CV toxicity risk • Symptomatic or asymptomatic moderate-to-severe CTRCD during treatment • Doxorubicin^b 250â399âmg/m² • High-risk HSCT^d
Late high risk	• RT > 15â25âGy MHD^c • RT 5â15âGy MHD^e + doxorubicin^b 100âmg/m² • Poorly controlled CVRF
Moderate risk	• Moderate baseline CV toxicity risk • Doxorubicin^b 100â249âmg/m² • RT 5â15âGy MHD^e • RT < 5âGy MHD^f + doxorubicin^b 100âmg/m²

Low risk	• Low baseline CV toxicity risk and normal end-of-therapy cardiac assessment • Mild CTRCD during therapy but recovered by the end of cancer therapy • RT < 5â^{FADNEL}Gy MHD^f • Doxorubicin^b < 100â^{FADNEL}mg/m²
-----------------	---

Risk category^a	Patient characteristics
Very high risk	• Very high baseline CV toxicity risk pre-treatment • Doxorubicin^b â^{HTJ} 400â^{FADNEL}mg/m² • RT > 25â^{FADNEL}Gy MHD^c • RT > 15â^{FADSTS}25â^{FADNEL}Gy MHD^c + doxorubicin^b â^{HTJ} 100â^{FADNEL}mg/m²
Early high risk (<5 years after therapy)	• High baseline CV toxicity risk • Symptomatic or asymptomatic moderate-to-severe CTRCD during treatment • Doxorubicin^b 250â^{FADSTS}399â^{FADNEL}mg/m² • High-risk HSCT^d
Late high risk	• RT > 15â^{FADSTS}25â^{FADNEL}Gy MHD^c • RT 5â^{FADSTS}15â^{FADNEL}Gy MHD^e + doxorubicin^b â^{HTJ} 100â^{FADNEL}mg/m² • Poorly controlled CVRF
Moderate risk	• Moderate baseline CV toxicity risk • Doxorubicin^b 100â^{FADSTS}249â^{FADNEL}mg/m² • RT 5â^{FADSTS}15â^{FADNEL}Gy MHD^e • RT < 5â^{FADNEL}Gy MHD^f + doxorubicin^b â^{HTJ} 100â^{FADNEL}mg/m²
Low risk	• Low baseline CV toxicity risk and normal end-of-therapy cardiac assessment • Mild CTRCD during therapy but recovered by the end of cancer therapy • RT < 5â^{FADNEL}Gy MHD^f

- Doxorubicin^b < 100^a mg/m²

CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; GVHD, graft vs. host disease; Gy, Gray; HSCT, haematopoietic stem cell transplantation; MHD, mean heart dose; RT, radiotherapy. References: [397](#), [399](#), [400](#), [673](#), [674](#)

a
RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category in case only a small part of the heart is exposed to a relatively high prescribed dose.

b
Or equivalent.

c
Or prescribed RT ^a35^a Gy to a volume exposing the heart if MHD is not available. Note that in this case, the limited information about cardiac exposure does not allow one to distinguish between high- and very high-risk categories.

d
High-risk HSCT patients: allogeneic HSCT; pre-existing CVD or multiple uncontrolled CVRF; cancer treatment history (mediastinal or mantle field radiation, alkylating agents, >250^a mg/m² doxorubicin or equivalent); conditioning schemes (total body irradiation, alkylating agents); development of GVHD.

e
Or prescribed RT 15^a34^a Gy to a volume exposing the heart if MHD is not available.

f
Or prescribed RT < 15^a Gy to a volume exposing the heart if MHD is not available.

Table 12

[Open in new tab](#)

Risk categories for asymptomatic adult cancer survivors

Risk category ^a	Patient characteristics
Very high risk	• Very high baseline CV toxicity risk pre-treatment • Doxorubicin^b ^a400^a mg/m² • RT > 25^a Gy MHD^c • RT > 15^a25^a Gy MHD^c + doxorubicin^b ^a100^a mg/m²

Early high risk (<5 years after therapy)	• High baseline CV toxicity risk • Symptomatic or asymptomatic moderate-to-severe CTRCD during treatment • Doxorubicin^b 250â€“399â€”mg/m² • High-risk HSCT^d
Late high risk	• RT > 15â€”25â€”Gy MHD^c • RT 5â€”15â€”Gy MHD^e + doxorubicin^b â€”100â€”mg/m² • Poorly controlled CVRF
Moderate risk	• Moderate baseline CV toxicity risk • Doxorubicin^b 100â€”249â€”mg/m² • RT 5â€”15â€”Gy MHD^e • RT < 5â€”Gy MHD^f + doxorubicin^b â€”100â€”mg/m²
Low risk	• Low baseline CV toxicity risk and normal end-of-therapy cardiac assessment • Mild CTRCD during therapy but recovered by the end of cancer therapy • RT < 5â€”Gy MHD^f • Doxorubicin^b < 100â€”mg/m²

Risk category ^a	Patient characteristics
Very high risk	• Very high baseline CV toxicity risk pre-treatment • Doxorubicin^b â€”400â€”mg/m² • RT > 25â€”Gy MHD^c • RT > 15â€”25â€”Gy MHD^c + doxorubicin^b â€”100â€”mg/m²
Early high risk (<5 years after therapy)	• High baseline CV toxicity risk • Symptomatic or asymptomatic moderate-to-severe CTRCD during treatment • Doxorubicin^b 250â€”399â€”mg/m²

	High-risk HSCT^d
Late high risk	- RT > 15^a–25^a Gy MHD^c - RT 5^a–15^a Gy MHD^e + doxorubicin^b > 100^a mg/m² - Poorly controlled CVRF
Moderate risk	- Moderate baseline CV toxicity risk - Doxorubicin^b 100^a–249^a mg/m² - RT 5^a–15^a Gy MHD^e - RT < 5^a Gy MHD^f + doxorubicin^b > 100^a mg/m²
Low risk	- Low baseline CV toxicity risk and normal end-of-therapy cardiac assessment - Mild CTRCD during therapy but recovered by the end of cancer therapy - RT < 5^a Gy MHD^f - Doxorubicin^b < 100^a mg/m²

CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; GVHD, graft vs. host disease; Gy, Gray; HSCT, haematopoietic stem cell transplantation; MHD, mean heart dose; RT, radiotherapy. References: [397](#), [399](#), [400](#), [673](#), [674](#)

a
RT risk categorization based on MHD is recommended over categorization based on prescribed dose, which may not accurately reflect cardiac radiation exposure. Depending on dose distribution and exposure of specific cardiac substructures (as well as clinical risk factors), the treatment team may judge the patient to belong to a higher risk category. In addition, a patient may be judged to belong to a lower risk category in case only a small part of the heart is exposed to a relatively high prescribed dose.

b
Or equivalent.

c
Or prescribed RT > 35^a Gy to a volume exposing the heart if MHD is not available. Note that in this case, the limited information about cardiac exposure does not allow one to distinguish between high- and very high-risk categories.

d

High-risk HSCT patients: allogenic HSCT; pre-existing CVD or multiple uncontrolled CVRF; cancer treatment history (mediastinal or mantle field radiation, alkylating agents, $>250\text{ mg/m}^2$ doxorubicin or equivalent); conditioning schemes (total body irradiation, alkylating agents); development of GVHD.

e

Or prescribed RT $15\text{--}34\text{ Gy}$ to a volume exposing the heart if MHD is not available.

f

Or prescribed RT $< 15\text{ Gy}$ to a volume exposing the heart if MHD is not available.

CS with a high or very high baseline risk and patients with abnormal LV function at the end-of-therapy assessment have a high or very high early risk, particularly in the first 2 years.^{61,667,668} Annual CV assessment with clinical examination, ECG, and NP measurement is recommended in CS. TTE should be considered at years 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter in asymptomatic very high- and early high-risk adult CS.

In adult CS with late high CTR-CVT risk (e.g. young adults with Hodgkin lymphoma or sarcomas who received a high total cumulative anthracycline dose or patients treated with high-dose radiation to a field involving the heart, e.g. Mantle RT) there is a progressive risk of CTRCD.^{661,669} Annual CV assessment with clinical examination, ECG, and NP measurement is recommended, starting 5 years after the end of treatment, provided the end-of-therapy assessment at 12 months is normal. TTE should also be considered every 5 years, as well as non-invasive screening for CAD ([Section 8.3](#)) and carotid disease ([Section 8.5](#)) according to local protocols.⁶⁷⁰

The long-term effects of CTRCD caused by trastuzumab and other targeted cancer therapies (e.g. TKI) beyond 10 years are unknown. Currently, there is no recommendation for lifelong surveillance in these CS unless they have another indication.

CV assessment with clinical examination, ECG, echocardiography, and NP measurement every 5 years should be considered in asymptomatic adult CS at moderate risk of future CTR-CVT and a normal end-of-therapy CV assessment.

Recommendation Table 41

[Open in new tab](#)

Recommendations for cardiovascular surveillance in asymptomatic adult cancer survivors

Recommendations	Class ^a	Level ^b
Annual CV risk assessment, ^c including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT. ^{d,631–633,671,672}	I	B
CV toxicity risk restratification ^e is recommended 5 years after therapy to organize long-term follow-up.	I	C
Echocardiography at years 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter should be considered in asymptomatic very high- and early high-risk adult CS. ^f	IIa	C
Echocardiography should be considered in asymptomatic late high-risk adult CS ^f starting at 5 years after radiation to a volume including the heart and then every 5 years.	IIa	C
Echocardiography may be considered every 5 years in asymptomatic moderate-risk adult CS. ^f	IIb	C
Non-invasive screening for CAD ^g should be considered every 5–10 years in asymptomatic patients who received >15 Gy MHD, ^d starting at 5 years after radiation.	IIa	C
Carotid ultrasound imaging should be considered every 5 years in asymptomatic patients with a history of head/neck RT, starting at 5 years after radiation and every 5–10 years thereafter.	IIa	C

Renal artery ultrasound should be considered in patients with a history of abdominal and pelvic radiation who present with worsening renal function and/or systemic hypertension.

IIa

C

© ESC 2022

BP, blood pressure; CAD, coronary artery disease; CMR, cardiac magnetic resonance; CS, cancer survivors; CT, computed tomography; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; HbA1c, glycated haemoglobin; MHD, mean heart dose; NP, natriuretic peptides; RT, radiotherapy.

^aClass of recommendation.

^bLevel of evidence.

^cClinical review, BP, lipid profile, HbA1c.

^dRT risk categorization based on MHD is recommended over categorization based on prescribed dose (≥ 35 Gy to a volume exposing the heart if MHD is not available).

^eRestratification includes evaluation of new or pre-existing CVRF and CVD (including CTR-CVT).

^fSee [Table 12](#).

^gStress echocardiography, cardiac CT, stress CMR, single-photon emission CT stress test, according to local protocol.²³⁴

Recommendations	Class ^a	Level ^b
Annual CV risk assessment, ^c including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT. ^{d,631–633,671,672}	I	B
CV toxicity risk restratification ^e is recommended 5 years after therapy to organize long-term follow-up.	I	C
Echocardiography at years 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter should be considered in asymptomatic very high- and early high-risk adult CS. ^f	IIa	C
Echocardiography should be considered in asymptomatic late high-risk adult CS ^f starting at 5 years after radiation to a volume including the heart and then every 5 years.	IIa	C
Echocardiography may be considered every 5 years in asymptomatic moderate-risk adult CS. ^f	IIb	C
Non-invasive screening for CAD ^g should be considered every 5–10 years in asymptomatic patients who received >15 Gy MHD, ^d starting at 5 years after radiation.	IIa	C
Carotid ultrasound imaging should be considered every 5 years in asymptomatic patients with a history of head/neck RT, starting at 5 years after radiation and every 5–10 years thereafter.	IIa	C

Renal artery ultrasound should be considered in patients with a history of abdominal and pelvic radiation who present with worsening renal function and/or systemic hypertension.

IIa

C

© ESC 2022

BP, blood pressure; CAD, coronary artery disease; CMR, cardiac magnetic resonance; CS, cancer survivors; CT, computed tomography; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; HbA1c, glycated haemoglobin; MHD, mean heart dose; NP, natriuretic peptides; RT, radiotherapy.

^aClass of recommendation.

^bLevel of evidence.

^cClinical review, BP, lipid profile, HbA1c.

^dRT risk categorization based on MHD is recommended over categorization based on prescribed dose (≥ 35 Gy to a volume exposing the heart if MHD is not available).

^eRestratification includes evaluation of new or pre-existing CVRF and CVD (including CTR-CVT).

^fSee [Table 12](#).

^gStress echocardiography, cardiac CT, stress CMR, single-photon emission CT stress test, according to local protocol.²³⁴

Recommendation Table 41

[Open in new tab](#)

Recommendations for cardiovascular surveillance in asymptomatic adult cancer survivors

Recommendations	Class ^a	Level ^b
Annual CV risk assessment, ^c including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT. ^{d,631–633,671,672}	I	B
CV toxicity risk restratification ^e is recommended 5 years after therapy to organize long-term follow-up.	I	C
Echocardiography at years 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter should be considered in asymptomatic very high- and early high-risk adult CS. ^f	IIa	C
Echocardiography should be considered in asymptomatic late high-risk adult CS ^f starting at 5 years after radiation to a volume including the heart and then every 5 years.	IIa	C
Echocardiography may be considered every 5 years in asymptomatic moderate-risk adult CS. ^f	IIb	C
Non-invasive screening for CAD ^g should be considered every 5–10 years in asymptomatic patients who received >15 Gy MHD, ^d starting at 5 years after radiation.	IIa	C
Carotid ultrasound imaging should be considered every 5 years in asymptomatic patients with a history of head/neck RT, starting at 5 years after radiation and every 5–10 years thereafter.	IIa	C

Renal artery ultrasound should be considered in patients with a history of abdominal and pelvic radiation who present with worsening renal function and/or systemic hypertension.

IIa

C

© ESC 2022

BP, blood pressure; CAD, coronary artery disease; CMR, cardiac magnetic resonance; CS, cancer survivors; CT, computed tomography; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; HbA1c, glycated haemoglobin; MHD, mean heart dose; NP, natriuretic peptides; RT, radiotherapy.

^aClass of recommendation.

^bLevel of evidence.

^cClinical review, BP, lipid profile, HbA1c.

^dRT risk categorization based on MHD is recommended over categorization based on prescribed dose (≥ 35 Gy to a volume exposing the heart if MHD is not available).

^eRestratification includes evaluation of new or pre-existing CVRF and CVD (including CTR-CVT).

^fSee [Table 12](#).

^gStress echocardiography, cardiac CT, stress CMR, single-photon emission CT stress test, according to local protocol.²³⁴

Recommendations	Class ^a	Level ^b
Annual CV risk assessment, ^c including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT. ^{d,631–633,671,672}	I	B
CV toxicity risk restratification ^e is recommended 5 years after therapy to organize long-term follow-up.	I	C
Echocardiography at years 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter should be considered in asymptomatic very high- and early high-risk adult CS. ^f	IIa	C
Echocardiography should be considered in asymptomatic late high-risk adult CS ^f starting at 5 years after radiation to a volume including the heart and then every 5 years.	IIa	C
Echocardiography may be considered every 5 years in asymptomatic moderate-risk adult CS. ^f	IIb	C
Non-invasive screening for CAD ^g should be considered every 5–10 years in asymptomatic patients who received >15 Gy MHD, ^d starting at 5 years after radiation.	IIa	C
Carotid ultrasound imaging should be considered every 5 years in asymptomatic patients with a history of head/neck RT, starting at 5 years after radiation and every 5–10 years thereafter.	IIa	C

Renal artery ultrasound should be considered in patients with a history of abdominal and pelvic radiation who present with worsening renal function and/or systemic hypertension.

IIa

C

© ESC 2022

BP, blood pressure; CAD, coronary artery disease; CMR, cardiac magnetic resonance; CS, cancer survivors; CT, computed tomography; CTR-CVT, cancer therapy-related cardiovascular toxicity; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; HbA1c, glycated haemoglobin; MHD, mean heart dose; NP, natriuretic peptides; RT, radiotherapy.

^aClass of recommendation.

^bLevel of evidence.

^cClinical review, BP, lipid profile, HbA1c.

^dRT risk categorization based on MHD is recommended over categorization based on prescribed dose (≥ 35 Gy to a volume exposing the heart if MHD is not available).

^eRestratification includes evaluation of new or pre-existing CVRF and CVD (including CTR-CVT).

^fSee [Table 12](#).

^gStress echocardiography, cardiac CT, stress CMR, single-photon emission CT stress test, according to local protocol.²³⁴

8.2. Myocardial dysfunction and heart failure

HF treatment in CS should follow the current 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF.¹⁴ Treatment with ACE-I/ARB and/or beta-blockers is recommended for both symptomatic and asymptomatic CS who have LVEF < 50% detected on CV assessment.^{14,61,208,675} In CS with mild asymptomatic CTRCD detected on CV assessment (LVEF > 50% but new fall in GLS and/or cardiac serum biomarker increase), treatment with ACE-I/ARB and/or beta-blockers may be considered.

Recommendation Table 42

[Open in new tab](#)

Recommendations for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy

Recommendations	Class ^a	Level ^b
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD. ^{c,208,425,675–678}	I	C
ACE-I/ARB and/or beta-blockers may be considered in adult CS with mild asymptomatic CTRCD. ^d	IIb	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cNew LVEF reduction by ≥ 10 percentage points to an LVEF of 40–49% OR new LVEF reduction by < 10 percentage points to an LVEF of 40–49% AND either new relative decline in GLS by $> 15\%$ from baseline OR new rise in cardiac biomarkers.

^dLVEF $\geq 50\%$ and new relative decline in GLS by $> 15\%$ from baseline AND/OR new rise in cardiac biomarkers.

Recommendations	Class ^a	Level ^b
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD. ^{c,208,425,675–678}	I	C
ACE-I/ARB and/or beta-blockers may be considered in adult CS with mild asymptomatic CTRCD. ^d	IIb	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cNew LVEF reduction by ≥ 10 percentage points to an LVEF of 40–49% OR new LVEF reduction by < 10 percentage points to an LVEF of 40–49% AND either new relative decline in GLS by $> 15\%$ from baseline OR new rise in cardiac biomarkers.

^dLVEF $\geq 50\%$ and new relative decline in GLS by $> 15\%$ from baseline AND/OR new rise in cardiac biomarkers.

Recommendation Table 42

[Open in new tab](#)

Recommendations for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy

Recommendations	Class ^a	Level ^b
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD. ^{c,208,425,675–678}	I	C
ACE-I/ARB and/or beta-blockers may be considered in adult CS with mild asymptomatic CTRCD. ^d	IIb	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cNew LVEF reduction by ≥ 10 percentage points to an LVEF of 40–49% OR new LVEF reduction by < 10 percentage points to an LVEF of 40–49% AND either new relative decline in GLS by $> 15\%$ from baseline OR new rise in cardiac biomarkers.

^dLVEF $\geq 50\%$ and new relative decline in GLS by $> 15\%$ from baseline AND/OR new rise in cardiac biomarkers.

Recommendations	Class ^a	Level ^b
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD. ^{c,208,425,675–678}	I	C
ACE-I/ARB and/or beta-blockers may be considered in adult CS with mild asymptomatic CTRCD. ^d	IIb	C

© ESC 2022

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; LVEF, left ventricular ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

^cNew LVEF reduction by ≥ 10 percentage points to an LVEF of 40–49% OR new LVEF reduction by < 10 percentage points to an LVEF of 40–49% AND either new relative decline in GLS by $> 15\%$ from baseline OR new rise in cardiac biomarkers.

^dLVEF $\geq 50\%$ and new relative decline in GLS by $> 15\%$ from baseline AND/OR new rise in cardiac biomarkers.

8.3. Coronary artery disease

Any vascular location within the RT treatment volume is at increased risk for both accelerated atherosclerosis and RT-related vasculopathy.^{173,392} RT to the chest (e.g. treatment of Hodgkin lymphoma, early-stage BC, lung and oesophageal cancer, and for some patients receiving infradiaphragmatic irradiation if the apex of the heart is within the treatment volume) increases the risk of CAD. The latency between RT and the appearance of CAD varies from a few years to several decades, depending upon the presence or absence of pre-existing atherosclerosis and the age of the patient at the time of RT. This is a serious complication for young CS with a

good prognosis and long-life expectancy (e.g. BC and Hodgkin lymphoma).^{389,390} Patients treated for mediastinal Hodgkin lymphoma have shown an increased risk of CAD as a first cardiac event.⁴⁰⁰ RT-induced CAD depends on the location of the RT treatment volume and most commonly affects either the proximal left anterior descending or the right coronary arteries. RT-related vasculopathy is progressive and typically manifests in severe, diffuse, long, smooth and concentric angiographic lesions.^{679,680}

The risk and severity of CAD increases with radiation dose, larger volume exposed, younger age at time of treatment (<25 years),³⁹⁰ time from treatment, smoking,⁴⁰⁰ the presence of other typical CVRF, type of radiation source, and concurrent metabolic risk factors.⁴⁹³ RT accelerates pre-existing atherosclerosis leading to increased ACS risk within 10 years of treatment.⁶⁸¹

Patients with RT-induced CAD undergoing PCI with bare-metal stent or balloon angioplasty have an increased risk of all-cause and CV mortality.⁶⁸² Conversely, after PCI with a drug-eluting stent, there is no difference in target lesion revascularization or cardiac mortality between patients with and without prior chest RT.⁶⁸³

Surgical revascularization in patients with prior RT may be complicated by poor tissue healing (skin and sternum), RT-induced injury to the left and right internal mammary arteries (LIMA and RIMA, respectively), inadequate target coronary vessels, and increased sternotomy-related pain.⁶⁸⁴ Pre-operative assessment of internal mammary artery viability, venous access, and sternal wound healing is recommended in CS with RT-induced CAD where CABG is considered. PCI with drug-eluting stents may be considered over CABG in CS with RT-induced severe left main or three-vessel disease, with a high SYNTAX (SYNergy between percutaneous coronary intervention with TAXus and cardiac surgery) score (>22), in whom the planned PCI is technically feasible given the increased complications associated with CABG after mediastinal RT.

Screening for CAD should be considered in high-risk patients who have received chest RT to a treatment volume including the heart. Screening should take the form of functional imaging and/or CCTA beginning at 5 years post-RT.^{234,484} The natural history of RT-related vasculopathy is different to atherosclerosis and may accelerate rapidly.¹⁷³ Functional cardiac imaging should be considered in asymptomatic CS with pre-existing CAD or when new significant CAD is detected on anatomical imaging. In asymptomatic patients with inducible ischaemia secondary to RT-induced CAD, a MDT is recommended to discuss revascularization needs according to the location of the RT-induced CAD, the ischaemia burden, LV function, arrhythmia burden, time since treatment, time since previous normal review (if available), concomitant valvular disease, risks of surgical or percutaneous revascularization, medical options, and patient preference.¹⁷³

Platinum-based chemotherapies are now recognized to cause CAD in CS. Cisplatin-based chemotherapy for testicular cancer is associated with a 1.5⁶⁸⁵7-fold increased risk of developing CAD.^{421,493,685} Testicular CS who received platinum-based chemotherapy should have their CVRF tightly controlled and be educated to report any new chest pain or cardiac symptoms to their doctor promptly. The role of screening for CAD in patients who received platinum-based chemotherapy is unknown.

Aggressive risk-factor modification and CV diagnostic work-up strongly enhance survival.^{5,672} Medical therapy with aspirin and statins for primary/secondary prevention, and beta-blockers and nitrates for symptom control, are recommended in CS.^{686,687}

Recommendations	Class ^a	Level ^b
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing ^c is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management. ^{635,688}	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C
Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
PCI may be considered in CS with radiation-induced CAD with severe left main or three-vessel disease with a high SYNTAX score (>22) in whom the procedure is technically feasible. ^{682,689,690}	IIb	B

CABG, coronary artery bypass graft; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CS, cancer survivors; LIMA, left internal mammary artery; MDT, multidisciplinary team; PCI, percutaneous coronary intervention; RIMA, right internal mammary artery; SYNTAX, SYnergy between percutaneous coronary intervention with TAXus and cardiac surgery.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to local protocols and Non-invasive imaging in coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance recommendations.²³⁴

© ESC 2022

Recommendations	Class ^a	Level ^b
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing ^c is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management. ^{635,688}	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C
Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
PCI may be considered in CS with radiation-induced CAD with severe left main or three-vessel disease with a high SYNTAX score (>22) in whom the procedure is technically feasible. ^{682,689,690}	IIb	B

CABG, coronary artery bypass graft; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CS, cancer survivors; LIMA, left internal mammary artery; MDT, multidisciplinary team; PCI, percutaneous coronary intervention; RIMA, right internal mammary artery; SYNTAX, SYnergy between percutaneous coronary intervention with TAXus and cardiac surgery.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to local protocols and Non-invasive imaging in coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance recommendations.²³⁴

© ESC 2022

Recommendation Table 43

[Open in new tab](#)

Recommendations for adult cancer survivors with coronary artery disease

Recommendations	Class ^a	Level ^b
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing ^c is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management. ^{635,688}	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C
Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
PCI may be considered in CS with radiation-induced CAD with severe left main or three-vessel disease with a high SYNTAX score (>22) in whom the procedure is technically feasible. ^{682,689,690}	IIb	B

CABG, coronary artery bypass graft; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CS, cancer survivors; LIMA, left internal mammary artery; MDT, multidisciplinary team; PCI, percutaneous coronary intervention; RIMA, right internal mammary artery; SYNTAX, SYNERgy between percutaneous coronary intervention with TAXus and cardiac surgery.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to local protocols and Non-invasive imaging in coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance recommendations.²³⁴

Recommendations	Class ^a	Level ^b
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing ^c is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management. ^{635,688}	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C
Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
PCI may be considered in CS with radiation-induced CAD with severe left main or three-vessel disease with a high SYNTAX score (>22) in whom the procedure is technically feasible. ^{682,689,690}	IIb	B

© ESC 2022

CABG, coronary artery bypass graft; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CS, cancer survivors; LIMA, left internal mammary artery; MDT, multidisciplinary team; PCI, percutaneous coronary intervention; RIMA, right internal mammary artery; SYNTAX, SYnergy between percutaneous coronary intervention with TAXus and cardiac surgery.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to local protocols and Non-invasive imaging in coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance recommendations.²³⁴

8.4. Valvular heart disease

VHD can appear in CS at any point in time but typically occurs 10 or more years after cancer treatment.⁶⁹¹ Chest RT is the main risk factor in CS, in particular at higher dose ranges, which can cause either stenosis or regurgitation, or both.³⁹¹ The reported incidences of valvular regurgitation are up to 40% of CS survivors who received high-dose chest RT to a volume involving the heart, with <10% presenting with clinically significant VHD.⁶⁷⁰

Prognosis and management depend on the extent and severity of VHD, as it does in patients without cancer.⁶⁹² TAVI should be considered for patients with symptomatic RT-induced severe aortic stenosis at intermediate surgical risk.^{504,506,693,694} Similar strategies with percutaneous mitral valve repair or replacement can be considered.⁶⁹⁵ Importantly, commonly used calculators such as STS PROM (Society of Thoracic Surgeons–Predicted Risk of Mortality) or EuroSCORE (European System for Cardiac Operative Risk Evaluation) II⁵⁰⁷ may underestimate the surgery-related risk in CS, and especially those who develop RT-induced VHD, due to additional RT-related risk factors such as pericardial calcification, aortic calcification, increased bleeding risk, impaired skin healing, and RT-related pulmonary fibrosis. A Heart Team with cardiac surgeons, interventional cardiologists, and cardio-oncology specialists should review each case to guide appropriate treatment. The Heart Team recommendation should be discussed with the patient, who can then make an informed treatment choice.

Recommendation Table 44

[Open in new tab](#)

Recommendations for adult cancer survivors with valvular heart disease

Recommendations	Class ^a	Level ^b
A MDT approach is recommended to discuss and define the surgical risk ^c in CS with severe VHD.	I	C
TAVI should be considered for patients with symptomatic severe aortic stenosis caused by radiation at intermediate surgical risk. ^{504,506,693,694,696,697}	IIa	B

© ESC 2022

CS, cancer survivors; EuroSCORE, European System for Cardiac Operative Risk Evaluation; MDT, multidisciplinary team; STS PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality; TAVI, transcatheter aortic valve implantation; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cSurgical risks include: vascular access, sternal and skin wound healing, concomitant cardiac disease, radiation-induced lung and thoracic vessels disease, aortic calcification, STS PROM/EuroSCORE II.

Recommendations	Class ^a	Level ^b
A MDT approach is recommended to discuss and define the surgical risk ^c in CS with severe VHD.	I	C

TAVI should be considered for patients with symptomatic severe aortic stenosis caused by radiation at intermediate surgical risk.^{504,506,693,694,696,697}

IIa

B

© ESC 2022

CS, cancer survivors; EuroSCORE, European System for Cardiac Operative Risk Evaluation; MDT, multidisciplinary team; STS PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality; TAVI, transcatheter aortic valve implantation; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cSurgical risks include: vascular access, sternal and skin wound healing, concomitant cardiac disease, radiation-induced lung and thoracic vessels disease, aortic calcification, STS PROM/EuroSCORE II.

Recommendation Table 44

[Open in new tab](#)

Recommendations for adult cancer survivors with valvular heart disease

Recommendations

Class^a

Level^b

A MDT approach is recommended to discuss and define the surgical risk^c in CS with severe VHD.

I

C

TAVI should be considered for patients with symptomatic severe aortic stenosis caused by radiation at intermediate surgical risk.^{504,506,693,694,696,697}

IIa

B

© ESC 2022

CS, cancer survivors; EuroSCORE, European System for Cardiac Operative Risk Evaluation; MDT, multidisciplinary team; STS PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality; TAVI, transcatheter aortic valve implantation; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cSurgical risks include: vascular access, sternal and skin wound healing, concomitant cardiac disease, radiation-induced lung and thoracic vessels disease, aortic calcification, STS PROM/EuroSCORE II.

Recommendations

Class^a

Level^b

A MDT approach is recommended to discuss and define the surgical risk^c in CS with severe VHD.

I

C

TAVI should be considered for patients with symptomatic severe aortic stenosis caused by radiation at intermediate surgical risk. [504,506,693,694,696,697](#)

IIa

B

© ESC 2022

CS, cancer survivors; EuroSCORE, European System for Cardiac Operative Risk Evaluation; MDT, multidisciplinary team; STS PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality; TAVI, transcatheter aortic valve implantation; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cSurgical risks include: vascular access, sternal and skin wound healing, concomitant cardiac disease, radiation-induced lung and thoracic vessels disease, aortic calcification, STS PROM/EuroSCORE II.

8.5. Peripheral artery disease and stroke

Peripheral arterial and cerebrovascular disease in CS can be due to the continuum of vascular disease pre-existing before, or developing during or after cancer therapy. Cancer therapies such as cisplatin, BCR-ABL inhibitors, and RT can have a direct long-lasting effect on the vasculature. Approximately 30% of CML patients on nilotinib may develop PAD, which is clinically recognized 2 to 4 years after the start of therapy.^{[698](#)} The disease process may progress even after discontinuation of nilotinib. Long-term vascular effects, generally associated with vascular reactivity, can also be seen in patients treated with ponatinib, cisplatin, and bleomycin.^{[699,700](#)} Accelerated vascular aging, inflammation, fibrosis, and atherosclerosis are characteristic consequences of RT.^{[701](#)} Up to 30% of patients may develop significant carotid artery stenoses (>70%) after head/neck radiation.^{[702,703](#)}

Vascular disease can also be an indirect consequence of cancer and its therapy, e.g. via reduction in physical activity, hyperlipidaemia, DM, obesity, hypothyroidism, and/or kidney disease. These CVRF-related effects are mostly additive to the direct treatment-related effects. Promoting vascular health and preventing vascular disease in CS is recommended.^{[672](#)} This should be in line with the 2021 ESC Guidelines on CVD prevention in clinical practice.^{[19](#)}

8.6. Pericardial complications

The risk of long-term pericardial complications after cancer drug-induced acute pericarditis, caused by anthracyclines, cyclophosphamide, cytarabine, and bleomycin, is unknown but generally considered low. Long-term dasatinib treatment may lead to pericardial effusion and pericarditis. The incidence of long-term ICI-associated pericardial complications is low.^{[10](#)}

RT-induced chronic pericardial diseases can appear months to decades after the initial RT and constrictive pericarditis is the most serious.^{[173,392](#)} Incidence is difficult to determine, and many cases are initially asymptomatic.^{[704](#)} Five-yearly echocardiographic surveillance for pericardial constriction in CS following RT-induced acute pericarditis may be considered. The absolute risk is considerably reduced with modern radiation

protocols,⁷⁰⁴ but a high rate of pericardial effusion has still been reported in patients with lung (grade â€” 2, >40%⁷⁰⁵) and oesophageal cancer (>25%⁷⁰⁶) treated with RT.

Pericardial disease has been less investigated than other RT-induced CVD, and clear protocols for post-therapy surveillance are lacking.^{707,708} In CS with chronic pericardial effusions following RT, cardiac imaging can assess for evidence of inflammation, constriction, or tamponade.⁷⁰⁹ Percutaneous balloon pericardiotomy or pericardial window creation should be used in selected cases for large or growing chronic effusions if haemodynamic compromise develops. Management of these conditions should follow general guideline recommendations.^{14,444}

Recommendation Table 45

[Open in new tab](#)

Recommendation for adult cancer survivors with pericardial complications

Recommendation	Class ^a	Level ^b	© ESC 2022
Patients with acute pericarditis during RT to a volume including the heart are at higher risk of developing chronic constrictive pericarditis, hence echocardiography surveillance every 5 years may be considered.	IIb	C	

RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendation	Class ^a	Level ^b	© ESC 2022
Patients with acute pericarditis during RT to a volume including the heart are at higher risk of developing chronic constrictive pericarditis, hence echocardiography surveillance every 5 years may be considered.	IIb	C	

RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendation Table 45

[Open in new tab](#)

Recommendation for adult cancer survivors with pericardial complications

Recommendation	Class ^a	Level ^b
Patients with acute pericarditis during RT to a volume including the heart are at higher risk of developing chronic constrictive pericarditis, hence echocardiography surveillance every 5 years may be considered.	IIb	C

RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

© ESC 2022

Recommendation	Class ^a	Level ^b
Patients with acute pericarditis during RT to a volume including the heart are at higher risk of developing chronic constrictive pericarditis, hence echocardiography surveillance every 5 years may be considered.	IIb	C

RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

© ESC 2022

8.7. Arrhythmias and autonomic disease

Arrhythmias, conduction disease, and autonomic disease are common complications in CS. Conduction disease after thoracic RT is typically associated with other CTR-CVT.⁷¹⁰ It may include AV block, bundle branch block, and sick sinus syndrome that should be monitored and treated according to the 2021 ESC Guidelines on cardiac pacing and cardiac resynchronization therapy.⁴⁴³ Patients who require valve replacement after thoracic RT have a high risk of post-operative AV block requiring permanent pacemaker therapy.⁷¹¹ Supraventricular and ventricular arrhythmias are more common in patients after thoracic RT,⁷¹² possibly due to RT-induced myocardial fibrosis. A common long-term complication after HSCT is supraventricular arrhythmia including AF and atrial flutter,⁴⁵⁷ particularly in CS treated with anthracyclines or with new CVRF or CV toxicity.

Autonomic dysfunction is an emerging but poorly understood complication observed in CS, and is most frequently observed as a late complication after thoracic RT. Orthostatic hypotension, postural orthostatic tachycardia syndrome, inappropriate sinus tachycardia, and loss of circadian heart rate variability can occur.^{713,714} Physicians caring for these patients should consider referral for autonomic evaluation. In addition, the perception of angina pain may be impaired,⁷¹⁴ making the diagnosis of post-radiation CAD challenging. Evidence-based pharmacological treatment strategies are based on studies of patients with other autonomic dysfunction aetiologies (e.g. DM or infiltrative diseases) and the reported effectiveness is generally poor.⁷¹⁴

8.8. Metabolic syndrome, lipid abnormalities, diabetes mellitus, and hypertension

There is a growing understanding about shared CVRF that may be responsible for cancer development or progression and premature CV morbi-mortality.³⁴ Modifiable CVRF continue to be underdiagnosed and undertreated in CS, especially hypertension,⁷¹⁵ obesity, DM, metabolic syndrome,⁷¹⁶ and dyslipidaemia. Early diagnosis via standardized risk-based screening and management of these conditions according to general ESC Guidelines is recommended¹⁹ to improve long-term outcomes in CS.⁶⁷²

Increasing numbers of patients with cancer are already overweight or obese at cancer diagnosis,⁷¹⁷ and additional weight gain is a frequent complication of anticancer treatments.⁷¹⁸ Obesity is associated with metabolic syndrome, worsening CVRF, and cancer. Increasing evidence indicates that being overweight increases the risk of cancer recurrence and reduces the likelihood of disease-free survival and overall survival among those diagnosed with cancer.^{719aPACST:724} There is also growing evidence to support intentional weight loss post-treatment in CS, which may result in improved prognosis and survival.⁷¹⁹ Dietary patterns characterized by a high intake of vegetables/fruits and whole grains has been shown to be associated with reduced mortality and cancer recurrence when compared with a high intake of refined grains, processed and red meats, and high-fat dairy products.^{725aPACST:727}

The identification and treatment of hyperlipidaemia in CS is associated with a profound impact on outcomes.^{182,183} There is a benefit for CS from an all-cause mortality perspective as well as for decreasing cancer recurrence.^{728aPACST:730}

Several studies have demonstrated the therapeutic benefits of exercise during primary anticancer treatment,^{731,732} and exercise is recommended during and after anticancer treatment.^{11,733} For CS,⁷³⁴ aerobic exercise results in improved survival.⁷³⁵ Based on current guidelines, patients undergoing anticancer therapy and long-term CS should be encouraged to exercise for at least 150aPACST:736 min per week.⁷³⁶

8.9. Pregnancy in cancer survivors

Improvements in the treatment of cancer have led to an increasing number of female paediatric and adolescent CS who experience pregnancy many years after their oncological treatments. Approximately 60% of them will have been previously exposed to anthracycline chemotherapy or chest RT and they have a 15-fold increase in their lifetime risk of developing HF.⁷³⁷ As young CS enter their reproductive years and contemplate pregnancy, it is important to understand the impact of cancer and its treatment on fertility, pregnancy outcomes, and CV health. There are limited data available regarding CV risk in pregnancy following cancer treatments. The overall incidence of LVD or HF associated with pregnancy in female adult CS varies according to the studied population. In a single institution report including 337 female CS treated with cardiotoxic therapies, 58 (17%) had a subsequent pregnancy.⁷³⁸ Cardiac events, defined as LVEF < 50% on two TTE or new CAD, were identified in 17 patients.⁷³⁸ Patients with cardiac events were likely to be younger at cancer diagnosis, received a higher cumulative dose of anthracycline, and had a longer delay (in years) from cancer treatment to first pregnancy compared with pregnant women with no cardiac event.⁷³⁸ In a recent meta-analysis of six studies, the weighted risk of pregnancy-associated LVD or HF in CS treated with

anthracyclines was 1.7% with no reported maternal cardiac deaths.⁷³⁹ Major risk factors for CV events during pregnancy in CS include CTRCD (incidence 28%; 47.4-fold higher odds),⁷³⁹ younger age at cancer diagnosis,^{738,740} longer time from cancer treatment to first pregnancy, and cumulative anthracycline dose.⁷³⁸

Management by an expert MDT (the pregnancy heart team) is recommended for all CS with CTRCD who are considering pregnancy.^{739,741,742} The risk of HF in CS without CTRCD is low, although vigilance remains important for potential maternal cardiac complications.

Recommendation Table 46

[Open in new tab](#)

Recommendations for cardiovascular monitoring in cancer survivors during pregnancy

Recommendations	Class ^a	Level ^b
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A baseline CV evaluation including history, physical examination, ECG, and echocardiography should be considered in all female CS who received potentially cardiotoxic cancer therapy and are considering pregnancy.	IIa	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
A second CV evaluation including echocardiography should be considered at 20 weeks of pregnancy in high-risk female CS ^c who received potentially cardiotoxic cancer therapy.	IIa	C

© ESC 2022

CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECG, electrocardiogram, NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Tables 11 and 12](#).

Recommendations	Class ^a	Level ^b
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A baseline CV evaluation including history, physical examination, ECG, and echocardiography should be considered in all female CS who received potentially cardiotoxic cancer therapy and are considering pregnancy.	IIa	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
A second CV evaluation including echocardiography should be considered at 20 weeks of pregnancy in high-risk female CS ^c who received potentially cardiotoxic cancer therapy.	IIa	C

CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECG, electrocardiogram, NP, natriuretic peptides.
^aClass of recommendation.
^bLevel of evidence.
^cSee [Tables 11 and 12](#).

© ESC 2022

Recommendation Table 46

[Open in new tab](#)

Recommendations for cardiovascular monitoring in cancer survivors during pregnancy

Recommendations	Class ^a	Level ^b
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A baseline CV evaluation including history, physical examination, ECG, and echocardiography should be considered in all female CS who received potentially cardiotoxic cancer therapy and are considering pregnancy.	IIa	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
A second CV evaluation including echocardiography should be considered at 20 weeks of pregnancy in high-risk female CS ^c who received potentially cardiotoxic cancer therapy.	IIa	C

CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECG, electrocardiogram, NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Tables 11 and 12](#).

Recommendations	Class ^a	Level ^b
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A baseline CV evaluation including history, physical examination, ECG, and echocardiography should be considered in all female CS who received potentially cardiotoxic cancer therapy and are considering pregnancy.	IIa	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
A second CV evaluation including echocardiography should be considered at 20 weeks of pregnancy in high-risk female CS ^c who received potentially cardiotoxic cancer therapy.	IIa	C

CS, cancer survivors; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; ECG, electrocardiogram, NP, natriuretic peptides.

^aClass of recommendation.

^bLevel of evidence.

^cSee [Tables 11 and 12](#).

© ESC 2022

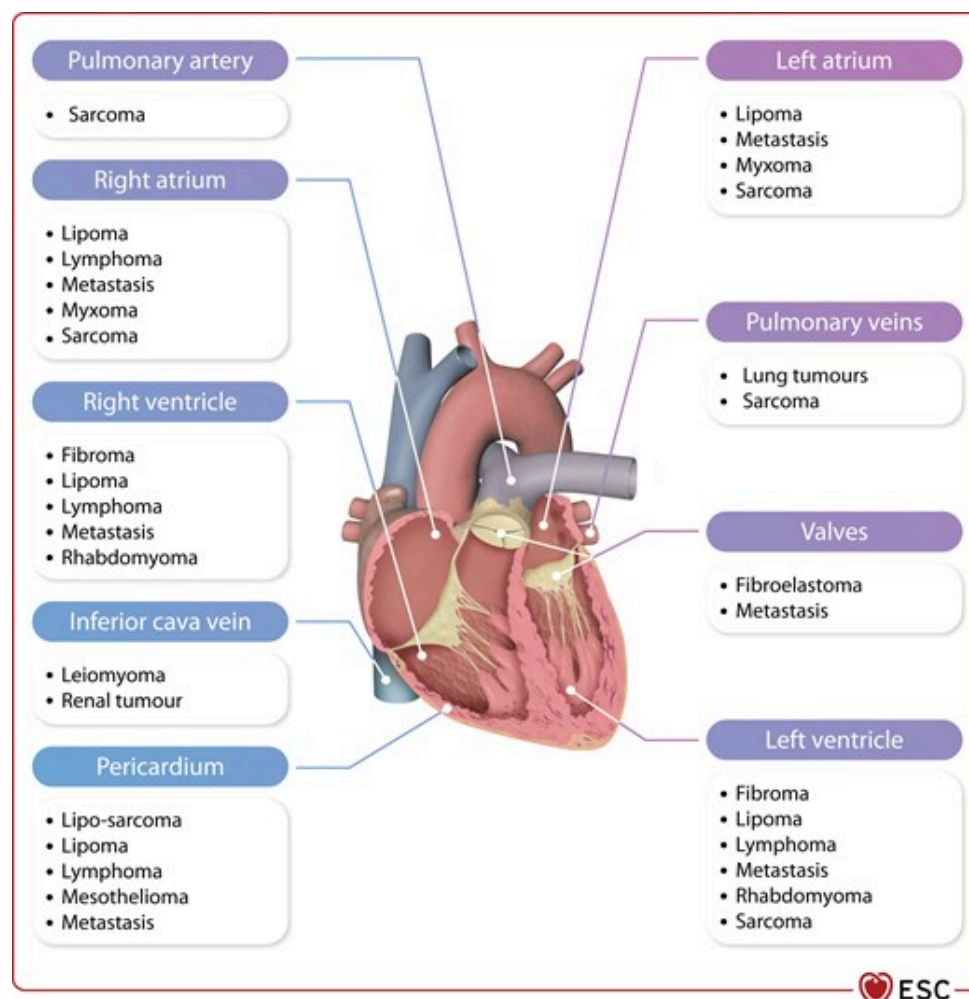
8.10. Pulmonary hypertension

Long-term clinical evaluation may be considered in patients who develop PH during therapy ([Section 6](#)). In patients with new exertional dyspnoea, fatigue, or angina, a TTE is recommended to assess the probability of PH. As TTE alone is not enough to confirm the diagnosis of PH, CS diagnosed with high PH probability require a right-heart catheterization to confirm the diagnosis. PH should be treated according to general guidelines with referral to a specialist PH service.⁶²⁰

9. Special populations

9.1. Cardiac tumours

Cardiac tumours are classified as either benign or malignant.⁷⁴³ Over 90% of primary cardiac tumours are benign (myxomas are predominant in adults, rhabdomyomas in children).⁷⁴⁴ Malignant primary tumours most commonly consist of sarcomas (approximately 65%) or lymphomas (approximately 25%).⁷⁴⁵ Cardiac metastases (from melanoma, lymphoma, leukaemia, breast, lung, and oesophageal cancers) are much more common than primary cardiac tumours ([FigureÂ 39](#)).⁷⁴⁶ Presenting symptoms are paraneoplastic (fever, weakness, fatigue), thromboembolic, haemodynamic (due to compression or obstruction from the tumour) or arrhythmic.^{747,748}



FigureÂ 39

Location of primary and secondary cardiac tumours.

[Open in new tab](#)[Download slide](#)

The diagnostic pathway should be based on knowledge about tumour type epidemiology, imaging features, and usually the requirement for a histopathological diagnosis. This topic has been extensively reviewed in



ESC CardioMed,⁷⁴⁹ and here we summarize the main recommendations for differential diagnosis and management. Differential diagnosis should exclude cardiac thrombi or the presence of chemotherapy catheters. Imaging must assess the possibilities of cardiac surgery, and may include: (1) echocardiography (initial approach using TTE or transoesophageal echocardiography)^{748,750}; (2) CMR (for cardiac tumour tissue characterization)^{751,752}; and (3) CT and PET (to distinguish malignant from benign lesions and assess for non-cardiac metastatic disease or primary cancers) ([FigureÂ 40](#)).^{753,754}

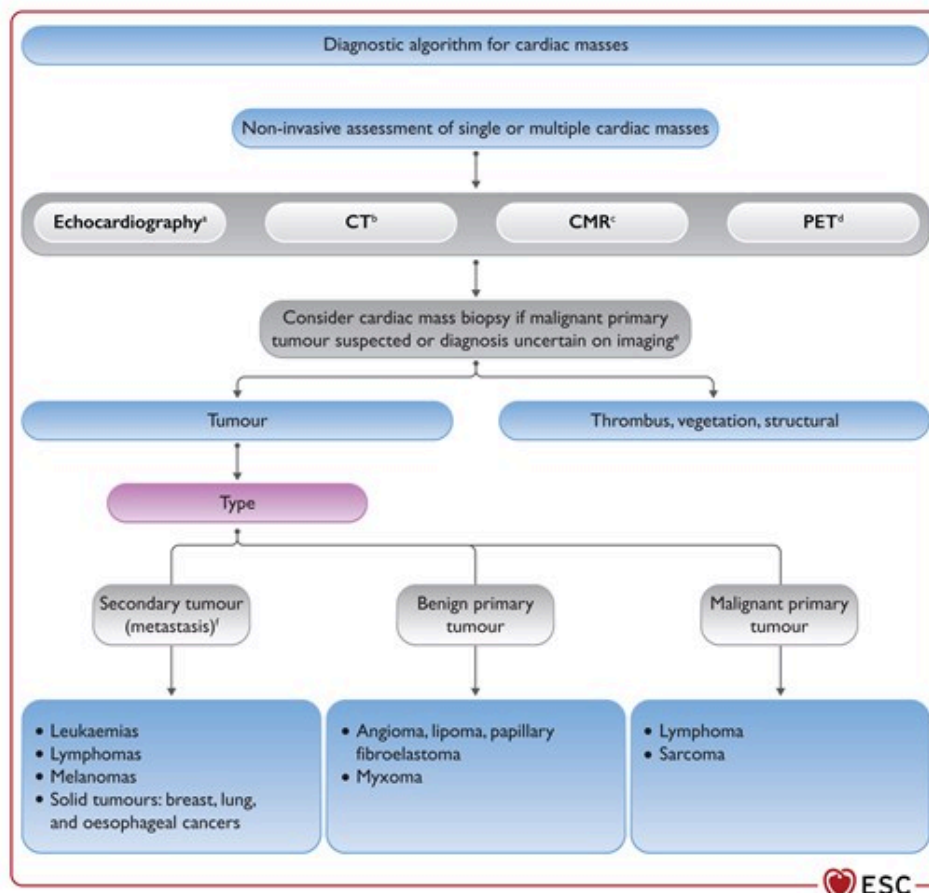


Figure 40

Diagnostic algorithm for cardiac masses.

CMR, cardiac magnetic resonance; CT, computed tomography; PET, positron emission tomography; TTE, transthoracic echocardiography. ^aTTE/transoesophageal echocardiography: location, size, and haemodynamic disturbances. Contrast echocardiography to assess vascularization. ^bIdentify primary extra-cardiac malignancy. Reveal extra-cardiac changes. Staging of malignant lesions. ^cTissue characterization (fat infiltration, necrosis, haemorrhage, calcification, and vascularization). Exclude thrombus. ^dDistinguish malignant vs. benign lesions. Staging of malignant lesions. ^eMass biopsy of suspected primary malignant cardiac tumours and/or biopsy of extracardiac masses if detected and safer to biopsy. ^f20a ^{20b}30 times more likely than primary tumours.

[Open in new tab](#)[Download slide](#)

Myxomas are primarily treated with surgery with a good prognosis. Malignant tumours are associated with a poor prognosis and evidence of the best treatment is lacking. Complete surgical resection is often impossible and adjuvant RT, systemic chemotherapy, and/or debulking palliative surgery are needed.⁷⁵⁵ Cardiac aggressive B-cell lymphomas require histopathological diagnosis (often obtained via analysis of pericardial effusion, EMB, or direct surgical biopsy) and are treated with chemotherapy, possibly followed by RT (Table 13).⁷⁵⁶

Table 13

[Open in new tab](#)

Management strategies and surgery indications for symptomatic and asymptomatic patients with benign and malignant cardiac tumours

Classification		Management strategies	Surgery indications
Benign tumours	Asymptomatic	MDT discussion is required considering: tumour type, location, size, growth rate, and likelihood of embolism. Anticoagulation should be considered for left-sided tumours or right-sided tumours associated with an intracardiac shunt, according to the individual's embolic and bleeding risk	If left-sided and endocardial: even if small and incidental, a MDT is needed to consider the indication for surgical removal due to the embolic risk
	Symptomatic	Non-surgical management for: • Rhabdomyomas (possible spontaneous regression) • Intramural haemangioma (possible response to corticosteroids) • Unresectable cases: if antiarrhythmic therapy is sufficient	Surgical resection is indicated in all other cases. For large, benign, unresectable, symptomatic cardiac tumours (obstruction, severe HF, or malignant arrhythmias), heart transplantation may be indicated in some cases
Malignant tumours	Asymptomatic	Histopathological diagnosis is required	If primary cardiac sarcoma, a complete surgical resection may increase survival
	Symptomatic	Chemotherapy and/or RT are the only therapeutic options for secondary cardiac tumours. If primary cardiac lymphoma: chemotherapy	Secondary cardiac tumours may also be treated with palliative cardiac surgery

Classification		Management strategies	Surgery indications
Benign tumours	Asymptomatic	MDT discussion is required considering: tumour type, location, size, growth rate, and likelihood of embolism. Anticoagulation should be considered for left-sided tumours or right-sided	If left-sided and endocardial: even if small and incidental, a MDT is needed to consider the indication for surgical removal due to the embolic risk

		tumours associated with an intracardiac shunt, according to the individual's embolic and bleeding risk	
	Symptomatic	Non-surgical management for: • Rhabdomyomas (possible spontaneous regression) • Intramural haemangioma (possible response to corticosteroids) • Unresectable cases: if antiarrhythmic therapy is sufficient	Surgical resection is indicated in all other cases. For large, benign, unresectable, symptomatic cardiac tumours (obstruction, severe HF, or malignant arrhythmias), heart transplantation may be indicated in some cases
Malignant tumours	Asymptomatic	Histopathological diagnosis is required	If primary cardiac sarcoma, a complete surgical resection may increase survival
	Symptomatic	Chemotherapy and/or RT are the only therapeutic options for secondary cardiac tumours. If primary cardiac lymphoma: chemotherapy	Secondary cardiac tumours may also be treated with palliative cardiac surgery

HF, heart failure; MDT, multidisciplinary team; RT, radiotherapy.

Table 13

[Open in new tab](#)

Management strategies and surgery indications for symptomatic and asymptomatic patients with benign and malignant cardiac tumours

Classification		Management strategies	Surgery indications
Benign tumours	Asymptomatic	MDT discussion is required considering: tumour type, location, size, growth rate, and likelihood of embolism. Anticoagulation should be considered for left-sided tumours or right-sided tumours associated with an intracardiac shunt, according to the individual's embolic and bleeding risk	If left-sided and endocardial: even if small and incidental, a MDT is needed to consider the indication for surgical removal due to the embolic risk

	Symptomatic	Non-surgical management for: • Rhabdomyomas (possible spontaneous regression) • Intramural haemangioma (possible response to corticosteroids) • Unresectable cases: if antiarrhythmic therapy is sufficient	Surgical resection is indicated in all other cases. For large, benign, unresectable, symptomatic cardiac tumours (obstruction, severe HF, or malignant arrhythmias), heart transplantation may be indicated in some cases
Malignant tumours	Asymptomatic	Histopathological diagnosis is required	If primary cardiac sarcoma, a complete surgical resection may increase survival
	Symptomatic	Chemotherapy and/or RT are the only therapeutic options for secondary cardiac tumours. If primary cardiac lymphoma: chemotherapy	Secondary cardiac tumours may also be treated with palliative cardiac surgery

Classification		Management strategies	Surgery indications
Benign tumours	Asymptomatic	MDT discussion is required considering: tumour type, location, size, growth rate, and likelihood of embolism. Anticoagulation should be considered for left-sided tumours or right-sided tumours associated with an intracardiac shunt, according to the individual's embolic and bleeding risk	If left-sided and endocardial: even if small and incidental, a MDT is needed to consider the indication for surgical removal due to the embolic risk
	Symptomatic	Non-surgical management for: • Rhabdomyomas (possible spontaneous regression) • Intramural haemangioma (possible response to corticosteroids) • Unresectable cases: if antiarrhythmic therapy is sufficient	Surgical resection is indicated in all other cases. For large, benign, unresectable, symptomatic cardiac tumours (obstruction, severe HF, or malignant arrhythmias), heart transplantation may be indicated in some cases

Malignant tumours	Asymptomatic	Histopathological diagnosis is required	If primary cardiac sarcoma, a complete surgical resection may increase survival
	Symptomatic	Chemotherapy and/or RT are the only therapeutic options for secondary cardiac tumours. If primary cardiac lymphoma: chemotherapy	Secondary cardiac tumours may also be treated with palliative cardiac surgery

HF, heart failure; MDT, multidisciplinary team; RT, radiotherapy.

9.2. Pregnant patients with cancer

The diagnosis of cancer during pregnancy is uncommon (1 in every 1000 pregnant women is diagnosed with cancer), with BC, melanoma, and cervical cancer being the most frequent diagnoses.⁷⁵⁷ Chemotherapy is generally not applied during the first trimester due to the high risk of foetal congenital abnormalities (up to 20%) and cytotoxic chemotherapies have different risk profiles during the second or third trimesters.^{758,759} Furthermore, chemotherapy administration is usually not given beyond week 34 of gestation to provide a 3-week window between the last cycle and delivery.⁷⁵⁷ ^{760,761} [Supplementary data, Table S19](#) summarizes the chemotherapies for pregnant patients with cancer.

Cardiac assessment prior to chemotherapy in pregnant women with cancer should consist of clinical history, physical examination, ECG, cardiac biomarker assessment and TTE ([Figure 41](#)).⁷⁴¹ Baseline and follow-up TTE should be interpreted in the context of physiological haemodynamic alterations during pregnancy. In normal pregnancy, increase in stroke volume, heart rate, and pre-load blood volume, and decrease in systemic vascular resistance, lead to an increase in cardiac output from the first trimester to 80–85% above baseline by the third trimester.^{762,764} An increase in LV mass and LV and RV volumes is observed in the third trimester. During normal pregnancy, LVEF is usually unchanged and can be used for CTRCD monitoring. Although NP and cTn may be slightly elevated during normal pregnancy (NT-proBNP < 300 ng/L, BNP < 100 pg/mL,¹⁴ and hs-cTnT^{765,766}), serial evaluation may be useful for close CTRCD monitoring during cancer treatment with the higher cut-off NP levels for pregnancy.

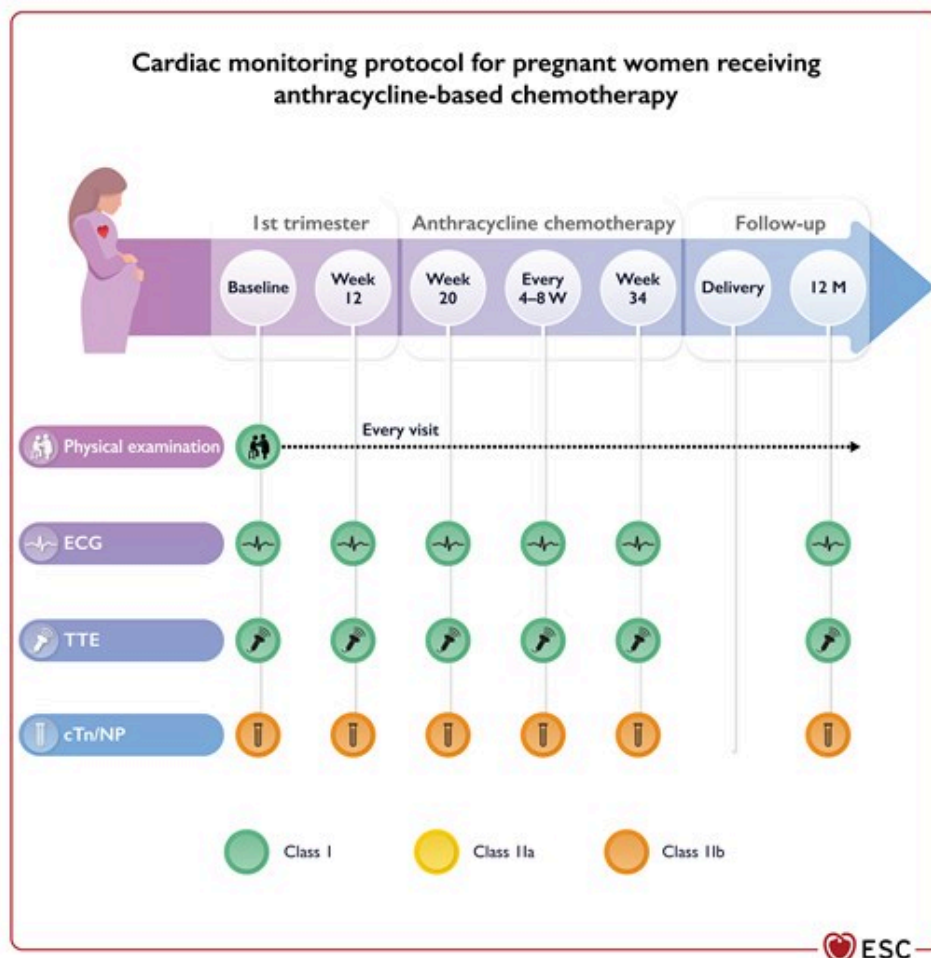


Figure 41

Cardiac monitoring protocol for pregnant women receiving anthracycline-based chemotherapy.

cTn, cardiac troponin; ECG, electrocardiogram; M, months; NP, natriuretic peptides; TTE, transoesophageal echocardiography; W, week.

[Open in new tab](#) [Download slide](#)

The topic of CVD during pregnancy has been extensively reviewed in the 2018 ESC Guidelines for the management of CVD during pregnancy.⁷⁴¹ Here we focus on specific recommendations in pregnant women with cancer receiving anthracycline chemotherapy.

9.2.1. Left ventricular dysfunction and heart failure

Medical history evaluating signs and symptoms of HF should be performed at every clinical visit of pregnant women with cancer receiving anthracycline chemotherapy. More frequent CV evaluations with TTE during treatments with potential CTRCD risk should be advised (e.g. every 4-8 weeks or every two cycles for a 3-weekly anthracycline chemotherapy cycle). The management of clinical HF or asymptomatic LVD during pregnancy is fully described in the 2018 ESC Guidelines for the management of CVD during pregnancy.⁷⁴¹

9.2.2. Venous thromboembolism and pulmonary embolism

Pregnant patients with cancer have an increased risk of developing VTE, especially when hospitalized.⁷⁶⁷⁻⁷⁶⁹ Identified risks for VTE in pregnant patients include having a history of BC or previous chemotherapy in the past 6 months. Recommendations for the diagnosis and treatment of PE during

pregnancy are the same as in the general 2018 ESC Guidelines for the management of CVD during pregnancy⁷⁴¹ and 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism.⁵⁶⁶

Determination of VTE risk score and the use of thromboprophylaxis protocols may be useful to prevent maternal morbidity and/or mortality due to VTE.⁷⁷⁰ LMWH have become the drug of choice for the prophylaxis and treatment of VTE in pregnant patients.⁷⁴¹ The recommendation for thromboprophylaxis should be individualized, weighing the risks of bleeding vs. thromboembolism in pregnant patients with cancer.

Recommendation Table 47

[Open in new tab](#)

Recommendations for cardiovascular assessment and monitoring of pregnant women with cancer

Recommendations	Class ^a	Level ^b
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy. ⁷⁴¹	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography. ⁷⁴¹	I	C
Monthly or bimonthly CV evaluation, including TTE, should be considered during cardiotoxic cancer therapy ^c in pregnant women with cancer.	IIa	C
cTn may be considered at baseline and during anthracycline chemotherapy in pregnant women with cancer.	IIb	C

© ESC 2022

cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; MDT, multidisciplinary team; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cFor patients receiving anthracycline-based chemotherapy.

Recommendations	Class ^a	Level ^b
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy. ⁷⁴¹	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography. ⁷⁴¹	I	C
Monthly or bimonthly CV evaluation, including TTE, should be considered during cardiotoxic cancer therapy ^c in pregnant women with cancer.	IIa	C
cTn may be considered at baseline and during anthracycline chemotherapy in pregnant women with cancer.	IIb	C

© ESC 2022

cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; MDT, multidisciplinary team; TTE, transthoracic echocardiography.
^aClass of recommendation.
^bLevel of evidence.
^cFor patients receiving anthracycline-based chemotherapy.

Recommendation Table 47

[Open in new tab](#)

Recommendations for cardiovascular assessment and monitoring of pregnant women with cancer

Recommendations	Class ^a	Level ^b
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy. ⁷⁴¹	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography. ⁷⁴¹	I	C
Monthly or bimonthly CV evaluation, including TTE, should be considered during cardiotoxic cancer therapy ^c in pregnant women with cancer.	IIa	C
cTn may be considered at baseline and during anthracycline chemotherapy in pregnant women with cancer.	IIb	C

© ESC 2022

cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; MDT, multidisciplinary team; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cFor patients receiving anthracycline-based chemotherapy.

Recommendations	Class ^a	Level ^b
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy. ⁷⁴¹	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography. ⁷⁴¹	I	C
Monthly or bimonthly CV evaluation, including TTE, should be considered during cardiotoxic cancer therapy ^c in pregnant women with cancer.	IIa	C
cTn may be considered at baseline and during anthracycline chemotherapy in pregnant women with cancer.	IIb	C

© ESC 2022

cTn, cardiac troponin; CV, cardiovascular; ECG, electrocardiogram; MDT, multidisciplinary team; TTE, transthoracic echocardiography.

^aClass of recommendation.

^bLevel of evidence.

^cFor patients receiving anthracycline-based chemotherapy.

9.3. Carcinoid valvular heart disease

Carcinoid tumours represent rare neuroendocrine malignancies originating from the enterochromaffin cells (Figure 42).⁷⁷¹ Carcinoid syndrome is a rare cause of acquired VHD including mainly right-sided valvular lesions, but also left-sided involvement, pericardial effusion, and myocardial metastases.⁷⁷² Coronary artery vasospasm and paroxysmal atrial or ventricular tachycardias may rarely occur due to sympathetic stimulation. Cardiac metastases are reported with an incidence of 3.8% on the ventricles, confirmed by PET-CT scans.^{773,774} Data from the SEER (Surveillance, Epidemiology, and End Results) registry identified that approximately 20% of patients with neuroendocrine malignancies develop carcinoid syndrome (7.6%–32.4%), which is associated with shorter survival (4.7 years compared with 7.1 years in patients without carcinoid syndrome) and poor quality of life.⁷⁷⁵ It is estimated that 20%–50% of these patients present cardiac involvement, especially of the right-sided cardiac valves.⁷⁷¹ In the presence of a patent foramen ovale, interatrial shunt, primary bronchial neuroendocrine tumour, or extensive liver metastases, humoral substances directly enter the systemic circulation, causing left-sided valvular involvement in up to one-third of cases.⁷⁷⁶

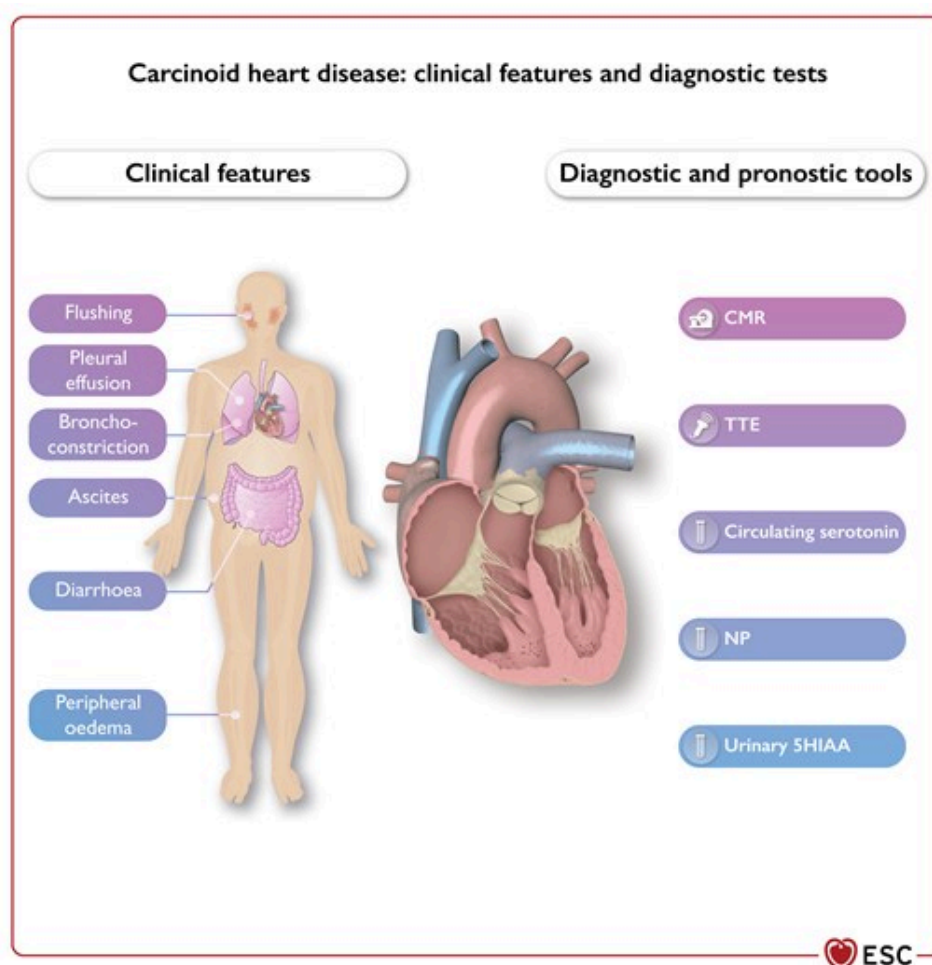


Figure 42

Carcinoid heart disease: clinical features and diagnostic tests.

5HIAA, 5-hydroxyindoleacetic acid; CMR, cardiac magnetic resonance; NP, natriuretic peptides; TTE, transthoracic echocardiography.

[Open in new tab](#)[Download slide](#)

NP should be considered for screening and surveillance of patients at risk of carcinoid cardiac involvement and TTE is recommended in patients with NT-proBNP > 260â€”pg/mL or clinical signs or symptoms.^{777,780} In asymptomatic patients with NT-proBNP < 260â€”pg/mL, repeat clinical and NP assessment should be considered every 6 months.

Survival has improved in carcinoid tumours, with the use of somatostatin analogues and surgical techniques in liver metastasis. However, right HF still represents a major cause of death.^{781,782} Many patients with severe tricuspid regurgitation due to carcinoid syndrome require both tricuspid and pulmonary valve surgery.⁷⁸³ Administration of i.v. somatostatin analogues (e.g. octreotide) is recommended to avoid a peri-operative carcinoid crisis. The infusion should be started on the morning of the procedure (up to 12â€”h pre-operatively), continued throughout the procedure (surgery, pre-operative coronary angiography, pacemaker implantation), and post-operatively for at least 48â€”h following valve surgery or until stable if a carcinoid crisis is triggered post-operatively.⁷⁷²

The optimal choice of valve prosthesis is still a matter of debate due to the balance of risk of both accelerated bioprosthetic valve degeneration vs. bleeding risks in patients with extensive liver metastases requiring therapeutic anticoagulation for mechanical valves.^{784,785} Complications include AV block, requiring pacemaker implantation in 25% of patients.⁷⁸⁶ Frequently, the reduced RV function does not improve despite tricuspid valve replacement and HF persists.⁷⁸⁷ Thrombus formation on the tricuspid bioprosthesis can occur, especially during the first 3 months post-operatively, and oral anticoagulation with VKA may be considered. Persistent serotonin elevation can cause recurrent bioprosthesis valve fibrosis. Valve-in-valve transcatheter intervention has been reported in bioprosthetic valve failure in metastatic carcinoid heart disease; however, future research is needed to define its role.^{783,788,789}

In patients with left-sided carcinoid valvular involvement, closure of interatrial shunts should be considered, although only sparse data exist for this approach.

Recommendation Table 48

[Open in new tab](#)

Recommendations for carcinoid valvular heart diseases

Recommendations	Class ^a	Level ^b
Echocardiography ^c is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status. ^{772,790,791}	I	B
NP should be considered for screening and surveillance of carcinoid heart disease every 6 months. ^{777–780}	IIa	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months. ^{d,783,785}	I	C
Valve replacement surgery should be considered in patients with asymptomatic severe carcinoid tricuspid or pulmonary VHD, progressive RV dysfunction/dilatation, and an expected survival ≥ 12 months. ^{d,772}	IIa	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months. ^{783,785}	I	C

MDT, multidisciplinary team; NP, natriuretic peptides; RV, right ventricular; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding saline contrast infusion at baseline to rule out patent foramen ovale.

^dWith controlled serotonin concentrations.

Recommendations	Class ^a	Level ^b
Echocardiography ^c is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status. ^{772,790,791}	I	B
NP should be considered for screening and surveillance of carcinoid heart disease every 6 months. ^{777–780}	IIa	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months. ^{d,783,785}	I	C
Valve replacement surgery should be considered in patients with asymptomatic severe carcinoid tricuspid or pulmonary VHD, progressive RV dysfunction/dilatation, and an expected survival ≥ 12 months. ^{d,772}	IIa	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months. ^{783,785}	I	C

MDT, multidisciplinary team; NP, natriuretic peptides; RV, right ventricular; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding saline contrast infusion at baseline to rule out patent foramen ovale.

^dWith controlled serotonin concentrations.

© ESC 2022

Recommendation Table 48

[Open in new tab](#)

Recommendations for carcinoid valvular heart diseases

Recommendations	Class ^a	Level ^b
Echocardiography ^c is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status. ^{772,790,791}	I	B
NP should be considered for screening and surveillance of carcinoid heart disease every 6 months. ^{777–780}	IIa	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months. ^{d,783,785}	I	C
Valve replacement surgery should be considered in patients with asymptomatic severe carcinoid tricuspid or pulmonary VHD, progressive RV dysfunction/dilatation, and an expected survival ≥ 12 months. ^{d,772}	IIa	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months. ^{783,785}	I	C

MDT, multidisciplinary team; NP, natriuretic peptides; RV, right ventricular; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding saline contrast infusion at baseline to rule out patent foramen ovale.

^dWith controlled serotonin concentrations.

Recommendations	Class ^a	Level ^b
Echocardiography ^c is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status. ^{772,790,791}	I	B
NP should be considered for screening and surveillance of carcinoid heart disease every 6 months. ^{777–780}	IIa	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months. ^{d,783,785}	I	C
Valve replacement surgery should be considered in patients with asymptomatic severe carcinoid tricuspid or pulmonary VHD, progressive RV dysfunction/dilatation, and an expected survival ≥ 12 months. ^{d,772}	IIa	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months. ^{783,785}	I	C

MDT, multidisciplinary team; NP, natriuretic peptides; RV, right ventricular; VHD, valvular heart disease.

^aClass of recommendation.

^bLevel of evidence.

^cIncluding saline contrast infusion at baseline to rule out patent foramen ovale.

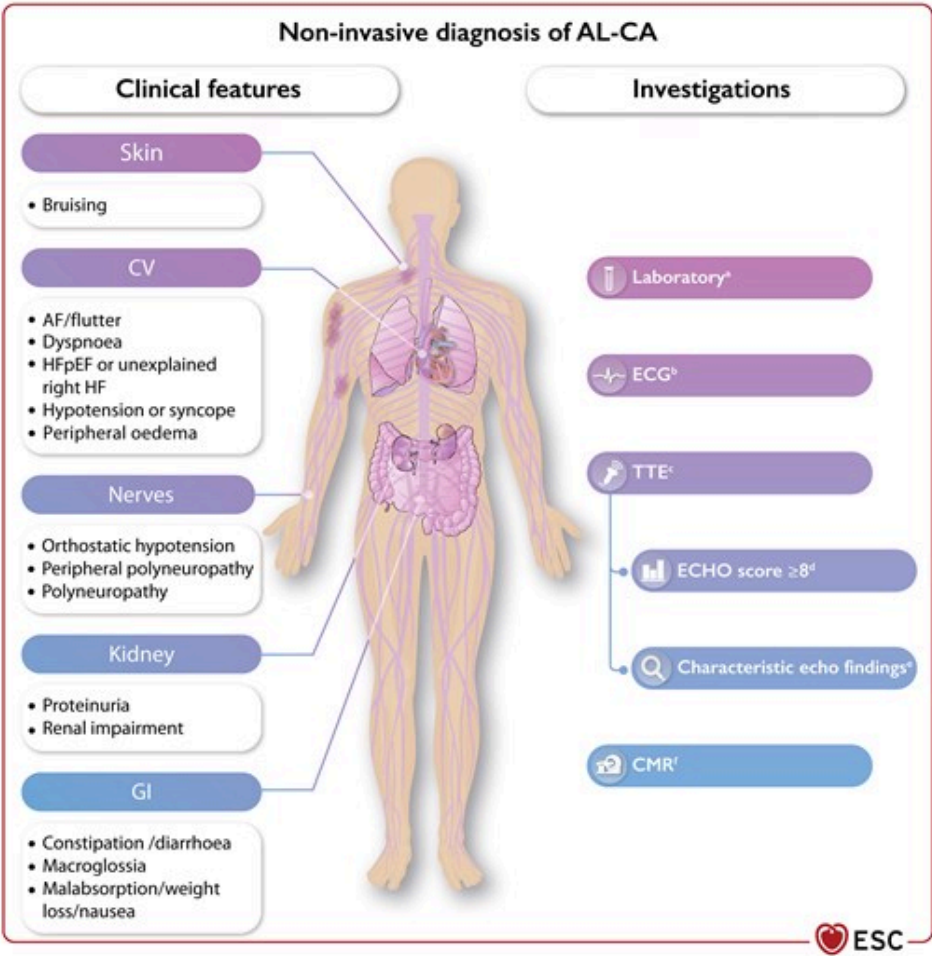
^dWith controlled serotonin concentrations.

© ESC 2022

9.4. Amyloid light-chain cardiac amyloidosis

Amyloid light-chain amyloidosis is a plasma cell dyscrasia, which is typically treated with therapies very similar to those used in MM, including PI-based therapy.⁷⁹² It can occur in conjunction with myeloma or independently as a light-chain protein-producing disorder. Amyloid light-chain amyloidosis is a systemic

disease ^{793,794} and it is critical to have a high degree of suspicion for the diagnosis of cardiac involvement (amyloid light-chain cardiac amyloidosis [AL-CA]) because a combination of specialized tests is needed to make an accurate diagnosis ([FigureÂ 43](#)).^{290,793,795,796} Cardiac serum biomarkers are an essential step in the diagnostic and prognostic assessments for these patients.^{797a}^{798,799} AL-CA has been extensively reviewed in a recent position paper from the Working Group on Myocardial and Pericardial Diseases.²⁹⁰



FigureÂ 43
Non-invasive diagnosis of amyloid light-chain cardiac amyloidosis.

^a $\bar{a}a_{PAD2}$, late diastolic velocity of mitral annulus obtained by tissue Doppler imaging; AF, atrial fibrillation; AL-CA, amyloid light-chain cardiac amyloidosis; CMR, cardiac magnetic resonance; CV, cardiovascular; E, mitral inflow early diastolic velocity obtained by pulsed wave; $\bar{e}a_{PAD2}$, early diastolic velocity of mitral annulus obtained by tissue doppler imaging; ECG, electrocardiogram; echo, echocardiography; ECV, extracellular volume fraction; GI, gastrointestinal; GLS, global longitudinal strain; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; IVS, interventricular septum; LGE, late gadolinium enhancement; LV, left ventricular; LVEDD, left ventricular end diastolic diameter; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PW, left ventricular posterior wall; $\bar{s}a_{PAD2}$, systolic velocity of tricuspid annulus obtained by Doppler tissue imaging; SPEP, serum protein electrophoresis; TAPSE, tricuspid annular plane systolic excursion; TTE, transthoracic echocardiography; UPEP, urine protein electrophoresis. Individually, the clinical manifestations and findings on cardiac testing for AL-CA are non-specific. Integration of all clinical and diagnostic findings is necessary when assessing the likelihood of the diagnosis. ^aDisproportionately high NT-proBNP; persisting elevated troponin levels; abnormal free light-chain levels (AL-CA); positive SPEP and/or UPEP (AL-CA).

^bDisproportionally low QRS voltage; early conduction system disease; pseudo-infarct pattern. ^cUnexplained LV thickness ≥ 12 mm + 1 or 2 characteristic echo findings or ECHO score ≥ 8 ; idiopathic pericardial effusion. ^dECHO score: relative LV wall thickness (IVS + PW/LVEDD) > 0.6 (3 points), Doppler E/e' > 11 (1 point); TAPSE ≥ 19 mm (2 points); GLS $\geq -13\%$ (1 point); systolic longitudinal strain apex to base ratio > 2.9 (3 points). ^eCharacteristic echocardiography findings: grade ≥ 2 diastolic dysfunction; reduced s_a , e_a , and a_a velocities (< 5 cm/s); decreased GLS to $\geq -15\%$. ^fDiffuse subendocardial or transmural LGE; elevated native T1 values; abnormal gadolinium kinetics (myocardial nulling preceding or coinciding with the blood pool); ECV $\geq 0.40\%$ (strongly supportive).

[Open in new tab](#)[Download slide](#)

The classical non-invasive definition of AL-CA is based on clinical suspicion, biomarkers, TTE, CMR, and nuclear scintigraphy criteria ([Figure 43](#)). Persistent troponin elevation and disproportionately high NT-proBNP (generally > 300 ng/L in the absence of renal failure or AF) to ventricular function parameters on TTE is a characteristic red flag for AL-CA.⁸⁰⁰ A decrease in GLS with a distinctive apical sparing pattern (preserved GLS values in the LV apical region) is considered specific for cardiac amyloidosis, although it is not helpful to distinguish between amyloid light-chain and transthyretin amyloidosis.⁸⁰¹ Additionally, GLS $\geq -15\%$ may serve as an independent prognostic factor of poor overall survival in patients with AL-CA.⁸⁰² CMR with LGE and parametric imaging has emerged as a new non-invasive gold-standard for diagnosis ([Figure 43](#)).^{803,804} Nuclear scintigraphy can differentiate transthyretin amyloidosis from AL-CA supported by the presence of monoclonal protein.²⁹⁰ EMB should be considered in patients with suspected AL-CA involvement if CMR is not diagnostic.²⁹⁰ A rare condition that may coexist with AL-CA is light-chain deposition disease, which frequently associates extensive renal involvement and poor prognosis.⁷⁹⁹

Recently, a staging system for AL-CA has demonstrated the prognostic impact of cTnT and NT-proBNP levels.⁷⁹⁷ Heart progression criteria are defined by NT-proBNP progression ($> 30\%$ and > 300 ng/L increase), cTnT progression ($\geq 33\%$ increase) or ejection fraction decrease ($\geq 10\%$ decrease).^{805,806,807} However, evaluating a cardiac response to treatment using a decrease in NT-proBNP levels and New York Heart Association class improvement is still challenging.

AL-CA frequently results in HF, major cardiac arrhythmias, orthostatic hypotension, sudden cardiac death, and an increased risk of arterial and venous thrombosis.^{808,809,810} Beta-blockers, ACE-I, ARB, or angiotensin receptor-neprilysin inhibitor may not be well tolerated because of hypotension.²⁹⁰ The management of AF is very complex in this population. Amiodarone is the preferred antiarrhythmic treatment and digitalis should be used with caution. Anticoagulation is recommended in all AL-CA patients with AF independent of the CHA₂DS₂-VASc score due to the high prothrombotic risk unless there is a contraindication.²⁹⁰ Currently, the guidelines for implanted devices, including pacemakers and ICDs, do not provide specific recommendations for AL-CA and decisions should be individualized after a MDT discussion.⁸¹¹

Optimal systemic therapy for AL-CA is rapidly changing, and the efficacy of certain combination therapies continues to improve.^{812,813} Autologous HSCT for AL-CA is not universally utilized but is a viable treatment option.⁸¹⁴ Therapies for AL-CA are evolving, and daratumumab and PI show promise for improved

outcomes.^{792,815a,817} Clinical observations, but no RCT evidence, suggest the potential role of doxycycline to improve survival in patients with AL-CA.^{818,819}

Recommendation Table 49

[Open in new tab](#)

Recommendations for amyloid light-chain cardiac amyloidosis diagnosis and monitoring

Recommendations	Class ^a	Level ^b
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia. ^{290,820–822}	I	B
CMR is recommended in patients with suspected AL-CA. ^{290,803,804}	I	A
EMB should be considered in patients with suspected AL-CA involvement if CMR is not diagnostic. ²⁹⁰	IIa	C
Admission with inpatient ECG monitoring should be considered for high-risk patients with AL-CA requiring PI during their first cycle of therapy. ^{c,808,811}	IIa	C

© ESC 2022

AL-CA, amyloid light-chain cardiac amyloidosis; CMR, cardiac magnetic resonance; cTn, cardiac troponin; ECG, electrocardiogram; EMB, endomyocardial biopsy; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; NP, natriuretic peptides; PI, proteasome inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to baseline evaluation using HFA-ICOS PI risk assessment tools (see [Section 4](#)).

Recommendations	Class ^a	Level ^b
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia. ^{290,820–822}	I	B
CMR is recommended in patients with suspected AL-CA. ^{290,803,804}	I	A
EMB should be considered in patients with suspected AL-CA involvement if CMR is not diagnostic. ²⁹⁰	IIa	C
Admission with inpatient ECG monitoring should be considered for high-risk patients with AL-CA requiring PI during their first cycle of therapy. ^{c,808,811}	IIa	C

AL-CA, amyloid light-chain cardiac amyloidosis; CMR, cardiac magnetic resonance; cTn, cardiac troponin; ECG, electrocardiogram; EMB, endomyocardial biopsy; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; NP, natriuretic peptides; PI, proteasome inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to baseline evaluation using HFA-ICOS PI risk assessment tools (see [Section 4](#)).

© ESC 2022

Recommendation Table 49

[Open in new tab](#)

Recommendations for amyloid light-chain cardiac amyloidosis diagnosis and monitoring

Recommendations	Class ^a	Level ^b
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia. ^{290,820–822}	I	B
CMR is recommended in patients with suspected AL-CA. ^{290,803,804}	I	A
EMB should be considered in patients with suspected AL-CA involvement if CMR is not diagnostic. ²⁹⁰	IIa	C
Admission with inpatient ECG monitoring should be considered for high-risk patients with AL-CA requiring PI during their first cycle of therapy. ^{c,808,811}	IIa	C

© ESC 2022

AL-CA, amyloid light-chain cardiac amyloidosis; CMR, cardiac magnetic resonance; cTn, cardiac troponin; ECG, electrocardiogram; EMB, endomyocardial biopsy; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; NP, natriuretic peptides; PI, proteasome inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to baseline evaluation using HFA-ICOS PI risk assessment tools (see [Section 4](#)).

Recommendations	Class ^a	Level ^b
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia. ^{290,820–822}	I	B
CMR is recommended in patients with suspected AL-CA. ^{290,803,804}	I	A
EMB should be considered in patients with suspected AL-CA involvement if CMR is not diagnostic. ²⁹⁰	IIa	C
Admission with inpatient ECG monitoring should be considered for high-risk patients with AL-CA requiring PI during their first cycle of therapy. ^{c,808,811}	IIa	C

© ESC 2022

AL-CA, amyloid light-chain cardiac amyloidosis; CMR, cardiac magnetic resonance; cTn, cardiac troponin; ECG, electrocardiogram; EMB, endomyocardial biopsy; HFA, Heart Failure Association; ICOS, International Cardio-Oncology Society; NP, natriuretic peptides; PI, proteasome inhibitors.

^aClass of recommendation.

^bLevel of evidence.

^cAccording to baseline evaluation using HFA-ICOS PI risk assessment tools (see [Section 4](#)).

9.5. Cardiac implantable electronic devices

RT can cause malfunction of cardiac implantable electronic devices (CIEDs).^{443,823} The risk of RT-induced CIED malfunction generally increases with the radiation dose,^{824,825} although the strongest predictor of malfunction is the magnitude of exposure to neutron emission from high-energy photon RT, conventionally defined as a beam energy >10 megavolts (MV).^{824,826,827} Non-neutron-producing treatment is therefore preferable in patients with a CIED.⁸²⁶

RT-induced CIED malfunction can manifest in: (1) transient interference, with inappropriate triggering during the irradiation only; (2) a reset, reverting to backup settings, recoverable with device reprogramming; and, rarely (3) permanent damage to the device due to direct CIED irradiation.^{826,827}

The clinical consequences of a CIED malfunction include the inhibition of pacing and inappropriate pacing at maximum sensor rate.⁸²⁶ The clinical effects of device malfunction are greatest when the patient is pacing-dependent. Theoretically, oversensing might lead to inappropriate ICD shocks, although this has not been reported in the literature.⁸²⁶

More recent registries have reported minimal or no adverse effects of RT on CIED malfunction.^{827,828} Nevertheless, as it is not possible to predict the behaviour of a CIED within or close to an RT treatment volume, general recommendations should be followed to minimize patient risk ([Figures 44-46](#)).^{188,824,825}

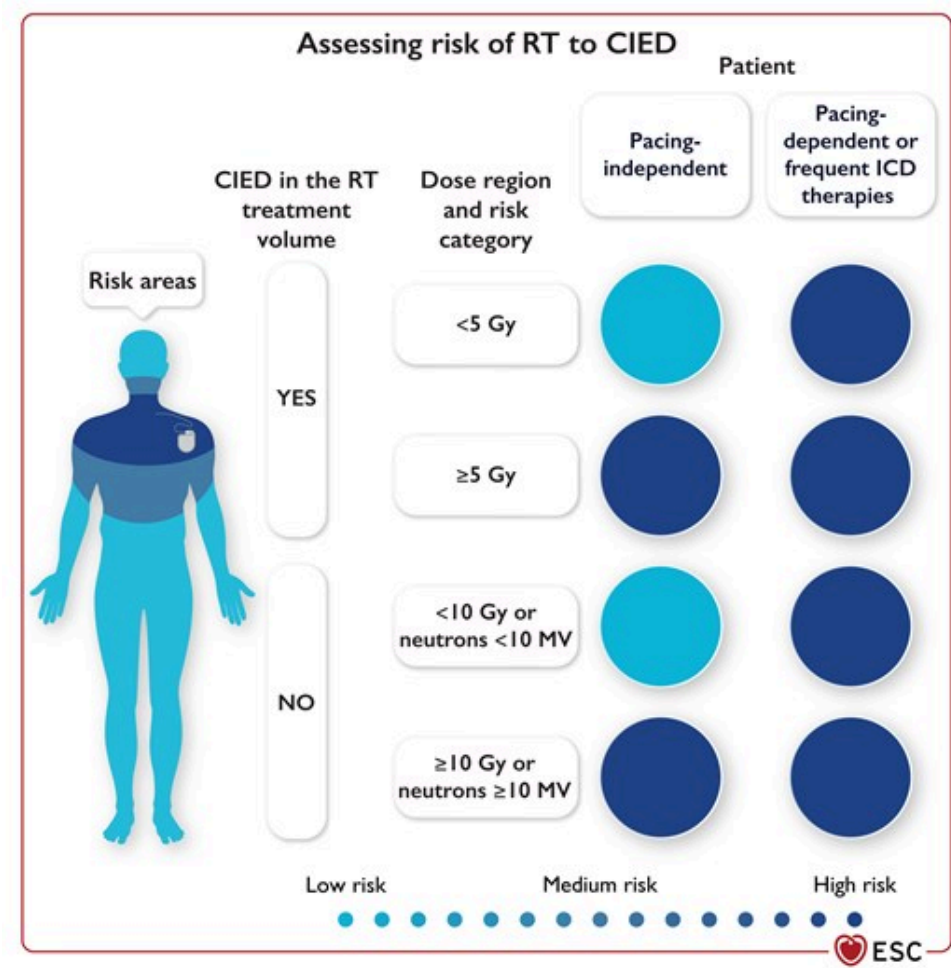


Figure 44

Risk stratification in patients with a cardiac implantable electronic device undergoing radiotherapy.

CIED, cardiac implantable electronic device; Gy, Gray; ICD, implantable cardioverter defibrillator; MV, megavolt; RT, radiotherapy.

[Open in new tab](#)[Download slide](#)

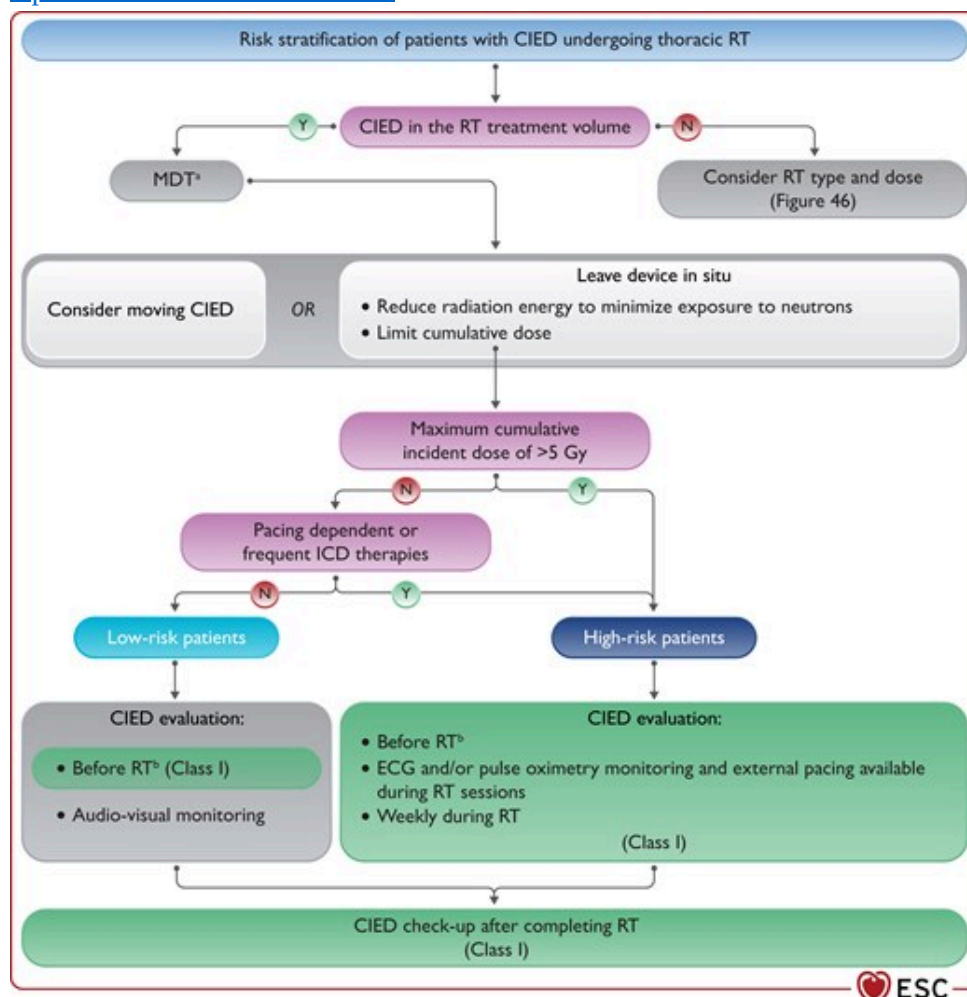


Figure 45

Management of patients with a cardiac implantable electronic device located in the radiotherapy treatment beam.

CIED, cardiac implantable electronic device; ECG, electrocardiogram; Gy, Gray; ICD, implantable cardioverter defibrillator; MDT, multidisciplinary team; N, no; RT, radiotherapy; Y, yes. ^a**Multidisciplinary discussion must consider:** (1) whether the CIED is interfering with the RT dose delivered to the tumour; (2) whether the RT is interfering with CIED function (aim to not exceed 2 Gy to permanent pacemaker and 1 Gy to ICD); (3) risks of moving the CIED: infection (especially in immunocompromised patients), procedural complications (e.g. bleeding with thrombocytopaenia); for younger patients with good prognosis, consider long-term effects of losing an access site (lead extraction/RT-induced thrombosis). ^bIf last CIED check >3 months earlier.

[Open in new tab](#)[Download slide](#)

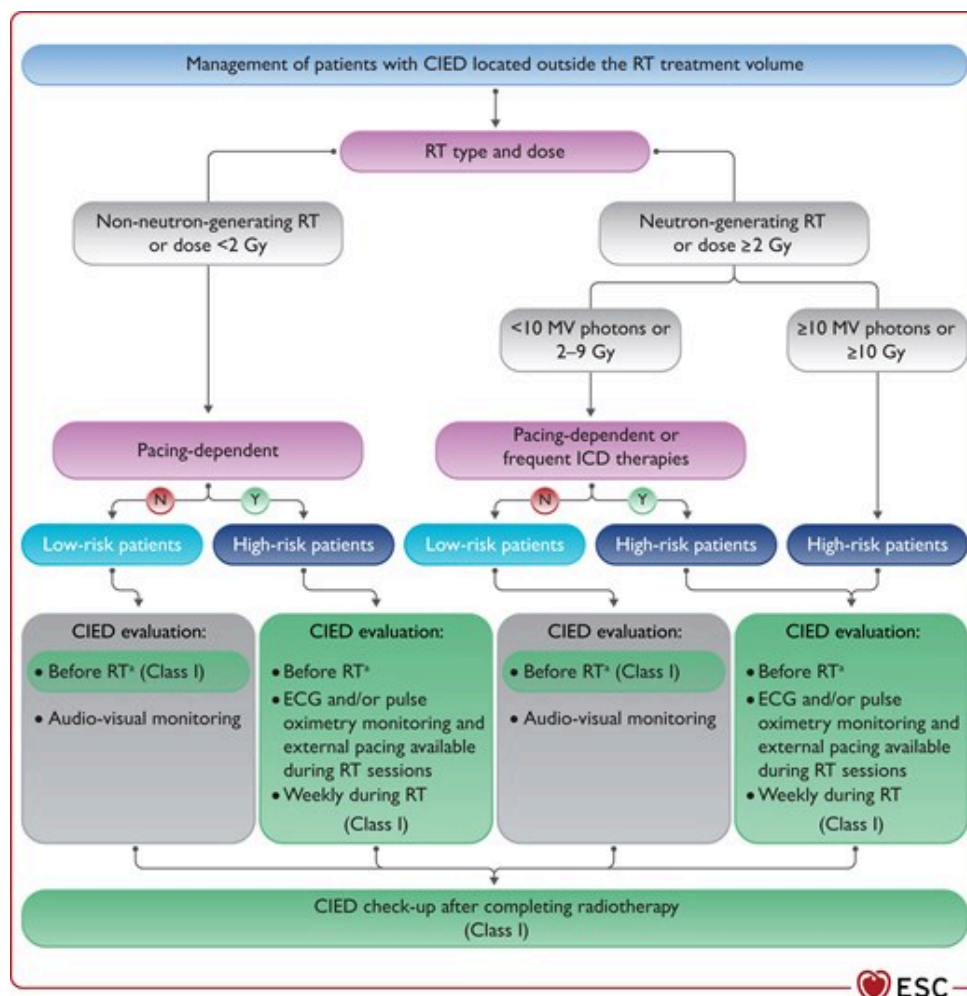


Figure 46

Management of patients with a cardiac implantable electronic device located outside the radiotherapy treatment volume.

CIED, cardiac implantable electronic device; ECG, electrocardiogram; Gy, Gray; ICD, implantable cardioverter defibrillator; MV, megavolt; N, no; RT, radiotherapy; Y, yes. ^aIf last CIED check >3 months earlier.

[Open in new tab](#)[Download slide](#)

Patients with a CIED should be reviewed by their cardiologist/electrophysiologist to assess the risk of CIED malfunction and patients should be informed of the potential risks of RT.⁴⁴³ For patients with rate-adaptive pacemakers, consideration should be given to temporary deactivation of the sensor during RT. Although inactivation of antitachycardia therapies in patients with ICDs is recommended in several publications, by either reprogramming or application of a magnet to ICDs, it is infrequently performed in clinical practice.⁸²⁶

CIEDs should not be placed directly in the RT treatment volume and the cumulative dose should not exceed 2 Gy to a pacemaker or 1 Gy to an ICD.⁸²⁷ If the CIED is situated in the path of the planned radiation beam, it could also interfere with adequate tumour treatment. The photon beam energy should be kept <10 MV as the risk of device malfunction/damage increases above this threshold. If higher doses are needed or if the CIED cannot be kept out of the beam, consideration should be given to removing and relocating the CIED away from the beam, although this will only very rarely be necessary. The main reason for device relocation is to allow adequate RT treatment of the tumour, but consideration should also be given

to possible RT-induced CIED malfunction/damage with consequent need for CIED replacement.⁸²⁶ However, CIED explant and resiting carries significant risks, including the risk of infection, which may be of particular importance in patients receiving chemotherapy or those who are immunosuppressed. For most patients in whom definitive tumour treatment is planned, the risk/benefit ratio will usually favour device relocation, whereas for patients receiving palliative RT or with significant comorbidities, relocation could be avoided.⁸²⁶ These decisions should be made by a MDT in conjunction with the patient. Device relocation is not recommended for CIEDs receiving a maximum cumulative incident dose of <5aGy, where the risk is considered negligible.^{826,828}

There should be continuous visual and voice contact with the patient during each treatment fraction. CIEDs should be periodically checked in patients with ICDs, especially those receiving >10 MV photon beam energy.^{827,829} For patients receiving electron or kV photon beam RT, CIED evaluation appears largely unnecessary.⁸²⁷ For patients treated with proton beam RT, special consideration should be paid to the neutron component of the beam, as the risk of CIED reset is potentially significant.^{824,830} The CIED should be rechecked within 2 weeks of completion of RT treatment. Systematic remote CIED monitoring may be helpful to optimize the patient's surveillance.⁸³¹

Recommendation Table 50

[Open in new tab](#)

Recommendations for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy

Recommendations	Class ^a	Level ^b
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment. ^{824,825,827,828}	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk. ^{824,826}	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session. ^{827,829,831}	I	C

© ESC 2022

CIED, cardiac implantable electronic device; ECG, electrocardiogram; RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendations	Class ^a	Level ^b
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment. ^{824,825,827,828}	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk. ^{824,826}	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session. ^{827,829,831}	I	C

© ESC 2022

CIED, cardiac implantable electronic device; ECG, electrocardiogram; RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendation Table 50

[Open in new tab](#)

Recommendations for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy

Recommendations	Class ^a	Level ^b
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment. ^{824,825,827,828}	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk. ^{824,826}	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session. ^{827,829,831}	I	C

© ESC 2022

CIED, cardiac implantable electronic device; ECG, electrocardiogram; RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

Recommendations	Class ^a	Level ^b
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment. ^{824,825,827,828}	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk. ^{824,826}	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session. ^{827,829,831}	I	C

© ESC 2022

CIED, cardiac implantable electronic device; ECG, electrocardiogram; RT, radiotherapy.
^aClass of recommendation.
^bLevel of evidence.

10. Patient information, communication, and self-management

Collaboration between different healthcare professionals and patients is of paramount importance for the most effective management of patients with cancer and CVD. Appropriate language and communication should be used to allow patients to receive clear and accurate information about their condition, and play an active role in managing their treatment.¹¹

The first goal of this process is to raise the patient's awareness of the possible presence or development of a CVD, either during cancer or after having some oncological therapy. Patients should understand that cancer and CVD share many CVRF and reducing risk is vital for the prevention of cancer, cancer relapse, and the development or worsening of a CVD during or after treatment. Patients should be informed at the end of chemotherapy that a personalized follow-up plan and regular CV controls are needed to detect potential reversible stages of CV toxicities. Education, counselling, and support to promote healthy lifestyle and to treat modifiable CVRF should be offered to patients with cancer, in order to reduce the burden of complications during and after anticancer therapy. Patients should receive guidance to recognize and to report signs and symptoms of CVD, in order to receive prompt and effective treatment, ideally without interfering with their cancer treatment. Patients should also be advised not to stop cardioprotective therapies without medical guidance, even if they recover their cardiac function. To help in this complex task, leaflets specifically designed for this context may be used,^{832,833} eventually with the aid of digital tools ([Figure 47](#)).

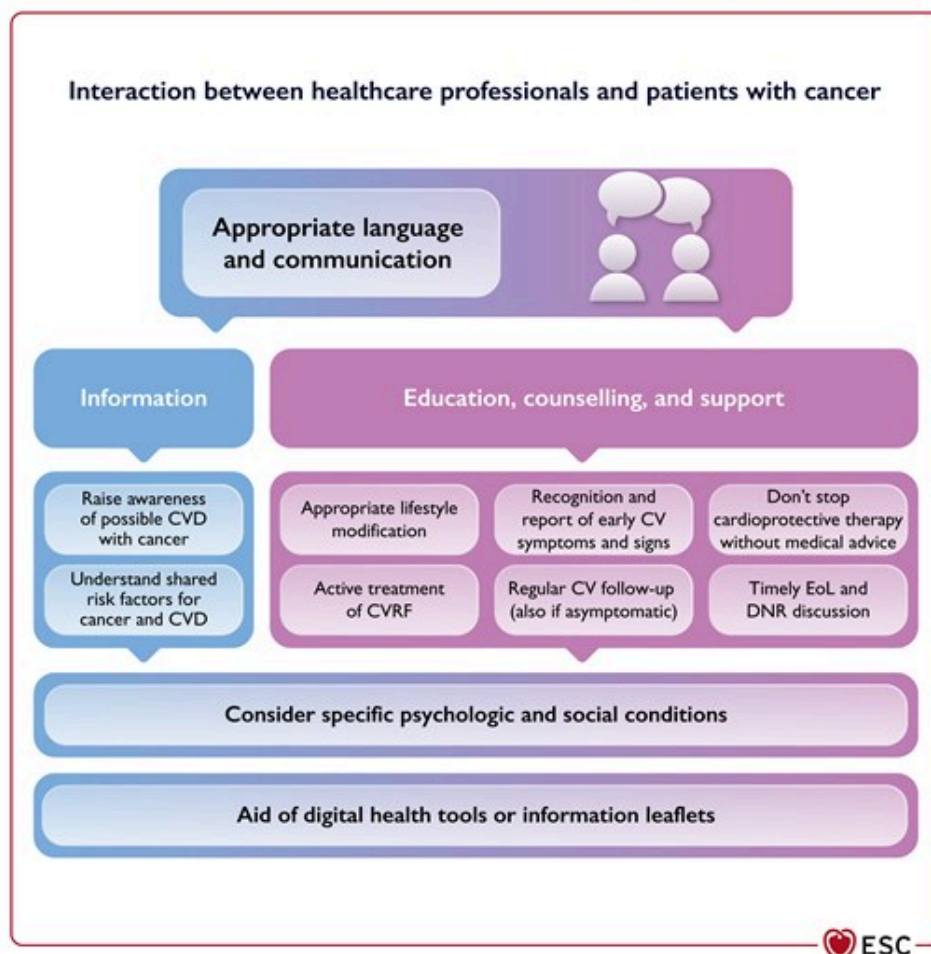


Figure 47

Patient information, communication, and self-management.

CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DNR, do not resuscitate; EoL, end of life.

[Open in new tab](#)[Download slide](#)

11. The role of scientific societies in the promotion and development of cardio-oncology in modern medicine

Cardio-oncology is a subspecialty that has seen huge development and growth in recent years with the formation of cardiology in almost all national and international societies and of cardio-oncology working groups. Moreover, cancer and medical associations have also developed an increasing interest in cardio-oncology. Important roles of these scientific societies are clinical research, education, and advocacy. The ESC-CCO strategic plan and mission include improvement of prevention, diagnosis, treatment, and management of CTR-CVT and enhancement of the standard of care for patients with cancer ([Figure 48](#)).

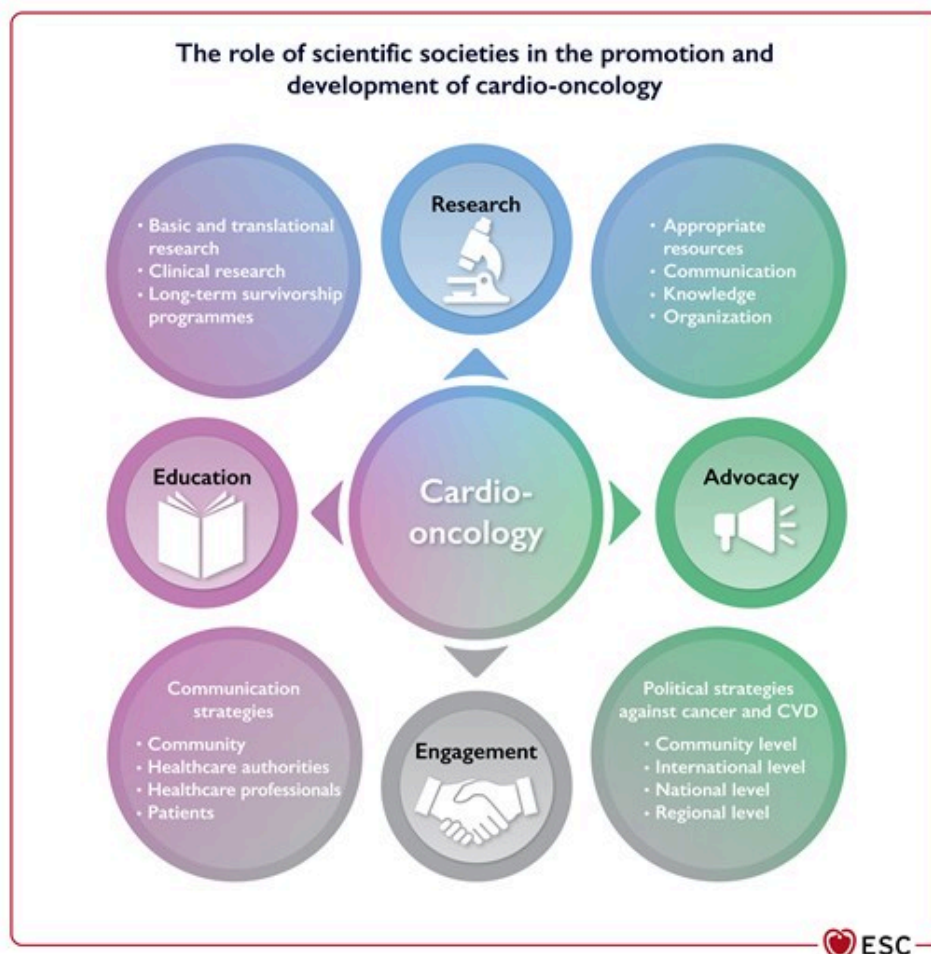


Figure 48

The role of scientific societies in the promotion and development of cardio-oncology.

CVD, cardiovascular disease.

[Open in new tab](#)[Download slide](#)

12. Key messages

This is the first ESC cardio-oncology Guideline and contains 272 new recommendations. The key messages from this guideline are:

- A guiding principle of cardio-oncology is integration, and cardio-oncology providers must have knowledge of the broad scope of cardiology, oncology, and haematology. Communication between different healthcare professionals is critical to optimize the care of patients with cancer and CVD.
- Cardio-oncology programmes facilitate cancer treatment by minimizing unnecessary cancer therapy interruptions and CTR-CVT across the entire continuum of cancer care. In patients who develop CTR-CVT, a MDT discussion is required to balance the risk/benefit of cancer treatment discontinuation.
- There is a new international definition of CTR-CVT ([Table 3](#)).
- CV toxicity risk is a dynamic variable. This guideline is structured to provide a personalized approach to care based upon the baseline CV toxicity risk. A baseline CV risk assessment is recommended for all patients with

cancer scheduled to receive a potentially cardiotoxic anticancer therapy. This enables the oncology team to consider CV risk while making cancer treatment choices, educating patients regarding their CV risk, and personalizing CV surveillance and follow-up strategy.

- Primary prevention of CV toxicity from cancer therapy aims to avoid or minimize the development of CTR-CVT in patients without CVD.
- Secondary prevention refers to interventions in patients with pre-existing CVD, including prior or new CTR-CVT. A MDT is recommended when patients with cancer have complex CVD that may impact on their cancer treatment.
- Defining and delivering an appropriate prevention and surveillance plan for potential CV complications is recommended. Optimal management of CVRF and pre-existing CVD is mandatory to facilitate cancer therapy and to improve patients' prognosis.
- Detailed monitoring pathways during cancer therapy including 3D echocardiography, GLS, and cardiac biomarkers are provided to detect CV toxicity based upon specific cancer therapies and baseline CV toxicity risk.
- Treatment recommendations for CTRCD during and after cancer therapy depend upon CTRCD severity and symptoms. New guidance on continuing trastuzumab in BC patients who develop asymptomatic moderate CTRCD (LVEF 40–49%) while starting cardioprotective medication is provided.
- Use of a structured algorithm to guide decisions regarding anticoagulation management in patients with cancer presenting with AF or VTE encompassing the TBIP assessment is encouraged.
- After cancer treatment is completed, the focus of the cardio-oncology team shifts to coordination of long-term follow-up. This starts with an end-of-treatment assessment in the first year after treatment, reviewing patients with cancer who have received cardiotoxic anticancer therapies to reassess their CV toxicity risk and guide long-term surveillance planning.
- A new algorithm ([Figure 37](#)) is provided to guide weaning off of CV medication in CS.
- Patients with cancer, CS, and the patient's family/carers should receive guidance to promote healthy lifestyle and recognize and report signs and symptoms of CVD, to receive prompt and effective treatment, without interfering with their cancer treatment.
- Patients must receive psychological support when needed and clear and accurate information about their condition to play an active role in managing their treatment and increase adherence to cancer and CV treatments.

13. Future needs

There are a low number of dedicated cardio-oncology services and most patients are reviewed in general cardiology clinics in Europe and worldwide. Strategic investments in cardio-oncology care networks and cardio-oncology services provision are needed to meet the projected increased clinical demand in the near future,⁸³⁴ and to facilitate research, training, and educational activities. A dedicated training core curriculum

for a minimum of 1-year medical training is urgently needed. It may include: (1) knowledge of the broad scope of cardiology, oncology, and haematology; (2) CV competencies for CTR-CVT prevention, surveillance, and management of patients with cancer in dedicated outpatientsâcardio-oncology clinics; (3) inpatient consultative services; and (4) dedicated time to achieve competences in CV imaging, HF, and vascular cardiology.

Collaboration between healthcare providers, clinical and basic investigators, healthcare authorities, regulatory bodies, advocacy groups, and patientsâassociations is needed to address future needs (see [Section 11](#)).

As this Guideline was developed, it became clear that there is a significant lack of RCT to guide decision-making, with many recommendations supported by level of evidence C. This is complicated by the fast-moving pace of new oncology treatment developments against a background of dynamic CV toxicity likelihood. Therefore, large numbers of patients and longer follow-up are required to provide sufficient statistical power and definitive answers. In the future, the following strategies and areas of research are priorities:

- New trial designs focusing upon the âat-riskâ cancer patient populations.
- Validating current HFA-ICOS risk assessment tools and surveillance algorithms.
- Assessment of new technologies for the detection of early CTRCD, broadening the biomarker panel and recognizing the specific patterns in early myocardial damage.
- Refining CV risk scores (e.g. EuroSCORE II, SCORE2, SCORE2-OP, CHA₂DS₂-VASc, HAS-BLED, SYNTAX) for application in cancer populations.
- Optimal treatment of steroid-resistant ICI CV toxicity and long-term CV effects of ICI therapy.
- Selection criteria for modern percutaneous structural (TAVI, Mitraclip, LAA occluder devices) and electrophysiological (ablation) CV therapies in patients with active cancer.
- Patient-specific predictive algorithms for QTc prolongation with cancer drugs.
- Assessment of genetic profiles in more specific CTRCD risk prediction.
- Identification of the cancer patient populations with mild or moderate CTRCD during treatment who can safely wean off long-term CV medication.
- Optimal modalities for screening long-term survivor populations for the complications of anthracycline chemotherapy and mediastinal radiation.
- Creation of large cardio-oncology registries to collect âbig dataâ on large patient populations.
- Application of artificial intelligence and other new data analytics to identify new patients with cancer at risk and new parameters that can predict risk of CTR-CVT, response to specific cardioprotective interventions, and long-term risk and safety to wean off CV therapies initiated during cancer treatment.

14. Gaps in evidence

Cancer and CVD are the two major public health problems with great economic and social impact. In addition, CTR-CVT are associated with an excess of both CV and oncological mortality, especially when they limit patients' ability to complete effective treatments. However, the intersection of cancer and CVD has only recently gained wider interest and many areas with lack of evidence need to be addressed in future research.

Role of cardio-oncology services and cardio-oncology care networks

- Robust evidence on the impact of dedicated cardio-oncology programmes and cardio-oncology rehabilitation on the prognosis of patients with cancer and survivors.
- Specification of roles of different healthcare professionals (including nurses and pharmacists) in cardio-oncology teams.
- Cardio-oncology care networks to improve the management of patients with cancer and to discuss difficult cases.
- Cardio-oncology team support and involvement in oncology trials design (including patients' representatives).
- Understand how to engage patients with cancer in their own CV care (inclusion of digital tools).

Research, education, and training in cardio-oncology

- Consensus about CV toxicity definitions used in oncology trials.
- Define standards for CV toxicity monitoring in oncology trials to avoid unexpected CV toxicities when new drugs are approved for clinical use.
- Relevant model systems to allow high-throughput screening of new cancer treatments for CV toxicity.
- Improved knowledge on CV toxicity mechanisms of new targeted cancer therapies and ICI and optimal treatment of CV toxicities.
- Improved knowledge on the effects of radiation to specific cardiac substructures and the interactions between cardiotoxic systemic therapy and RT.
- Further research into the underlying mechanisms that connect CVD and cancer, such as a genetic predisposition to CV toxicity.
- Personalized medicine and use of big data and artificial intelligence tools.

Cardiovascular toxicity risk stratification

- Development of CV toxicity risk prediction tools including both treatment- and patient-related risk factors.
- Validated prospective CV toxicity risk scores based on clinical outcomes.
- Further research on the role of genetics in CV toxicity risk stratification.

- Validation of CPET parameters for CV outcomes in patients with cancer.

Prevention, diagnosis, and management of CTR-CVT

- Raise awareness of the benefits of minimizing CV risk in patients with cancer in order to reduce the risk of CTR-CVT.
- More data on new technologies (biomarkers, advanced echocardiography, CMR, etc.) and genetic profiles for the detection of early CV toxicity.
- Prospective studies showing the impact on outcomes and/or quality of life (and frailty) of early CTR-CVT diagnosis and treatment.
- Further evidence from prospective RCTs to define when cardioprotective medications improve patients' outcomes.
- Further research on the potential for aerobic exercise to reduce CTR-CVT.
- RCTs of (new) CV therapies in patients with different types of CTR-CVT.

Long-term cancer survivorship programmes

- Development of optimal CV follow-up programmes after treatment for cancer (research on risk stratification, efficacy, and frequency of screening protocols).
- Best screening strategies for RT-induced CAD.
- Further research on CV preventive strategies for long-term CS.

15. What to do and what not to do messages from the Guidelines

[Open in new tab](#)

Recommendations	Class ^a	Level ^b
Recommendation Table 1 for a general approach to cardiovascular toxicity risk categorization		
CV toxicity risk stratification before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer.	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
It is recommended that patients categorized as low CV toxicity risk should proceed with anticancer therapy without delay.	I	C
Cardiology referral is recommended in high-risk and very high-risk patients before anticancer therapy.	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment who require potentially cardiotoxic anticancer therapy.	I	C

Recommendation Table 2 for electrocardiogram baseline assessment		
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, referral to a cardiologist is recommended.	I	C
Recommendation Table 3 for cardiac biomarker assessment prior to potentially cardiotoxic therapies		
Baseline measurement of NP and/or cTn is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD.	I	C
Recommendation Table 4 for cardiac imaging modalities in patients with cancer		
General		
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer.	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF.	I	B
GLS is recommended in all patients with cancer having echocardiography, if available.	I	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy.	I	C
Recommendation Table 5 for primary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 6 for secondary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVD according to applicable ESC Guidelines is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 7 for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended in all patients with cancer before anthracycline chemotherapy.	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment.	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment.	I	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy.	I	B
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients.	I	B
Recommendation Table 8 for baseline risk assessment and monitoring during HER2-targeted therapies and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended before HER2-targeted therapies in all patients.	I	B
In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment.	I	B
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment.	I	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies.	I	C

Recommendation Table 9 for baseline risk assessment and monitoring during fluoropyrimidine therapy

Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP or equivalent is recommended before starting fluoropyrimidines.

I**C**

A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.

I**C****Recommendation Table 10 for baseline risk assessment and monitoring during VEGFi****BP monitoring**

BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.

I**C**

Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.

I**C****ECG monitoring**

In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter.

I**C****Echocardiography**

Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.

I**C****Recommendation Table 11 for baseline risk assessment and monitoring during second- and third-generation BCR-ABL tyrosine kinase inhibitors**

Baseline CV risk assessment is recommended in patients who require second- or third-generation BCR-ABL TKI.

I**C**

In patients treated with nilotinib or ponatinib, CV risk assessment is recommended every 3 months during the first year and every 6–12 months thereafter.

I**C**

Baseline echocardiography is recommended in patients scheduled to receive dasatinib.

I**C****Recommendation Table 12 for baseline risk assessment and monitoring Bruton tyrosine kinase inhibitor therapy****BP monitoring and management**

BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit.

I**B****Echocardiography**

Baseline echocardiography is recommended in high-risk patients scheduled to receive BTK inhibitors.

I**C**

TTE is recommended in all patients who develop AF during BTK inhibitor therapy.

I**C****AF**

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.

I**C****Recommendation Table 13 for baseline risk assessment and monitoring during multiple myeloma therapies****BP monitoring**

BP measurement is recommended for patients treated with PI at every clinical visit.

I**C****Cardiac serum biomarkers**

Measurement of NP is recommended prior to PI in high- and very high-risk patients.

I**C**

NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA.

I**B****TTE**

Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.

I**C**

VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE.	I	B
Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors (excluding previous VTE) at least during the first 6 months of therapy.	I	A
Recommendation Table 14 for baseline risk assessment and monitoring during combined RAF and MEK inhibitor therapy		
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter.	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Recommendation Table 15 for baseline risk assessment and monitoring during immunotherapy		
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy.	I	B
Baseline echocardiography is recommended in high-risk patients before starting ICI therapy.	I	B
CV assessment is recommended every 6–12 months in high-risk patients who require long-term (>12 months) ICI treatment.	I	C
Recommendation Table 16 for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in patients treated with ADT without pre-existing CVD.	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy.	I	B
Annual CV risk assessment is recommended during ADT.	I	B
Recommendation Table 17 for baseline risk assessment and monitoring during endocrine therapy for breast cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in BC patients receiving endocrine therapies without pre-existing CVD.	I	C
Annual CV risk assessment is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP.	I	C
Recommendation Table 18 for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy		
QTc monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib.	I	A
QTc monitoring is recommended in patients treated with ribociclib with any dose increase.	I	B
Recommendation Table 19 for baseline risk assessment and monitoring during ALK and EGFR inhibitors		
Baseline CV risk assessment is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib.	I	B
Recommendation Table 20 for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies		
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies.	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies.	I	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 .	I	C

Recommendation Table 21 for baseline risk assessment of patients before radiotherapy to a volume including the heart

Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended.

I**B****Recommendation Table 22 for baseline risk assessment in haematopoietic stem cell transplantation patients**

Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.

I**C**

Echocardiography is recommended in all patients before HSCT.

I**C****Recommendation Table 23 for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment**

A specialist CV assessment is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment.

I**C****Recommendation Table 24 for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy****Anthracycline chemotherapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy.

I**B**

Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD.

I**C**

Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD and a multidisciplinary approach regarding the decision to restart is recommended.

I**C**

A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD.

I**C****Anthracycline chemotherapy-induced asymptomatic CTRCD**

Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD.

I**C**

Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq$ 50% and who have developed a significant fall in GLS or a troponin or a NP elevation $>$ ULN.

I**C****Recommendation Table 25 for the management of cancer treatment-related cardiac dysfunction during HER2-targeted therapies****HER2-targeted therapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF $<$ 50% during HER2-targeted treatment.

I**B**

Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved.

I**C**

In patients who develop mild symptomatic CTRCD, HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended.

I**C****HER2-targeted therapy-induced asymptomatic CTRCD**

Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD.

I**C**

Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF $\geq$ 50%) CTRCD⁶ with more frequent cardiac monitoring.

I**C**

ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD⁶ during HER2-targeted treatment.

I**C****Recommendation Table 26 for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis**

cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis.

I**B**

In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.

I**C**

Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids are recommended in patients with cancer and confirmed ICI-associated myocarditis.	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction.	I	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C
Recommendation Table 27 for the diagnosis and management of Takotsubo syndrome in patients with cancer		
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI.	I	B
QT-prolonging drugs are not recommended during the acute TTS phase.	III	C
Recommendation Table 28 for the management of acute coronary syndromes in patients receiving anticancer treatment		
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause.	I	C
In patients with cancer, thrombocytopenia and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Recommendation Table 29 for the management of chronic coronary syndromes in patients receiving anticancer treatment		
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment.	I	C
Recommendation Table 30 for the management of valvular heart disease in patients receiving anticancer treatment		
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences.	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C
Recommendation Table 31 for the management of atrial fibrillation in patients receiving anticancer treatment		
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥ 2 (men) or ≥ 3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation.	I	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF.	I	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer.	III	C
Recommendation Table 32 for the management of long QTc and ventricular arrhythmias in patients receiving anticancer treatment		
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment.	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation.	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors is recommended in patients with cancer who develop QTcF ≥ 500 ms.	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy.	I	C

A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy.	I	C
Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.	I	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.	I	C
Recommendation Table 33 for the management of arterial hypertension in patients receiving anticancer treatment		
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C
Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs recommended for BP management in patients with cancer.	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.	III	C
Recommendation Table 34 for the management of venous thromboembolism in patients receiving anticancer treatment		
Apixaban, edoxaban, or rivaroxaban are recommended for the treatment of symptomatic or incidental VTE in patients with cancer without contraindications.	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count $>50\,000/\mu\text{L}$.	I	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains in situ.	I	C
Recommendation Table 35 for venous thromboembolism prophylaxis during anticancer treatment		
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk.	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bed rest or reduced mobility in the absence of bleeding or other contraindications.	I	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C
Recommendation Table 36 for management of peripheral artery disease during anticancer treatment		
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy is recommended.	I	C
Recommendation Table 37 for the management of pulmonary hypertension during anticancer treatment		
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV > 3.4 m/s.	I	C
In patients with confirmed dasatinib-induced PAH or new asymptomatic peak TRV > 3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C
Recommendation Table 38 for the management of pericardial diseases in patients receiving anticancer treatment		
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy.	I	C

Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis.	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C
Recommendation Table 39 for end-of-cancer therapy cardiovascular risk assessment		
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended.	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice.	I	B
In asymptomatic high-risk patients, echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy.	I	B
Cardiology referral is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment.	I	C
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed TKI-mediated hypertension during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy.	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy.	I	C
Recommendation Table 40 for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors		
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended.	I	B
Annual screening for modifiable CVRF is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Recommendation Table 41 for cardiovascular surveillance in asymptomatic adult cancer survivors		
Annual CV risk assessment, including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT.	I	B
CV toxicity risk re-stratification is recommended 5 years after therapy to organize long-term follow-up.	I	C
Recommendation Table 42 for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy		
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD.	I	C
Recommendation Table 43 for adult cancer survivors with coronary artery disease		
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management.	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C

Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
Recommendation Table 44 for adult cancer survivors with valvular heart disease		
A MDT approach is recommended to discuss and define the surgical risk in CS with severe VHD.	I	C
Recommendation Table 46 for cardiovascular monitoring in cancer survivors during pregnancy		
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
Recommendation Table 47 for cardiovascular assessment and monitoring of pregnant women with cancer		
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy.	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography.	I	C
Recommendation Table 48 for carcinoid valvular heart diseases		
Echocardiography is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status.	I	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months.	I	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months.	I	C
Recommendation Table 49 for amyloid light-chain cardiac amyloidosis diagnosis and monitoring		
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia.	I	B
CMR is recommended in patients with suspected AL-CA.	I	A
Recommendation Table 50 for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy		
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment.	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk.	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session.	I	C

© ESC 2022

3D, three-dimensional; ACE-I, angiotensin-converting enzyme inhibitors; ACS, acute coronary syndromes; ADT, androgen deprivation therapy; AF, atrial fibrillation; AL-CA, amyloid light-chain cardiac amyloidosis; ALK, anaplastic lymphoma kinase; ARB, angiotensin receptor blocker; ASTCT, American Society for Transplantation and Cellular; AV, atrioventricular; BC, breast cancer; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BP, blood pressure; BTK, Bruton tyrosine kinase; CABG, coronary artery bypass graft; CAD, coronary artery disease; CAR-T, chimeric antigen receptor T cell; CCB, calcium channel blockers; CCS, chronic coronary syndromes; CCTA, coronary computed tomography angiography; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; CRS, cytokine release syndrome; CT, computed tomography; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CS, cancer survivors; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DAPT, dual antiplatelet therapy; EACTS, European Association for Cardio-Thoracic Surgery; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; GLS, global longitudinal strain; HbA_{1c}, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICU, intensive care unit; iv, intravenous; LIMA, left internal mammary artery; LMWH, low-molecular-weight heparins; LQTS, long QT syndrome; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides; NSTE-ACS, non-ST-segment elevation acute coronary syndromes; PAD, peripheral artery disease; PAH, pulmonary arterial hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; QTcf, corrected QT interval using Fridericia correction; RAF, rapidly accelerated fibrosarcoma; RIMA, right internal mammary artery; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons; STEMI, ST-segment elevation myocardial infarction; TdP, torsade de pointes; TIL, tumour-infiltrating lymphocytes; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography; TTS, Takotsubo syndrome; TRV, tricuspid regurgitation velocity; ULN, upper limit of normal; VEGF, vascular endothelial growth factor inhibitors; VHD, valvular heart disease; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

Recommendations	Class ^a	Level ^b
Recommendation Table 1 for a general approach to cardiovascular toxicity risk categorization		
CV toxicity risk stratification before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer.	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
It is recommended that patients categorized as low CV toxicity risk should proceed with anticancer therapy without delay.	I	C
Cardiology referral is recommended in high-risk and very high-risk patients before anticancer therapy.	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment who require potentially cardiotoxic anticancer therapy.	I	C

Recommendation Table 2 for electrocardiogram baseline assessment		
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, referral to a cardiologist is recommended.	I	C
Recommendation Table 3 for cardiac biomarker assessment prior to potentially cardiotoxic therapies		
Baseline measurement of NP and/or cTn is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD.	I	C
Recommendation Table 4 for cardiac imaging modalities in patients with cancer		
General		
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer.	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF.	I	B
GLS is recommended in all patients with cancer having echocardiography, if available.	I	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy.	I	C
Recommendation Table 5 for primary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 6 for secondary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVD according to applicable ESC Guidelines is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 7 for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended in all patients with cancer before anthracycline chemotherapy.	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment.	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment.	I	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy.	I	B
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients.	I	B
Recommendation Table 8 for baseline risk assessment and monitoring during HER2-targeted therapies and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended before HER2-targeted therapies in all patients.	I	B
In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment.	I	B
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment.	I	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies.	I	C

Recommendation Table 9 for baseline risk assessment and monitoring during fluoropyrimidine therapy

Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP or equivalent is recommended before starting fluoropyrimidines.

I**C**

A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.

I**C****Recommendation Table 10 for baseline risk assessment and monitoring during VEGFi****BP monitoring**

BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.

I**C**

Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.

I**C****ECG monitoring**

In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter.

I**C****Echocardiography**

Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.

I**C****Recommendation Table 11 for baseline risk assessment and monitoring during second- and third-generation BCR-ABL tyrosine kinase inhibitors**

Baseline CV risk assessment is recommended in patients who require second- or third-generation BCR-ABL TKI.

I**C**

In patients treated with nilotinib or ponatinib, CV risk assessment is recommended every 3 months during the first year and every 6–12 months thereafter.

I**C**

Baseline echocardiography is recommended in patients scheduled to receive dasatinib.

I**C****Recommendation Table 12 for baseline risk assessment and monitoring Bruton tyrosine kinase inhibitor therapy****BP monitoring and management**

BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit.

I**B****Echocardiography**

Baseline echocardiography is recommended in high-risk patients scheduled to receive BTK inhibitors.

I**C**

TTE is recommended in all patients who develop AF during BTK inhibitor therapy.

I**C****AF**

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.

I**C****Recommendation Table 13 for baseline risk assessment and monitoring during multiple myeloma therapies****BP monitoring**

BP measurement is recommended for patients treated with PI at every clinical visit.

I**C****Cardiac serum biomarkers**

Measurement of NP is recommended prior to PI in high- and very high-risk patients.

I**C**

NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA.

I**B****TTE**

Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.

I**C**

VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE.	I	B
Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors (excluding previous VTE) at least during the first 6 months of therapy.	I	A
Recommendation Table 14 for baseline risk assessment and monitoring during combined RAF and MEK inhibitor therapy		
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter.	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Recommendation Table 15 for baseline risk assessment and monitoring during immunotherapy		
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy.	I	B
Baseline echocardiography is recommended in high-risk patients before starting ICI therapy.	I	B
CV assessment is recommended every 6–12 months in high-risk patients who require long-term (>12 months) ICI treatment.	I	C
Recommendation Table 16 for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in patients treated with ADT without pre-existing CVD.	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy.	I	B
Annual CV risk assessment is recommended during ADT.	I	B
Recommendation Table 17 for baseline risk assessment and monitoring during endocrine therapy for breast cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in BC patients receiving endocrine therapies without pre-existing CVD.	I	C
Annual CV risk assessment is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP.	I	C
Recommendation Table 18 for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy		
QTc monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib.	I	A
QTc monitoring is recommended in patients treated with ribociclib with any dose increase.	I	B
Recommendation Table 19 for baseline risk assessment and monitoring during ALK and EGFR inhibitors		
Baseline CV risk assessment is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib.	I	B
Recommendation Table 20 for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies		
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies.	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies.	I	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 .	I	C

Recommendation Table 21 for baseline risk assessment of patients before radiotherapy to a volume including the heart

Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended.

I**B****Recommendation Table 22 for baseline risk assessment in haematopoietic stem cell transplantation patients**

Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.

I**C**

Echocardiography is recommended in all patients before HSCT.

I**C****Recommendation Table 23 for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment**

A specialist CV assessment is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment.

I**C****Recommendation Table 24 for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy****Anthracycline chemotherapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy.

I**B**

Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD.

I**C**

Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD and a multidisciplinary approach regarding the decision to restart is recommended.

I**C**

A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD.

I**C****Anthracycline chemotherapy-induced asymptomatic CTRCD**

Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD.

I**C**

Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq$ 50% and who have developed a significant fall in GLS or a troponin or a NP elevation $>$ ULN.

I**C****Recommendation Table 25 for the management of cancer treatment-related cardiac dysfunction during HER2-targeted therapies****HER2-targeted therapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF $<$ 50% during HER2-targeted treatment.

I**B**

Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved.

I**C**

In patients who develop mild symptomatic CTRCD, HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended.

I**C****HER2-targeted therapy-induced asymptomatic CTRCD**

Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD.

I**C**

Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF $\geq$ 50%) CTRCD⁶ with more frequent cardiac monitoring.

I**C**

ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD⁶ during HER2-targeted treatment.

I**C****Recommendation Table 26 for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis**

cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis.

I**B**

In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.

I**C**

Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids are recommended in patients with cancer and confirmed ICI-associated myocarditis.	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction.	I	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C
Recommendation Table 27 for the diagnosis and management of Takotsubo syndrome in patients with cancer		
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI.	I	B
QT-prolonging drugs are not recommended during the acute TTS phase.	III	C
Recommendation Table 28 for the management of acute coronary syndromes in patients receiving anticancer treatment		
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause.	I	C
In patients with cancer, thrombocytopenia and ACS, aspirin is not recommended if platelets $<10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia and ACS, clopidogrel is not recommended if platelets $<30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $<50\,000/\mu\text{L}$.	III	C
Recommendation Table 29 for the management of chronic coronary syndromes in patients receiving anticancer treatment		
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment.	I	C
Recommendation Table 30 for the management of valvular heart disease in patients receiving anticancer treatment		
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences.	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C
Recommendation Table 31 for the management of atrial fibrillation in patients receiving anticancer treatment		
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥ 2 (men) or ≥ 3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation.	I	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF.	I	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer.	III	C
Recommendation Table 32 for the management of long QTc and ventricular arrhythmias in patients receiving anticancer treatment		
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment.	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation.	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors is recommended in patients with cancer who develop QTcF ≥ 500 ms.	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy.	I	C

A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy.	I	C
Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.	I	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.	I	C
Recommendation Table 33 for the management of arterial hypertension in patients receiving anticancer treatment		
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C
Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs recommended for BP management in patients with cancer.	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.	III	C
Recommendation Table 34 for the management of venous thromboembolism in patients receiving anticancer treatment		
Apixaban, edoxaban, or rivaroxaban are recommended for the treatment of symptomatic or incidental VTE in patients with cancer without contraindications.	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count $>50\,000/\mu\text{L}$.	I	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains in situ.	I	C
Recommendation Table 35 for venous thromboembolism prophylaxis during anticancer treatment		
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk.	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bed rest or reduced mobility in the absence of bleeding or other contraindications.	I	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C
Recommendation Table 36 for management of peripheral artery disease during anticancer treatment		
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy is recommended.	I	C
Recommendation Table 37 for the management of pulmonary hypertension during anticancer treatment		
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV > 3.4 m/s.	I	C
In patients with confirmed dasatinib-induced PAH or new asymptomatic peak TRV > 3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C
Recommendation Table 38 for the management of pericardial diseases in patients receiving anticancer treatment		
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy.	I	C

Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis.	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C
Recommendation Table 39 for end-of-cancer therapy cardiovascular risk assessment		
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended.	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice.	I	B
In asymptomatic high-risk patients, echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy.	I	B
Cardiology referral is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment.	I	C
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed TKI-mediated hypertension during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy.	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy.	I	C
Recommendation Table 40 for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors		
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended.	I	B
Annual screening for modifiable CVRF is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Recommendation Table 41 for cardiovascular surveillance in asymptomatic adult cancer survivors		
Annual CV risk assessment, including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT.	I	B
CV toxicity risk re-stratification is recommended 5 years after therapy to organize long-term follow-up.	I	C
Recommendation Table 42 for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy		
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD.	I	C
Recommendation Table 43 for adult cancer survivors with coronary artery disease		
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management.	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C

Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
Recommendation Table 44 for adult cancer survivors with valvular heart disease		
A MDT approach is recommended to discuss and define the surgical risk in CS with severe VHD.	I	C
Recommendation Table 46 for cardiovascular monitoring in cancer survivors during pregnancy		
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
Recommendation Table 47 for cardiovascular assessment and monitoring of pregnant women with cancer		
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy.	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography.	I	C
Recommendation Table 48 for carcinoid valvular heart diseases		
Echocardiography is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status.	I	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months.	I	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months.	I	C
Recommendation Table 49 for amyloid light-chain cardiac amyloidosis diagnosis and monitoring		
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia.	I	B
CMR is recommended in patients with suspected AL-CA.	I	A
Recommendation Table 50 for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy		
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment.	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk.	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session.	I	C

© ESC 2022

3D, three-dimensional; ACE-I, angiotensin-converting enzyme inhibitors; ACS, acute coronary syndromes; ADT, androgen deprivation therapy; AF, atrial fibrillation; AL-CA, amyloid light-chain cardiac amyloidosis; ALK, anaplastic lymphoma kinase; ARB, angiotensin receptor blocker; ASTCT, American Society for Transplantation and Cellular; AV, atrioventricular; BC, breast cancer; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BP, blood pressure; BTK, Bruton tyrosine kinase; CABG, coronary artery bypass graft; CAD, coronary artery disease; CAR-T, chimeric antigen receptor T cell; CCB, calcium channel blockers; CCS, chronic coronary syndromes; CCTA, coronary computed tomography angiography; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; CRS, cytokine release syndrome; CT, computed tomography; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CS, cancer survivors; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DAPT, dual antiplatelet therapy; EACTS, European Association for Cardio-Thoracic Surgery; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICU, intensive care unit; iv, intravenous; LIMA, left internal mammary artery; LMWH, low-molecular-weight heparins; LQTS, long QT syndrome; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides; NSTE-ACS, non-ST-segment elevation acute coronary syndromes; PAD, peripheral artery disease; PAH, pulmonary arterial hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; QTcf, corrected QT interval using Fridericia correction; RAF, rapidly accelerated fibrosarcoma; RIMA, right internal mammary artery; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons; STEMI, ST-segment elevation myocardial infarction; TdP, torsade de pointes; TIL, tumour-infiltrating lymphocytes; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography; TTS, Takotsubo syndrome; TRV, tricuspid regurgitation velocity; ULN, upper limit of normal; VEGF, vascular endothelial growth factor inhibitors; VHD, valvular heart disease; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

[Open in new tab](#)

Recommendations	Class ^a	Level ^b
Recommendation Table 1 for a general approach to cardiovascular toxicity risk categorization		
CV toxicity risk stratification before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer.	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
It is recommended that patients categorized as low CV toxicity risk should proceed with anticancer therapy without delay.	I	C
Cardiology referral is recommended in high-risk and very high-risk patients before anticancer therapy.	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment who require potentially cardiotoxic anticancer therapy.	I	C

Recommendation Table 2 for electrocardiogram baseline assessment		
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, referral to a cardiologist is recommended.	I	C
Recommendation Table 3 for cardiac biomarker assessment prior to potentially cardiotoxic therapies		
Baseline measurement of NP and/or cTn is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD.	I	C
Recommendation Table 4 for cardiac imaging modalities in patients with cancer		
General		
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer.	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF.	I	B
GLS is recommended in all patients with cancer having echocardiography, if available.	I	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy.	I	C
Recommendation Table 5 for primary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 6 for secondary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVD according to applicable ESC Guidelines is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 7 for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended in all patients with cancer before anthracycline chemotherapy.	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment.	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment.	I	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy.	I	B
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients.	I	B
Recommendation Table 8 for baseline risk assessment and monitoring during HER2-targeted therapies and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended before HER2-targeted therapies in all patients.	I	B
In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment.	I	B
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment.	I	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies.	I	C

Recommendation Table 9 for baseline risk assessment and monitoring during fluoropyrimidine therapy

Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP or equivalent is recommended before starting fluoropyrimidines.

I**C**

A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.

I**C****Recommendation Table 10 for baseline risk assessment and monitoring during VEGFi****BP monitoring**

BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.

I**C**

Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.

I**C****ECG monitoring**

In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter.

I**C****Echocardiography**

Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.

I**C****Recommendation Table 11 for baseline risk assessment and monitoring during second- and third-generation BCR-ABL tyrosine kinase inhibitors**

Baseline CV risk assessment is recommended in patients who require second- or third-generation BCR-ABL TKI.

I**C**

In patients treated with nilotinib or ponatinib, CV risk assessment is recommended every 3 months during the first year and every 6–12 months thereafter.

I**C**

Baseline echocardiography is recommended in patients scheduled to receive dasatinib.

I**C****Recommendation Table 12 for baseline risk assessment and monitoring Bruton tyrosine kinase inhibitor therapy****BP monitoring and management**

BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit.

I**B****Echocardiography**

Baseline echocardiography is recommended in high-risk patients scheduled to receive BTK inhibitors.

I**C**

TTE is recommended in all patients who develop AF during BTK inhibitor therapy.

I**C****AF**

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.

I**C****Recommendation Table 13 for baseline risk assessment and monitoring during multiple myeloma therapies****BP monitoring**

BP measurement is recommended for patients treated with PI at every clinical visit.

I**C****Cardiac serum biomarkers**

Measurement of NP is recommended prior to PI in high- and very high-risk patients.

I**C**

NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA.

I**B****TTE**

Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.

I**C**

VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE.	I	B
Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors (excluding previous VTE) at least during the first 6 months of therapy.	I	A
Recommendation Table 14 for baseline risk assessment and monitoring during combined RAF and MEK inhibitor therapy		
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter.	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Recommendation Table 15 for baseline risk assessment and monitoring during immunotherapy		
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy.	I	B
Baseline echocardiography is recommended in high-risk patients before starting ICI therapy.	I	B
CV assessment is recommended every 6–12 months in high-risk patients who require long-term (>12 months) ICI treatment.	I	C
Recommendation Table 16 for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in patients treated with ADT without pre-existing CVD.	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy.	I	B
Annual CV risk assessment is recommended during ADT.	I	B
Recommendation Table 17 for baseline risk assessment and monitoring during endocrine therapy for breast cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in BC patients receiving endocrine therapies without pre-existing CVD.	I	C
Annual CV risk assessment is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP.	I	C
Recommendation Table 18 for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy		
QTc monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib.	I	A
QTc monitoring is recommended in patients treated with ribociclib with any dose increase.	I	B
Recommendation Table 19 for baseline risk assessment and monitoring during ALK and EGFR inhibitors		
Baseline CV risk assessment is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib.	I	B
Recommendation Table 20 for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies		
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies.	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies.	I	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 .	I	C

Recommendation Table 21 for baseline risk assessment of patients before radiotherapy to a volume including the heart

Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended.

I**B****Recommendation Table 22 for baseline risk assessment in haematopoietic stem cell transplantation patients**

Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.

I**C**

Echocardiography is recommended in all patients before HSCT.

I**C****Recommendation Table 23 for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment**

A specialist CV assessment is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment.

I**C****Recommendation Table 24 for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy****Anthracycline chemotherapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy.

I**B**

Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD.

I**C**

Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD and a multidisciplinary approach regarding the decision to restart is recommended.

I**C**

A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD.

I**C****Anthracycline chemotherapy-induced asymptomatic CTRCD**

Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD.

I**C**

Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq$ 50% and who have developed a significant fall in GLS or a troponin or a NP elevation $>$ ULN.

I**C****Recommendation Table 25 for the management of cancer treatment-related cardiac dysfunction during HER2-targeted therapies****HER2-targeted therapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF $<$ 50% during HER2-targeted treatment.

I**B**

Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved.

I**C**

In patients who develop mild symptomatic CTRCD, HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended.

I**C****HER2-targeted therapy-induced asymptomatic CTRCD**

Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD.

I**C**

Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF $\geq$ 50%) CTRCD⁶ with more frequent cardiac monitoring.

I**C**

ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD⁶ during HER2-targeted treatment.

I**C****Recommendation Table 26 for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis**

cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis.

I**B**

In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.

I**C**

Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids are recommended in patients with cancer and confirmed ICI-associated myocarditis.	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction.	I	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C
Recommendation Table 27 for the diagnosis and management of Takotsubo syndrome in patients with cancer		
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI.	I	B
QT-prolonging drugs are not recommended during the acute TTS phase.	III	C
Recommendation Table 28 for the management of acute coronary syndromes in patients receiving anticancer treatment		
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI/ACS with life expectancy ≥ 6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause.	I	C
In patients with cancer, thrombocytopenia and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Recommendation Table 29 for the management of chronic coronary syndromes in patients receiving anticancer treatment		
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment.	I	C
Recommendation Table 30 for the management of valvular heart disease in patients receiving anticancer treatment		
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences.	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C
Recommendation Table 31 for the management of atrial fibrillation in patients receiving anticancer treatment		
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥ 2 (men) or ≥ 3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation.	I	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF.	I	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer.	III	C
Recommendation Table 32 for the management of long QTc and ventricular arrhythmias in patients receiving anticancer treatment		
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment.	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation.	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors is recommended in patients with cancer who develop QTcF ≥ 500 ms.	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy.	I	C

A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy.	I	C
Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.	I	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.	I	C
Recommendation Table 33 for the management of arterial hypertension in patients receiving anticancer treatment		
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C
Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs recommended for BP management in patients with cancer.	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.	III	C
Recommendation Table 34 for the management of venous thromboembolism in patients receiving anticancer treatment		
Apixaban, edoxaban, or rivaroxaban are recommended for the treatment of symptomatic or incidental VTE in patients with cancer without contraindications.	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count $>50\,000/\mu\text{L}$.	I	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains in situ.	I	C
Recommendation Table 35 for venous thromboembolism prophylaxis during anticancer treatment		
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk.	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bed rest or reduced mobility in the absence of bleeding or other contraindications.	I	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C
Recommendation Table 36 for management of peripheral artery disease during anticancer treatment		
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy is recommended.	I	C
Recommendation Table 37 for the management of pulmonary hypertension during anticancer treatment		
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV > 3.4 m/s.	I	C
In patients with confirmed dasatinib-induced PAH or new asymptomatic peak TRV > 3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C
Recommendation Table 38 for the management of pericardial diseases in patients receiving anticancer treatment		
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy.	I	C

Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis.	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C
Recommendation Table 39 for end-of-cancer therapy cardiovascular risk assessment		
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended.	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice.	I	B
In asymptomatic high-risk patients, echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy.	I	B
Cardiology referral is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment.	I	C
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed TKI-mediated hypertension during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy.	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy.	I	C
Recommendation Table 40 for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors		
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended.	I	B
Annual screening for modifiable CVRF is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Recommendation Table 41 for cardiovascular surveillance in asymptomatic adult cancer survivors		
Annual CV risk assessment, including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT.	I	B
CV toxicity risk re-stratification is recommended 5 years after therapy to organize long-term follow-up.	I	C
Recommendation Table 42 for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy		
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD.	I	C
Recommendation Table 43 for adult cancer survivors with coronary artery disease		
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management.	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C

Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
Recommendation Table 44 for adult cancer survivors with valvular heart disease		
A MDT approach is recommended to discuss and define the surgical risk in CS with severe VHD.	I	C
Recommendation Table 46 for cardiovascular monitoring in cancer survivors during pregnancy		
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
Recommendation Table 47 for cardiovascular assessment and monitoring of pregnant women with cancer		
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy.	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography.	I	C
Recommendation Table 48 for carcinoid valvular heart diseases		
Echocardiography is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status.	I	B
A MDT discussion for optimal medical management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months.	I	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months.	I	C
Recommendation Table 49 for amyloid light-chain cardiac amyloidosis diagnosis and monitoring		
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia.	I	B
CMR is recommended in patients with suspected AL-CA.	I	A
Recommendation Table 50 for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy		
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment.	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk.	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session.	I	C

© ESC 2022

3D, three-dimensional; ACE-I, angiotensin-converting enzyme inhibitors; ACS, acute coronary syndromes; ADT, androgen deprivation therapy; AF, atrial fibrillation; AL-CA, amyloid light-chain cardiac amyloidosis; ALK, anaplastic lymphoma kinase; ARB, angiotensin receptor blocker; ASTCT, American Society for Transplantation and Cellular; AV, atrioventricular; BC, breast cancer; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BP, blood pressure; BTK, Bruton tyrosine kinase; CABG, coronary artery bypass graft; CAD, coronary artery disease; CAR-T, chimeric antigen receptor T cell; CCB, calcium channel blockers; CCS, chronic coronary syndromes; CCTA, coronary computed tomography angiography; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; CRS, cytokine release syndrome; CT, computed tomography; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CS, cancer survivors; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DAPT, dual antiplatelet therapy; EACTS, European Association for Cardio-Thoracic Surgery; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; GLS, global longitudinal strain; HbA1c, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICU, intensive care unit; iv, intravenous; LIMA, left internal mammary artery; LMWH, low-molecular-weight heparins; LQTS, long QT syndrome; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides; NSTE-ACS, non-ST-segment elevation acute coronary syndromes; PAD, peripheral artery disease; PAH, pulmonary arterial hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; QTcf, corrected QT interval using Fridericia correction; RAF, rapidly accelerated fibrosarcoma; RIMA, right internal mammary artery; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons; STEMI, ST-segment elevation myocardial infarction; TdP, torsade de pointes; TIL, tumour-infiltrating lymphocytes; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography; TTS, Takotsubo syndrome; TRV, tricuspid regurgitation velocity; ULN, upper limit of normal; VEGF, vascular endothelial growth factor inhibitors; VHD, valvular heart disease; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

Recommendations	Class ^a	Level ^b
Recommendation Table 1 for a general approach to cardiovascular toxicity risk categorization		
CV toxicity risk stratification before starting potentially cardiotoxic anticancer therapy is recommended in all patients with cancer.	I	B
Communicating the results of the CV toxicity risk assessment to the patient and other appropriate healthcare professionals is recommended.	I	C
It is recommended that patients categorized as low CV toxicity risk should proceed with anticancer therapy without delay.	I	C
Cardiology referral is recommended in high-risk and very high-risk patients before anticancer therapy.	I	C
Discussion of the risk/benefit balance of cardiotoxic anticancer treatment in high- and very high-risk patients in a multidisciplinary approach prior to starting treatment is recommended.	I	C
Cardiology referral is recommended for patients with cancer and pre-existing CVD or abnormal findings at baseline CV toxicity risk assessment who require potentially cardiotoxic anticancer therapy.	I	C

Recommendation Table 2 for electrocardiogram baseline assessment		
An ECG is recommended in all patients starting cancer therapy as part of their baseline CV risk assessment.	I	C
In patients with an abnormal baseline ECG, referral to a cardiologist is recommended.	I	C
Recommendation Table 3 for cardiac biomarker assessment prior to potentially cardiotoxic therapies		
Baseline measurement of NP and/or cTn is recommended in all patients with cancer at risk of CTRCD if these biomarkers are going to be measured during treatment to detect CTRCD.	I	C
Recommendation Table 4 for cardiac imaging modalities in patients with cancer		
General		
Echocardiography is recommended as the first-line modality for the assessment of cardiac function in patients with cancer.	I	C
3D echocardiography is recommended as the preferred echocardiographic modality to measure LVEF.	I	B
GLS is recommended in all patients with cancer having echocardiography, if available.	I	C
Baseline cardiac imaging prior to potentially cardiotoxic therapies		
Baseline comprehensive TTE is recommended in all patients with cancer at high risk and very high risk of CV toxicity before starting anticancer therapy.	I	C
Recommendation Table 5 for primary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVRF according to the 2021 ESC Guidelines on CVD prevention in clinical practice is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 6 for secondary prevention of cancer therapy-related cardiovascular toxicity		
Management of CVD according to applicable ESC Guidelines is recommended before, during, and after cancer therapy.	I	C
Recommendation Table 7 for baseline risk assessment and monitoring during anthracycline chemotherapy and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended in all patients with cancer before anthracycline chemotherapy.	I	B
In all adults receiving anthracycline chemotherapy, an echocardiogram is recommended within 12 months after completing treatment.	I	B
In high- and very high-risk patients, echocardiography is recommended every two cycles and within 3 months after completing treatment.	I	C
Cardiac serum biomarkers		
Baseline measurement of NP and cTn is recommended in high- and very high-risk patients prior to anthracycline chemotherapy.	I	B
cTn and NP monitoring before every cycle during anthracycline chemotherapy and 3 and 12 months after therapy completion is recommended in high- and very high-risk patients.	I	B
Recommendation Table 8 for baseline risk assessment and monitoring during HER2-targeted therapies and in the first 12 months after therapy		
TTE		
Baseline echocardiography is recommended before HER2-targeted therapies in all patients.	I	B
In patients receiving neoadjuvant or adjuvant HER2-targeted therapies, echocardiography is recommended every 3 months and within 12 months after completing treatment.	I	B
In metastatic HER2+ disease, echocardiography is recommended every 3 months during the first year; if the patient remains asymptomatic without CV toxicity, then surveillance can be reduced to every 6 months during future treatment.	I	C
Cardiac biomarkers		
Baseline NP and cTn measurement are recommended in high- and very high-risk patients prior to anti-HER2-targeted therapies.	I	C

Recommendation Table 9 for baseline risk assessment and monitoring during fluoropyrimidine therapy

Baseline CV risk assessment and evaluation including BP measurement, ECG, lipid profile, HbA1c measurement, and SCORE2/SCORE2-OP or equivalent is recommended before starting fluoropyrimidines.

I**C**

A baseline echocardiogram is recommended in patients with a history of symptomatic CVD before starting fluoropyrimidines.

I**C****Recommendation Table 10 for baseline risk assessment and monitoring during VEGFi****BP monitoring**

BP measurement is recommended for patients treated with VEGFi, bevacizumab, or ramucirumab at every clinical visit.

I**C**

Daily home monitoring of BP for patients treated with VEGFi during the first cycle, after each increase of VEGFi dose, and every 2–3 weeks thereafter is recommended.

I**C****ECG monitoring**

In patients treated with VEGFi at moderate or high risk of QTc prolongation, QTc monitoring is recommended monthly during the first 3 months and every 3–6 months thereafter.

I**C****Echocardiography**

Baseline echocardiography is recommended in high- and very high-risk patients treated with VEGFi or bevacizumab.

I**C****Recommendation Table 11 for baseline risk assessment and monitoring during second- and third-generation BCR-ABL tyrosine kinase inhibitors**

Baseline CV risk assessment is recommended in patients who require second- or third-generation BCR-ABL TKI.

I**C**

In patients treated with nilotinib or ponatinib, CV risk assessment is recommended every 3 months during the first year and every 6–12 months thereafter.

I**C**

Baseline echocardiography is recommended in patients scheduled to receive dasatinib.

I**C****Recommendation Table 12 for baseline risk assessment and monitoring Bruton tyrosine kinase inhibitor therapy****BP monitoring and management**

BP measurement is recommended for patients treated with BTK inhibitors at every clinical visit.

I**B****Echocardiography**

Baseline echocardiography is recommended in high-risk patients scheduled to receive BTK inhibitors.

I**C**

TTE is recommended in all patients who develop AF during BTK inhibitor therapy.

I**C****AF**

Opportunistic screening for AF by pulse-taking or ECG rhythm strip is recommended at every clinical visit during BTK inhibitor therapy.

I**C****Recommendation Table 13 for baseline risk assessment and monitoring during multiple myeloma therapies****BP monitoring**

BP measurement is recommended for patients treated with PI at every clinical visit.

I**C****Cardiac serum biomarkers**

Measurement of NP is recommended prior to PI in high- and very high-risk patients.

I**C**

NP and cTn measurements are recommended at baseline and every 3–6 months in patients with AL-CA.

I**B****TTE**

Baseline echocardiography, including assessment for AL-CA, is recommended in all patients with MM scheduled to receive PI.

I**C**

VTE prophylaxis		
Therapeutic doses of LMWH are recommended in patients with MM with previous VTE.	I	B
Prophylactic doses of LMWH are recommended in patients with MM with VTE-related risk factors (excluding previous VTE) at least during the first 6 months of therapy.	I	A
Recommendation Table 14 for baseline risk assessment and monitoring during combined RAF and MEK inhibitor therapy		
BP monitoring at each clinical visit and weekly outpatient monitoring during the first 3 months of treatment and monthly thereafter is recommended.	I	C
In patients treated with cobimetinib/vemurafenib, an ECG is recommended at 2 and 4 weeks after initiation of treatment and every 3 months thereafter.	I	C
Baseline echocardiography is recommended in all high- and very high-risk patients scheduled to receive combined RAF and MEK inhibitors.	I	C
Recommendation Table 15 for baseline risk assessment and monitoring during immunotherapy		
ECG, NP, and cTn measurements are recommended in all patients before starting ICI therapy.	I	B
Baseline echocardiography is recommended in high-risk patients before starting ICI therapy.	I	B
CV assessment is recommended every 6–12 months in high-risk patients who require long-term (>12 months) ICI treatment.	I	C
Recommendation Table 16 for baseline risk assessment and monitoring during androgen deprivation therapy for prostate cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in patients treated with ADT without pre-existing CVD.	I	B
Baseline and serial ECGs are recommended in patients at risk of QTc prolongation during ADT therapy.	I	B
Annual CV risk assessment is recommended during ADT.	I	B
Recommendation Table 17 for baseline risk assessment and monitoring during endocrine therapy for breast cancer		
Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended in BC patients receiving endocrine therapies without pre-existing CVD.	I	C
Annual CV risk assessment is recommended during endocrine therapy in BC patients with high 10-year risk of (fatal and non-fatal) CV events according to SCORE2/SCORE2-OP.	I	C
Recommendation Table 18 for baseline risk assessment and monitoring during cyclin-dependent kinase 4/6 inhibitor therapy		
QTc monitoring is recommended at baseline and 14 and 28 days in all patients with cancer receiving ribociclib.	I	A
QTc monitoring is recommended in patients treated with ribociclib with any dose increase.	I	B
Recommendation Table 19 for baseline risk assessment and monitoring during ALK and EGFR inhibitors		
Baseline CV risk assessment is recommended in patients before ALK inhibitors and EGFR inhibitors.	I	C
Baseline echocardiography is recommended in all patients with cancer before starting osimertinib.	I	B
Recommendation Table 20 for baseline risk assessment and monitoring in patients receiving chimeric antigen receptor T cell and tumour-infiltrating lymphocytes therapies		
Baseline ECG, NP, and cTn are recommended in all patients with cancer before starting CAR-T and TIL therapies.	I	C
A baseline echocardiography is recommended in patients with pre-existing CVD before starting CAR-T and TIL therapies.	I	C
Measurement of NP, cTn, and echocardiography are recommended in patients who develop CRS of ASTCT ≥ 2 .	I	C

Recommendation Table 21 for baseline risk assessment of patients before radiotherapy to a volume including the heart

Baseline CV risk assessment and estimation of 10-year fatal and non-fatal CVD risk with SCORE2 or SCORE2-OP is recommended.

I**B****Recommendation Table 22 for baseline risk assessment in haematopoietic stem cell transplantation patients**

Baseline and serial CV risk assessment (3 and 12 months, then yearly) including BP measurement, ECG, lipid measurement, and HbA1c is recommended in HSCT patients.

I**C**

Echocardiography is recommended in all patients before HSCT.

I**C****Recommendation Table 23 for the management of cardiovascular disease and cancer therapy-related cardiovascular toxicity in patients receiving anticancer treatment**

A specialist CV assessment is recommended for optimal diagnostic workup and management of patients with cancer who present with new CV toxicity during and after cancer treatment.

I**C****Recommendation Table 24 for the management of cancer treatment-related cardiac dysfunction during anthracycline chemotherapy****Anthracycline chemotherapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic CTRCD during anthracycline chemotherapy.

I**B**

Discontinuation of anthracycline chemotherapy is recommended in patients who develop symptomatic severe CTRCD.

I**C**

Temporary interruption of anthracycline chemotherapy is recommended in patients who develop symptomatic moderate CTRCD and a multidisciplinary approach regarding the decision to restart is recommended.

I**C**

A multidisciplinary approach regarding interruption vs. continuation of anthracycline chemotherapy is recommended in patients who develop mild symptomatic CTRCD.

I**C****Anthracycline chemotherapy-induced asymptomatic CTRCD**

Temporary interruption of anthracycline chemotherapy and initiation of HF therapy is recommended in patients who develop asymptomatic moderate or severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision when to restart is recommended in all patients with moderate or severe asymptomatic CTRCD.

I**C**

Continuation of anthracycline chemotherapy is recommended in asymptomatic patients who have LVEF $\geq$ 50% and who have developed a significant fall in GLS or a troponin or a NP elevation $>$ ULN.

I**C****Recommendation Table 25 for the management of cancer treatment-related cardiac dysfunction during HER2-targeted therapies****HER2-targeted therapy-induced symptomatic CTRCD**

HF therapy is recommended for patients who develop symptomatic moderate-to-severe CTRCD with LVEF $<$ 50% during HER2-targeted treatment.

I**B**

Temporary interruption of HER2-targeted treatment is recommended in patients who develop moderate or severe symptomatic CTRCD and the decision to restart should be based on a multidisciplinary approach after improvement of LV function and symptoms resolved.

I**C**

In patients who develop mild symptomatic CTRCD, HF therapy and a multidisciplinary approach regarding the decision to continue vs. interrupt HER2-targeted therapy are recommended.

I**C****HER2-targeted therapy-induced asymptomatic CTRCD**

Temporary interruption of HER2-targeted therapy and initiation of HF therapy is recommended in patients who develop asymptomatic severe CTRCD.

I**C**

A multidisciplinary approach regarding the decision to restart HER2-targeted treatment is recommended in patients with severe asymptomatic CTRCD.

I**C**

Continuation of HER2-targeted therapy is recommended in patients who develop asymptomatic mild (LVEF $\geq$ 50%) CTRCD⁶ with more frequent cardiac monitoring.

I**C**

ACE-I/ARB and beta-blockers are recommended in patients who develop asymptomatic moderate (LVEF 40–49%) CTRCD⁶ during HER2-targeted treatment.

I**C****Recommendation Table 26 for the diagnosis and management of immune checkpoint inhibitor-associated myocarditis**

cTn, ECG, and CV imaging (echocardiography and CMR) are recommended to diagnose ICI-associated myocarditis.

I**B**

In patients with suspected ICI-associated myocarditis, temporary interruption of ICI treatment is recommended until the diagnosis is confirmed or refuted.

I**C**

Interruption of ICI treatment is recommended in patients with confirmed ICI-associated myocarditis.	I	C
Continuous ECG monitoring to assess for new AV block and tachyarrhythmias during the acute phase is recommended for all patients with symptomatic ICI-associated myocarditis.	I	C
Early high-dose corticosteroids are recommended in patients with cancer and confirmed ICI-associated myocarditis.	I	C
Continuation of high-dose corticosteroids is recommended for the treatment of ICI-associated myocarditis until resolution of symptoms, LV systolic dysfunction, conduction abnormalities, and significant cTn reduction.	I	C
Admission to ICU (level 3), treatment with i.v. methylprednisolone, and optimal CV treatment including mechanical support (when indicated) is recommended for patients with ICI-associated fulminant myocarditis.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment in selected patients with previous uncomplicated ICI-associated myocarditis.	I	C
Recommendation Table 27 for the diagnosis and management of Takotsubo syndrome in patients with cancer		
Coronary angiography (invasive or CCTA) is recommended to exclude ACS.	I	C
CMR is recommended to exclude myocarditis and MI.	I	B
QT-prolonging drugs are not recommended during the acute TTS phase.	III	C
Recommendation Table 28 for the management of acute coronary syndromes in patients receiving anticancer treatment		
An invasive strategy is recommended in patients with cancer presenting with STEMI or high-risk NSTEMI-ACS with life expectancy ≥ 6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients where the cancer therapy is suspected as a contributing cause.	I	C
In patients with cancer, thrombocytopenia and ACS, aspirin is not recommended if platelets $< 10\,000/\mu\text{L}$.	III	C
In patients with cancer, thrombocytopenia and ACS, clopidogrel is not recommended if platelets $< 30\,000/\mu\text{L}$ and prasugrel or ticagrelor are not recommended if platelets $< 50\,000/\mu\text{L}$.	III	C
Recommendation Table 29 for the management of chronic coronary syndromes in patients receiving anticancer treatment		
Individualized duration of DAPT is recommended in patients with cancer with CCS, following revascularization, based upon thrombotic/ ischaemic and bleeding risk, type and stage of cancer, and current cancer treatment.	I	C
Recommendation Table 30 for the management of valvular heart disease in patients receiving anticancer treatment		
In patients with cancer and pre-existing severe VHD, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient preferences.	I	C
In patients with cancer developing new VHD during cancer therapy, management according to the 2021 ESC/EACTS Guidelines for the management of VHD is recommended, taking into consideration cancer prognosis and patient comorbidities.	I	C
Recommendation Table 31 for the management of atrial fibrillation in patients receiving anticancer treatment		
Long-term anticoagulation is recommended for stroke/systemic thromboembolism prevention in patients with cancer with AF and a CHA ₂ DS ₂ -VASc score ≥ 2 (men) or ≥ 3 (women) as per the 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation.	I	C
Thromboembolic and bleeding risk reassessment is recommended during follow-up in patients with cancer with AF.	I	C
Antiplatelet therapy or prophylactic LMWH are not recommended for stroke or systemic thromboembolism prevention in AF with cancer.	III	C
Recommendation Table 32 for the management of long QTc and ventricular arrhythmias in patients receiving anticancer treatment		
How to manage QTc prolongation in patients with cancer		
Discontinuation of QTc-prolonging cancer therapy is recommended in patients who develop TdP or sustained ventricular tachyarrhythmias during treatment.	I	C
Temporary interruption of QTc-prolonging cancer therapy is recommended in patients who develop asymptomatic QTcF ≥ 500 ms and an ECG should be repeated every 24 h until resolution of the QTcF prolongation.	I	C
Immediate withdrawal of any offending drug and correction of electrolyte abnormalities and other risk factors is recommended in patients with cancer who develop QTcF ≥ 500 ms.	I	C
Weekly ECG monitoring is recommended in asymptomatic patients with cancer with QTcF 480–500 ms who are treated with a QTc-prolonging cancer therapy.	I	C

A 12-lead ECG is recommended after any dose increase of QTc-prolonging cancer therapy.	I	C
Restarting QTc-prolonging cancer therapy		
A multidisciplinary discussion is recommended before restarting QTc-prolonging drugs in patients who have developed significant QTcF prolongation, to discuss alternative cancer treatments.	I	C
Weekly ECG monitoring during the first 4–6 weeks and then monthly thereafter is recommended in patients with cancer after restarting QTc-prolonging cancer therapy.	I	C
Recommendation Table 33 for the management of arterial hypertension in patients receiving anticancer treatment		
General		
Effective treatment of cancer therapy-induced arterial hypertension to prevent cancer treatment interruption and CV complications is recommended.	I	C
A BP target <140 mmHg systolic and <90 mmHg diastolic is recommended during cancer therapy.	I	C
The competing cancer and CV risk evaluation is recommended if the systolic BP is ≥ 180 mmHg or diastolic BP ≥ 110 mmHg, and any cancer therapy associated with hypertension should be deferred or temporarily withheld until the BP is controlled to values <160 mmHg (systolic) and <100 mmHg (diastolic).	I	C
Cancer therapy-induced arterial hypertension treatment		
ACE-I or ARB are the first-line antihypertensive drugs recommended for BP management in patients with cancer.	I	B
Dihydropyridine CCB are recommended as second-line antihypertensive drugs for patients with cancer with uncontrolled BP.	I	C
Combination therapy with ACE-I or ARB and dihydropyridine CCB is recommended in patients with cancer with systolic BP ≥ 160 mmHg and diastolic BP ≥ 100 mmHg.	I	C
Diltiazem and verapamil are not recommended to treat arterial hypertension in patients with cancer due to their drug–drug interactions.	III	C
Recommendation Table 34 for the management of venous thromboembolism in patients receiving anticancer treatment		
Apixaban, edoxaban, or rivaroxaban are recommended for the treatment of symptomatic or incidental VTE in patients with cancer without contraindications.	I	A
LMWH are recommended for the treatment of symptomatic or incidental VTE in patients with cancer with platelet count $>50\,000/\mu\text{L}$.	I	A
Catheter-associated VTE		
Duration of anticoagulation in patients with cancer with a catheter-associated VTE is recommended for a minimum of 3 months and continuing longer if the catheter remains in situ.	I	C
Recommendation Table 35 for venous thromboembolism prophylaxis during anticancer treatment		
Extended prophylaxis with LMWH for 4 weeks post-operatively is recommended for patients with cancer undergoing major open or laparoscopic abdominal or pelvic surgery with low bleeding risk and high VTE risk.	I	B
Prophylactic LMWH for the primary prevention of VTE is indicated in hospitalized patients with cancer or those with prolonged bed rest or reduced mobility in the absence of bleeding or other contraindications.	I	B
A discussion with the patient about the relative benefits and harms, cancer prognosis, drug cost, and duration of treatment is recommended prior to prophylactic anticoagulation for the primary prevention of VTE.	I	C
Recommendation Table 36 for management of peripheral artery disease during anticancer treatment		
In patients who develop new symptomatic PAD, a multidisciplinary approach regarding the decision to continue vs. interrupt culprit cancer therapy is recommended.	I	C
Recommendation Table 37 for the management of pulmonary hypertension during anticancer treatment		
Right-heart catheterization and discontinuation of dasatinib is recommended in patients who develop symptomatic or asymptomatic increase in peak TRV > 3.4 m/s.	I	C
In patients with confirmed dasatinib-induced PAH or new asymptomatic peak TRV > 3.4 m/s, an alternative BCR-ABL inhibitor is recommended after peak TRV recovery to <2.8 m/s.	I	C
Recommendation Table 38 for the management of pericardial diseases in patients receiving anticancer treatment		
General		
Diagnosis and management of acute pericarditis in patients with cancer based on the 2015 ESC Guidelines for the diagnosis and management of pericardial diseases is recommended and a multidisciplinary discussion is needed before interrupting cancer therapy.	I	C

Diagnosis and management of ICI-associated pericarditis		
Multimodality CV imaging (echocardiography, CMR ± CT), ECG and measurement of cardiac biomarkers are recommended to confirm the diagnosis, assess the haemodynamic consequences of pericardial disease, and rule out associated myocarditis.	I	C
Prednisolone and colchicine are recommended for patients with ICI-associated pericarditis.	I	C
Interruption of ICI treatment in patients with confirmed ICI-associated pericarditis with moderate-to-severe pericardial effusion is recommended.	I	C
A multidisciplinary discussion is recommended before restarting ICI treatment.	I	C
Recommendation Table 39 for end-of-cancer therapy cardiovascular risk assessment		
Educating and supporting patients with cancer to make appropriate healthy lifestyle choices is recommended.	I	C
Education is recommended for patients with cancer regarding recognition for early signs and symptoms of CVD.	I	C
CVRF assessment is recommended during the first year after cancer therapy and thereafter according to the 2021 ESC Guidelines on CVD prevention in clinical practice.	I	B
In asymptomatic high-risk patients, echocardiography and cardiac serum biomarkers are recommended at 3 and 12 months after completion of cancer therapy.	I	B
Cardiology referral is recommended in patients with cancer with new cardiac symptoms or new asymptomatic abnormalities in echocardiography and/or cardiac serum biomarkers at the end of therapy assessment.	I	C
Long-term continuation of cardiac medication is recommended in patients who develop severe CTRCD during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed TKI-mediated hypertension during cancer therapy.	I	C
CV follow-up and treatment optimization is recommended in patients who developed vascular toxicities during cancer therapy.	I	C
ECG follow-up is recommended in patients who developed QT lengthening or LQTS during cancer therapy.	I	C
Recommendation Table 40 for cardiovascular surveillance in asymptomatic adults who are childhood and adolescent cancer survivors		
Education of adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart and their healthcare providers regarding their increased CV risk is recommended.	I	B
Annual screening for modifiable CVRF is recommended in adults who are childhood and adolescent CS treated with anthracyclines, mitoxantrone, and/or RT to a volume including the heart.	I	C
CV assessment is recommended in female childhood and adolescent CS prior to pregnancy or in the first trimester.	I	C
Recommendation Table 41 for cardiovascular surveillance in asymptomatic adult cancer survivors		
Annual CV risk assessment, including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or RT.	I	B
CV toxicity risk re-stratification is recommended 5 years after therapy to organize long-term follow-up.	I	C
Recommendation Table 42 for adult cancer survivors who develop cancer therapy-related cardiac dysfunction late after cardiotoxic cancer therapy		
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic CTRCD.	I	C
Recommendation Table 43 for adult cancer survivors with coronary artery disease		
Asymptomatic radiation-induced CAD detected during surveillance		
Non-invasive stress testing is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischaemia-directed management.	I	C
A MDT discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischaemia or severe left main CAD.	I	C

Symptomatic CAD		
Pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended in CS with radiation-induced CAD where CABG is considered.	I	C
Recommendation Table 44 for adult cancer survivors with valvular heart disease		
A MDT approach is recommended to discuss and define the surgical risk in CS with severe VHD.	I	C
Recommendation Table 46 for cardiovascular monitoring in cancer survivors during pregnancy		
In high-risk female CS, pre-pregnancy counselling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	I	C
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	I	C
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	I	C
Recommendation Table 47 for cardiovascular assessment and monitoring of pregnant women with cancer		
Management by an expert MDT (the pregnancy heart team) in an expert centre is recommended for pregnant women with cancer who require cardiotoxic cancer therapy.	I	C
Cardiac assessment prior to cardiotoxic cancer therapy in pregnant women is recommended and consists of clinical history, physical examination, ECG, and echocardiography.	I	C
Recommendation Table 48 for carcinoid valvular heart diseases		
Echocardiography is recommended for the detection of carcinoid cardiac involvement in all patients with carcinoid syndrome and elevated NP levels and/or clinical signs of carcinoid heart disease, and for surveillance every 3 or 6 months depending on the severity of cardiac involvement and clinical status.	I	B
A MDT discussion for optimal medial management to prevent carcinoid crisis is recommended before any invasive or surgical cardiac procedure.	I	C
Valve replacement surgery is recommended in symptomatic patients with severe carcinoid tricuspid or pulmonary VHD and an expected survival ≥ 12 months.	I	C
Valve replacement or repair surgery is recommended in symptomatic patients with severe carcinoid mitral or aortic VHD and an expected survival ≥ 12 months.	I	C
Recommendation Table 49 for amyloid light-chain cardiac amyloidosis diagnosis and monitoring		
Echocardiography, NP, and cTn are recommended for the diagnosis of AL-CA in patients with plasma cell dyscrasia.	I	B
CMR is recommended in patients with suspected AL-CA.	I	A
Recommendation Table 50 for risk stratification and monitoring for patients with cardiac implantable electronic devices undergoing radiotherapy		
Risk stratification including planned radiation type and energy, dose to CIED, the patient's device type, and pacing dependence is recommended prior to starting treatment.	I	C
In patients undergoing RT, a CIED check is recommended in all patients before and after completing RT, and during RT according to individual risk.	I	C
In patients with a CIED undergoing RT at high risk of arrhythmia and/or device dysfunction, ECG monitoring and/or pulse oximetry are recommended during every RT session.	I	C

© ESC 2022

3D, three-dimensional; ACE-I, angiotensin-converting enzyme inhibitors; ACS, acute coronary syndromes; ADT, androgen deprivation therapy; AF, atrial fibrillation; AL-CA, amyloid light-chain cardiac amyloidosis; ALK, anaplastic lymphoma kinase; ARB, angiotensin receptor blocker; ASTCT, American Society for Transplantation and Cellular; AV, atrioventricular; BC, breast cancer; BCR-ABL, breakpoint cluster region-Abelson oncogene locus; BP, blood pressure; BTK, Bruton tyrosine kinase; CABG, coronary artery bypass graft; CAD, coronary artery disease; CAR-T, chimeric antigen receptor T cell; CCB, calcium channel blockers; CCS, chronic coronary syndromes; CCTA, coronary computed tomography angiography; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age >75 years (2 points), Diabetes mellitus, Stroke (2 points)—Vascular disease, Age 65–74 years, Sex category (female); CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; CRS, cytokine release syndrome; CT, computed tomography; cTn, cardiac troponin; CTRCD, cancer therapy-related cardiac dysfunction; CS, cancer survivors; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; DAPT, dual antiplatelet therapy; EACTS, European Association for Cardio-Thoracic Surgery; ECG, electrocardiogram; EGFR, epidermal growth factor receptor; ESC, European Society of Cardiology; GLS, global longitudinal strain; HbA_{1c}, glycated haemoglobin; HER2, human epidermal receptor 2; HF, heart failure; HSCT, haematopoietic stem cell transplantation; ICI, immune checkpoint inhibitors; ICU, intensive care unit; iv, intravenous; LIMA, left internal mammary artery; LMWH, low-molecular-weight heparins; LQTS, long QT syndrome; LV, left ventricular; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; MEK, mitogen-activated extracellular signal-regulated kinase; MI, myocardial infarction; MM, multiple myeloma; NP, natriuretic peptides; NSTEMI-ACS, non-ST-segment elevation acute coronary syndromes; PAD, peripheral artery disease; PAH, pulmonary arterial hypertension; PI, proteasome inhibitors; QTc, corrected QT interval; QTcf, corrected QT interval using Fridericia correction; RAF, rapidly accelerated fibrosarcoma; RIMA, right internal mammary artery; RT, radiotherapy; SCORE2, Systematic Coronary Risk Estimation 2; SCORE2-OP, Systematic Coronary Risk Estimation 2—Older Persons; STEMI, ST-segment elevation myocardial infarction; TdP, torsade de pointes; TIL, tumour-infiltrating lymphocytes; TKI, tyrosine kinase inhibitors; TTE, transthoracic echocardiography; TTS, Takotsubo syndrome; TRV, tricuspid regurgitation velocity; ULN, upper limit of normal; VEGF, vascular endothelial growth factor inhibitors; VHD, valvular heart disease; VTE, venous thromboembolism.

^aClass of recommendation.

^bLevel of evidence.

16. Quality indicators for cardio-oncology

Quality indicators (QIs) are tools that may be used to evaluate care quality, including structural, process, and outcomes of care.⁸³⁵ They may also serve as a mechanism for enhancing adherence to guideline recommendations, through associated quality improvement initiatives and the benchmarking of care providers.^{836,837} As such, the role of QIs in improving care and outcomes for CVD is increasingly recognized by healthcare authorities, professional organizations, payers, and the public.⁸³⁵

The ESC understands the need for measuring and reporting quality and outcomes of CV care and has established methods for the development of the ESC QIs for the quantification of care and outcomes for CVD.⁸³⁵ These methods were used to develop QIs pertinent to cardio-oncology in parallel with the writing of this Clinical Practice Guideline document and through the collaboration with patient representatives and

domain experts. The QIs, alongside their measurement specifications and development process will be published separately.

17. Supplementary data

[Supplementary data](#) is available at *European Heart Journal* online.

18. Data availability statement

No new data were generated or analysed in support of this research.

19. Author information

Author/Task Force Member Affiliations: **Liam S. Couch**, King's College London BHF Centre, the Rayne Institute, St Thomas's Hospital, King's College London, London, United Kingdom; **Riccardo Asteggiano**, Cardiology, LARC (Laboratorio Analisi Ricerca Clinica), Turin, Italy, and School of Medicine, Insubria University, Varese, Italy; **Marianne C. Aznar**, Division of Cancer Sciences, Faculty of Biology, Medicine and Health, University of Manchester, Manchester, United Kingdom; **Jutta Bergler-Klein**, Department of Cardiology, Medical University of Vienna, Vienna, Austria; **Giuseppe Boriani**, Cardiology Division, Department of Biomedical, Metabolic and Neural Sciences, University of Modena and Reggio Emilia, Policlinico di Modena, Modena, Italy; **Daniela Cardinale**, Cardioncology Unit, European Institute of Oncology I.R.C.C.S., Milan, Italy; **Raul Cordoba**, Department of Hematology, Fundacion Jimenez Diaz University Hospital, Madrid, Spain, and Cancer Research Group, Health Research Institute IIS-FJD, Madrid, Spain; **Bernard Cosyns**, Cardiology, Centrum voor Hart en Vaatziekten (CHVZ), Universitair Ziekenhuis Brussel (UZB), Brussels, Belgium, and In Vivo Molecular and Cellular (ICMI) Center, Vrije Universiteit Brussel, Brussels, Belgium; **David J. Cutter**, Nuffield Department of Population Health, University of Oxford, Oxford, United Kingdom, and Oxford Cancer Centre, Oxford University Hospitals NHS Foundation Trust, Oxford, United Kingdom; **Evandro de Azambuja**, Medical Oncology Department, Institut Jules Bordet, Brussels, Belgium; **Rudolf A. de Boer**, Cardiology, University Medical Center Groningen, Groningen, Netherlands; **Susan F. Dent**, Department of Medicine, Duke Cancer Institute, Durham, NC, United States of America; **Dimitrios Farmakis**, Medical School, University of Cyprus, Nicosia, Cyprus; **Sofie A. Gevaert**, Cardiology, Ghent University Hospital, Ghent, Belgium; **Diana A. Gorog**, Postgraduate Medicine, University of Hertfordshire, Hatfield, United Kingdom, and National Heart and Lung Institute, Imperial College, London, United Kingdom, and Cardiology Department, East and North Hertfordshire NHS Trust, Stevenage, United Kingdom; **Joerg Herrmann**, Department of Cardiovascular Diseases, Mayo Clinic, Rochester, MN, United States of America; **Daniel Lenihan**, Cardio-Oncology, International Cardio-Oncology Society, Tampa, FL, United States of America, and Cardiology, Saint Francis Healthcare, Cape Girardeau, MO, United States of America; **Javid Moslehi**, Section of Cardio-Oncology & Immunology, Cardiovascular Research Institute, University of California, San Francisco (UCSF), San Francisco, CA, United States of America; **Brenda Moura**, Cardiology Department, Armed Forces Hospital, Porto, Portugal, and Faculty of Medicine, University of Porto, Porto, Portugal; **Sonja S. Salinger**, Clinic for Cardiovascular Disease, University Clinical Center, Nis, Serbia, and Medical Faculty, University of Nis, Nis, Serbia; **Richard Stephens** (United Kingdom), ESC

Patient Forum, Sophia Antipolis, France; **Thomas M. Suter**, Department of Cardiology, Inselspital, Bern University Hospital, University of Bern, Bern, Switzerland; **Sebastian Szmit**, Department of Pulmonary Circulation, Thromboembolic Diseases and Cardiology, Centre of Postgraduate Medical Education, Otwock, Poland, and Institute of Hematology and Transfusion Medicine, Transfusion Medicine, Warsaw, Poland; **Juan Tamargo**, Pharmacology and Toxicology, Universidad Complutense, Madrid, Spain; **Paaladinesh Thavendiranathan**, Department of Medicine, Division of Cardiology, Ted Rogers Program in Cardiotoxicity Prevention, Peter Munk Cardiac Center, Toronto General Hospital, University Health Network, University of Toronto, Toronto, Canada; **Carlo G. Tocchetti**, Cardio-Oncology Unit, Department of Translational Medical Sciences, Federico II University, Naples, Italy, and Center for Basic and Clinical Immunology Research (CISI), Federico II University, Naples, Italy, and Interdepartmental Center for Clinical and Translational Research (CIRCET), Federico II University, Naples, Italy; **Peter van der Meer**, Cardiology, University Medical Center Groningen, Groningen, Netherlands; **Helena J.H. van der Pal**, Princess Máxima Center for Pediatric Oncology, Princess Máxima Center for Pediatric Oncology, Utrecht, Netherlands.

20. Appendix

ESC Scientific Document Group

Includes Document Reviewers and ESC National Cardiac Societies.

Document Reviewers: Patrizio Lancellotti (CPG Review Coordinator) (Belgium), Franck Thuny (CPG Review Coordinator) (France), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Berthe Aleman (Netherlands), Joachim Alexandre (France), Ana Barac (United States of America), Michael A. Borger (Germany), Ruben Casado-Arroyo (Belgium), Jennifer Cautela (France), Jolanta Alekšienė (Lithuania), Maja Cikes (Croatia), Alain Cohen-Solal (France), Kreena Dhiman (United Kingdom), Stéphane Ederhy (France), Thor Edvardsen (Norway), Laurent Fauchier (France), Michael Fradley (United States of America), Julia Grapsa (United Kingdom), Sigrun Halvorsen (Norway), Michael Heuser (Germany), Marc Humbert (France), Tiny Jaarsma (Sweden), Thomas Kahan (Sweden), Aleksandra Konradi (Russian Federation), Konstantinos C. Koskinas (Switzerland), Dipak Kotecha (United Kingdom), Bonnie Ky (United States of America), Ulf Landmesser (Germany), Basil S. Lewis (Israel), Ales Linhart (Czech Republic), Gregory Y. H. Lip (United Kingdom), Maja-Lisa Løchen (Norway), Katarzyna Malaczynska-Rajpold (United Kingdom), Marco Metra (Italy), Richard Mindham (United Kingdom), Marie Moonen (Belgium), Tomas G. Neilan (United States of America), Jens Cosedis Nielsen (Denmark), Anna-Sonia Petronio (Italy), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Joe-Elie Salem (France), Gianluigi Savarese (Sweden), Marta Sitges (Spain), Jurrien ten Berg (Netherlands), Rhian M. Touyz (Canada/United Kingdom), Agnieszka Tycinska (Poland), Matthias Wilhelm (Switzerland), Jose Luis Zamorano (Spain)

ESC National Cardiac Societies actively involved in the review process of the 2022 ESC Guidelines on cardio-oncology: **Algeria:** Algerian Society of Cardiology, Nadia Laredj; **Armenia:** Armenian Cardiologists Association, Parounak Zelveian; **Austria:** Austrian Society of Cardiology, Peter P. Rainer; **Azerbaijan:** Azerbaijan Society of Cardiology, Fuad Samadov; **Belarus:** Belorussian Scientific Society of Cardiologists, Uladzimir Andrushchuk; **Belgium:** Belgian Society of Cardiology, Bernhard L. Gerber; **Bosnia and Herzegovina:** Association of Cardiologists of Bosnia and Herzegovina, Mirsad Selimović; **Bulgaria:** Bulgarian Society of Cardiology, Elena Kinova; **Croatia:** Croatian Cardiac Society, Jure Samardžić; **Cyprus:**

Cyprus Society of Cardiology, Evagoras Economides; **Czechia:** Czech Society of Cardiology, Radek Pudil; **Denmark:** Danish Society of Cardiology, Kirsten M. Nielsen; **Egypt:** Egyptian Society of Cardiology, Tarek A. Kafafy; **Estonia:** Estonian Society of Cardiology, Riina Vettus; **Finland:** Finnish Cardiac Society, Suvi Tuohinen; **France:** French Society of Cardiology, Stéphanie Ederhy; **Georgia:** Georgian Society of Cardiology, Zurab Pagava; **Germany:** German Cardiac Society, Tienush Rassaf; **Greece:** Hellenic Society of Cardiology, Alexandros Briasoulis; **Hungary:** Hungarian Society of Cardiology, Dániel Czuriga; **Iceland:** Icelandic Society of Cardiology, Karl K. Andersen; **Ireland:** Irish Cardiac Society, Yvonne Smyth; **Israel:** Israel Heart Society, Zaza Iakobishvili; **Italy:** Italian Federation of Cardiology, Iris Parrini; **Kazakhstan:** Association of Cardiologists of Kazakhstan, Amina Rakisheva; **Kosovo (Republic of):** Kosovo Society of Cardiology, Edita Pllana Pruthi; **Kyrgyzstan:** Kyrgyz Society of Cardiology, Erkin Mirrakhimov; **Latvia:** Latvian Society of Cardiology, Oskars Kalejs; **Lebanon:** Lebanese Society of Cardiology, Hadi Skouri; **Libya:** Libyan Cardiac Society, Hisham Benlamin; **Lithuania:** Lithuanian Society of Cardiology, Diana Aliaduonytė; **Luxembourg:** Luxembourg Society of Cardiology, Alessandra Iovino; **Malta:** Maltese Cardiac Society, Alice M. Moore; **Moldova (Republic of):** Moldavian Society of Cardiology, Daniela Bursacovschi; **Morocco:** Moroccan Society of Cardiology, Aatif Benyass; **Netherlands:** Netherlands Society of Cardiology, Olivier Manintveld; **North Macedonia:** North Macedonian Society of Cardiology, Marijan Bosevski; **Norway:** Norwegian Society of Cardiology, Geeta Gulati; **Poland:** Polish Cardiac Society, Przemysław Leszek; **Portugal:** Portuguese Society of Cardiology, Manuela Fiuza; **Romania:** Romanian Society of Cardiology, Ruxandra Jurcut; **Russian Federation:** Russian Society of Cardiology, Yury Vasyuk; **San Marino:** San Marino Society of Cardiology, Marina Foscoli; **Serbia:** Cardiology Society of Serbia, Dragan Simic; **Slovakia:** Slovak Society of Cardiology, Miroslav Slanina; **Slovenia:** Slovenian Society of Cardiology, Luka Lipar; **Spain:** Spanish Society of Cardiology, Ana Martin-Garcia; **Sweden:** Swedish Society of Cardiology, Laila Häbbert; **Switzerland:** Swiss Society of Cardiology, Reto Kurmann; **Syrian Arab Republic:** Syrian Cardiovascular Association, Ahmad Alayed; **Tunisia:** Tunisian Society of Cardiology and Cardio-Vascular Surgery, Leila Abid; **Turkey:** Turkish Society of Cardiology, Cafer Zorkun; **Ukraine:** Ukrainian Association of Cardiology, Elena Nesukay; **United Kingdom of Great Britain and Northern Ireland:** British Cardiovascular Society, Charlotte Manisty; **Uzbekistan:** Association of Cardiologists of Uzbekistan, Nigora Srojidinova.

ESC Clinical Practice Guidelines (CPG) Committee: Colin Baigent (Chairperson) (United Kingdom), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Sotiris Antoniou (United Kingdom), Elena Arbelo (Spain), Riccardo Asteggiano (Italy), Andreas Baumbach (United Kingdom), Michael A. Borger (Germany), Jelena Uelutkienė (Lithuania), Maja Cikes (Croatia), Jean-Philippe Collet (France), Volkmar Falk (Germany), Laurent Fauchier (France), Chris P. Gale (United Kingdom), Sigrun Halvorsen (Norway), Bernard Iung (France), Tiny Jaarsma (Sweden), Aleksandra Konradi (Russian Federation), Konstantinos C. Koskinas (Switzerland), Dipak Kotecha (United Kingdom), Ulf Landmesser (Germany), Basil S. Lewis (Israel), Ales Linhart (Czech Republic), Maja-Lisa Løchen (Norway), Richard Mindham (United Kingdom), Jens Cosedis Nielsen (Denmark), Steffen E. Petersen (United Kingdom), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Marta Sitges (Spain), Rhian M. Touyz (Canada/United Kingdom).

References

1Herrmann J, Lenihan D, Armenian S, Barac A, Blaes A, Cardinale D, et al. Defining cardiovascular toxicities of cancer therapies: an International Cardio-Oncology Society (IC-OS) consensus statement. *Eur Heart J* 2022;43:2802-299.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

2Herrmann J, Lerman A, Sandhu NP, Villarraga HR, Mulvagh SL, Kohli M. Evaluation and management of patients with heart disease and cancer: Cardio-oncology. *Mayo Clin Proc* 2014;89:1287-1306.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

3Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2021;71:209-249.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)

4Zamorano JL, Lancellotti P, Rodriguez Muñoz D, Aboyans V, Asteggiano R, Galderisi M, et al. 2016 ESC Position Paper on cancer treatments and cardiovascular toxicity developed under the auspices of the ESC Committee for Practice Guidelines. *Eur Heart J* 2016;37:2768-2801.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

5Lancellotti P, Suter TM, López-Fernández T, Galderisi M, Lyon AR, Van Der Meer P, et al. Cardio-oncology services: rationale, organization, and implementation. *Eur Heart J* 2019;40:1756-1763.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

6Michel G, Mulder RL, van der Pal HJH, Skinner R, Bardi E, Brown MC, et al. Evidence-based recommendations for the organization of long-term follow-up care for childhood and adolescent cancer survivors: a report from the PanCareSurFup Guidelines Working Group. *J Cancer Surviv* 2019;13:759-772.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

7Armenian SH, Hudson MM, Mulder RL, Chen MH, Constine LS, Dwyer M, et al. Recommendations for cardiomyopathy surveillance for survivors of childhood cancer: a report from the International Late Effects of Childhood Cancer Guideline Harmonization Group. *Lancet Oncol* 2015;16:e123-e136.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

8van Kalsbeek RJ, Mulder RLRL, Skinner R, Kremer LCMCM. The concept of cancer survivorship and models for long-term follow-up. *Front Horm Res* 2021;54:1-15.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

9Herrmann J. Adverse cardiac effects of cancer therapies: cardiotoxicity and arrhythmia. *Nat Rev Cardiol* 2020;17:474-502.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

10Herrmann J. Vascular toxic effects of cancer therapies. *Nat Rev Cardiol* 2020;17:503-522.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 11 Gilchrist SC, Barac A, Ades PA, Alfano CM, Franklin BA, Jones LW, et al. Cardio-oncology rehabilitation to manage cardiovascular outcomes in cancer patients and survivors: a scientific statement from the American Heart Association. *Circulation* 2019;139:e997âe1012.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 12 Lyon AR, Dent S, Stanway S, Earl H, Brezden-Masley C, Cohen-Solal A, et al. Baseline cardiovascular risk assessment in cancer patients scheduled to receive cardiotoxic cancer therapies: a position statement and new risk assessment tools from the Cardio-Oncology Study Group of the Heart Failure Association of the European Society. *Eur J Heart Fail* 2020;22:1945â1960.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 13 Battisti NML, Andres MS, Lee KA, Ramalingam S, Nash T, Mappouridou S, et al. Incidence of cardiotoxicity and validation of the Heart Failure Association International Cardio-Oncology Society risk stratification tool in patients treated with trastuzumab for HER2-positive early breast cancer. *Breast Cancer Res Treat* 2021;188:149â163.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 14 McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;42:3599â3726.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 15 Jaffe AS, Vasile VC, Milone M, Saenger AK, Olson KN, Apple FS. Diseased skeletal muscle: a noncardiac source of increased circulating concentrations of cardiac troponin T. *J Am Coll Cardiol* 2011;58:1819â1824.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 16 Schmid J, Liesinger L, Birner-Gruenberger R, Stojakovic T, Scharnagl H, Dieplinger B, et al. Elevated cardiac troponin T in patients with skeletal myopathies. *J Am Coll Cardiol* 2018;71:1540â1549.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 17 Delombaerde D, Vervloet D, Franssen C, Croes L, Gremontprez F, Prenen H, et al. Clinical implications of isolated troponinemia following immune checkpoint inhibitor therapy. *ESMO Open* 2021;6:100216.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 18 Ferreira VM, Schulz-Menger J, Holmvang G, Kramer CM, Carbone I, Sechtem U, et al. Cardiovascular magnetic resonance in nonischemic myocardial inflammation: expert recommendations. *J Am Coll Cardiol* 2018;72:3158â3176.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 19 Visseren FLJ, Mach F, Smulders YM, Carballo D, Koskinas KC, Bäck M, et al. 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice. *Eur Heart J* 2021;42:3227â3337.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 20 Vandenberg B, Vandael E, Robyns T, Vandenberghe J, Garweg C, Foulon V, et al. Which QT correction formulae to use for QT monitoring? *J Am Heart Assoc* 2016;5:e003264.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

21Armenian ^ SH, Lacchetti ^ C, Barac ^ A, Carver ^ J, Constine ^ LS, Denduluri ^ N, et al.
^ Prevention and monitoring of cardiac dysfunction in survivors of adult cancers: American Society of
Clinical Oncology clinical practice guideline. J Clin Oncol ^ 2017;35:893a^1911.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

22Curigliano ^ G, Lenihan ^ D, Fradley ^ M, Ganatra ^ S, Barac ^ A, Blaes ^ A, et al. ^ Management
of cardiac disease in cancer patients throughout oncological treatment: ESMO consensus
recommendations. Ann Oncol ^ 2020;31:171a^190.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

23Rossello ^ X, Dorresteijn ^ JAN, Janssen ^ A, Lambrinou ^ E, Scherrenberg ^ M, Bonnefoy-Cudraz
^ E, et al. ^ Risk prediction tools in cardiovascular disease prevention: a report from the ESC
Prevention of CVD Programme led by the European Association of Preventive Cardiology (EAPC) in
collaboration with the Acute Cardiovascular Care Association (ACCA) and the Association of
Cardiovascular Nursing and Allied Professions (ACNAP). Eur J Cardiovasc Nurs
^ 2019;18:534a^544.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

24Romond ^ EH, Jeong ^ JH, Rastogi ^ P, Swain ^ SM, Geyer ^ CE, Ewer ^ MS, et al. ^ Seven-year
follow-up assessment of cardiac function in NSABP B-31, a randomized trial comparing doxorubicin and
cyclophosphamide followed by paclitaxel (ACP) with ACP plus trastuzumab as adjuvant therapy for
patients with node-positive, human epidermal growth factor receptor 2-positive breast cancer. J Clin
Oncol ^ 2012;30:3792a^3799.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

25Ezaz ^ G, Long ^ JB, Gross ^ CP, Chen ^ J. Risk prediction model for heart failure and
cardiomyopathy after adjuvant trastuzumab therapy for breast cancer. J Am Heart Assoc
^ 2014;3:e000472.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

26Rushton ^ M, Johnson ^ C, Dent ^ S. Trastuzumab-induced cardiotoxicity: testing a clinical risk score in
a real-world cardio-oncology population. Curr Oncol ^ 2017;24:176a^180.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

27Dranitsaris ^ G, Rayson ^ D, Vincent ^ M, Chang ^ J, Gelmon ^ K, Sandor ^ D, et al. ^ The
development of a predictive model to estimate cardiotoxic risk for patients with metastatic breast cancer
receiving anthracyclines. Breast Cancer Res Treat ^ 2008;107:443a^450.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

28Abdel-Qadir ^ H, Thavendiranathan ^ P, Austin ^ PC, Lee ^ DS, Amir ^ E, Tu J ^ V, et al.
^ Development and validation of a multivariable prediction model for major adverse cardiovascular
events after early stage breast cancer: a population-based cohort study. Eur Heart J
^ 2019;40:3913a^3920.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

29Kang ^ Y, Assuncao ^ BL, Denduluri ^ S, McCurdy ^ S, Luger ^ S, Lefebvre ^ B, et al.
^ Symptomatic heart failure in acute leukemia patients treated with anthracyclines. JACC

CardioOncology 2019;1:208â217.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

30Mart n Garc a A, Mitroi C, Maz n Ramos P, Garc a Sanz R, Virizuela JA, Arenas M, et al. Stratification and management of cardiovascular risk in cancer patients. A consensus document of the SEC, FEC, SEOM, SEOR, SEHH, SEMG, AEEMT, AECC, and AECC. Rev Espa ola Cardiol (English Ed) 2021;74:438â448.

[Google Scholar](#) [Crossref](#) [Search ADS](#)

31Caro-Cod n J, L pez-Fern ndez T, H lvarez-Ortega C, Zamora Au n P, Rodr guez IR, G mez Prieto P, et al. Cardiovascular risk factors during cancer treatment. Prevalence and prognostic relevance: insights from the CARDIOTOX registry. Eur J Prev Cardiol 2022;29

(6)

:859â868.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)

32Feijen EAM, Leisenring WM, Stratton KL, Ness KK, Van Der Pal HJH, Van Dalen EC, et al. Derivation of anthracycline and anthraquinone equivalence ratios to doxorubicin for late-onset cardiotoxicity. JAMA Oncol 2019;5:864â871.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

33Pareek N, Cevallos J, Moliner P, Shah M, Tan LL, Chambers V, et al. Activity and outcomes of a cardio-oncology service in the United Kingdom: a five-year experience. Eur J Heart Fail 2018;20:1721â1731.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

34Koene RJ, Prizment AE, Blaes A, Konety SH. Shared risk factors in cardiovascular disease and cancer. Circulation 2016;133:1104â1114.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

35Caocci G, Mulas O, Abruzzese E, Luciano L, Iurlo A, Attolico I, et al. Arterial occlusive events in chronic myeloid leukemia patients treated with ponatinib in the real-life practice are predicted by the Systematic Coronary Risk Evaluation (SCORE) chart. Hematol Oncol 2019;37:296â302.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

36Libby P, Sidlow R, Lin AE, Gupta D, Jones LW, Moslehi J, et al. Clonal hematopoiesis: crossroads of aging, cardiovascular disease, and cancer: JACC review topic of the week. J Am Coll Cardiol 2019;74:567â577.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

37Garcia-Pavia P, Kim Y, Restrepo-Cordoba MA, Lunde IG, Wakimoto H, Smith AM, et al. Genetic variants associated with cancer therapy-induced cardiomyopathy. Circulation 2019;140:31â41.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

38Bhatia S. Genetics of anthracycline cardiomyopathy in cancer survivors. JACC CardioOncology 2020;2:539â552.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

39Katzke A VA, Kaaks A R, Kühn A T. Lifestyle and cancer risk. *Cancer J* 2015;21:104â110.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

40Sharifi-Rad A J, Rodrigues A CF, Sharopov A F, Docea A AO, Karaca A AC, Sharifi-Rad A M, et al.
Diet, lifestyle and cardiovascular diseases: linking pathophysiology to cardioprotective effects of
natural bioactive compounds. *Int J Environ Res Public Health* 2020;17:2326.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

41Turner A MC, Andersen A ZJ, Baccarelli A A, Diver A WR, Gapstur A SM, Pope A CA, et al.
Outdoor air pollution and cancer: an overview of the current evidence and public health
recommendations. *CA Cancer J Clin* 2020;70 :460â479.
[Google Scholar](#) [OpenURL Placeholder Text](#)

42Lind A PM, Lind A L. Are persistent organic pollutants linked to lipid abnormalities, atherosclerosis and
cardiovascular disease? A review. *J Lipid Atheroscler* 2020;9:334â348.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

43Zamorano A JL, Gottfridsson A C, Asteggiano A R, Atar A D, Badimon A L, Bax A JJ, et al. The
cancer patient and cardiology. *Eur J Heart Fail* 2020;22:2290â2309.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

44Fradley A MG, Moslehi A J. QT prolongation and oncology drug development. *Card Electrophysiol Clin*
2015;7:341â355.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

45Porta-Sánchez A A, Gilbert A C, Spears A D, Amir A E, Chan A J, Nanthakumar A K, et al.
Incidence, diagnosis, and management of QT prolongation induced by cancer therapies: a systematic
review. *J Am Heart Assoc* 2017;6:e007724.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

46Curigliano A G, Spitaleri A G, Fingert A HJ, de Braud A F, Sessa A C, Loh A E, et al. Drug-induced
QTc interval prolongation: a proposal towards an efficient and safe anticancer drug development. *Eur J*
Cancer 2008;44:494â500.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

47Viganego A F, Singh A R, Fradley A MG. Arrhythmias and other electrophysiology issues in cancer
patients receiving chemotherapy or radiation. *Curr Cardiol Rep* 2016;18:52.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

48Curigliano A G, Spitaleri A G, De Braud A F, Cardinale A D, Cipolla A C, Civelli A M, et al. QTc
prolongation assessment in anticancer drug development: clinical and methodological issues.
Ecancermedicalscience 2009;3:130.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

49Salem A J-E, Nguyen A LS, Moslehi A JJ, Ederhy A S, Lebrun-Vignes A B, Roden A DM, et al.
Anticancer drug-induced life-threatening ventricular arrhythmias: a World Health Organization
pharmacovigilance study. *Eur Heart J* 2021;42:3915â3928.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 50Lentz R, Feinglass J, Ma S, Akhter N. Risk factors for the development of atrial fibrillation on ibrutinib treatment. *Leuk Lymphoma* 2019;60:1447â1453.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 51Mato AR, Clasen S, Pickens P, Gashonia L, Rhodes J, Svoboda J, et al. Left atrial abnormality (LAA) as a predictor of ibrutinib-associated atrial fibrillation in patients with chronic lymphocytic leukemia. *Cancer Biol Ther* 2018;19:1â2.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 52Singla A, Hogan WJ, Ansell SM, Buadi FK, Dingli D, Dispenzieri A, et al. Incidence of supraventricular arrhythmias during autologous peripheral blood stem cell transplantation. *Biol Blood Marrow Transplant* 2013;19:1233â1237.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 53Pudil R, Mueller C, AlutkienÄ J, Henriksen PA, Lenihan D, Dent S, et al. Role of serum biomarkers in cancer patients receiving cardiotoxic cancer therapies: a position statement from the Cardio-Oncology Study Group of the Heart Failure Association and the Cardio-Oncology Council of the European Society of Cardiology. *Eur J Heart Fail* 2020;22:1966â1983.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 54AlutkienÄ J, Pudil R, L3pez-Fern3ndez T, Grapsa J, Nihoyannopoulos P, Bergler-Klein J, et al. Role of cardiovascular imaging in cancer patients receiving cardiotoxic therapies: a position statement on behalf of the Heart Failure Association (HFA), the European Association of Cardiovascular Imaging (EACVI) and the Cardio-Oncology Council of the European Society of Cardiology (ESC). *Eur J Heart Fail* 2020;22:1504â1524.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 55Michel L, Mincu RI, Mahabadi AA, Settellemeier S, Al-Rashid F, Rassaf T, et al. Troponins and brain natriuretic peptides for the prediction of cardiotoxicity in cancer patients: a meta-analysis. *Eur J Heart Fail* 2020;22:350â361.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 56Lipshultz SE, Miller TL, Scully RE, Lipsitz SR, Rifai N, Silverman LB, et al. Changes in cardiac biomarkers during doxorubicin treatment of pediatric patients with high-risk acute lymphoblastic leukemia: associations with long-term echocardiographic outcomes. *J Clin Oncol* 2012;30:1042â1049.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 57Lipshultz SE, Rifai N, Dalton VM, Levy DE, Silverman LB, Lipsitz SR, et al. The effect of dexrazoxane on myocardial injury in doxorubicin-treated children with acute lymphoblastic leukemia. *N Engl J Med* 2004;351:145â153.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 58Xue K, Gu JJ, Zhang Q, Liu X, Wang J, Li XQ, et al. Cardiotoxicity as indicated by LVEF and troponin T sensitivity following two anthracycline-based regimens in lymphoma: results from a randomized prospective clinical trial. *Oncotarget* 2016;7:32519â32531.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 59Cardinale Â D, Sandri Â MT, Colombo Â A, Colombo Â N, Boeri Â M, Lamantia Â G, et al.
Â Prognostic value of troponin I in cardiac risk stratification of cancer patients undergoing high-dose chemotherapy. *Circulation* Â 2004;109:2749â^{PADSTS}2754.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 60Putt Â M, Hahn Â VS, Januzzi Â JL, Sawaya Â H, Sebag Â IA, Plana Â JC, et al. Â Longitudinal changes in multiple biomarkers are associated with cardiotoxicity in breast cancer patients treated with doxorubicin, taxanes, and trastuzumab. *Clin Chem* Â 2015;61:1164â^{PADSTS}1172.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 61Cardinale Â D, Colombo Â A, Torrisi Â R, Sandri Â MT, Civelli Â M, Salvatici Â M, et al.
Â Trastuzumab-induced cardiotoxicity: clinical and prognostic implications of troponin I evaluation. *J Clin Oncol* Â 2010;28:3910â^{PADSTS}3916.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 62Zardavas Â D, Suter Â TM, Van Veldhuisen Â DJ, Steinseifer Â J, Noe Â J, Lauer Â S, et al. Â Role of troponins I and T and N-terminal prohormone of brain natriuretic peptide in monitoring cardiac safety of patients with early-stage human epidermal growth factor receptor 2-positive breast cancer receiving trastuzumab: a herceptin adjuvant study cardiac marker substudy. *J Clin Oncol* Â 2017;35:878â^{PADSTS}884.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 63Mueller Â C, McDonald Â K, de Boer Â RA, Maisel Â A, Cleland Â JGF, Kozhuharov Â N, et al.
Â Heart Failure Association of the European Society of Cardiology practical guidance on the use of natriuretic peptide concentrations. *Eur J Heart Fail* Â 2019;21:715â^{PADSTS}731.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 64Feola Â M, Garrone Â O, Occelli Â M, Francini Â A, Biggi Â A, Visconti Â G, et al. Â Cardiotoxicity after anthracycline chemotherapy in breast carcinoma: effects on left ventricular ejection fraction, troponin I and brain natriuretic peptide. *Int J Cardiol* Â 2011;148:194â^{PADSTS}198.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 65Demissei Â BG, Hubbard Â RA, Zhang Â L, Smith Â AM, Sheline Â K, McDonald Â C, et al.
Â Changes in cardiovascular biomarkers with breast cancer therapy and associations with cardiac dysfunction. *J Am Heart Assoc* Â 2020;9:e014708.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 66Cornell Â RF, Ky Â B, Weiss Â BM, Dahm Â CN, Gupta Â DK, Du Â L, et al. Â Prospective study of cardiac events during proteasome inhibitor therapy for relapsed multiple myeloma. *J Clin Oncol* Â 2019;37:1946â^{PADSTS}1955.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 67Pavo Â N, Raderer Â M, HÃ¼lsmann Â M, Neuhold Â S, Adlbrecht Â C, Strunk Â G, et al.
Â Cardiovascular biomarkers in patients with cancer and their association with all-cause mortality. *Heart* Â 2015;101:1874â^{PADSTS}1880.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 68LÃ³pez-SendÃ³n Â J, Ãlvarez-Ortega Â C, Zamora AuÃ±on Â P, BuÃ±o Soto Â A, Lyon Â AR, Farmakis Â D, et al. Â Classification, prevalence, and outcomes of anticancer therapy-induced

cardiotoxicity: the CARDIOTOX registry. *Eur Heart J* 2020;41:1720â1729.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

69Ky B, Putt M, Sawaya H, French B, Januzzi JL, Sebag IA, et al. Early increases in multiple biomarkers predict subsequent cardiotoxicity in patients with breast cancer treated with doxorubicin, taxanes, and trastuzumab. *J Am Coll Cardiol* 2014;63:809â816.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

70Giofr  S, Chiesa M, Cardinale DM, Ricci V, Vavassori C, Cipolla CM, et al. Circulating microRNAs as potential predictors of anthracycline-induced troponin elevation in breast cancer patients: diverging effects of doxorubicin and epirubicin. *J Clin Med* 2020;9:1418.

[Google Scholar](#) [OpenURL Placeholder Text](#)

71Beer LA, Kossenkova AV, Liu Q, Luning Prak E, Domchek S, Speicher DW, et al. Baseline immunoglobulin e levels as a marker of doxorubicin- and trastuzumab-associated cardiac dysfunction. *Circ Res* 2016;119:1135â1144.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

72Plana JC, Thavendiranathan P, Bucciarelli-Ducci C, Lancellotti P. Multi-modality imaging in the assessment of cardiovascular toxicity in the cancer patient. *JACC Cardiovasc Imaging* 2018;11:1173â1186.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

73Galderisi M, Cosyns B, Edvardsen T, Cardim N, Delgado V, Di Salvo G, et al. Standardization of adult transthoracic echocardiography reporting in agreement with recent chamber quantification, diastolic function, and heart valve disease recommendations: an expert consensus document of the European Association of Cardiovascular Imaging. *Eur Heart J Cardiovasc Imaging* 2017;18:1301â1310.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

74De Azambuja E, Procter MJ, Van Veldhuisen DJ, Agbor-Tarh D, Metzger-Filho O, Steinseifer J, et al. Trastuzumab-associated cardiac events at 8 years of median follow-up in the herceptin adjuvant trial (BIG 1-01). *J Clin Oncol* 2014;32:2159â2165.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

75Santoro C, Arpino G, Esposito R, Lembo M, Paciolla I, Cardalesi C, et al. 2D and 3D strain for detection of subclinical anthracycline cardiotoxicity in breast cancer patients: A balance with feasibility. *Eur Heart J Cardiovasc Imaging* 2017;18:930â936.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

76Zhang KW, Finkelman BS, Gulati G, Narayan HK, Upshaw J, Narayan V, et al. Abnormalities in 3-dimensional left ventricular mechanics with anthracycline chemotherapy are associated with systolic and diastolic dysfunction. *JACC Cardiovasc Imaging* 2018;11:1059â1068.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

77Lang RM, Badano LP, Mor-Avi V, Afilalo J, Armstrong A, Ernande L, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *Eur*

Heart J Cardiovasc Imaging 2015;16:233a-270.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

78Thavendiranathan P, Grant AD, Negishi T, Plana JC, Popović ZB, Marwick TH. Reproducibility of echocardiographic techniques for sequential assessment of left ventricular ejection fraction and volumes: application to patients undergoing cancer chemotherapy. J Am Coll Cardiol 2013;61:77a-84.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

79Dorosz JL, Lezotte DC, Weitzenkamp DA, Allen LA, Salcedo EE. Performance of 3-dimensional echocardiography in measuring left ventricular volumes and ejection fraction: a systematic review and meta-analysis. J Am Coll Cardiol 2012;59:1799a-1808.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

80Hoffmann R, Barletta G, Von Bardeleben S, Vanoverschelde JL, Kasprzak J, Greis C, et al. Analysis of left ventricular volumes and function: a multicenter comparison of cardiac magnetic resonance imaging, cine ventriculography, and unenhanced and contrast-enhanced two-dimensional and three-dimensional echocardiography. J Am Soc Echocardiogr 2014;27:292a-301.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

81Jenkins C, Moir S, Chan J, Rakhit D, Haluska B, Marwick TH. Left ventricular volume measurement with echocardiography: a comparison of left ventricular opacification, three-dimensional echocardiography, or both with magnetic resonance imaging. Eur Heart J 2009;30:98a-106.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

82Porter TR, Mulvagh SL, Abdelmoneim SS, Becher H, Belcik JT, Bierig M, et al. Clinical applications of ultrasonic enhancing agents in echocardiography: 2018 American Society of Echocardiography Guidelines Update. J Am Soc Echocardiogr 2018;31:241a-274.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

83Houbois CP, Nolan M, Somerset E, Shalmon T, Esmaeilzadeh M, Lamacie MM, et al. Serial cardiovascular magnetic resonance strain measurements to identify cardiotoxicity in breast cancer: comparison with echocardiography. JACC Cardiovasc Imaging 2021;14:962a-974.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

84Dobbin SJH, Mangion K, Berry C, Roditi G, Basak S, Sourbron S, et al. Cardiotoxicity and myocardial hypoperfusion associated with anti-vascular endothelial growth factor therapies: prospective cardiac magnetic resonance imaging in patients with cancer. Eur J Heart Fail 2020;22:1276a-1277.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

85Mousavi N, Tan TC, Ali M, Halpern EF, Wang L, Scherrer-Crosbie M. Echocardiographic parameters of left ventricular size and function as predictors of symptomatic heart failure in patients with a left ventricular ejection fraction of 50a-59% treated with anthracyclines. Eur Heart J Cardiovasc Imaging 2015;16:977a-984.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)

86Åelutkienė J, Plymen CM, Flachskampf FA, de Boer RA, Grapsa J, Manka R, et al. Innovative imaging methods in heart failure: a shifting paradigm in cardiac assessment. Position

statement on behalf of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2018;20:1615â1633.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

87Kalam K, Otahal P, Marwick TH. Prognostic implications of global LV dysfunction: a systematic review and meta-analysis of global longitudinal strain and ejection fraction. *Heart* 2014;100:1673â1680.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

88Baron T, Berglund L, Hedin EM, Flachskampf FA. Test-retest reliability of new and conventional echocardiographic parameters of left ventricular systolic function. *Clin Res Cardiol* 2019;108:355â365.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

89Lambert J, Lamacie M, Thampinathan B, Altaha MA, Esmailzadeh M, Nolan M, et al. Variability in echocardiography and MRI for detection of cancer therapy cardiotoxicity. *Heart* 2020;106:817â823.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

90Thavendiranathan P, Negishi T, Côté MA, Penicka M, Massey R, Cho GY, et al. Single versus standard multiview assessment of global longitudinal strain for the diagnosis of cardiotoxicity during cancer therapy. *JACC Cardiovasc Imaging* 2018;11:1109â1118.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

91Negishi K, Negishi T, Haluska BA, Hare JL, Plana JC, Marwick TH. Use of speckle strain to assess left ventricular responses to cardiotoxic chemotherapy and cardioprotection. *Eur Heart J Cardiovasc Imaging* 2014;15:324â331.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

92Thavendiranathan P, Poulin F, Lim KD, Plana JC, Woo A, Marwick TH. Use of myocardial strain imaging by echocardiography for the early detection of cardiotoxicity in patients during and after cancer chemotherapy: a systematic review. *J Am Coll Cardiol* 2014;63:2751â2768.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

93Oikonomou EK, Kokkinidis DG, Kampaktsis PN, Amir EA, Marwick TH, Gupta D, et al. Assessment of prognostic value of left ventricular global longitudinal strain for early prediction of chemotherapy-induced cardiotoxicity: a systematic review and meta-analysis. *JAMA Cardiol* 2019;4:1007â1018.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

94Plana JC, Galderisi M, Barac A, Ewer MS, Ky B, Scherrer-Crosbie M, et al. Expert consensus for multimodality imaging evaluation of adult patients during and after cancer therapy: a report from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *Eur Heart J Cardiovasc Imaging* 2014;15:1063â1093.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

95Farsalinos KE, Daraban AM, Anagnostou S, Thomas JD, Badano LP, Voigt JU. Head-to-head comparison of global longitudinal strain measurements among nine different vendors: the

EACVI/ASE inter-vendor comparison study. *J Am Soc Echocardiogr* 2015;28:1171-1181.e2.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

96Narayan HK, French B, Khan AM, Plappert T, Hyman D, Bajulaiye A, et al. Noninvasive measures of ventricular arterial coupling and circumferential strain predict cancer therapeutics-related cardiac dysfunction. *JACC Cardiovasc Imaging* 2016;9:1131-1141.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

97Nagiub M, Nixon JV, Kontos MC. Ability of nonstrain diastolic parameters to predict doxorubicin-induced cardiomyopathy: a systematic review with meta-analysis. *Cardiol Rev* 2018;26:29-34.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

98Upshaw JN, Finkelman B, Hubbard RA, Smith AM, Narayan HK, Arndt L, et al. Comprehensive assessment of changes in left ventricular diastolic function with contemporary breast cancer therapy. *JACC Cardiovasc Imaging* 2020;13:198-210.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

99Phillips WJ, Johnson C, Law A, Turek M, Small AR, Dent S, et al. Comparison of Framingham risk score and chest-CT identified coronary artery calcification in breast cancer patients to predict cardiovascular events. *Int J Cardiol* 2019;289:138-143.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

100Knuuti J, Wijns W, Saraste A, Capodanno D, Barbato E, Funck-Brentano C, et al. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes: The Task Force for the diagnosis and management of chronic coronary syndromes of the European Society of Cardiology (ESC). *Eur Heart J* 2020;41:407-477.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

101Lopez-Mattei JC, Yang EH, Ferencik M, Baldassarre LA, Dent S, Budoff MJ. Cardiac computed tomography in cardio-oncology: *JACC: CardioOncology* primer. *Cardio Oncol* 2021;3:635-649.

[Google Scholar](#) [OpenURL Placeholder Text](#)

102Thavendiranathan P, Negishi T, Somerset E, Negishi K, Penicka M, Lemieux J, et al. Strain-guided management of potentially cardiotoxic cancer therapy. *J Am Coll Cardiol* 2021;77:392-401.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

103Esmaeilzadeh M, Fresno CMU, Somerset E, Shalmon T, Amir E, Fan CPS, et al. A combined echocardiography approach for the diagnosis of cancer therapy-related cardiac dysfunction in women with early-stage breast cancer. *JAMA Cardiol* 2022;7:330-340.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

104Drafts BC, Twomley KM, D'Agostino R, Lawrence J, Avis N, Ellis LR, et al. Low to moderate dose anthracycline-based chemotherapy is associated with early noninvasive imaging evidence of subclinical cardiovascular disease. *JACC Cardiovasc Imaging* 2013;6:877-885.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 105Giusca Â S, Korosoglou Â G, Montenbruck Â M, GerÅak Â B, Schwarz Â AK, Esch Â S, et al. Â Multiparametric early detection and prediction of cardiotoxicity using myocardial strain, T1 and T2 mapping, and biochemical markers: a longitudinal cardiac resonance imaging study during 2 years of follow-up. *Circ Cardiovasc Imaging* Â 2021;14:e012459.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 106Dhir Â V, Yan Â AT, Nisenbaum Â R, Sloninko Â J, Connelly Â KA, Barfett Â J, et al. Â Assessment of left ventricular function by CMR versus MUGA scans in breast cancer patients receiving trastuzumab: a prospective observational study. *Int J Cardiovasc Imaging* Â 2019;35:2085â2093.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 107Walker Â J, Bhullar Â N, Fallah-Rad Â N, Lytwyn Â M, Golian Â M, Fang Â T, et al. Â Role of three-dimensional echocardiography in breast cancer: comparison with two-dimensional echocardiography, multiple-gated acquisition scans, and cardiac magnetic resonance imaging. *J Clin Oncol* Â 2010;28:3429â3436.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 108Huang Â H, Nijjar Â PS, Misialek Â JR, Blaes Â A, Derrico Â NP, Kazmirczak Â F, et al. Â Accuracy of left ventricular ejection fraction by contemporary multiple gated acquisition scanning in patients with cancer: comparison with cardiovascular magnetic resonance. *J Cardiovasc Magn Reson* Â 2017;19:34.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 109Hoppeler Â H, Weibel Â ER. Limits for oxygen and substrate transport in mammals. *J Exp Biol* Â 1998;201:1051â1064.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 110Kaminsky Â LA, Imboden Â MT, Arena Â R, Myers Â J. Reference standards for cardiorespiratory fitness measured with cardiopulmonary exercise testing using cycle ergometry: data from the Fitness Registry and the Importance of Exercise National Database (FRIEND) registry. *Mayo Clin Proc* Â 2017;92:228â233.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 111Imboden Â MT, Harber Â MP, Whaley Â MH, Finch Â WH, Bishop Â DL, Kaminsky Â LA. Cardiorespiratory fitness and mortality in healthy men and women. *J Am Coll Cardiol* Â 2018;72:2283â2292.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 112JettÃ© Â M, Sidney Â K, BlÃ¼mchen Â G. Metabolic equivalents (METs) in exercise testing, exercise prescription, and evaluation of functional capacity. *Clin Cardiol* Â 1990;13:555â565.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 113Ross Â R, Blair Â SN, Arena Â R, Church Â TS, DesprÃ©s Â JP, Franklin Â BA, et al. Â Importance of assessing cardiorespiratory fitness in clinical practice: a case for fitness as a clinical vital sign: a scientific statement from the American Heart Association. *Circulation* Â 2016;134:e653âe699.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 114Schmid Â D, Leitzmann Â MF. Cardiorespiratory fitness as predictor of cancer mortality: a systematic review and meta-analysis. *Ann Oncol* Â 2015;26:272â278.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 115Gupta Â S, Rohatgi Â A, Ayers Â CR, Willis Â BL, Haskell Â WL, Khera Â A, et al.
Â Cardiorespiratory fitness and classification of risk of cardiovascular disease mortality. *Circulation*
Â 2011;123:1377â1383.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 116Holtermann Â A, Marott Â JL, Gyntelberg Â F, SÃ¸gaard Â K, Mortensen Â OS, Prescott Â E, et al.
Â Self-reported cardiorespiratory fitness: prediction and classification of risk of cardiovascular disease
mortality and longevityâa prospective investigation in the Copenhagen City Heart Study. *J Am Heart
Assoc* Â 2015;4:e001495.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 117Stamatakis Â E, Hamer Â M, OâDonovan Â G, Batty Â GD, Kivimaki Â M. A non-exercise
testing method for estimating cardiorespiratory fitness: associations with all-cause and cardiovascular
mortality in a pooled analysis of eight population-based cohorts. *Eur Heart J* Â 2013;34:750â758.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 118Wickramasinghe Â CD, Ayers Â CR, Das Â S, De Lemos Â JA, Willis Â BL, Berry Â JD. Prediction
of 30-year risk for cardiovascular mortality by fitness and risk factor levels: the Cooper Center
Longitudinal Study. *Circ Cardiovasc Qual Outcomes* Â 2014;7:597â602.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 119Fardman Â A, Banschick Â GD, Rabia Â R, Percik Â R, Fourey Â D, Segev Â S, et al.
Â Cardiorespiratory fitness and survival following cancer diagnosis. *Eur J Prev Cardiol*
Â 2021;28:1242â1249.
[Google Scholar](#) [OpenURL Placeholder Text](#) Â
- 120Groarke Â JD, Payne Â DL, Claggett Â B, Mehra Â MR, Gong Â J, Caron Â J, et al. Â Association of
post-diagnosis cardiorespiratory fitness with cause-specific mortality in cancer. *Eur Heart J Qual Care
Clin Outcomes* Â 2020;6:315â322.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 121Jones Â LM, Stoner Â L, Brown Â C, Baldi Â JC, McLaren Â B. Cardiorespiratory fitness predicts
cardiovascular health in breast cancer survivors, independent of body composition, age and time post-
treatment completion. *Breast Cancer* Â 2019;26:729â737.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 122Ha Â D, Mazzone Â PJ, Ries Â AL, Malhotra Â A, Fuster Â M. The utility of exercise testing in
patients with lung cancer. *J Thorac Oncol* Â 2016;11:1397â1410.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 123West Â MA, Lythgoe Â D, Barben Â CP, Noble Â L, Kemp Â GJ, Jack Â S, et al. Â Cardiopulmonary
exercise variables are associated with postoperative morbidity after major colonic surgery: a prospective
blinded observational study. *Br J Anaesth* Â 2014;112:665â671.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 124West Â MA, Parry Â MG, Lythgoe Â D, Barben Â CP, Kemp Â GJ, Grocott Â MPW, et al.
Â Cardiopulmonary exercise testing for the prediction of morbidity risk after rectal cancer surgery. *Br J
Surg* Â 2014;101:1166â1172.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 125Halvorsen S, Mehilli J, Cassese S, Hall TS, Abdelhamid M, Barbato E, *et al.* 2022 ESC guidelines on cardiovascular assessment and management of patients undergoing non-cardiac surgery. *Eur Heart J* 2022;43:3826â€¦3924.
[Crossref](#) [Search ADS](#) [PubMed](#) 125
- 126Aminkeng 1 F, Bhavsar 1 AP, Visscher 1 H, Rassekh 1 SR, Li 1 Y, Lee 1 JW, *et al.* 1 A coding variant in RARG confers susceptibility to anthracycline-induced cardiotoxicity in childhood cancer. *Nat Genet* 2015;47:1079â€¦1084.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 126
- 127Blanco 1 JG, Sun 1 CL, Landier 1 W, Chen 1 L, Esparza-Duran 1 D, Leisenring 1 W, *et al.* 1 Anthracycline-related cardiomyopathy after childhood cancer: role of polymorphisms in carbonyl reductase genesâ€¦a report from the Childrenâ€¦s Oncology Group. *J Clin Oncol* 2012;30:1415â€¦1421.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 127
- 128Wojnowski 1 L, Kulle 1 B, Schirmer 1 M, SchlÃ¼ter 1 G, Schmidt 1 A, Rosenberger 1 A, *et al.* 1 NAD(P)H oxidase and multidrug resistance protein genetic polymorphisms are associated with doxorubicin-induced cardiotoxicity. *Circulation* 2005;112:3754â€¦3762.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 128
- 129Johnson 1 DB, Balko 1 JM, Compton 1 ML, Chalkias 1 S, Gorham 1 J, Xu 1 Y, *et al.* 1 Fulminant myocarditis with combination immune checkpoint blockade. *N Engl J Med* 2016;375:1749â€¦1755.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 129
- 130European Society of Cardiology. ESC CardioMed. (3rd ed) . Oxford: Oxford University Press; 2018.
[Google Scholar](#) [OpenURL Placeholder Text](#) 130
- 131Moslehi 1 JJ. Cardiovascular toxic effects of targeted cancer therapies. *N Engl J Med* 2016;375:1457â€¦1467.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 131
- 132Ameri 1 P, Canepa 1 M, Anker 1 MS, Belenkov 1 Y, Bergler-Klein 1 J, Cohen-Solal 1 A, *et al.* 1 Cancer diagnosis in patients with heart failure: epidemiology, clinical implications and gaps in knowledge. *Eur J Heart Fail* 2018;20:879â€¦887.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 132
- 133Dâ€¦Ascenzi 1 F, Anselmi 1 F, Fiorentini 1 C, Mannucci 1 R, Bonifazi 1 M, Mondillo 1 S. The benefits of exercise in cancer patients and the criteria for exercise prescription in cardio-oncology. *Eur J Prev Cardiol* 2021;28:725â€¦735.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 133
- 134Ogunmoroti 1 O, Allen 1 NB, Cushman 1 M, Michos 1 ED, Rundek 1 T, Rana 1 JS, *et al.* 1 Association between lifeâ€¦s simple 7 and noncardiovascular disease: the multi-ethnic study of atherosclerosis. *J Am Heart Assoc* 2016;5:e003954.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) 134
- 135Michos 1 ED, Marshall 1 CH. Healthy lifestyle benefits both cancer and cardiovascular disease: more bang for the buck. *JACC CardioOncology* 2021;3:675â€¦677.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

136Murray ^ J, Bennett ^ H, Bezak ^ E, Perry ^ R. The role of exercise in the prevention of cancer therapy-related cardiac dysfunction in breast cancer patients undergoing chemotherapy: systematic review. *Eur J Prev Cardiol* ^ 2022;29:463a^472.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

137Scott ^ JM, Nilsen ^ TS, Gupta ^ D, Jones ^ LW. Exercise therapy and cardiovascular toxicity in cancer. *Circulation* ^ 2018;137:1176a^1191.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

138Williams ^ B, Mancia ^ G, Spiering ^ W, Rosei ^ EA, Azizi ^ M, Burnier ^ M, et al. ^ 2018 ESC/ESH Guidelines for the management of arterial hypertension. *Eur Heart J* ^ 2018;39:3021a^3104.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

139Cosentino ^ F, Grant ^ PJ, Aboyans ^ V, Bailey ^ CJ, Ceriello ^ A, Delgado ^ V, et al. ^ 2019 ESC Guidelines on diabetes, pre-diabetes, and cardiovascular diseases developed in collaboration with the EASD. *Eur Heart J* ^ 2020;41:255a^323.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

140Mach ^ F, Baigent ^ C, Catapano ^ AL, Koskinas ^ KC, Casula ^ M, Badimon ^ L, et al. ^ 2019 ESC/EAS Guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *Eur Heart J* ^ 2020;41:111a^188.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

141Beavers ^ CJ, Rodgers ^ JE, Bagnola ^ AJ, Beckie ^ TM, Campia ^ U, Di Palo ^ KE, et al. ^ Cardio-oncology drug interactions: a scientific statement from the American Heart Association. *Circulation* ^ 2022;145:e811a^e838.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

142Ferdinandy ^ P, Baczko ^ I, Bencsik ^ P, Giricz ^ Z, Gál ^ A, Pacher ^ P, et al. ^ Definition of hidden drug cardiotoxicity: paradigm change in cardiac safety testing and its clinical implications. *Eur Heart J* ^ 2019;40:1771a^1777.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

143Kalam ^ K, Marwick ^ TH. Role of cardioprotective therapy for prevention of cardiotoxicity with chemotherapy: a systematic review and meta-analysis. *Eur J Cancer* ^ 2013;49:2900a^2909.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

144Chen ^ J, Long ^ JB, Hurria ^ A, Owusu ^ C, Steingart ^ RM, Gross ^ CP. Incidence of heart failure or cardiomyopathy after adjuvant trastuzumab therapy for breast cancer. *J Am Coll Cardiol* ^ 2012;60:2504a^2512.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

145Bosch ^ X, Rovira ^ M, Sitges ^ M, Domínguez ^ A, Ortiz-Pérez ^ JT, De Caralt ^ TM, et al. ^ Enalapril and carvedilol for preventing chemotherapy-induced left ventricular systolic dysfunction in patients with malignant hemopathies: the OVERCOME trial (preventiOn of left Ventricular dysfunction with Enalapril and caRvedilol in patients submitted to intensive ChemOtherapy for the treatment of

Malignant hEmopathies). J Am Coll Cardiol Â 2013;61:2355â2362.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

146Avila Â MS, Ayub-Ferreira Â SM, de Barros Wanderley Â MR, das Dores Cruz Â F, GonÃ§alves BrandÃ£o Â SM, Rigaud Â VOC, et al. Â Carvedilol for prevention of chemotherapy-related cardiotoxicity: the CECCY trial. J Am Coll Cardiol Â 2018;71:2281â2290.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

147Cardinale Â D, Ciceri Â F, Latini Â R, Franzosi Â MG, Sandri Â MT, Civelli Â M, et al. Â Anthracycline-induced cardiotoxicity: a multicenter randomised trial comparing two strategies for guiding prevention with enalapril: The International CardioOncology Society-one trial. Eur J Cancer Â 2018;94:126â137.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

148Akpek Â M, Ozdogru Â I, Sahin Â O, Inanc Â M, Dogan Â A, Yazici Â C, et al. Â Protective effects of spironolactone against anthracycline-induced cardiomyopathy. Eur J Heart Fail Â 2015;17:81â89.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

149Acar Â Z, Kale Â A, Turgut Â M, Demircan Â S, Durna Â K, Demir Â S, et al. Â Efficiency of atorvastatin in the protection of anthracycline-induced cardiomyopathy. J Am Coll Cardiol Â 2011;58:988â989.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

150Gulati Â G, Heck Â SL, Ree Â AH, Hoffmann Â P, Schulz-Menger Â J, Fagerland Â MW, et al. Â Prevention of cardiac dysfunction during adjuvant breast cancer therapy (PRADA): a 2 Â 2 factorial, randomized, placebo-controlled, double-blind clinical trial of candesartan and metoprolol. Eur Heart J Â 2016;37:1671â1680.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

151Gulati Â G, Heck Â SL, RÃsjÃ, Â H, Ree Â AH, Hoffmann Â P, Hagne Â TA, et al. Â Neurohormonal blockade and circulating cardiovascular biomarkers during anthracycline therapy in breast cancer patients: results from the PRADA (Prevention of Cardiac Dysfunction During Adjuvant Breast Cancer Therapy) study. J Am Heart Assoc Â 2017;6:e006513.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

152Guglin Â M, Krischer Â J, Tamura Â R, Fink Â A, Bello-Matricaria Â L, McCaskill-Stevens Â W, et al. Â Randomized trial of lisinopril versus carvedilol to prevent trastuzumab cardiotoxicity in patients with breast cancer. J Am Coll Cardiol Â 2019;73:2859â2868.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

153Pituskin Â E, Mackey Â JR, Koshman Â S, Jassal Â D, Pitz Â M, Haykowsky Â MJ, et al. Â Multidisciplinary approach to novel therapies in cardio-oncology research (MANTICORE 101-Breast): a randomized trial for the prevention of trastuzumab-associated cardiotoxicity. J Clin Oncol Â 2017;35:870â877.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

154Boekhout Â AH, Gietema Â JA, Kerklaan Â BM, VanWerkhoven Â ED, Altena Â R, Honkoop Â A, et al. Â Angiotensin II receptor inhibition with candesartan to prevent trastuzumab-related cardiotoxic effects in patients with early breast cancer a randomized clinical trial. JAMA Oncol

Â 2016;2:1030â1037.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

155Caspani Â F, Tralongo Â AC, Campiotti Â L, Asteggiano Â R, Guasti Â L, Squizzato Â A. Prevention of anthracycline-induced cardiotoxicity: a systematic review and meta-analysis. *Intern Emerg Med* Â 2021;16:477â486.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

156Huang Â S, Zhao Â Q, Yang Â ZG, Diao Â KY, He Â Y, Shi Â K, et al. Â Protective role of beta-blockers in chemotherapy-induced cardiotoxicityâa systematic review and meta-analysis of carvedilol. *Heart Fail Rev* Â 2019;24:325â333.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

157Vaduganathan Â M, Hirji Â SA, Qamar Â A, Bajaj Â N, Gupta Â A, Zaha Â VG, et al. Â Efficacy of neurohormonal therapies in preventing cardiotoxicity in patients with cancer undergoing chemotherapy. *JACC CardioOncology* Â 2019;1:54â65.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

158Macedo Â AVS, Hajjar Â LA, Lyon Â AR, Nascimento Â BR, Putzu Â A, Rossi Â L, et al. Â Efficacy of dexrazoxane in preventing anthracycline cardiotoxicity in breast cancer. *JACC CardioOncology* Â 2019;1:68â79.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

159Li Â X, Li Â Y, Zhang Â T, Xiong Â X, Liu Â N, Pang Â B, et al. Â Role of cardioprotective agents on chemotherapy-induced heart failure: a systematic review and network meta-analysis of randomized controlled trials. *Pharmacol Res* Â 2020;151:104577.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

160Fang Â K, Zhang Â Y, Liu Â W, He Â C. Effects of angiotensin-converting enzyme inhibitor/angiotensin receptor blocker use on cancer therapy-related cardiac dysfunction: a meta-analysis of randomized controlled trials. *Heart Fail Rev* Â 2021;26:101â109.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

161Shapira Â J, Gotfried Â M, Lishner Â M, Ravid Â M. Reduced cardiotoxicity of doxorubicin by a 6-hour infusion regimen. A prospective randomized evaluation. *Cancer* Â 1990;65:870â873.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

162Hensley Â ML, Hagerty Â KL, Kewalramani Â T, Green Â DM, Meropol Â NJ, Wasserman Â TH, et al. Â American Society of Clinical Oncology 2008 clinical practice guideline update: use of chemotherapy and radiation therapy protectants. *J Clin Oncol* Â 2009;27:127â145.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

163European Medicines Agency. Savene: EPARâProduct Information (Internet) 2008 (updated 2019).

164Vejpongsa Â P, Yeh Â ETH. Prevention of anthracycline-induced cardiotoxicity: challenges and opportunities. *J Am Coll Cardiol* Â 2014;64:938â945.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 165 Yamaguchi Â N, Fujii Â T, Aoi Â S, Kozuch Â PS, Hortobagyi Â GN, Blum Â RH. Comparison of cardiac events associated with liposomal doxorubicin, epirubicin and doxorubicin in breast cancer: a Bayesian network meta-analysis. *Eur J Cancer* Â 2015;51:2314â2320.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 166 Swain Â SM, Whaley Â FS, Gerber Â MC, Ewer Â MS, Bianchini Â JR, Gams Â RA. Delayed administration of dexrazoxane provides cardioprotection for patients with advanced breast cancer treated with doxorubicin-containing therapy. *J Clin Oncol* Â 1997;15:1333â1340.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 167 van Dalen Â EC, Caron Â HN, Dickinson Â HO, Kremer Â LCM. Cardioprotective interventions for cancer patients receiving anthracyclines. *Cochrane Database Syst Rev* Â 2011;2011:CD003917.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 168 Waterhouse Â DN, Tardi Â PG, Mayer Â LD, Bally Â MB. A comparison of liposomal formulations of doxorubicin with drug administered in free form: changing toxicity profiles. *Drug Saf* Â 2001;24:903â920.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 169 Verellen Â D, De Ridder Â M, Linthout Â N, Tournel Â K, Soete Â G, Storme Â G. Innovations in image-guided radiotherapy. *Nat Rev Cancer* Â 2007;7:949â960.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 170 De Ruyscher Â D, Niedermann Â G, Burnet Â NG, Siva Â S, Lee Â AWM, Hegi-Johnson Â F. Â Radiotherapy toxicity. *Nat Rev Dis Prim* Â 2019;5:13.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 171 Meattini Â I, Poortmans Â PM, Aznar Â MC, Becherini Â C, Bonzano Â E, Cardinale Â D, et al. Â Association of breast cancer irradiation with cardiac toxic effects: a narrative review. *JAMA Oncol* Â 2021;7:924â932.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 172 Banfill Â K, Giuliani Â M, Aznar Â M, Franks Â K, McWilliam Â A, Schmitt Â M, et al. Â Cardiac toxicity of thoracic radiotherapy: existing evidence and future directions. *J Thorac Oncol* Â 2021;16:216â227.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 173 Mitchell Â JD, Cehic Â DA, Morgia Â M, Bergrom Â C, Toohey Â J, Guerrero Â PA, et al. Â Cardiovascular manifestations from therapeutic radiation: a multidisciplinary expert consensus statement from the International Cardio-Oncology Society. *JACC CardioOncology* Â 2021;3:360â380.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 174 Kirwan Â CC, Coles Â CE, Bliss Â J, Kirwan Â C, Kilburn Â L, Fox Â L, et al. Â Itâs PRIMETIME. Postoperative avoidance of radiotherapy: biomarker selection of women at very low risk of local recurrence. *Clin Oncol* Â 2016;28:594â596.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

- 175Cardinale Â D, Colombo Â A, Sandri Â MT, Lamantia Â G, Colombo Â N, Civelli Â M, et al. Â Prevention of high-dose chemotherapy-induced cardiotoxicity in high-risk patients by angiotensin-converting enzyme inhibition. *Circulation* Â 2006;114:2474â2481.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 176Abdel-Qadir Â H, Bobrowski Â D, Zhou Â L, Austin Â PC, Calvillo-ArgÃ¼elles Â O, Amir Â E, et al. Â Statin exposure and risk of heart failure after anthracycline-or trastuzumab-based chemotherapy for early breast cancer: a propensity scoreâmatched cohort study. *J Am Heart Assoc* Â 2021;10:e018393.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 177Calvillo-ArgÃ¼elles Â O, Abdel-Qadir Â H, Michalowska Â M, Billia Â F, Suntheralingam Â S, Amir Â E, et al. Â Cardioprotective effect of statins in patients with HER2-positive breast cancer receiving trastuzumab therapy. *Can J Cardiol* Â 2019;35:153â159.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 178Seicean Â S, Seicean Â A, Plana Â JC, Budd Â GT, Marwick Â TH. Effect of statin therapy on the risk for incident heart failure in patients with breast cancer receiving anthracycline chemotherapy: an observational clinical cohort study. *J Am Coll Cardiol* Â 2012;60:2384â2390.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 179Chotenimitkhun Â R, D'Agostino Â R, Lawrence Â JA, Hamilton Â CA, Jordan Â JH, Vasu Â S, et al. Â Chronic statin administration may attenuate early anthracycline-associated declines in left ventricular ejection function. *Can J Cardiol* Â 2015;31:302â307.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 180Obasi Â M, Abovich Â A, Vo Â JB, Gao Â Y, Papatheodorou Â SI, Nohria Â A, et al. Â Statins to mitigate cardiotoxicity in cancer patients treated with anthracyclines and/or trastuzumab: a systematic review and meta-analysis. *Cancer Causes Control* Â 2021;32:1395â1405.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 181Kim Â J, Nishimura Â Y, Kewcharoen Â J, Yess Â J. Statin use can attenuate the decline in left ventricular ejection fraction and the incidence of cardiomyopathy in cardiotoxic chemotherapy recipients: a systematic review and meta-analysis. *J Clin Med* Â 2021;10:3731.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 182Sanfilippo Â KM, Keller Â J, Gage Â BF, Luo Â S, Wang Â TF, Moskowitz Â G, et al. Â Statins are associated with reduced mortality in multiple myeloma. *J Clin Oncol* Â 2016;34:4008â4014.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 183Afzal Â A, Fiala Â MA, Gage Â BF, Wildes Â TM, Sanfilippo Â K. Statins reduce mortality in multiple myeloma: a population-based US study. *Clin Lymphoma Myeloma Leuk* Â 2020;20:e937âe943.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 184Nabati Â M, Janbabai Â G, Esmailian Â J, Yazdani Â J. Effect of rosuvastatin in preventing chemotherapy-induced cardiotoxicity in women with breast cancer: a randomized, single-blind, placebo-controlled trial. *J Cardiovasc Pharmacol Ther* Â 2019;24:233â241.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 185 Shahid Â I, Yamani Â N, Ali Â A, Kumar Â P, Figueredo Â V, Unzek Â S, et al. Â Meta-analysis evaluating the use of statins to attenuate cardiotoxicity in cancer patients receiving anthracyclines and trastuzumab-based chemotherapy. *Am J Cardiol* Â 2021;156:142â^{PADSTS}145.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 186 Nabati Â M, Janbabai Â G, Baghyari Â S, Esmaili Â K, Yazdani Â J. Cardioprotective effects of carvedilol in inhibiting doxorubicin-induced cardiotoxicity. *J Cardiovasc Pharmacol* Â 2017;69:279â^{PADSTS}285.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 187 Warrington Â L, Absolom Â K, Conner Â M, Kellar Â I, Clayton Â B, Ayres Â M, et al. Â Electronic systems for patients to report and manage side effects of cancer treatment: systematic review. *J Med Internet Res* Â 2019;21:e10875.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 188 LÃ³pez-FernÃ¡ndez Â T, MartÃ¡n GarcÃ­a Â A, Santaballa BeltrÃ¡n Â A, Montero Luis Â Ã^{HOF}, GarcÃ­a Sanz Â R, MazÃ¡n Ramos Â P, et al. Â Cardio-onco-hematology in clinical practice. Position paper and recommendations. *Rev EspaÃ±ola Cardiol (English Ed)* Â 2017;70:474â^{PADSTS}486.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 189 Lynce Â F, Barac Â A, Geng Â X, Dang Â C, Yu Â AF, Smith Â KL, et al. Â Prospective evaluation of the cardiac safety of HER2-targeted therapies in patients with HER2-positive breast cancer and compromised heart function: the SAFE-HEaRt study. *Breast Cancer Res Treat* Â 2019;175:595â^{PADSTS}603.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 190 Fornaro Â A, Olivotto Â I, Rigacci Â L, Ciaccheri Â M, Tomberli Â B, Ferrantini Â C, et al. Â Comparison of long-term outcome in anthracycline-related versus idiopathic dilated cardiomyopathy: a single centre experience. *Eur J Heart Fail* Â 2018;20:898â^{PADSTS}906.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 191 Chow Â SL, Maisel Â AS, Anand Â I, Bozkurt Â B, De Boer Â RA, Felker Â GM, et al. Â Role of biomarkers for the prevention, assessment, and management of heart failure: a scientific statement from the American Heart Association. *Circulation* Â 2017;135:e1054â^{PADSTS}e1091. Erratum in: *Circulation* 2017;136:e345.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 192 De Boer Â RA, Daniels Â LB, Maisel Â AS, Januzzi Â JL. State of the art: newer biomarkers in heart failure. *Eur J Heart Fail* Â 2015;17:559â^{PADSTS}569.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 193 Piek Â A, Du Â W, de Boer Â RA, SilljÃ© Â HHW. Novel heart failure biomarkers: why do we fail to exploit their potential? Â *Crit Rev Clin Lab Sci* Â 2018;55:246â^{PADSTS}263.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 194 Du Â W, Piek Â A, Marloes Schouten Â E, van de Kolk Â CWA, Mueller Â C, Mebazaa Â A, et al. Â Plasma levels of heart failure biomarkers are primarily a reflection of extracardiac production. *Theranostics* Â 2018;8:4155â^{PADSTS}4169.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 195Suthahar A N, Lau ES, Blaha MJ, Paniagua SM, Larson MG, Psaty BM, et al. Sex-specific associations of cardiovascular risk factors and biomarkers with incident heart failure. *J Am Coll Cardiol* 2020;76:1455â€”1465.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 196Bracun V, Aboumsallem JP, van der Meer P, de Boer RA. Cardiac biomarkers in patients with cancer: considerations, clinical implications, and future avenues. *Curr Oncol Rep* 2020;22:67.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 197Ananthan K, Lyon AR. The role of biomarkers in cardio-oncology. *J Cardiovasc Transl Res* 2020;13:431â€”450.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 198L3pez-Fern3ndez T, Thavendiranathan P. Emerging cardiac imaging modalities for the early detection of cardiotoxicity due to anticancer therapies. *Rev Espa3ola Cardiol (English Ed)* 2017;70:487â€”495.
[Google Scholar](#) [Crossref](#) [Search ADS](#)
- 199Eschenhagen T, Force T, Ewer MS, De Keulenaer GW, Suter TM, Anker SD, et al. Cardiovascular side effects of cancer therapies: a position statement from the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2011;13:1â€”10.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 200Keramida K, Farmakis D, L3pez Fern3ndez T, Lancellotti P. Focused echocardiography in cardio-oncology. *Echocardiography* 2020;37:1149â€”1158.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 201Liu JE, Barac A, Thavendiranathan P, Scherrer-Crosbie M. Strain imaging in cardio-oncology. *JACC CardioOncology* 2020;2:677â€”689.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 202Negishi K, Negishi T, Hare JL, Haluska BA, Plana JC, Marwick TH. Independent and incremental value of deformation indices for prediction of trastuzumab-induced cardiotoxicity. *J Am Soc Echocardiogr* 2013;26:493â€”498.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 203Negishi T, Thavendiranathan P, Penicka M, Lemieux J, Aakhus S, Miyazaki S, et al. Cardioprotection using strain-guided management of potentially cardiotoxic cancer therapy: 1 year results of the SUCCOUR trial. *Eur Heart J* 2020;41:ehaa946.3282.
[Google Scholar](#) [Crossref](#) [Search ADS](#)
- 204Vallabhaneni S, Zhang KW, Alvarez JA, Joshua C, Henning DM, Woodard PK, et al. Role of cardiovascular magnetic resonance in early detection and treatment of cardiac dysfunction in oncology patients. *Int J Cardiovasc Imaging* 2021;37:3003â€”3017.
[Google Scholar](#) [OpenURL Placeholder Text](#)
- 205Lapinskas T, Hireche-Chikaoui H, Zieschang V, Erley J, Stehning C, Gebker R, et al. Effect of comprehensive initial training on the variability of left ventricular measures using fast-SENCE cardiac magnetic resonance imaging. *Sci Rep* 2019;9:12223.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

206Korosoglou ^ G, Giusca ^ S, Montenbruck ^ M, Patel ^ AR, Lapinskas ^ T, GÄttze ^ C, et al.
^ Fast strain-encoded cardiac magnetic resonance for diagnostic classification and risk stratification of heart failure patients. *JACC Cardiovasc Imaging* ^ 2021;14:1177^PABST^1188.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

207Yancy ^ CW, Jessup ^ M, Bozkurt ^ B, Butler ^ J, Casey ^ DE, Colvin ^ MM, et al. ^ 2017
ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure:
a report of the American College of Cardiology/American Heart Association Task Force on Clinical
Practice Guidelines and the Heart Failure Society of America. *Circulation* ^ 2017;136:e137^PABST^e161.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

208Cardinale ^ D, Colombo ^ A, Bacchiani ^ G, Tedeschi ^ I, Meroni ^ CA, Veglia ^ F, et al. ^ Early
detection of anthracycline cardiotoxicity and improvement with heart failure therapy. *Circulation*
^ 2015;131:1981^PABST^1988.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

209Wang ^ L, Tan ^ TC, Halpern ^ EF, Neilan ^ TG, Francis ^ SA, Picard ^ MH, et al. ^ Major
cardiac events and the value of echocardiographic evaluation in patients receiving anthracycline-based
chemotherapy. *Am J Cardiol* ^ 2015;116:442^PABST^446.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

210Abu-Khalaf ^ MM, Safonov ^ A, Stratton ^ J, Wang ^ S, Hatzis ^ C, Park ^ E, et al. ^ Examining
the cost-effectiveness of baseline left ventricular function assessment among breast cancer patients
undergoing anthracycline-based therapy. *Breast Cancer Res Treat* ^ 2019;176:261^PABST^270.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

211Lenihan ^ DJ, Stevens ^ PL, Massey ^ M, Plana ^ JC, Araujo ^ DM, Fanale ^ MA, et al. ^ The
utility of point-of-care biomarkers to detect cardiotoxicity during anthracycline chemotherapy: a
feasibility study. *J Card Fail* ^ 2016;22:433^PABST^438.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

212Meessen ^ JMTA, Cardinale ^ D, Ciceri ^ F, Sandri ^ MT, Civelli ^ M, Bottazzi ^ B, et al.
^ Circulating biomarkers and cardiac function over 3 years after chemotherapy with anthracyclines: the
ICOS-ONE trial. *ESC Heart Fail* ^ 2020;7:1452^PABST^1466.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

213Jones ^ M, OâPABST^Gorman ^ P, Kelly ^ C, Mahon ^ N, Fitzgibbon ^ MC. High-sensitive cardiac
troponin-I facilitates timely detection of subclinical anthracycline-mediated cardiac injury. *Ann Clin
Biochem* ^ 2017;54:149^PABST^157.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

214BrandÃ£o ^ M, PondÃ© ^ NF, Poggio ^ F, Kotecki ^ N, Salis ^ M, Lambertini ^ M, et al.
^ Combination therapies for the treatment of HER2-positive breast cancer: current and future prospects.
Expert Rev Anticancer Ther ^ 2018;18:629^PABST^649.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 215Yarden A Y, Sliwkowski A MX. Untangling the ErbB signalling network. *Nat Rev Mol Cell Biol* A 2001;2:127aFAFSTSI137.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 216Belmonte A F, Das A S, Sysa-Shah A P, Sivakumaran A V, Stanley A B, Guo A X, et al. A ErbB2 overexpression upregulates antioxidant enzymes, reduces basal levels of reactive oxygen species, and protects against doxorubicin cardiotoxicity. *Am J Physiol Heart Circ Physiol* A 2015;309:H1271aFAFSTSIH1280.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 217Martel A S, Maurer A C, Lambertini A M, PondA© A N, De Azambuja A E. Breast cancer treatment-induced cardiotoxicity. *Expert Opin Drug Saf* A 2017;16:1021aFAFSTSI1038.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 218de Azambuja A E, Ponde A N, Procter A M, Rastogi A P, Cecchini A RS, Lambertini A M, et al. A A pooled analysis of the cardiac events in the trastuzumab adjuvant trials. *Breast Cancer Res Treat* A 2020;179:161aFAFSTSI171.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 219Eiger A D, PondA© A NF, Agbor-Tarh A D, Moreno-Aspitia A A, Piccart A M, Hilbers A FS, et al. A Long-term cardiac outcomes of patients with HER2-positive breast cancer treated in the adjuvant lapatinib and/or trastuzumab treatment optimization trial. *Br J Cancer* A 2020;122:1453aFAFSTSI1460.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 220Tan-Chiu A E, Yothers A G, Romond A E, Geyer A CE, Ewer A M, Keefe A D, et al. A Assessment of cardiac dysfunction in a randomized trial comparing doxorubicin and cyclophosphamide followed by paclitaxel, with or without trastuzumab as adjuvant therapy in node-positive, HER2-overexpressing breast cancer. *J Clin Oncol* A 2005;23:7811aFAFSTSI7819.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 221Martin A M, Press A M, Ph A D, Mackey A J, Glaspy A J, Chan A A, et al. A Adjuvant trastuzumab in HER2-positive breast cancer. *NEJM* A 2011;365:1273aFAFSTSI1283.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) A
- 222Smith A I, Procter A M, Gelber A RD, Guillaume A S, Feyereislova A A, Dowsett A M, et al. A 2-year follow-up of trastuzumab after adjuvant chemotherapy in HER2-positive breast cancer: a randomised controlled trial. *Lancet* A 2007;369:29aFAFSTSI36.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 223Piccart-Gebhart A MJ, Procter A M, Leyland-Jones A B, Goldhirsch A A, Untch A M, Smith A I, et al. A Trastuzumab after adjuvant chemotherapy in HER2-positive breast cancer. *N Engl J Med* A 2005;353:1659aFAFSTSI1672.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A
- 224Romond A EH, Perez A EA, Bryant A J, Suman A VJ, Geyer A CE, Davidson A NE, et al. A Trastuzumab plus adjuvant chemotherapy for operable HER2-positive breast cancer. *N Engl J Med* A 2005;353:1673aFAFSTSI1684.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) A

225Cameron Â D, Piccart-Gebhart Â MJ, Gelber Â RD, Procter Â M, Goldhirsch Â A, de Azambuja Â E, et al. Â 11 yearsâ follow-up of trastuzumab after adjuvant chemotherapy in HER2-positive early breast cancer: final analysis of the HERceptin Adjuvant (HERA) trial. *Lancet* Â 2017;389:1195â1205.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

226Tan Â TC, Bouras Â S, Sawaya Â H, Sebag Â IA, Cohen Â V, Picard Â MH, et al. Â Time trends of left ventricular ejection fraction and myocardial deformation indices in a cohort of women with breast cancer treated with anthracyclines, taxanes, and trastuzumab. *J Am Soc Echocardiogr* Â 2015;28:509â514.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

227Ben Kridis Â W, Sghaier Â S, Charfeddine Â S, Toumi Â N, Daoud Â J, Kammoun Â S, et al. Â A prospective study about trastuzumab-induced cardiotoxicity in HER2-positive breast cancer. *Am J Clin Oncol Cancer Clin Trials* Â 2020;43:510â516.

[Google Scholar](#) [OpenURL Placeholder Text](#) Â

228Yang Â ZY, Wang Â W, Wang Â X, Qin Â ZQ. Cardiotoxicity of epidermal growth factor receptor 2-targeted drugs for breast cancer. *Front Pharmacol* Â 2021;12:741451.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

229Task Force Members, Montalescot Â G, Sechtem Â U, Achenbach Â S, Andreotti Â F, Arden Â C, et al. Â 2013 ESC guidelines on the management of stable coronary artery disease: the Task Force on the management of stable coronary artery disease of the European Society of Cardiology. *Eur Heart J* Â 2013;34:2949â3003.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

230Padegimas Â A, Carver Â JR. How to diagnose and manage patients with fluoropyrimidine-induced chest pain: a single center approach. *JACC CardioOncology* Â 2020;2:650â654.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

231Kwakman Â JJM, Simkens Â LHJ, Mol Â L, Kok Â WEM, Koopman Â M, Punt Â CJA. Incidence of capecitabine-related cardiotoxicity in different treatment schedules of metastatic colorectal cancer: a retrospective analysis of the CAIRO studies of the Dutch Colorectal Cancer Group. *Eur J Cancer* Â 2017;76:93â99.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

232Frickhofen Â N, Beck Â FJ, Jung Â B, Fuhr Â HG, Andrasch Â H, Sigmund Â M. Capecitabine can induce acute coronary syndrome similar to 5-fluorouracil. *Ann Oncol* Â 2002;13:797â801.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

233Polk Â A, Vistisen Â K, Vaage-Nilsen Â M, Nielsen Â DL. A systematic review of the pathophysiology of 5-fluorouracil-induced cardiotoxicity. *BMC Pharmacol Toxicol* Â 2014;15:47.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

234Edvardsen Â T, Asch Â FM, Davidson Â B, Delgado Â V, DeMaria Â A, Dilsizian Â V, et al. Â Non-invasive imaging in coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society

for Cardiovascular Magnetic Resonance. Eur Heart J Cardiovasc Imaging 2022;23:e633.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

235Sara JD, Kaur J, Khodadadi R, Rehman M, Lobo R, Chakrabarti S, et al. 5-fluorouracil and cardiotoxicity: a review. Ther Adv Med Oncol 2018;10:1758835918780140.

[Google Scholar](#) [OpenURL Placeholder Text](#)

236Bair SM, Choueiri TK, Moslehi J. Cardiovascular complications associated with novel angiogenesis inhibitors: emerging evidence and evolving perspectives. Trends Cardiovasc Med 2013;23:104-113.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

237Choueiri TK, Schutz FAB, Je Y, Rosenberg JE, Bellmunt J. Risk of arterial thromboembolic events with sunitinib and sorafenib: a systematic review and meta-analysis of clinical trials. J Clin Oncol 2010;28:2280-2285.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

238Shah RR, Morganroth J, Shah DR. Cardiovascular safety of tyrosine kinase inhibitors: With a special focus on cardiac repolarisation (QT interval). Drug Saf 2013;36:295-316.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

239Abdel-Qadir H, Ethier JL, Lee DS, Thavendiranathan P, Amir E. Cardiovascular toxicity of angiogenesis inhibitors in treatment of malignancy: a systematic review and meta-analysis. Cancer Treat Rev 2017;53:120-127.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

240Desai A, Noor A, Joshi S, Kim AS. Takotsubo cardiomyopathy in cancer patients. Cardio-Oncology 2019;5:7.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

241Nazer B, Humphreys BD, Moslehi J. Effects of novel angiogenesis inhibitors for the treatment of cancer on the cardiovascular system: focus on hypertension. Circulation 2011;124:1687-1691.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

242Shah CP, Moreb JS. Cardiotoxicity due to targeted anticancer agents: a growing challenge. Ther Adv Cardiovasc Dis 2019;13:1753944719843435.

[Google Scholar](#) [OpenURL Placeholder Text](#)

243Azizi M, Chedid A, Oudard S. Home blood-pressure monitoring in patients receiving sunitinib. N Engl J Med 2008;358:95-97.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

244Hamnvik OPR, Choueiri TK, Turchin A, McKay RR, Goyal L, Davis M, et al. Clinical risk factors for the development of hypertension in patients treated with inhibitors of the VEGF signaling pathway. Cancer 2015;121:311-319.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

245Ghatalia P, Morgan CJ, Je Y, Nguyen PL, Trinh QD, Choueiri TK, et al. Congestive heart failure with vascular endothelial growth factor receptor tyrosine kinase inhibitors. Crit Rev Oncol

Hematol 2015;94:228â237.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

246Nhola LF, Abdelmoneim SS, Villarraga HR, Kohli M, Grothey A, Bordun KA, et al. Echocardiographic assessment for the detection of cardiotoxicity due to vascular endothelial growth factor inhibitor therapy in metastatic renal cell and colorectal cancers. *J Am Soc Echocardiogr* 2019;32:267â276.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

247Uraizee I, Cheng S, Moslehi J. Reversible cardiomyopathy associated with sunitinib and sorafenib. *N Engl J Med* 2011;365:1649â1650.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

248Pandey AK, Singhi EK, Arroyo JP, Ikizler TA, Gould ER, Brown J, et al. Mechanisms of VEGF (vascular endothelial growth factor) inhibitor-associated hypertension and vascular disease. *Hypertension* 2018;71:E1âE8.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

249Zang J, Wu S, Tang L, Xu X, Bai J, Ding C, et al. Incidence and risk of QTc interval prolongation among cancer patients treated with vandetanib: a systematic review and meta-analysis. *PLoS One* 2012;7:e30353.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

250Ghatalia P, Je Y, Kaymakcalan MD, Sonpavde G, Choueiri TK. QTc interval prolongation with vascular endothelial growth factor receptor tyrosine kinase inhibitors. *Br J Cancer* 2015;112:296â305.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

251Alexandre J, Salem JE, Moslehi J, Sassier M, Ropert C, Cautela J, et al. Identification of anticancer drugs associated with atrial fibrillation: analysis of the WHO pharmacovigilance database. *Eur Heart J Cardiovasc Pharmacother* 2021;7:312â320.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

252EMA. Medicines | European Medicines Agency. *Eur Med Agency Sci Med Heal* n.d.

253FDA. Drugs@FDA: FDA-Approved Drugs. *FDA US Food Drug Adm* n.d.

254Steingart RM, Bakris GL, Chen HX, Chen MH, Force T, Ivy SP, et al. Management of cardiac toxicity in patients receiving vascular endothelial growth factor signaling pathway inhibitors. *Am Heart J* 2012;163:156â163.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

255Maitland ML, Bakris GL, Black HR, Chen HX, Durand JB, Elliott WJ, et al. Initial assessment, surveillance, and management of blood pressure in patients receiving vascular endothelial growth factor signaling pathway inhibitors. *J Natl Cancer Inst* 2010;102:596â604.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

256Barber MC, Mauro MJ, Moslehi J. Cardiovascular care of patients with chronic myeloid leukemia (CML) on tyrosine kinase inhibitor (TKI) therapy. *Hematology* 2017;2017:110â114.

257Moslehi ^ JJ, Deininger ^ M. Tyrosine kinase inhibitor-associated cardiovascular toxicity in chronic myeloid leukemia. *J Clin Oncol* ^ 2015;33:4210a^4218.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

258Li ^ W, Croce ^ K, Steensma ^ DP, McDermott ^ DF, Ben-Yehuda ^ O, Moslehi ^ J. Vascular and metabolic implications of novel targeted cancer therapies: focus on kinase inhibitors. *J Am Coll Cardiol* ^ 2015;66:1160a^1178.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

259Buza ^ V, Rajagopalan ^ B, Curtis ^ AB. Cancer treatment-induced arrhythmias: focus on chemotherapy and targeted therapies. *Circ Arrhythmia Electrophysiol* ^ 2017;10:e005443.

[Google Scholar](#) [Crossref](#) [Search ADS](#) ^

260Herrmann ^ J, Yang ^ EH, Iliescu ^ CA, Cilingiroglu ^ M, Charitakis ^ K, Hakeem ^ A, et al. ^ Vascular toxicities of cancer therapies: the old and the newa^an evolving avenue. *Circulation* ^ 2016;133:1272a^1289.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

261Cirimi ^ S, El Abd ^ A, Letinier ^ L, Navarra ^ M, Salvo ^ F. Cardiovascular toxicity of tyrosine kinase inhibitors used in chronic myeloid leukemia: an analysis of the FDA adverse event reporting system database (FAERS). *Cancers (Basel)* ^ 2020;12:826.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

262Gribben ^ JG, Bosch ^ F, Cymbalista ^ F, Geisler ^ CH, Ghia ^ P, Hillmen ^ P, et al. ^ Optimising outcomes for patients with chronic lymphocytic leukaemia on ibrutinib therapy: European recommendations for clinical practice. *Br J Haematol* ^ 2018;180:666a^679.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

263Shanafelt ^ TD, Parikh ^ SA, Noseworthy ^ PA, Goede ^ V, Chaffee ^ KG, Bahlo ^ J, et al. ^ Atrial fibrillation in patients with chronic lymphocytic leukemia (CLL). *Leuk Lymphoma* ^ 2017;58:1630a^1639.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

264Dickerson ^ T, Wiczer ^ T, Waller ^ A, Philippon ^ J, Porter ^ K, Haddad ^ D, et al. ^ Hypertension and incident cardiovascular events following ibrutinib initiation. *Blood* ^ 2019;134:1919a^1928.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

265Brown ^ JR, Moslehi ^ J, Oa^Brien ^ S, Ghia ^ P, Hillmen ^ P, Cymbalista ^ F, et al. ^ Characterization of atrial fibrillation adverse events reported in ibrutinib randomized controlled registration trials. *Haematologica* ^ 2017;102:1796a^1805.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

266LÃpez-FernÃndez, T, Canales ^ M, Farmakis ^ D, GarcÃa-Sanz ^ R, Bosch ^ F, Loscertales ^ J, et al. ^ Ibrutinib-associated atrial fibrillation: a practical approach. *Ann Hematol Oncol* ^ 2018;5:1203.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

- 267 Salem Â JE, Manouchehri Â A, Bretagne Â M, Lebrun-Vignes Â B, Groarke Â JD, Johnson Â DB, et al. Â Cardiovascular toxicities associated with ibrutinib. *J Am Coll Cardiol* Â 2019;74:1667â1678.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 268 Abdel-Qadir Â H, Sabrie Â N, Leong Â D, Pang Â A, Austin Â PC, Prica Â A, et al. Â Cardiovascular risk associated with ibrutinib use in chronic lymphocytic leukemia: a population-based cohort study. *J Clin Oncol* Â 2021;39:3453â3462.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 269 Byrd Â JC, Hillmen Â P, Ghia Â P, Kater Â AP, Chanan-Khan Â A, Furman Â RR, et al. Â Acalabrutinib versus ibrutinib in previously treated chronic lymphocytic leukemia: results of the first randomized phase III trial. *J Clin Oncol* Â 2021;39:3441â3452.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 270 Common Terminology Criteria for Adverse Events (CTCAE) v6.0. National Cancer Institute, 2022.
https://ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm.
- 271 Brown Â JR, Byrd Â JC, Ghia Â P, Sharman Â JP, Hillmen Â P, Stephens Â DM, et al. Â Cardiovascular adverse events in patients with chronic lymphocytic leukemia receiving acalabrutinib monotherapy: pooled analysis of 762 patients. *Haematologica* Â 2022;107:1335â1346.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 272 Svennberg Â E, Tjong Â F, Goette Â A, Akoum Â N, Di Biase Â L, Bordachar Â P, et al. Â How to use digital devices to detect and manage arrhythmias: an EHRA practical guide. *Europace*. 2022;24:979â1005.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 273 Hindricks Â G, Potpara Â T, Dagres Â N, Arbelo Â E, Bax Â JJ, Blomstr m-Lundqvist Â C, et al. Â 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* Â 2021;42:373â498.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 274 McMullen Â JR, Boey Â EJ, Ooi Â JYY, Seymour Â JF, Keating Â MJ, Tam Â CS. Ibrutinib increases the risk of atrial fibrillation, potentially through inhibition of cardiac PI3K-Akt signaling. *Blood* Â 2014;124:3829â3830.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 275 Lee Â HJ, Chihara Â D, Wang Â M, Mouhayer Â E, Kim Â P. Ibrutinib-related atrial fibrillation in patients with mantle cell lymphoma. *Leuk Lymphoma* Â 2016;57:2914â2916.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 276 Rajkumar Â SV. Â Multiple myeloma: 2020 update on diagnosis, risk-stratification and management. *Am J Hematol* Â 2020;95:548â567.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 277 Kumar Â SK, Dispenzieri Â A, Lacy Â MQ, Gertz Â MA, Buadi Â FK, Pandey Â S, et al. Â Continued improvement in survival in multiple myeloma: changes in early mortality and outcomes in older patients. *Leukemia* Â 2014;28:1122â1128.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

278Willis ^ MS, Patterson ^ C. Proteotoxicity and cardiac dysfunction ^ Alzheimer's disease of the heart? ^ N Engl J Med ^ 2013;368:455^464.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

279Stewart ^ AK, Rajkumar ^ SV, Dimopoulos ^ MA, Masszi ^ T, ^ piÄ^ka ^ I, Oriol ^ A, et al. ^ Carfilzomib, lenalidomide, and dexamethasone for relapsed multiple myeloma. N Engl J Med ^ 2015;372:142^152.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

280Chari ^ A, Keith Stewart ^ A, Russell ^ SD, Moreau ^ P, Herrmann ^ J, Banchs ^ J, et al. ^ Analysis of carfilzomib cardiovascular safety profile across relapsed and/or refractory multiple myeloma clinical trials. Blood Adv ^ 2018;2:1633^1644.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

281Waxman ^ AJ, Clasen ^ S, Hwang ^ WT, Garfall ^ A, Vogl ^ DT, Carver ^ J, et al. ^ Carfilzomib-associated cardiovascular adverse events: a systematic review and meta-analysis. JAMA Oncol ^ 2018;4:e174519.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

282Mauri ^ L, Elmariah ^ S, Yeh ^ RW, Cutlip ^ DE, Steg ^ PG, Windecker ^ S, et al. ^ Causes of late mortality with dual antiplatelet therapy after coronary stents. Eur Heart J ^ 2016;37:378^385.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) ^

283Gavazzoni ^ M, Lombardi ^ CM, Vizzardi ^ E, Gorga ^ E, Sciatti ^ E, Rossi ^ L, et al. ^ Irreversible proteasome inhibition with carfilzomib as first line therapy in patients with newly diagnosed multiple myeloma: early *in vivo* cardiovascular effects. Eur J Pharmacol ^ 2018;838:85^90.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

284Siegel ^ D, Martin ^ T, Nooka ^ A, Harvey ^ RD, Vij ^ R, Niesvizky ^ R, et al. ^ Integrated safety profile of single-agent carfilzomib: experience from 526 patients enrolled in 4 phase II clinical studies. Haematologica ^ 2013;98:1753^1761.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

285Ridolfi ^ RL, Bulkley ^ BH, Hutchins ^ GM. The conduction system in cardiac amyloidosis. Clinical and pathologic features of 23 patients. Am J Med ^ 1977;62:677^686.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

286Fakhri ^ B, Fiala ^ MA, Shah ^ N, Vij ^ R, Wildes ^ TM. Measuring cardiopulmonary complications of carfilzomib treatment and associated risk factors using the SEER-Medicare database. Cancer ^ 2020;126:808^813.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

287Feng ^ DL, Edwards ^ WD, Oh ^ JK, Chandrasekaran ^ K, Grogan ^ M, Martinez ^ MW, et al. ^ Intracardiac thrombosis and embolism in patients with cardiac amyloidosis. Circulation ^ 2007;116:2420^2426.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 288 Mancuso Â S, Carlisi Â M, Sarocchi Â M, Napolitano Â M, Siragusa Â S. Cardio-oncology in multiple myeloma: is it time for a specific focus? Â Leuk Lymphoma Â 2018;59:1764â1766.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 289 Danhof Â S, Schreder Â M, Rasche Â L, Striffler Â S, Einsele Â H, Knop Â S. âReal-lifeâ experience of preapproval carfilzomib-based therapy in myelomaâ analysis of cardiac toxicity and predisposing factors. Eur J Haematol Â 2016;97:25â32.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 290 Garcia-Pavia Â P, Rapezzi Â C, Adler Â Y, Arad Â M, Basso Â C, Brucato Â A, et al. Â Diagnosis and treatment of cardiac amyloidosis. A position statement of the European Society of Cardiology Working Group on Myocardial and Pericardial Diseases. Eur J Heart Fail Â 2021;23:512â526.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 291 Feng Â DL, Syed Â IS, Martinez Â M, Oh Â JK, Jaffe Â AS, Grogan Â M, et al. Â Intracardiac thrombosis and anticoagulation therapy in cardiac amyloidosis. Circulation Â 2009;119:2490â2497.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 292 Sonneveld Â P, Asselbergs Â E, Zweegman Â S, Van Der Holt Â B, Kersten Â MJ, Vellenga Â E, et al. Â Phase 2 study of carfilzomib, thalidomide, and dexamethasone as induction/consolidation therapy for newly diagnosed multiple myeloma. Blood Â 2015;125:449â456.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 293 Wang Â M, Martin Â T, Bensinger Â W, Alsina Â M, Siegel Â DS, Kavalierchik Â E, et al. Â Phase 2 dose-expansion study (PX-171-006) of carfilzomib, lenalidomide, and low-dose dexamethasone in relapsed or progressive multiple myeloma. Blood Â 2013;122:3122â3128.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 294 Fradley Â MG, Groarke Â JD, Laubach Â J, Alsina Â M, Lenihan Â DJ, Cornell Â RF, et al. Â Recurrent cardiotoxicity potentiated by the interaction of proteasome inhibitor and immunomodulatory therapy for the treatment of multiple myeloma. Br J Haematol Â 2018;180:271â275.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 295 Li Â W, Garcia Â D, Cornell Â RF, Gailani Â D, Laubach Â J, Maglio Â ME, et al. Â Cardiovascular and thrombotic complications of novel multiple myeloma therapies: a review. JAMA Oncol Â 2017;3:980â988.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 296 Palumbo Â A, Rajkumar Â SV, Dimopoulos Â MA, Richardson Â PG, San Miguel Â J, Barlogie Â B, et al. Â Prevention of thalidomide- and lenalidomide-associated thrombosis in myeloma. Leukemia Â 2008;22:414â423.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 297 Martinez-Naharro Â A, Gonzalez-Lopez Â E, Corovic Â A, Mirelis Â JG, Baksi Â AJ, Moon Â JC, et al. Â High prevalence of intracardiac thrombi in cardiac amyloidosis. J Am Coll Cardiol Â 2019;73:1733â1734.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 298Lyman Â GH, Carrier Â M, Ay Â C, Di Nisio Â M, Hicks Â LK, Khorana Â AA, et al. Â American Society of Hematology 2021 guidelines for management of venous thromboembolism: prevention and treatment in patients with cancer. *Blood Adv* Â 2021;5:927â974.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 299Key Â NS, Khorana Â AA, Kuderer Â NM, Bohlke Â K, Lee Â AYY, Arcelus Â JI, et al. Â Venous thromboembolism prophylaxis and treatment in patients with cancer: ASCO clinical practice guideline update. *J Clin Oncol* Â 2020;38:496â520.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 300Piedra Â K, Peterson Â T, Tan Â C, Orozco Â J, Hultcrantz Â M, Hassoun Â H, et al. Â Comparison of venous thromboembolism incidence in newly diagnosed multiple myeloma patients receiving bortezomib, lenalidomide, dexamethasone (RVD) or carfilzomib, lenalidomide, dexamethasone (KRD) with aspirin or rivaroxaban thromboprophylaxis. *Br J Haematol* Â 2022;196:105â109.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 301Cornell Â RF, Goldhaber Â SZ, Engelhardt Â BG, Moslehi Â J, Jagasia Â M, Harrell Â S, et al. Â Primary prevention of venous thromboembolism with apixaban for multiple myeloma patients receiving immunomodulatory agents. *Br J Haematol* Â 2020;190:555â561.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 302Swan Â D, Rocci Â A, Bradbury Â C, Thachil Â J. Venous thromboembolism in multiple myelomaâchoice of prophylaxis, role of direct oral anticoagulants and special considerations. *Br J Haematol* Â 2018;183:538â556.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 303Lendvai Â N, Tsakos Â I, Devlin Â SM, Schaffer Â WL, Hassoun Â H, Lesokhin Â AM, et al. Â Predictive biomarkers and practical considerations in the management of carfilzomib-associated cardiotoxicity. *Leuk Lymphoma* Â 2018;59:1981â1985.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 304Palumbo Â A, Cavo Â M, Bringhen Â S, Zamagni Â E, Romano Â A, Patriarca Â F, et al. Â Aspirin, warfarin, or enoxaparin thromboprophylaxis in patients with multiple myeloma treated with thalidomide: a phase III, open-label, randomized trial. *J Clin Oncol* Â 2011;29:986â993.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 305Larocca Â A, Cavallo Â F, Bringhen Â S, Di Raimondo Â F, Falanga Â A, Evangelista Â A, et al. Â Aspirin or enoxaparin thromboprophylaxis for patients with newly diagnosed multiple myeloma treated with lenalidomide. *Blood* Â 2012;119:933â939.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 306Chalayer Â E, Bourmaud Â A, Tinquaut Â F, Chauvin Â F, Tardy Â B. Cost-effectiveness analysis of low-molecular-weight heparin versus aspirin thromboprophylaxis in patients newly diagnosed with multiple myeloma. *Thromb Res* Â 2016;145:119â125.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 307Zoppellaro Â G, Veronese Â N, Granziera Â S, Gobbi Â L, Stubbs Â B, Cohen Â AT. Primary thromboembolic prevention in multiple myeloma patients: an exploratory meta-analysis on aspirin use.

Semin Hematol 2018;55:182â184.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

308Mincu RI, Mahabadi AA, Michel L, Mrotzek SM, Schadendorf D, Rassaf T, et al. Cardiovascular adverse events associated with BRAF and MEK inhibitors: a systematic review and meta-analysis. JAMA Netw Open 2019;2:e198890.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

309Glen C, Tan YY, Waterston A, Evans TRJ, Jones RJ, Petrie MC, et al. Mechanistic and clinical overview cardiovascular toxicity of BRAF and MEK inhibitors: JACC: CardioOncology state-of-the-art review. Cardio Oncol 2022;4:1â18.

[Google Scholar](#) [OpenURL Placeholder Text](#)

310Flaherty KT, Robert C, Hersey P, Nathan P, Garbe C, Milhem M, et al. Improved survival with MEK inhibition in BRAF-mutated melanoma. N Engl J Med 2012;367:107â114.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

311Robert C, Karaszewska B, Schachter J, Rutkowski P, Mackiewicz A, Stroiakovski D, et al. Improved overall survival in melanoma with combined dabrafenib and trametinib. N Engl J Med 2015;372:30â39.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

312Long GV, Stroyakovskiy D, Gogas H, Levchenko E, De Braud F, Larkin J, et al. Dabrafenib and trametinib versus dabrafenib and placebo for Val600 BRAF-mutant melanoma: a multicentre, double-blind, phase 3 randomised controlled trial. Lancet 2015;386:444â451.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

313Bronte E, Bronte G, Novo G, Rinaldi G, Bronte F, Passiglia F, et al. Cardiotoxicity mechanisms of the combination of BRAF-inhibitors and MEK-inhibitors. Pharmacol Ther 2018;192:65â73.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

314Gogas HJ, Flaherty KT, Dummer R, Ascierto PA, Arance A, Mandala M, et al. Adverse events associated with encorafenib plus binimetinib in the COLUMBUS study: incidence, course and management. Eur J Cancer 2019;119:97â106.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

315Banks M, Crowell K, Proctor A, Jensen BC. Cardiovascular effects of the MEK inhibitor, trametinib: a case report, literature review, and consideration of mechanism. Cardiovasc Toxicol 2017;17:487â493.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

316Hu JR, Florido R, Lipson EJ, Naidoo J, Ardehali R, Tocchetti CG, et al. Cardiovascular toxicities associated with immune checkpoint inhibitors. Cardiovasc Res 2019;115:854â868.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

317Postow MA, Sidlow R, Hellmann MD. Immune-related adverse events associated with immune checkpoint blockade. N Engl J Med 2018;378:158â168.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

318Drobní ZD, Alvi RM, Taron J, Zafar A, Murphy SP, Rambarat PK, et al. Association between immune checkpoint inhibitors with cardiovascular events and atherosclerotic plaque. *Circulation* 2020;142:2299-2311.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

319Rubio-Infante N, Ramírez-Flores YA, Castillo EC, Lozano O, García-Rivas G, Torre-Amione G. Cardiotoxicity associated with immune checkpoint inhibitor therapy: a meta-analysis. *Eur J Heart Fail* 2021;23:1739-1747.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

320Salem JE, Manouchehri A, Moey M, Lebrun-Vignes B, Bastarache L, Pariente A, et al. Cardiovascular toxicities associated with immune checkpoint inhibitors: an observational, retrospective, pharmacovigilance study. *Lancet Oncol* 2018;19:1579-1589.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

321Dolladille C, Ederhy S, Allouche S, Dupas Q, Gervais R, Madeline J, et al. Late cardiac adverse events in patients with cancer treated with immune checkpoint inhibitors. *J Immunother Cancer* 2020;8:e000261.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

322D'Souza M, Nielsen D, Svane IM, Iversen K, Rasmussen PV, Madelaire C, et al. The risk of cardiac events in patients receiving immune checkpoint inhibitors: a nationwide Danish study. *Eur Heart J* 2021;42:1621-1631.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

323Dolladille C, Akroun J, Morice P-M, Domp Martin A, Ezine E, Sassier M, et al. Cardiovascular immunotoxicities associated with immune checkpoint inhibitors: a safety meta-analysis. *Eur Heart J* 2021;42:4964-4977.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

324Zamami Y, Niimura T, Okada N, Koyama T, Fukushima K, Izawa-Ishizawa Y, et al. Factors associated with immune checkpoint inhibitor-related myocarditis. *JAMA Oncol* 2019;5:1635-1637.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

325Zhang L, Reynolds KL, Lyon AR, Palaskas N, Neilan TG. The evolving immunotherapy landscape and the epidemiology, diagnosis, and management of cardiotoxicity: *JACC: CardioOncology* primer. *JACC CardioOncology* 2021;3:35-47.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

326Lyon AR, Yousaf N, Battisti NML, Moslehi J, Larkin J. Immune checkpoint inhibitors and cardiovascular toxicity. *Lancet Oncol* 2018;19:e447-e458.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

327Schiffer WB, Deych E, Lenihan DJ, Zhang KW. Coronary and aortic calcification are associated with cardiovascular events on immune checkpoint inhibitor therapy. *Int J Cardiol* 2021;322:177-182.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

328Naing ^ A, Infante ^ J, Goel ^ S, Burris ^ H, Black ^ C, Marshall ^ S, et al. ^ Anti-PD-1 monoclonal antibody MEDI0680 in a phase I study of patients with advanced solid malignancies. *J Immunother Cancer* ^ 2019;7:225.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

329Allenbach ^ Y, Anquetil ^ C, Manouchehri ^ A, Benveniste ^ O, Lambotte ^ O, Lebrun-Vignes ^ B, et al. ^ Immune checkpoint inhibitor-induced myositis, the earliest and most lethal complication among rheumatic and musculoskeletal toxicities. *Autoimmun Rev* ^ 2020;19:102586.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

330Anquetil ^ C, Salem ^ JE, Lebrun-Vignes ^ B, Johnson ^ DB, Mammen ^ AL, Stenzel ^ W, et al. ^ Immune checkpoint inhibitor-associated myositis: expanding the spectrum of cardiac complications of the immunotherapy revolution. *Circulation* ^ 2018;138:743^745.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

331Bonaca ^ MP, Olenchock ^ BA, Salem ^ JE, Wiviott ^ SD, Ederhy ^ S, Cohen ^ A, et al. ^ Myocarditis in the setting of cancer therapeutics: proposed case definitions for emerging clinical syndromes in cardio-oncology. *Circulation* ^ 2019;140:80^91.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

332Lehmann ^ LH, Cautela ^ J, Palaskas ^ N, Baik ^ AH, Meijers ^ WC, Allenbach ^ Y, et al. ^ Clinical strategy for the diagnosis and treatment of immune checkpoint inhibitor-associated myocarditis: a narrative review. *JAMA Cardiol* ^ 2021;6:1329^1337.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

333Rini ^ B, Moslehi ^ JJ, Bonaca ^ M, Schmidinger ^ M, Albiges ^ L, Choueiri ^ TK, et al. ^ Prospective cardiovascular surveillance of immune checkpoint inhibitor-based combination therapy in patients with advanced renal cell cancer: data from the phase 3 JAVELIN Renal 101 trial. *J Clin Oncol* ^ 2022;40:1929^1938.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

334Awadalla ^ M, Mahmood ^ SS, Groarke ^ JD, Hassan ^ MZO, Nohria ^ A, Rokicki ^ A, et al. ^ Global longitudinal strain and cardiac events in patients with immune checkpoint inhibitor-related myocarditis. *J Am Coll Cardiol* ^ 2020;75:467^478.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

335Kondapalli ^ L, Bottinor ^ W, Lenneman ^ C. By releasing the brakes with immunotherapy, are we accelerating atherosclerosis? ^ *Circulation* ^ 2020;142:2312^2315.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

336Johnson ^ DB, Nebhan ^ CA, Moslehi ^ JJ, Balko ^ JM. Immune-checkpoint inhibitors: long-term implications of toxicity. *Nat Rev Clin Oncol* ^ 2022;19:254^267.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

337Zhao ^ J, Zhu ^ S, Sun ^ L, Meng ^ F, Zhao ^ L, Zhao ^ Y, et al. ^ Androgen deprivation therapy for prostate cancer is associated with cardiovascular morbidity and mortality: a meta-analysis of population-based observational studies. *PLoS One* ^ 2014;9:e107516.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

338Bhatia ^ N, Santos ^ M, Jones ^ LW, Beckman ^ JA, Penson ^ DF, Morgans ^ AK, et al.
^ Cardiovascular effects of androgen deprivation therapy for the treatment of prostate cancer: ABCDE
Steps to reduce cardiovascular disease in patients with prostate cancer. *Circulation*
^ 2016;133:537a^541.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

339Okwuosa ^ TM, Morgans ^ A, Rhee ^ J-W, Reding ^ KW, Maliski ^ S, Plana ^ J-C, et al. ^ Impact
of hormonal therapies for treatment of hormone-dependent cancers (breast and prostate) on the
cardiovascular system: effects and modifications: a scientific statement from the American Heart
Association. *Circ Genomic Precis Med* ^ 2021;14:e000082.

[Google Scholar](#) [Crossref](#) [Search ADS](#) ^

340Barber ^ M, Nguyen ^ L, Wassermann ^ J, Spano ^ J, Funck-Brentano ^ C, Salem ^ J. Cardiac
arrhythmia considerations of hormone cancer therapies. *Cardiovasc Res* ^ 2019;115:878a^894.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

341Shore ^ ND, Saad ^ F, Cookson ^ MS, George ^ DJ, Saltzstein ^ DR, Tutrone ^ R, et al. ^ Oral
relugolix for androgen-deprivation therapy in advanced prostate cancer. *N Engl J Med*
^ 2020;382:2187a^2196.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

342Abufaraj ^ M, Iwata ^ T, Kimura ^ S, Haddad ^ A, Al-Ani ^ H, Abusubaih ^ L, et al. ^ Differential
impact of gonadotropin-releasing hormone antagonist versus agonist on clinical safety and oncologic
outcomes on patients with metastatic prostate cancer: a meta-analysis of randomized controlled trials. *Eur Urol* ^ 2021;79:44a^53.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

343Lopes ^ RD, Higano ^ CS, Slovin ^ SF, Nelson ^ AJ, Bigelow ^ R, SÃ,rensen ^ PS, et al.
^ Cardiovascular safety of degarelix versus leuprolide in patients with prostate cancer: the primary
results of the PRONOUNCE randomized trial. *Circulation* ^ 2021;144:1295a^1307.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

344Wilk ^ M, WaÅ,ko-Grabowska ^ A, Skoneczna ^ I, Szmit ^ S. Angiotensin system inhibitors may
improve outcomes of patients with castration-resistant prostate cancer during abiraterone acetate
treatment^a cardio-oncology study. *Front Oncol* ^ 2021;11:664741.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

345Salem ^ JE, Waintraub ^ X, Courtillot ^ C, Shaffer ^ CM, Gandjbakhch ^ E, Maupain ^ C, et al.
^ Hypogonadism as a reversible cause of Torsades de Pointes in men. *Circulation*
^ 2018;138:110a^113.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

346Salem ^ JE, Alexandre ^ J, Bachelot ^ A, Funck-Brentano ^ C. Influence of steroid hormones on
ventricular repolarization. *Pharmacol Ther* ^ 2016;167:38a^47.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 347Salem Â JE, Yang Â T, Moslehi Â JJ, Waintraub Â X, Gandjbakhch Â E, Bachelot Â A, et al. Â Androgenic effects on ventricular repolarization a translational study from the international pharmacovigilance database to iPSC-cardiomyocytes. *Circulation* Â 2019;140:1070â1080.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 348Hasegawa Â K, Ito Â H, Kaseno Â K, Miyazaki Â S, Shiomi Â Y, Tama Â N, et al. Â Impact of medical castration on malignant arrhythmias in patients with prostate cancer. *J Am Heart Assoc* Â 2021;10:e017267.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 349Fradley Â MG, Beckie Â TM, Brown Â SA, Cheng Â RK, Dent Â SF, Nohria Â A, et al. Â Recognition, prevention, and management of arrhythmias and autonomic disorders in cardio-oncology: a scientific statement from the American Heart Association. *Circulation* Â 2021;144:E41âE55.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 350Olsson Â H, Petri Â N, Erichsen Â L, Malmberg Â A, Grundemar Â L. Effect of degarelix, a gonadotropin-releasing hormone receptor antagonist for the treatment of prostate cancer, on cardiac repolarisation in a randomised, placebo and active comparator controlled thorough QT/QTc trial in healthy men. *Clin Drug Investig* Â 2017;37:873â879.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 351Baum Â M, Buzdar Â AU, Cuzick Â J, Forbes Â J, Houghton Â J, Klijn Â JGM, et al. Â Anastrozole alone or in combination with tamoxifen versus tamoxifen alone for adjuvant treatment of postmenopausal women with early breast cancer: first results of the ATAC randomised trial. *Lancet* Â 2002;359:2131â2139.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 352ARIMIDEXÂ® (anastrozole). Highlights of Prescribing Information. Wilmington, DE: Â AstraZeneca Pharmaceuticals LP, 2014.
[Google Scholar](#) [OpenURL Placeholder Text](#) Â
- 353Breast International Group (BIG) 1-98 Collaborative Group, Thürlimann Â B, Keshaviah Â A, Coates Â AS, Mouridsen Â H, Mauriac Â L, et al. Â A comparison of letrozole and tamoxifen in postmenopausal women with early breast cancer. *N Engl J Med* Â 2005;353:2747â2757.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 354Amir Â E, Seruga Â B, Niraula Â S, Carlsson Â L, Ocaña Â A. Toxicity of adjuvant endocrine therapy in postmenopausal breast cancer patients: a systematic review and meta-analysis. *J Natl Cancer Inst* Â 2011;103:1299â1309.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 355Goldvaser Â H, Barnes Â TA, Seruga Â B, Cescon Â DW, Ocaña Â A, Ribnikar Â D, et al. Â Toxicity of extended adjuvant therapy with aromatase inhibitors in early breast cancer: a systematic review and meta-analysis. *J Natl Cancer Inst* Â 2018;110:31â39.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 356Curigliano Â G, Azambuja Â E, Lenihan Â D, Calabrese Â MG, Cardinale Â D, Cipolla Â CM. Prevention, monitoring, and management of cardiac dysfunction in patients with metastatic breast cancer.

Oncologist 2019;24:e1034âe1043.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

357KISQALI (ribociclib). Highlights of Prescribing Information. East Hanover, NJ: Novartis Pharmaceuticals Corporation, 2018.

[Google Scholar](#) [OpenURL Placeholder Text](#)

358IBRANCE (palbociclib). Highlights of Prescribing Information. New York, NY: Pfizer, 2018.

[Google Scholar](#) [OpenURL Placeholder Text](#)

359VERZENIO (abemaciclib). Highlights of Prescribing Information. Indianapolis, IN: Eli Lilly and Company, 2018.

[Google Scholar](#) [OpenURL Placeholder Text](#)

360Santoni M, Occhipinti G, Romagnoli E, Miccini F, Scoccia L, Giulietti M, et al. Different cardiotoxicity of palbociclib and ribociclib in breast cancer: gene expression and pharmacological data analyses, biological basis, and therapeutic implications. *BioDrugs* 2019;33:613â620.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

361Hortobagyi GN, Stemmer SM, Burris HA, Yap Y-S, Sonke GS, Paluch-Shimon S, et al. Ribociclib as first-line therapy for HR-positive, advanced breast cancer. *N Engl J Med* 2016;375:1738â1748.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

362Tripathy D, Im SA, Colleoni M, Franke F, Bardia A, Harbeck N, et al. Ribociclib plus endocrine therapy for premenopausal women with hormone-receptor-positive, advanced breast cancer (MONALEESA-7): a randomised phase 3 trial. *Lancet Oncol* 2018;19:904â915.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

363Finn RS, Martin M, Rugo HS, Jones S, Im S-A, Gelmon K, et al. Palbociclib and letrozole in advanced breast cancer. *N Engl J Med* 2016;375:1925â1936.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

364Cristofanilli M, Turner NC, Bondarenko I, Ro J, Im SA, Masuda N, et al. Fulvestrant plus palbociclib versus fulvestrant plus placebo for treatment of hormone-receptor-positive, HER2-negative metastatic breast cancer that progressed on previous endocrine therapy (PALOMA-3): final analysis of the multicentre, double-blind, phase 3. *Lancet Oncol* 2016;17:425â439.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

365Slamon DJ, Neven P, Chia S, Fasching PA, De Laurentiis M, Im SA, et al. Phase III randomized study of ribociclib and fulvestrant in hormone receptor-positive, human epidermal growth factor receptor 2-negative advanced breast cancer: MONALEESA-3. *J Clin Oncol* 2018;36:2465â2472.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

366Durairaj C, Ruiz-Garcia A, Gauthier ER, Huang X, Lu DR, Hoffman JT, et al. Palbociclib has no clinically relevant effect on the QTc interval in patients with advanced breast cancer. *Anticancer Drugs* 2018;29:271â280.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

367Im ^ S-A, Lu ^ Y-S, Bardia ^ A, Harbeck ^ N, Colleoni ^ M, Franke ^ F, et al. ^ Overall survival with ribociclib plus endocrine therapy in breast cancer. *N Engl J Med* ^ 2019;381:307^PAGESTS:316.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

368Slamon ^ DJ, Neven ^ P, Chia ^ S, Fasching ^ PA, De Laurentiis ^ M, Im ^ S-A, et al. ^ Overall survival with ribociclib plus fulvestrant in advanced breast cancer. *N Engl J Med* ^ 2020;382:514^PAGESTS:524.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

369Tamargo ^ J, Caballero ^ R, Delp^3n ^ E. Cancer chemotherapy and cardiac arrhythmias: a review. *Drug Saf* ^ 2015;38:129^PAGESTS:152.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

370Wang ^ L, Wang ^ W. Safety and efficacy of anaplastic lymphoma kinase tyrosine kinase inhibitors in non-small cell lung cancer (Review). *Oncol Rep* ^ 2021;45:13^PAGESTS:28.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) ^

371Rao ^ VU, Reeves ^ DJ, Chugh ^ AR, O^PAGESTS:Quinn ^ R, Fradley ^ MG, Raghavendra ^ M, et al. ^ Clinical approach to cardiovascular toxicity of oral antineoplastic agents: JACC state-of-the-art review. *J Am Coll Cardiol* ^ 2021;77:2693^PAGESTS:2716.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

372Shaw ^ AT, Bauer ^ TM, de Marinis ^ F, Felip ^ E, Goto ^ Y, Liu ^ G, et al. ^ First-line lorlatinib or crizotinib in advanced ALK-positive lung cancer. *N Engl J Med* ^ 2020;383:2018^PAGESTS:2029.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

373Thein ^ KZ, Swarup ^ S, Ball ^ S, Quirch ^ M, Vorakunthada ^ Y, Htwe ^ KK, et al. ^ Incidence of cardiac toxicities in patients with advanced non-small cell lung cancer treated with osimertinib: a combined analysis of two phase III randomized controlled trials. *Ann Oncol* ^ 2018;29:viii500.

[Google Scholar](#) [Crossref](#) [Search ADS](#) ^

374Anand ^ K, Ensor ^ J, Trachtenberg ^ B, Bernicker ^ EH. Osimertinib-induced cardiotoxicity: a retrospective review of the FDA Adverse Events Reporting System (FAERS). *JACC CardioOncology* ^ 2019;1:172^PAGESTS:178.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

375Kunimasa ^ K, Kamada ^ R, Oka ^ T, Oboshi ^ M, Kimura ^ M, Inoue ^ T, et al. ^ Cardiac adverse events in EGFR-mutated non-small cell lung cancer treated with osimertinib. *JACC CardioOncology* ^ 2020;2:1^PAGESTS:10.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

376Ewer ^ MS, Tekumalla ^ SH, Walding ^ A, Atuah ^ KN. Cardiac safety of osimertinib: a review of data. *J Clin Oncol* ^ 2021;39:328^PAGESTS:337.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

377Frey ^ N, Porter ^ D. Cytokine release syndrome with chimeric antigen receptor T cell therapy. *Biol Blood Marrow Transplant* ^ 2019;25:e123^PAGESTS:e127.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

378Alvi ^ RM, Frigault ^ MJ, Fradley ^ MG, Jain ^ MD, Mahmood ^ SS, Awadalla ^ M, et al.
^ Cardiovascular events among adults treated with chimeric antigen receptor T-cells (CAR-T). J Am Coll
Cardiol ^ 2019;74:3099^PACS:3108.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

379Goldman ^ A, Maor ^ E, Bomze ^ D, Liu ^ JE, Herrmann ^ J, Fein ^ J, et al. ^ Adverse
cardiovascular and pulmonary events associated with chimeric antigen receptor T-cell therapy. J Am Coll
Cardiol ^ 2021;78:1800^PACS:1813.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

380Fradley ^ MG, Damrongwatanasuk ^ R, Chandrasekhar ^ S, Alomar ^ M, Kip ^ KE, Sarnaik ^ AA.
Cardiovascular toxicity and mortality associated with adoptive cell therapy and tumor-infiltrating
lymphocytes for advanced stage melanoma. J Immunother ^ 2021;44:86^PACS:89.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

381Ghosh ^ AK, Chen ^ DH, Guha ^ A, Mackenzie ^ S, Walker ^ JM, Roddie ^ C. CAR T cell
therapy-related cardiovascular outcomes and management: systemic disease or direct cardiotoxicity?
^ JACC CardioOncology ^ 2020;2:97^PACS:109.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

382Lefebvre ^ B, Kang ^ Y, Smith ^ AM, Frey ^ NV, Carver ^ JR, Scherrer-Crosbie ^ M.
Cardiovascular effects of CAR T cell therapy: a retrospective study. JACC CardioOncology
^ 2020;2:193^PACS:203.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

383Salem ^ JE, Ederhy ^ S, Lebrun-Vignes ^ B, Moslehi ^ JJ. Cardiac events associated with chimeric
antigen receptor T-cells (CAR-T): a VigiBase perspective. J Am Coll Cardiol ^ 2020;75:2521^PACS:2523.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

384Ganatra ^ S, Redd ^ R, Hayek ^ SS, Parikh ^ R, Azam ^ T, Yanik ^ GA, et al. ^ Chimeric antigen
receptor T-cell therapy-associated cardiomyopathy in patients with refractory or relapsed non-Hodgkin
lymphoma. Circulation ^ 2020;142:1687^PACS:1690.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

385Lee ^ JB, Vasic ^ D, Kang ^ H, Fang ^ KKL, Zhang ^ L. State-of-art of cellular therapy for acute
leukemia. Int J Mol Sci ^ 2021;22:4590.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

386Brahmer ^ JR, Abu-Sbeih ^ H, Ascierto ^ PA, Brufsky ^ J, Cappelli ^ LC, Cortazar ^ FB, et al.
^ Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immune checkpoint
inhibitor-related adverse events. J Immunother Cancer ^ 2021;9:e002435.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

387Ragoonanan ^ D, Khazal ^ SJ, Abdel-Azim ^ H, McCall ^ D, Cuglievan ^ B, Tambaro ^ FP, et al.
^ Diagnosis, grading and management of toxicities from immunotherapies in children, adolescents and
young adults with cancer. Nat Rev Clin Oncol ^ 2021;18:435^PACS:453.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

388Maus M Â V, Alexander Â S, Bishop Â MR, Brudno Â JN, Callahan Â C, Davila Â ML, et al.

Â Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immune effector cell-related adverse events. *J Immunother Cancer* Â 2020;8:e001511.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

389Darby Â SC, Ewertz Â M, McGale Â P, Bennet Â AM, Blom-Goldman Â U, BrÃ,nnum Â D, et al.

Â Risk of ischemic heart disease in women after radiotherapy for breast cancer. *N Engl J Med* Â 2013;368:987â998.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

390Van Nimwegen Â FA, Schaapveld Â M, Cutter Â DJ, Janus Â CPM, Krol Â ADG, Hauptmann Â M,

et al. Â Radiation dose-response relationship for risk of coronary heart disease in survivors of Hodgkin lymphoma. *J Clin Oncol* Â 2016;34:235â243.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

391Cutter Â DJ, Schaapveld Â M, Darby Â SC, Hauptmann Â M, Van Nimwegen Â FA, Krol Â ADG, et

al. Â Risk for valvular heart disease after treatment for Hodgkin lymphoma. *J Natl Cancer Inst* Â 2015;107:djv008.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

392Bergom Â C, Bradley Â JA, Ng Â AK, Samson Â P, Robinson Â C, Lopez-Mattei Â J, et al. Â Past,

present, and future of radiation-induced cardiotoxicity: refinements in targeting, surveillance, and risk stratification. *JACC CardioOncology* Â 2021;3:343â359.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

393Belzile-Dugas Â E, Eisenberg Â MJ. Radiation-induced cardiovascular disease: review of an

underrecognized pathology. *J Am Heart Assoc* Â 2021;10:e021686.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

394Carlson Â LE, Watt Â GP, Tonorezos Â ES, Chow Â EJ, Yu Â AF, Woods Â M, et al. Â Coronary

artery disease in young women after radiation therapy for breast cancer. *JACC CardioOncology* Â 2021;3:381â392.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

395Jacob Â S, Camilleri Â J, Derreumaux Â S, Walker Â V, Lairez Â O, Lapeyre Â M, et al. Â Is mean

heart dose a relevant surrogate parameter of left ventricle and coronary arteries exposure during breast cancer radiotherapy: a dosimetric evaluation based on individually-determined radiation dose (BACCARAT study). *Radiat Oncol* Â 2019;14:29.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

396Hoppe Â BS, Bates Â JE, Mendenhall Â NP, Morris Â CG, Louis Â D, Ho Â MW, et al. Â The

meaningless meaning of mean heart dose in mediastinal lymphoma in the modern radiation therapy era. *Pract Radiat Oncol* Â 2020;10:e147âe154.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

397Maraldo M Â V, Giusti Â F, Vogelius Â IR, Lundemann Â M, Van der Kaaij Â MAE, Ramadan Â S,

et al. Â Cardiovascular disease after treatment for Hodgkinâs lymphoma: an analysis of nine collaborative EORTC-LYSA trials. *Lancet Haematol* Â 2015;2:e492âe502.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

398 Atkins Â KM, Bitterman Â DS, Chaunzwa Â TL, Kozono Â DE, Baldini Â EH, Aerts Â HJWL, et al. Â Mean heart dose is an inadequate surrogate for left anterior descending coronary artery dose and the risk of major adverse cardiac events in lung cancer radiation therapy. *Int J Radiat Oncol* Â 2021;110:1473â1479.

[Google Scholar](#) [OpenURL Placeholder Text](#) Â

399 Mulrooney Â DA, Yeazel Â MW, Kawashima Â T, Mertens Â AC, Mitby Â P, Stovall Â M, et al. Â Cardiac outcomes in a cohort of adult survivors of childhood and adolescent cancer: retrospective analysis of the childhood cancer survivor study cohort. *BMJ* Â 2009;339:b4606.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

400 Van Nimwegen Â FA, Schaapveld Â M, Janus Â CPM, Krol Â ADG, Petersen Â EJ, Raemaekers Â JMM, et al. Â Cardiovascular disease after Hodgkin lymphoma treatment 40-year disease risk. *JAMA Intern Med* Â 2015;175:1007â1017.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

401 Jacobse Â JN, Duane Â FK, Boekel Â NB, Schaapveld Â M, Hauptmann Â M, Hooning Â MJ, et al. Â Radiation dose-response for risk of myocardial infarction in breast cancer survivors. *Int J Radiat Oncol Biol Phys* Â 2019;103:595â604.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

402 Fuchs Â M, Goergen Â H, Kobe Â C, Kuhnert Â G, Lohri Â A, Greil Â R, et al. Â Positron emission tomography-guided treatment in early-stage favorable Hodgkin lymphoma: final results of the international, randomized phase III HD16 trial by the German Hodgkin Study Group. *J Clin Oncol* Â 2019;37:2835â2845.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

403 Radford Â J, Illidge Â T, Counsell Â N, Hancock Â B, Pettengell Â R, Johnson Â P, et al. Â Results of a trial of PET-directed therapy for early-stage Hodgkin's lymphoma. *N Engl J Med* Â 2015;372:1598â1607.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

404 Kunkler Â IH, Williams Â LJ, Jack Â WJL, Cameron Â DA, Dixon Â JM. Breast-conserving surgery with or without irradiation in women aged 65 years or older with early breast cancer (PRIME II): a randomised controlled trial. *Lancet Oncol* Â 2015;16:266â273.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

405 Song Â AJ, Manukian Â G, Taylor Â AK, Anne Â PR, Simone Â NL. Concerns for active breathing control (ABC) with breast cancer in the era of COVID-19: maximizing infection control while minimizing heart dose. *Adv Radiat Oncol* Â 2020;5:573â574.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

406 Petersen Â PM, Aznar Â MC, Berthelsen Â AK, Loft Â A, Schut Â DA, Maraldo Â M, et al. Â Prospective phase II trial of image-guided radiotherapy in Hodgkin lymphoma: benefit of deep inspiration breath-hold. *Acta Oncol (Madr)* Â 2015;54:60â66.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

407 Persson Â GF, Scherman Rydh Â J, Josipovic Â M, Maraldo M Â V, Nygård Â L, Costa Â J, et al. Â Deep inspiration breath-hold volumetric modulated arc radiotherapy decreases dose to mediastinal

structures in locally advanced lung cancer. *Acta Oncol (Madr)* 2016;55:1053â1056.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

408Dabaja Â BS, Hoppe Â BS, Plastaras Â JP, Newhauser Â W, Rosolova Â K, Flampouri Â S, et al. Â Proton therapy for adults with mediastinal lymphomas: the International Lymphoma Radiation Oncology Group guidelines. *Blood* 2018;132:1635â1646. Erratum in: *Blood* 2019;133:1384â1385.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

409Rotz Â SJ, Ryan Â TD, Hayek Â SS. Cardiovascular disease and its management in children and adults undergoing hematopoietic stem cell transplantation. *J Thromb Thrombolysis* 2021;51:854â869.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

410Oliveira Â GH, Al-Kindi Â SG, Guha Â A, Dey Â AK, Rhea Â IB, DeLima Â MJ. Cardiovascular risk assessment and management of patients undergoing hematopoietic cell transplantation. *Bone Marrow Transplant* 2021;56:544â551.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

411Sorrer Â ML, Maris Â MB, Storb Â R, Baron Â F, Sandmaier Â BM, Maloney Â DG, et al. Â Hematopoietic cell transplantation (HCT)-specific comorbidity index: a new tool for risk assessment before allogeneic HCT. *Blood* 2005;106:2912â2919.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

412Tichelli Â A, Bucher Â C, RovÃ³ Â A, Stussi Â G, Stern Â M, Paulussen Â M, et al. Â Premature cardiovascular disease after allogeneic hematopoietic stem-cell transplantation. *Blood* 2007;110:3463â3471.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

413Alblooshi Â R, Kanfar Â S, Lord Â B, Atenafu Â EG, Michelis Â FV, Pasic Â I, et al. Â Clinical prevalence and outcome of cardiovascular events in the first 100 days postallogeneic hematopoietic stem cell transplant. *Eur J Haematol* 2021;106:32â39.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

414DulÃ©ry Â R, Mohty Â R, Labopin Â M, Sestili Â S, Malard Â F, Brissot Â E, et al. Â Early cardiac toxicity associated with post-transplant cyclophosphamide in allogeneic stem cell transplantation. *JACC CardioOncology* 2021;3:250â259.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

415LÃ³pez-FernÃ¡ndez Â T, Vadillo Â IS, de la GuÃ¡a Â AL, Barbier Â KH. Cardiovascular issues in hematopoietic stem cell transplantation (HSCT). *Curr Treat Options Oncol* 2021;22:51.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

416Ohmoto Â A, Fuji Â S. Cardiac complications associated with hematopoietic stem-cell transplantation. *Bone Marrow Transplant* 2021;56:2637â2643.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

417Takatsuka Â H, Nakajima Â T, Nomura Â K, Okikawa Â Y, Wakae Â T, Toda Â A, et al. Â Prognosis value of atrial natriuretic peptide and brain natriuretic peptide for heart failure in patients undergoing allogeneic bone marrow transplantation. *Hematology* 2006;11:351â354.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

418Snowden ^ JA, Hill ^ GR, Hunt ^ P, Carnoutsos ^ S, Spearing ^ RL, Espiner ^ E, et al.
^ Assessment of cardiotoxicity during haemopoietic stem cell transplantation with plasma brain
natriuretic peptide. Bone Marrow Transplant ^ 2000;26:309^313.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

419Alvarez-Cardona ^ JA, Zhang ^ KW, Mitchell ^ JD, Zaha ^ VG, Fisch ^ MJ, Lenihan ^ DJ.
Cardiac biomarkers during cancer therapy: practical applications for cardio-oncology. JACC
CardioOncology ^ 2020;2:791^794.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

420Mohammed ^ J, Smith ^ SR, Burns ^ L, Basak ^ G, Aljurf ^ M, Savani ^ BN, et al. ^ Role of
physical therapy before and after hematopoietic stem cell transplantation: white paper report. Biol Blood
Marrow Transplant ^ 2019;25:e191^e198.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

421Haugnes ^ HS, Wethal ^ T, Aass ^ N, Dahl ^ O, Klepp ^ O, Langberg ^ CW, et al.
^ Cardiovascular risk factors and morbidity in long-term survivors of testicular cancer: a 20-year follow-
up study. J Clin Oncol ^ 2010;28:4649^4657.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

422Herrmann ^ J. Cardiovascular toxicity with cisplatin in patients with testicular cancer: looking for
something heavier than heavy metal. JACC CardioOncology ^ 2020;2:456^459.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

423Cerchione ^ C, Peleteiro RaÃndo ^ A, Mosquera Orgueira ^ A, Mosquera Torre ^ A, Bao PÃ©rez ^
L, Marconi ^ G, et al. ^ Safety of FLT3 inhibitors in patients with acute myeloid leukemia. Expert Rev
Hematol ^ 2021;14:851^865.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

424Dong ^ H, Yao ^ L, Wang ^ M, Wang ^ M, Li ^ X, Sun ^ X, et al. ^ Can ACEI/ARB prevent the
cardiotoxicity caused by chemotherapy in early-stage breast cancer? ^ A meta-analysis of randomized
controlled trials. Transl Cancer Res ^ 2020;11:7034^7043.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

425Cardinale ^ D, Colombo ^ A, Lamantia ^ G, Colombo ^ N, Civelli ^ M, De Giacomi ^ G, et al.
^ Anthracycline-induced cardiomyopathy: clinical relevance and response to pharmacologic therapy. J
Am Coll Cardiol ^ 2010;55:213^220.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

426Modi ^ S, Saura ^ C, Yamashita ^ T, Park ^ YH, Kim ^ S-B, Tamura ^ K, et al. ^ Trastuzumab
deruxtecan in previously treated HER2-positive breast cancer. N Engl J Med ^ 2020;382:610^621.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

427Hussain ^ Y, Drill ^ E, Dang ^ CT, Liu ^ JE, Steingart ^ RM, Yu ^ AF. Cardiac outcomes of
trastuzumab therapy in patients with HER2-positive breast cancer and reduced left ventricular ejection
fraction. Breast Cancer Res Treat ^ 2019;175:239^246.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 428Leong Â DP, Cosman Â T, Alhussein Â MM, Kumar Tyagi Â N, Karampatos Â S, Barron Â CC, et al. Â Safety of continuing trastuzumab despite mild cardiotoxicity: a phase I trial. *JACC CardioOncology* Â 2019;1:1âFADSTS:10.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 429Omland Â T, Heck Â SL, Gulati Â G. The role of cardioprotection in cancer therapy cardiotoxicity: *JACC: CardioOncology* state-of-the-art review. *Cardio Oncol* Â 2022;4:19âFADSTS:37.
[Google Scholar](#) [OpenURL Placeholder Text](#) Â
- 430Russell Â SD, Blackwell Â KL, Lawrence Â J, Pippen Â JE, Roe Â MT, Wood Â F, et al. Â Independent adjudication of symptomatic heart failure with the use of doxorubicin and cyclophosphamide followed by trastuzumab adjuvant therapy: A combined review of cardiac data from the National Surgical Adjuvant Breast and Bowel Project B-31 and the North Central Cancer Treatment Group N9831 clinical trials. *J Clin Oncol* Â 2010;28:3416âFADSTS:3421.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 431Ewer Â MS, Vooletich Â MT, Durand Â JB, Woods Â ML, Davis Â JR, Valero Â V, et al. Â Reversibility of trastuzumab-related cardiotoxicity: new insights based on clinical course and response to medical treatment. *J Clin Oncol* Â 2005;23:7820âFADSTS:7826.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 432Guarneri Â V, Lenihan Â DJ, Valero Â V, Durand Â JB, Broglio Â K, Hess Â KR, et al. Â Long-term cardiac tolerability of trastuzumab in metastatic breast cancer: the M.D. Anderson Cancer Center experience. *J Clin Oncol* Â 2006;24:4107âFADSTS:4115.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 433Khoury Â K, Lynce Â F, Barac Â A, Geng Â X, Dang Â C, Yu Â AF, et al. Â Long-term follow-up assessment of cardiac safety in SAFE-HEaRt, a clinical trial evaluating the use of HER2-targeted therapies in patients with breast cancer and compromised heart function. *Breast Cancer Res Treat* Â 2021;185:863âFADSTS:868.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 434Mahmood Â SS, Fradley Â MG, Cohen J Â V, Nohria Â A, Reynolds Â KL, Heinzerling Â LM, et al. Â Myocarditis in patients treated with immune checkpoint inhibitors. *J Am Coll Cardiol* Â 2018;71:1755âFADSTS:1764.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 435Power Â JR, Alexandre Â J, Choudhary Â A, Ozbay Â B, Hayek Â S, Asnani Â A, et al. Â Electrocardiographic manifestations of immune checkpoint inhibitor myocarditis. *Circulation* Â 2021;144:1521âFADSTS:1523. Erratum in: *Circulation* 2021;144:e490.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 436Zhang Â L, Zlotoff Â DA, Awadalla Â M, Mahmood Â SS, Nohria Â A, Hassan Â MZO, et al. Â Major adverse cardiovascular events and the timing and dose of corticosteroids in immune checkpoint inhibitor-associated myocarditis. *Circulation* Â 2020;141:2031âFADSTS:2034.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 437Boughdad Â S, Latifyan Â S, Fenwick Â C, Bouchaab Â H, Suffiotti Â M, Moslehi Â JJ, et al. Â 68Ga-DOTATOC PET/CT to detect immune checkpoint inhibitor-related myocarditis. *J Immunother*

Cancer 2021;9:e003594.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

438Finke D, Heckmann MB, Herpel E, Katus HA, Haberkorn U, Leuschner F, et al. Early detection of checkpoint inhibitor-associated myocarditis using 68Ga-FAPI PET/CT. *Front Cardiovasc Med* 2021;8:614997.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

439Chen Y, Jia Y, Liu Q, Shen Y, Zhu H, Dong X, et al. Myocarditis related to immune checkpoint inhibitors treatment: two case reports and literature review. *Ann Palliat Med* 2021;10:8512â8517.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

440Palaskas NL, Segura A, Lelenwa L, Siddiqui BA, Subudhi SK, Lopez-Mattei J, et al. Immune checkpoint inhibitor myocarditis: elucidating the spectrum of disease through endomyocardial biopsy. *Eur J Heart Fail* 2021;23:1725â1735.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

441Brugada J, Katritsis DG, Arbelo E, Arribas F, Bax JJ, Blomstrom-Lundqvist C, et al. 2019 ESC Guidelines for the management of patients with supraventricular tachycardia. *Eur Heart J* 2020;41:655â720.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

442Zeppenfeld K, Tfelt-Hansen J, de Riva M, Winkel BG, Behr ER, Blom NA, et al. 2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death. *Eur Heart J* 2022;43:3997â4126.

[Crossref](#) [Search ADS](#) [PubMed](#)

443Gliksan M, Nielsen JC, Kronborg MB, Michowitz Y, Auricchio A, Barbash IM, et al. 2021 ESC Guidelines on cardiac pacing and cardiac resynchronization therapy. *Eur Heart J* 2021;42:3427â3520.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

444Adler Y, Charron P, Imazio M, Badano L, BarÃ³n-Esquivias G, Bogaert J, et al. 2015 ESC Guidelines for the diagnosis and management of pericardial diseases: the Task Force for the Diagnosis and Management of Pericardial Diseases of the European Society of Cardiology (ESC) Endorsed by: The European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* 2015;36:2921â2964.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

445Ammirati E, Frigerio M, Adler ED, Basso C, Birnie DH, Brambatti M, et al. Management of acute myocarditis and chronic inflammatory cardiomyopathy: an expert consensus document. *Circ Heart Fail* 2020;13:e007405.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

446Thuny F, Alexandre J, Salem JE, Mirabel M, Dolladille C, Cohen-Solal A, et al. Management of immune checkpoint inhibitorâinduced myocarditis: the French Working GroupâPlea for a pragmatic approach. *JACC CardioOncology* 2021;3:157â161.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

447Chung Â ES, Packer Â M, Lo Â KH, Fasanmade Â AA, Willerson Â JT. Randomized, double-blind, placebo-controlled, pilot trial of infliximab, a chimeric monoclonal antibody to tumor necrosis factor-Î±, in patients with moderate-to-severe heart failure: Results of the Anti-TNF Therapy Against Congestive Heart Failure (ATTACH) trial. *Circulation* Â 2003;107:3133â3140.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

448Cautela Â J, Zeriouh Â S, Gaubert Â M, Bonello Â L, Laine Â M, Peyrol Â M, et al. Â Intensified immunosuppressive therapy in patients with immune checkpoint inhibitor-induced myocarditis. *J Immunother Cancer* Â 2020;8:e001887.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

449Dolladille Â C, Ederhy Â S, Sassier Â M, Cautela Â J, Thuny Â F, Cohen Â AA, et al. Â Immune checkpoint inhibitor rechallenge after immune-related adverse events in patients with cancer. *JAMA Oncol* Â 2020;6:865â871.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

450Roth Â ME, Muluneh Â B, Jensen Â BC, Madamanchi Â C, Lee Â CB. Left ventricular dysfunction after treatment with ipilimumab for metastatic melanoma. *Am J Ther* Â 2016;23:e1925âe1928.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

451Weinstock Â C, Khozin Â S, Suzman Â D, Zhang Â L, Tang Â S, Wahby Â S, et al. Â U.S. Food and Drug Administration approval summary: atezolizumab for metastatic nonâsmall cell lung cancer. *Clin Cancer Res* Â 2017;23:4534â4539.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

452Ball Â S, Ghosh Â RK, Wongsasengsak Â S, Bandyopadhyay Â D, Ghosh Â GC, Aronow Â WS, et al. Â Cardiovascular toxicities of immune checkpoint inhibitors: JACC review topic of the week. *J Am Coll Cardiol* Â 2019;74:1714â1727.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

453Thavendiranathan Â P, Zhang Â L, Zafar Â A, Drobni Â ZD, Mahmood Â SS, Cabral Â M, et al. Â Myocardial T1 and T2 mapping by magnetic resonance in patients with immune checkpoint inhibitorâassociated myocarditis. *J Am Coll Cardiol* Â 2021;77:1503â1516.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

454Brahmer Â JR, Lacchetti Â C, Schneider Â BJ, Atkins Â MB, Brassil Â KJ, Caterino Â JM, et al. Â Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: American Society of Clinical Oncology clinical practice guideline. *J Clin Oncol* Â 2018;36:1714â1768.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

455Guha Â A, Addison Â D, Jain Â P, Gutierrez Â JM, Ghosh Â A, Roddie Â C, et al. Â Cardiovascular events associated with chimeric antigen receptor T cell therapy: cross-sectional FDA adverse events reporting system analysis: cardiovascular events with CAR-T therapy. *Biol Blood Marrow Transplant* Â 2020;26:2211â2216.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

456Kupari Â M, Volin Â L, Suokas Â A, Hekali Â P, Ruutu Â T. Cardiac involvement in bone marrow transplantation: serial changes in left ventricular size, mass and performance. *J Intern Med*

Â 1990;227:259â266.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

457Tonorezos Â ES, Stillwell Â EE, Calloway Â JJ, Glew Â T, Wessler Â JD, Rebolledo Â BJ, et al.
Â Arrhythmias in the setting of hematopoietic cell transplants. Bone Marrow Transplant
Â 2015;50:1212â1216.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

458Collet Â J-P, Thiele Â H, Barbato Â E, BarthÃ©lÃ©my Â O, Bauersachs Â J, Bhatt Â DL, et al.
Â 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without
persistent ST-segment elevation. Eur Heart J Â 2021;42:1289â1367.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

459Abo Â S, Ritchie Â D, Denehy Â L, Panek-Hudson Â Y, Irving Â L, Granger Â CL. A hospital and
home-based exercise program to address functional decline in people following allogeneic stem cell
transplantation. Support Care Cancer Â 2018;26:1727â1736.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

460Squires Â RW, Shultz Â AM, Herrmann Â J. Exercise training and cardiovascular health in cancer
patients. Curr Oncol Rep Â 2018;20:27.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

461Keen Â C, Skilbeck Â J, Ross Â H, Smith Â L, Collins Â K, Dixey Â J, et al. Â Is it feasible to
conduct a randomised controlled trial of pretransplant exercise (prehabilitation) for patients with multiple
myeloma awaiting autologous haematopoietic stem cell transplantation? Protocol for the PREeMPT
study. BMJ Open Â 2018;8:e021333.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

462Desai Â R, Desai Â A, Abbas Â SA, Patel Â U, Bansod Â S, Damarlapally Â N, et al. Â National
prevalence, trends and outcomes of takotsubo syndrome in hospitalizations with prior history of
mediastinal/intrathoracic cancer and radiation therapy. Int J Cardiol Â 2020;309:14â18.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

463Sattler Â K, El-Battrawy Â I, Lang Â S, Zhou Â X, Schramm Â K, TÃ¼rmen Â E, et al.
Â Prevalence of cancer in Takotsubo cardiomyopathy: short and long-term outcome. Int J Cardiol
Â 2017;238:159â165.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

464Omerovic Â E, Citro Â R, Bossone Â E, Redfors Â B, Backs Â J, Bruns Â B, et al. Â Pathophysiology
of Takotsubo syndromeâa joint scientific statement from the HFA TTS and Myocardial Function
Working Group of the ESCâPart 2: vascular pathophysiology, gender and sex hormones, genetics,
chronic cardiovascular problems and clinical implications. Eur J Heart Fail Â 2022;24:274â286.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

465Omerovic Â E, Citro Â R, Bossone Â E, Redfors Â B, Backs Â J, Bruns Â B, et al. Â Pathophysiology
of Takotsubo syndromeâa joint scientific statement from the HFA TTS Study Group and Myocardial
Function Working Group of the ESCâPart 1: overview and the central role for catecholamines and
sympathetic nervous system. Eur J Heart Fail Â 2022;24:257â273.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 466Couch Â LS, Fiedler Â J, Chick Â G, Clayton Â R, Dries Â E, Wienecke Â LM, et al. Â Circulating microRNAs predispose to Takotsubo syndrome following high-dose adrenaline exposure. *Cardiovasc Res* Â 2022;118:1758â1770.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 467Ghadri Â J-R, Wittstein Â IS, Prasad Â A, Sharkey Â S, Dote Â K, Akashi Â YJ, et al. Â International expert consensus document on Takotsubo syndrome (part II): diagnostic workup, outcome, and management. *Eur Heart J* Â 2018;39:2047â2062.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 468Ghadri Â J-R, Wittstein Â IS, Prasad Â A, Sharkey Â S, Dote Â K, Akashi Â YJ, et al. Â International expert consensus document on Takotsubo syndrome (part I): clinical characteristics, diagnostic criteria, and pathophysiology. *Eur Heart J* Â 2018;39:2032â2046.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 469Lyon Â AR, Bossone Â E, Schneider Â B, Sechtem Â U, Citro Â R, Underwood Â SR, et al. Â Current state of knowledge on Takotsubo syndrome: a Position Statement from the Taskforce on Takotsubo Syndrome of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* Â 2016;18:8â27.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 470Demers Â M, Krause Â DS, Schatzberg Â D, Martinod Â K, Voorhees Â JR, Fuchs Â TA, et al. Â Cancers predispose neutrophils to release extracellular DNA traps that contribute to cancer-associated thrombosis. *Proc Natl Acad Sci U S A* Â 2012;109:13076â13081.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 471Mrotzek Â SM, Lena Â A, Hadzibegovic Â S, Ludwig Â R, Al-Rashid Â F, Mahabadi Â AA, et al. Â Assessment of coronary artery disease during hospitalization for cancer treatment. *Clin Res Cardiol* Â 2021;110:200â210.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 472Nykl Â R, Fischer Â O, Vykoupil Â K, Taborsky Â M. A unique reason for coronary spasm causing temporary ST elevation myocardial infarction (inferior STEMI)âsystemic inflammatory response syndrome after use of pembrolizumab. *Arch Med Sci Atheroscler Dis* Â 2017;2:100â102.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 473Ferreira Â M, Pichon Â E, Carmier Â D, Bouquet Â E, Pageot Â C, Bejan-Angoulvant Â T, et al. Â Coronary toxicities of anti-PD-1 and anti-PD-L1 immunotherapies: a case report and review of the literature and international registries. *Target Oncol* Â 2018;13:509â515.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 474Iannaccone Â M, D'Ascenzo Â F, VadalÃ Â P, Wilton Â SB, Noussan Â P, Colombo Â F, et al. Â Prevalence and outcome of patients with cancer and acute coronary syndrome undergoing percutaneous coronary intervention: a BleeMACS substudy. *Eur Heart Journal Acute Cardiovasc Care* Â 2018;7:631â638.
[Google Scholar](#) [OpenURL Placeholder Text](#) Â
- 475Bharadwaj Â A, Potts Â J, Mohamed Â MO, Parwani Â P, Swamy Â P, Lopez-Mattei Â JC, et al. Â Acute myocardial infarction treatments and outcomes in 6.5 million patients with a current or historical

diagnosis of cancer in the USA. *Eur Heart J* 2020;41:2183â2193.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

476Velders Â MA, Boden Â H, Hofma Â SH, Osanto Â S, Van Der Hoeven Â BL, Heestermans Â AACM, et al. Â Outcome after ST elevation myocardial infarction in patients with cancer treated with primary percutaneous coronary intervention. *Am J Cardiol* 2013;112:1867â1872.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

477Ueki Â Y, VÃlgeli Â B, Karagiannis Â A, Zanchin Â T, Zanchin Â C, Rhyner Â D, et al. Â Ischemia and bleeding in cancer patients undergoing percutaneous coronary intervention. *JACC CardioOncology* 2019;1:145â155.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

478Potts Â JE, Iliescu Â CA, Lopez Mattei Â JC, Martinez Â SC, Holmvang Â L, Ludman Â P, et al. Â Percutaneous coronary intervention in cancer patients: a report of the prevalence and outcomes in the United States. *Eur Heart J* 2019;40:1790â1800.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

479Guddati Â AK, Joy Â PS, Kumar Â G. Analysis of outcomes of percutaneous coronary intervention in metastatic cancer patients with acute coronary syndrome over a 10-year period. *J Cancer Res Clin Oncol* 2016;142:471â479.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

480Pothineni Â NV, Shah Â NN, Rochlani Â Y, Saad Â M, Kovelamudi Â S, Marmagkiolis Â K, et al. Â Temporal trends and outcomes of acute myocardial infarction in patients with cancer. *Ann Transl Med* 2017;5:482.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

481Yusuf Â SW, Daraban Â N, Abbasi Â N, Lei Â X, Durand Â JB, Daher Â IN. Treatment and outcomes of acute coronary syndrome in the cancer population. *Clin Cardiol* 2012;35:443â450.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

482Gevaert Â SA, Halvorsen Â S, Sinnaeve Â PR, Sambola Â A, Gulati Â G, Lancellotti Â P, et al. Â Evaluation and management of cancer patients presenting with acute cardiovascular disease: a Consensus Document of the Acute CardioVascular Care (ACVC) association and the ESC council of Cardio-OncologyâPart 1: acute coronary syndromes and acute pericardial diseases. *Eur Heart J Acute Cardiovasc Care* 2021;10:947â959.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

483Mohamed Â MO, Van Spall Â HGC, Kontopantelis Â E, Alkhouli Â M, Barac Â A, Elgendy Â IY, et al. Â Effect of primary percutaneous coronary intervention on in-hospital outcomes among active cancer patients presenting with ST-elevation myocardial infarction: a propensity score matching analysis. *Eur Heart J Acute Cardiovasc Care* 2021;10:829â839.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

484Iliescu Â CA, Grines Â CL, Herrmann Â J, Yang Â EH, Cilingiroglu Â M, Charitakis Â K, et al. Â SCAI Expert consensus statement: Evaluation, management, and special considerations of cardio-oncology patients in the cardiac catheterization laboratory (endorsed by the Cardiological Society of India, and Sociedad Latino Americana de CardiologÃa Intervencionista). *Catheter Cardiovasc Interv*

Â 2016;87:E202âE223.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

485Iliescu Â CA, Cilingiroglu Â M, Giza Â DE, Rosales Â O, Lebeau Â J, Guerrero-Mantilla Â I, et al. Â Bringing on the lightâ in a complex clinical scenario: optical coherence tomographyâguided discontinuation of antiplatelet therapy in cancer patients with coronary artery disease (PROTECT-OCT registry). *Am Heart J* Â 2017;194:83â91.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

486Iliescu Â C, Balanescu Â DV, Donisan Â T, Giza Â DE, MuÃ±oz Gonzalez Â ED, Cilingiroglu Â M, et al. Â Safety of diagnostic and therapeutic cardiac catheterization in cancer patients with acute coronary syndrome and chronic thrombocytopenia. *Am J Cardiol* Â 2018;122:1465â1470.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

487Cianci Â G, Morelli Â MF, Cannita Â K, Morese Â R, Ricevuto Â E, Di Rocco Â ZC, et al. Â Prophylactic options in patients with 5-fluorouracil-associated cardiotoxicity. *Br J Cancer* Â 2003;88:1507â1509.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

488Ambrosy Â AP, Kunz Â PL, Fisher Â GA, Witteles Â RM. Capecitabine-induced chest pain relieved by diltiazem. *Am J Cardiol* Â 2012;110:1623â1626.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

489Akpek Â G, Hartshorn Â KL. Failure of oral nitrate and calcium channel blocker therapy to prevent 5-fluorouracil-related myocardial ischemia: a case report. *Cancer Chemother Pharmacol* Â 1999;43:157â161.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

490Chang Â HM, Okwuosa Â TM, Scarabelli Â T, Moudgil Â R, Yeh Â ETH. Cardiovascular complications of cancer therapy: best practices in diagnosis, prevention, and management: part 2. *J Am Coll Cardiol* Â 2017;70:2552â2565.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

491Campia Â U, Moslehi Â JJ, Amiri-Kordestani Â L, Barac Â A, Beckman Â JA, Chism Â DD, et al. Â Cardio-oncology: vascular and metabolic perspectives: a scientific statement from the American Heart Association. *Circulation* Â 2019;139:e579âe602.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

492Yeh Â ETH, Bickford Â CL. Cardiovascular complications of cancer therapy: incidence, pathogenesis, diagnosis, and management. *J Am Coll Cardiol* Â 2009;53:2231â2247.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

493Lenneman Â CG, Sawyer Â DB. Cardio-oncology: an update on cardiotoxicity of cancer-related treatment. *Circ Res* Â 2016;118:1008â1020.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

494Novo Â G, Di Lisi Â D, Bronte Â E, MacAione Â F, Accurso Â V, Badalamenti Â G, et al. Â Cardiovascular toxicity in cancer patients treated with tyrosine kinase inhibitors: a real-world single-center experience. *Oncology* Â 2020;98:445â451.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

495Bharadwaj ^ AS, Swamy ^ PM, Mamas ^ MA. Outcomes of percutaneous coronary interventions in cancer patients. *Expert Rev Cardiovasc Ther* ^ 2020;18:25^^{PADSTS}32.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

496Kwok ^ CS, Wong ^ CW, Kontopantelis ^ E, Barac ^ A, Brown ^ SA, Velagapudi ^ P, et al. ^ Percutaneous coronary intervention in patients with cancer and readmissions within 90 days for acute myocardial infarction and bleeding in the USA. *Eur Heart J* ^ 2021;42:1019^^{PADSTS}1034.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

497van Werkum ^ JW, Heestermaans ^ AA, Zomer ^ AC, Kelder ^ JC, Sutorp ^ MJ, Rensing ^ BJ, et al. ^ Predictors of coronary stent thrombosis. The Dutch Stent Thrombosis Registry. *J Am Coll Cardiol* ^ 2009;53:1399^^{PADSTS}1409.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

498Urban ^ P, Mehran ^ R, Collieran ^ R, Angiolillo ^ DJ, Byrne ^ RA, Capodanno ^ D, et al. ^ Defining high bleeding risk in patients undergoing percutaneous coronary intervention. *Circulation* ^ 2019;140:240^^{PADSTS}261.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

499Stewart ^ MH, Jahangir ^ E, Polin ^ NM. Valvular heart disease in cancer patients: etiology, diagnosis, and management. *Curr Treat Options Cardiovasc Med* ^ 2017;19:53.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

500Hull ^ MC, Morris ^ CG, Pepine ^ CJ, Mendenhall ^ NP. Valvular dysfunction and carotid, subclavian, and coronary artery disease in survivors of Hodgkin lymphoma treated with radiation therapy. *J Am Med Assoc* ^ 2003;290:2831^^{PADSTS}2837.

[Google Scholar](#) [Crossref](#) [Search ADS](#) ^

501Salz ^ T, Zabor ^ EC, de Nully Brown ^ P, Dalton ^ SO, Raghunathan ^ NJ, Matasar ^ MJ, et al. ^ Preexisting cardiovascular risk and subsequent heart failure among non-Hodgkin lymphoma survivors. *J Clin Oncol* ^ 2017;35:3837^^{PADSTS}3843.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

502Serrano ^ C, Cort^s ^ J, De Mattos-Arruda ^ L, Bellet ^ M, G^mez ^ P, Saura ^ C, et al. ^ Trastuzumab-related cardiotoxicity in the elderly: a role for cardiovascular risk factors. *Ann Oncol* ^ 2012;23:897^^{PADSTS}902.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

503Sato ^ A, Yoshihisa ^ A, Miyata-Tatsumi ^ M, Oikawa ^ M, Kobayashi ^ A, Ishida ^ T, et al. ^ Valvular heart disease as a possible predictor of trastuzumab-induced cardiotoxicity in patients with breast cancer. *Mol Clin Oncol* ^ 2019;10:37^^{PADSTS}42.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) ^

504Watanabe ^ Y, Kozuma ^ K, Hioki ^ H, Kawashima ^ H, Nara ^ Y, Kataoka ^ A, et al. ^ Comparison of results of transcatheter aortic valve implantation in patients with versus without active cancer. *Am J Cardiol* ^ 2016;118:572^^{PADSTS}577.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 505 Nagata H, Kanzaki R, Kanou T, Ose N, Funaki S, Shintani Y, et al. Two cases of lobectomy for lung cancer after transcatheter aortic valve implantation. *Surg Case Reports* 2018;4:139.
[Google Scholar](#) [OpenURL Placeholder Text](#)
- 506 Landes U, Iakobishvili Z, Vronsky D, Zusman O, Barsheshet A, Jaffe R, et al. Transcatheter aortic valve replacement in oncology patients with severe aortic stenosis. *JACC Cardiovasc Interv* 2019;12:786-788.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 507 Vahanian A, Beyersdorf F, Praz F, Milojevic M, Baldus S, Bauersachs J, et al. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J* 2022;43:561-632.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 508 Lpez-Fernndez T, Martn-García A, Roldn Rabadn I, Mitroi C, Mazn Ramos P, DÍez-Villanueva P, et al. Atrial fibrillation in active cancer patients: expert position paper and recommendations. *Rev Española Cardiol (English Ed)* 2019;72:749-759.
[Google Scholar](#) [Crossref](#) [Search ADS](#)
- 509 Farmakis D, Parissis J, Filippatos G. Insights into onco-cardiology: atrial fibrillation in cancer. *J Am Coll Cardiol* 2014;63:945-953.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 510 Yun JP, Choi EK, Do Han K, Park SH, Jung JH, Park SH, et al. Risk of atrial fibrillation according to cancer type: a nationwide population-based study. *JACC CardioOncology* 2021;3:221-232.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 511 Guha A, Fradley MG, Dent SF, Weintraub NL, Lustberg MB, Alonso A, et al. Incidence, risk factors, and mortality of atrial fibrillation in breast cancer: a SEER-Medicare analysis. *Eur Heart J* 2021;43:300-312.
[Google Scholar](#) [Crossref](#) [Search ADS](#)
- 512 Hu YF, Liu CJ, Chang PMH, Tsao HM, Lin YJ, Chang SL, et al. Incident thromboembolism and heart failure associated with new-onset atrial fibrillation in cancer patients. *Int J Cardiol* 2013;165:355-357.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 513 Mosarla RC, Vaduganathan M, Qamar A, Moslehi J, Piazza G, Giugliano RP. Anticoagulation strategies in patients with cancer: JACC review topic of the week. *J Am Coll Cardiol* 2019;73:1336-1349.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 514 Malavasi VL, Fantecchi E, Gianolio L, Pesce F, Longo G, Marietta M, et al. Atrial fibrillation in patients with active malignancy and use of anticoagulants: under-prescription but no adverse impact on all-cause mortality. *Eur J Intern Med* 2019;59:27-33.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 515Al-Kindi Â SG, Oliveira Â GH. Prevalence of preexisting cardiovascular disease in patients with different types of cancer the unmet need for onco-cardiology. *Mayo Clin Proc* Â 2016;91:81â83.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 516Alexandre Â J, Moslehi Â JJ, Bersell Â KR, Funck-Brentano Â C, Roden Â DM, Salem Â JE. Anticancer drug-induced cardiac rhythm disorders: current knowledge and basic underlying mechanisms. *Pharmacol Ther* Â 2018;189:89â103.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 517Boriani Â G, Corradini Â P, Cuneo Â A, Falanga Â A, FoÃ Â R, Gaidano Â G, et al. Â Practical management of ibrutinib in the real life: focus on atrial fibrillation and bleeding. *Hematol Oncol* Â 2018;36:624â632.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 518Tang Â CPS, Lip Â GYH, McCormack Â T, Lyon Â AR, Hillmen Â P, Iyengar Â S, et al. Â Management of cardiovascular complications of Bruton tyrosine kinase inhibitors. *Br J Haematol* Â 2022;196:70â78.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 519Pastori Â D, Marang Â A, Bisson Â A, Menichelli Â D, Herbert Â J, Lip Â GYH, et al. Â Thromboembolism, mortality, and bleeding in 2,435,541 atrial fibrillation patients with and without cancer: a nationwide cohort study. *Cancer* Â 2021;127:2122â2129.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 520Potpara Â TS, Lip Â GYH, Blomstrom-Lundqvist Â C, Boriani Â G, Van Gelder Â IC, Heidbuchel Â H, et al. Â The 4S-AF scheme (Stroke Risk; Symptoms; Severity of Burden; Substrate): a novel approach to in-depth characterization (rather than classification) of atrial fibrillation. *Thromb Haemost* Â 2021;121:270â278.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 521Boriani Â G, Bonini Â N, Albin Â A, Venturelli Â A, Imberti Â JF, Vitolo Â M. Cardioversion of recent-onset atrial fibrillation: current evidence, practical considerations, and controversies in a complex clinical scenario. *Kardiol Pol* Â 2020;78:1088â1098.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 522Kanmanthareddy Â A, Vallakati Â A, Reddy Yeruva Â M, Dixit Â S, Di Biase Â L, Mansour Â M, et al. Â Pulmonary vein isolation for atrial fibrillation in the postpneumonectomy population: a feasibility, safety, and outcomes study. *J Cardiovasc Electrophysiol* Â 2015;26:385â389.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 523Lip Â GYH, Banerjee Â A, Boriani Â G, En Chiang Â C, Fargo Â R, Freedman Â B, et al. Â Antithrombotic therapy for atrial fibrillation: CHEST guideline and expert panel report. *Chest* Â 2018;154:1121â1201.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 524Proietti Â M, Lane Â DA, Boriani Â G, Lip Â GYH. Stroke prevention, evaluation of bleeding risk, and anticoagulant treatment management in atrial fibrillation contemporary international guidelines. *Can J Cardiol* Â 2019;35:619â633.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

- 525Boriani Â G, Lee Â G, Parrini Â I, Lopez-Fernandez Â T, Lyon Â AR, Suter Â T, et al.
 Â Anticoagulation in patients with atrial fibrillation and active cancer: an international survey on patient management. *Eur J Prev Cardiol* Â 2021;28:611â^{PADSTS}621.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 526Dâ^{PADSGC}Souza Â M, Carlson Â N, FosbÃ,l Â E, Lamberts Â M, Smedegaard Â L, Nielsen Â D, et al.
 Â CHA₂DS₂-VASC score and risk of thromboembolism and bleeding in patients with atrial fibrillation and recent cancer. *Eur J Prev Cardiol* Â 2018;25:651â^{PADSTS}658.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 527Farmakis Â D. Anticoagulation for atrial fibrillation in active cancer: what the cardiologists think. *Eur J Prev Cardiol* Â 2021;28:608â^{PADSTS}610.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 528Cohen Â A, Donal Â E, Delgado Â V, Pepi Â M, Tsang Â T, Gerber Â B, et al. Â EACVI recommendations on cardiovascular imaging for the detection of embolic sources: endorsed by the Canadian Society of Echocardiography. *Eur Heart J Cardiovasc Imaging* Â 2021;22:E24â^{PADSTS}E57.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 529Schulman Â S, Anger Â SU, Bergqvist Â D, Eriksson Â B, Lassen Â MR, Fisher Â W. Definition of major bleeding in clinical investigations of antihemostatic medicinal products in surgical patients. *J Thromb Haemost* Â 2010;8:202â^{PADSTS}204.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 530Steffel Â J, Collins Â R, Antz Â M, Cornu Â P, Desteghe Â L, Haeusler Â KG, et al. Â 2021 European Heart Rhythm Association practical guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation. *Europace* Â 2021;23:1612â^{PADSTS}1676.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 531Chen Â ST, Hellkamp Â AS, Becker Â RC, Berkowitz Â SD, Breithardt Â G, Fox Â KAA, et al.
 Â Efficacy and safety of rivaroxaban vs. warfarin in patients with non-valvular atrial fibrillation and a history of cancer: Observations from ROCKET AF. *Eur Heart J Qual Care Clin Outcomes*
 Â 2019;5:145â^{PADSTS}152.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 532Melloni Â C, Dunning Â A, Granger Â CB, Thomas Â L, Khouri Â MG, Garcia Â DA, et al.
 Â Efficacy and safety of apixaban versus warfarin in patients with atrial fibrillation and a history of cancer: insights from the ARISTOTLE trial. *Am J Med* Â 2017;130:1440â^{PADSTS}1448.e1.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 533Fanola Â CL, Ruff Â CT, Murphy Â SA, Jin Â J, Duggal Â A, Babilonia Â NA, et al. Â Efficacy and safety of edoxaban in patients with active malignancy and atrial fibrillation: analysis of the engage AF-TIMI 48 trial. *J Am Heart Assoc* Â 2018;7:e008987.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 534Sawant Â AC, Kumar Â A, McCray Â W, Tetewsky Â S, Parone Â L, Sridhara Â S, et al. Â Superior safety of direct oral anticoagulants compared to warfarin in patients with atrial fibrillation and underlying cancer: a national Veterans Affairs database study. *J Geriatr Cardiol* Â 2019;16:706â^{PADSTS}709.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

- 535Shah Â S, Norby Â FL, Datta Â YH, Lutsey Â PL, MacLehose Â RF, Chen Â LY, et al.
Â Comparative effectiveness of direct oral anticoagulants and warfarin in patients with cancer and atrial fibrillation. *Blood Adv* Â 2018;2:200â^{PADSTS}209.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 536Mariani Â MV, Magnocavallo Â M, Straito Â M, Piro Â A, Severino Â P, Iannucci Â G, et al.
Â Direct oral anticoagulants versus vitamin K antagonists in patients with atrial fibrillation and cancer a meta-analysis. *J Thromb Thrombolysis* Â 2021;51:419â^{PADSTS}429.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 537Deitelzweig Â S, Keshishian Â AV, Zhang Â Y, Kang Â A, Dhamane Â AD, Luo Â X, et al.
Â Effectiveness and safety of oral anticoagulants among nonvalvular atrial fibrillation patients with active cancer. *JACC CardioOncology* Â 2021;3:411â^{PADSTS}424.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 538Lin Â YS, Kuan Â FC, Chao Â TF, Wu Â M, Chen Â SW, Chen Â MC, et al. Â Mortality associated with the use of non-vitamin K antagonist oral anticoagulants in cancer patients: dabigatran versus rivaroxaban. *Cancer Med* Â 2021;10:7079â^{PADSTS}7088.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 539Isogai Â T, Saad Â AM, Abushouk Â AI, Shekhar Â S, Kuroda Â S, Gad Â MM, et al. Â Procedural and short-term outcomes of percutaneous left atrial appendage closure in patients with cancer. *Am J Cardiol* Â 2021;141:154â^{PADSTS}157.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 540Boriani Â G, Fauchier Â L, Aguinaga Â L, Beattie Â JM, Blomstrom Lundqvist Â C, Cohen Â A, et al. Â European Heart Rhythm Association (EHRA) consensus document on management of arrhythmias and cardiac electronic devices in the critically ill and post-surgery patient, endorsed by Heart Rhythm Society (HRS), Asia Pacific Heart Rhythm Society (APHRS), Cardiac Arrhythmia Society of Southern Africa (CASSA), and Latin American Heart Rhythm Society (LAHRS). *Europace* Â 2019;21:7â^{PADSTS}8.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 541Butt Â JH, Olesen Â JB, Havers-Borgersen Â E, Gundlund Â A, Andersson Â C, Gislason Â GH, et al.
Â Risk of thromboembolism associated with atrial fibrillation following noncardiac surgery. *J Am Coll Cardiol* Â 2018;72:2027â^{PADSTS}2036.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 542Enriquez Â A, Biagi Â J, Redfearn Â D, Boles Â U, Kamel Â D, Ali Â FS, et al. Â Increased incidence of ventricular arrhythmias in patients with advanced cancer and implantable cardioverter-defibrillators. *JACC Clin Electrophysiol* Â 2017;3:50â^{PADSTS}56.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 543Roden Â DM. Predicting drug-induced QT prolongation and torsades de pointes. *J Physiol* Â 2016;594:2459â^{PADSTS}2468.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 544Al-Khatib Â SM, Stevenson Â WG, Ackerman Â MJ, Bryant Â WJ, Callans Â DJ, Curtis Â AB, et al.
Â 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: Executive summary: a report of the American College of

Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Heart Rhythm 2018;15:e190â€“e252.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

545Schwartz PJ, Moss AJ, Vincent GM, Crampton RS. Diagnostic criteria for the long QT syndrome. An update. Circulation 1993;88:782â€“784.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

546Tisdale JE, Chung MK, Campbell KB, Hammad M, Joglar JA, Leclerc J, et al. Drug-induced arrhythmias: a scientific statement from the American Heart Association. Circulation 2020;142:E214â€“E233.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

547AZCERT. [CredibleMeds.org](https://www.crediblemeds.org/) n.d.

548Coppola C, Rienzo A, Piscopo G, Barbieri A, Arra C, Maurea N. Management of QT prolongation induced by anti-cancer drugs: target therapy and old agents. Different algorithms for different drugs. Cancer Treat Rev 2018;63:135â€“143.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

549European Medicines Agency. ICH guideline E14/S7B on clinical and nonclinical evaluation of QT/QTc interval prolongation and proarrhythmic potentialâ€”questions & answers. Sci Med Heal 2020:EMA/CHMP/ICH/415588/2020.

[OpenURL Placeholder Text](#)

550Atallah-Yunes SA, Kadado AJ, Kaufman GP, Hernandez-Montfort J. Immune checkpoint inhibitor therapy and myocarditis: a systematic review of reported cases. J Cancer Res Clin Oncol 2019;145:1527â€“1557.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

551Cirne F, Zhou S, Kappel C, El-Kadi A, Barron CC, Ellis PM, et al. ALK inhibitor-induced bradycardia: a systematic review and meta-analysis. Lung Cancer 2021;161:9â€“17.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

552Hassen LJ, Lenihan DJ, Baliga RR. Hypertension in the cardio-oncology clinic. Heart Fail Clin 2019;15:487â€“495.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

553Szmít S, Jurczak W, Zaucha JM, Drozd-Sokołowska J, Spycharowicz W, Joks M, et al. Pre-existing arterial hypertension as a risk factor for early left ventricular systolic dysfunction following (R)-CHOP chemotherapy in patients with lymphoma. J Am Soc Hypertens 2014;8:791â€“799.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

554Di Lorenzo G, Autorino R, Bruni G, Cartent G, Ricevuto E, Tudini M, et al. Cardiovascular toxicity following sunitinib therapy in metastatic renal cell carcinoma: a multicenter analysis. Ann Oncol 2009;20:1535â€“1542.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 555 Penttilä P, Rautiola J, Poussa T, Peltola K, Bono P. Angiotensin inhibitors as treatment of sunitinib/pazopanib-induced hypertension in metastatic renal cell carcinoma. *Clin Genitourin Cancer* 2017;15:384â€390.e3.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 556 Izzedine H, Derosa L, Le Teuff G, Albiges L, Escudier B. Hypertension and angiotensin system inhibitors: impact on outcome in sunitinib-treated patients for metastatic renal cell carcinoma. *Ann Oncol* 2015;26:1128â€1133.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 557 McKay RR, Rodriguez GE, Lin X, Kaymakcalan MD, Hamnvik OPR, Sabbisetti VS, et al. Angiotensin system inhibitors and survival outcomes in patients with metastatic renal cell carcinoma. *Clin Cancer Res* 2015;21:2471â€2479.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 558 Khorana AA, Francis CW, Culakova E, Kuderer NM, Lyman GH. Thromboembolism is a leading cause of death in cancer patients receiving outpatient chemotherapy. *J Thromb Haemost* 2007;5:632â€634.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 559 Puurunen MK, Gona PN, Larson MG, Murabito JM, Magnani JW, O'Donnell CJ. Epidemiology of venous thromboembolism in the Framingham Heart Study. *Thromb Res* 2016;145:27â€33.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 560 Walker AJ, Card TR, West J, Crooks C, Grainge MJ. Incidence of venous thromboembolism in patients with cancerâ€a cohort study using linked United Kingdom databases. *Eur J Cancer* 2013;49:1404â€1413.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 561 Lyman GH. Venous thromboembolism in the patient with cancer: focus on burden of disease and benefits of thromboprophylaxis. *Cancer* 2011;117:1334â€1349.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 562 Abdulla A, Davis WM, Ratnaweera N, Szefer E, Ballantyne Scott B, Lee AYY. A meta-analysis of case fatality rates of recurrent venous thromboembolism and major bleeding in patients with cancer. *Thromb Haemost* 2020;120:702â€713.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)
- 563 Mulder FI, Carrier M, van Doormaal F, Robin P, Otten HM, Salaun PY, et al. Risk scores for occult cancer in patients with unprovoked venous thromboembolism: results from an individual patient data meta-analysis. *J Thromb Haemost* 2020;18:2622â€2628.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 564 Khorana AA, Connolly GC. Assessing risk of venous thromboembolism in the patient with cancer. *J Clin Oncol* 2009;27:4839â€4847.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

565Li X, Hu Y, Lin P, Zhang J, Tang Y, Yi Q, et al. Comparison of different clinical prognostic scores in patients with pulmonary embolism and active cancer. *Thromb Haemost* 2021;121:834-844.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

566Konstantinides SV, Meyer G, Bueno H, Gali   N, Gibbs JSR, Agno W, et al. 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). *Eur Heart J* 2020;41:543-603.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

567Mazzolai L, Agno W, Alatri A, Bauersachs R, Becattini C, Brodmann M, et al. A Second consensus document on diagnosis and management of acute deep vein thrombosis: updated document elaborated by the ESC Working Group on aorta and peripheral vascular diseases and the ESC Working Group on pulmonary circulation and right ventricular function. *Eur J Prev Cardiol* 2022;29:1248-1263.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

568Navi BB, Reiner AS, Kamel H, Iadecola C, Okin PM, Elkind MS V, et al. Risk of arterial thromboembolism in patients with cancer. *J Am Coll Cardiol* 2017;70:926-938.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

569Mukai M, Oka T. Mechanism and management of cancer-associated thrombosis. *J Cardiol* 2018;72:89-93.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

570Armstrong GT, Plana JC, Zhang N, Srivastava D, Green DM, Ness KK, et al. Screening adult survivors of childhood cancer for cardiomyopathy: comparison of echocardiography and cardiac magnetic resonance imaging. *J Clin Oncol* 2012;30:2876-2884.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

571Hooks M, Okasha O, Velangi PS, Nijjar PS, Farzaneh-Far A, Shenoy C. Left ventricular thrombus on cardiovascular magnetic resonance imaging in non-ischaemic cardiomyopathy. *Eur Heart J Cardiovasc Imaging* 2020. doi:

[10.1093/ehjci/jeaa244](https://doi.org/10.1093/ehjci/jeaa244)

. Online ahead of print 7 October 2020.

[Google Scholar](#) [OpenURL Placeholder Text](#) [Crossref](#)

572Lee AYY, Levine MN, Baker RI, Bowden C, Kakkar AK, Prins M, et al. Low-molecular-weight heparin versus a coumarin for the prevention of recurrent venous thromboembolism in patients with cancer. *N Engl J Med* 2003;349:146-153.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

573Hull RD, Pineo GF, Brant RF, Mah AF, Burke N, Dear R, et al. Long-term low-molecular-weight heparin versus usual care in proximal-vein thrombosis patients with cancer. *Am J Med* 2006;119:1062-1072.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

574Deitcher SR, Kessler CM, Merli G, Rigas JR, Lyons RM, Fareed J. Secondary prevention of venous thromboembolic events in patients with active cancer: enoxaparin alone versus

initial enoxaparin followed by warfarin for a 180-day period. Clin Appl Thromb

Â 2006;12:389â^{PADSTS}396.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

575Meyer Â G, Marjanovic Â Z, Valcke Â J, Lorcerie Â B, Gruel Â Y, Solal-Celigny Â P, et al.

Â Comparison of low-molecular-weight heparin and warfarin for the secondary prevention of venous thromboembolism in patients with cancer: a randomized controlled study. Arch Intern Med

Â 2002;162:1729â^{PADSTS}1735.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

576Lee Â AYY, Kamphuisen Â PW, Meyer Â G, Bauersachs Â R, Janas Â MS, Jarner Â MF, et al.

Â Tinzaparin vs warfarin for treatment of acute venous thromboembolism in patients with active cancer: a randomized clinical trial. JAMA J Am Med Assoc Â 2015;314:677â^{PADSTS}686.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

577Lee Â YJ, Park Â JK, Uhm Â JS, Kim Â JY, Pak Â HN, Lee Â MH, et al. Â Bleeding risk and major adverse events in patients with cancer on oral anticoagulation therapy. Int J Cardiol

Â 2016;203:372â^{PADSTS}378.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

578Raskob Â GE, van Es Â N, Verhamme Â P, Carrier Â M, Di Nisio Â M, Garcia Â D, et al.

Â Edoxaban for the treatment of cancer-associated venous thromboembolism. N Engl J Med Â 2018;378:615â^{PADSTS}624.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

579Young Â AM, Marshall Â A, Thirlwall Â J, Chapman Â O, Lokare Â A, Hill Â C, et al. Â Comparison of an oral factor Xa inhibitor with low molecular weight heparin in patients with cancer with venous thromboembolism: results of a randomized trial (SELECT-D). J Clin Oncol Â 2018;36:2017â^{PADSTS}2023.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

580McBane Â II Â RD, Wysokinski Â WE, Le-Rademacher Â JG, Zemla Â T, Ashrani Â A, Tafur Â A, et al. Â Apixaban and dalteparin in active malignancy-associated venous thromboembolism: the ADAM VTE trial. J Thromb Haemost Â 2020;18:411â^{PADSTS}421.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

581Agnelli Â G, Becattini Â C, Meyer Â G, Muñoz Â A, Huisman M Â V, Connors Â JM, et al.

Â Apixaban for the treatment of venous thromboembolism associated with cancer. N Engl J Med Â 2020;382:1599â^{PADSTS}1607.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

582Ay Â C, Beyer-Westendorf Â J, Pabinger Â I. Treatment of cancer-associated venous thromboembolism in the age of direct oral anticoagulants. Ann Oncol Â 2019;30:897â^{PADSTS}907.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

583Ageno Â W, Vedovati Â MC, Cohen Â A, Huisman Â M, Bauersachs Â R, Gussoni Â G, et al.

Â Bleeding with apixaban and dalteparin in patients with cancer-associated venous thromboembolism: results from the Caravaggio study. Thromb Haemost Â 2021;121:616â^{PADSTS}624.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

584Cohen A, Keshishian A, Lee T, Wygant G, Rosenblatt L, Hlavacek P, et al.

Effectiveness and safety of apixaban, low-molecular-weight heparin, and warfarin among venous thromboembolism patients with active cancer: a US claims data analysis. *Thromb Haemost* 2021;121:383â395.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

585Giustozzi M, Agnelli G, Del Toro-Cervera J, Klok FA, Rosovsky RP, Martin AC, et al.

Direct oral anticoagulants for the treatment of acute venous thromboembolism associated with cancer: a systematic review and meta-analysis. *Thromb Haemost* 2020;120:1128â1136.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)

586Sabatino J, De Rosa S, Polimeni A, Sorrentino S, Indolfi C. Direct oral anticoagulants in

patients with active cancer: a systematic review and meta-analysis. *JACC CardioOncology* 2020;2:428â440.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

587Kraaijpoel N, Carrier M. How I treat cancer-associated venous thromboembolism. *Blood*

2019;133:291â298.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

588Den Exter PL, Hooijer J, Dekkers OM, Huisman M V. Risk of recurrent venous

thromboembolism and mortality in patients with cancer incidentally diagnosed with pulmonary embolism: a comparison with symptomatic patients. *J Clin Oncol* 2011;29:2405â2409.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

589Jara-Palomares L, Solier-Lopez A, Elias-Hernandez T, Asensio-Cruz M, Blasco-Esquivas A

I, Marin-Barrera L, et al. Tinzaparin in cancer associated thrombosis beyond 6 months: TiCAT study. *Thromb Res* 2017;157:90â96.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

590Francis CW, Kessler CM, Goldhaber SZ, Kovacs MJ, Monreal M, Huisman M V, et al.

Treatment of venous thromboembolism in cancer patients with dalteparin for up to 12 months: the DALTECAN Study. *J Thromb Haemost* 2015;13:1028â1035.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

591Otto CM, Nishimura RA, Bonow RO, Carabello BA, Erwin JP, Gentile F, et al. 2020

ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol* 2021;77:e25âe197.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

592Riondino S, Ferroni P, Del Monte G, Formica V, Guadagni F, Roselli M. Venous

thromboembolism in cancer patients on simultaneous and palliative care. *Cancers (Basel)* 2020;12:1167.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

593Xin Z, Liu F, Du Y, Mao F, Wang X, Xu P, et al. Primary prophylaxis for venous

thromboembolism in ambulatory cancer patients: a systematic review and network meta-analysis. *Ann*

Cardiothorac Surg Â 2020;9:2970â2981.

[Google Scholar](#) [OpenURL Placeholder Text](#) Â

594Farge Â D, Frere Â C, Connors Â JM, Ay Â C, Khorana Â AA, Munoz Â A, et al. Â 2019 international clinical practice guidelines for the treatment and prophylaxis of venous thromboembolism in patients with cancer. *Lancet Oncol* Â 2019;20:e566âe581.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

595Bergqvist Â D, Agnelli Â G, Cohen Â AT, Eldor Â A, Nilsson Â PE, Le Moigne-Amrani Â A, et al. Â Duration of prophylaxis against venous thromboembolism with enoxaparin after surgery for cancer. *N Engl J Med* Â 2002;346:975â980.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

596Khorana Â A, Kuderer Â NM, Culakova Â E, Lyman Â GH, Francis Â CW. Development and validation of a predictive model for chemotherapy-associated thrombosis. *Blood* Â 2008;111:4902â4907.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

597Gerotziafas Â GT, Taher Â A, Abdel-Razeq Â H, AboElnazar Â E, Spyropoulos Â AC, El Shemmari Â S, et al. Â A predictive score for thrombosis associated with breast, colorectal, lung, or ovarian cancer: the prospective COMPASSÂ cancer-associated thrombosis study. *Oncologist* Â 2017;22:1222â1231.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

598Agnelli Â G, Gussoni Â G, Bianchini Â C, Verso Â M, MandalÃ Â M, Cavanna Â L, et al. Â Nadroparin for the prevention of thromboembolic events in ambulatory patients with metastatic or locally advanced solid cancer receiving chemotherapy: a randomised, placebo-controlled, double-blind study. *Lancet Oncol* Â 2009;10:943â949.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

599Agnelli Â G, George Â DJ, Kakkar Â AK, Fisher Â W, Lassen Â MR, Mismetti Â P, et al. Â Semuloparin for thromboprophylaxis in patients receiving chemotherapy for cancer. *N Engl J Med* Â 2012;366:601â609.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

600Di Nisio Â M, Porreca Â E, Candeloro Â M, De Tursi Â M, Russi Â I, Rutjes Â AWSS. Primary prophylaxis for venous thromboembolism in ambulatory cancer patients receiving chemotherapy. *Cochrane Database Syst Rev* Â 2016;12:CD008500.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

601Carrier Â M, Abou-Nassar Â K, Mallick Â R, Tagalakis Â V, Shivakumar Â S, Schattner Â A, et al. Â Apixaban to prevent venous thromboembolism in patients with cancer. *N Engl J Med* Â 2019;380:711â719.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

602Khorana Â AA, Soff Â GA, Kakkar Â AK, Vadhan-Raj Â S, Riess Â H, Wun Â T, et al. Â Rivaroxaban for thromboprophylaxis in high-risk ambulatory patients with cancer. *N Engl J Med* Â 2019;380:720â728.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 603Angelini Â DE, Radivoyevitch Â T, McCrae Â KR, Khorana Â AA. Bleeding incidence and risk factors among cancer patients treated with anticoagulation. *Am J Hematol* Â 2019;94:780â785.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 604Menapace Â LA, McCrae Â KR, Khorana Â AA. Predictors of recurrent venous thromboembolism and bleeding on anticoagulation. *Thromb Res* Â 2016;140:S93âS98.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 605Roule Â V, Verdier Â L, Blanchart Â K, Ardouin Â P, Lemaitre Â A, Bignon Â M, et al. Â Systematic review and meta-analysis of the prognostic impact of cancer among patients with acute coronary syndrome and/or percutaneous coronary intervention. *BMC Cardiovasc Disord* Â 2020;20:38.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 606Freedberg Â DE, Kim Â LS, Yang Â YX. The risks and benefits of long-term use of proton pump inhibitors: expert review and best practice advice from the American Gastroenterological Association. *Gastroenterology* Â 2017;152:706â715.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 607Valgimigli Â M, Bueno Â H, Byrne Â RA, Collet Â JP, Costa Â F, Jeppsson Â A, et al. Â 2017 ESC focused update on dual antiplatelet therapy in coronary artery disease developed in collaboration with EACTS. *Eur Heart J* Â 2018;39:213â260.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 608Baran Â DA, Grines Â CL, Bailey Â S, Burkhoff Â D, Hall Â SA, Henry Â TD, et al. Â SCAI clinical expert consensus statement on the classification of cardiogenic shock: This document was endorsed by the American College of Cardiology (ACC), the American Heart Association (AHA), the Society of Critical Care Medicine (SCCM), and the Society of Thoracic Surgeons (STS) in April 2019. *Catheter Cardiovasc Interv* Â 2019;94:29â37.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 609Schiffer Â CA, Bohlke Â K, Anderson Â KC. Platelet transfusion for patients with cancer: American Society of Clinical Oncology Clinical Practice Guideline Update Summary. *J Oncol Pract* Â 2018;14:129â133.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 610Al-Samkari Â H, Connors Â JM. Managing the competing risks of thrombosis, bleeding, and anticoagulation in patients with malignancy. *Blood Adv* Â 2019;3:3770â3779.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 611Parr Â SK, Liang Â J, Schadler Â KL, Gilchrist Â SC, Steele Â CC, Ade Â CJ. Anticancer therapy-related increases in arterial stiffness: a systematic review and meta-analysis. *J Am Heart Assoc* Â 2020;9:e015598.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 612Gambichler Â T, Strutzmann Â S, Tannapfel Â A, Susok Â L. Paraneoplastic acral vascular syndrome in a patient with metastatic melanoma under immune checkpoint blockade. *BMC Cancer* Â 2017;17:327.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 613Aichberger Â KJ, Herndlhofer Â S, Schernthaner Â GH, Schillinger Â M, Mitterbauer-Hohendanner Â G, Sillaber Â C, et al. Â Progressive peripheral arterial occlusive disease and other vascular events during nilotinib therapy in CML. *Am J Hematol* Â 2011;86:533â539.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 614Dorer Â DJ, Knickerbocker Â RK, Baccarani Â M, Cortes Â JE, Hochhaus Â A, Talpaz Â M, et al. Â Impact of dose intensity of ponatinib on selected adverse events: multivariate analyses from a pooled population of clinical trial patients. *Leuk Res* Â 2016;48:84â91.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 615Valent Â P, Hadzijusufovic Â E, Schernthaner Â GH, Wolf Â D, Rea Â D, Le Coutre Â P. Vascular safety issues in CML patients treated with BCR/ABL1 kinase inhibitors. *Blood* Â 2015;125:901â906.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 616Montani Â D, Bergot Â E, GÃ¼nther Â S, Savale Â L, Bergeron Â A, Bourdin Â A, et al. Â Pulmonary arterial hypertension in patients treated by dasatinib. *Circulation* Â 2012;125:2128â2137.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 617Ranchoux Â B, GÃ¼nther Â S, Quarck Â R, Chaumais Â MC, DorfmanÃ¼ller Â P, Antigny Â F, et al. Â Chemotherapy-induced pulmonary hypertension: role of alkylating agents. *Am J Pathol* Â 2015;185:356â371.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 618Jevnikar Â M, Montani Â D, Savale Â L, Seferian Â A, Jutant Â EM, Boucly Â A, et al. Â Chronic thromboembolic pulmonary hypertension and totally implantable central venous access systems. *Eur Respir J* Â 2021;57:2002208.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 619Price Â LC, Seckl Â MJ, DorfmanÃ¼ller Â P, Wort Â SJ. Tumoral pulmonary hypertension. *Eur Respir Rev* Â 2019;28:180065.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 620Humbert Â M, Kovacs Â G, Hoeper Â MM, Badagliacca Â R, Berger Â RMF, Brida Â M, et al. Â 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J* Â 2022;43:3618â3731.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 621Kantarjian Â H, Shah Â NP, Hochhaus Â A, Cortes Â J, Shah Â S, Ayala Â M, et al. Â Dasatinib versus imatinib in newly diagnosed chronic-phase chronic myeloid leukemia. *N Engl J Med* Â 2010;362:2260â2270.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 622Imazio Â M, Colopi Â M, De Ferrari Â GM. Pericardial diseases in patients with cancer: Contemporary prevalence, management and outcomes. *Heart* Â 2020;106:569â574.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

623Kim Â SR, Kim Â EK, Cho Â J, Chang Â SA, Park Â SJ, Lee Â SC, et al. Â Effect of anti-inflammatory drugs on clinical outcomes in patients with malignant pericardial effusion. *J Am Coll Cardiol* Â 2020;76:1551â1561.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

624Gong Â J, Drobni Â ZD, Zafar Â A, Quinaglia Â T, Hartmann Â S, Gilman Â HK, et al. Â Pericardial disease in patients treated with immune checkpoint inhibitors. *J Immunother Cancer* Â 2021;9:e002771.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

625Inno Â A, Maurea Â N, Metro Â G, Carbone Â A, Russo Â A, Gori Â S. Immune checkpoint inhibitors-associated pericardial disease: a systematic review of case reports. *Cancer Immunol Immunother* Â 2021;70:3041â3053.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

626Sánchez-Enrique Â C, Nuñez-Gil Â IJ, Viana-Tejedor Â A, De Agustín Â A, Vivas Â D, Palacios-Rubio Â J, et al. Â Cause and long-term outcome of cardiac tamponade. *Am J Cardiol* Â 2016;117:664â669.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

627Saab Â J, Hoda Â RS, Narula Â N, Hoda Â SA, Geraghty Â BE, Nasar Â A, et al. Â Diagnostic yield of cytopathology in evaluating pericardial effusions: clinicopathologic analysis of 419 specimens. *Cancer Cytopathol* Â 2017;125:128â137.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

628Patel Â N, Rafique Â AM, Eshaghian Â S, Mendoza Â F, Biner Â S, Cercek Â B, et al. Â Retrospective comparison of outcomes, diagnostic value, and complications of percutaneous prolonged drainage versus surgical pericardiotomy of pericardial effusion associated with malignancy. *Am J Cardiol* Â 2013;112:1235â1239.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

629Palaskas Â N, Morgan Â J, Daigle Â T, Banchs Â J, Durand Â JB, Hong Â D, et al. Â Targeted cancer therapies with pericardial effusions requiring pericardiocentesis focusing on immune checkpoint inhibitors. *Am J Cardiol* Â 2019;123:1351â1357.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

630Shaheen Â S, Mirshahidi Â H, Nagaraj Â G, Hsueh Â CT. Conservative management of nivolumab-induced pericardial effusion: a case report and review of literature. *Exp Hematol Oncol* Â 2018;7:11.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

631Dixon Â SB, Howell Â CR, Lu Â L, Plana Â JC, Joshi Â VM, Luepker R Â V, et al. Â Cardiac biomarkers and association with subsequent cardiomyopathy and mortality among adult survivors of childhood cancer: a report from the St. Jude Lifetime Cohort. *Cancer* Â 2021;127:458â466.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

632Hershman Â DL, Till Â C, Shen Â S, Wright Â JD, Ramsey Â SD, Barlow Â WE, et al. Â Association of cardiovascular risk factors with cardiac events and survival outcomes among patients with breast cancer enrolled in SWOG clinical trials. *J Clin Oncol* Â 2018;36:2710â2717.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 633Nolan Â MT, Marwick Â TH, Plana Â JC, Li Â Z, Ness Â KK, Joshi Â VM, et al. Â Effect of traditional heart failure risk factors on myocardial dysfunction in adult survivors of childhood cancer. *JACC Cardiovasc Imaging* Â 2018;11:1202â^{PADSTS}1203.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 634Cho Â H, Lee Â S, Sim Â SH, Park Â IH, Lee Â KS, Kwak Â MH, et al. Â Cumulative incidence of chemotherapy-induced cardiotoxicity during a 2-year follow-up period in breast cancer patients. *Breast Cancer Res Treat* Â 2020;182:333â^{PADSTS}343.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 635Smarz Â K, Jaxa-Chamiec Â T, Chwyczko Â T, GÂ^{SPH}Â³wczyÂ^{IND}ska Â R, Jegier Â A, Niedozytko Â P, et al. Â Cardiopulmonary exercise testing in adult cardiology: expert opinion of the Working Group of Cardiac Rehabilitation and Exercise Physiology of the Polish Cardiac Society. *Kardiologia Pol* Â 2019;77:730â^{PADSTS}756.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 636Scott Â JM, Zabor Â EC, Schwitzer Â E, Koelwyn Â GJ, Adams Â SC, Nilsen Â TS, et al. Â Efficacy of exercise therapy on cardiorespiratory fitness in patients with cancer: a systematic review and meta-analysis. *J Clin Oncol* Â 2018;36:2297â^{PADSTS}2304.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 637Sasso Â JP, Eves Â ND, Christensen Â JF, Koelwyn Â GJ, Scott Â J, Jones Â LW. A framework for prescription in exercise-oncology research. *J Cachexia Sarcopenia Muscle* Â 2015;6:115â^{PADSTS}124.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 638Wallen Â MP, Hennessy Â D, Brown Â S, Evans Â L, Rawstorn Â JC, Wong Shee Â A, et al. Â High-intensity interval training improves cardiorespiratory fitness in cancer patients and survivors: a meta-analysis. *Eur J Cancer Care (Engl)* Â 2020;29:e13267.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 639Lee Â K, Tripathy Â D, Demark-Wahnefried Â W, Courneya Â KS, Sami Â N, Bernstein Â L, et al. Â Effect of aerobic and resistance exercise intervention on cardiovascular disease risk in women with early-stage breast cancer: a randomized clinical trial. *JAMA Oncol* Â 2019;5:710â^{PADSTS}714.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 640Adams Â SC, Delorey Â DS, Davenport Â MH, Fairey Â AS, North Â S, Courneya Â KS. Effects of high-intensity interval training on fatigue and quality of life in testicular cancer survivors. *Br J Cancer* Â 2018;118:1313â^{PADSTS}1321.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 641Mijwel Â S, Jervaeus Â A, Bolam Â KA, Norrbom Â J, Bergh Â J, Rundqvist Â H, et al. Â High-intensity exercise during chemotherapy induces beneficial effects 12 months into breast cancer survivorship. *J Cancer Surviv* Â 2019;13:244â^{PADSTS}256.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 642Marriott Â CFS, Petrella Â AFM, Marriott Â ECS, Boa Sorte Silva Â NC, Petrella Â RJ. Â High-intensity interval training in older adults: a scoping review. *Sport Med - Open* Â 2021;7:49.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

- 643Barac A, Murtagh G, Carver JR, Chen MH, Freeman AM, Herrmann J, et al. Cardiovascular health of patients with cancer and cancer survivors: a roadmap to the next level. *J Am Coll Cardiol* 2015;65:2739-2746.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 644Chu TF, Rupnick MA, Kerkela R, Dallabrida SM, Zurakowski D, Nguyen L, et al. Cardiotoxicity associated with tyrosine kinase inhibitor sunitinib. *Lancet* 2007;370:2011-2019.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 645Khakoo AY, Kassiotis CM, Tannir N, Plana JC, Halushka M, Bickford C, et al. Heart failure associated with sunitinib malate: a multitargeted receptor tyrosine kinase inhibitor. *Cancer* 2008;112:2500-2508.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 646Kim PY, Irizarry-Caro JA, Ramesh T, Iliescu C, Lopez-Mattei JC. How to diagnose and manage QT prolongation in cancer patients. *JACC CardioOncology* 2021;3:145-149.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 647Gatta G, Botta L, Rossi S, Aareleid T, Bielska-Lasota M, Clavel J, et al. Childhood cancer survival in Europe 1999-2007: results of EUROCare-5, a population-based study. *Lancet Oncol* 2014;15:35-47.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 648Geenen MM, Cardous-Ubbink MC, Kremer LCM, Van Den Bos C, Van Der Pal HJH, Heinen RC, et al. Medical assessment of adverse health outcomes in long-term survivors of childhood cancer. *J Am Med Assoc* 2007;297:2705-2715.
[Google Scholar](#) [Crossref](#) [Search ADS](#)
- 649Armenian SH, Armstrong GT, Aune G, Chow EJ, Ehrhardt MJ, Ky B, et al. Cardiovascular disease in survivors of childhood cancer: insights into epidemiology, pathophysiology, and prevention. *J Clin Oncol* 2018;36:2135-2144.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 650Fidler MM, Reulen RC, Henson K, Kelly J, Cutter D, Levitt GA, et al. Population-based long-term cardiac-specific mortality among 34 489 five-year survivors of childhood cancer in Great Britain. *Circulation* 2017;135:951-963.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 651Kremer LCM, van Dalen EC, Offringa M, VoÅ»te PA. Frequency and risk factors of anthracycline-induced clinical heart failure in children: a systematic review. *Ann Oncol* 2002;13:503-512.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 652Bates JE, Howell RM, Liu Q, Yasui Y, Mulrooney DA, Dhakal S, et al. Therapy-related cardiac risk in childhood cancer survivors: an analysis of the childhood cancer survivor study. *J Clin Oncol* 2019;37:1090-1101.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

- 653van Dalen EC, Mulder RL, Suh E, Ehrhardt MJ, Aune GJ, Bardi E, et al. Coronary artery disease surveillance among childhood, adolescent and young adult cancer survivors: a systematic review and recommendations from the International Late Effects of Childhood Cancer Guideline Harmonization Group. *Eur J Cancer* 2021;156:127â137.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 654Leerink JM, van der Pal HJH, Kremer LCM, Feijen EAM, Meregalli PG, Pourier MS, et al. Refining the 10-year prediction of left ventricular systolic dysfunction in long-term survivors of childhood cancer. *JACC CardioOncology* 2021;3:62â72.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 655Armstrong GT, Oeffinger KC, Chen Y, Kawashima T, Yasui Y, Leisenring W, et al. Modifiable risk factors and major cardiac events among adult survivors of childhood cancer. *J Clin Oncol* 2013;31:3673â3680.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 656Armenian SH, Sun CL, Vase T, Ness KK, Blum E, Francisco L, et al. Cardiovascular risk factors in hematopoietic cell transplantation survivors: role in development of subsequent cardiovascular disease. *Blood* 2012;120:4505â4512.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 657Carpenter K, Scavotto M, McGovern A, Ma C, Kenney LB, Mack JW, et al. Early parental knowledge of late effect risks in children with cancer. *Pediatr Blood Cancer* 2022;69:e29473.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 658Herrmann J. From trends to transformation: where cardio-oncology is to make a difference. *Eur Heart J* 2019;40:3898â3900.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 659Sturgeon KM, Deng L, Bluethmann SM, Zhou S, Trifiletti DM, Jiang C, et al. A population-based study of cardiovascular disease mortality risk in US cancer patients. *Eur Heart J* 2019;40:3889â3897.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 660Stoltzfus KC, Zhang Y, Sturgeon K, Sinoway LI, Trifiletti DM, Chinchilli VM, et al. Fatal heart disease among cancer patients. *Nat Commun* 2020;11:2011.
[Google Scholar](#) [OpenURL Placeholder Text](#)
- 661Banke A, Fosbøl EL, Møller JE, Gislason GH, Andersen M, Bernsdorf M, et al. Long-term effect of epirubicin on incidence of heart failure in women with breast cancer: insight from a randomized clinical trial. *Eur J Heart Fail* 2018;20:1447â1453.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 662Jacobse JN, Steggink LC, Sonke GS, Schaapveld M, Hummel YM, Steenbruggen TG, et al. Myocardial dysfunction in long-term breast cancer survivors treated at ages 40â50 years. *Eur J Heart Fail* 2020;22:338â346.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

663Boyne Â DJ, Mickle Â AT, Brenner Â DR, Friedenreich Â CM, Cheung Â WY, Tang Â KL, et al. Â Long-term risk of cardiovascular mortality in lymphoma survivors: a systematic review and meta-analysis. *Cancer Med* Â 2018;7:4801â4813.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

664Darby Â SC, McGale Â P, Taylor Â CW, Peto Â R. Long-term mortality from heart disease and lung cancer after radiotherapy for early breast cancer: prospective cohort study of about 300 000 women in US SEER cancer registries. *Lancet Oncol* Â 2005;6:557â565.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

665de Vries Â S, Schaapveld Â M, Janus Â CPM, Daniëls Â LA, Petersen Â EJ, van der Maazen Â RWM, et al. Â Long-term cause-specific mortality in Hodgkin lymphoma patients. *J Natl Cancer Inst* Â 2021;113:760â769.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

666Armenian Â SH, Yang Â D, Teh Â JB, Atencio Â LC, Gonzales Â A, Wong Â FL, et al. Â Prediction of cardiovascular disease among hematopoietic cell transplantation survivors. *Blood Adv* Â 2018;2:1756â1764.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

667Goldhirsch Â A, Gelber Â RD, Piccart-Gebhart Â MJ, De Azambuja Â E, Procter Â M, Suter Â TM, et al. Â 2 years versus 1 year of adjuvant trastuzumab for HER2-positive breast cancer (HERA): An open-label, randomised controlled trial. *Lancet* Â 2013;382:1021â1028.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

668Advani Â PP, Ballman Â KV, Dockter Â TJ, Colon-Otero Â G, Perez Â EA. Long-term cardiac safety analysis of NCCTG N9831 (Alliance) adjuvant trastuzumab trial. *J Clin Oncol* Â 2016;34:581â587.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

669Pinder Â MC, Duan Â Z, Goodwin Â JS, Hortobagyi Â GN, Giordano Â SH. Congestive heart failure in older women treated with adjuvant anthracycline chemotherapy for breast cancer. *J Clin Oncol* Â 2007;25:3808â3815.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

670Lancellotti Â P, Nkomo Â VT, Badano Â LP, Bergler Â J, Bogaert Â J, Davin Â L, et al. Â Expert Consensus for multi-modality imaging evaluation of cardiovascular complications of radiotherapy in adults: a report from the European Association of Cardiovascular Imaging and the American Society of Echocardiography. *J Am Soc Echocardiogr* Â 2013;26:1013â1032.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

671Tromp Â J, Boerman Â LM, Sama Â IE, Maass Â SWMC, Maduro Â JH, Hummel Â YM, et al. Â Long-term survivors of early breast cancer treated with chemotherapy are characterized by a pro-inflammatory biomarker profile compared to matched controls. *Eur J Heart Fail* Â 2020;22:1239â1246.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

672Cao Â Z, Xu Â C, Yang Â H, Li Â S, Wang Â Y. The role of healthy lifestyle in cancer incidence and temporal transitions to cardiometabolic disease. *JACC CardioOncology* Â 2021;3:663â674.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

673Limat Â S, Daguindau Â E, Cahn Â JY, Nerich Â V, Brion Â A, Perrin Â S, et al. Â Incidence and risk-factors of CHOP/R-CHOP-related cardiotoxicity in patients with aggressive non-Hodgkin's lymphoma. J Clin Pharm Ther Â 2014;39:168â174.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

674Hershman Â DL, McBride Â RB, Eisenberger Â A, Wei Â YT, Grann Â VR, Jacobson Â JS. Doxorubicin, cardiac risk factors, and cardiac toxicity in elderly patients with diffuse B-cell non-Hodgkin's lymphoma. J Clin Oncol Â 2008;26:3159â3165.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

675SOLVD Investigators, Yusuf Â S, Pitt Â B, Davis Â CE, Hood Â WB Â Jr, Cohn Â JN. Effect of enalapril on mortality and the development of heart failure in asymptomatic patients with reduced left ventricular ejection fractions. N Engl J Med Â 1992;327:685â691.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

676Pfeffer Â MA, Braunwald Â E, MoyÃ© Â LA, Basta Â L, Brown Â EJ, Cuddy Â TE, et al. Â Effect of captopril on mortality and morbidity in patients with left ventricular dysfunction after myocardial infarction. N Engl J Med Â 1992;327:669â677.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

677Dargie Â HJ. Effect of carvedilol on outcome after myocardial infarction in patients with left-ventricular dysfunction: the CAPRICORN randomised trial. Lancet Â 2001;357:1385â1390.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

678Abdel-Qadir Â H, Tai Â F, Croxford Â R, Austin Â PC, Amir Â E, Calvillo-ArgÃ¼elles Â O, et al. Â Characteristics and outcomes of women developing heart failure after early stage breast cancer chemotherapy: a population-based matched cohort study. Circ Heart Fail Â 2021;14:e008110.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

679Curigliano Â G, Cardinale Â D, Suter Â T, Plataniotis Â G, De Azambuja Â E, Sandri Â MT, et al. Â Cardiovascular toxicity induced by chemotherapy, targeted agents and radiotherapy: ESMO clinical practice guidelines. Ann Oncol Â 2012;23:vii155â166.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

680Adams Â MJ, Hardenbergh Â PH, Constine Â LS, Lipshultz Â SE. Radiation-associated cardiovascular disease. Crit Rev Oncol Hematol Â 2003;45:55â75.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

681Taylor Â C, Duane Â FK, Dodwell Â D, Gray Â R, Wang Â Z, Wang Â Y, et al. Â Estimating the risks of breast cancer radiotherapy: evidence from modern radiation doses to the lungs and heart and from previous randomized trials. J Clin Oncol Â 2017;35:1641â1649.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

682Reed Â GW, Masri Â A, Griffin Â BP, Kapadia Â SR, Ellis Â SG, Desai Â MY. Long-term mortality in patients with radiation-associated coronary artery disease treated with percutaneous coronary intervention. Circ Cardiovasc Interv Â 2016;9:e003483.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 683Liang Â JJ, Sio Â TT, Slusser Â JP, Lennon Â RJ, Miller Â RC, Sandhu Â G, et al. Â Outcomes after percutaneous coronary intervention with stents in patients treated with thoracic external beam radiation for cancer. *JACC Cardiovasc Interv* Â 2014;7:1412â1420.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 684Cuomo Â JR, Javaheri Â SP, Sharma Â GK, Kapoor Â D, Berman Â AE, Weintraub Â NL. How to prevent and manage radiation-induced coronary artery disease. *Heart* Â 2018;104:1647â1653.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 685Feldman Â DR, Schaffer Â WL, Steingart Â RM. Late cardiovascular toxicity following chemotherapy for germ cell tumors. *JNCCN J Natl Compr Cancer Netw* Â 2012;10:537â544.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 686Sudhakar Â R. Response to treatment and outcomes of acute coronary syndrome in the cancer population. *Clin Cardiol* Â 2012;35:646.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 687Nielsen Â SF, Nordestgaard Â BG, Bojesen Â SE. Statin use and reduced cancer-related mortality. *N Engl J Med* Â 2013;368:574â577.
[Google Scholar](#) [OpenURL Placeholder Text](#) Â
- 688Han Â X-J, Li Â J-Q, Khannanova Â Z, Li Â Y. Optimal management of coronary artery disease in cancer patients. *Chronic Dis Transl Med* Â 2019;5:221â233.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â
- 689Neumann Â FJ, Sousa-Uva Â M, Ahlsson Â A, Alfonso Â F, Banning Â AP, Benedetto Â U, et al. Â (2018 ESC/EACTS Guidelines on myocardial revascularization). *Kardiol Pol* Â 2018;76:1585â1664.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 690Wu Â W, Masri Â A, Popovic Â ZB, Smedira Â NG, Lytle Â BW, Marwick Â TH, et al. Â Long-term survival of patients with radiation heart disease undergoing cardiac surgery: a cohort study. *Circulation* Â 2013;127:1476â1484.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 691Gujral Â DM, Lloyd Â G, Bhattacharyya Â S. Radiation-induced valvular heart disease. *Heart* Â 2016;102:269â276.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 692Baumgartner Â H, Falk Â V, Bax Â JJ, De Bonis Â M, Hamm Â C, Holm Â PJ, et al. Â 2017 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J* Â 2017;38:2739â2791.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 693Lind Â A, Totzeck Â M, Mahabadi Â AA, Jãĩnosi Â RA, El Gabry Â M, Ruhparwar Â A, et al. Â Impact of cancer in patients undergoing transcatheter aortic valve replacement: a single-center study. *JACC CardioOncology* Â 2020;2:735â743.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 694 Yazdchi A F, Hirji SA, Nohria A, Percy E, Harloff M, Malarczyk A, et al. Transcatheter compared with surgical aortic valve replacement in patients with previous chest-directed radiation therapy. *JACC CardioOncology* 2021;3:397-407.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 695 Guha A, Dey AK, Omer S, Abraham WT, Attizzani G, Jneid H, et al. Contemporary trends and outcomes of percutaneous and surgical mitral valve replacement or repair in patients with cancer. *Am J Cardiol* 2020;125:1355-1360.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 696 Elbadawi A, Albaeni A, Elgendy IY, Ogunbayo GO, Jimenez E, Cornwell L, et al. Transcatheter versus surgical aortic valve replacement in patients with prior mediastinal radiation. *JACC Cardiovasc Interv* 2020;13:2658-2666.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 697 Zafar MR, Mustafa SF, Miller TW, Alkhwilani T, Sharma UC. Outcomes after transcatheter aortic valve replacement in cancer survivors with prior chest radiation therapy: a systematic review and meta-analysis. *Cardio-Oncology* 2020;6:8.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 698 Hadzijusufovic E, Albrecht-Schgoer K, Huber K, Hoermann G, Grebien F, Eisenwort G, et al. Nilotinib-induced vasculopathy: identification of vascular endothelial cells as a primary target site. *Leukemia* 2017;31:2388-2397.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 699 Berger CC, Bokemeyer C, Schneider M, Kuczyk MA, Schmoll HJ. Secondary Raynaud's phenomenon and other late vascular complications following chemotherapy for testicular cancer. *Eur J Cancer* 1995;31A:2229-2238.
[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)
- 700 Stelwagen J, Lubberts S, Stegink LC, Steursma G, Kruijff LM, Donkerbroek JW, et al. Vascular aging in long-term survivors of testicular cancer more than 20 years after treatment with cisplatin-based chemotherapy. *Br J Cancer* 2020;123:1599-1607.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 701 Andreassi MG, Piccaluga E, Gargani L, Sabatino L, Borghini A, Fatta F, et al. Subclinical carotid atherosclerosis and early vascular aging from long-term low-dose ionizing radiation exposure: a genetic, telomere, and vascular ultrasound study in cardiac catheterization laboratory staff. *JACC Cardiovasc Interv* 2015;8:616-627.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 702 Carmody BJ, Arora S, Avena R, Curry KM, Simpkins J, Cosby K, et al. Accelerated carotid artery disease after high-dose head and neck radiotherapy: is there a role for routine carotid duplex surveillance? *J Vasc Surg* 1999;30:1045-1051.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 703 Carpenter DJ, Mowery YM, Broadwater G, Rodrigues A, Wisdom AJ, Dorth JA, et al. The risk of carotid stenosis in head and neck cancer patients after radiation therapy. *Oral Oncol*

Â 2018;80:9â15.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

704Szpakowski Â N, Desai Â MY. Radiation-associated pericardial disease. Curr Cardiol Rep Â 2019;21:97.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

705Schultz-Hector Â S, Trott Â KR. Radiation-induced cardiovascular diseases: is the epidemiologic evidence compatible with the radiobiologic data? Â Int J Radiat Oncol Biol Phys Â 2007;67:10â18.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

706Ning Â MS, Tang Â L, Gomez Â DR, Xu Â T, Luo Â Y, Huo Â J, et al. Â Incidence and predictors of pericardial effusion after chemoradiation therapy for locally advanced non-small cell lung cancer. Int J Radiat Oncol Biol Phys Â 2017;99:70â79.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

707Wei Â X, Liu Â HH, Tucker Â SL, Wang Â S, Mohan Â R, Cox Â JD, et al. Â Risk factors for pericardial effusion in inoperable esophageal cancer patients treated with definitive chemoradiation therapy. Int J Radiat Oncol Biol Phys Â 2008;70:707â714.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

708Gagliardi Â G, Constine Â LS, Moiseenko Â V, Correa Â C, Pierce Â LJ, Allen Â AM, et al. Â Radiation dose-volume effects in the heart. Int J Radiat Oncol Biol Phys Â 2010;76:S77âS85.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

709Chiabrando Â JG, Bonaventura Â A, Vecchiâ Â A, Wohlford Â GF, Mauro Â AG, Jordan Â JH, et al. Â Management of acute and recurrent pericarditis: JACC state-of-the-art review. J Am Coll Cardiol Â 2020;75:76â92.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

710Donnellan Â E, Phelan Â D, Mccarthy Â CP, Collier Â P, Desai Â M, Griffin Â B. Radiation-induced heart disease: a practical guide to diagnosis and management. Cleve Clin J Med Â 2016;83:914â922.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

711Crestanello Â JA, McGregor Â CGA, Danielson Â GK, Daly Â RC, Dearani Â JA, Orszulak Â TA, et al. Â Mitral and tricuspid valve repair in patients with previous mediastinal radiation therapy. Ann Thorac Surg Â 2004;78:826â831.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

712Heidenreich Â PA, Kapoor Â JR. Radiation induced heart disease. Heart Â 2009;95:252â258.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

713Walsh Â D, Nelson Â KA. Autonomic nervous system dysfunction in advanced cancer. Support Care Cancer Â 2002;10:523â528.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

714Noor Â B, Akhavan Â S, Leuchter Â M, Yang Â EH, Ajijola Â OA. Quantitative assessment of cardiovascular autonomic impairment in cancer survivors: a single center case series. Cardio-Oncology Â 2020;6:11.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 715Gibson Â TM, Li Â Z, Green Â DM, Armstrong Â GT, Mulrooney Â DA, Srivastava Â DK, et al.
Â Blood pressure status in adult survivors of childhood cancer: a report from the St. Jude Lifetime Cohort Study. *Cancer Epidemiol Biomarkers Prev* Â 2017;26:1705â1713.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 716Friedman Â DN, Tonorezos Â ES, Cohen Â P. Diabetes and metabolic syndrome in survivors of childhood cancer. *Horm Res Paediatr* Â 2019;91:118â127.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 717Pekmezi Â DW, Demark-Wahnefried Â W. Updated evidence in support of diet and exercise interventions in cancer survivors. *Acta Oncol (Madr)* Â 2011;50:167â178.
[Google Scholar](#) [Crossref](#) [Search ADS](#) Â
- 718Chlebowski Â RT, Aiello Â E, McTiernan Â A. Weight loss in breast cancer patient management. *J Clin Oncol* Â 2002;20:1128â1143.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 719Protani Â M, Coory Â M, Martin Â JH. Effect of obesity on survival of women with breast cancer: systematic review and meta-analysis. *Breast Cancer Res Treat* Â 2010;123:627â635.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 720Meyerhardt Â JA, Ma Â J, Courneya Â KS. Energetics in colorectal and prostate cancer. *J Clin Oncol* Â 2010;28:4066â4073.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 721Wright Â ME, Chang Â SC, Schatzkin Â A, Albanes Â D, Kipnis Â V, Mouw Â T, et al.
Â Prospective study of adiposity and weight change in relation to prostate cancer incidence and mortality. *Cancer* Â 2007;109:675â684.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 722Siegel Â EM, Ulrich Â CM, Poole Â EM, Holmes Â RS, Jacobsen Â PB, Shibata Â D. The effects of obesity and obesity-related conditions on colorectal cancer prognosis. *Cancer Control* Â 2010;17:52â57.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 723Kroenke Â CH, Chen Â WY, Rosner Â B, Holmes Â MD. Weight, weight gain, and survival after breast cancer diagnosis. *J Clin Oncol* Â 2005;23:1370â1378.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 724Meyerhardt Â JA, Niedzwiecki Â D, Hollis Â D, Saltz Â LB, Mayer Â RJ, Nelson Â H, et al.
Â Impact of body mass index and weight change after treatment on cancer recurrence and survival in patients with stage III colon cancer: Findings from Cancer and Leukemia Group B 89803. *J Clin Oncol* Â 2008;26:4109â4115.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 725Kroenke Â CH, Fung Â TT, Hu Â FB, Holmes Â MD. Dietary patterns and survival after breast cancer diagnosis. *J Clin Oncol* Â 2005;23:9295â9303.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

726Kwan Â ML, Weltzien Â E, Kushi Â LH, Castillo Â A, Slattery Â ML, Caan Â BJ. Dietary patterns and breast cancer recurrence and survival among women with early-stage breast cancer. *J Clin Oncol* Â 2009;27:919â926.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

727Meyerhardt Â JA, Niedzwiecki Â D, Hollis Â D, Saltz Â LB, Hu Â FB, Mayer Â RJ, et al. Â Association of dietary patterns with cancer recurrence and survival in patients with stage III colon cancer. *J Am Med Assoc* Â 2007;298:754â764.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

728Yang Â J, Li Â C, Shen Â Y, Zhou Â H, Shao Â Y, Zhu Â W, et al. Â Impact of statin use on cancer-specific mortality and recurrence: a meta-analysis of 60 observational studies. *Medicine* Â 2020;99:e19596.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

729Kim Â J, Choi Â EA, Han Â YE, Woo Lee Â J, Seul Kim Â Y, Kim Â Y, et al. Â Association between statin use and all-cause mortality in cancer survivors, based on the Korean health insurance service between 2002 and 2015. *Nutr Metab Cardiovasc Dis* Â 2020;30:434â440.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

730Ren Â QW, Yu Â SY, Teng Â THK, Li Â X, Cheung Â KS, Wu Â MZ, et al. Â Statin associated lower cancer risk and related mortality in patients with heart failure. *Eur Heart J* Â 2021;42:3049â3059.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

731GalvÃ£o Â DA, Taaffe Â DR, Spry Â N, Joseph Â D, Newton Â RU. Combined resistance and aerobic exercise program reverses muscle loss in men undergoing androgen suppression therapy for prostate cancer without bone metastases: a randomized controlled trial. *J Clin Oncol* Â 2010;28:340â347.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

732Speck Â RM, Courneya Â KS, MÃsse Â LC, Duval Â S, Schmitz Â KH. An update of controlled physical activity trials in cancer survivors: a systematic review and meta-analysis. *J Cancer Surviv* Â 2010;4:87â100.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

733Campia Â U, Barac Â A. Exercise and aerobic fitness to reduce cancer-related cardiovascular toxicity. *Curr Treat Options Cardiovasc Med* Â 2016;18:44.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

734Jones Â LW, Liu Â Q, Armstrong Â GT, Ness Â KK, Yasui Â Y, Devine Â K, et al. Â Exercise and risk of major cardiovascular events in adult survivors of childhood Hodgkin lymphoma: a report from the childhood cancer survivor study. *J Clin Oncol* Â 2014;32:3643â3650.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

735Scott Â JM, Li Â N, Liu Â Q, Yasui Â Y, Leisenring Â W, Nathan Â PC, et al. Â Association of exercise with mortality in adult survivors of childhood cancer. *JAMA Oncol* Â 2018;4:1352â1358.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

736Rock Â CL, Doyle Â C, Demark-Wahnefried Â W, Meyerhardt Â J, Courneya Â KS, Schwartz Â AL, et al. Â Nutrition and physical activity guidelines for cancer survivors. *CA Cancer J Clin*

Â 2012;62:242â274.

[Google Scholar](#) [OpenURL Placeholder Text](#) Â

737CDC. Cancer Data and Statistics | Survival Information from the National Cancer Institute and Centers for Disease Control and Prevention. 2013.

738Thompson Â KA, Hildebrandt Â MAT, Ater Â JL. Cardiac outcomes with pregnancy after cardiotoxic therapy for childhood cancer. *J Am Coll Cardiol* Â 2017;69:594â595.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

739Nolan Â M, Oikonomou Â EK, Silversides Â CK, Hines Â MR, Thompson Â KA, Campbell Â BA, et al. Â Impact of cancer therapy-related cardiac dysfunction on risk of heart failure in pregnancy. *JACC CardioOncology* Â 2020;2:153â162.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

740Chait-Rubinek Â L, Mariani Â JA, Goroncy Â N, Herschtal Â A, Wheeler Â GC, Dwyer Â MK, et al. Â A retrospective evaluation of risk of peripartum cardiac dysfunction in survivors of childhood, adolescent and young adult malignancies. *Cancers (Basel)* Â 2019;11:1046.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

741Regitz-Zagrosek Â V, Roos-Hesselink Â JW, Bauersachs Â J, Blomstr  m-Lundqvist Â C, C  fkov  ; Â R, De Bonis Â M, et al. Â 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J* Â 2018;39:3165â3241.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

742Mehta Â LS, Warnes Â CA, Bradley Â E, Burton Â T, Economy Â K, Mehran Â R, et al. Â Cardiovascular considerations in caring for pregnant patients: a scientific statement from the American Heart Association. *Circulation* Â 2020;141:e884âe903.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

743Travis Â WD, Brambilla Â E, Burke Â AP, Marx Â A, Nicholson Â AG, eds. *WHO Classification of Tumours of the Lung, Pleura, Thymus and Heart*, (4th ed) . 2015. ISBN 978-92-832-2436-5

[Google Scholar](#) [OpenURL Placeholder Text](#) Â

744Burke Â A, Tavora Â F. Â The 2015 WHO Classification of tumors of the heart and pericardium. *J Thorac Oncol* Â 2016;11:441â452.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

745Cresti Â A, Chiavarelli Â M, Glauber Â M, Tanganelli Â P, Scalese Â M, Cesareo Â F, et al. Â Incidence rate of primary cardiac tumors: a 14-year population study. *J Cardiovasc Med* Â 2016;17:37â43.

[Google Scholar](#) [Crossref](#) [Search ADS](#) Â

746Maleszewski Â JJ, Bois Â MC, Bois Â JP, Young Â PM, Stulak Â JM, Klarich Â KW. Neoplasia and the heart: pathological review of effects with clinical and radiological correlation. *J Am Coll Cardiol* Â 2018;72:202â227.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

747Tyebally Â S, Chen Â S, Bhattacharyya Â S, Mughrabi Â S, Hussain Â Z, Manisty Â C, et al. Â Cardiac tumors: *JACC CardioOncology* state-of-the-art review. *JACC CardioOncology* Â 2020;2:293â311.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

748Kirkpatrick Â JN, Wong Â T, Bednarz Â JE, Spencer Â KT, Sugeng Â L, Ward Â RP, et al. Â Differential diagnosis of cardiac masses using contrast echocardiographic perfusion imaging. *J Am Coll Cardiol* Â 2004;43:1412â1419.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

749Thiene Â G, Rizzo Â S, Marra Â MP, Valente Â M, Basso Â C. Â Masses and cardiac tumours: classification and diagnosis. *ESC CardioMED*, 2018. DOI: [10.1093/med/9780198784906.003.0386](https://doi.org/10.1093/med/9780198784906.003.0386)

750Zaragosa-Macias Â E, Chen Â MA, Gill Â EA. Real time three-dimensional echocardiography evaluation of intracardiac masses. *Echocardiography* Â 2012;29:207â219.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

751Beroukhim Â RS, Prakash Â A, Valsangiacomo Buechel Â ER, Cava Â JR, Dorfman Â AL, Festa Â P, et al. Â Characterization of cardiac tumors in children by cardiovascular magnetic resonance imaging: a multicenter experience. *J Am Coll Cardiol* Â 2011;58:1044â1054.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

752Rahbar Â K, Seifarth Â H, SchÃ¶fers Â M, Stegger Â L, Hoffmeier Â A, Spieker Â T, et al. Â Differentiation of malignant and benign cardiac tumors using 18F-FDG PET/CT. *J Nucl Med* Â 2012;53:856â863.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

753Kassop Â D, Donovan Â MS, Cheezum Â MK, Nguyen Â BT, Gambill Â NB, Blankstein Â R, et al. Â Cardiac masses on cardiac CT: a review. *Curr Cardiovasc Imaging Rep* Â 2014;7:9281.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

754D'Angelo Â EC, Paolisso Â P, Vitale Â G, FoÃ Â A, Bergamaschi Â L, Magnani Â I, et al. Â Diagnostic accuracy of cardiac computed tomography and 18F-fluorodeoxyglucose with positron emission tomography in cardiac masses. *JACC Cardiovasc Imaging* Â 2020;13:2400â2411.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

755Butany Â J, Nair Â V, Naseemuddin Â A, Nair Â GM, Catton Â C, Yau Â T. Cardiac tumours: diagnosis and management. *Lancet Oncol* Â 2005;6:219â228.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

756Szmit Â S, Zagrodzka Â M, Kurzyna Â M, Opolski Â G, Szczylik Â C. Sunitinib malate, a receptor tyrosine kinase inhibitor, is effective in the treatment of restrictive heart failure due to heart metastases from renal cell carcinoma. *Cardiology* Â 2009;114:67â71.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

757Peccatori Â FA, Azim Â JA, Orecchia Â R, Hoekstra Â HJ, Pavlidis Â N, Kesic Â V, et al. Â Cancer, pregnancy and fertility: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol* Â 2013;24:vi160â70.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

758Cubillo ^ A, Morales ^ S, Goñi ^ E, Matute ^ F, Muñoz ^ JL, Páez-Díaz ^ D, et al.
^ Multidisciplinary consensus on cancer management during pregnancy. Clin Transl Oncol
^ 2021;23:1054^1066.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

759Silverstein ^ J, Post ^ AL, Chien ^ AJ, Olin ^ R, Tsai ^ KK, Ngo ^ Z, et al. ^ Multidisciplinary
management of cancer during pregnancy. JCO Oncol Pract ^ 2020;16:545^557.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

760Amant ^ F, Berveiller ^ P, Boere ^ IA, Cardonick ^ E, Fruscio ^ R, Fumagalli ^ M, et al.
^ Gynecologic cancers in pregnancy: guidelines based on a third international consensus meeting. Ann
Oncol ^ 2019;30:1601^1612.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

761Dunleavy ^ K, McLintock ^ C. How I treat lymphoma in pregnancy. Blood
^ 2020;136:2118^2124.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

762Ducas ^ RA, Elliott ^ JE, Melnyk ^ SF, Premecz ^ S, Dasilva ^ M, Cleverley ^ K, et al.
^ Cardiovascular magnetic resonance in pregnancy: insights from the Cardiac Hemodynamic Imaging
and Remodeling in Pregnancy (CHIRP) study. J Cardiovasc Magn Reson ^ 2014;16:1.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

763Savu ^ O, Jurcu ^ R, Giuc ^ S, Van Mieghem ^ T, Gussi ^ I, Popescu ^ BA, et al.
^ Morphological and functional adaptation of the maternal heart during pregnancy. Circ Cardiovasc
Imaging ^ 2012;5:289^297.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

764Narayanan ^ M, Elkayam ^ U, Naqvi ^ TZ. Echocardiography in pregnancy: part 2. Curr Cardiol Rep
^ 2016;18:90.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

765Furen ^ E, Eriksson ^ P, Wennerholm ^ UB, Dellborg ^ M. Pregnancy in a healthy population:
dynamics of NTproBNP and hs-cTroponin T. Open Heart ^ 2020;7:e001293.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

766Tanous ^ D, Siu ^ SC, Mason ^ J, Greutmann ^ M, Wald ^ RM, Parker ^ JD, et al. ^ B-type
natriuretic peptide in pregnant women with heart disease. J Am Coll Cardiol ^ 2010;56:1247^1253.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

767Heit ^ JA, Kobbervig ^ CE, James ^ AH, Petterson ^ TM, Bailey ^ KR, Melton ^ LJ ^ III. Trends
in the incidence of venous thromboembolism during pregnancy or postpartum: a 30-year population-
based study. Ann Intern Med ^ 2005;143:697^706.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

768Sultan ^ AA, West ^ J, Tata ^ LJ, Fleming ^ KM, Nelson-Piercy ^ C, Grainge ^ MJ. Risk of first
venous thromboembolism in pregnant women in hospital: population based cohort study from England.

BMJ 2013;347:f6099.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

769Sultan AA, West J, Tata LJ, Fleming KM, Nelson-Piercy C, Grainge MJ. Risk of first venous thromboembolism in and around pregnancy: a population-based cohort study. *Br J Haematol* 2012;156:366â373.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

770Hase EA, de Barros VIPVL, Igai AMK, Francisco RPV, Zugaib M. Risk assessment of venous thromboembolism and thromboprophylaxis in pregnant women hospitalized with cancer: preliminary results from a risk score. *Clinics (Sao Paulo)* 2018;73:e368.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

771Fox DJ, Khattar RS. Carcinoid heart disease: presentation, diagnosis, and management. *Heart* 2004;90:1224â1228.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

772Davar J, Connolly HM, Caplin ME, Pavel M, Zacks J, Bhattacharyya S, et al. Diagnosing and managing carcinoid heart disease in patients with neuroendocrine tumors: an expert statement. *J Am Coll Cardiol* 2017;69:1288â1304.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

773Dobson R, Burgess MI, Pritchard DM, Cuthbertson DJ. The clinical presentation and management of carcinoid heart disease. *Int J Cardiol* 2014;173:29â32.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

774Lichtenauer M, Pichler T, Eder S, Mirna M, Magnes T, Wernly B, et al. Carcinoid heart disease involving the left heart: a case report and biomarker analysis. *ESC Heart Fail* 2019;6:222â227.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

775Halperin DM, Shen C, Dasari A, Xu Y, Chu Y, Zhou S, et al. Frequency of carcinoid syndrome at neuroendocrine tumour diagnosis: a population-based study. *Lancet Oncol* 2017;18:525â534.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

776Bhattacharyya S, Davar J, Dreyfus G, Caplin ME. Carcinoid heart disease. *Circulation* 2007;116:2860â2865.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

777Zuetenhorst JM, Korse CM, Bonfrer JMG, Bakker RH, Taal BG. Role of natriuretic peptides in the diagnosis and treatment of patients with carcinoid heart disease. *Br J Cancer* 2004;90:2073â2079.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

778Dobson R, Burgess MI, Banks M, Pritchard DM, Vora J, Valle JW, et al. The association of a panel of biomarkers with the presence and severity of carcinoid heart disease: a cross-sectional study. *PLoS One* 2013;8:e73679.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

779Korse Â CM, Taal Â BG, De Groot Â CA, Bakker Â RH, Bonfrer Â JMG. Chromogranin-A and N-terminal pro-brain natriuretic peptide: an excellent pair of biomarkers for diagnostics in patients with neuroendocrine tumor. J Clin Oncol Â 2009;27:4293â4299.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

780Bhattacharyya Â S, Toumpanakis Â C, Caplin Â ME, Davar Â J. Usefulness of N-terminal pro-brain natriuretic peptide as a biomarker of the presence of carcinoid heart disease. Am J Cardiol Â 2008;102:938â942.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

781Kulke Â MH, HÃ¶rsch Â D, Caplin Â ME, Anthony Â LB, Bergsland Â E, Åberg Â K, et al. Â Telotristat ethyl, a tryptophan hydroxylase inhibitor for the treatment of carcinoid syndrome. J Clin Oncol Â 2017;35:14â23.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

782Dumoulein Â M, Verslype Â C, Van Cutsem Â E, Meuris Â B, Herijgers Â P, Flameng Â W, et al. Â Carcinoid heart disease: case and literature review. Acta Cardiol Â 2010;65:261â264.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

783Nguyen Â A, Schaff Â HV, Abel Â MD, Luis Â SA, Lahr Â BD, Halfdanarson Â TR, et al. Â Improving outcome of valve replacement for carcinoid heart disease. J Thorac Cardiovasc Surg Â 2019;158:99â107.e2.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

784Mabvuure Â N, Cumberworth Â A, Hindocha Â S. In patients with carcinoid syndrome undergoing valve replacement: will a biological valve have acceptable durability? Â Interact Cardiovasc Thorac Surg Â 2012;15:467â471.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

785Connolly Â HM, Schaff Â HV, Abel Â MD, Rubin Â J, Askew Â JW, Li Â Z, et al. Â Early and late outcomes of surgical treatment in carcinoid heart disease. J Am Coll Cardiol Â 2015;66:2189â2196.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

786Sunjic Â I, Shin Â D, Sunjic Â KM, Popat Â JV, Tran Â T, Chae Â SH, et al. Â Incidence of atrioventricular block after valve replacement in carcinoid heart disease. Cardiol Res Â 2020;11:56â60.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

787Kuntze Â T, Owais Â T, Secknus Â MA, Kaemmerer Â D, Baum Â R, Girdauskas Â E. Results of contemporary valve surgery in patients with carcinoid heart disease. J Heart Valve Dis Â 2016;25:356â363.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#) Â

788Luthra Â S, Olevano Â C, Richens Â T, Tsang Â GM. Percutaneous transcatheter valve-in-valve pulmonary and tricuspid replacement in carcinoid heart disease. JACC Case Reports Â 2020;2:533â536.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 789Zacks Â JS, Lavine Â R. Avoiding a repeat sternotomy in recurrent carcinoid heart disease. JACC Case Reports Â 2020;2:537â538.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 790Bhattacharyya Â S, Toumpanakis Â C, Burke Â M, Taylor Â AM, Caplin Â ME, Davar Â J. Features of carcinoid heart disease identified by 2- and 3-dimensional echocardiography and cardiac MRI. Circ Cardiovasc Imaging Â 2010;3:103â111.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 791Pellikka Â PA, Tajik Â AJ, Khandheria Â BK, Seward Â JB, Callahan Â JA, Pitot Â HC, et al. Â Carcinoid heart disease: clinical and echocardiographic spectrum in 74 patients. Circulation Â 1993;87:1188â1196.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 792Palladini Â G, Milani Â P, Merlini Â G. Management of AL amyloidosis in 2020. Blood Â 2020;136:2620â2627.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 793Witteles Â RM, Liedtke Â M. AL amyloidosis for the cardiologist and oncologist: epidemiology, diagnosis, and management. JACC CardioOncology Â 2019;1:117â130.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 794Merlini Â G, Wechalekar Â AD, Palladini Â G. Systemic light chain amyloidosis: an update for treating physicians. Blood Â 2013;121:5124â5130.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 795Maurer Â MS, Elliott Â P, Comenzo Â R, Semigran Â M, Rapezzi Â C. Addressing common questions encountered in the diagnosis and management of cardiac amyloidosis. Circulation Â 2017;135:1357â1377.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 796National Comprehensive Cancer Network (NCCN). NCCN Clinical Practice Guidelines in Oncology. Systemic Light Chain Amyloidosis Version 1. Natl Compr Cancer Netw Â 2021. <https://www.nccn.org/>.
[OpenURL Placeholder Text](#) Â
- 797Kumar Â S, Dispenzieri Â A, Lacy Â MQ, Hayman Â SR, Buadi Â FK, Colby Â C, et al. Â Revised prognostic staging system for light chain amyloidosis incorporating cardiac biomarkers and serum free light chain measurements. J Clin Oncol Â 2012;30:989â995.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 798Zhang Â KW, Miao Â J, Mitchell Â JD, Alvarez-Cardona Â J, Tomasek Â K, Su Â YR, et al. Â Plasma hepatocyte growth factor for diagnosis and prognosis in light chain and transthyretin cardiac amyloidosis. JACC CardioOncology Â 2020;2:56â66.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 799Mohan Â M, Buross Â A, Mathur Â P, Gokden Â N, Singh Â M, Susanibar Â S, et al. Â Clinical characteristics and prognostic factors in multiple myeloma patients with light chain deposition disease. Am J Hematol Â 2017;92:739â745.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â

- 800Grogan A M, Dispenzieri A, Gertz MA. Light-chain cardiac amyloidosis: strategies to promote early diagnosis and cardiac response. *Heart* 2017;103:1065â1072.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 801JurcuÅ£ R, Onciul S, Adam R, Stan C, Coriu D, Rapezzi C, et al. Multimodality imaging in cardiac amyloidosis: a primer for cardiologists. *Eur Heart J Cardiovasc Imaging* 2020;21:833â844.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 802Lee Chuy K, Drill E, Yang JC, Landau H, Hassoun H, Nahhas O, et al. Incremental value of global longitudinal strain for predicting survival in patients with advanced AL amyloidosis. *JACC CardioOncology* 2020;2:223â231.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 803Brownrigg J, Lorenzini M, Lumley M, Elliott P. Diagnostic performance of imaging investigations in detecting and differentiating cardiac amyloidosis: a systematic review and meta-analysis. *ESC Heart Fail* 2019;6:1041â1051.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 804Baggiano A, Boldrini M, Martinez-Naharro A, Kotecha T, Petrie A, Rezk T, et al. Noncontrast magnetic resonance for the diagnosis of cardiac amyloidosis. *JACC Cardiovasc Imaging* 2020;13:69â80.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 805Comenzo RL, Reece D, Palladini G, Seldin D, Santhorawala V, Landau H, et al. Consensus guidelines for the conduct and reporting of clinical trials in systemic light-chain amyloidosis. *Leukemia* 2012;26:2317â2325.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 806Eckhert E, Witteles R, Kaufman G, Lafayette R, Arai S, Schrier S, et al. Grading cardiac response in AL amyloidosis: implications for relapse and survival. *Br J Haematol* 2019;186:144â146.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 807Vaxman I, Gertz M. Recent advances in the diagnosis, risk stratification, and management of systemic light-chain amyloidosis. *Acta Haematol* 2019;141:93â106.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 808Giancaterino S, Urey MA, Darden D, Hsu JC. Management of arrhythmias in cardiac amyloidosis. *JACC Clin Electrophysiol* 2020;6:351â361.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 809El-Am EA, Dispenzieri A, Melduni RM, Ammash NM, White RD, Hodge DO, et al. Direct current cardioversion of atrial arrhythmias in adults with cardiac amyloidosis. *J Am Coll Cardiol* 2019;73:589â597.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)
- 810Tahir UA, Doros G, Kim JS, Connors LH, Seldin DC, Sam F. Predictors of mortality in light chain cardiac amyloidosis with heart failure. *Sci Rep* 2019;9:8552.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

811Khanna ^ S, Lo ^ P, Cho ^ K, Subbiah ^ R. Ventricular arrhythmias in cardiac amyloidosis: a review of current literature. Clin Med Insights Cardiol ^ 2020;14:1179546820963055.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

812Kastritis ^ E, Leleu ^ X, Arnulf ^ B, Zamagni ^ E, Cibeira ^ MT, Kwok ^ F, et al. ^ Bortezomib, melphalan, and dexamethasone for light-chain amyloidosis. J Clin Oncol ^ 2020;38:3252a^3260.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

813Palladini ^ G, Kastritis ^ E, Maurer ^ MS, Zonder ^ J, Minnema ^ MC, Wechalekar ^ AD, et al. ^ Daratumumab plus CyBorD for patients with newly diagnosed AL amyloidosis: safety run-in results of ANDROMEDA. Blood ^ 2020;136:71a^80.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

814Huang ^ X, Ren ^ G, Chen ^ W, Guo ^ J, Zhao ^ L, Zeng ^ C, et al. ^ The role of induction therapy before autologous stem cell transplantation in low disease burden AL amyloidosis patients. Amyloid ^ 2021;28:75a^83.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

815Barrett ^ CD, Dobos ^ K, Liedtke ^ M, Tuzovic ^ M, Haddad ^ F, Kobayashi ^ Y, et al. ^ A changing landscape of mortality for systemic light chain amyloidosis. JACC Heart Fail ^ 2019;7:958a^966.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

816Zhang ^ KW, Stockerl-Goldstein ^ KE, Lenihan ^ DJ. Emerging therapeutics for the treatment of light chain and transthyretin amyloidosis. JACC Basic Transl Sci ^ 2019;4:438a^448.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

817Kastritis ^ E, Palladini ^ G, Minnema ^ MC, Wechalekar ^ AD, Jaccard ^ A, Lee ^ HC, et al. ^ Daratumumab-based treatment for immunoglobulin light-chain amyloidosis. N Engl J Med ^ 2021;385:46a^58.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

818Wechalekar ^ AD, Whelan ^ C. Encouraging impact of doxycycline on early mortality in cardiac light chain (AL) amyloidosis. Blood Cancer J ^ 2017;7:e546.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

819Shen ^ K, Fu ^ W, Wu ^ Y, Dong ^ Y, Huang ^ Z, Wei ^ Y, et al. ^ Doxycycline combined with bortezomib^cyclophosphamide^dexamethasone chemotherapy for newly diagnosed cardiac light-chain amyloidosis: a multicenter randomized controlled trial. Circulation ^ 2021;145:8a^17.

[Google Scholar](#) [OpenURL Placeholder Text](#) ^

820Salinaro ^ F, Meier-Ewert ^ HK, Miller ^ EJ, Pandey ^ S, Sanchrawala ^ V, Berk ^ JL, et al. ^ Longitudinal systolic strain, cardiac function improvement, and survival following treatment of light-chain (AL) cardiac amyloidosis. Eur Heart J Cardiovasc Imaging ^ 2017;18:1057a^1064.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) ^

- 821 Palladini Â G, Foli Â A, Milani Â P, Russo Â P, Albertini Â R, Lavatelli Â F, et al. Â Best use of cardiac biomarkers in patients with AL amyloidosis and renal failure. *Am J Hematol* Â 2012;87:465â471.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 822 Dispenzieri Â A, Gertz Â MA, Kyle Â RA, Lacy Â MQ, Burritt Â MF, Therneau Â TM, et al. Â Serum cardiac troponins and N-terminal pro-brain natriuretic peptide: a staging system for primary systemic amyloidosis. *J Clin Oncol* Â 2004;22:3751â3757.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 823 Zaremba Â T, Jakobsen Â AR, SÃ¸gaard Â M, ThÃ¸gersen Â AM, Johansen Â MB, Madsen Â LÃ¸b, et al. Â Risk of device malfunction in cancer patients with implantable cardiac device undergoing radiotherapy: a population-based cohort study. *Pacing Clin Electrophysiol* Â 2015;38:343â356.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 824 Miften Â M, Mihailidis Â D, Kry Â SF, Reft Â C, Esquivel Â C, Farr Â J, et al. Â Management of radiotherapy patients with implanted cardiac pacemakers and defibrillators: a report of the AAPM TG-203. *Med Phys* Â 2019;46:e757âe788.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 825 Hurkmans Â CW, Knegjens Â JL, Oei Â BS, Maas Â AJJ, Uiterwaal Â GJ, van der Borden Â AJ, et al. Â Management of radiation oncology patients with a pacemaker or ICD: a new comprehensive practical guideline in The Netherlands. *Radiat Oncol* Â 2012;7:198.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 826 Indik Â JH, Gimbel Â JR, Abe Â H, Alkmim-Teixeira Â R, Birgersdotter-Green Â U, Clarke Â GD, et al. Â 2017 HRS expert consensus statement on magnetic resonance imaging and radiation exposure in patients with cardiovascular implantable electronic devices. *Heart Rhythm* Â 2017;14:e97âe153.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 827 Zaremba Â T, Jakobsen Â AR, SÃ¸gaard Â M, ThÃ¸gersen Â AM, Riahi Â S. Radiotherapy in patients with pacemakers and implantable cardioverter defibrillators: a literature review. *Europace* Â 2016;18:479â491.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 828 Grant Â JD, Jensen Â GL, Tang Â C, Pollard Â JM, Kry Â SF, Krishnan Â S, et al. Â Radiotherapy-induced malfunction in contemporary cardiovascular implantable electronic devices: clinical incidence and predictors. *JAMA Oncol* Â 2015;1:624â632.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 829 Zecchin Â M, Severgnini Â M, Fiorentino Â A, Malavasi Â VL, Menegotti Â L, Alongi Â F, et al. Â Management of patients with cardiac implantable electronic devices (CIED) undergoing radiotherapy: a consensus document from Associazione Italiana Aritmologia e Cardioritmo (AIAC), Associazione Italiana Radioterapia Oncologica (AIRO), Associazione Italiana Fisica Medica (AIFM). *Int J Cardiol* Â 2018;255:175â183.
[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#) Â
- 830 Gomez Â DR, Poenisch Â F, Pinnix Â CC, Sheu Â T, Chang Â JY, Memon Â N, et al. Â Malfunctions of implantable cardiac devices in patients receiving proton beam therapy: incidence and predictors. *Int J*

Radiat Oncol Biol Phys 2013;87:570–575.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

831Sharifzadehgan A, Laurans M, Thuillot M, Huertas A, Baudinaud P, Narayanan K, et al. Radiotherapy in patients with a cardiac implantable electronic device. *Am J Cardiol* 2020;128:196–201.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

832Stienen JJC, Ottevanger PB, Wennekes L, Dekker HM, van der Maazen RWM, Mandigers CMPW, et al. Development and evaluation of an educational E-tool to help patients with non-Hodgkin's lymphoma manage their personal care pathway. *JMIR Res Protoc* 2015;4:e6.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

833Murphy P, Levine A, Lerma T, Young S, Hwang J, Goldsby R. A portable survivorship care plan: a tool that helps educate and improve knowledge in childhood cancer survivors. *Support Care Cancer* 2021;29:169–177.

[Google Scholar](#) [PubMed](#) [OpenURL Placeholder Text](#)

834Asteggiano R, Aboyans V, Lee G, Salinger S, Richter D. Cardiology care delivered to cancer patients. *Eur Heart J* 2020;41:205–206.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

835Aktaa S, Batra G, Wallentin L, Baigent C, Erlinge D, James S, et al. European Society of Cardiology methodology for the development of quality indicators for the quantification of cardiovascular care and outcomes. *Eur Heart J Qual Care Clin Outcomes* 2022;8:4–13.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

836Minchin M, Roland M, Richardson J, Rowark S, Guthrie B. Quality of care in the United Kingdom after removal of financial incentives. *N Engl J Med* 2018;379:948–957.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

837Song Z, Ji Y, Safran DG, Chernew ME. Health care spending, utilization, and quality 8 years into global payment. *N Engl J Med* 2019;381:252–263.

[Google Scholar](#) [Crossref](#) [Search ADS](#) [PubMed](#)

Author notes



The two chairpersons contributed equally to the document and are joint corresponding authors.

Author/Task Force Member affiliations are listed in Author information.

¹Representing the European Society for Therapeutic Radiology and Oncology (ESTRO); ²representing the European Hematology Association (EHA); ³representing the International Cardio-Oncology Society (IC-OS).

ESC Clinical Practice Guidelines (CPG) Committee: listed in the Appendix.

ESC subspecialty communities having participated in the development of this document:

Associations: Association for Acute CardioVascular Care (ACVC), European Association of Cardiovascular Imaging (EACVI), European Association of Preventive Cardiology (EAPC), European Association of Percutaneous Cardiovascular Interventions (EAPCI), European Heart Rhythm Association (EHRA), Heart Failure Association (HFA).

Councils: Council of Cardio-Oncology, Council on Hypertension, Council on Valvular Heart Disease.

Working Groups: Aorta and Peripheral Vascular Diseases, Cardiovascular Pharmacotherapy, e-Cardiology, Myocardial Function, Pulmonary Circulation and Right Ventricular Function, Thrombosis.

Patient Forum

The content of these European Society of Cardiology (ESC) Guidelines has been published for personal and educational use only. No commercial use is authorized. No part of the ESC Guidelines may be translated or reproduced in any form without written permission from the ESC. Permission can be obtained upon submission of a written request to Oxford University Press, the publisher of the *European Heart Journal* and the party authorized to handle such permissions on behalf of the ESC (journals.permissions@oup.com).

Disclaimer: The ESC Guidelines represent the views of the ESC and were produced after careful consideration of the scientific and medical knowledge and the evidence available at the time of their publication. The ESC is not responsible in the event of any contradiction, discrepancy and/or ambiguity between the ESC Guidelines and any other official recommendations or guidelines issued by the relevant public health authorities, in particular in relation to good use of healthcare or therapeutic strategies. Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgement, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies; however, the ESC Guidelines do not override, in any way whatsoever, the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient and, where appropriate and/or necessary, the patient's caregiver. Nor do the ESC Guidelines exempt health professionals from taking into full and careful consideration the relevant official updated recommendations or guidelines issued by the competent public health authorities, in order to manage each patient's case in light of the scientifically accepted data pursuant to their respective ethical and professional obligations. It is also the health professional's responsibility to verify the applicable rules and regulations relating to drugs and medical devices at the time of prescription.

This article has been co-published with permission in the *European Heart Journal* and the *European Heart Journal* *Cardiovascular Imaging*. © The European Society of Cardiology 2022. All rights reserved. The articles are identical except for minor stylistic and spelling differences in keeping with each journal's style. Either citation can be used when citing this article. For permissions please e-mail: journals.permissions@oup.com.

All experts involved in the development of these Guidelines have submitted declarations of interest. These have been compiled in a report and published in a supplementary document simultaneously to the Guidelines. The report is also available on the ESC website www.escardio.org/Guidelines

© The European Society of Cardiology 2022

This article is published and distributed under the terms of the Oxford University Press, Standard Journals Publication Model

(https://academic.oup.com/journals/pages/open_access/funder_policies/chorus/standard_publication_model)